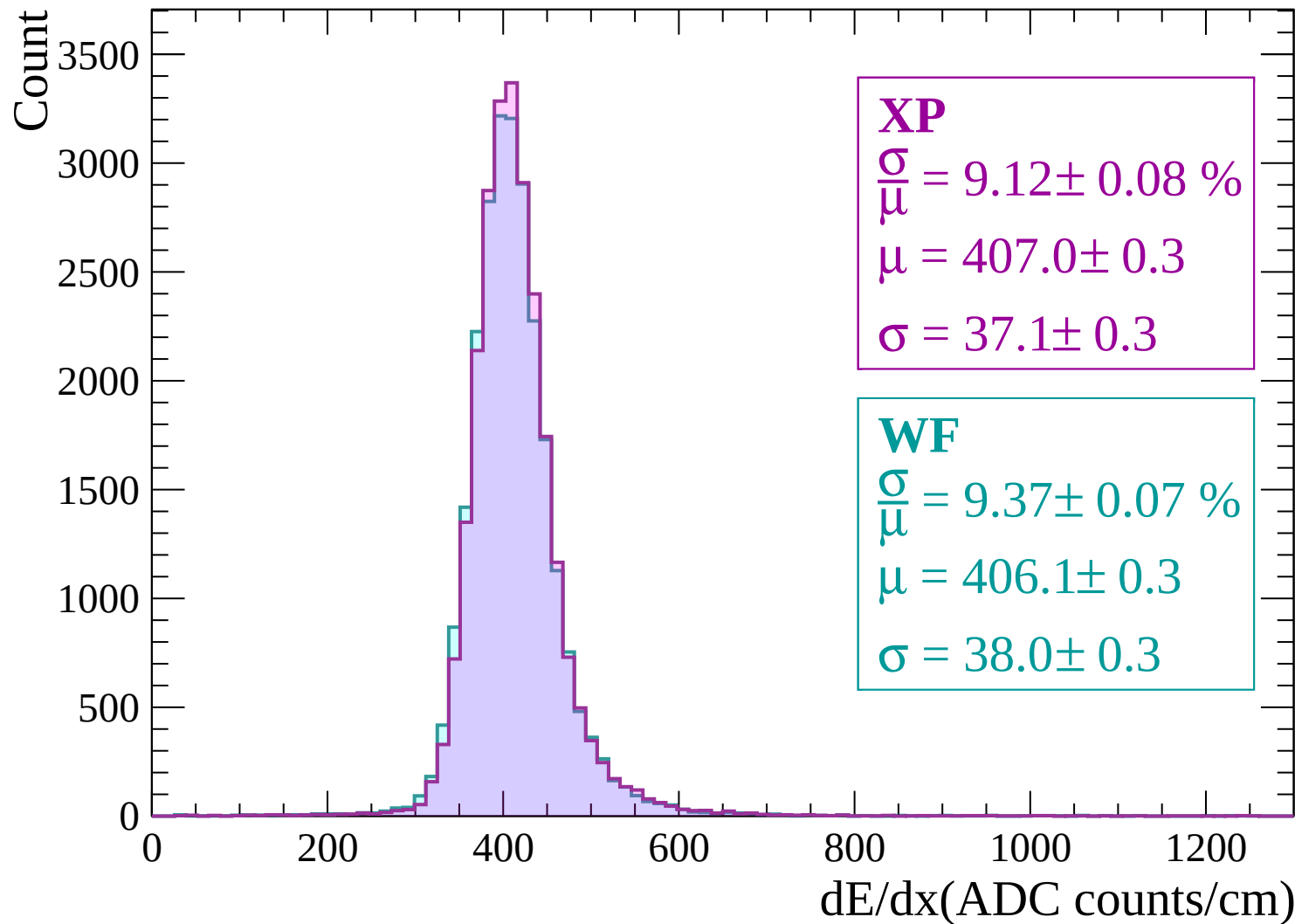
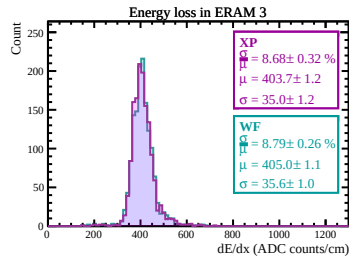
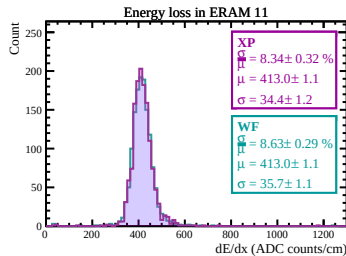
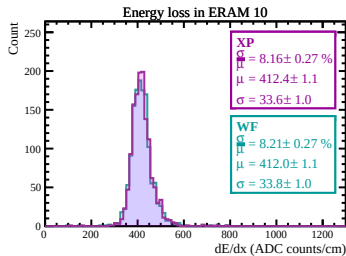
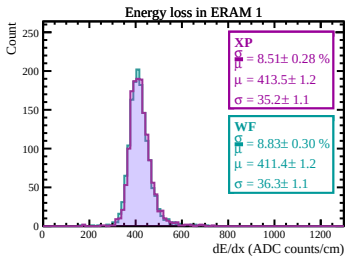
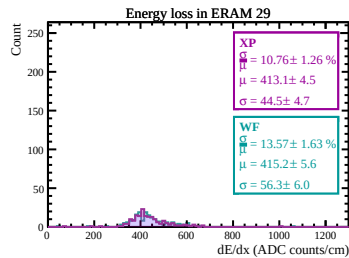
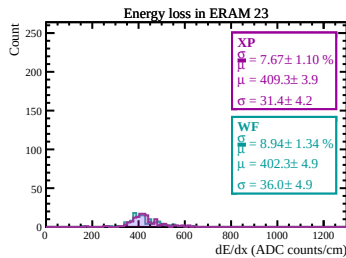
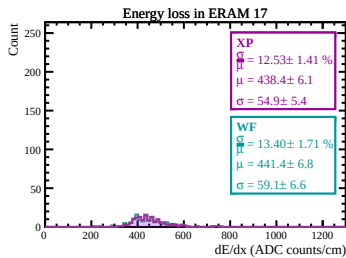
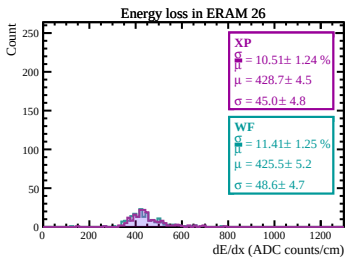
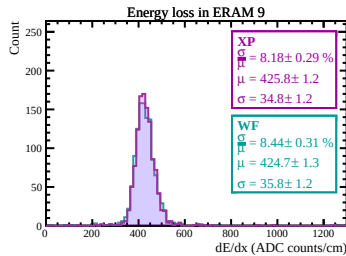
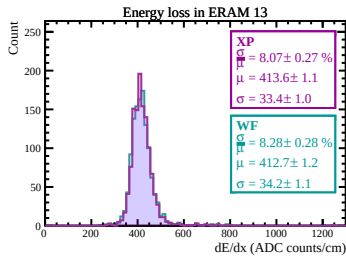
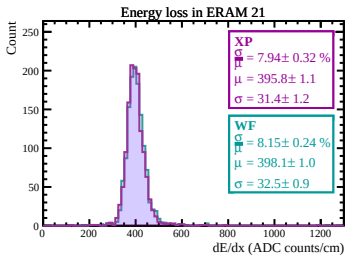
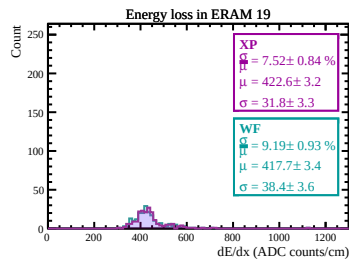
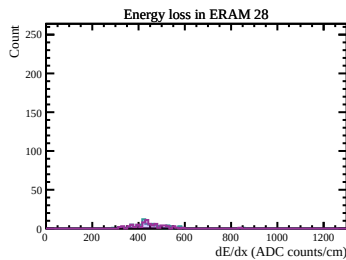
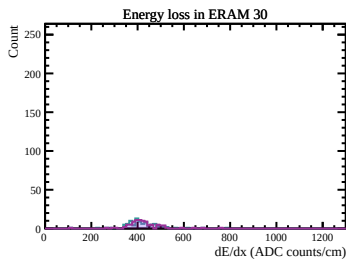
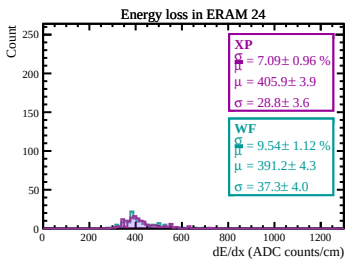
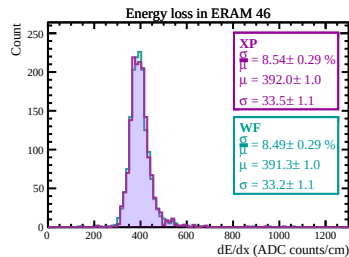
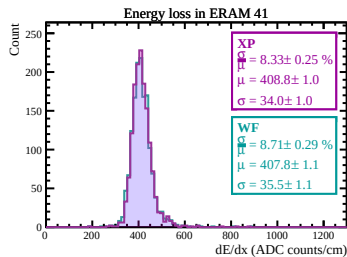
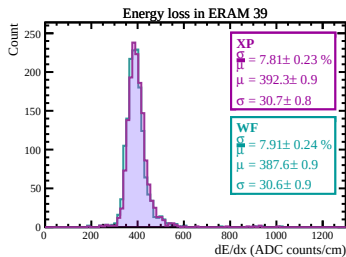
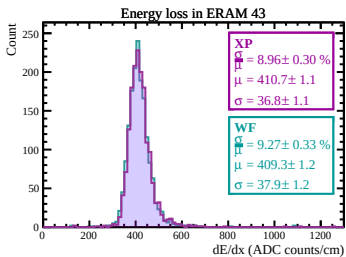
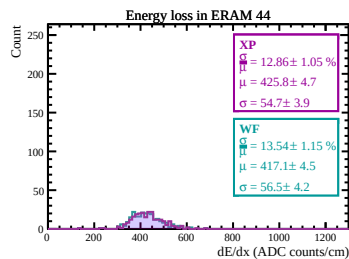
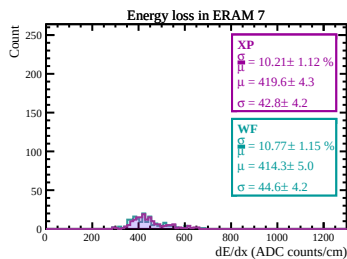
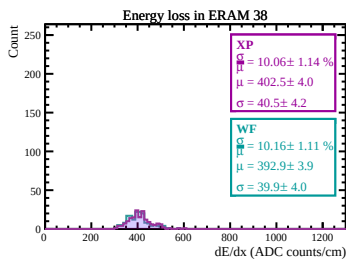
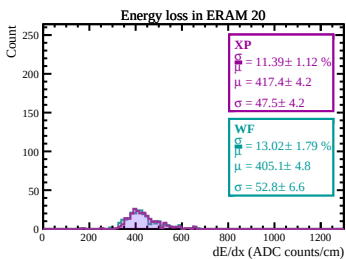
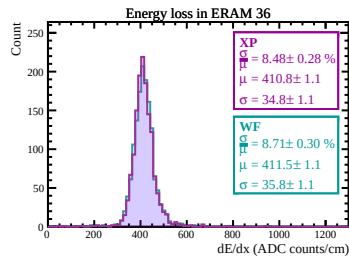
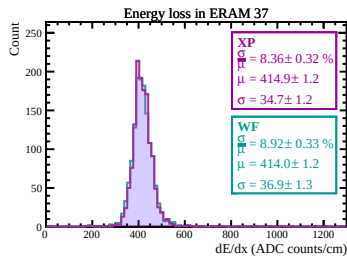
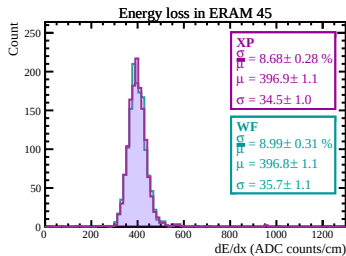
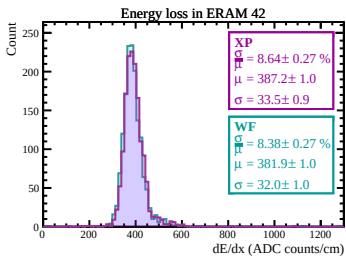
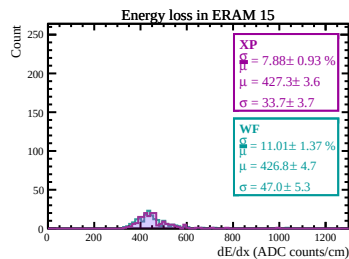
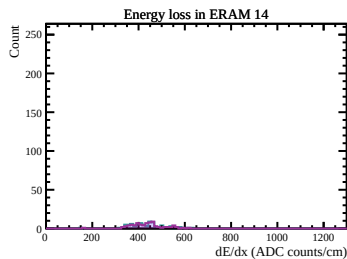
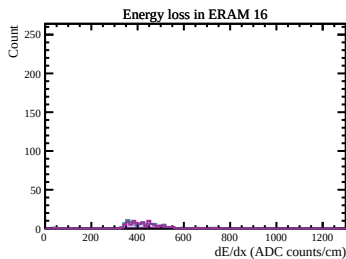
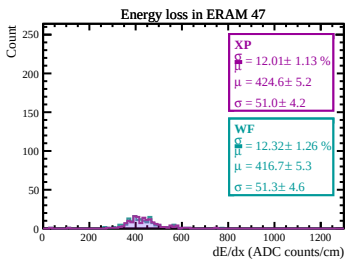
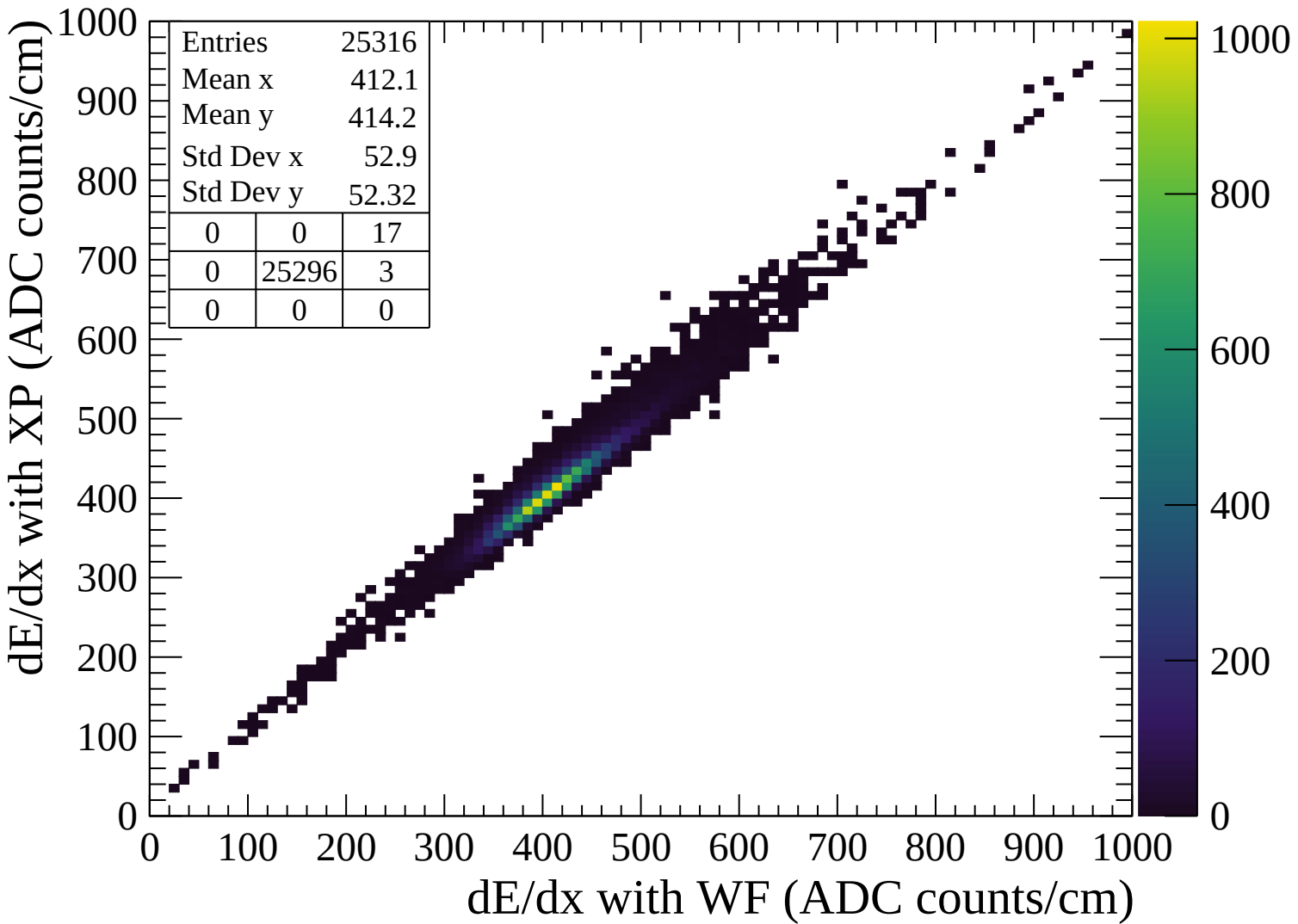
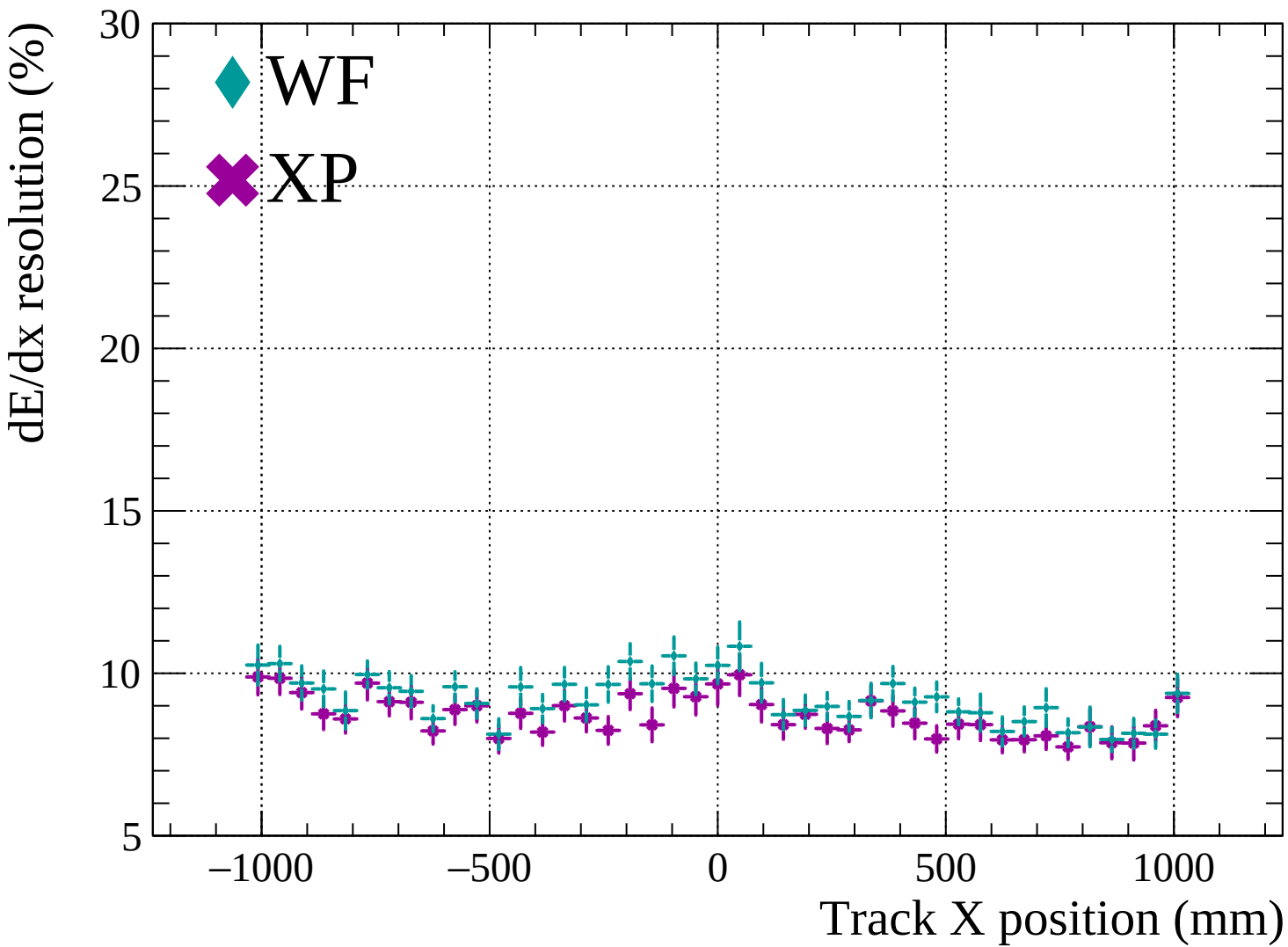


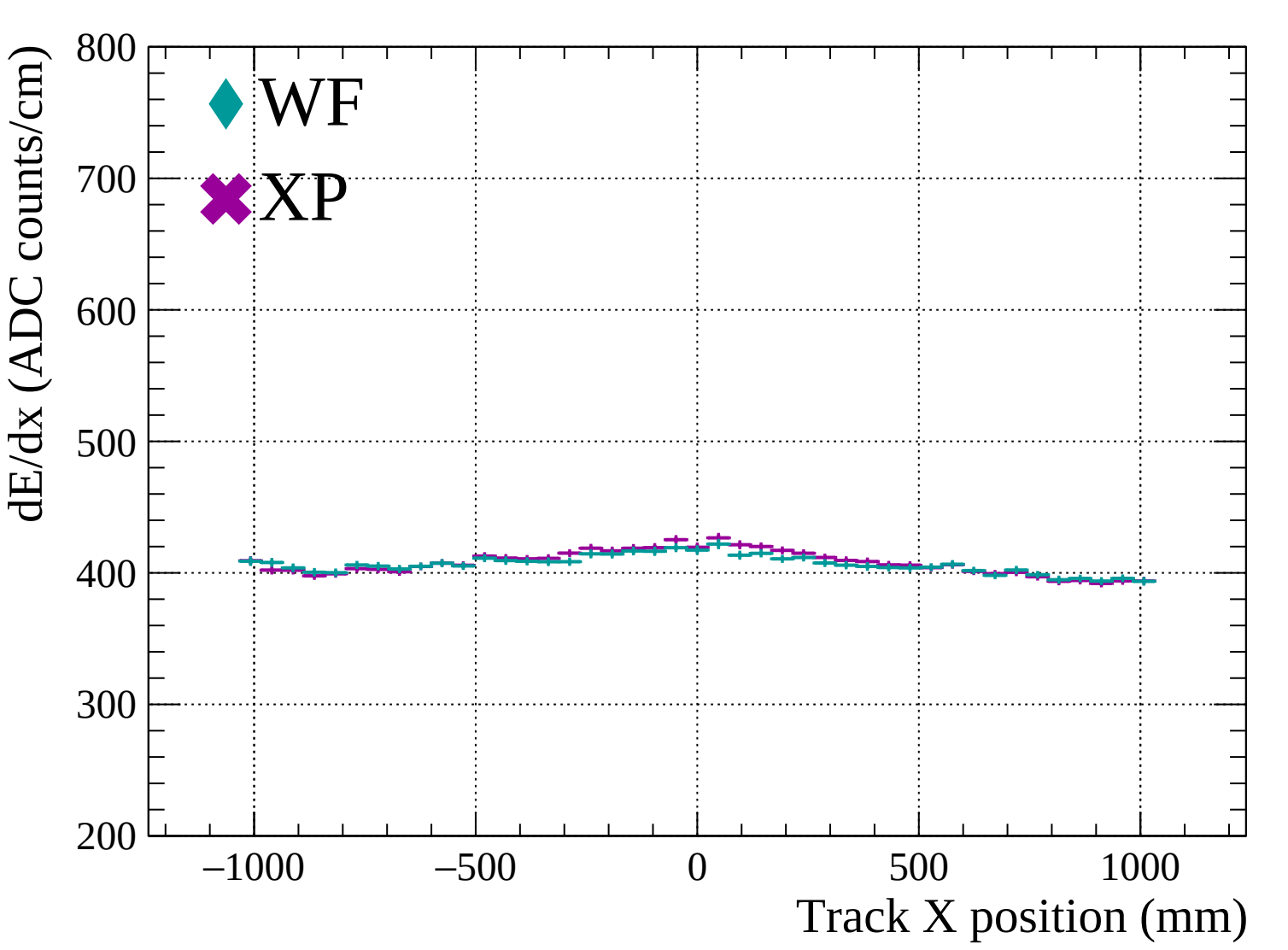


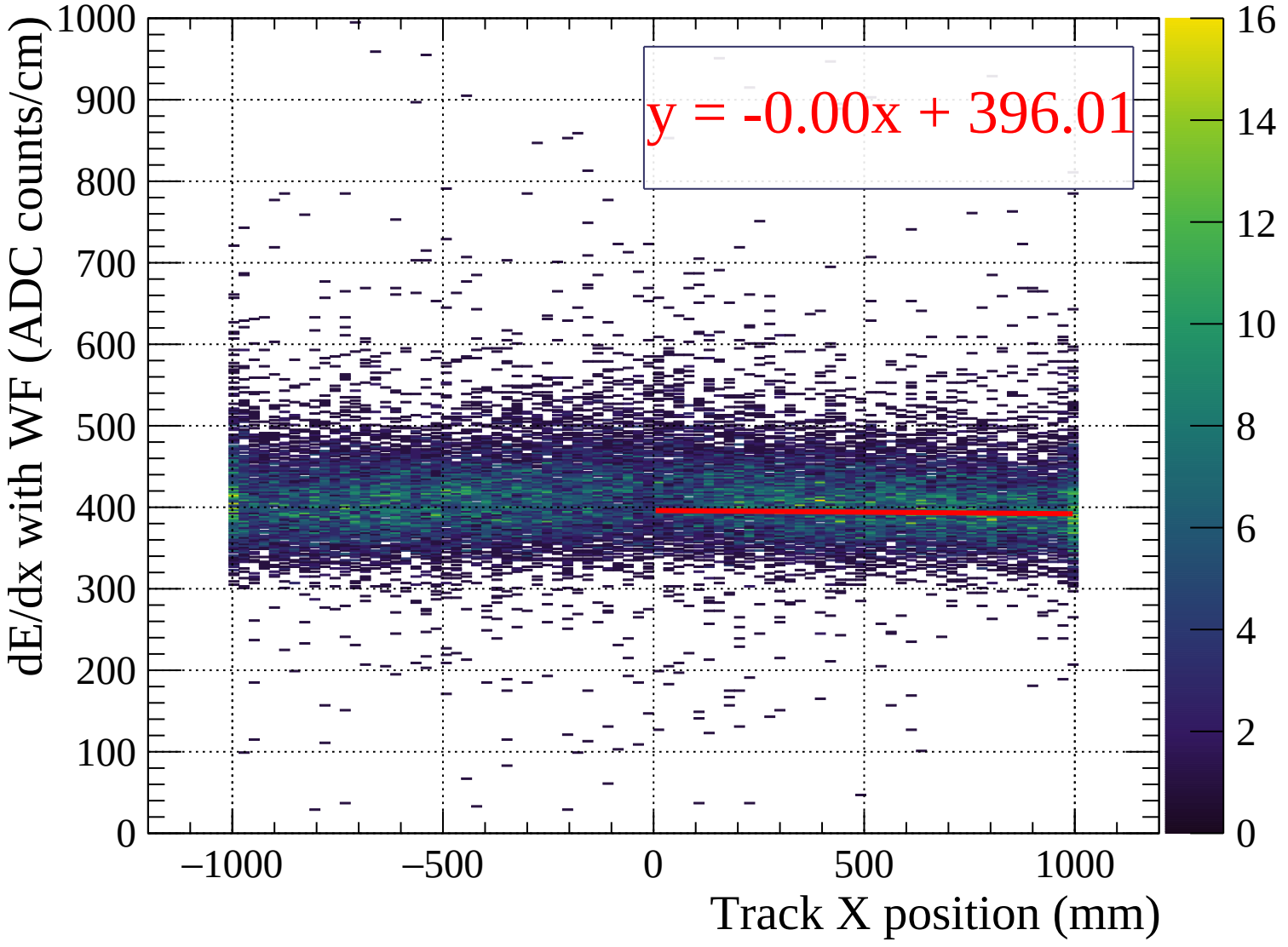


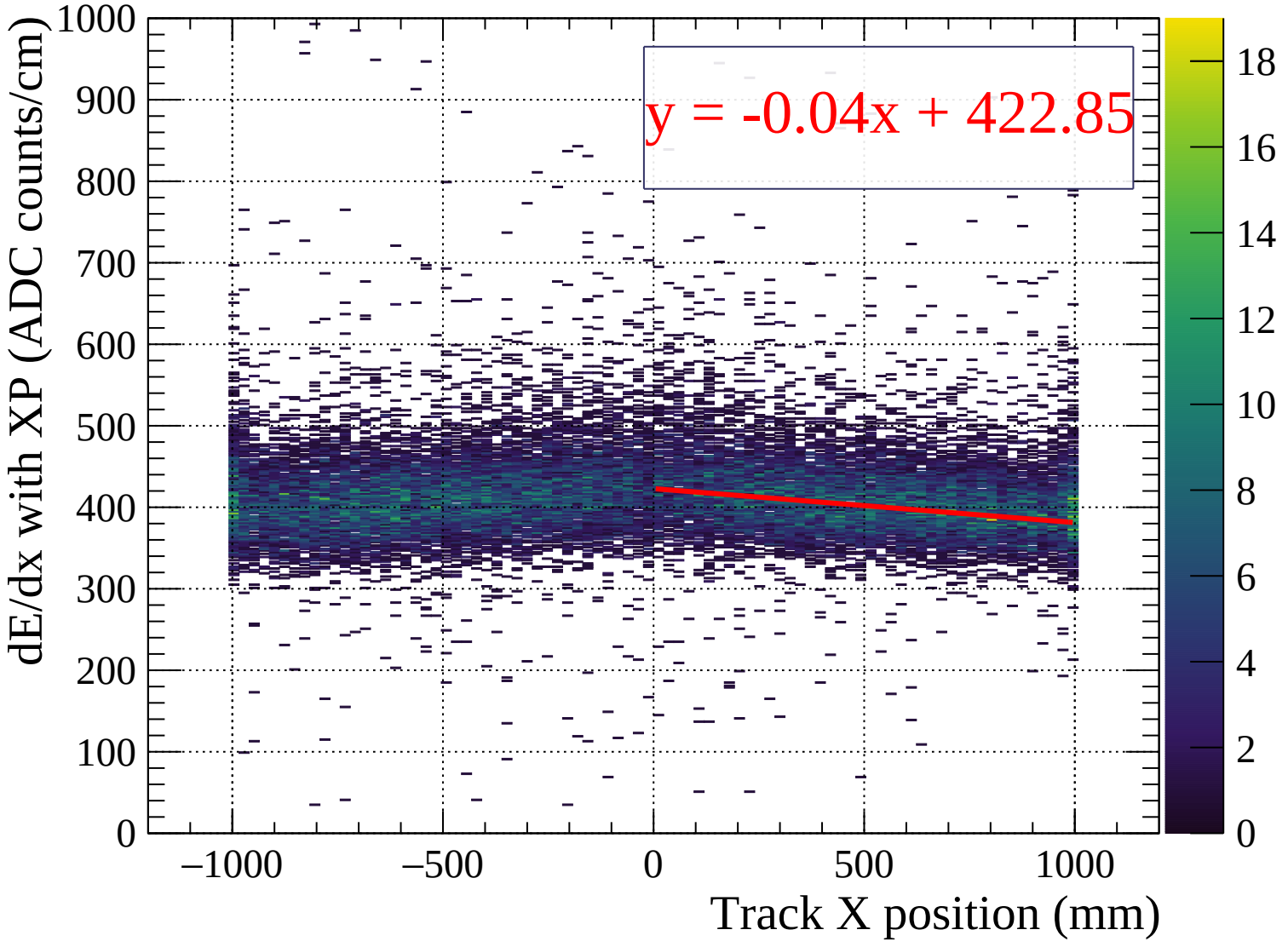


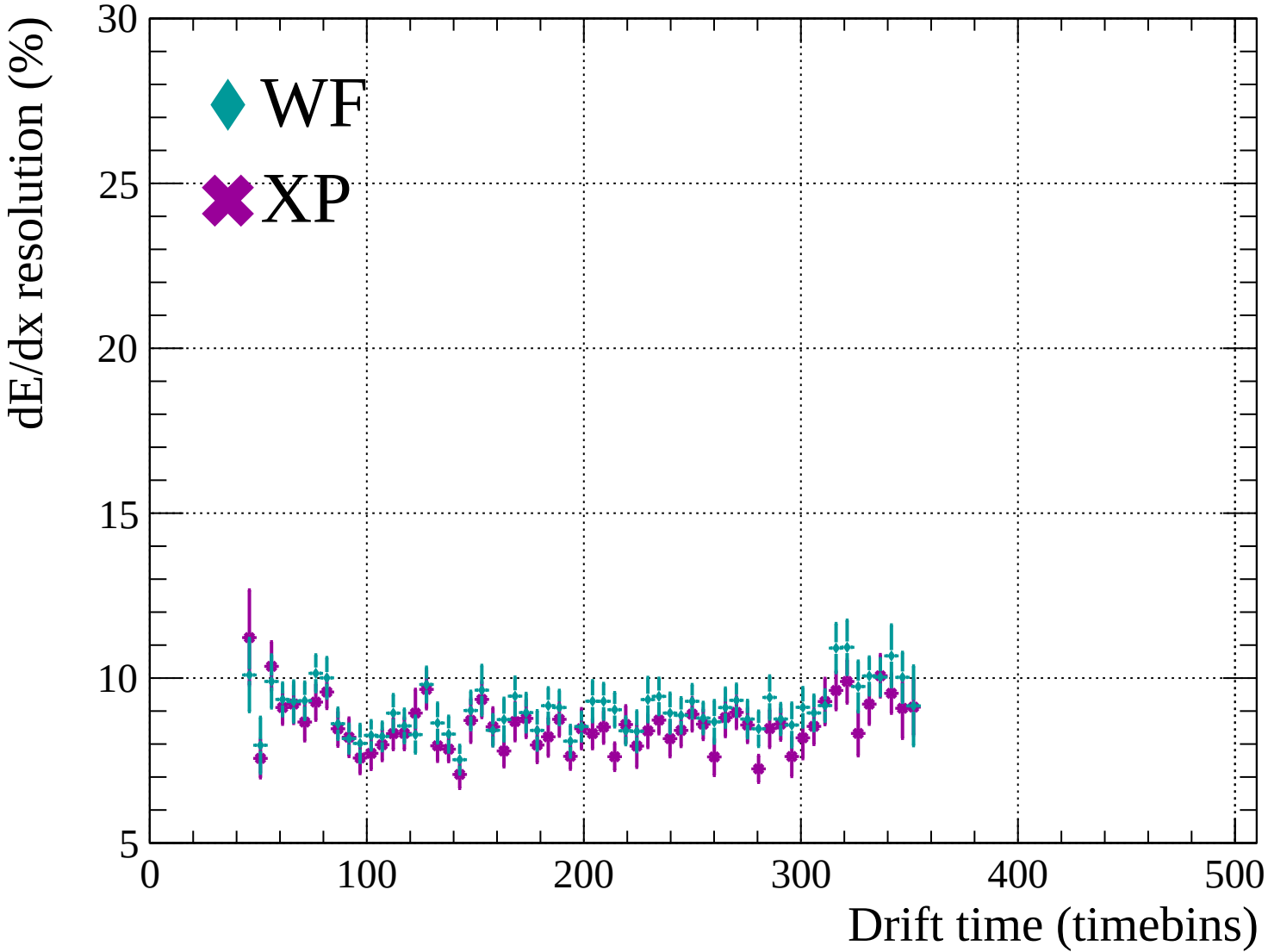


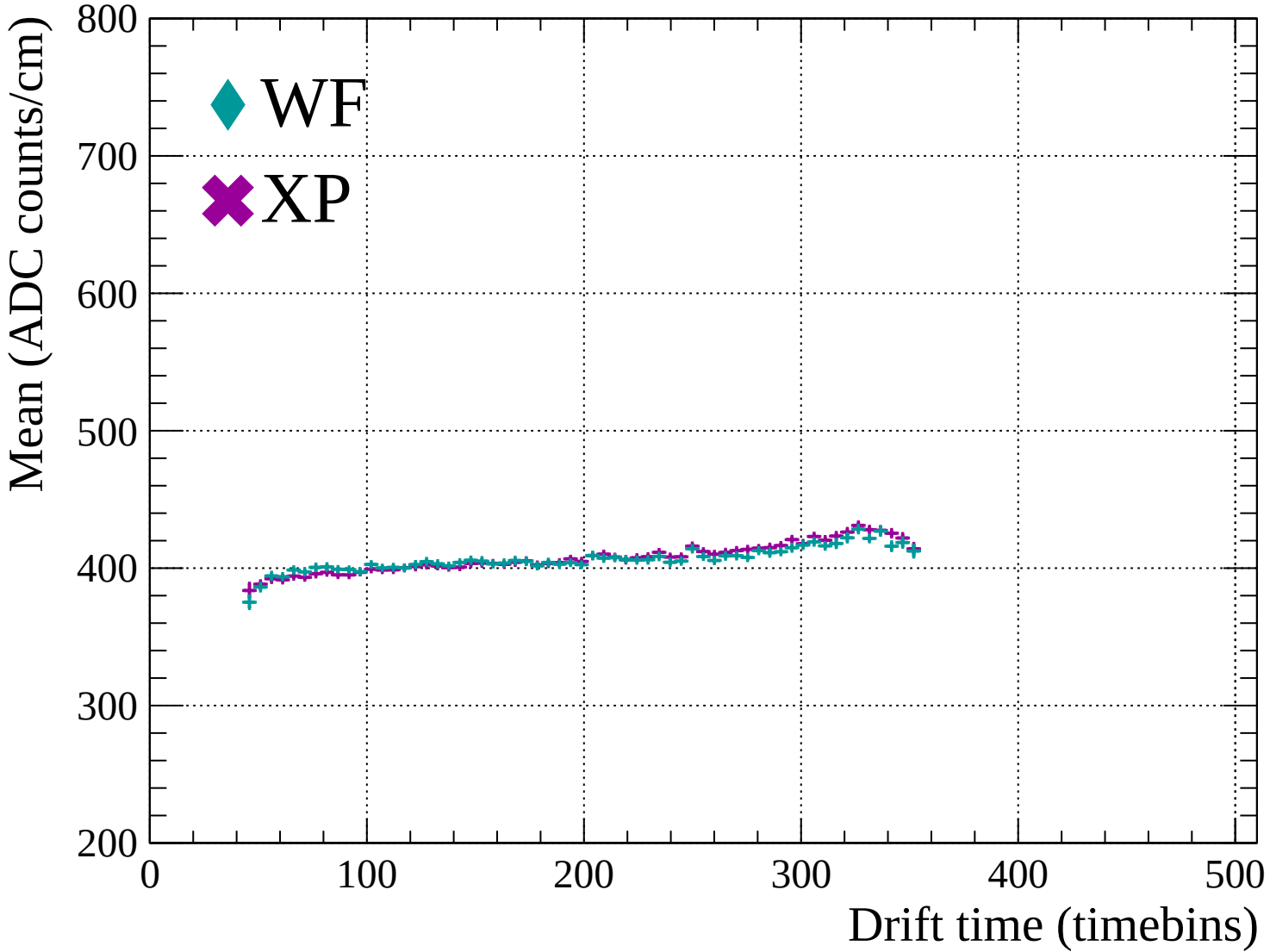


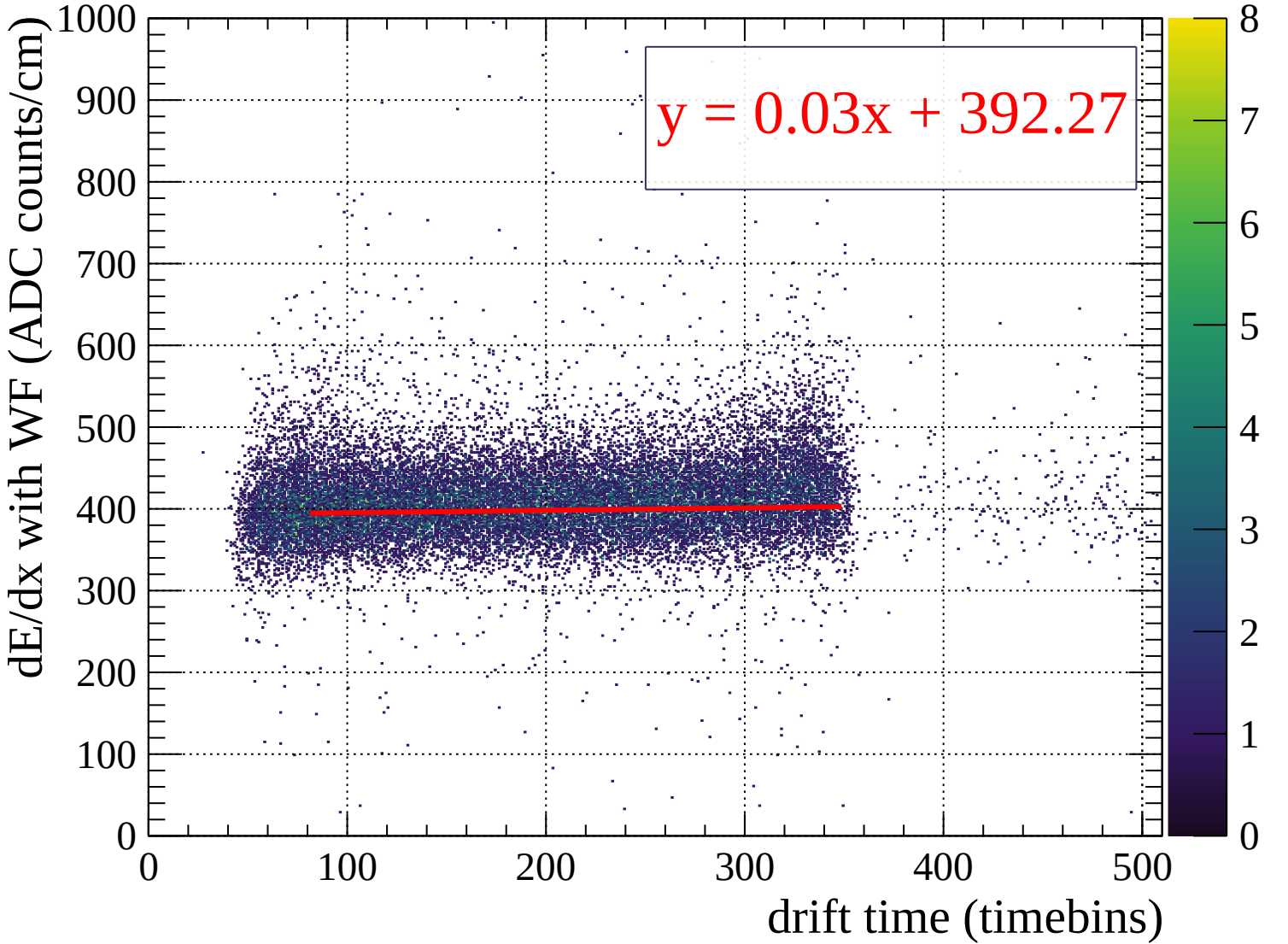


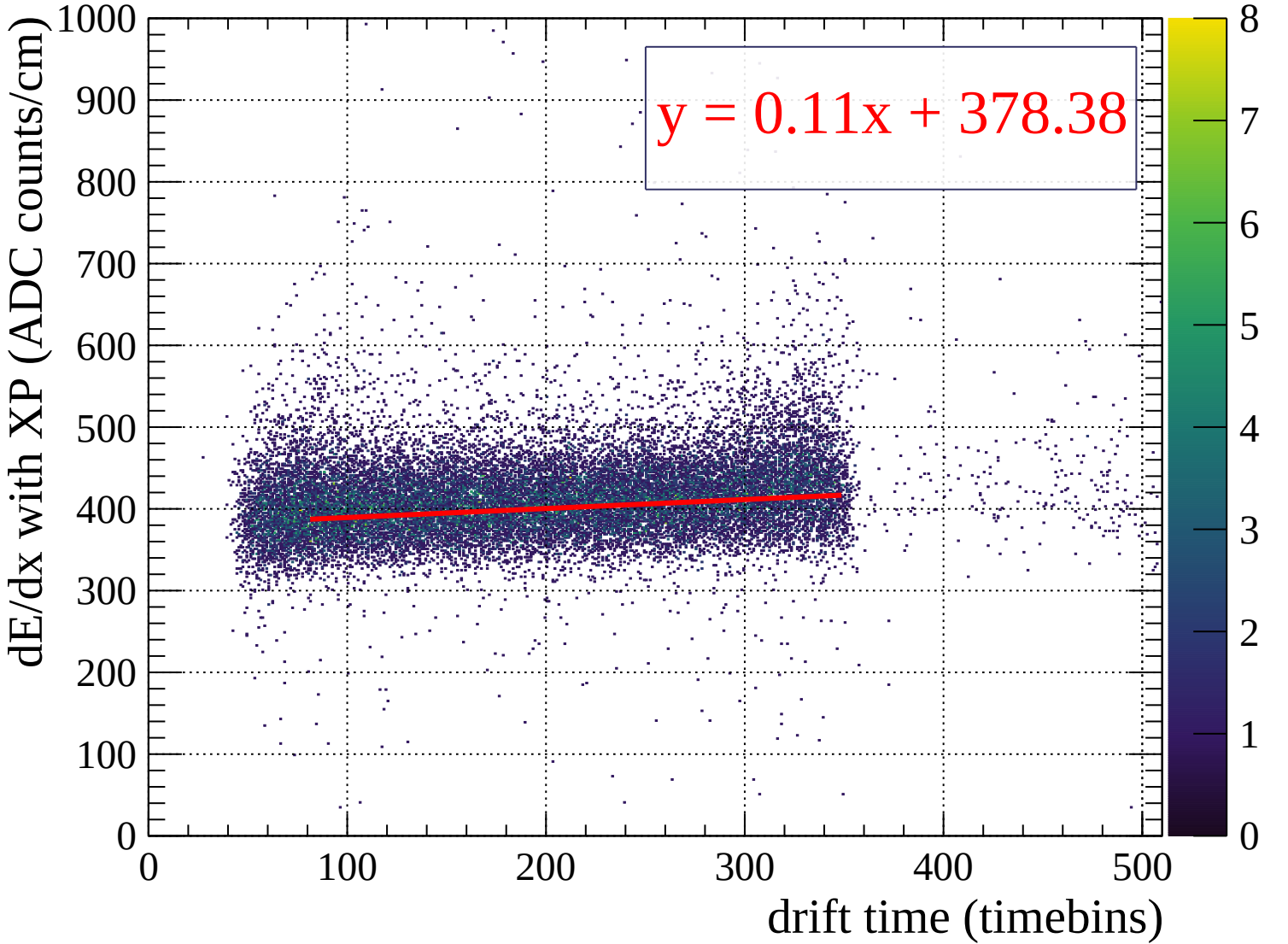


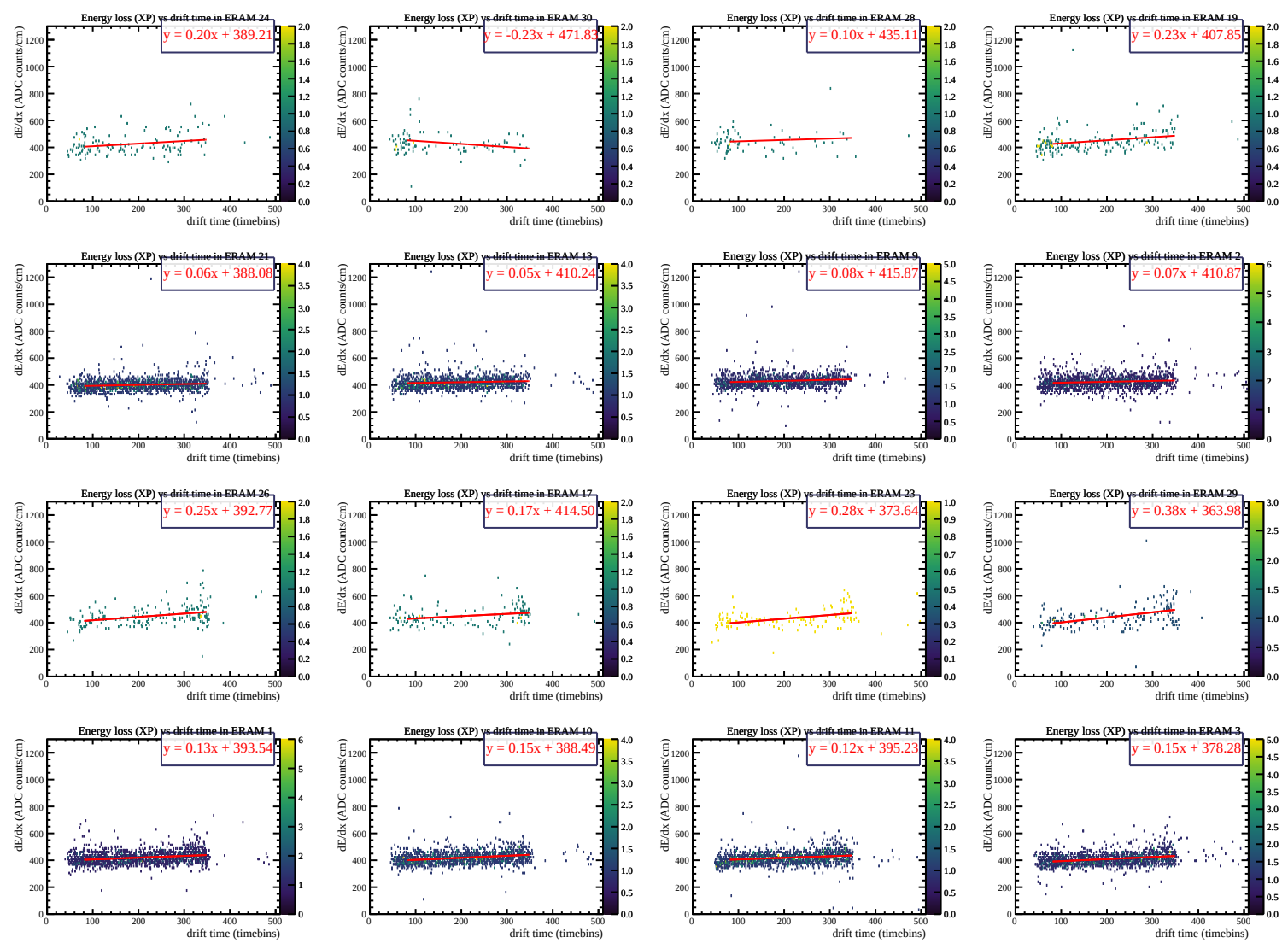


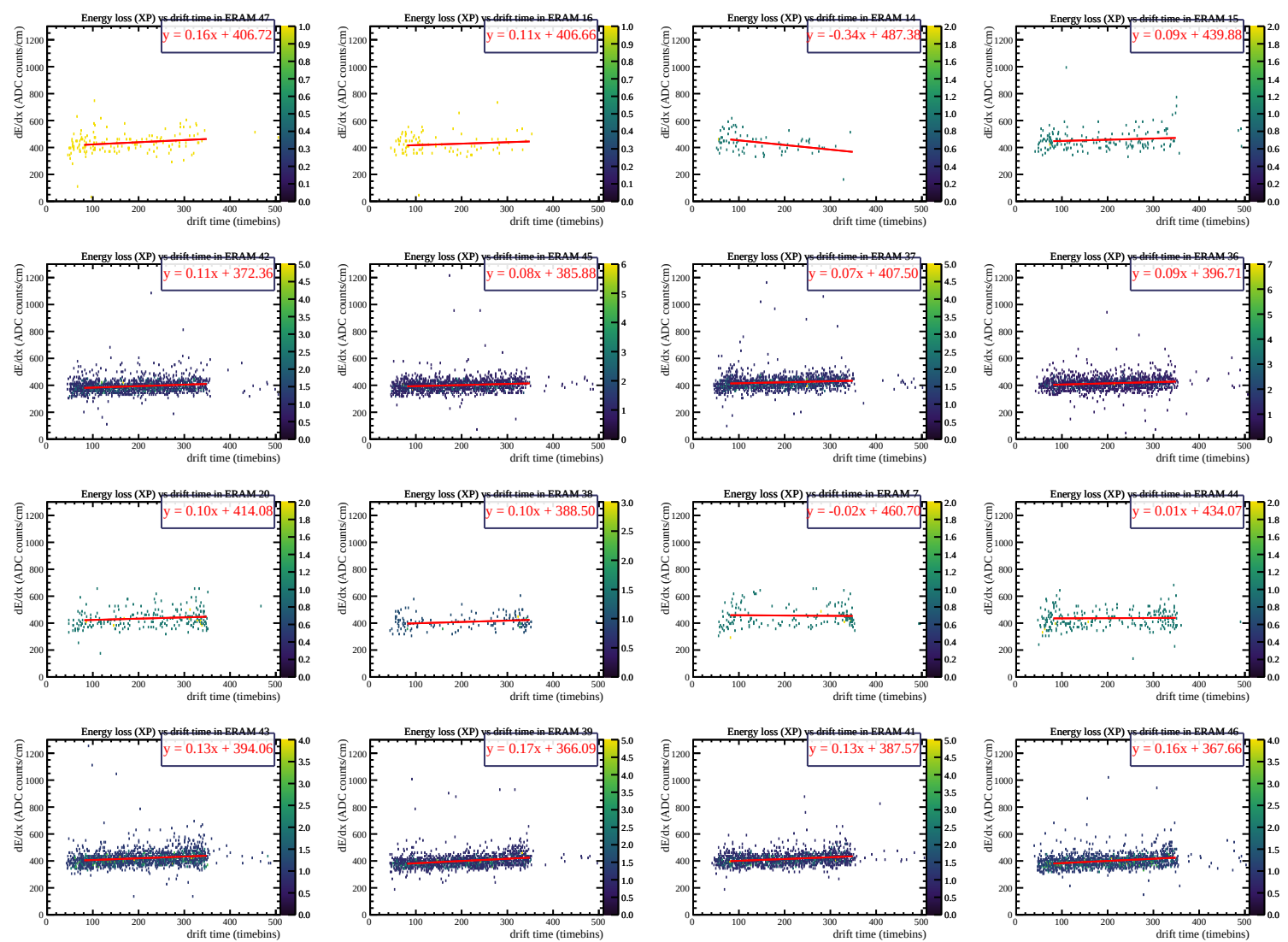


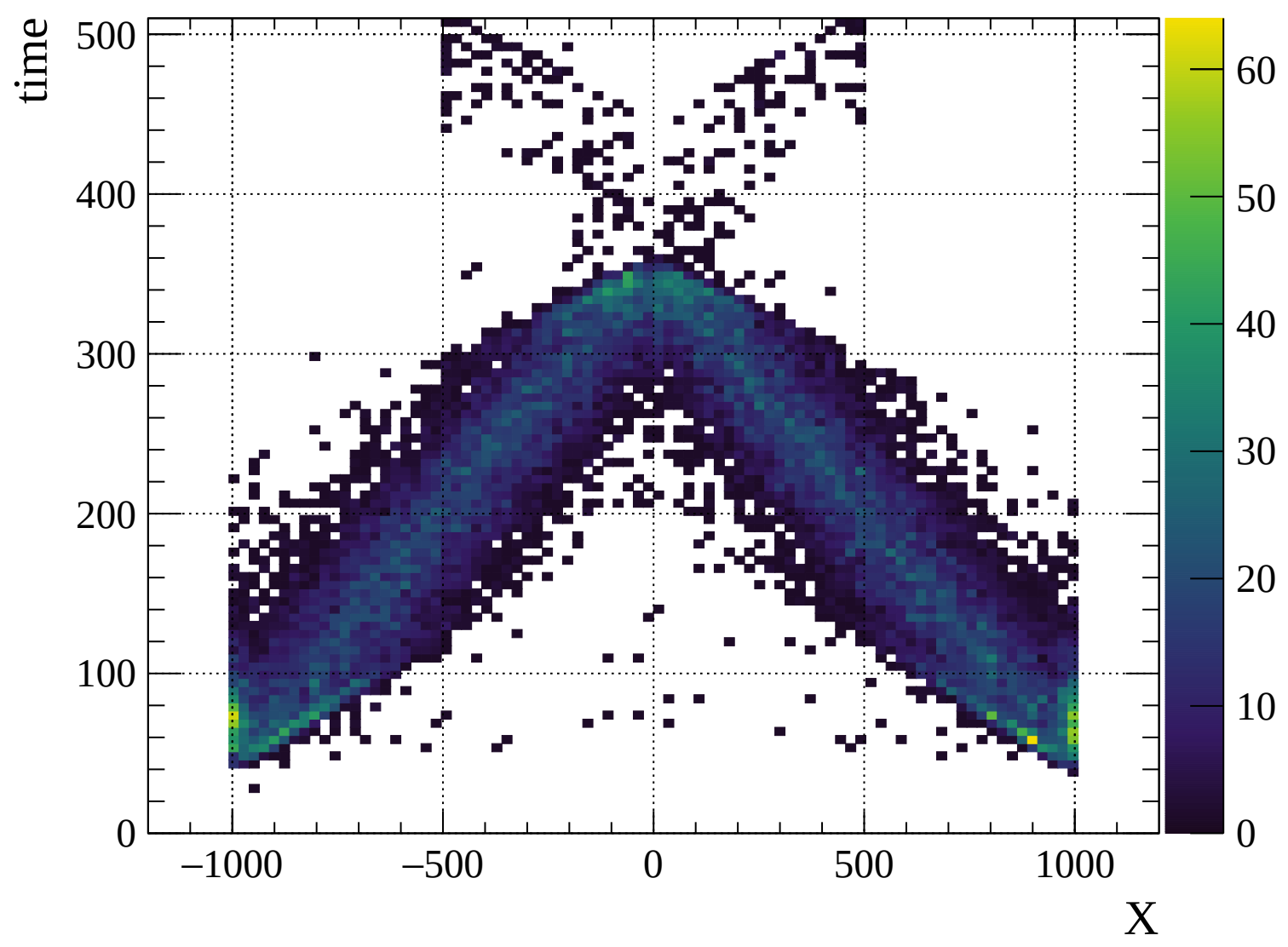


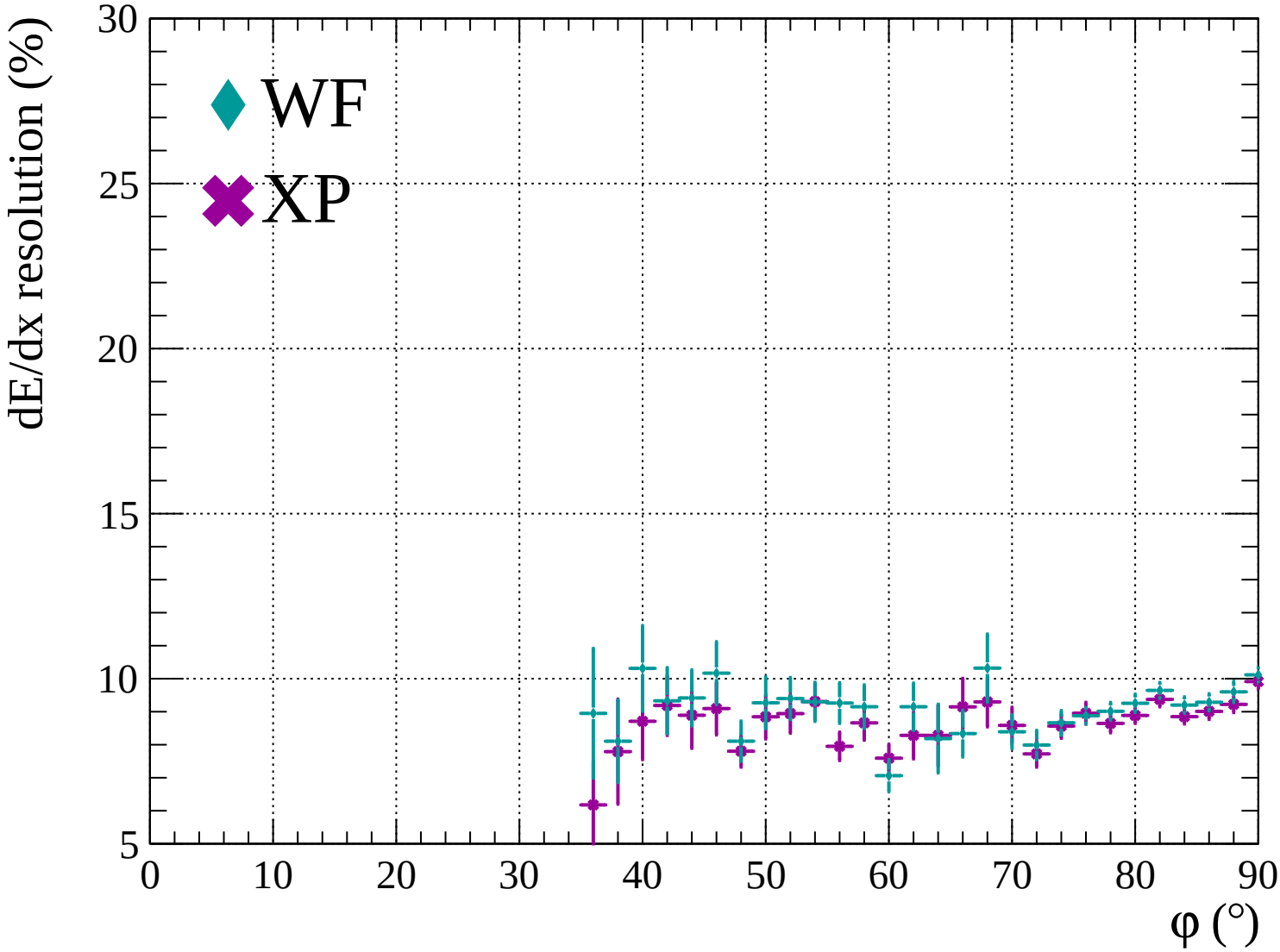


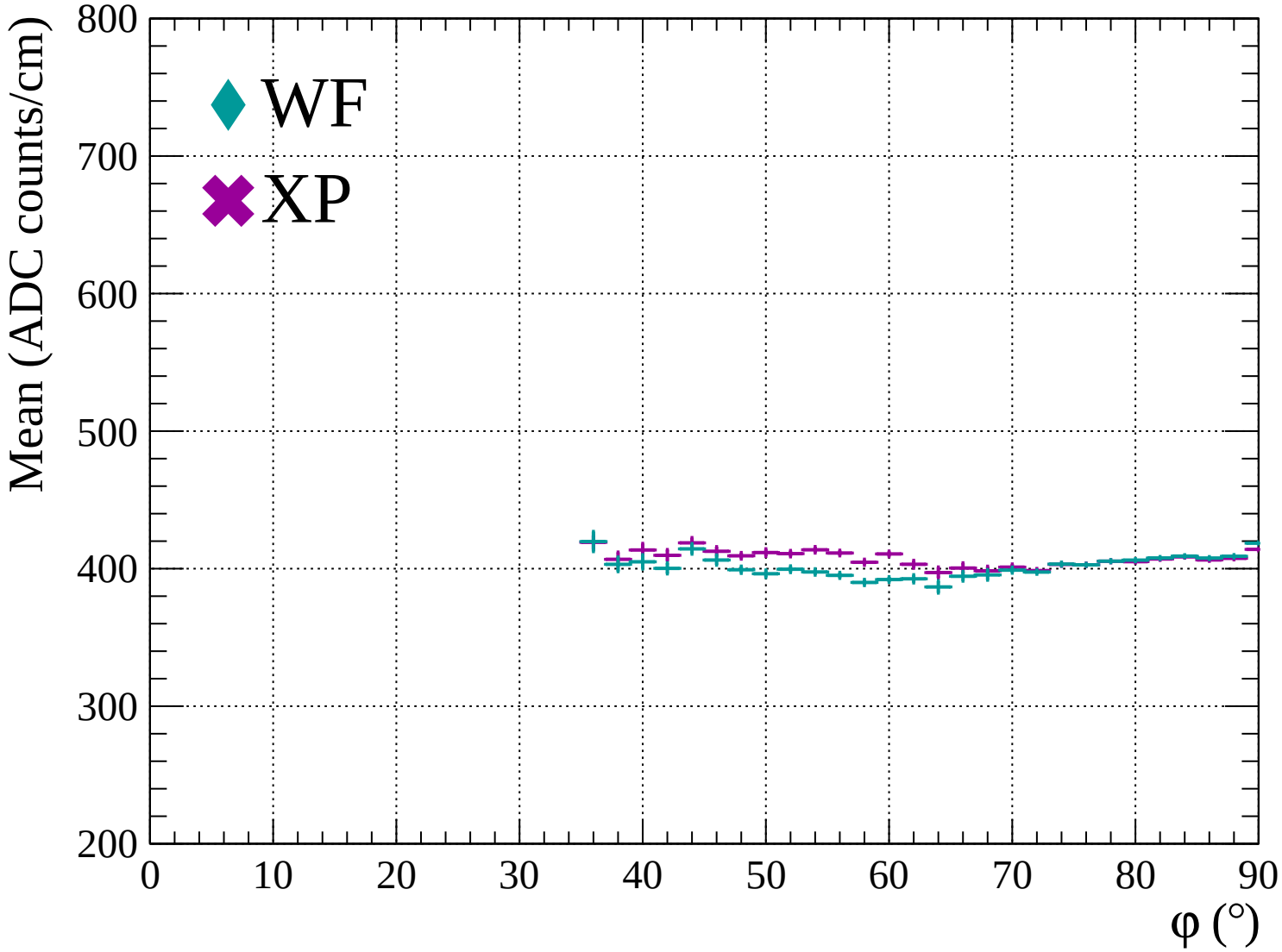


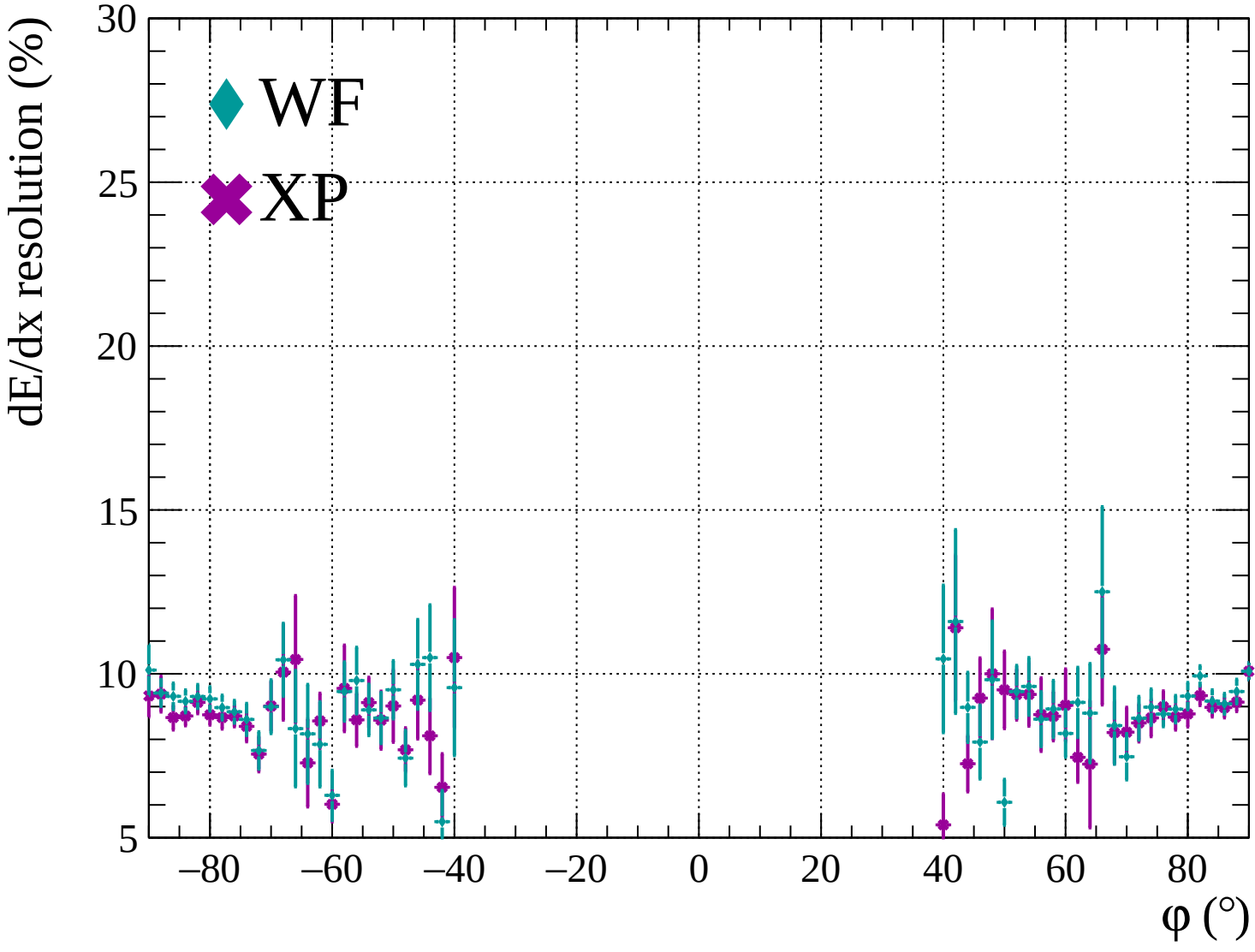


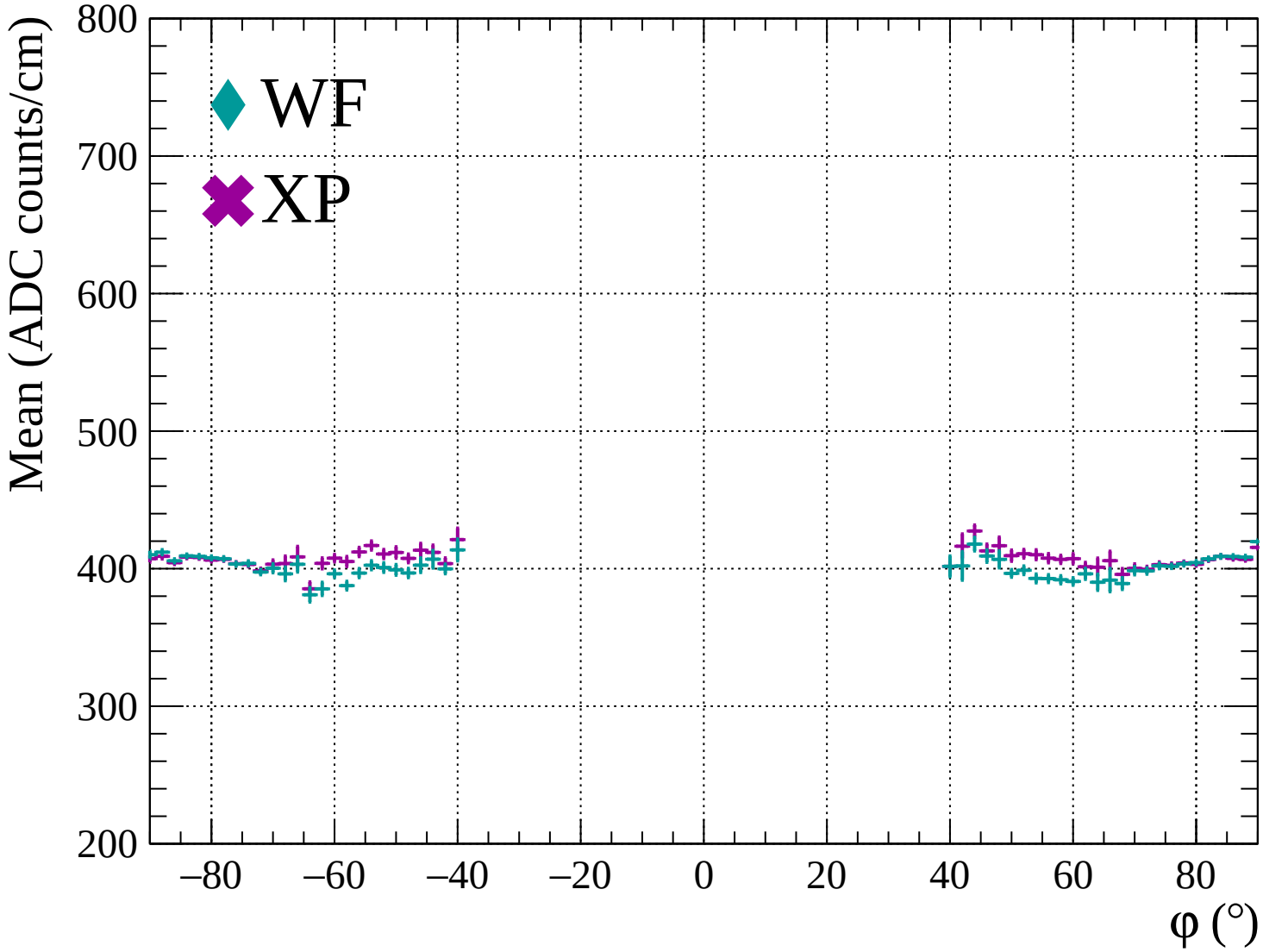


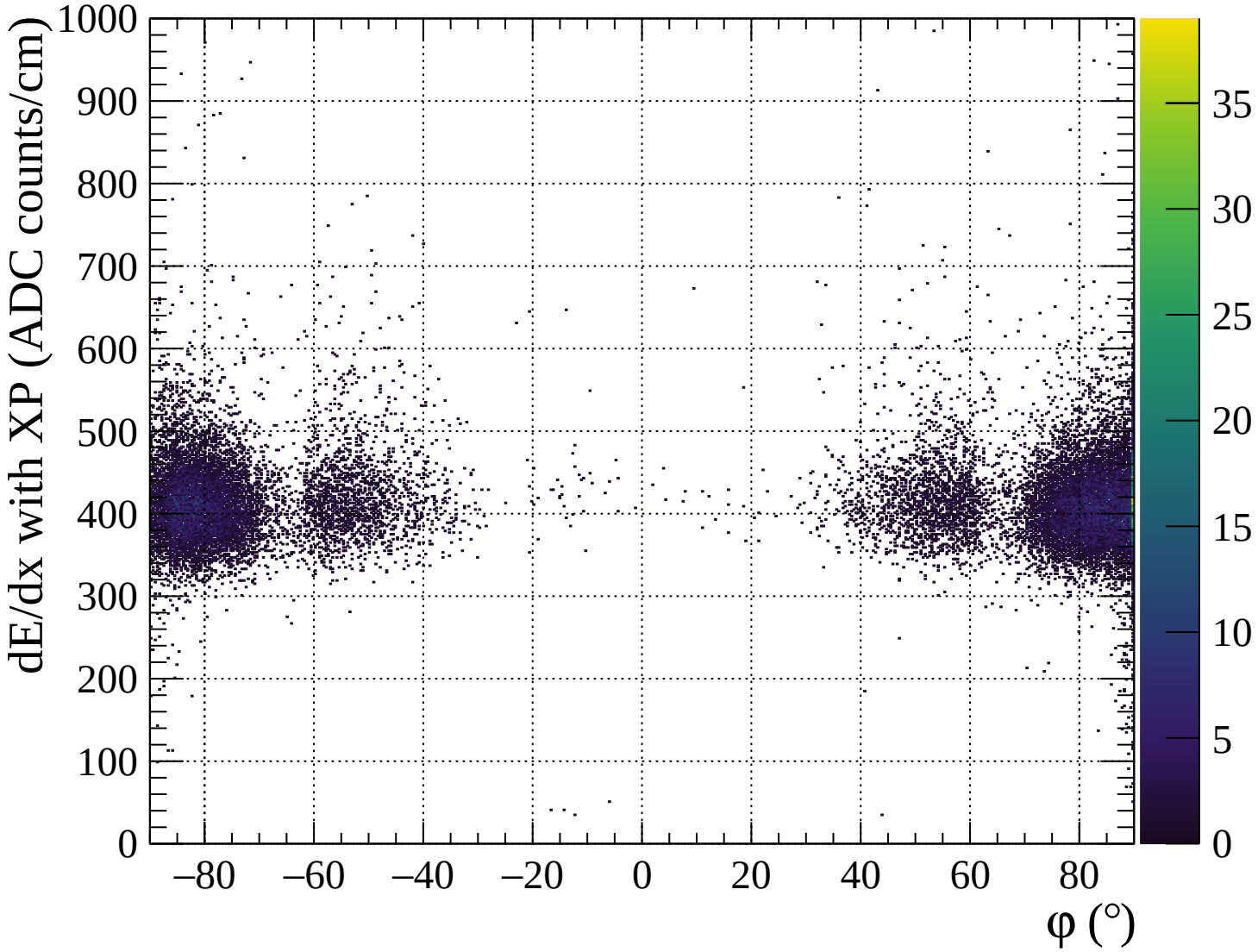


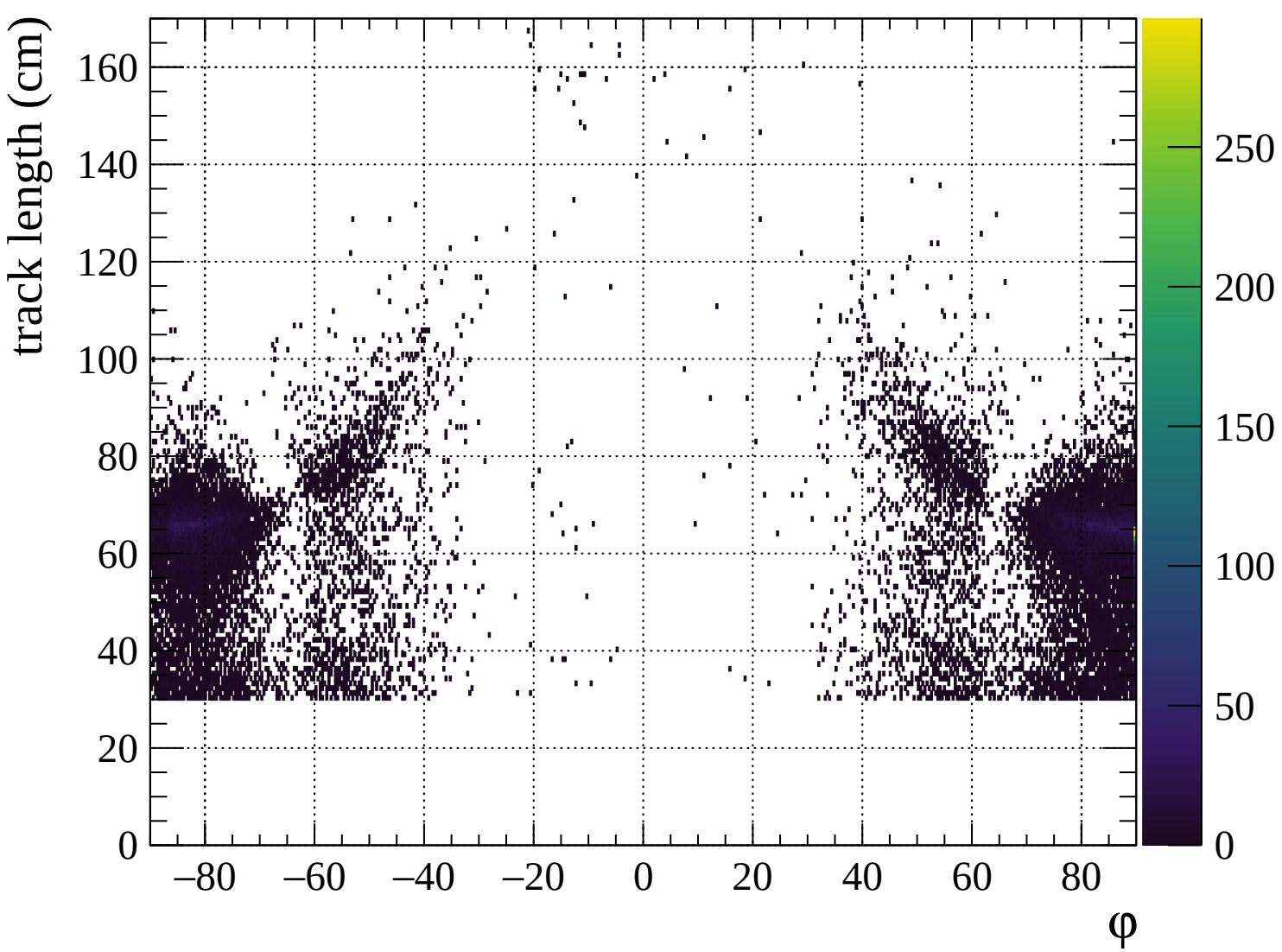


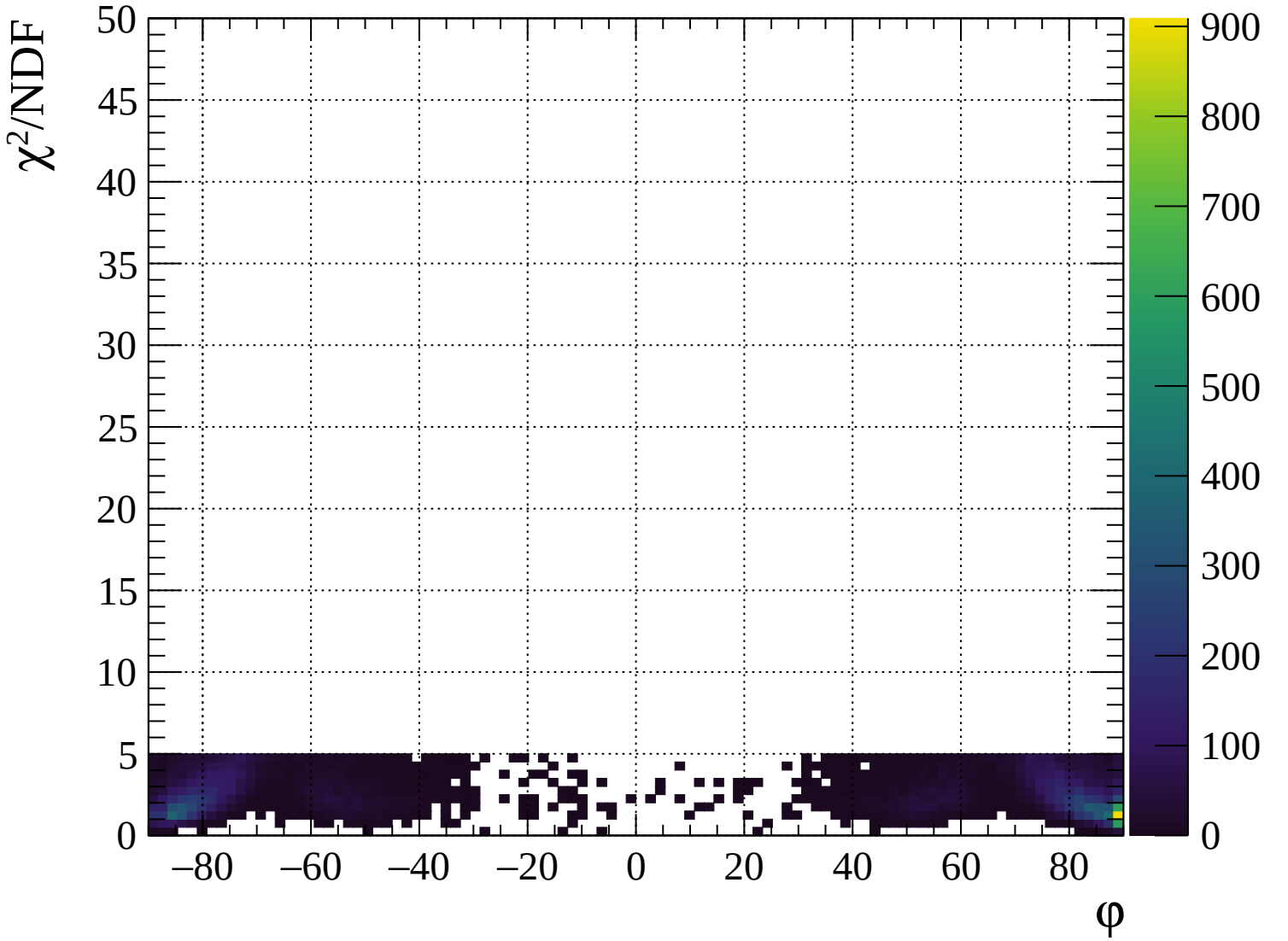


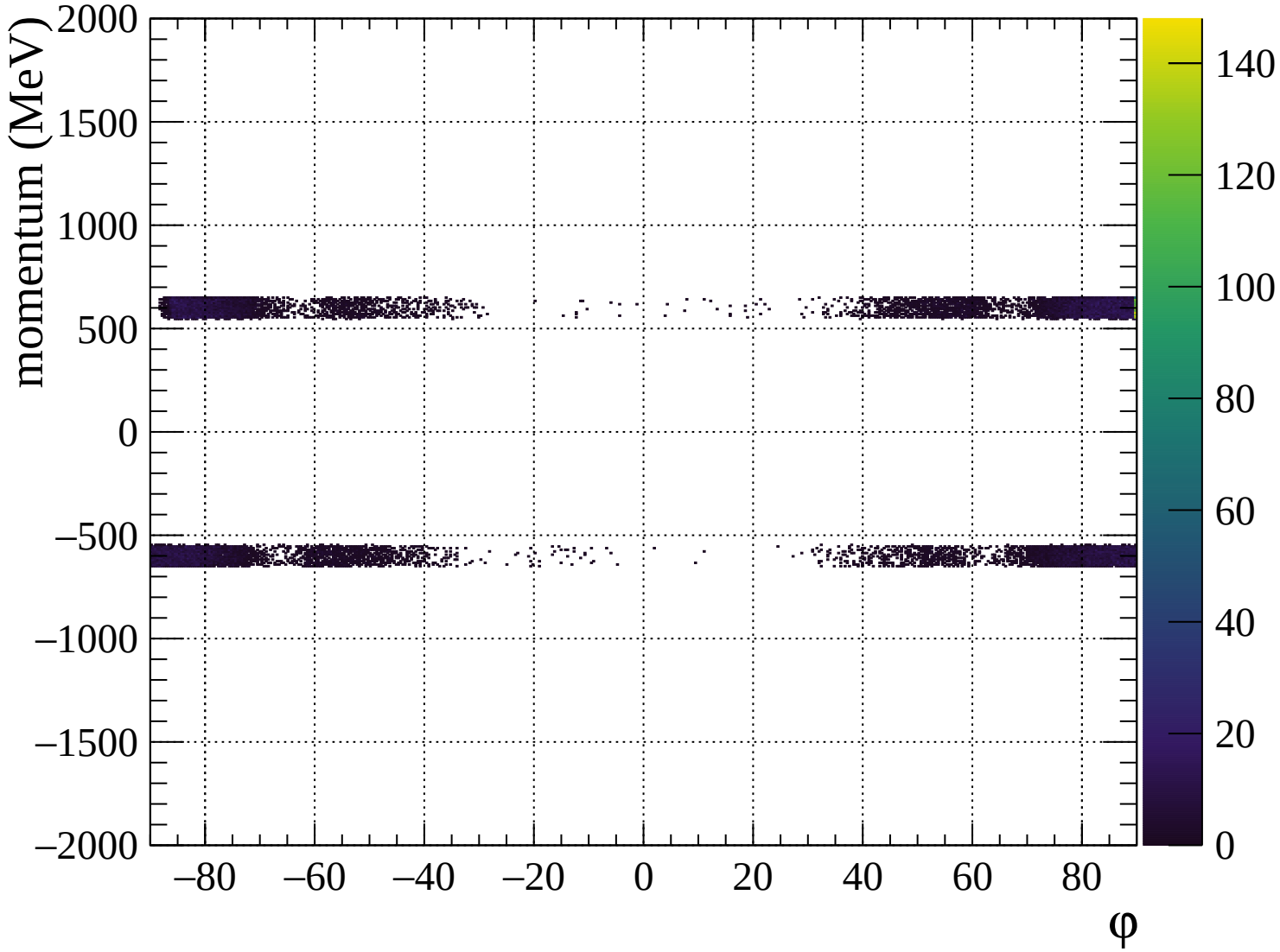


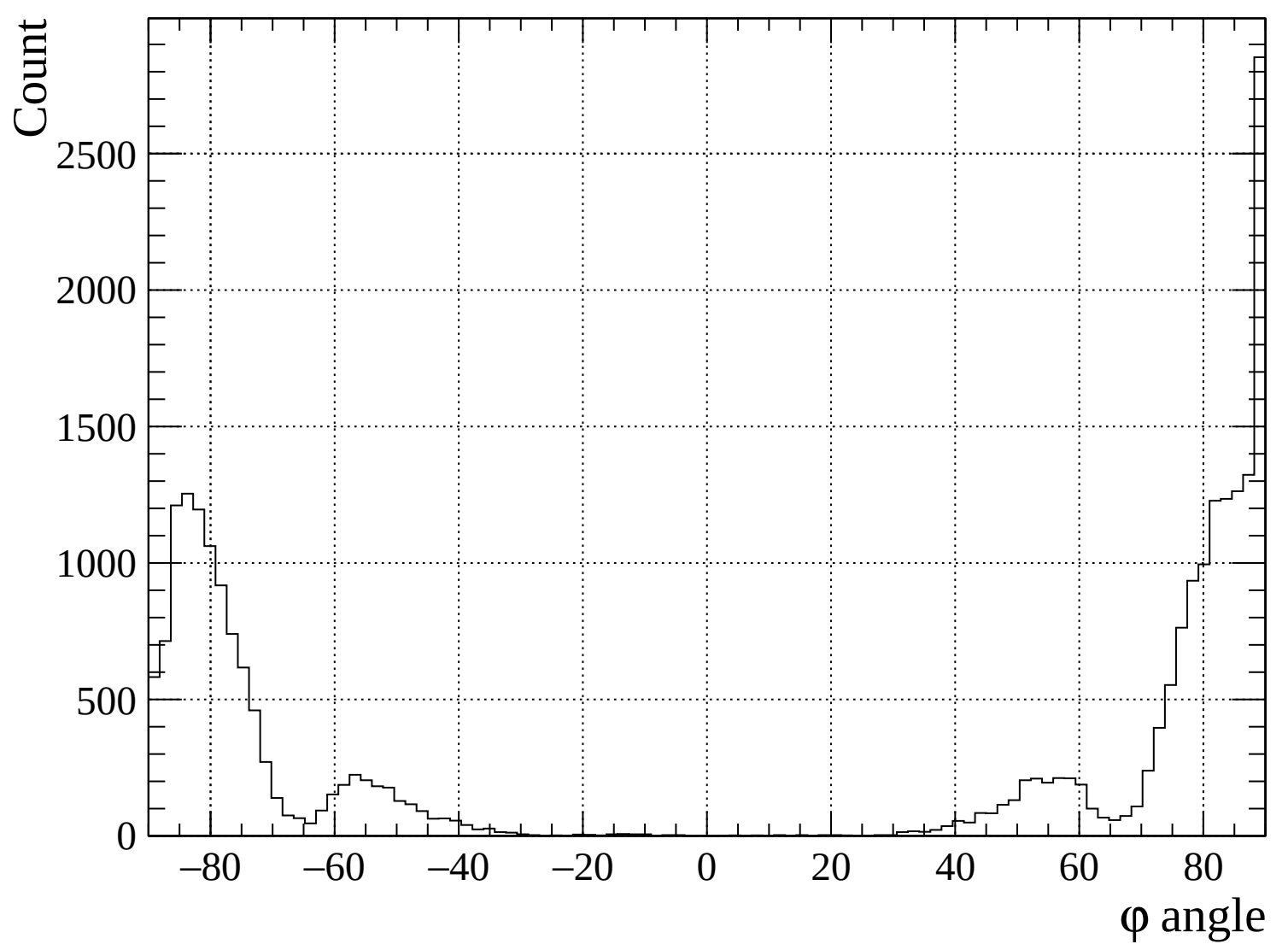


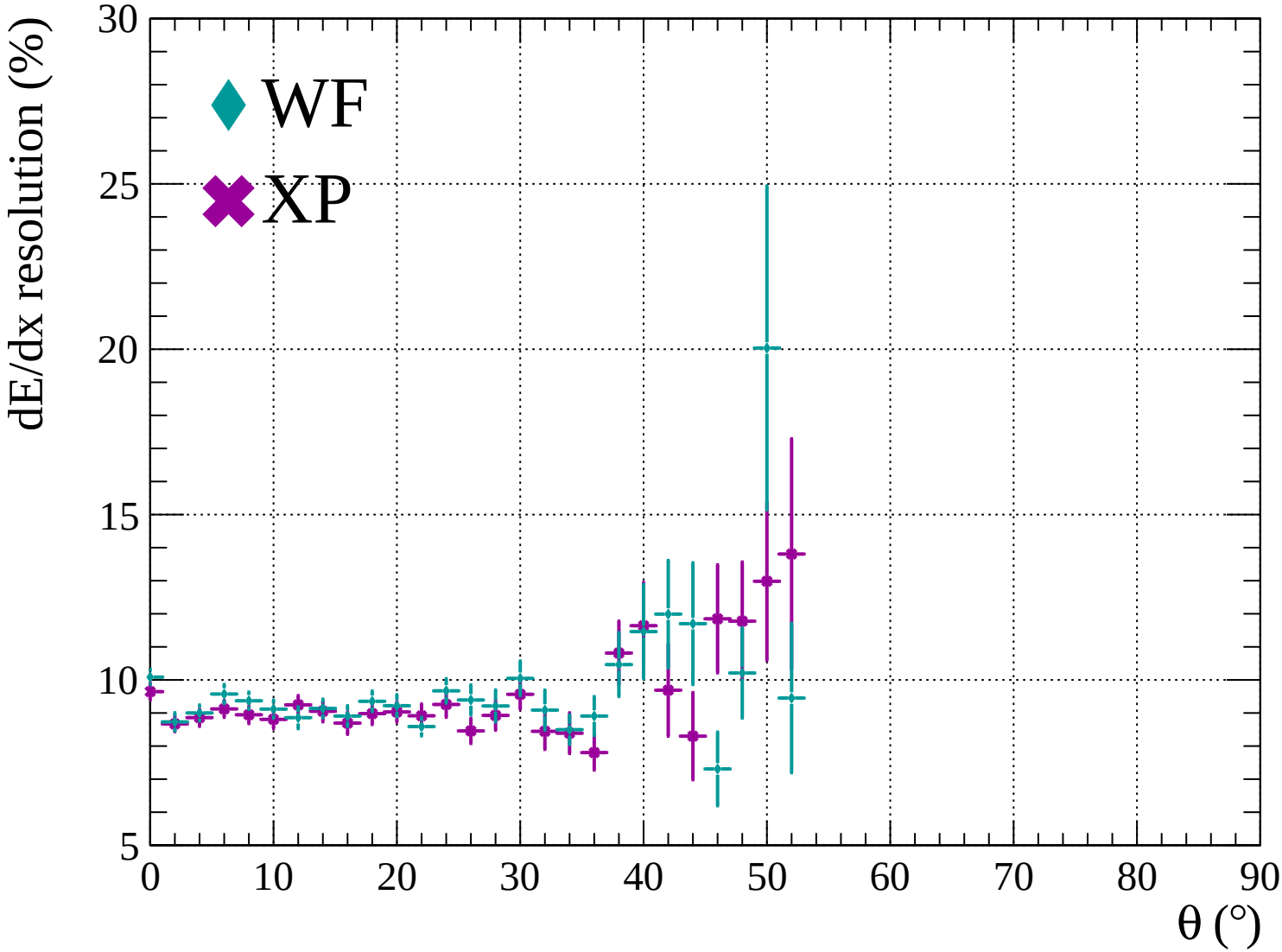


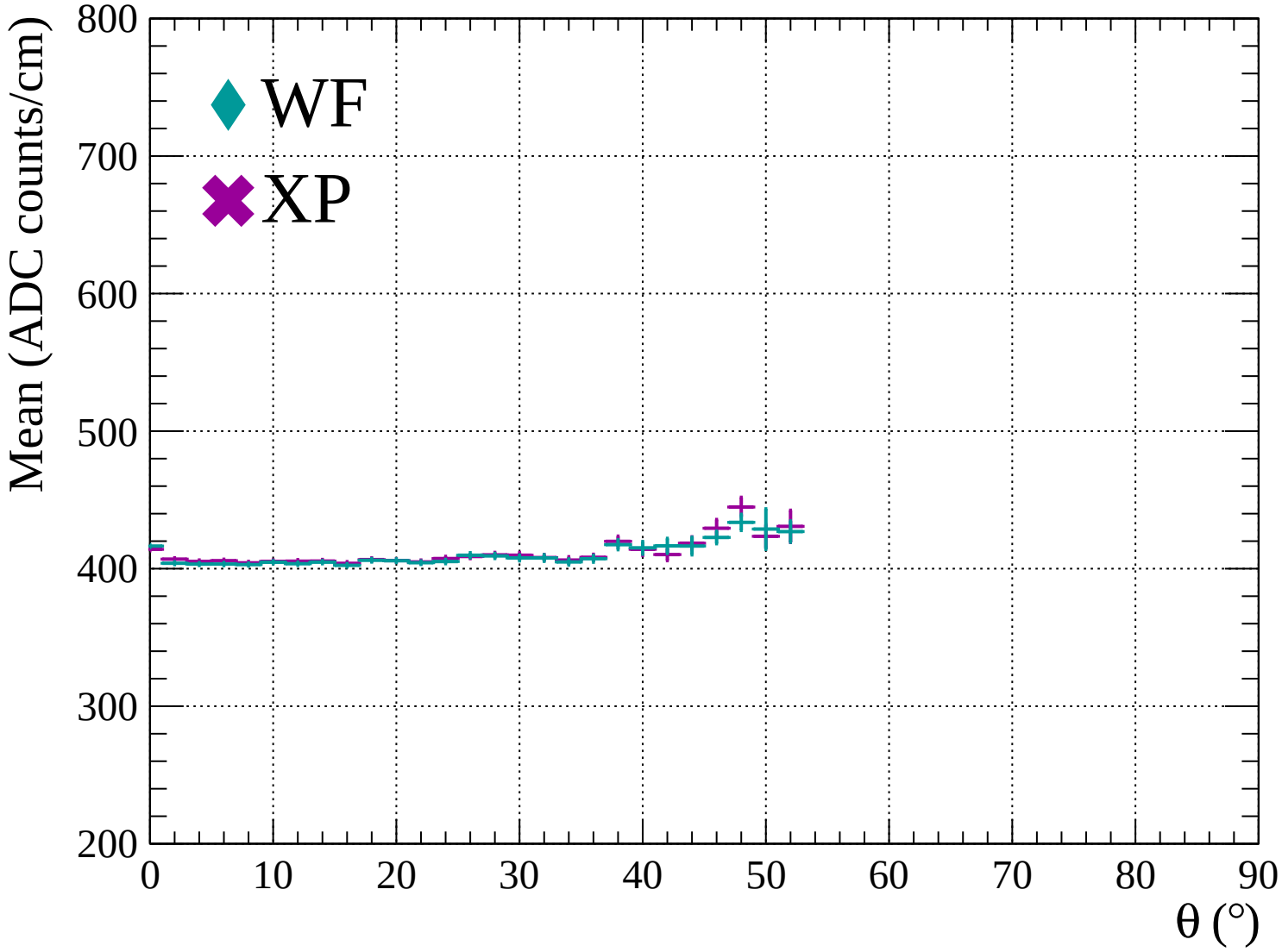


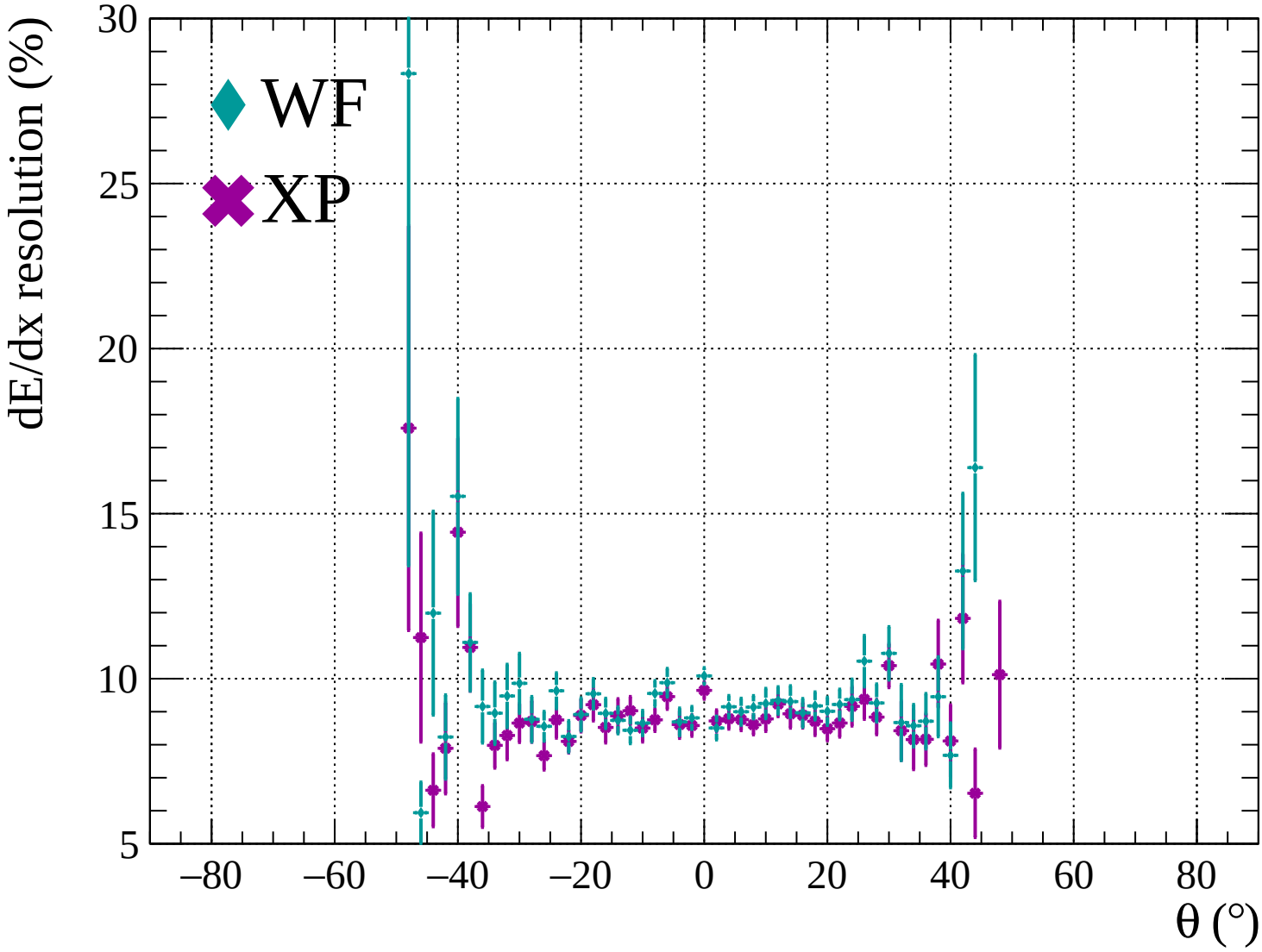


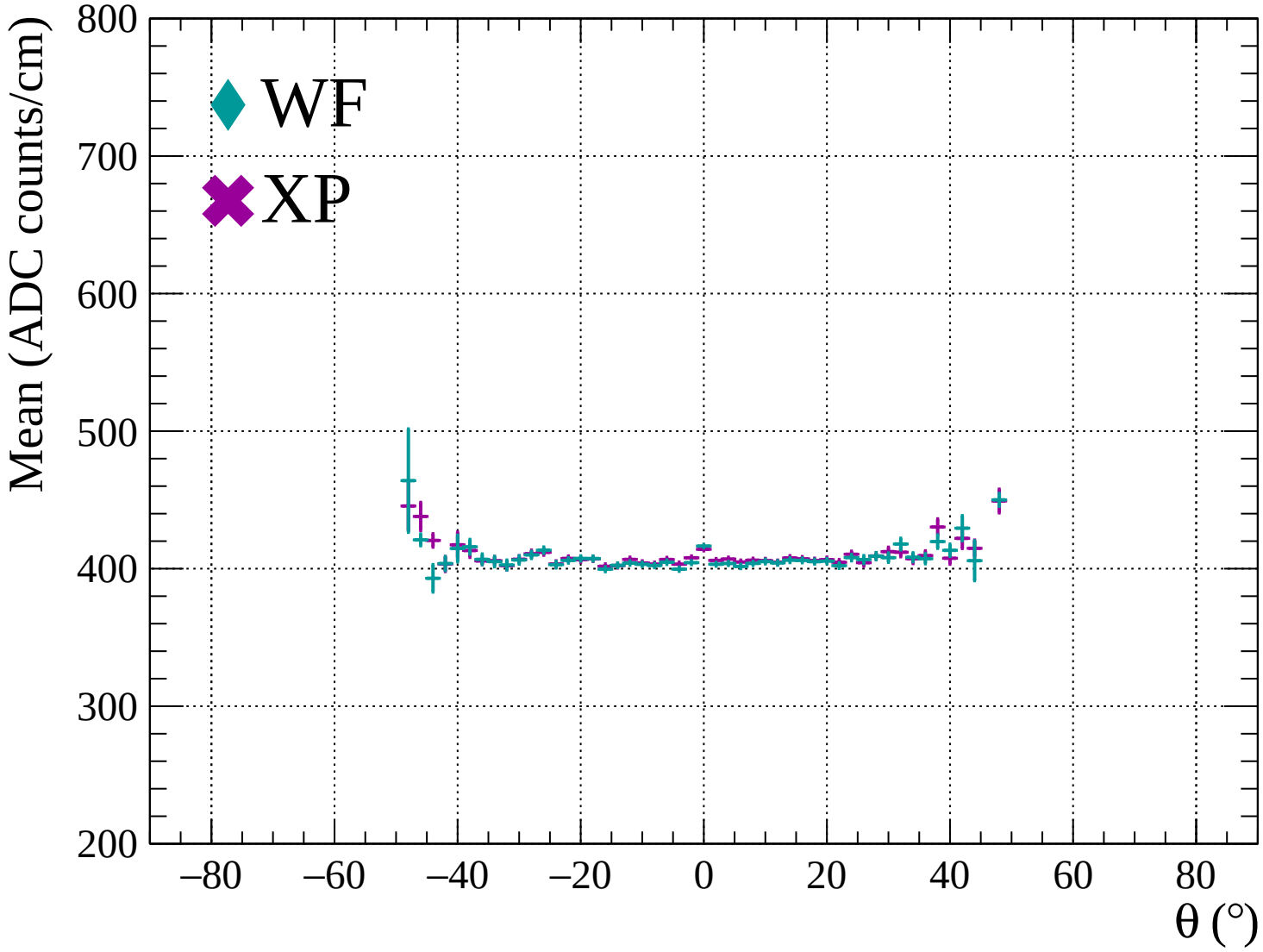


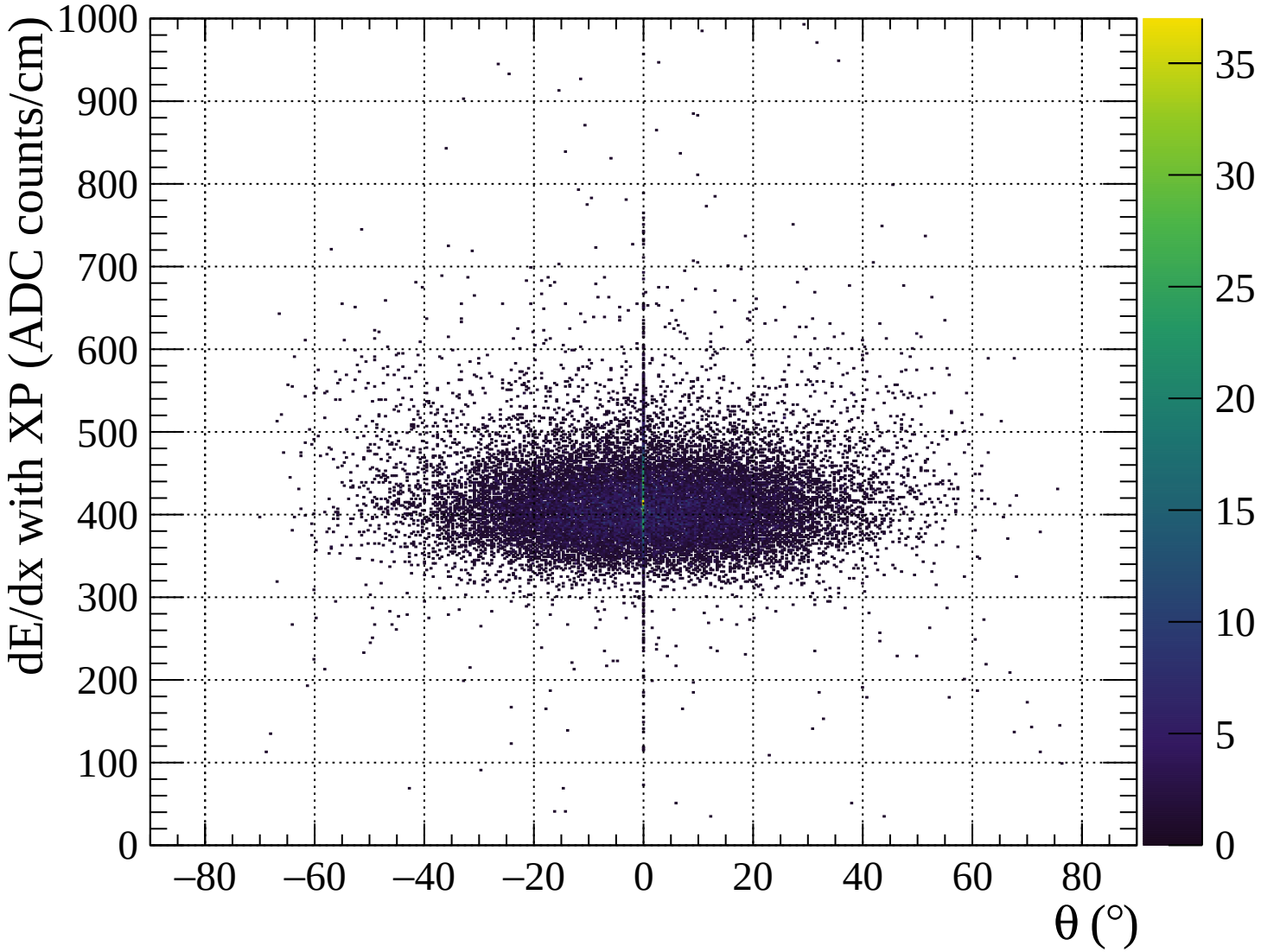


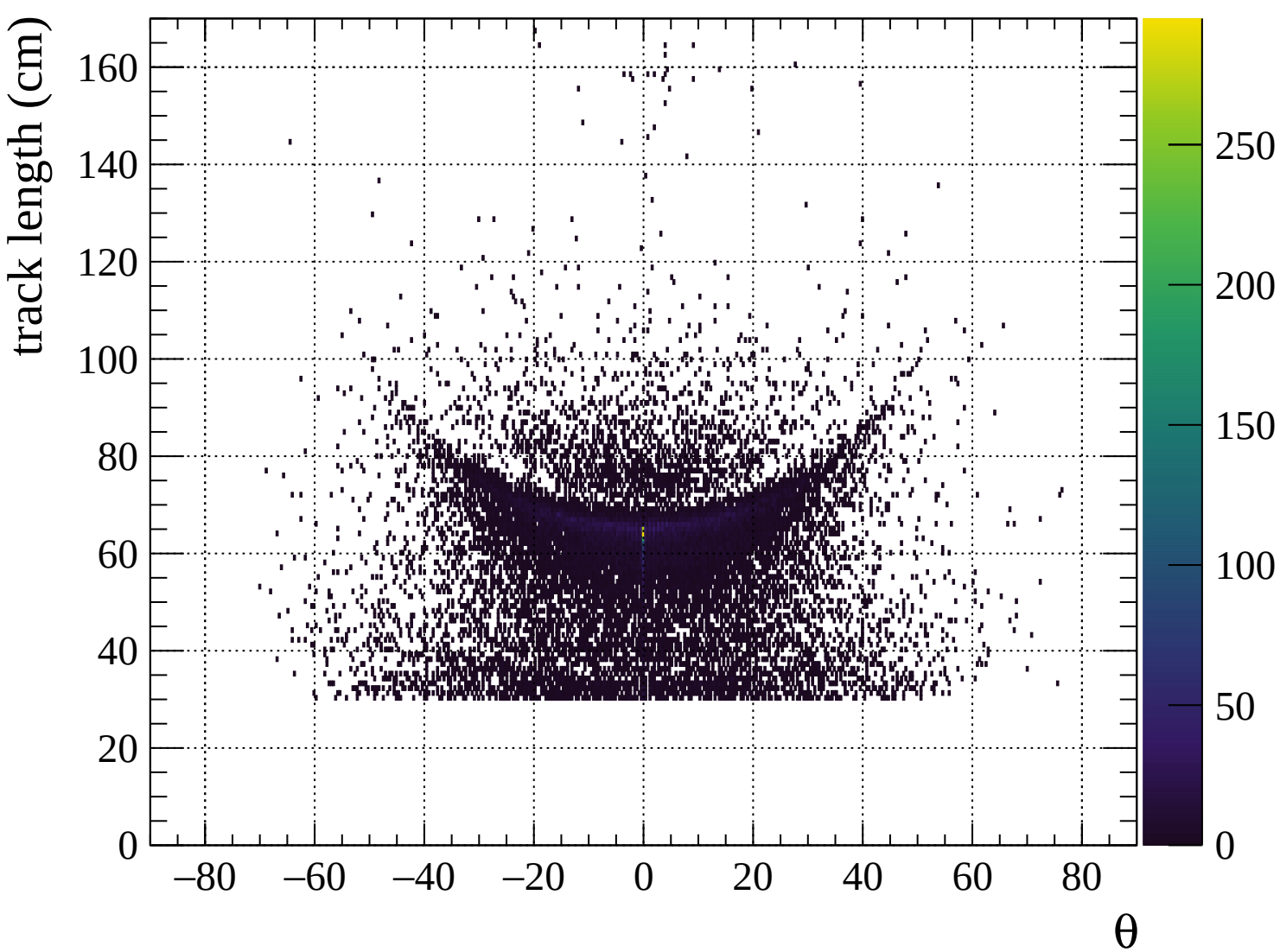


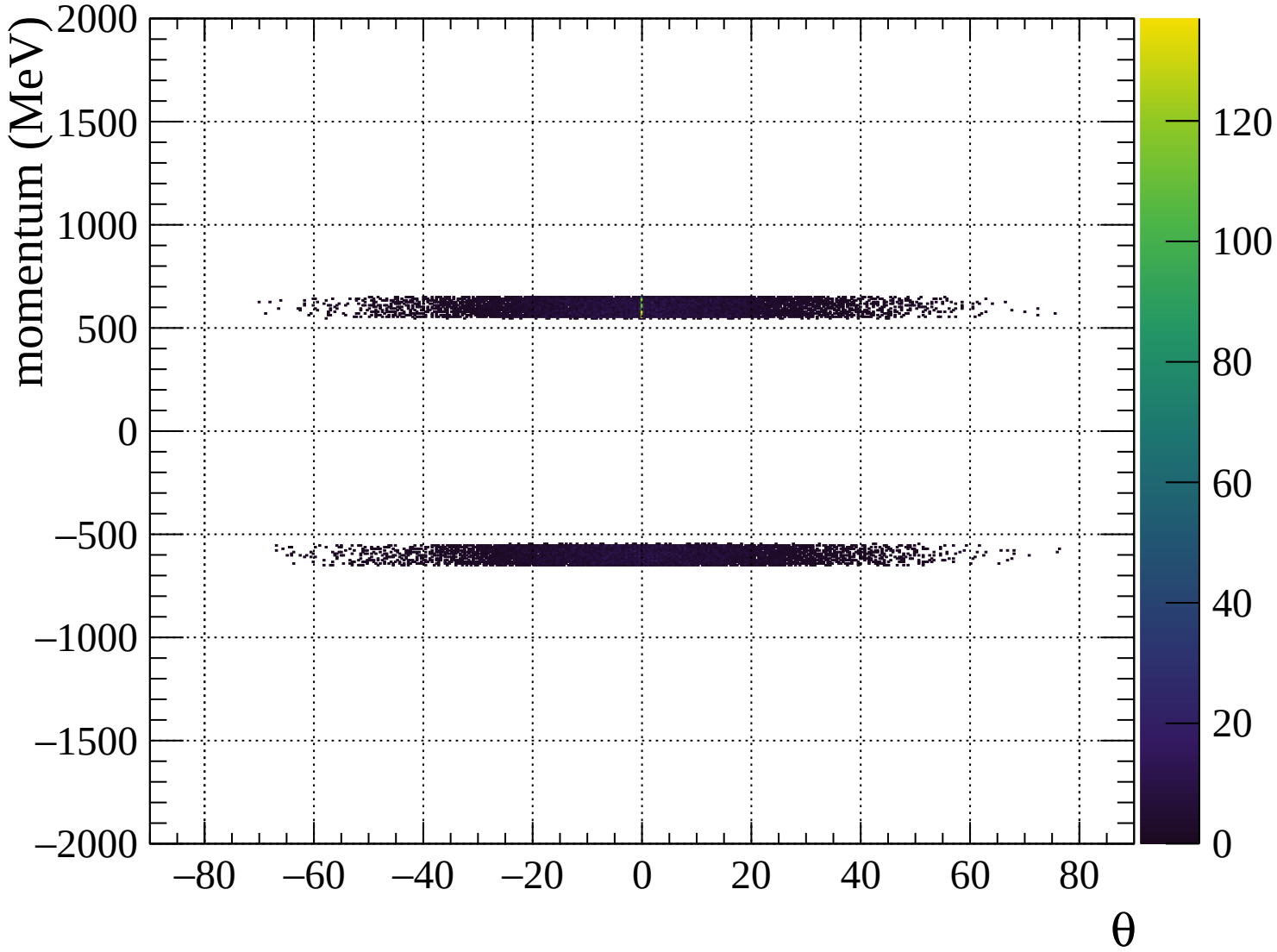


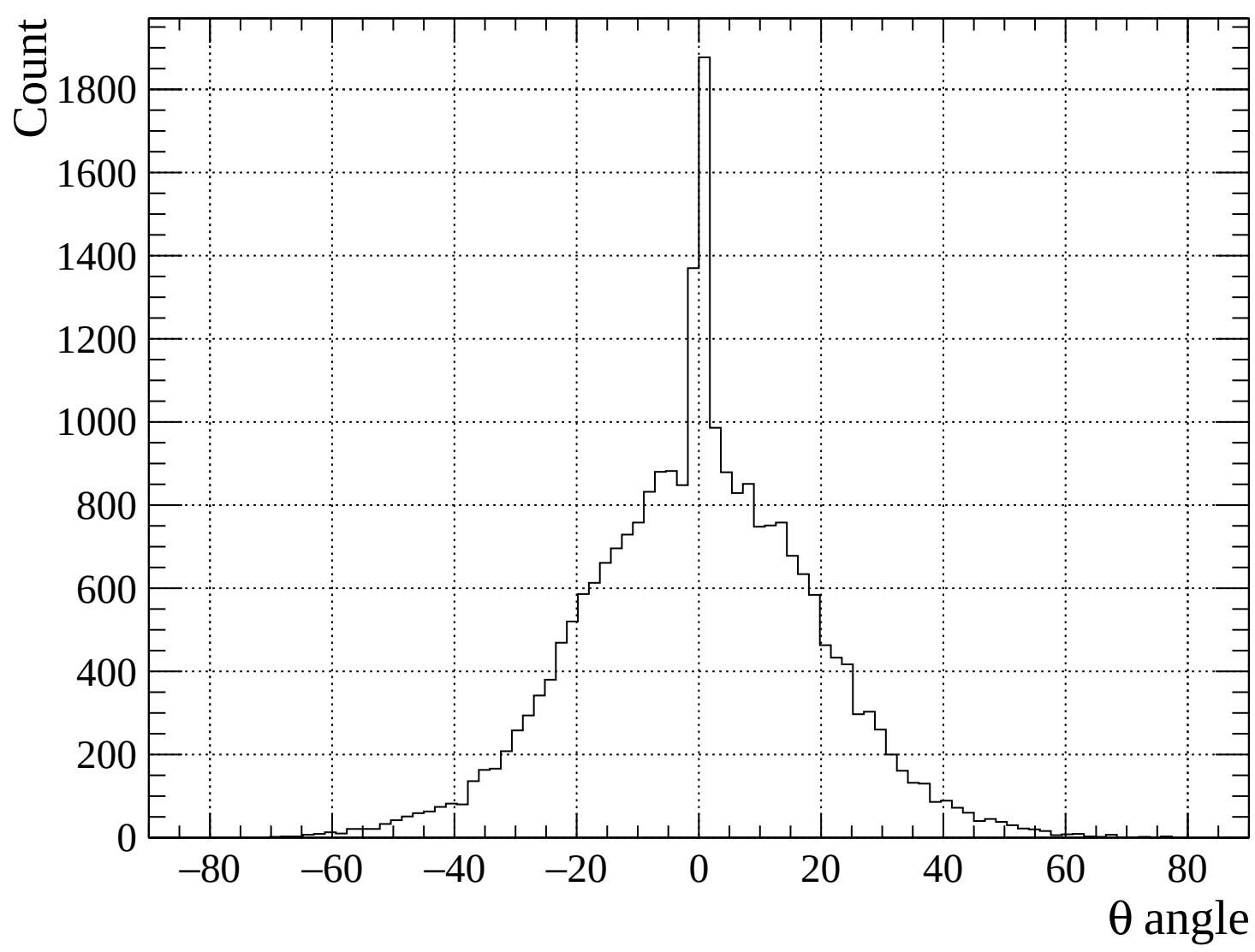


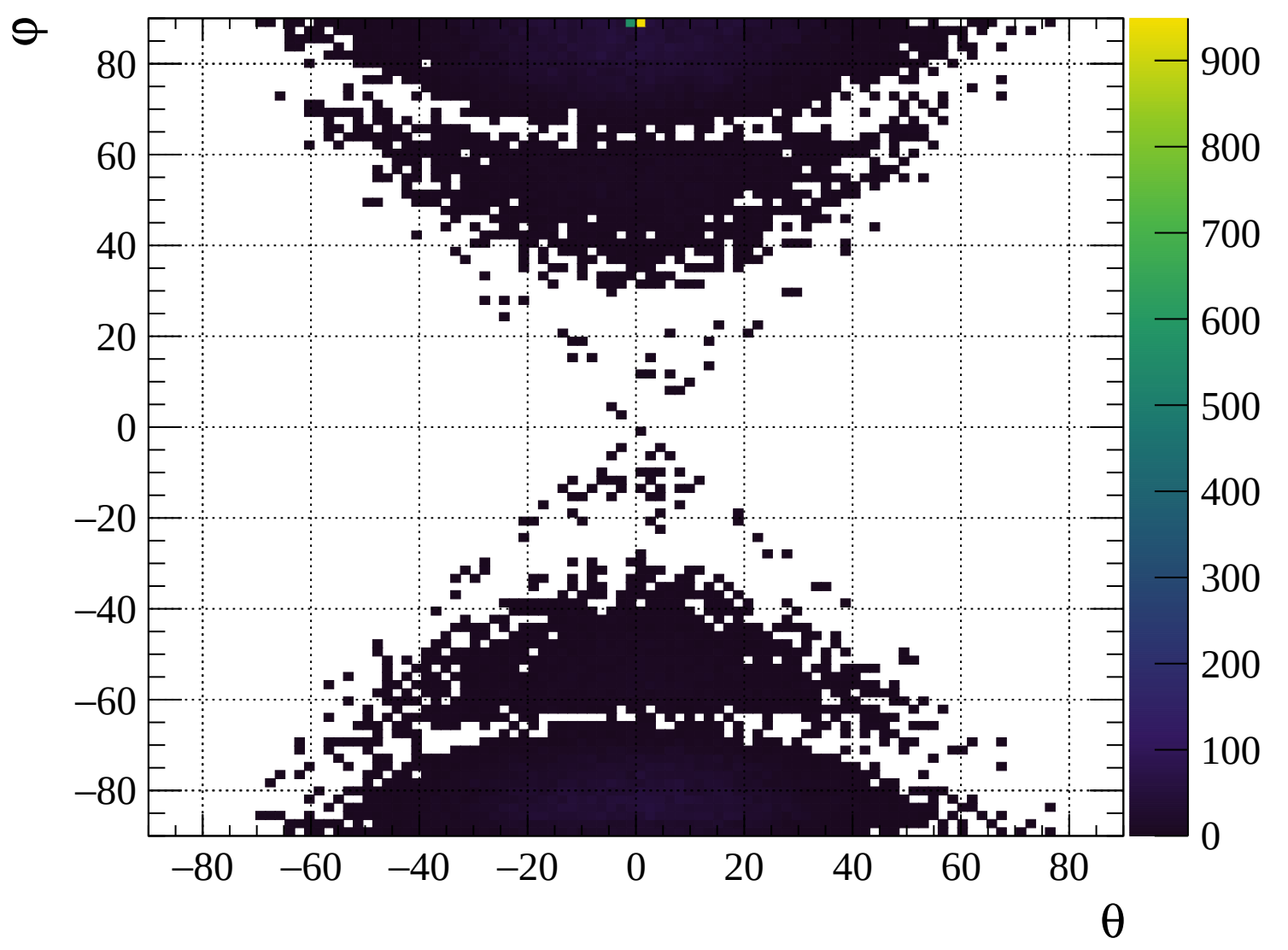


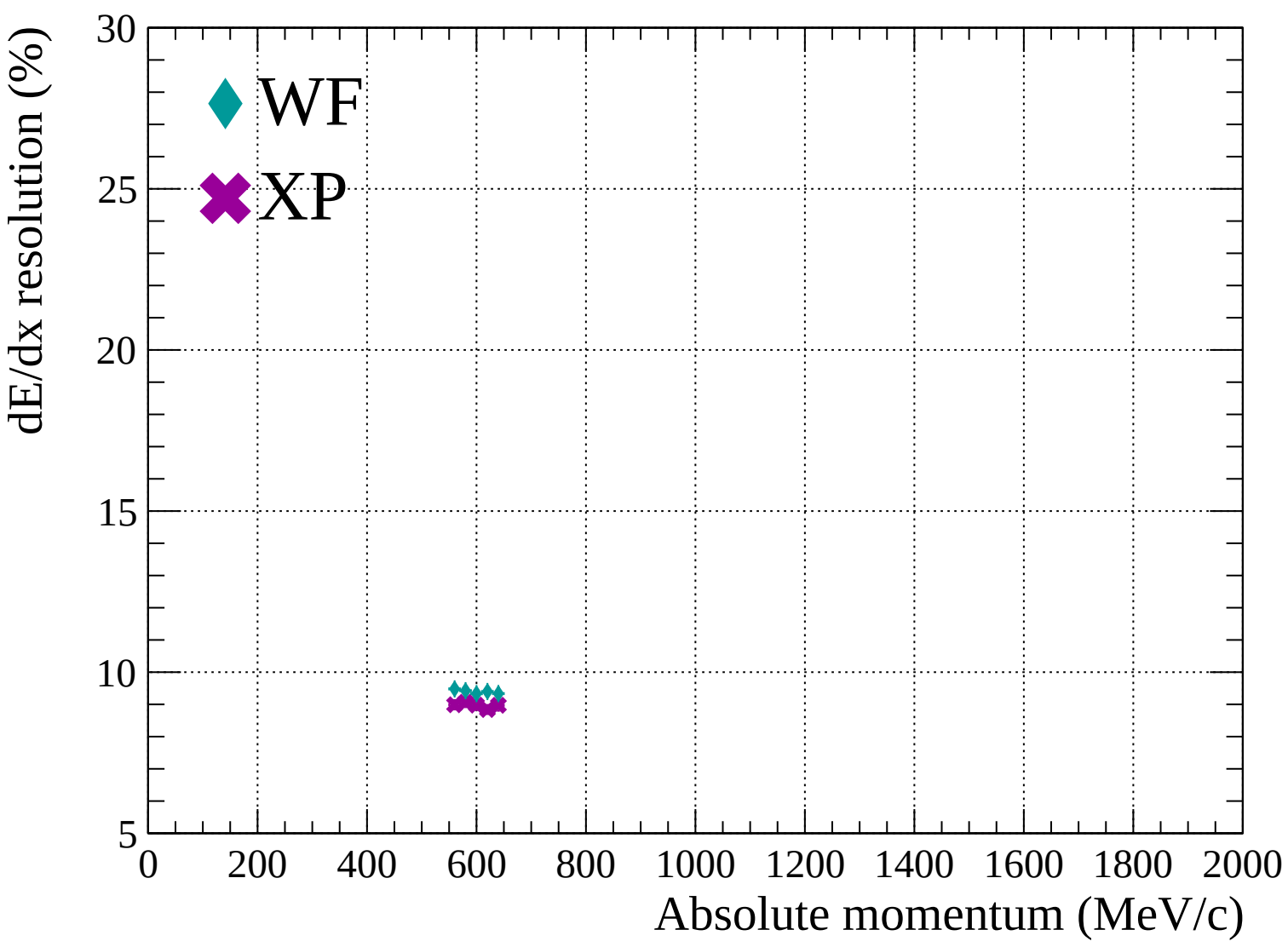


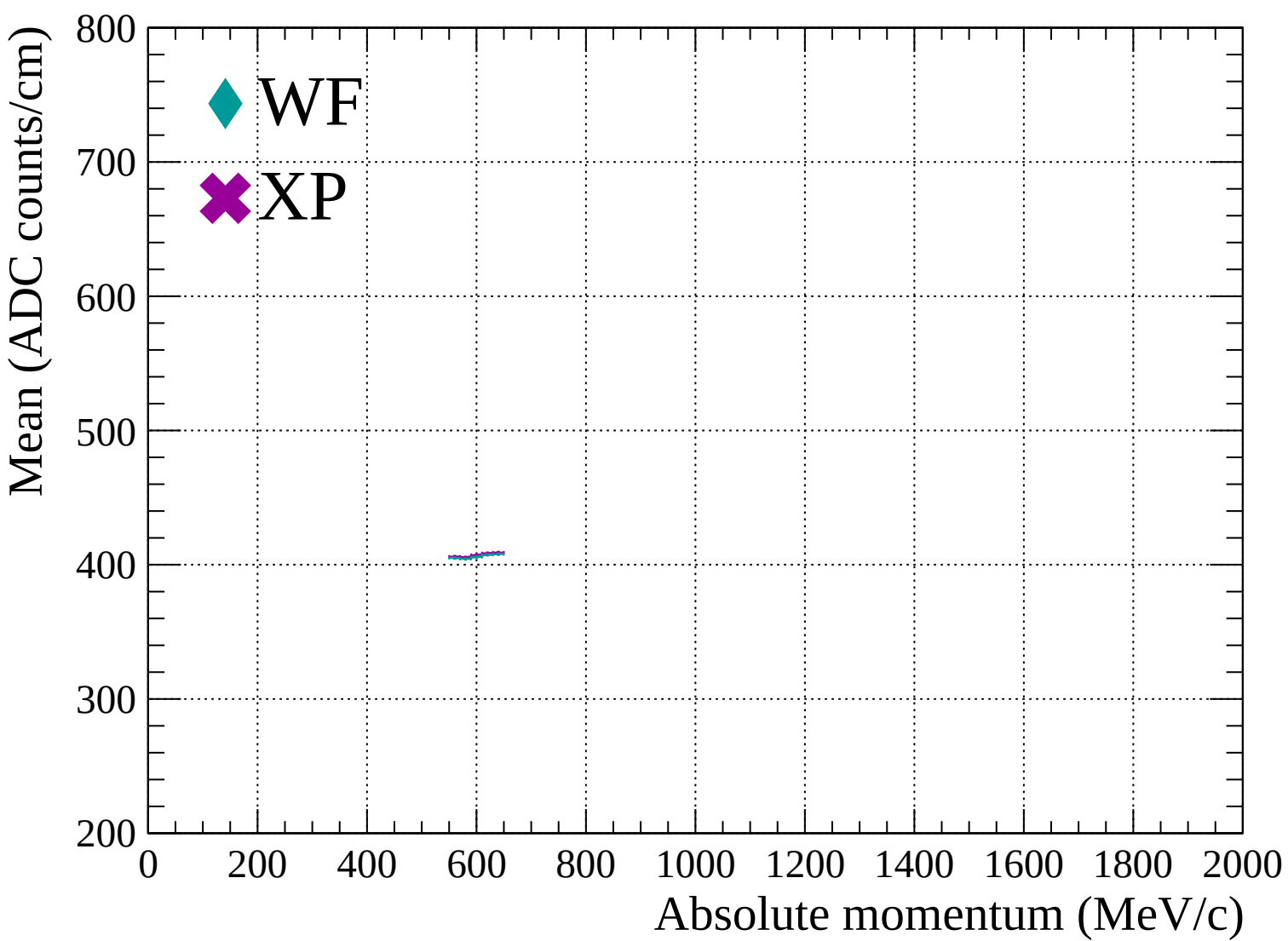


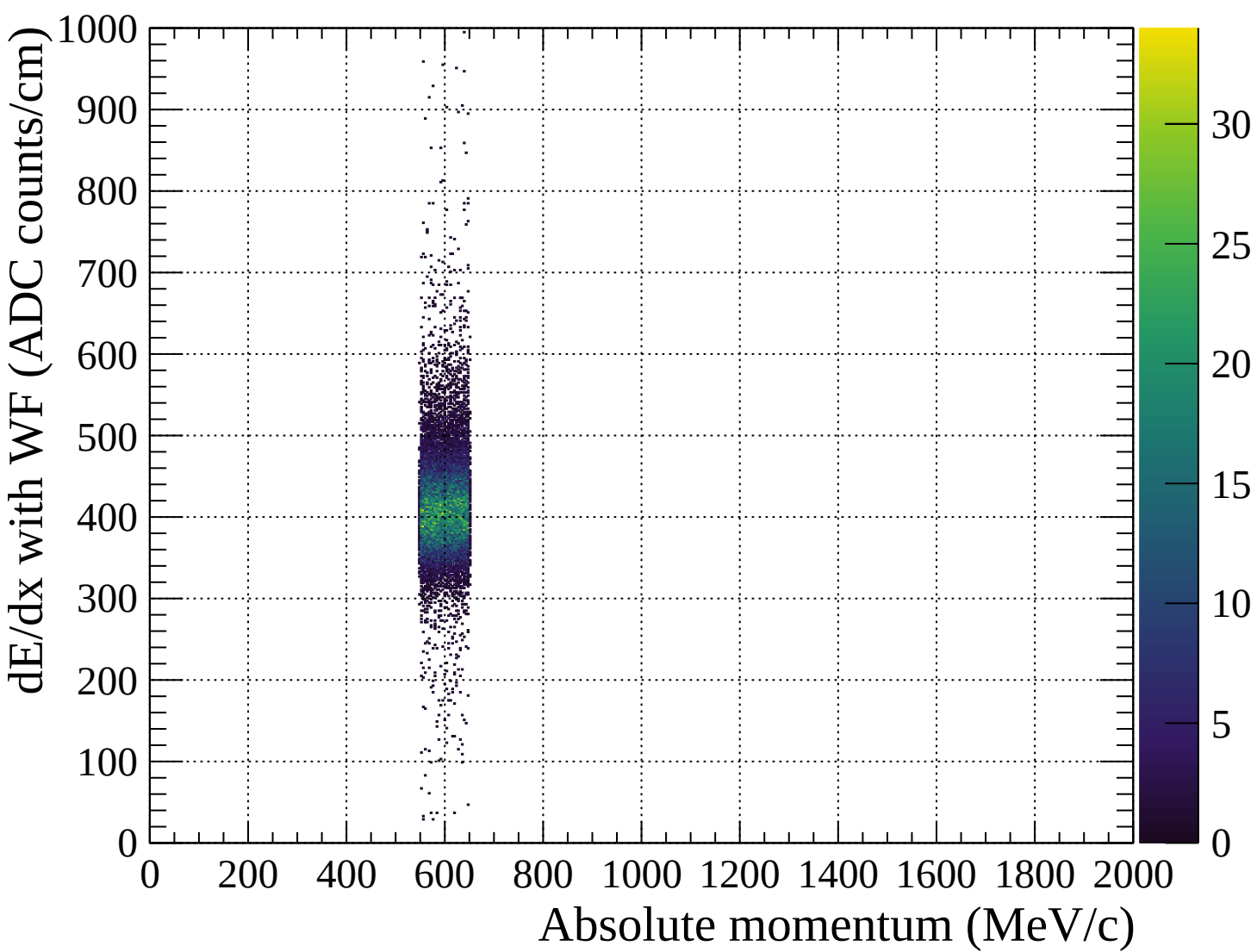


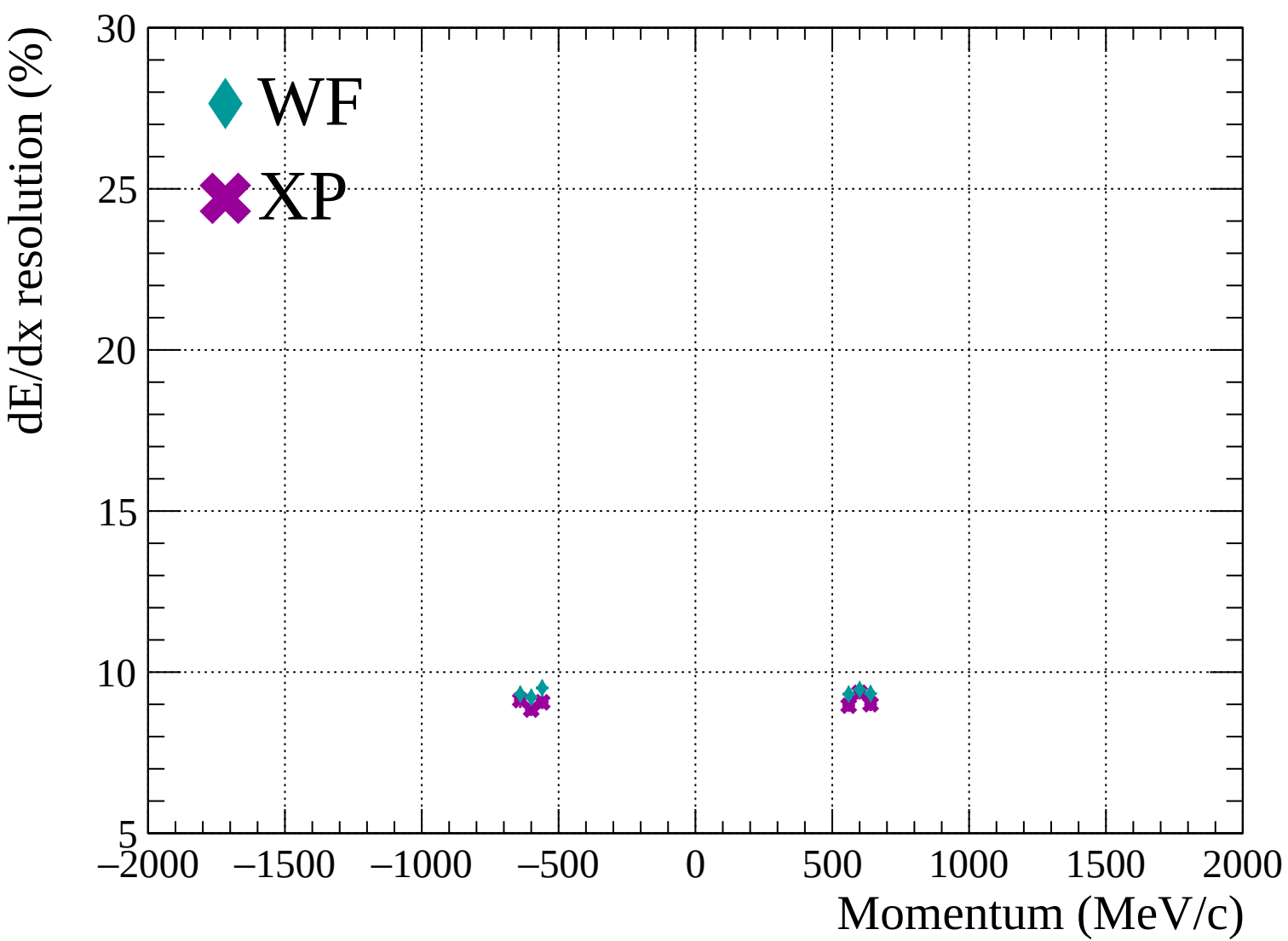


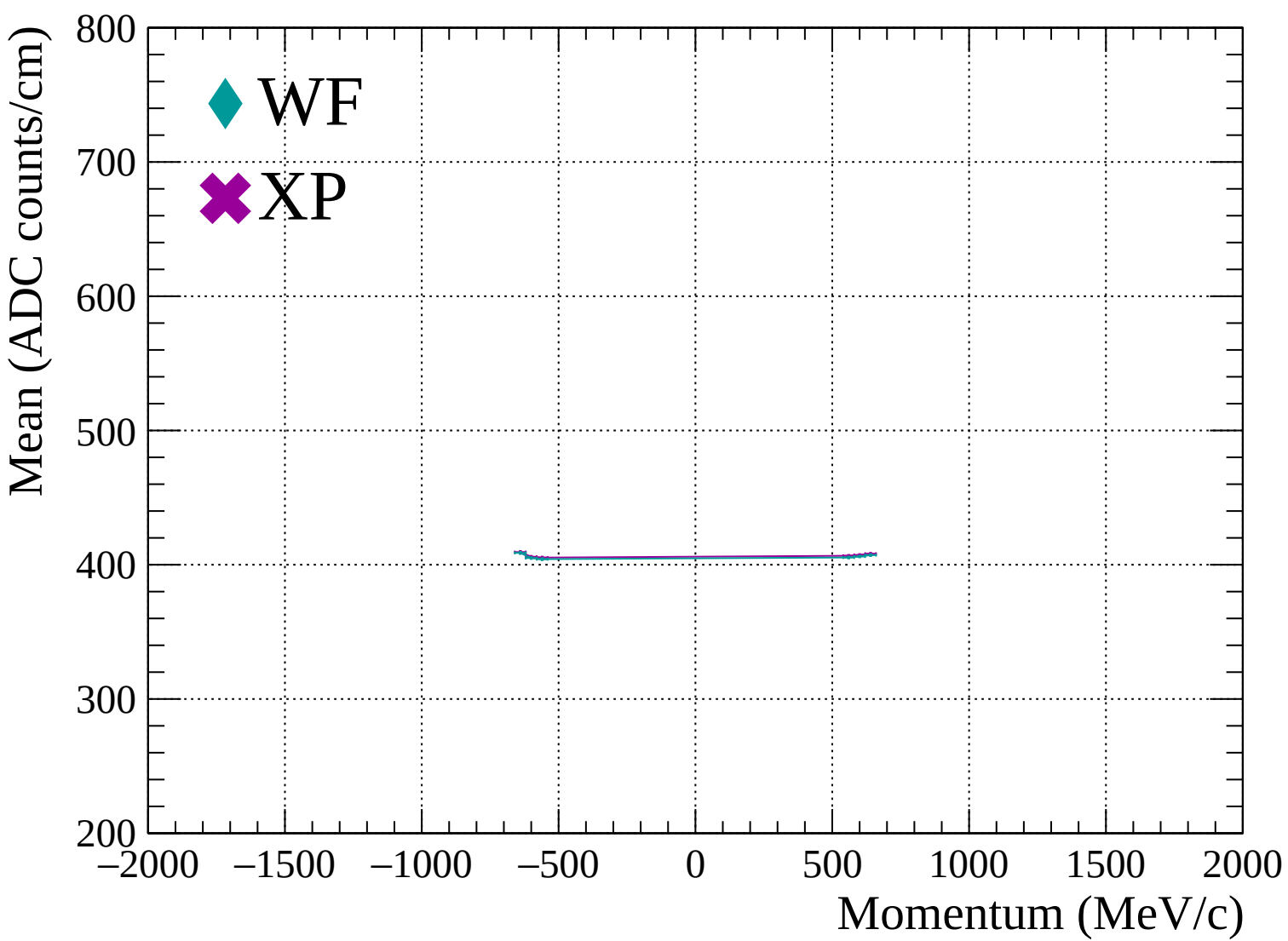


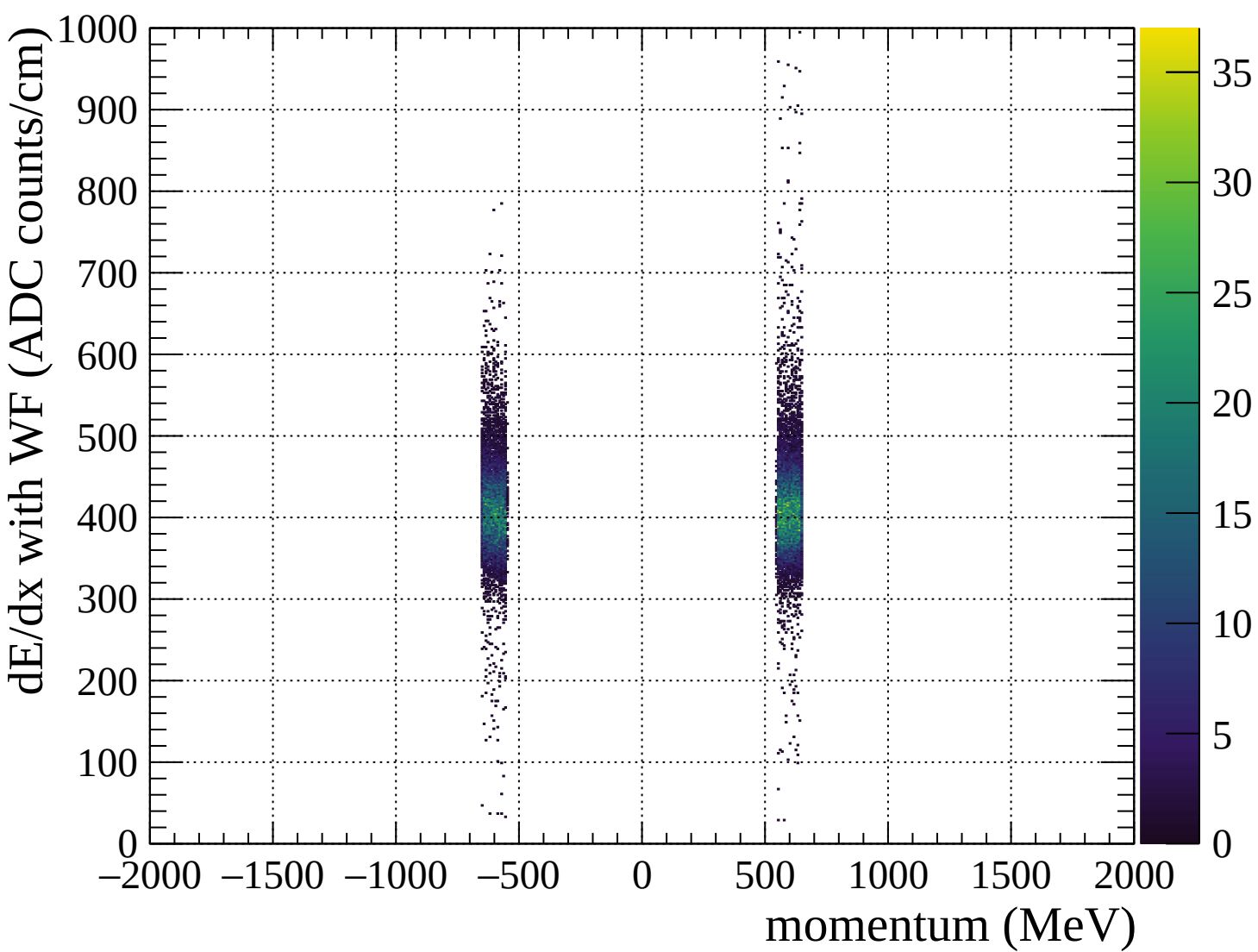


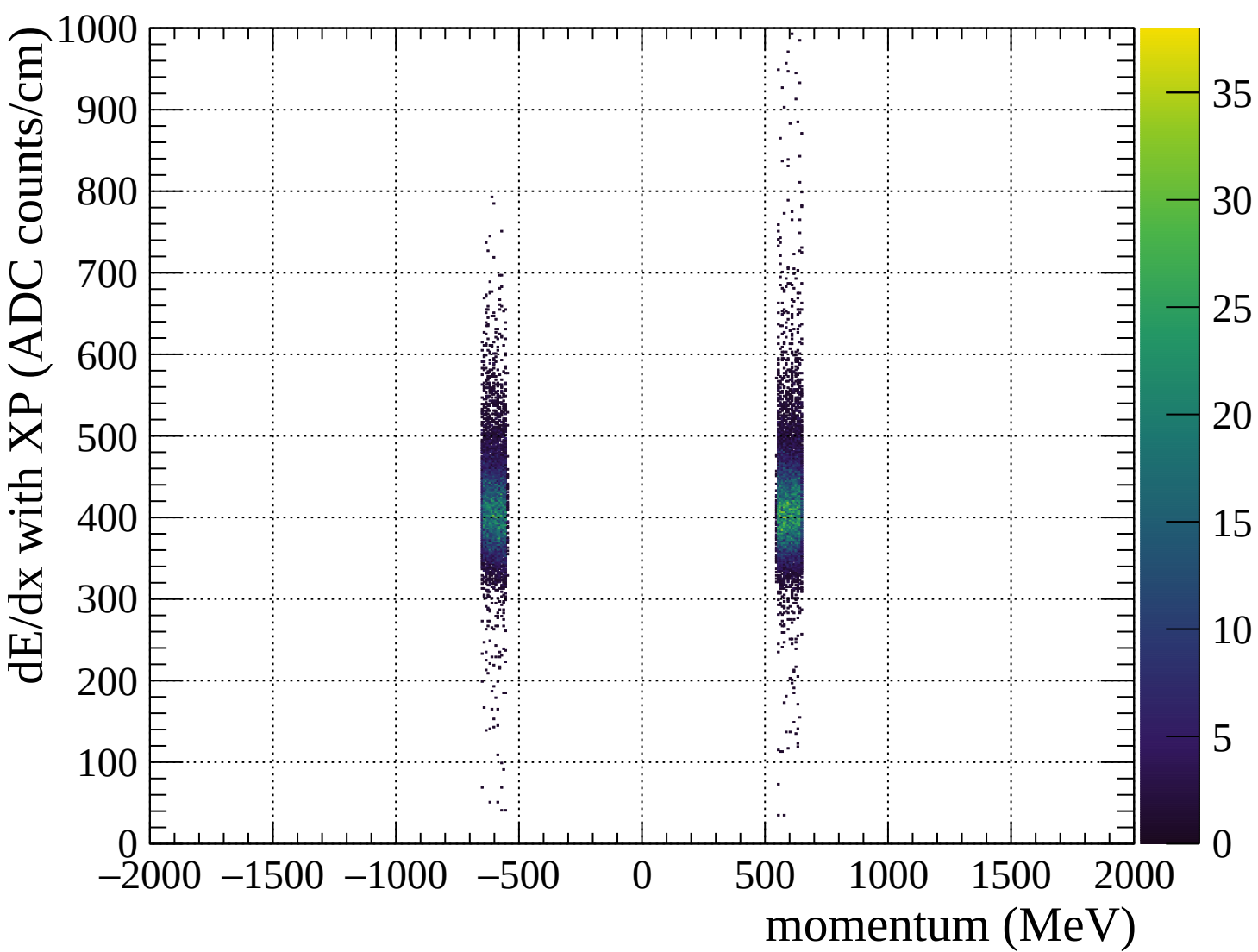


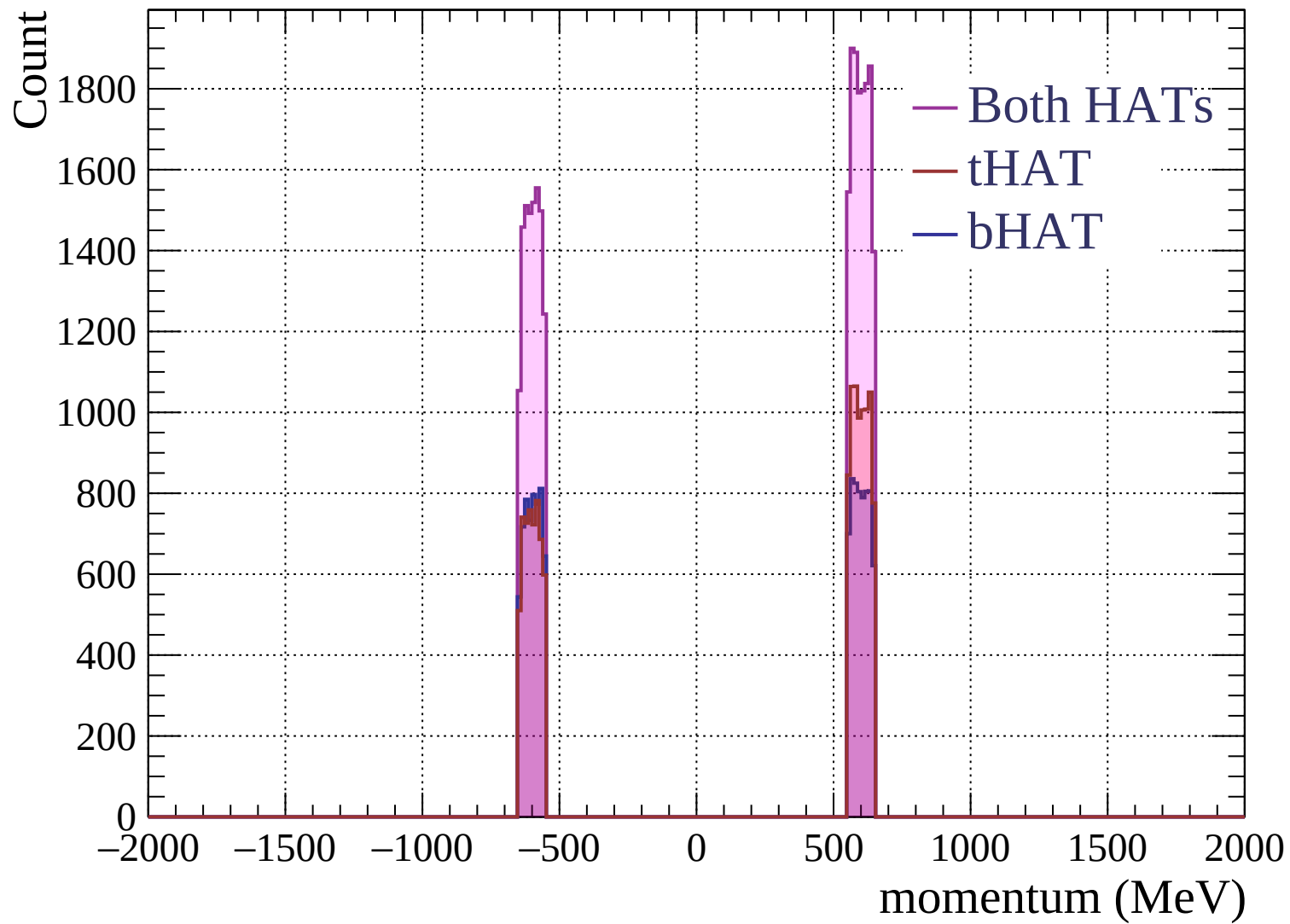


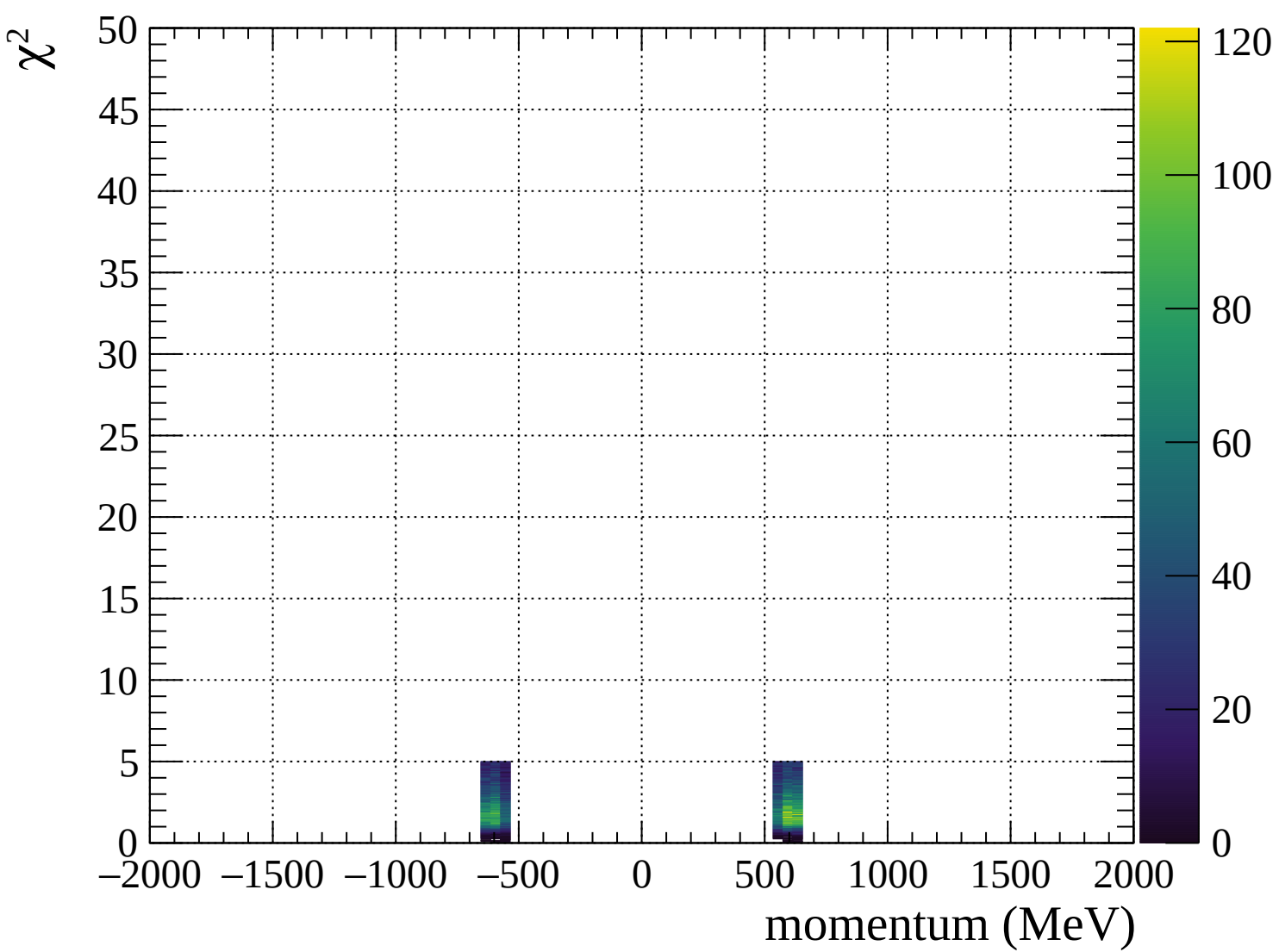




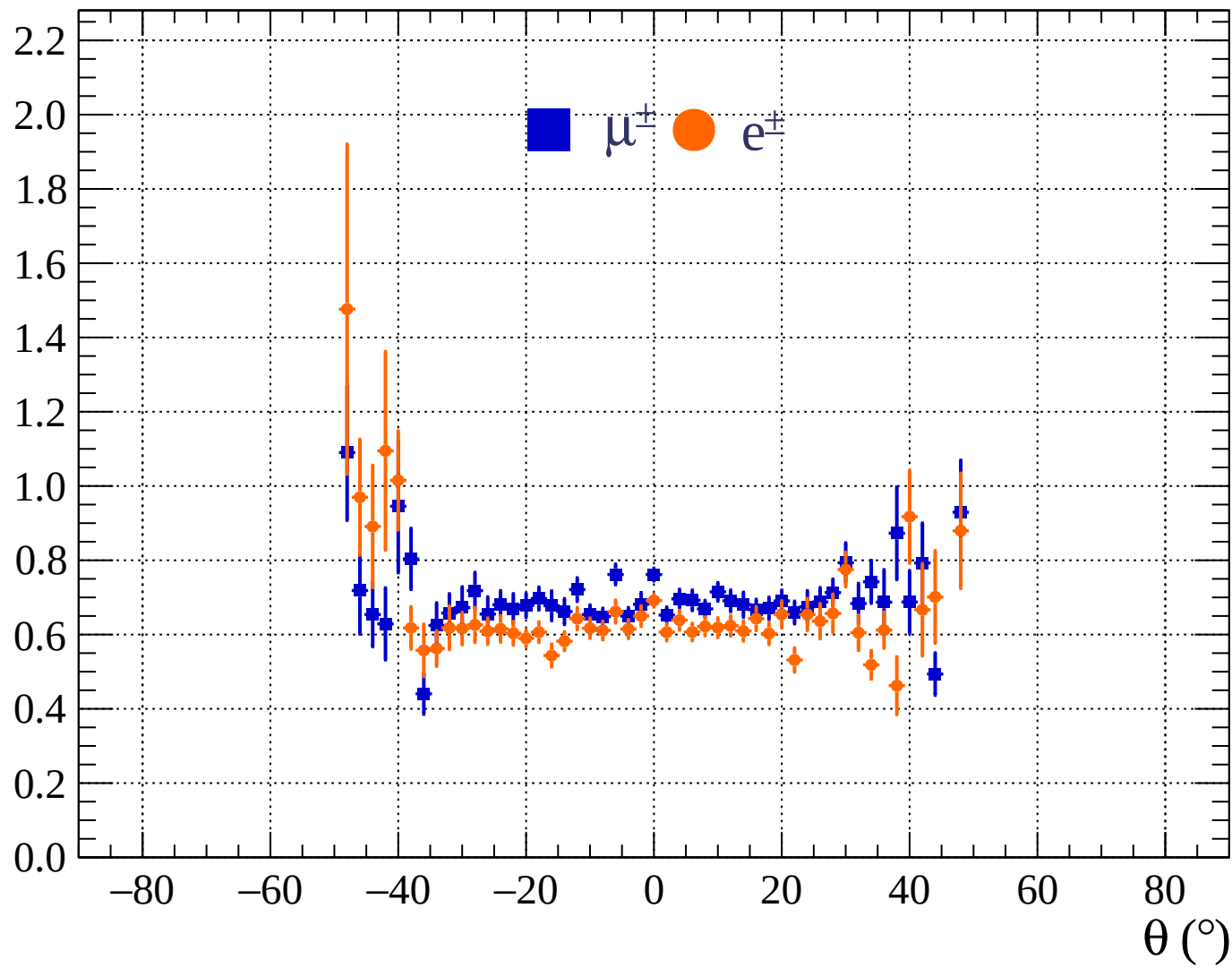


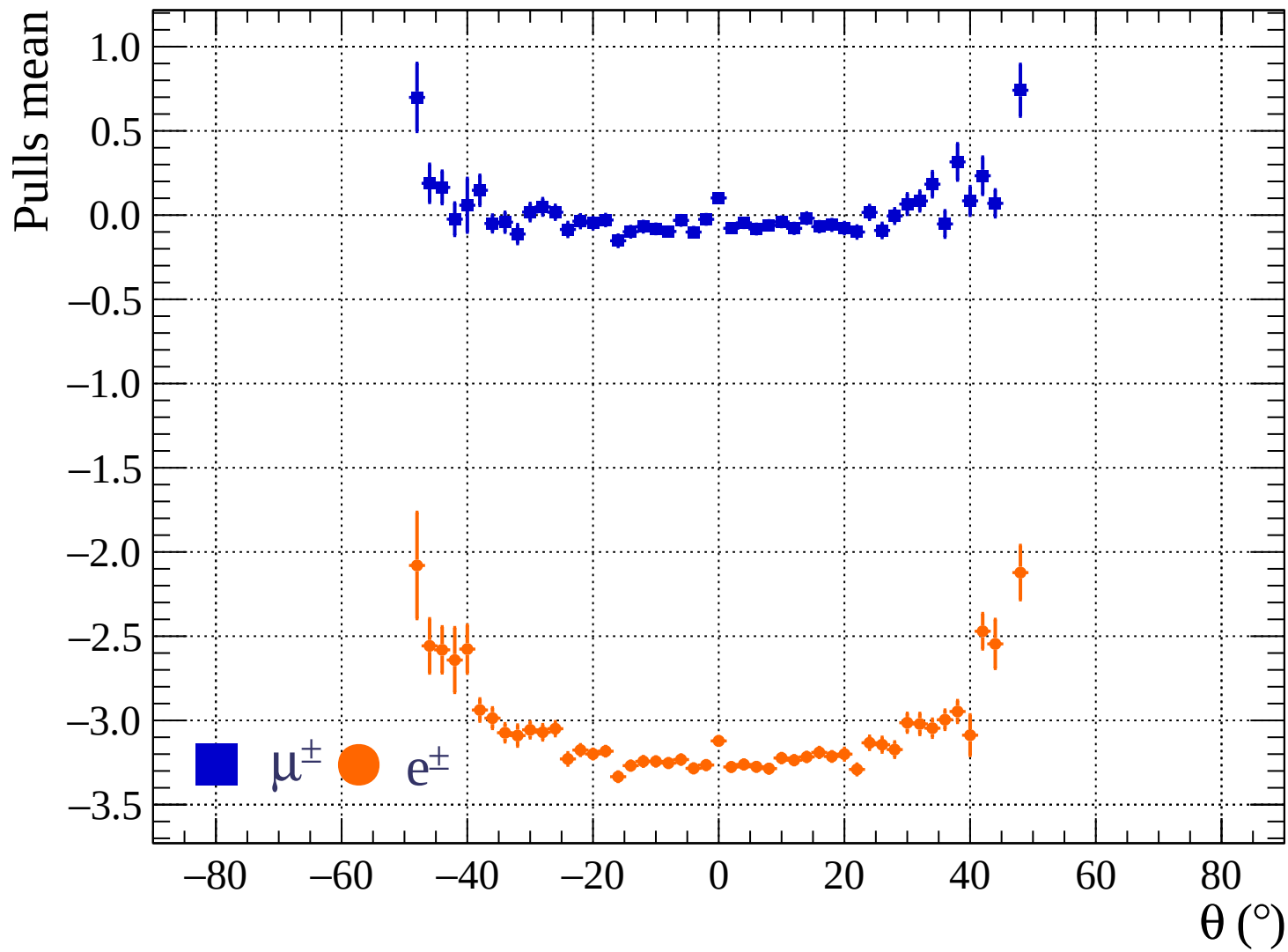






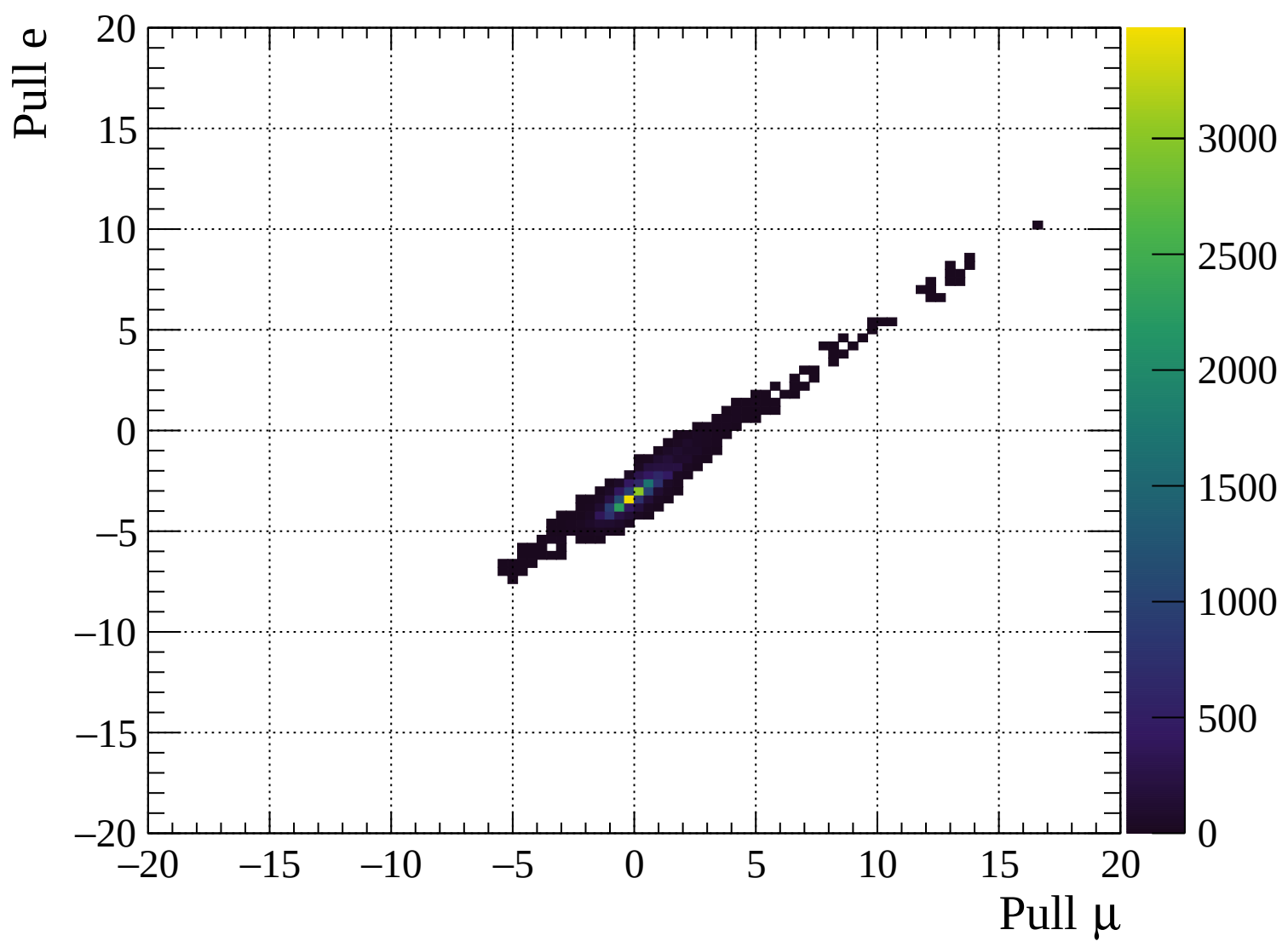
Pulls standard deviation

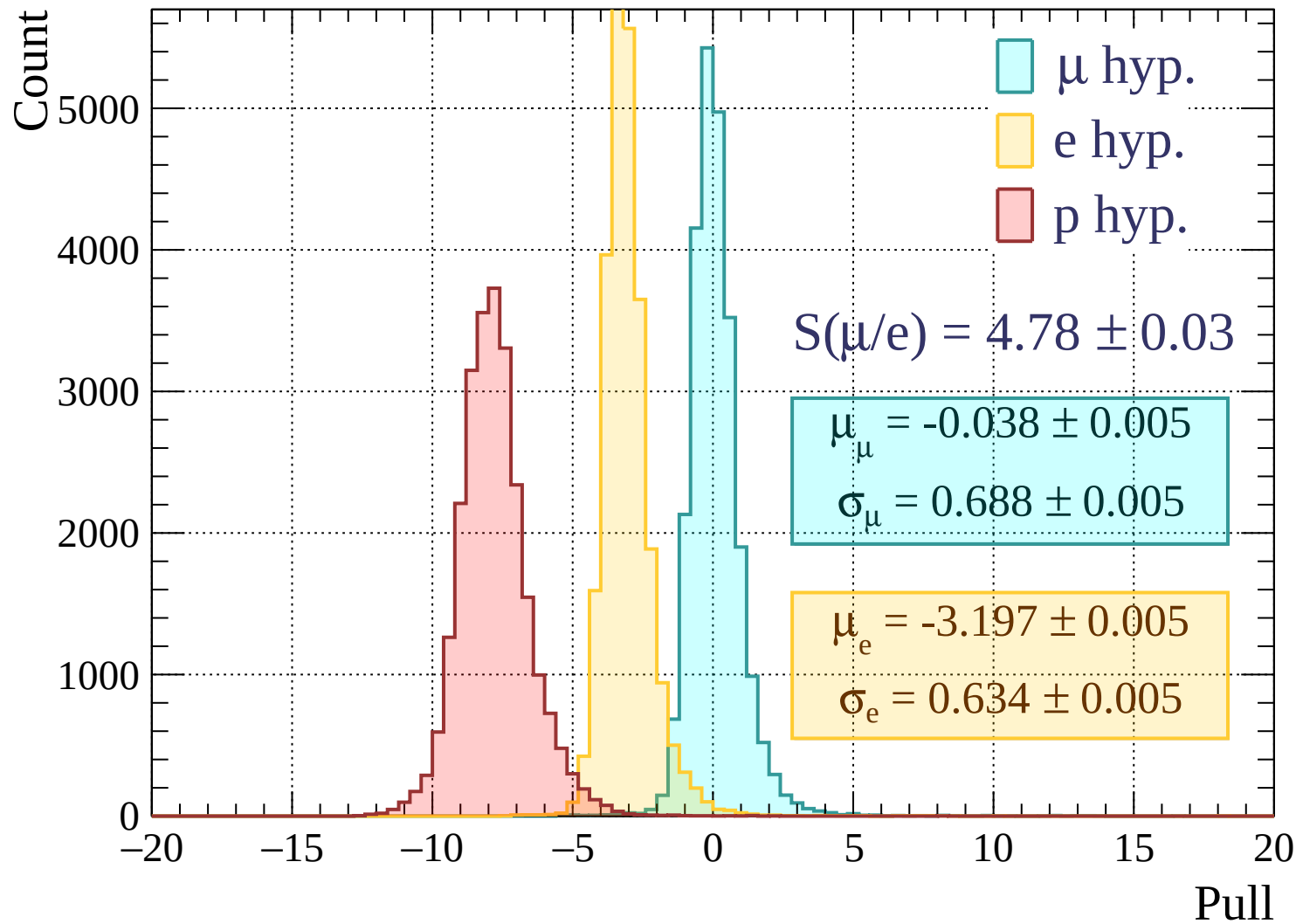


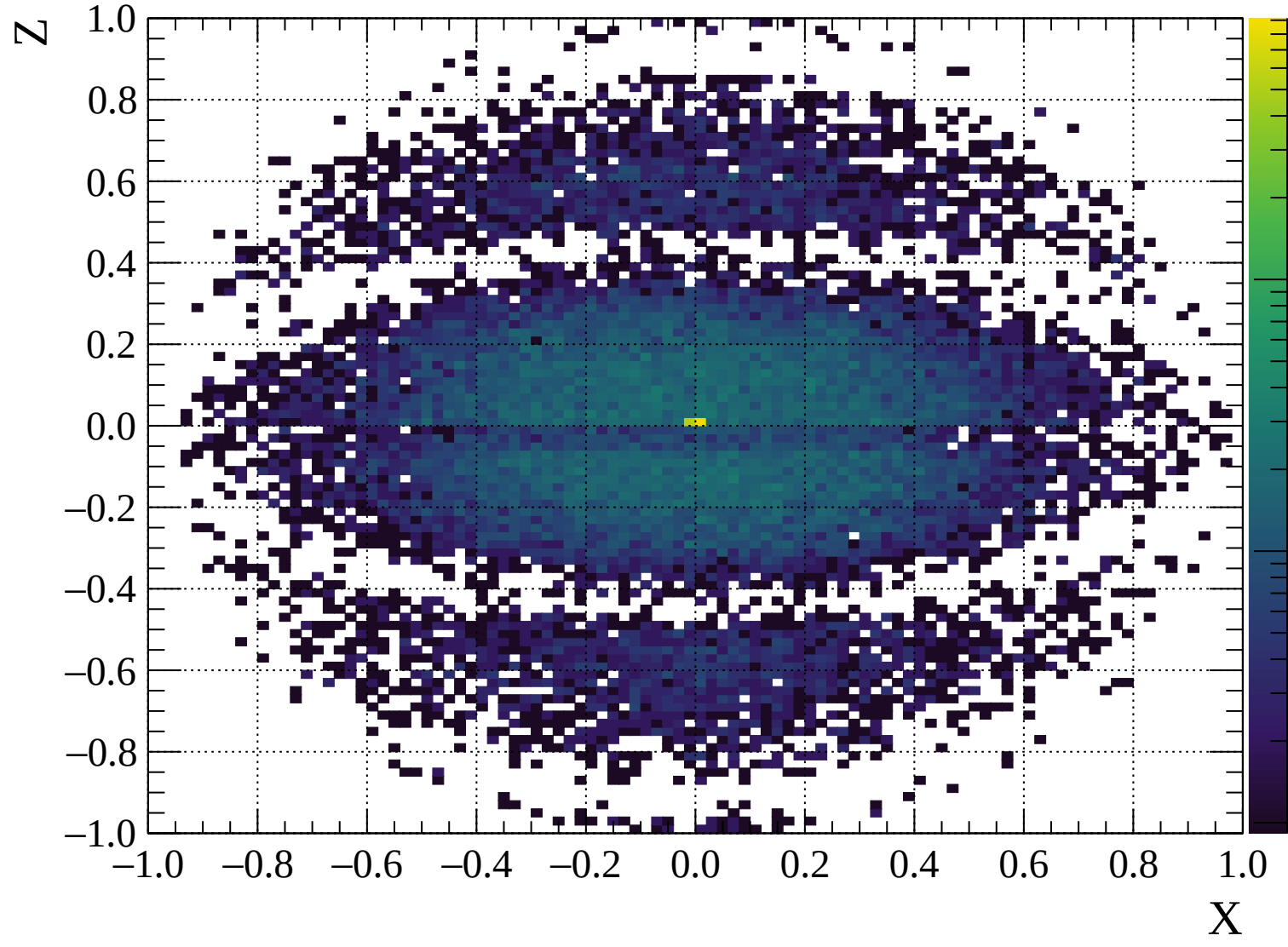


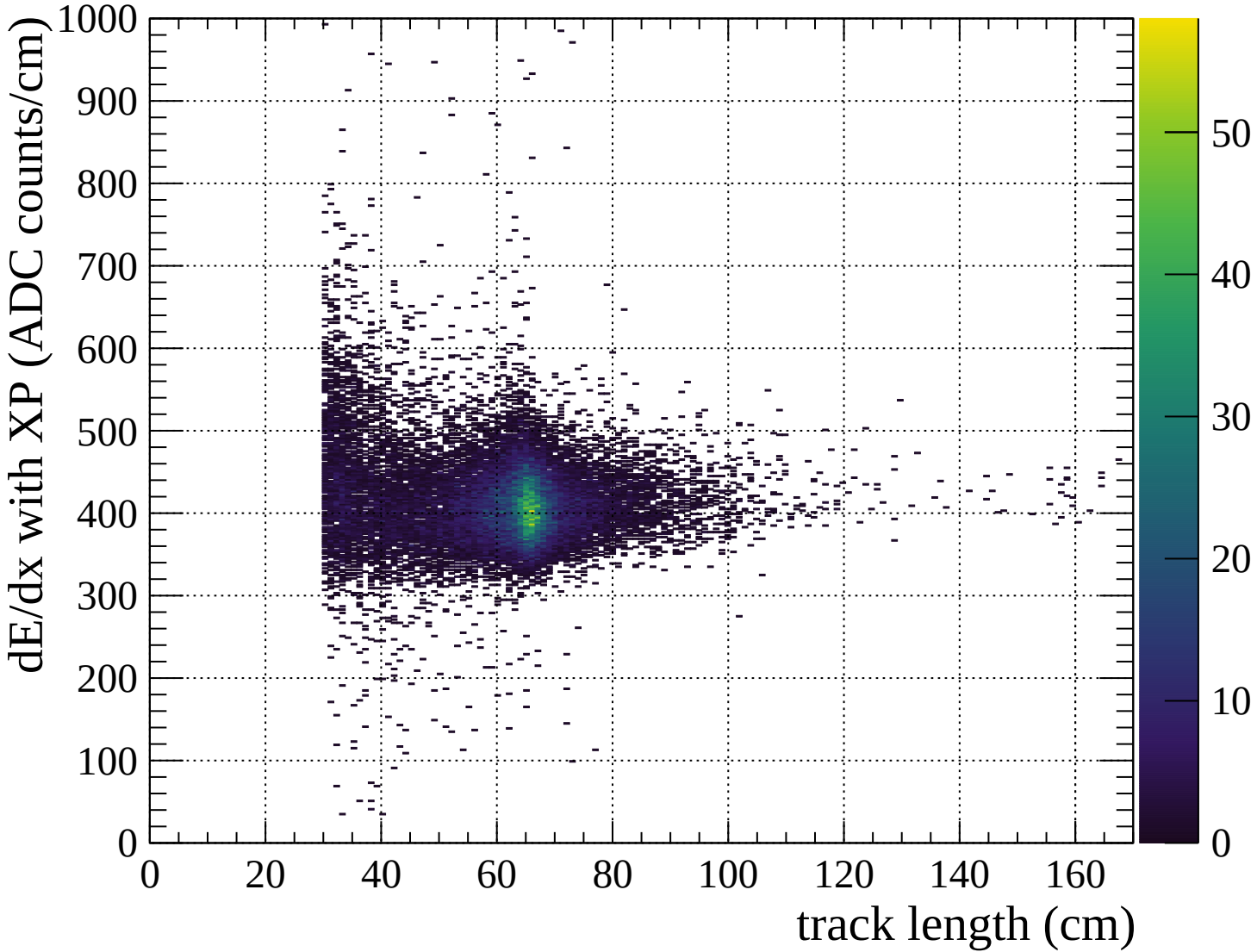
■ $\mu^\pm$ ● $e^\pm$

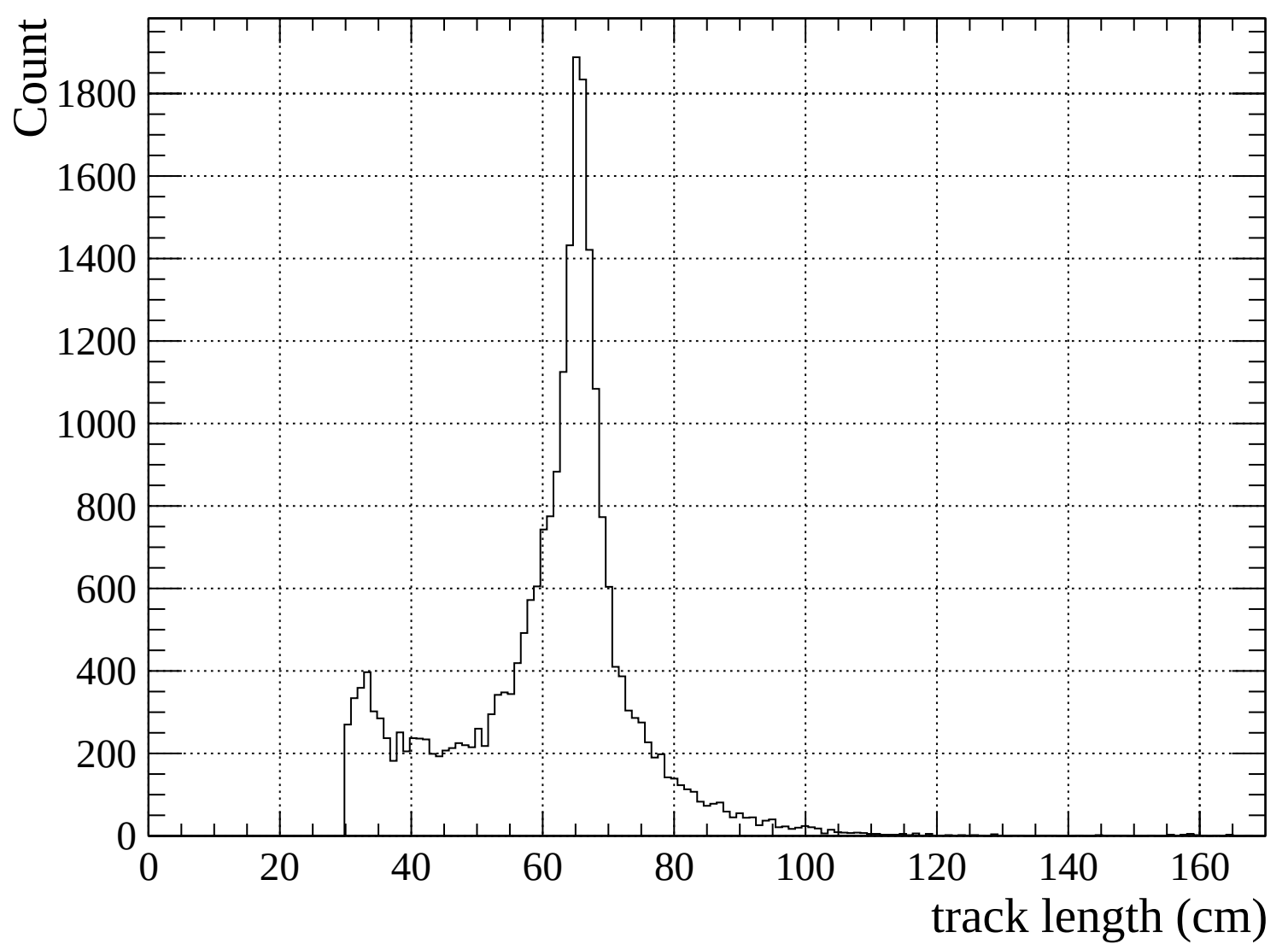
■ $\mu^\pm$ ● $e^\pm$

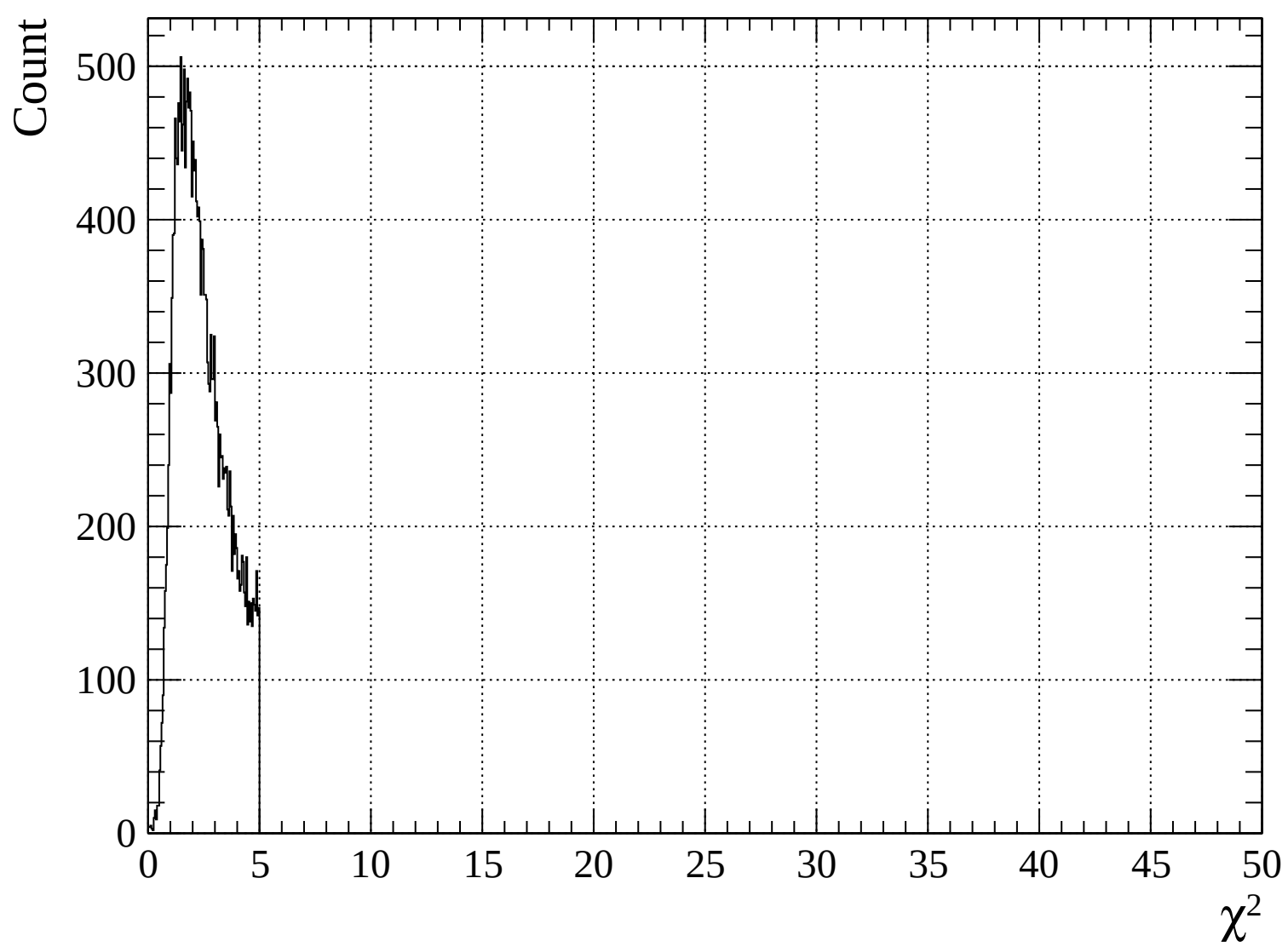


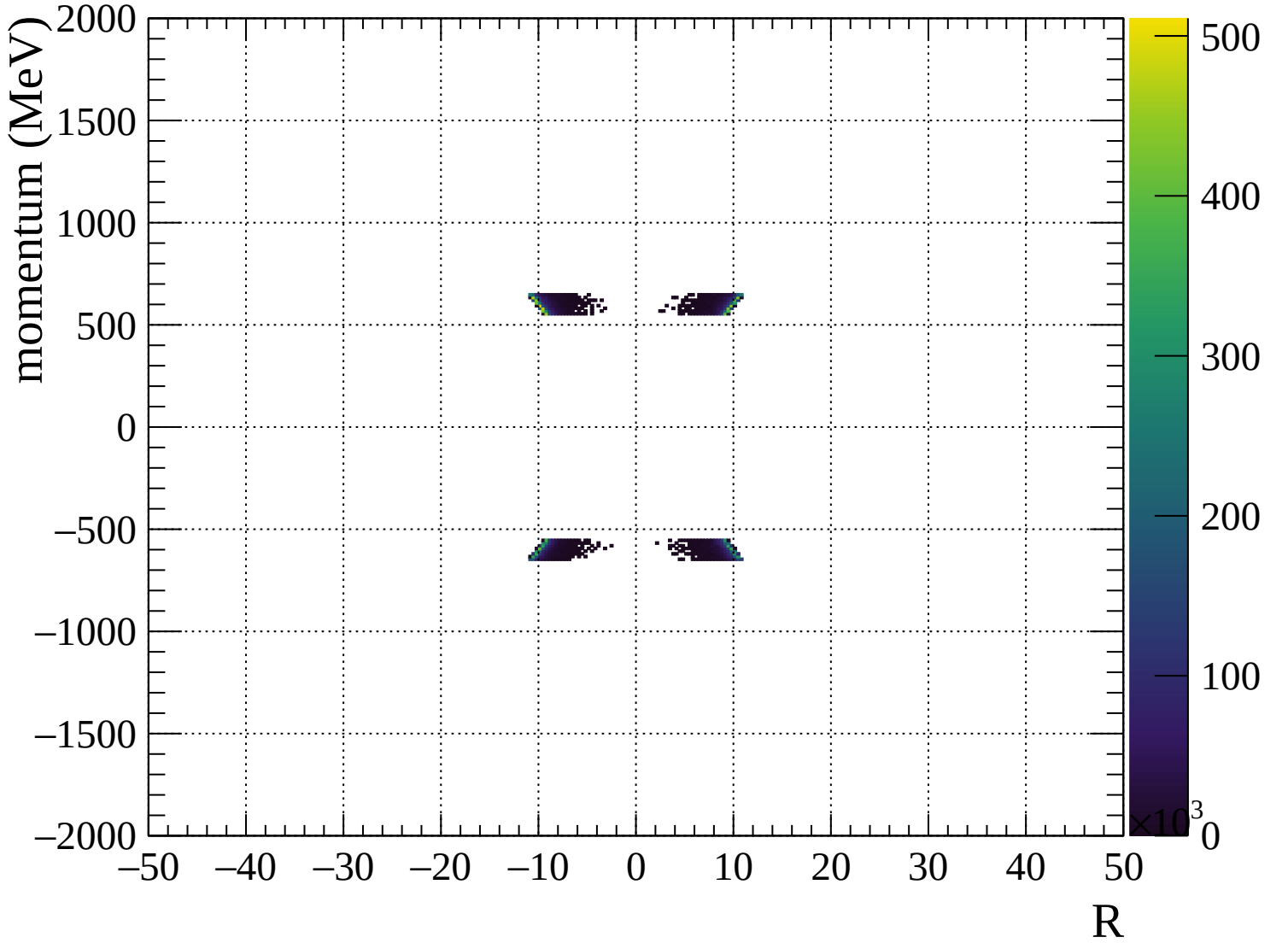


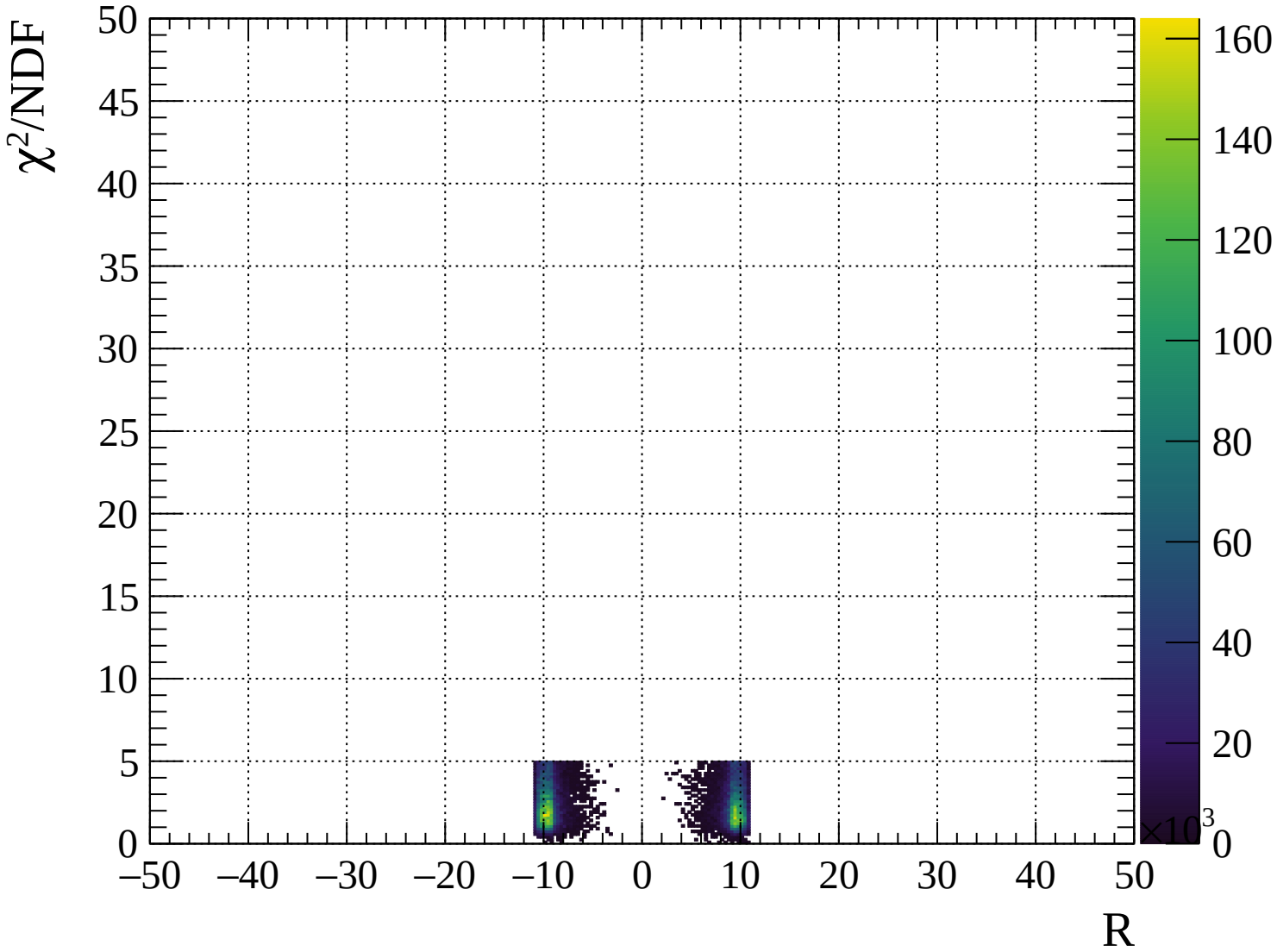






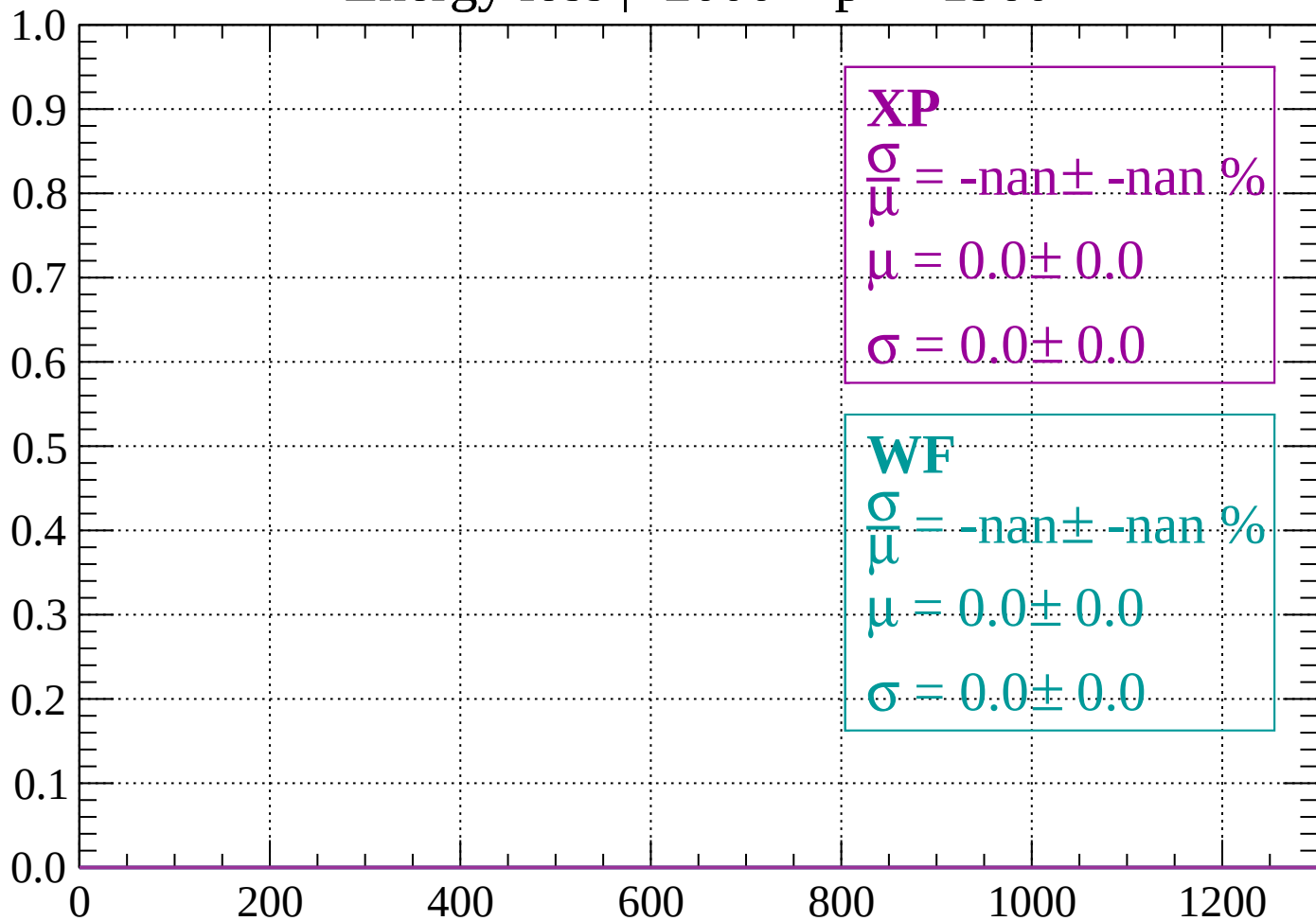






Energy loss | -2000 < p < -1960

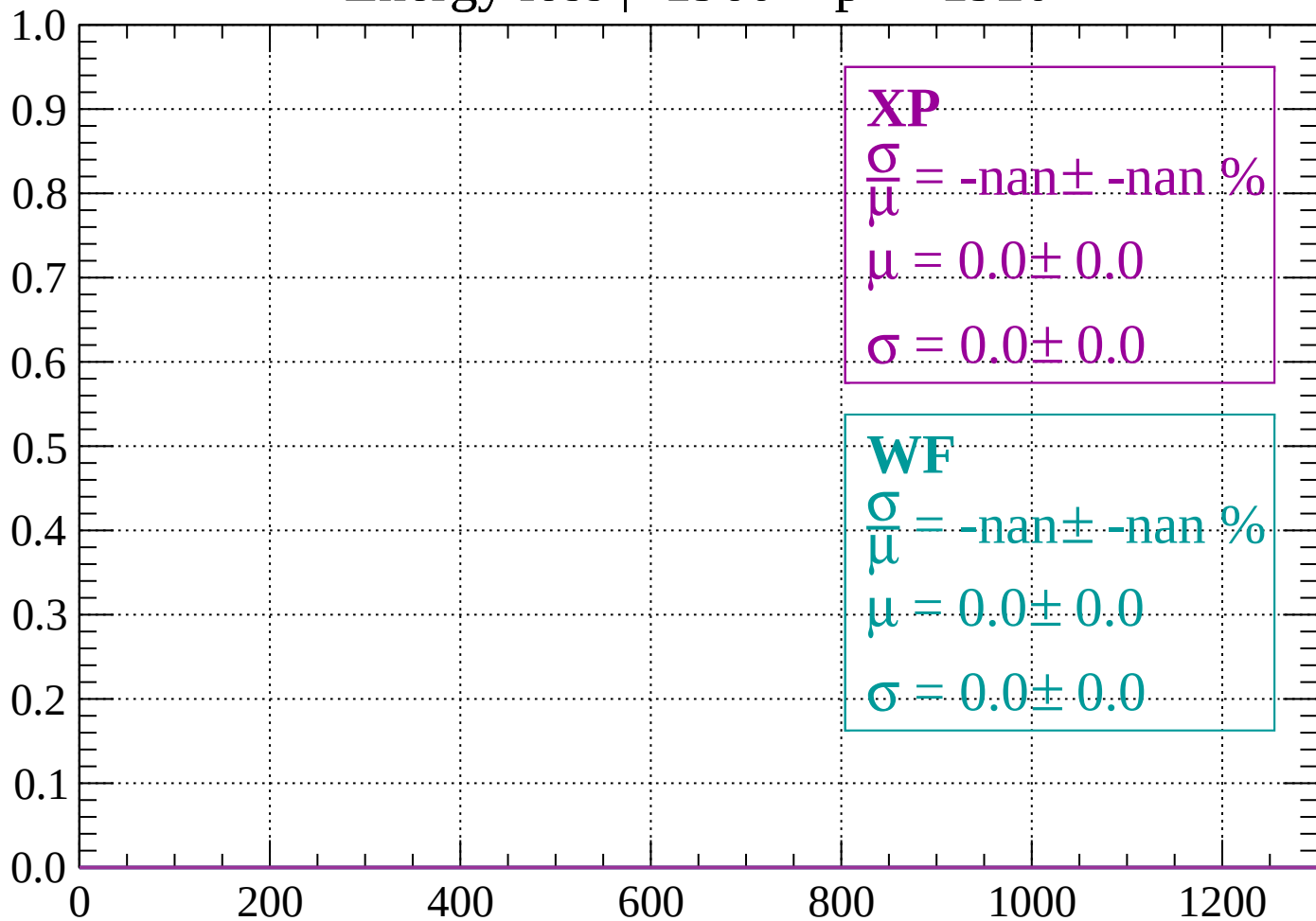
Count



dE/dx (ADC counts/cm)

Energy loss | -1960 < p < -1920

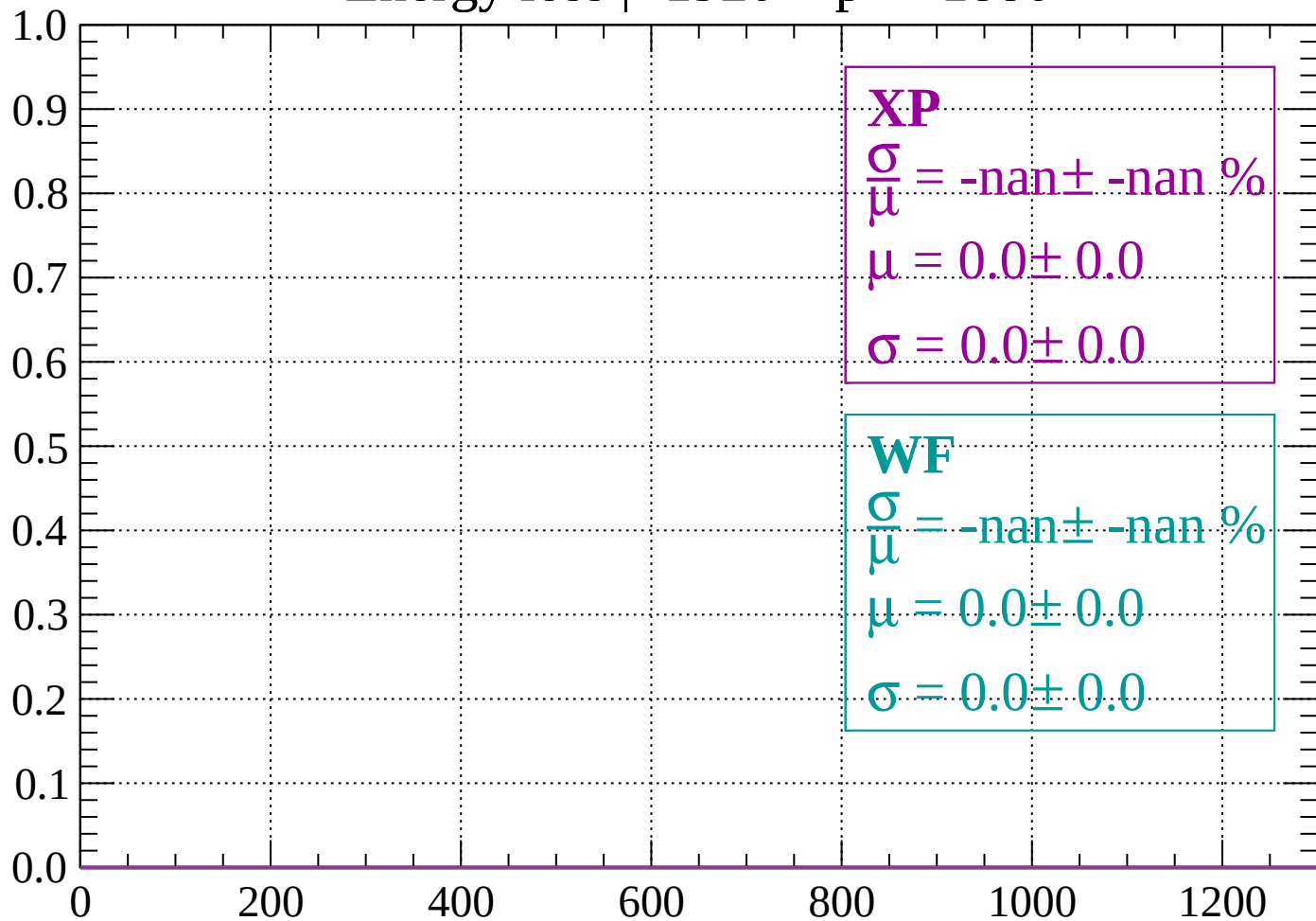
Count



dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1880

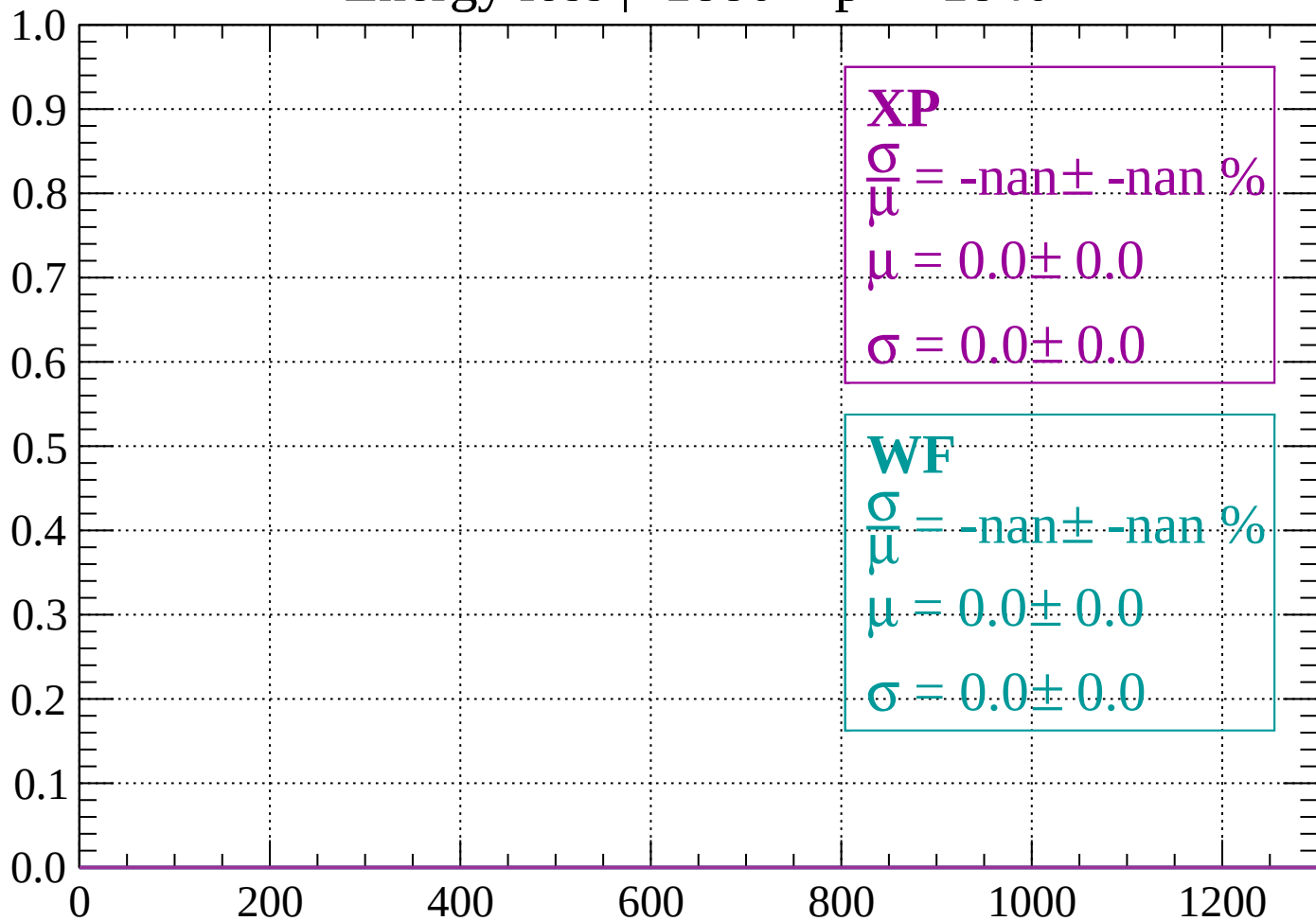
Count



dE/dx (ADC counts/cm)

Energy loss | -1880 < p < -1840

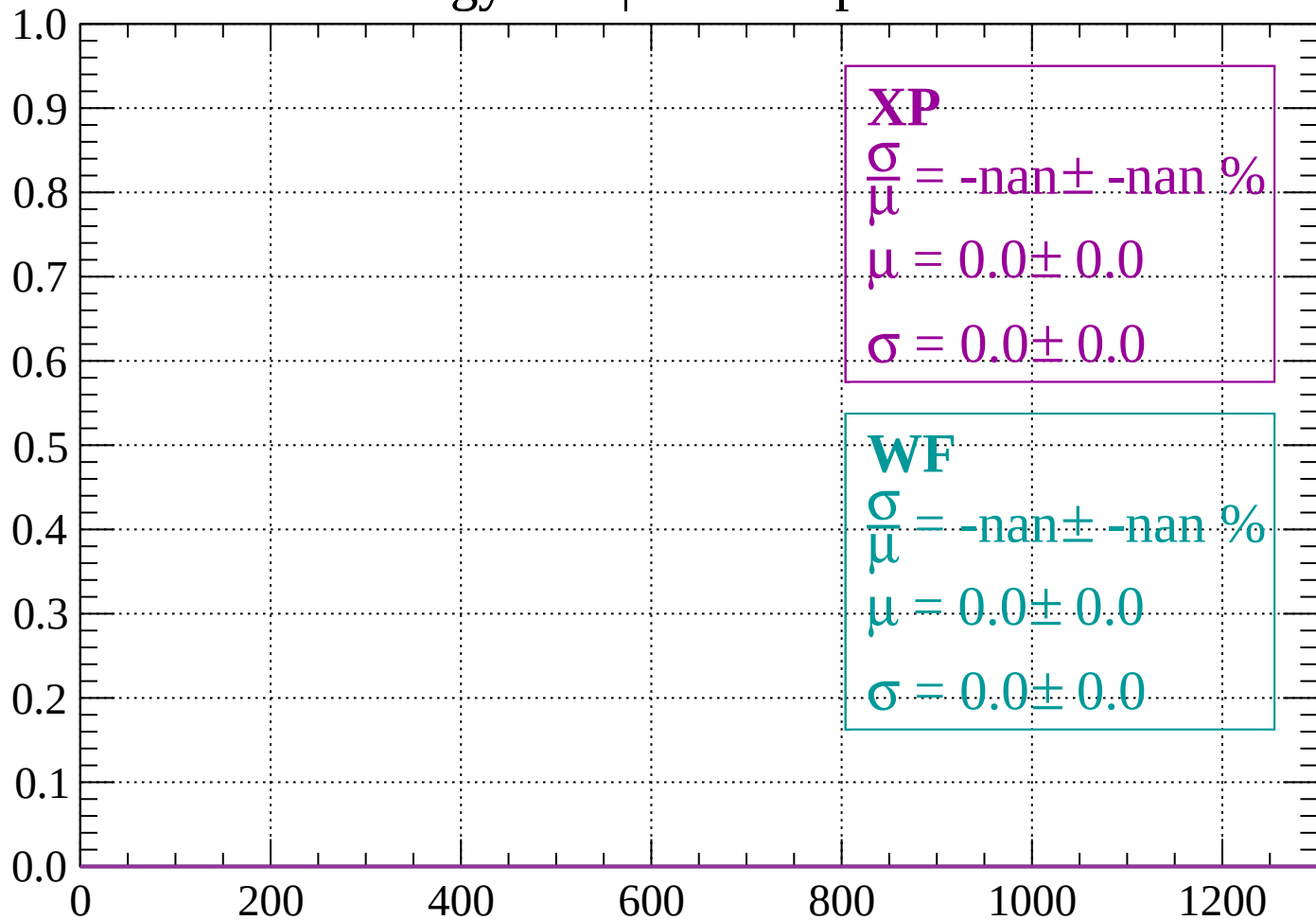
Count



dE/dx (ADC counts/cm)

Energy loss | -1840 < p < -1800

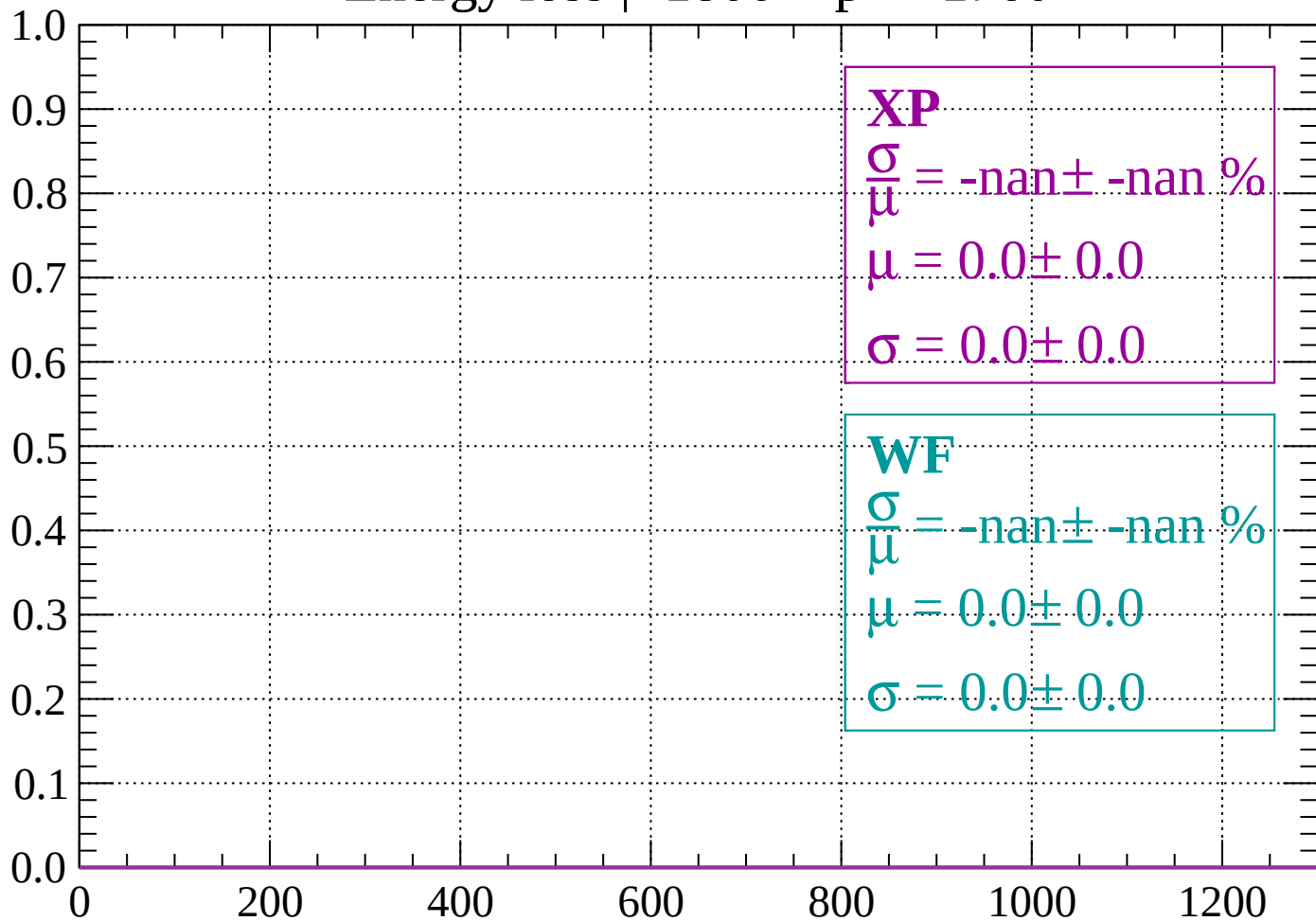
Count



dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

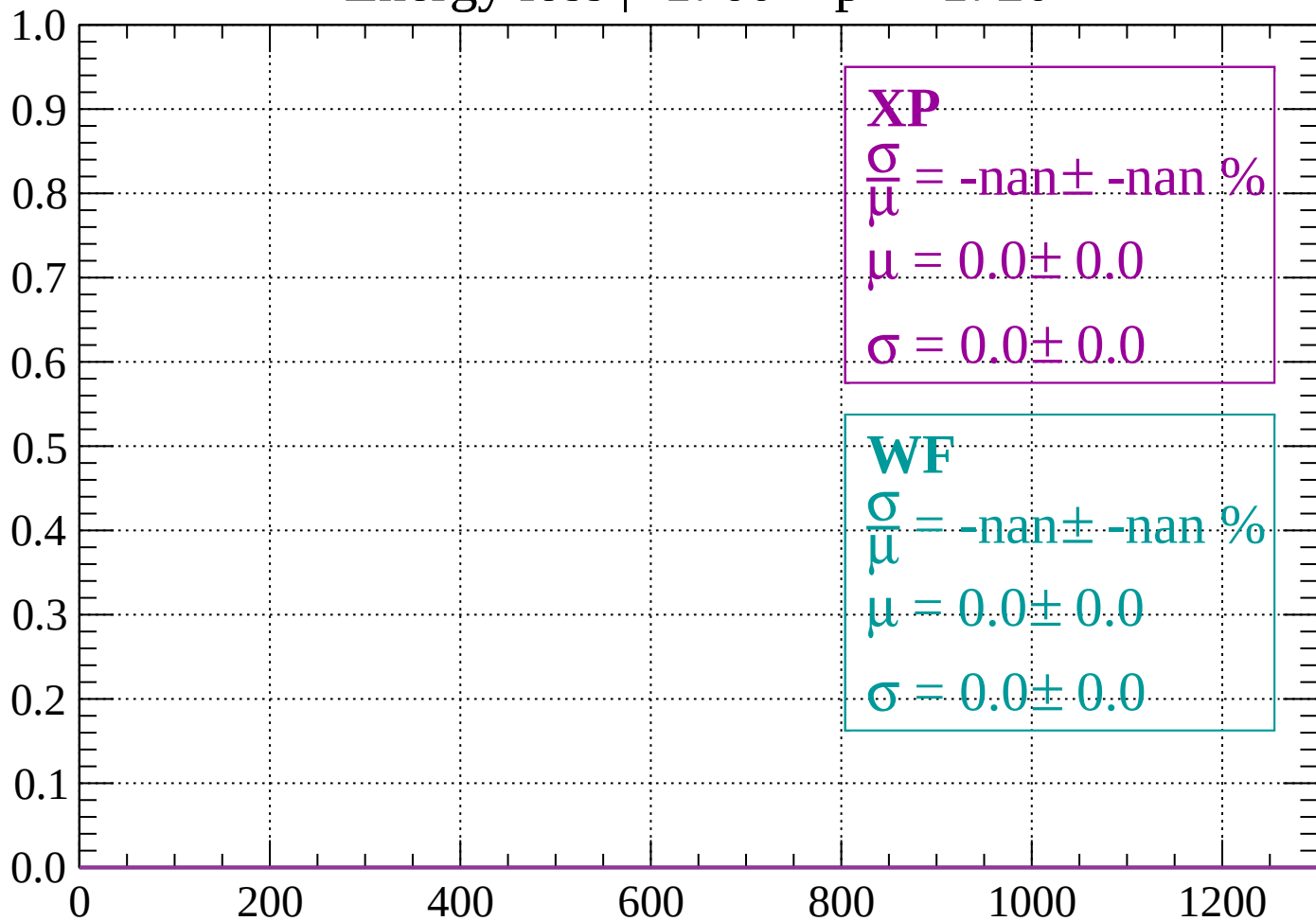
Count



dE/dx (ADC counts/cm)

Energy loss | $-1760 < p < -1720$

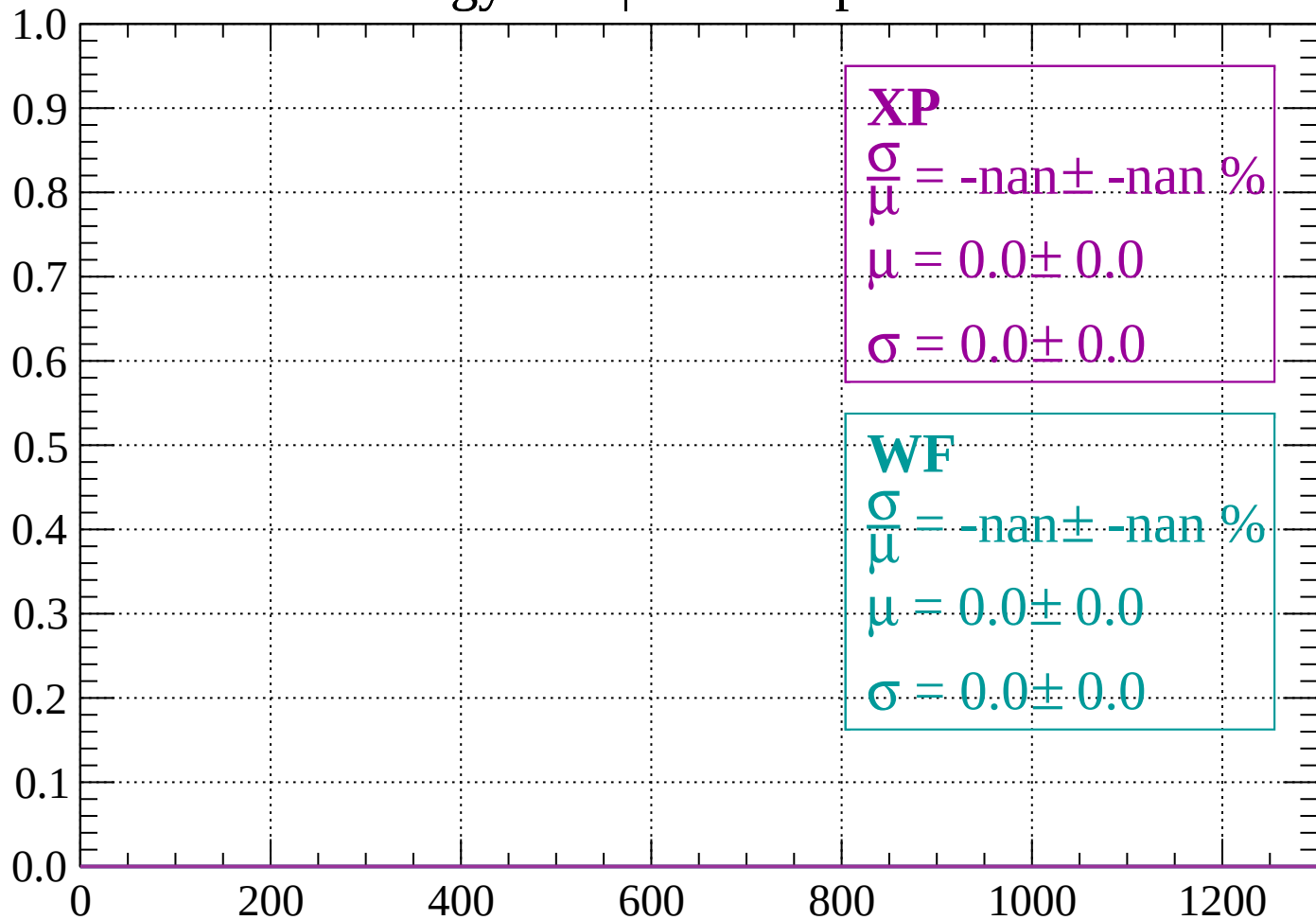
Count



dE/dx (ADC counts/cm)

Energy loss | -1720 < p < -1680

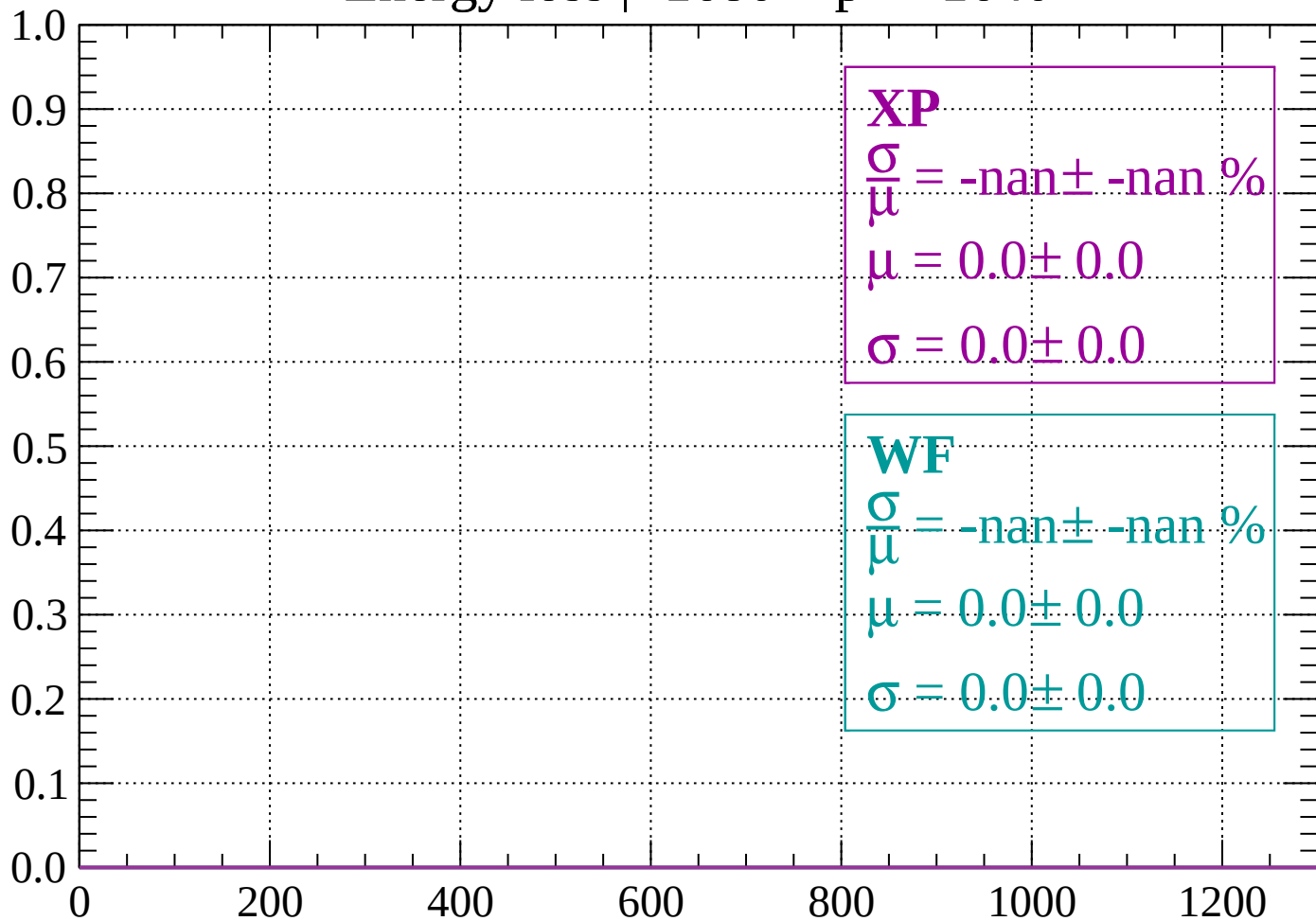
Count



dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1640

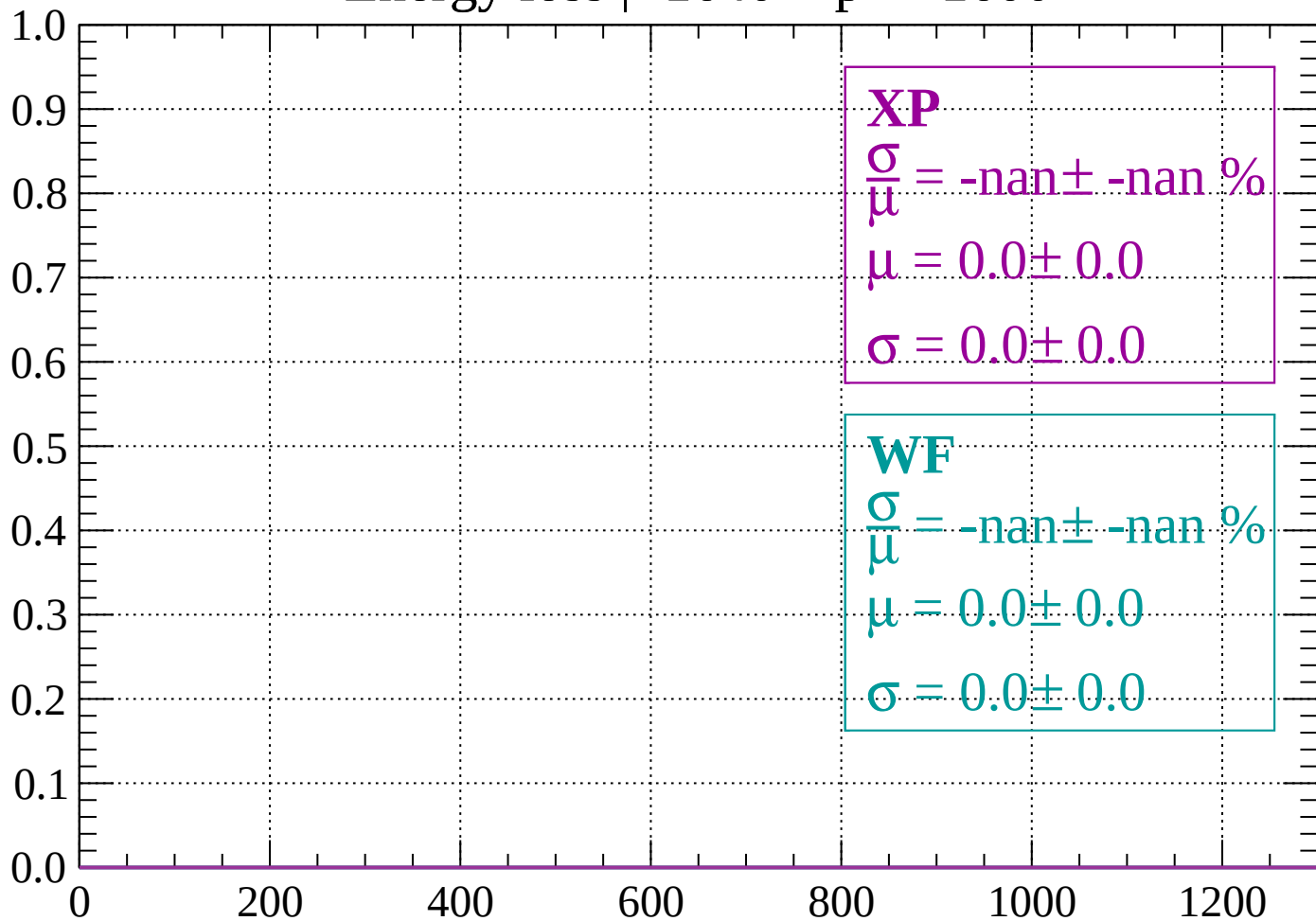
Count



dE/dx (ADC counts/cm)

Energy loss | -1640 < p < -1600

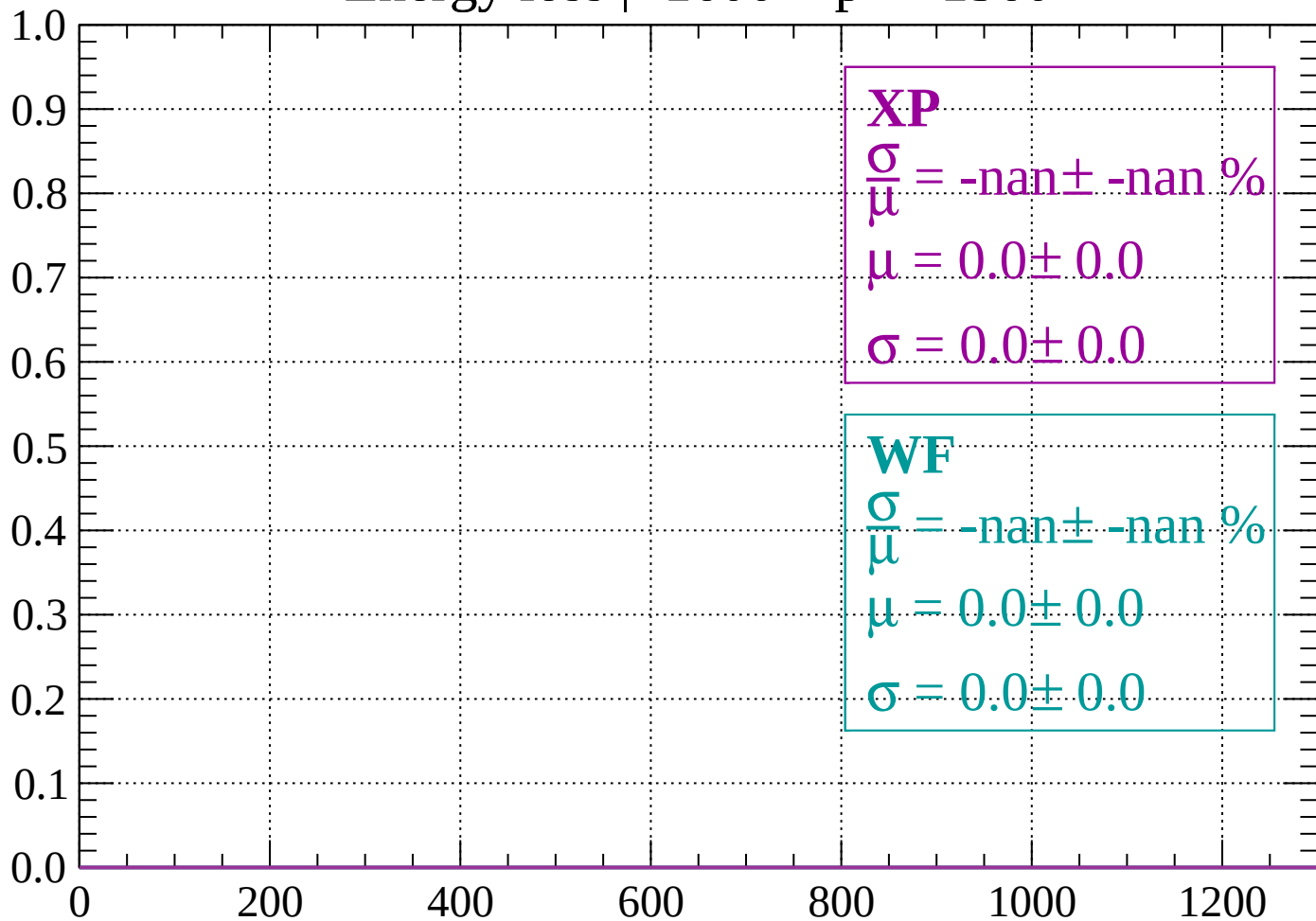
Count



dE/dx (ADC counts/cm)

Energy loss | -1600 < p < -1560

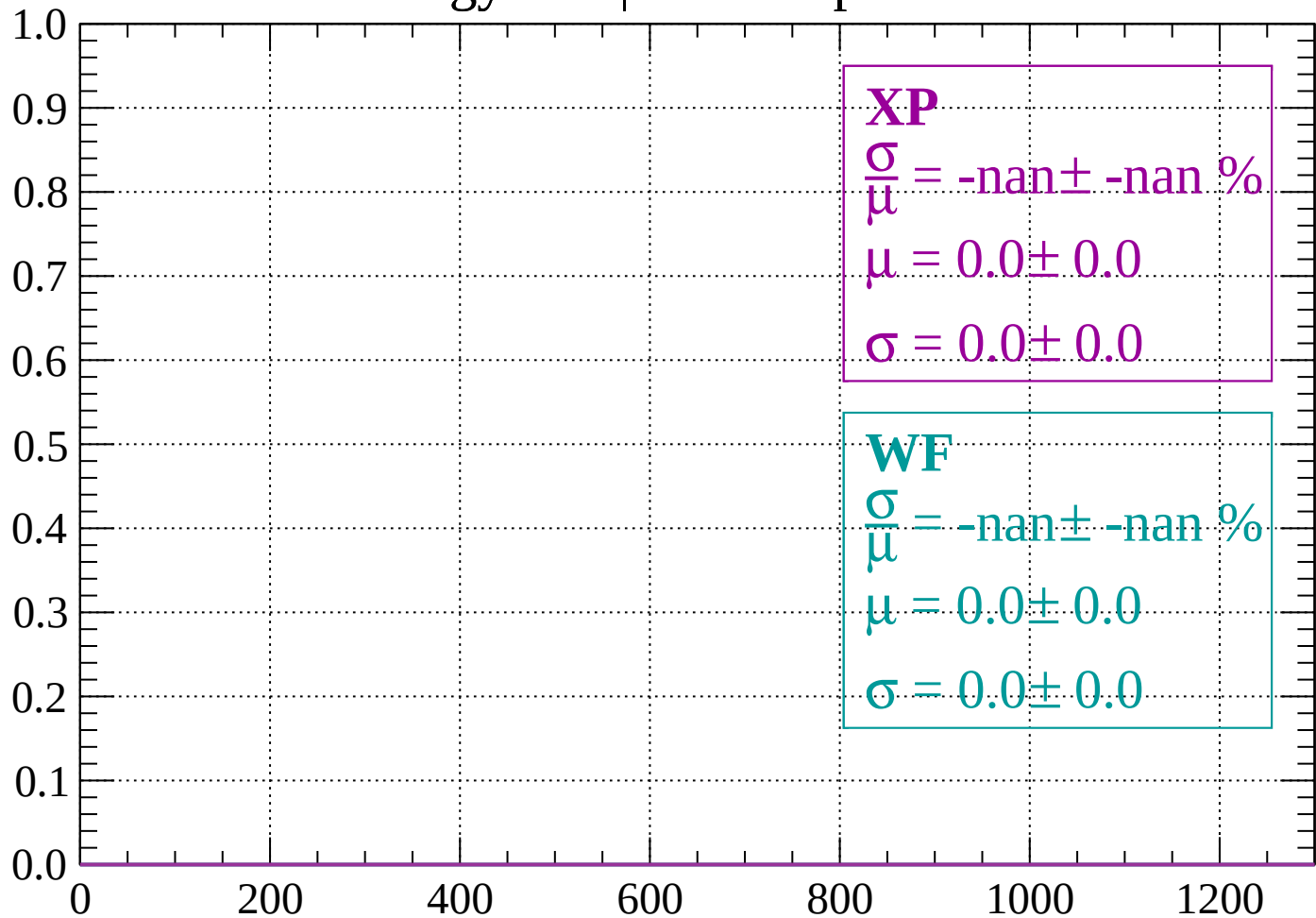
Count



dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1520

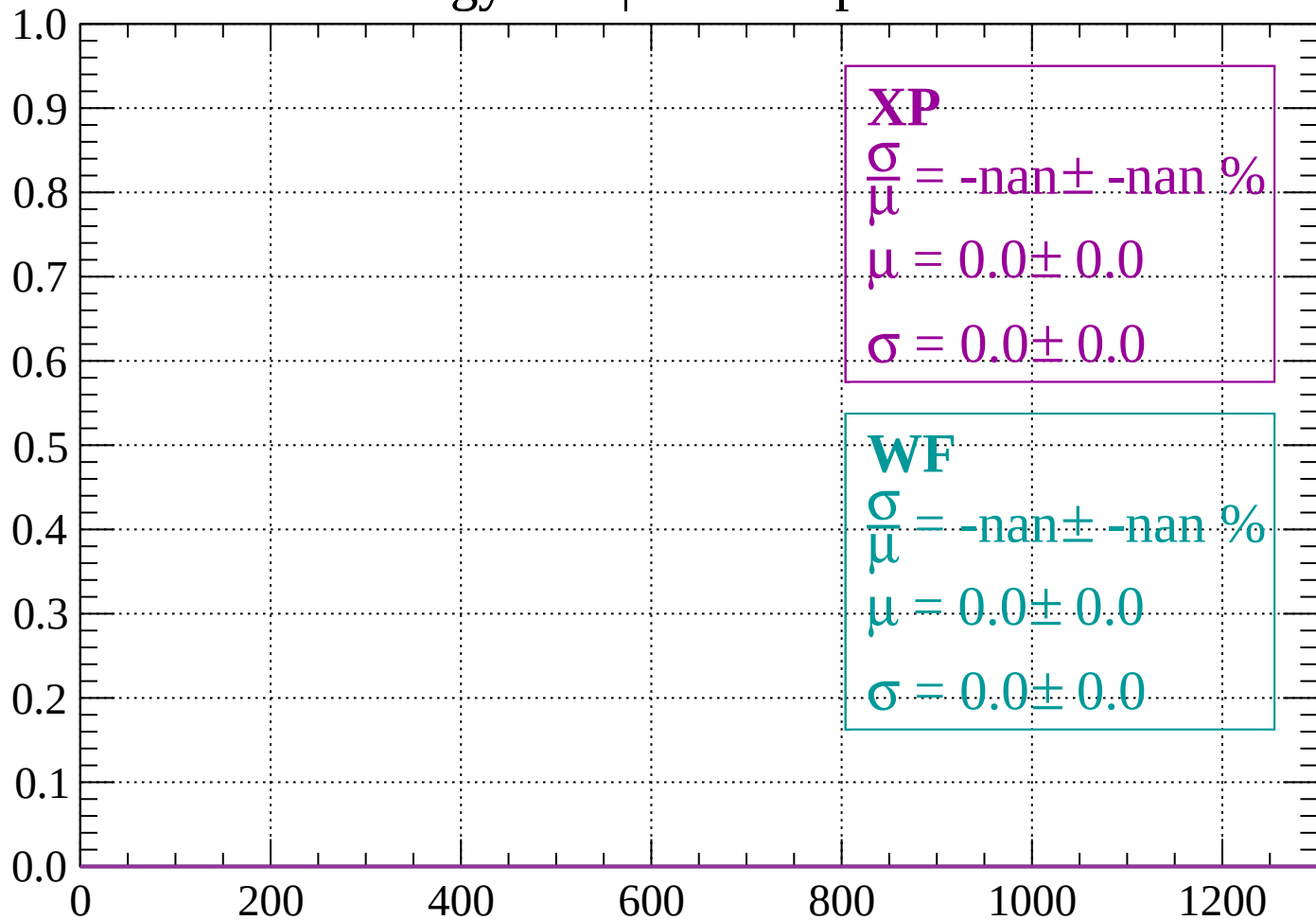
Count



dE/dx (ADC counts/cm)

Energy loss | -1520 < p < -1480

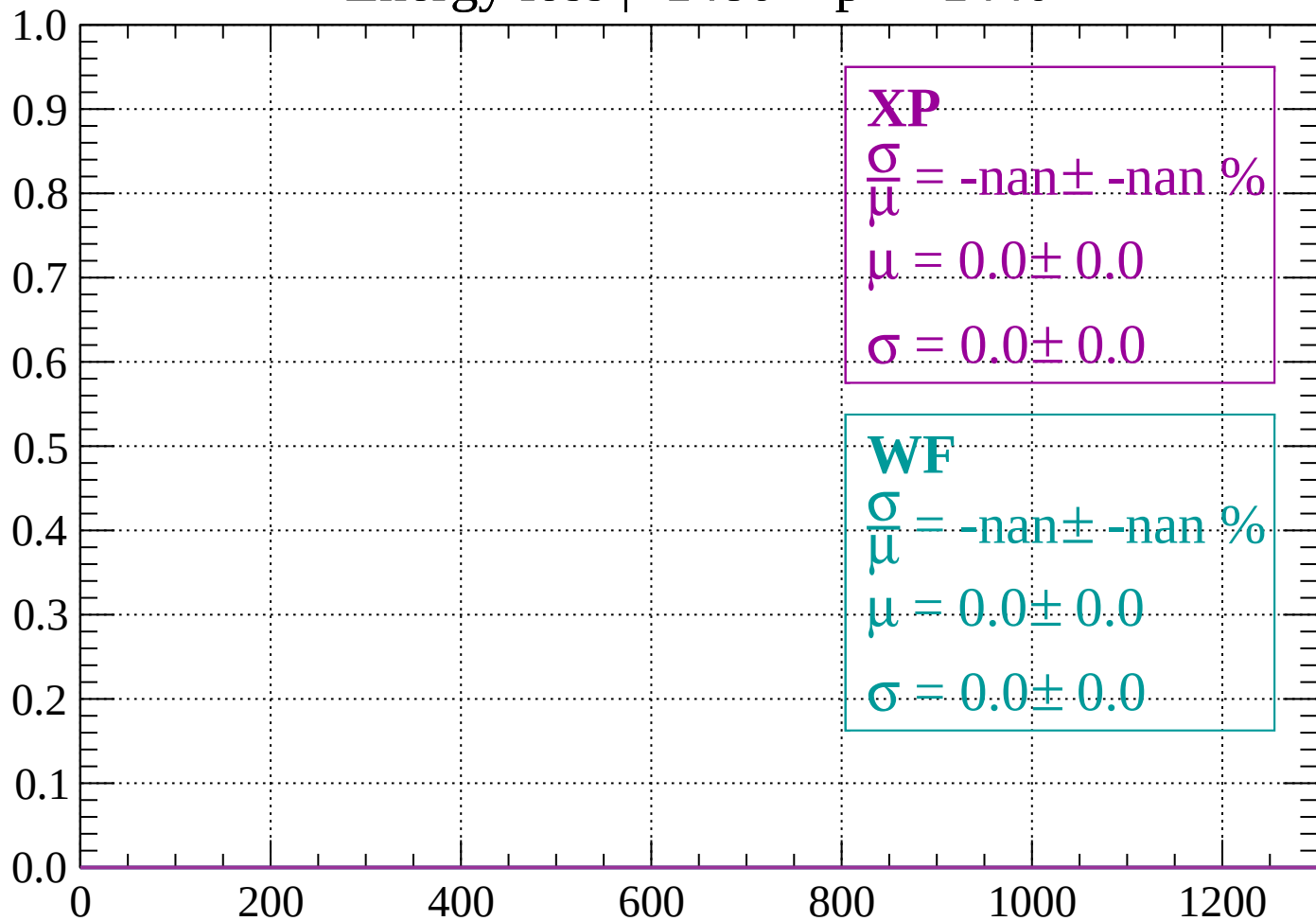
Count



dE/dx (ADC counts/cm)

Energy loss | -1480 < p < -1440

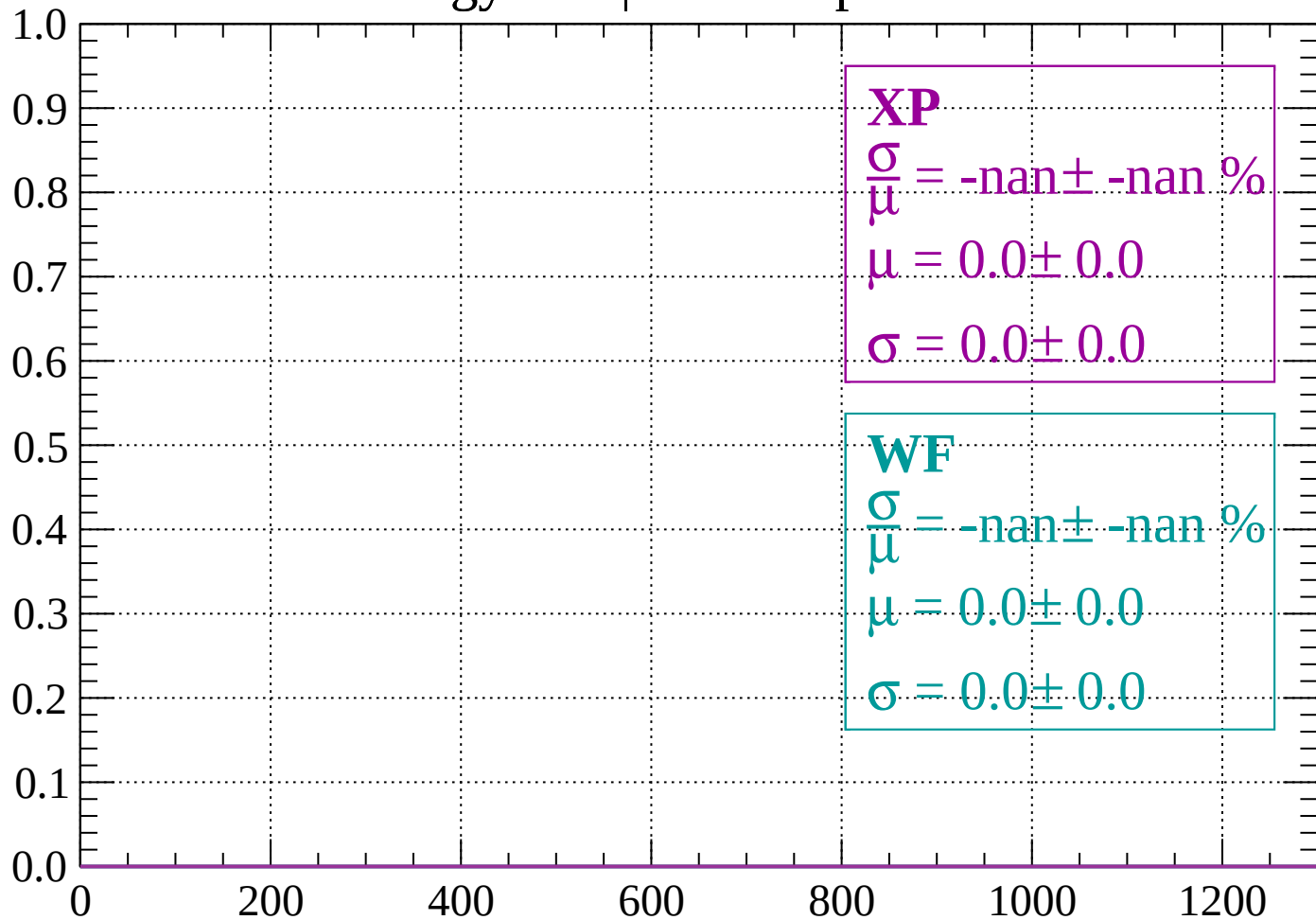
Count



dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$

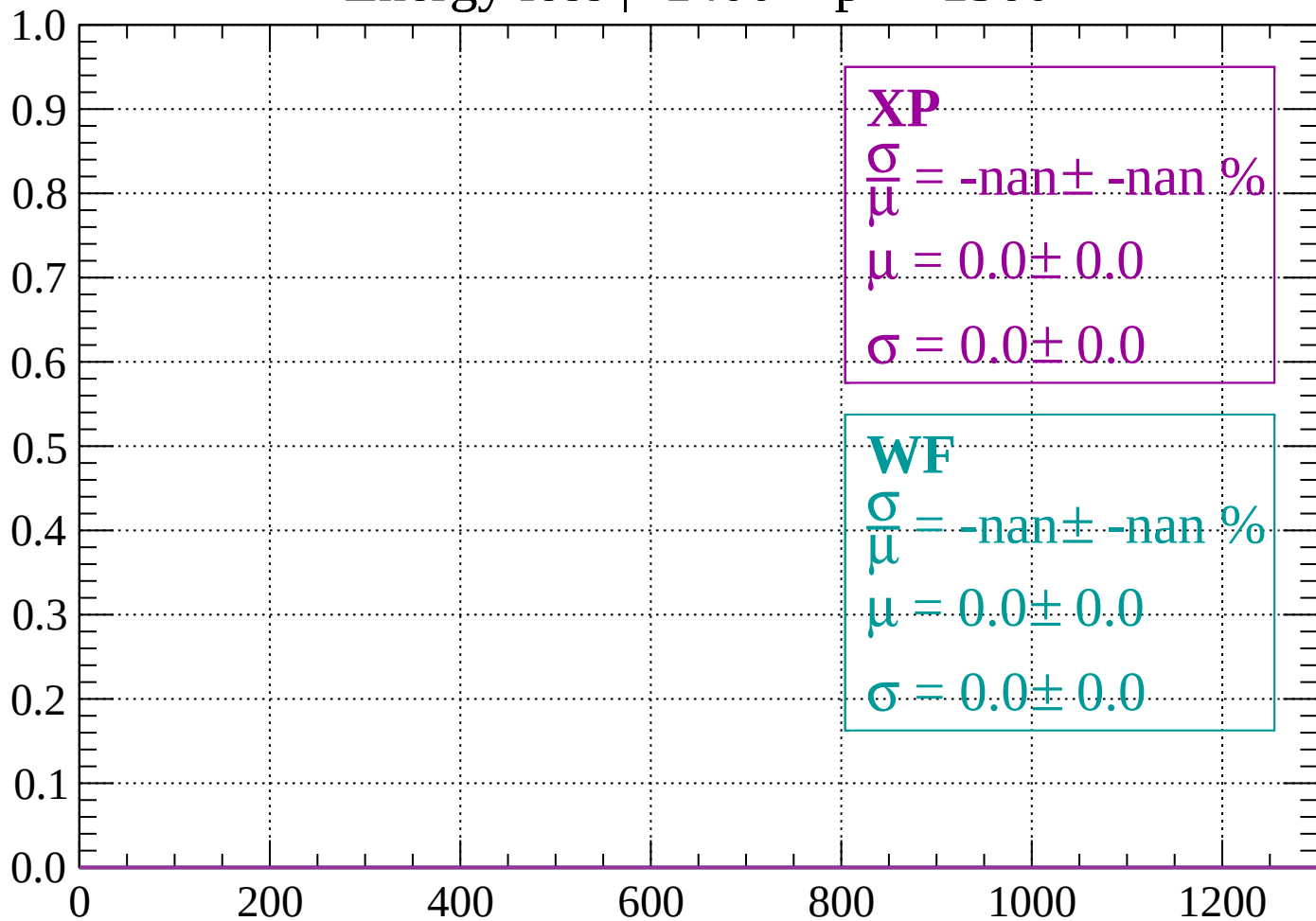
Count



dE/dx (ADC counts/cm)

Energy loss | -1400 < p < -1360

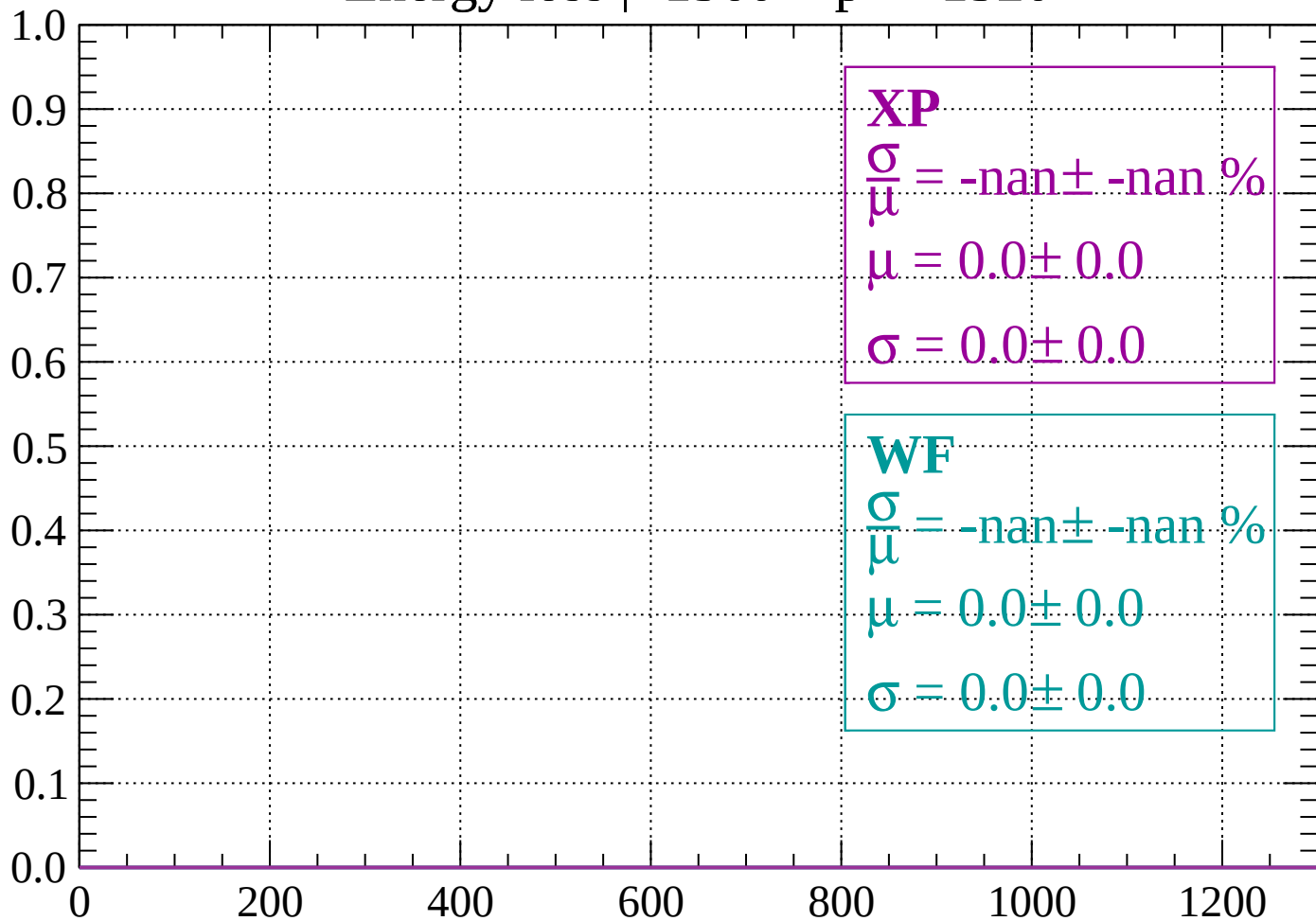
Count



dE/dx (ADC counts/cm)

Energy loss | -1360 < p < -1320

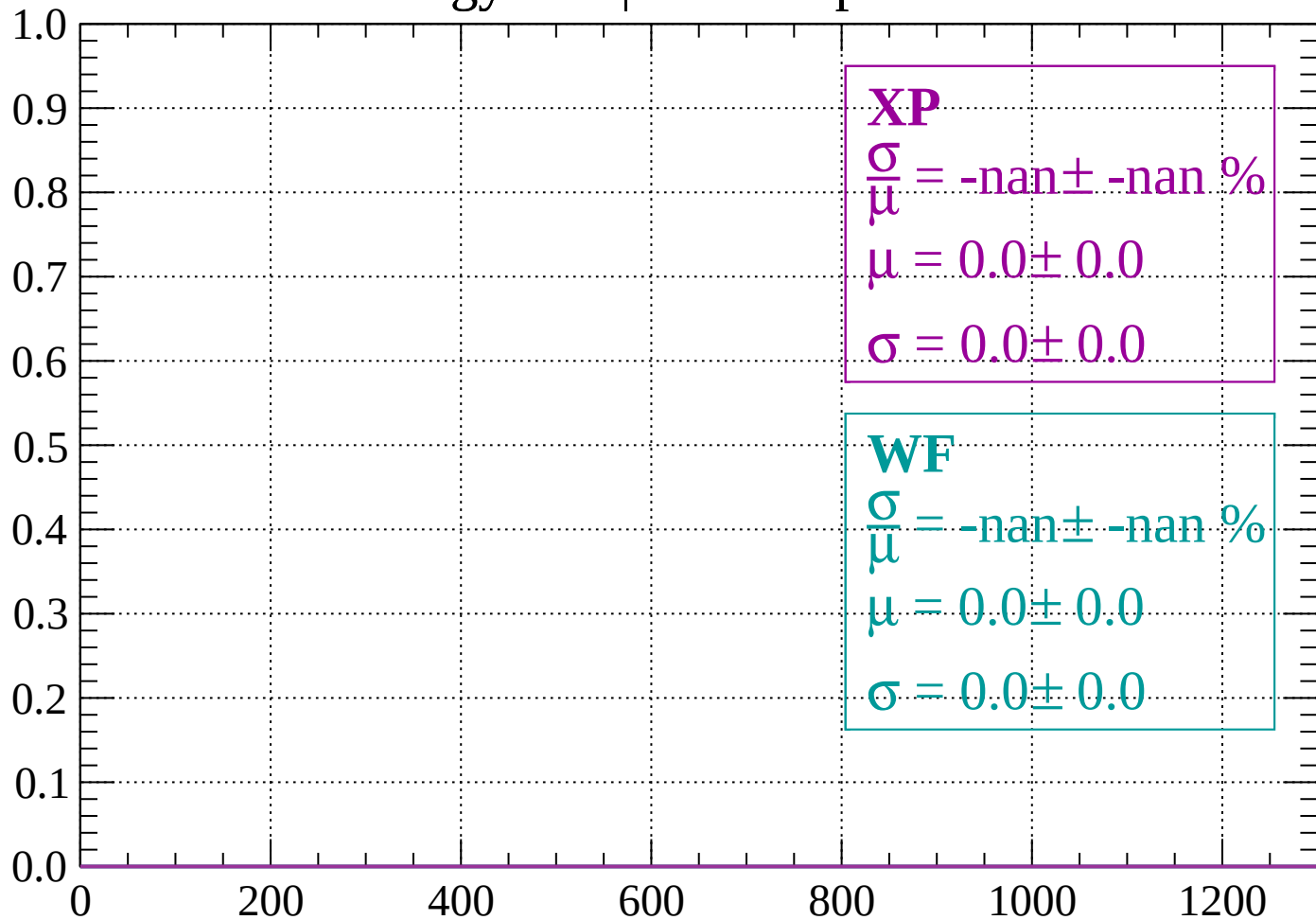
Count



dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1280

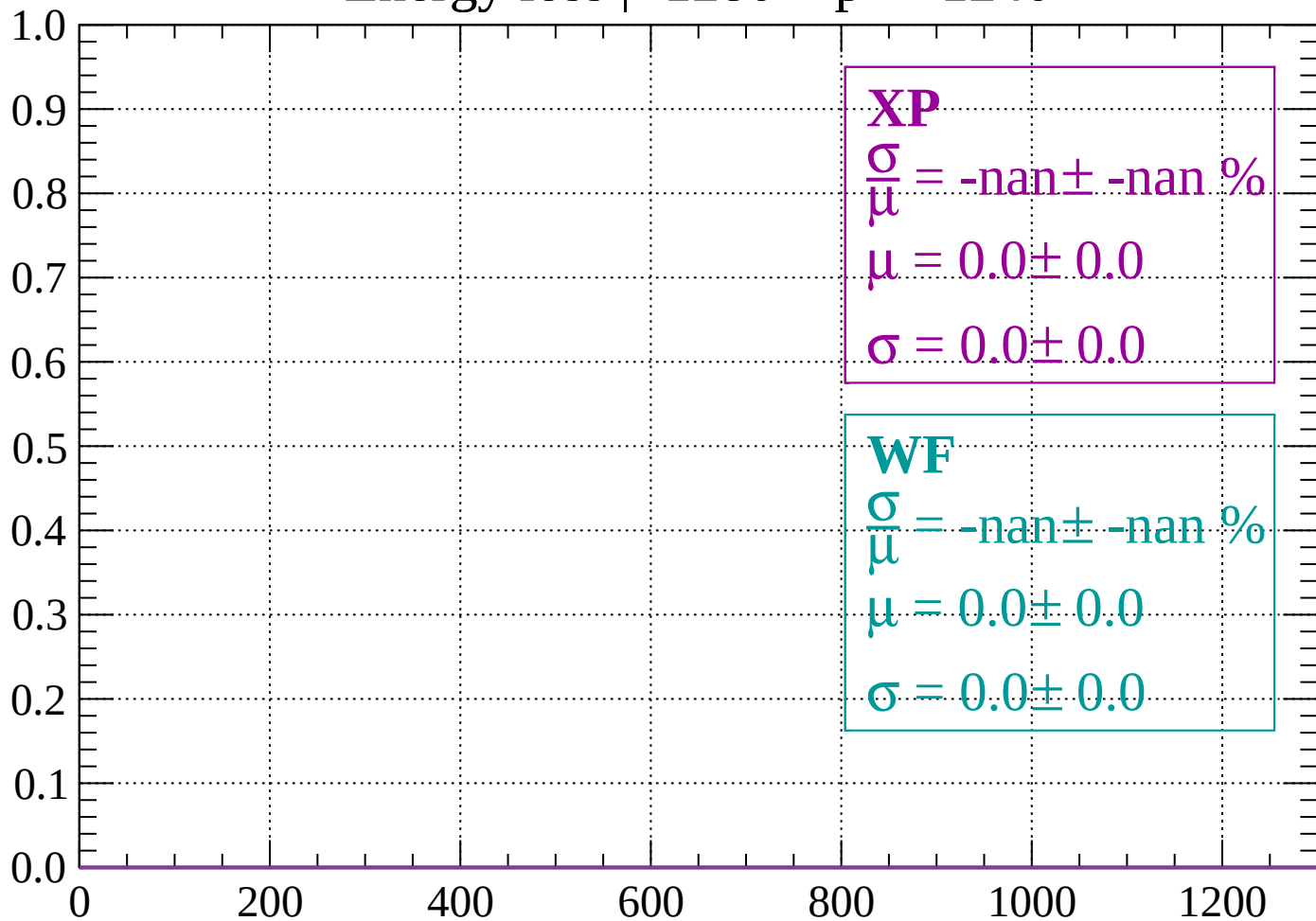
Count



dE/dx (ADC counts/cm)

Energy loss | -1280 < p < -1240

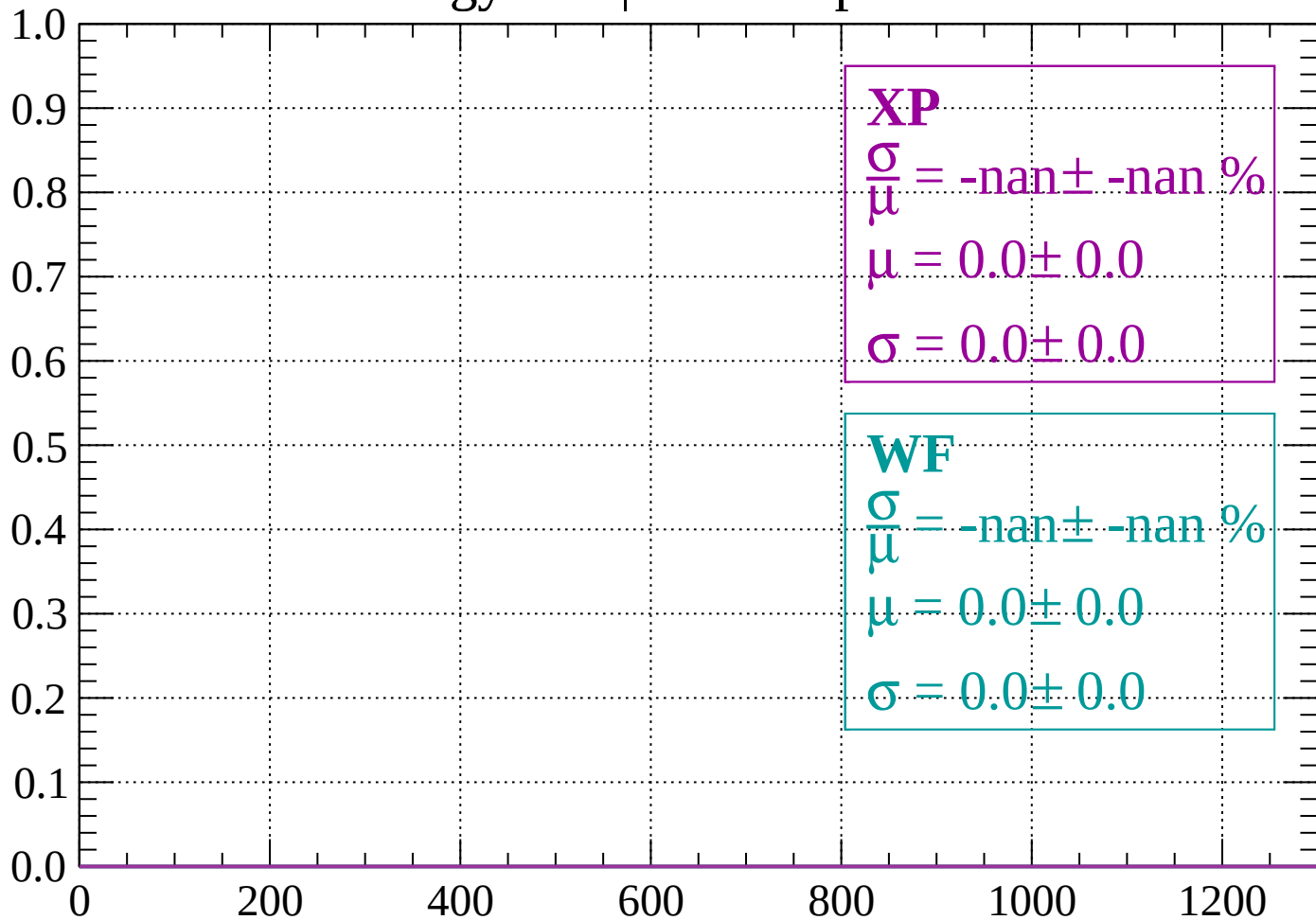
Count



dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

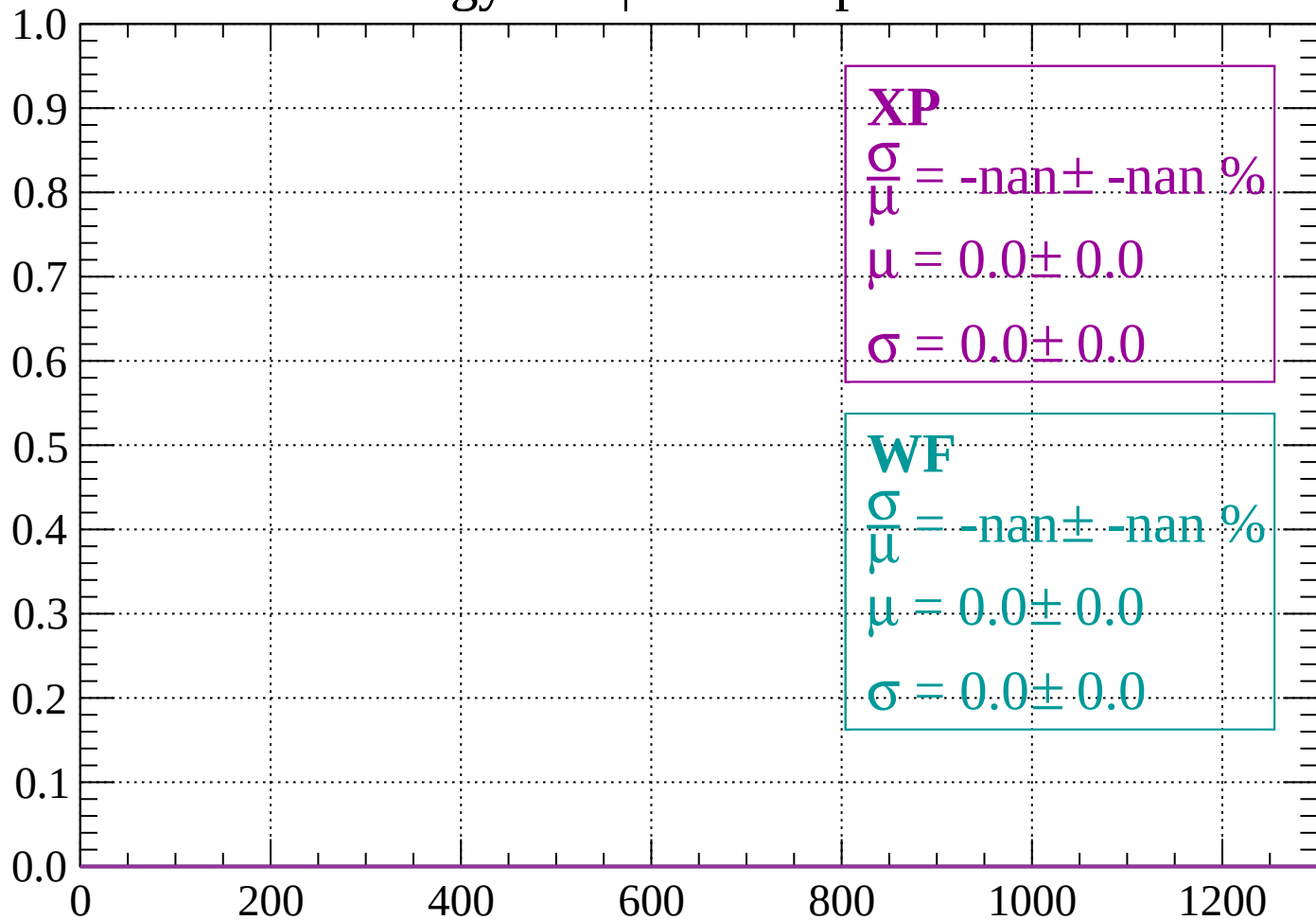
Count



dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1160$

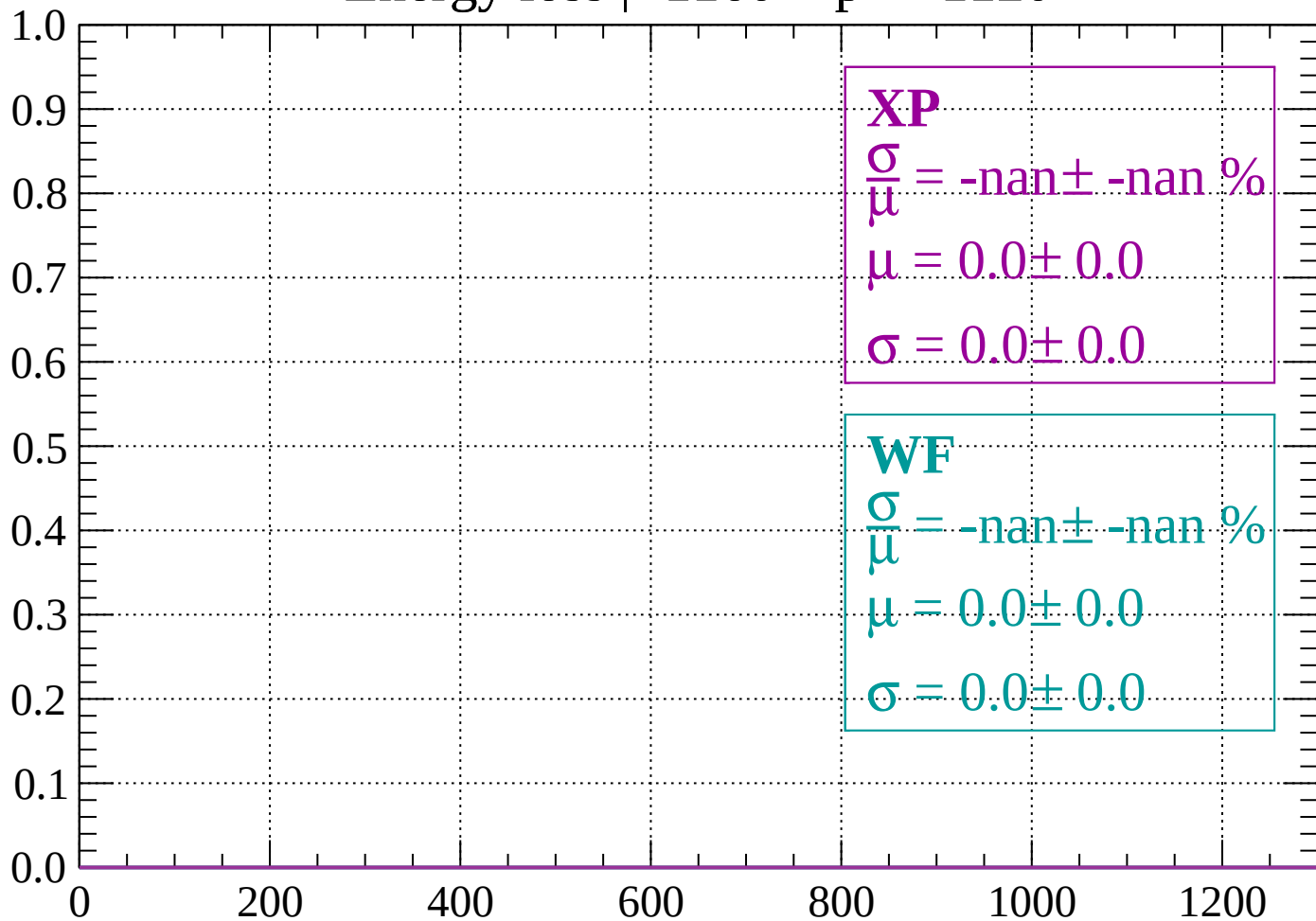
Count



dE/dx (ADC counts/cm)

Energy loss | $-1160 < p < -1120$

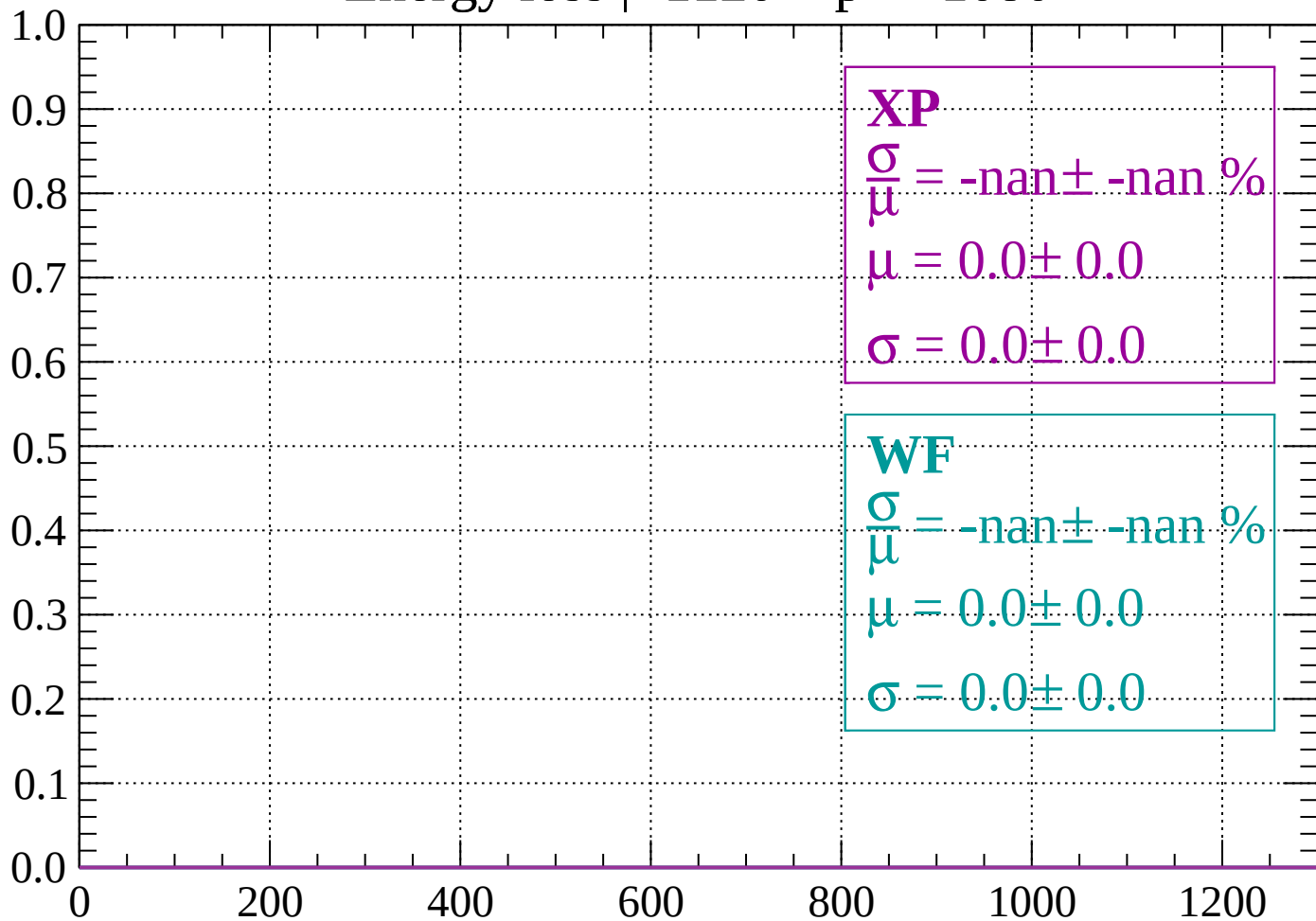
Count



dE/dx (ADC counts/cm)

Energy loss | -1120 < p < -1080

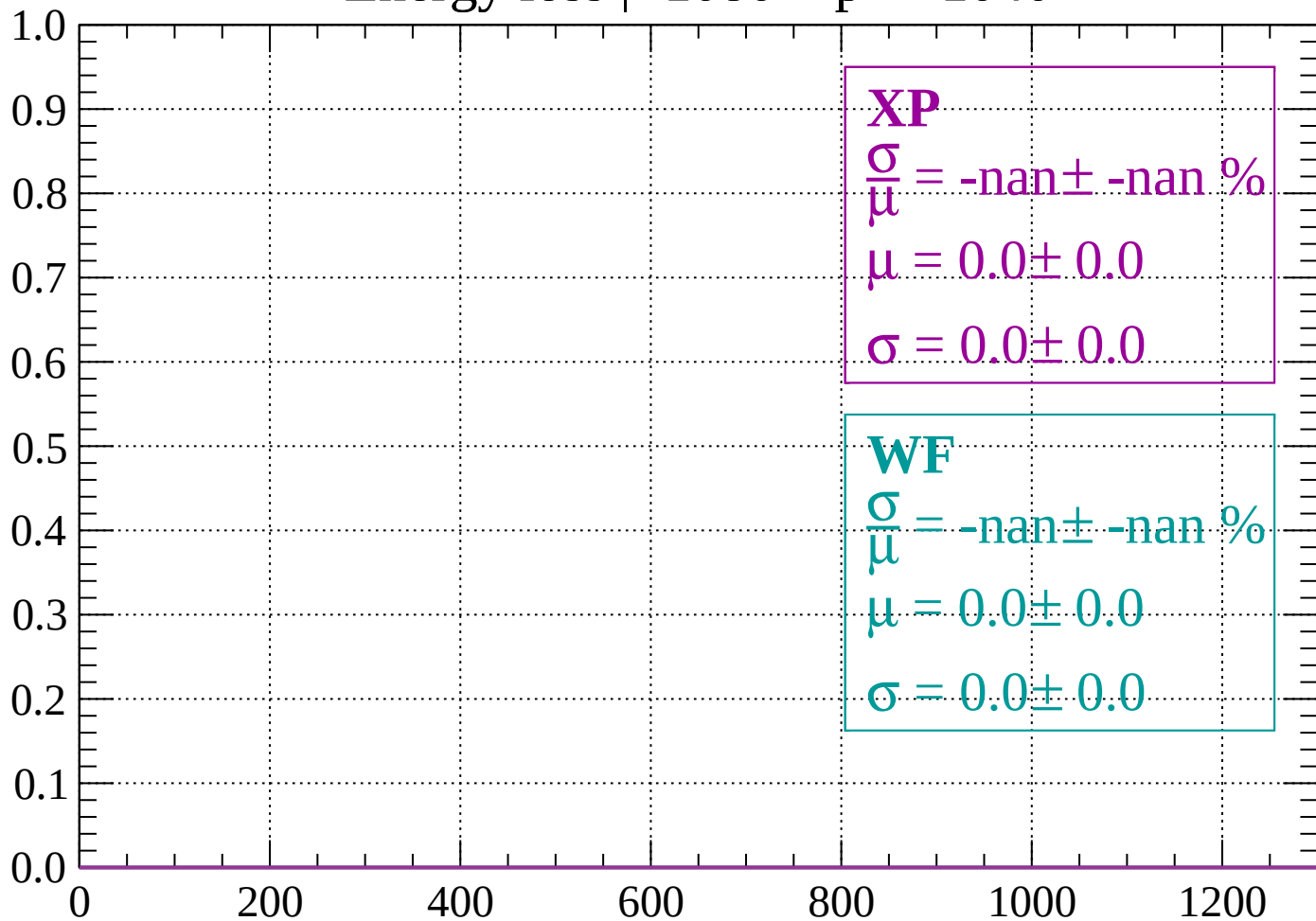
Count



dE/dx (ADC counts/cm)

Energy loss | -1080 < p < -1040

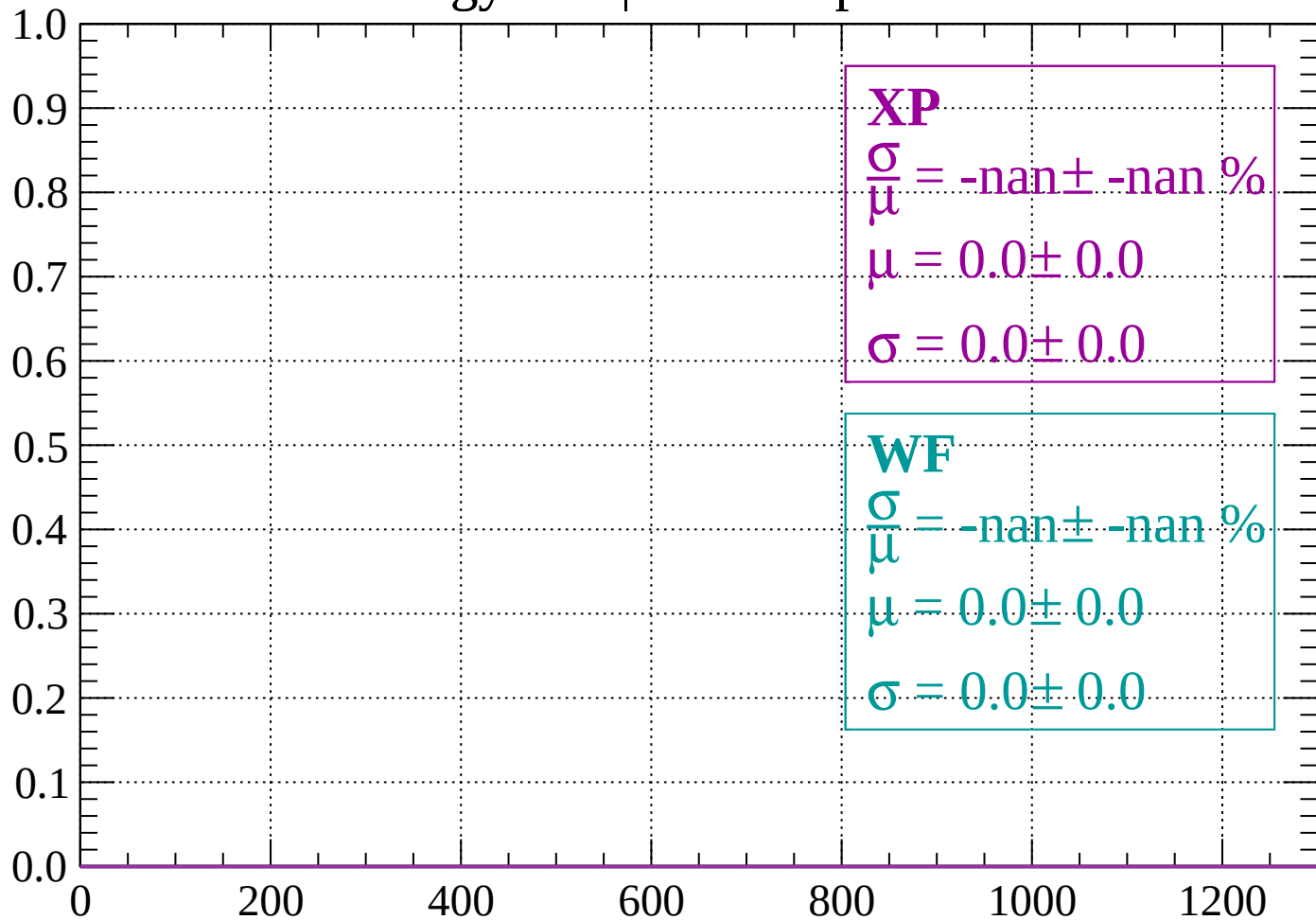
Count



dE/dx (ADC counts/cm)

Energy loss | -1040 < p < -1000

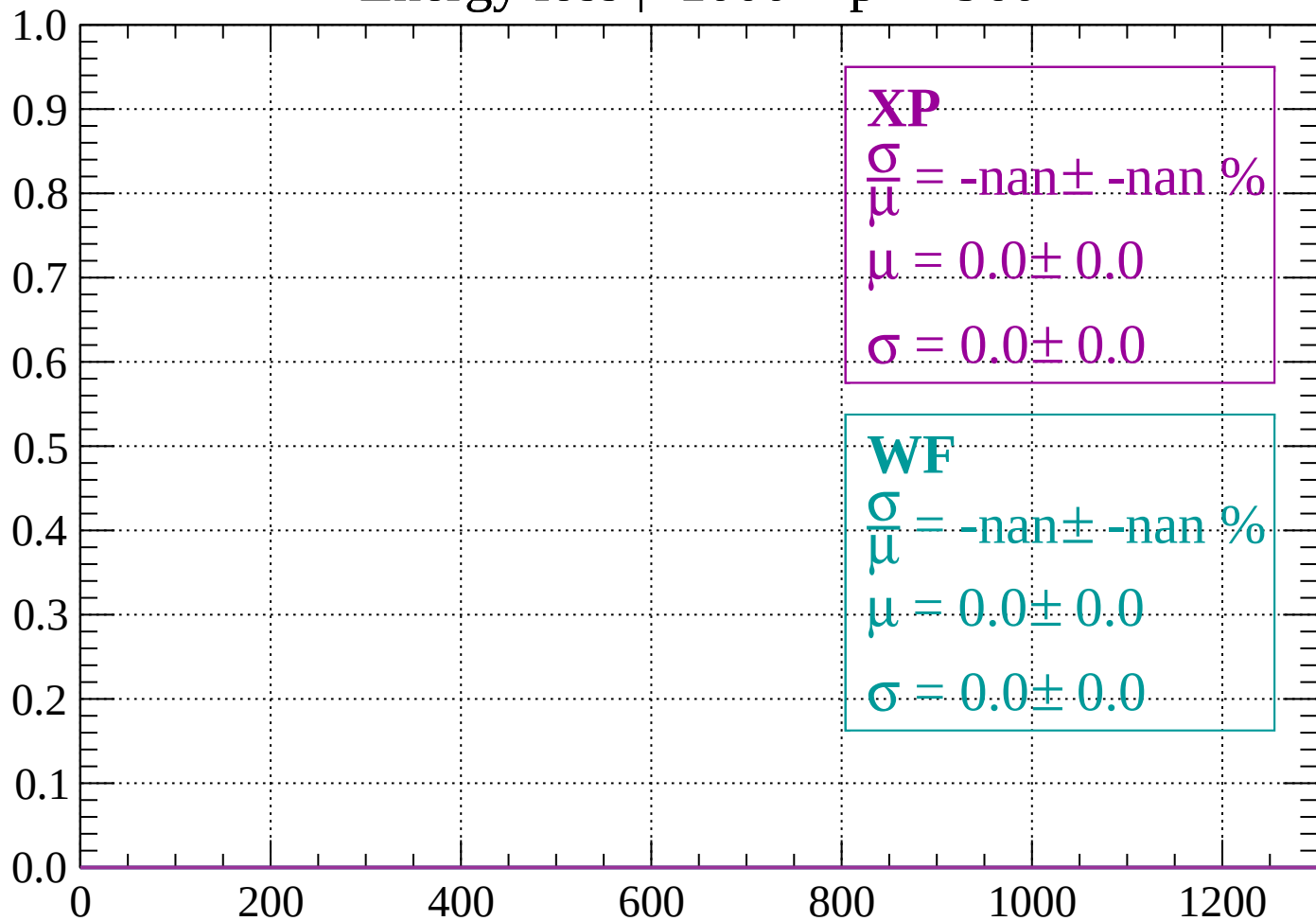
Count



dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

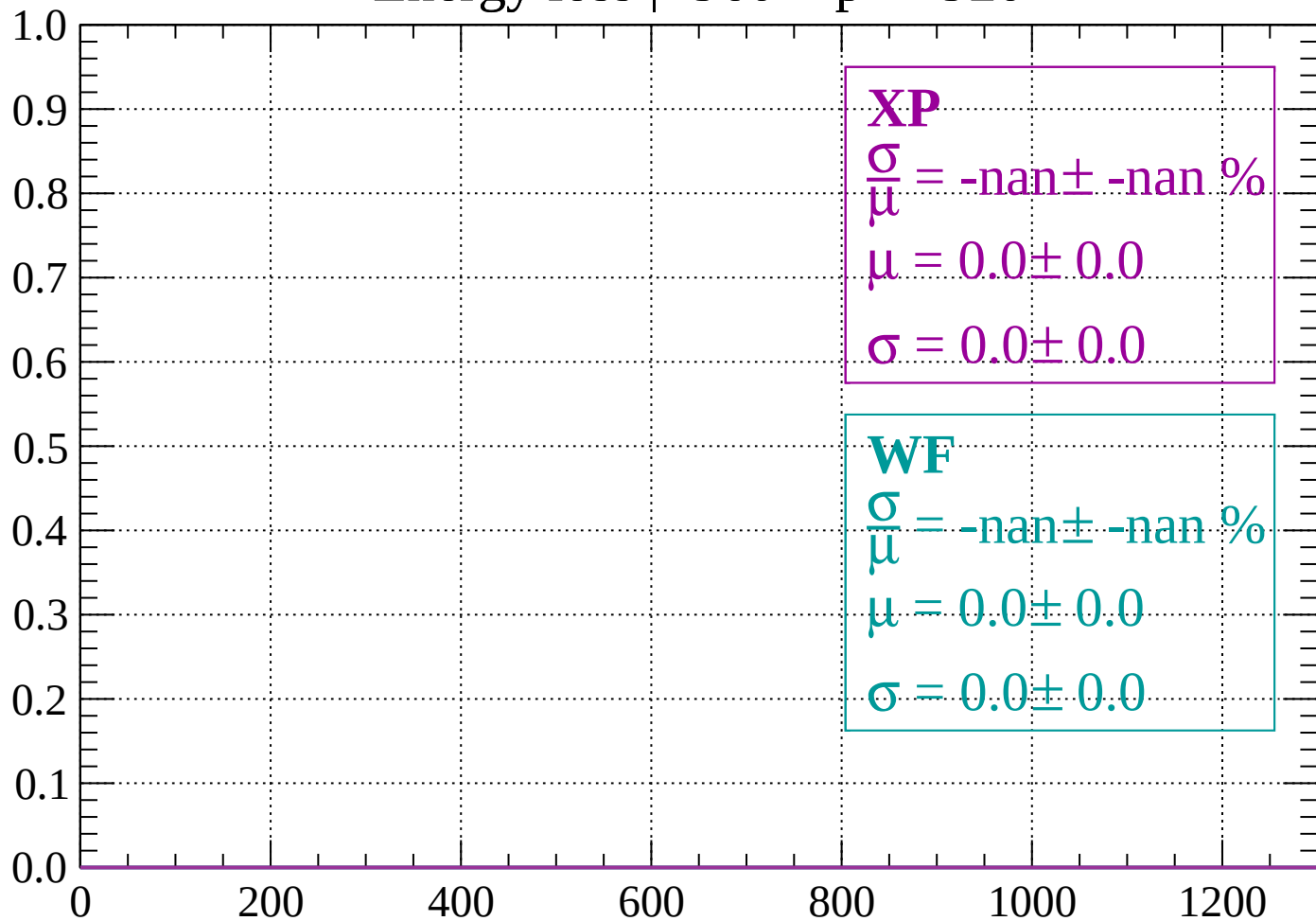
Count



dE/dx (ADC counts/cm)

Energy loss | -960 < p < -920

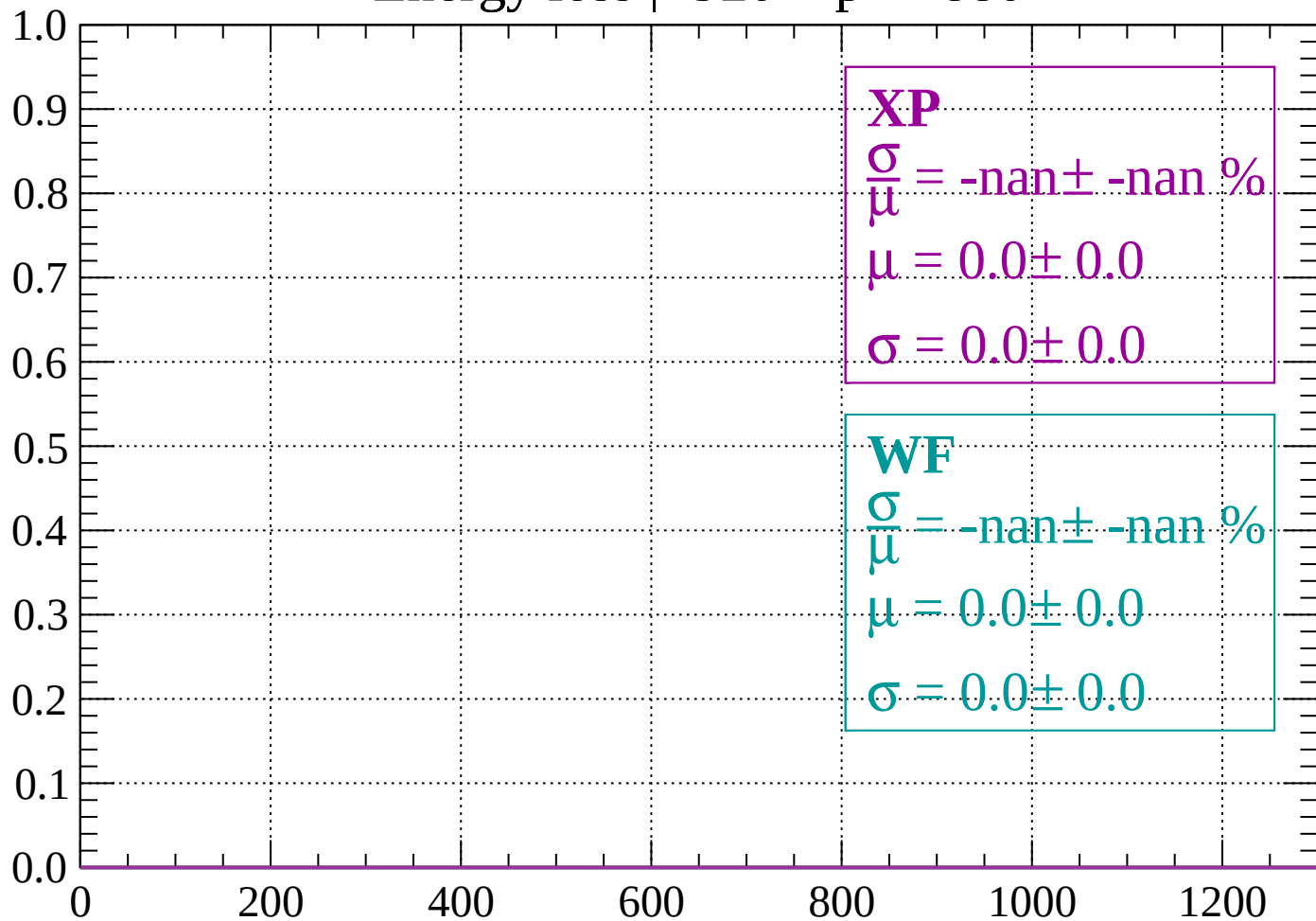
Count



dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

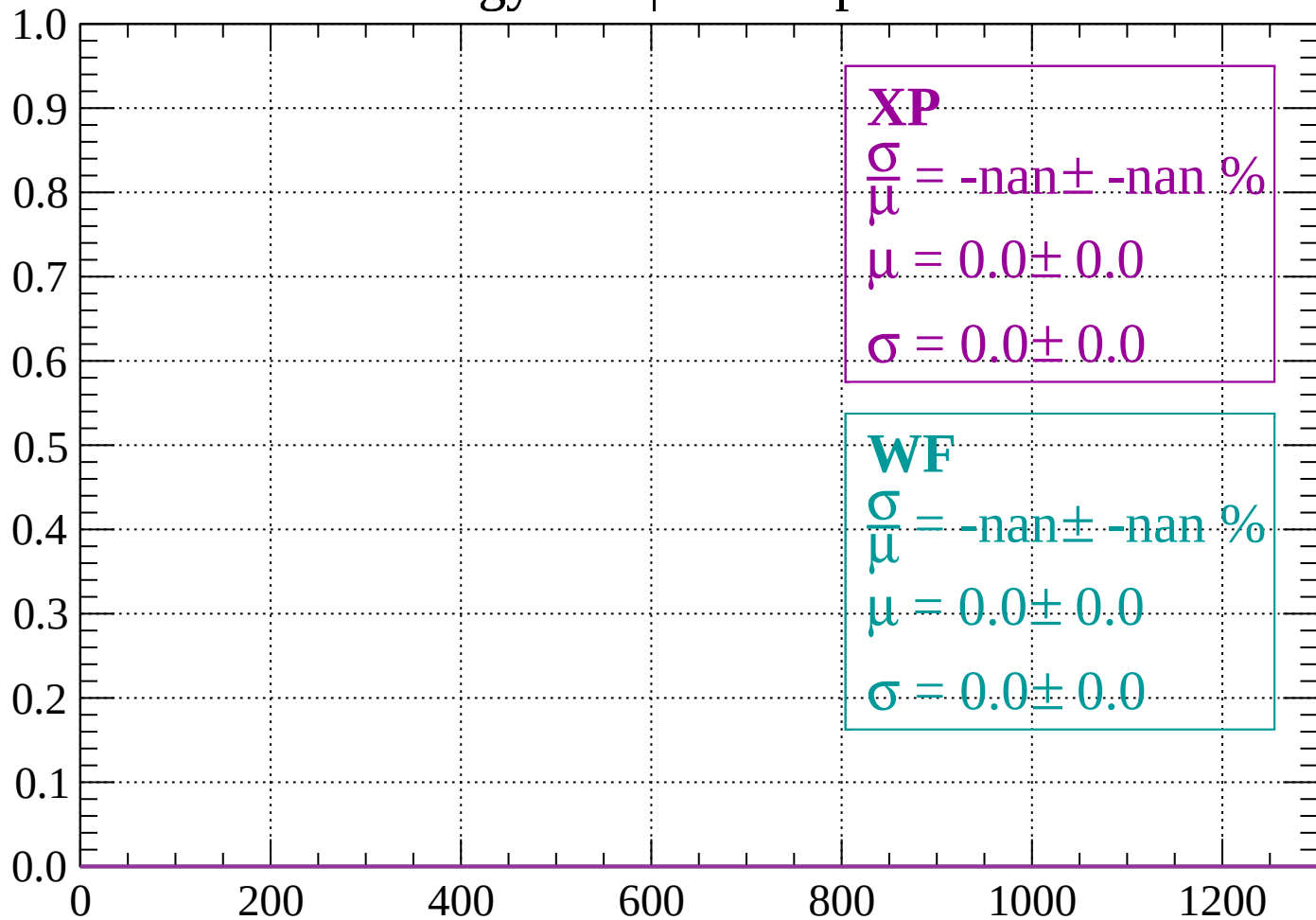
Count



dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

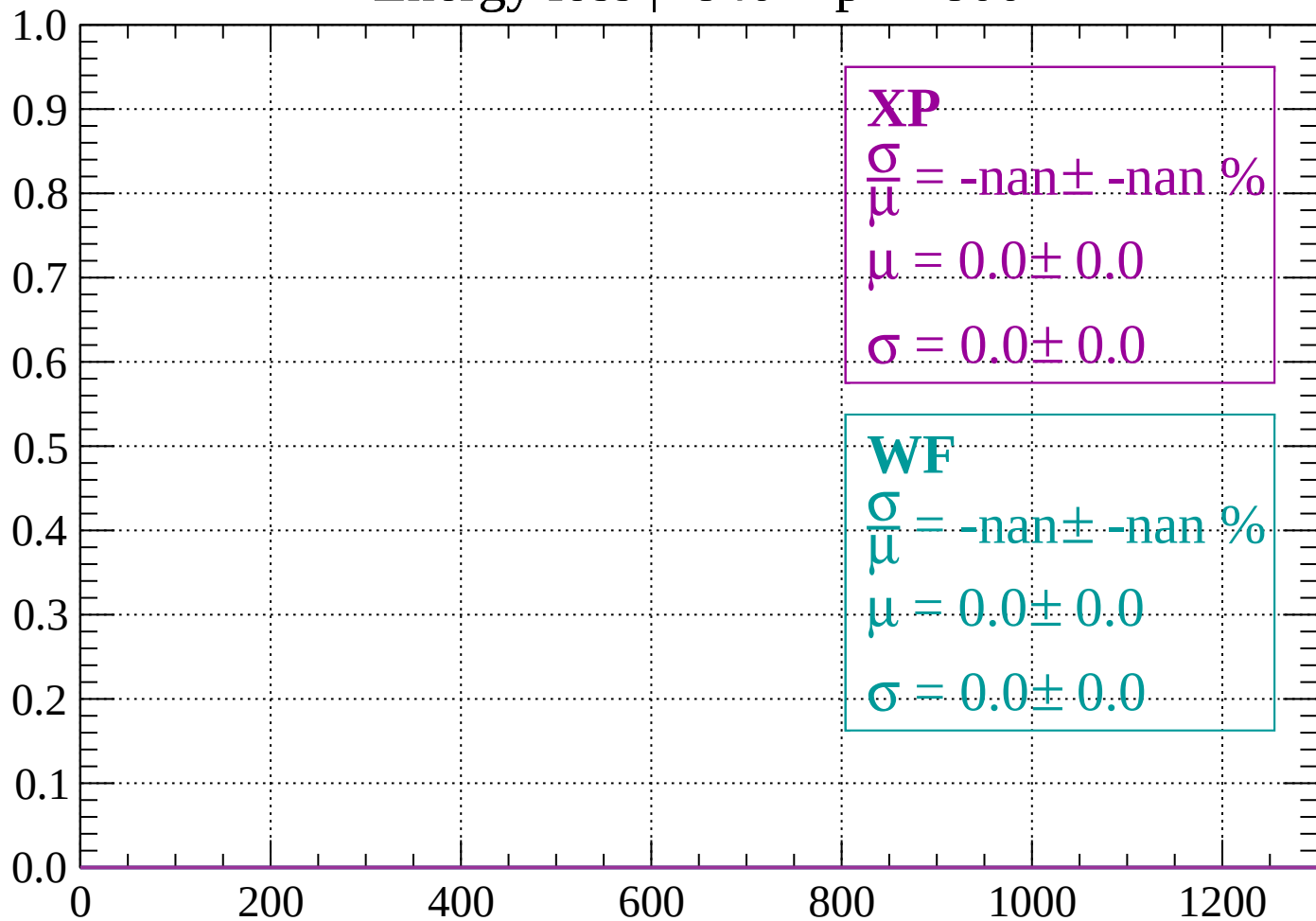
Count



dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

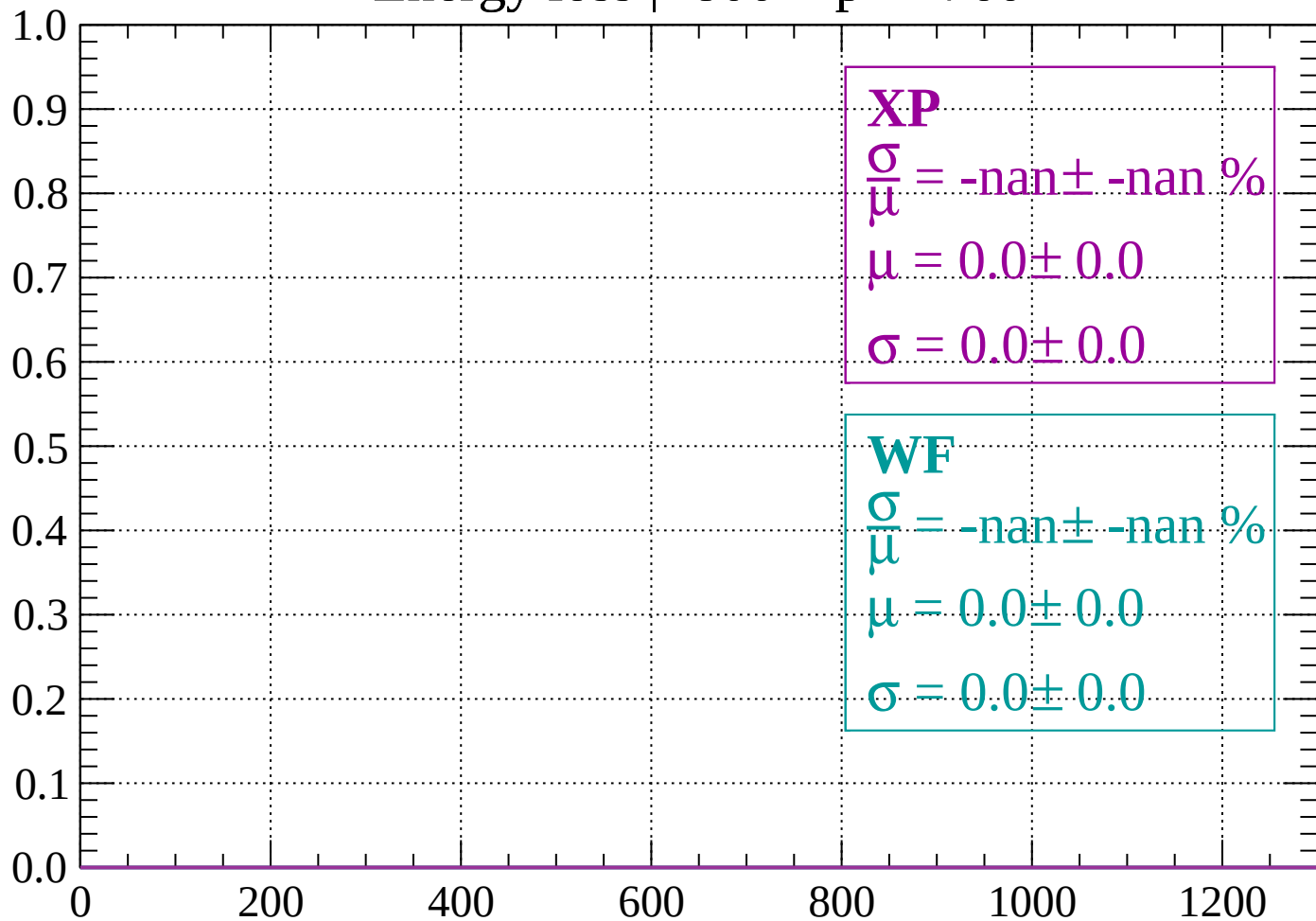
Count



dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

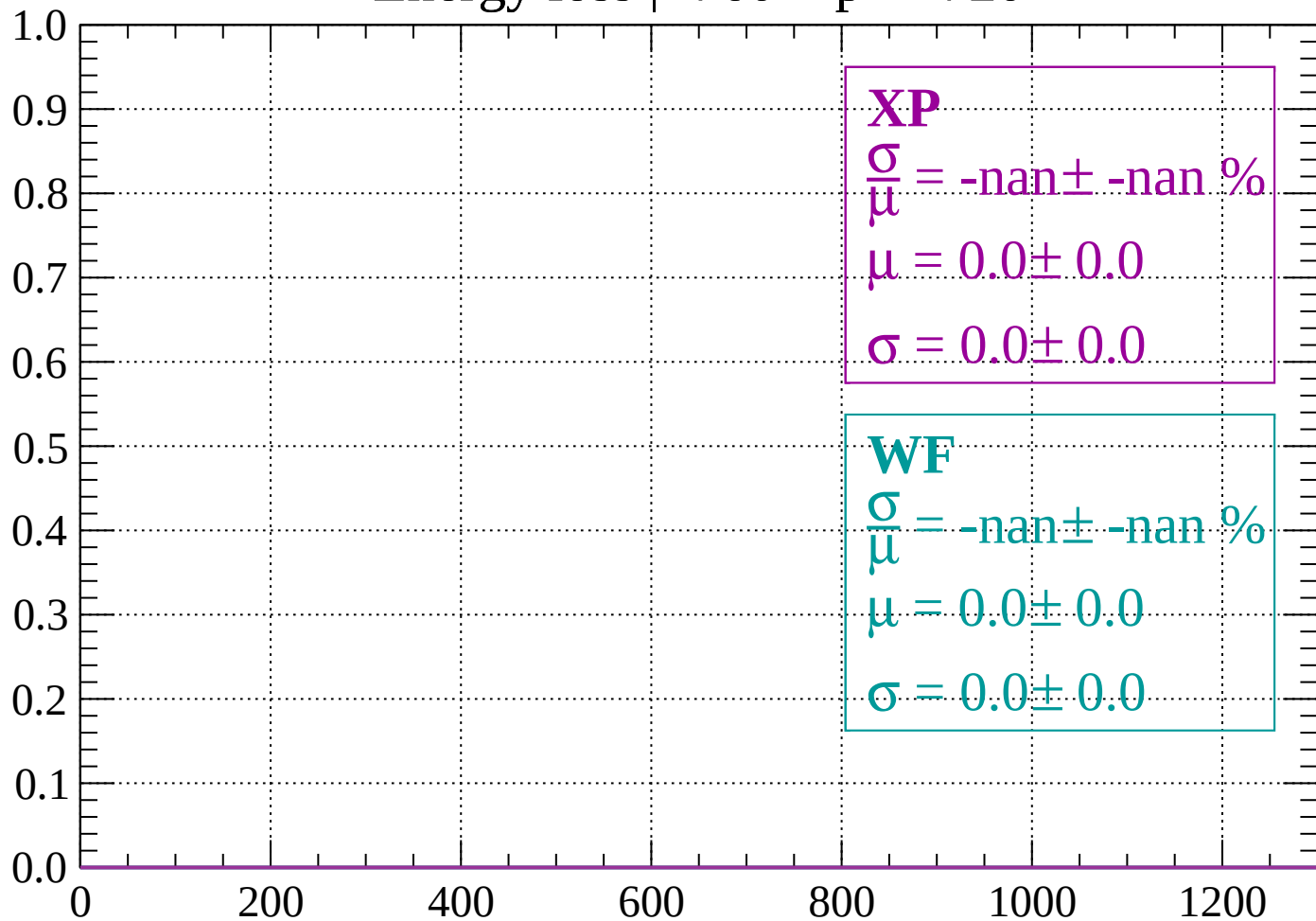
Count



dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

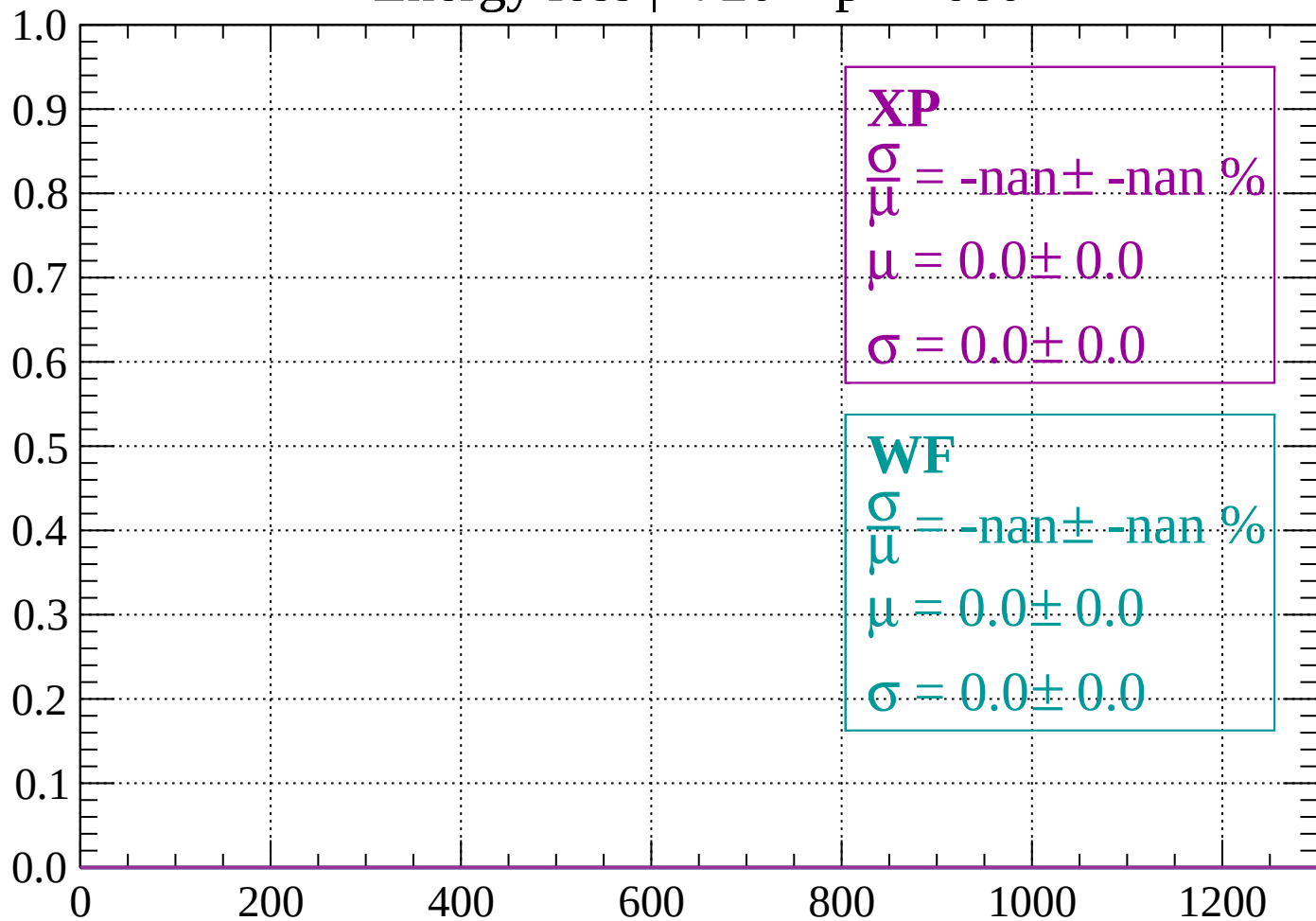
Count



dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

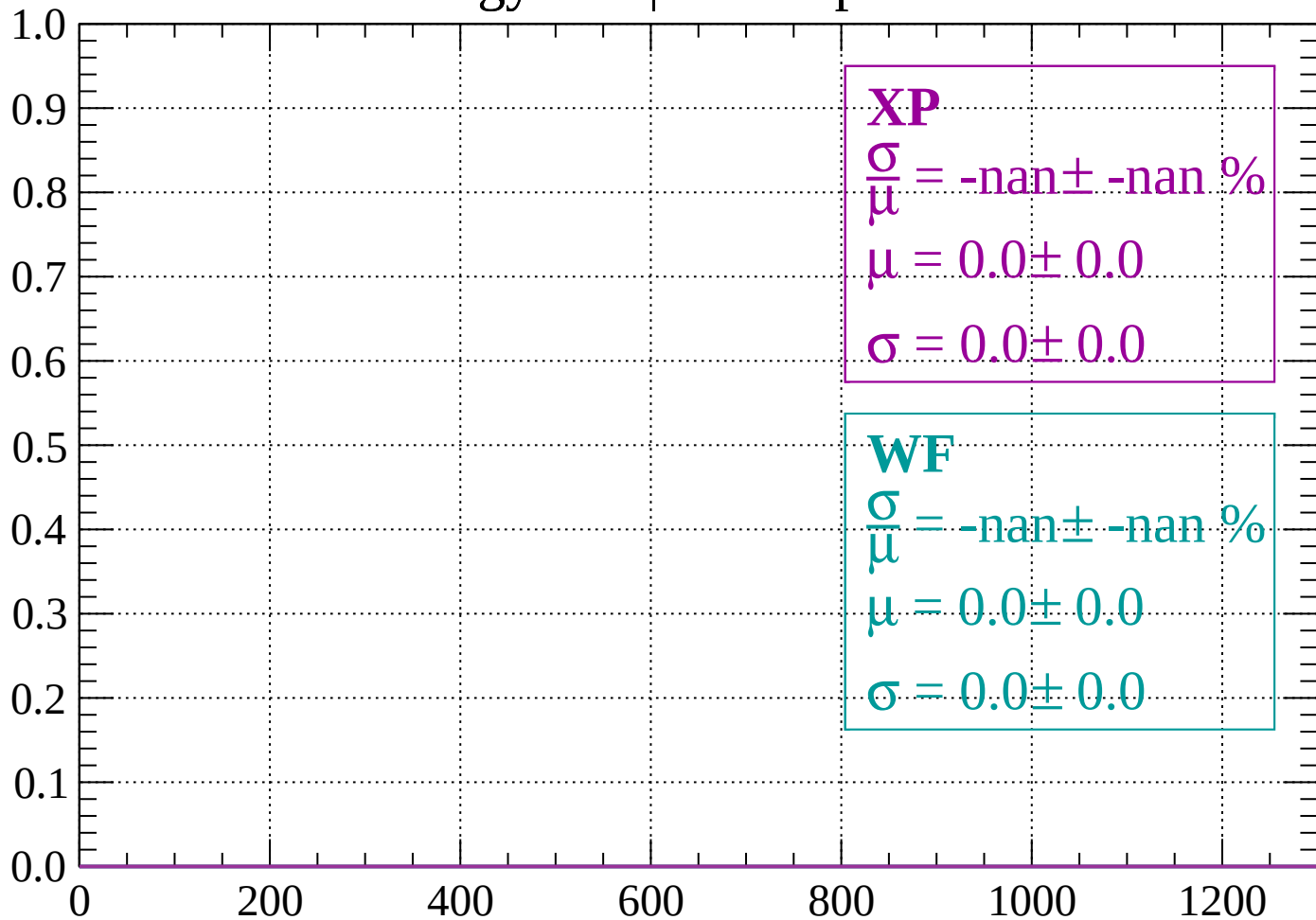
Count



dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

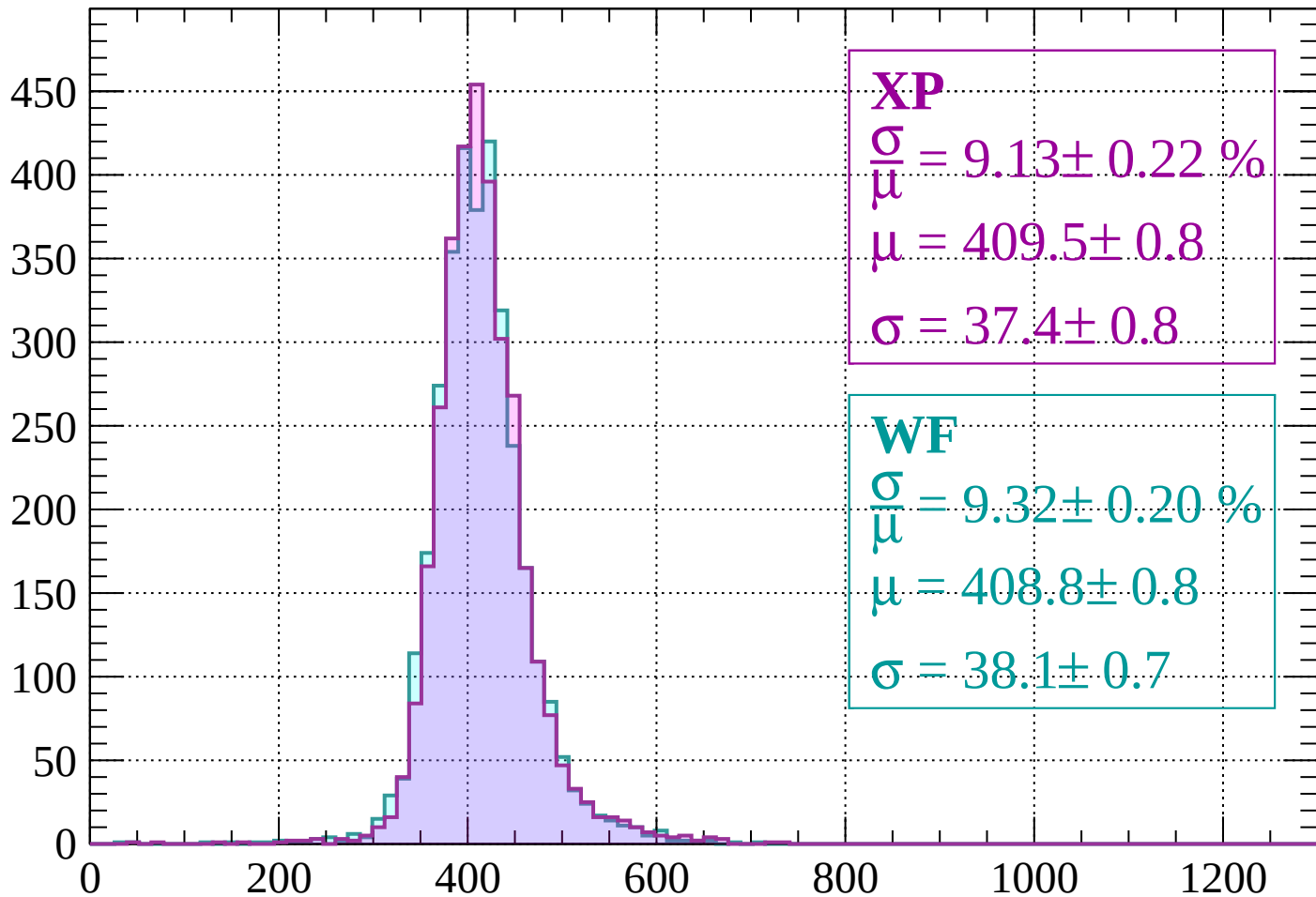
Count



dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.22 \%$$

$$\mu = 409.5 \pm 0.8$$

$$\sigma = 37.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.32 \pm 0.20 \%$$

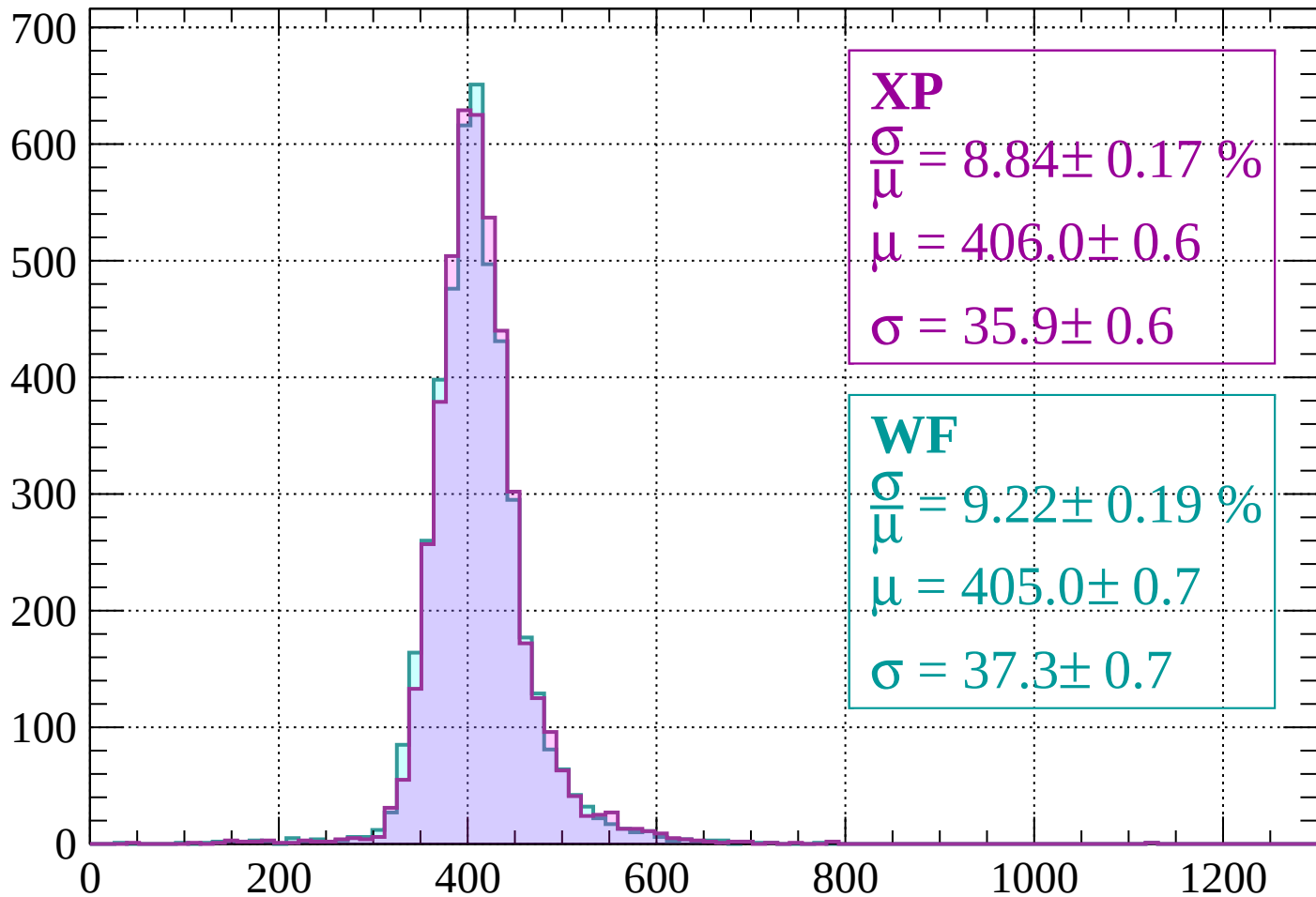
$$\mu = 408.8 \pm 0.8$$

$$\sigma = 38.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.84 \pm 0.17 \%$$

$$\mu = 406.0 \pm 0.6$$

$$\sigma = 35.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.22 \pm 0.19 \%$$

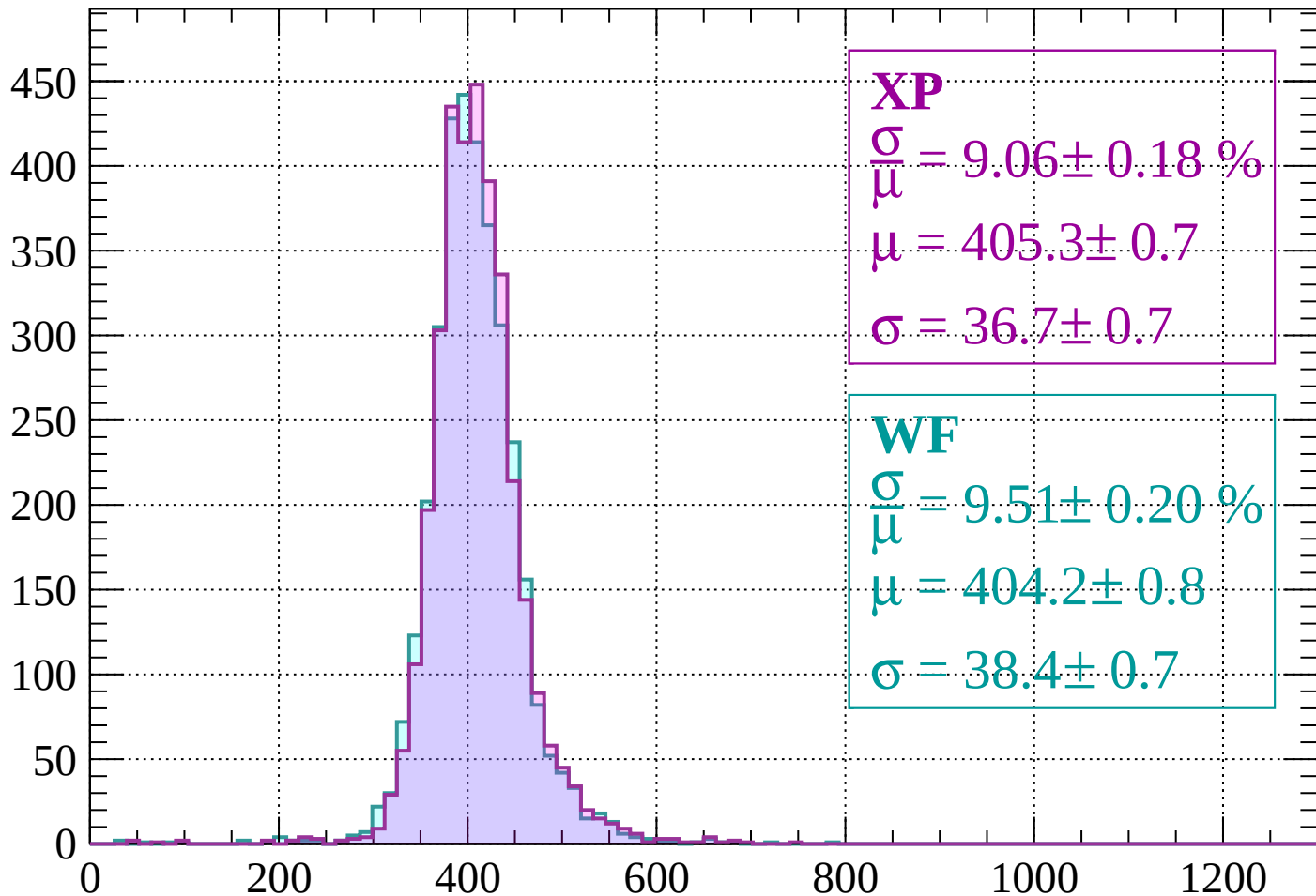
$$\mu = 405.0 \pm 0.7$$

$$\sigma = 37.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 9.06 \pm 0.18 \%$$

$$\mu = 405.3 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.51 \pm 0.20 \%$$

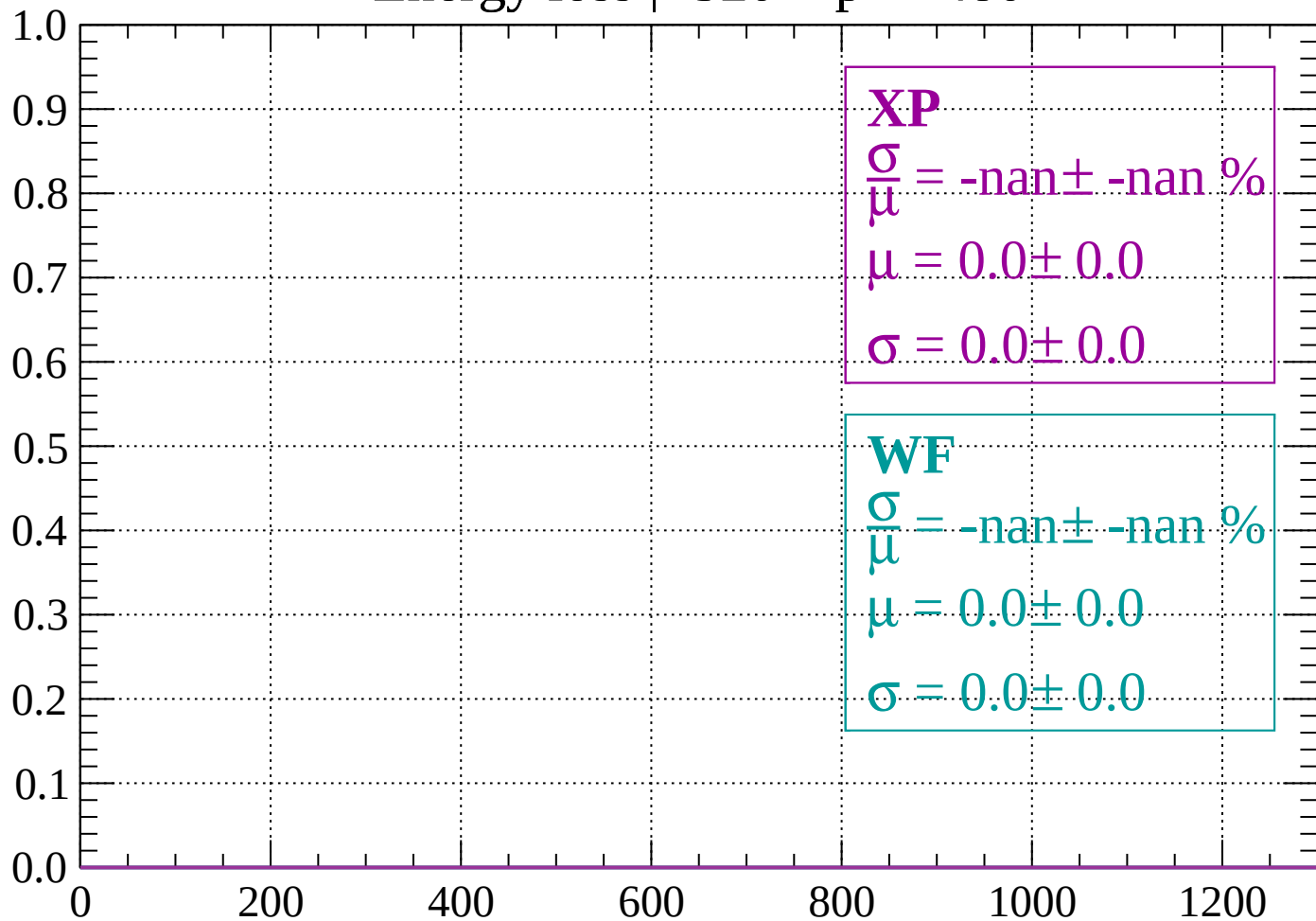
$$\mu = 404.2 \pm 0.8$$

$$\sigma = 38.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

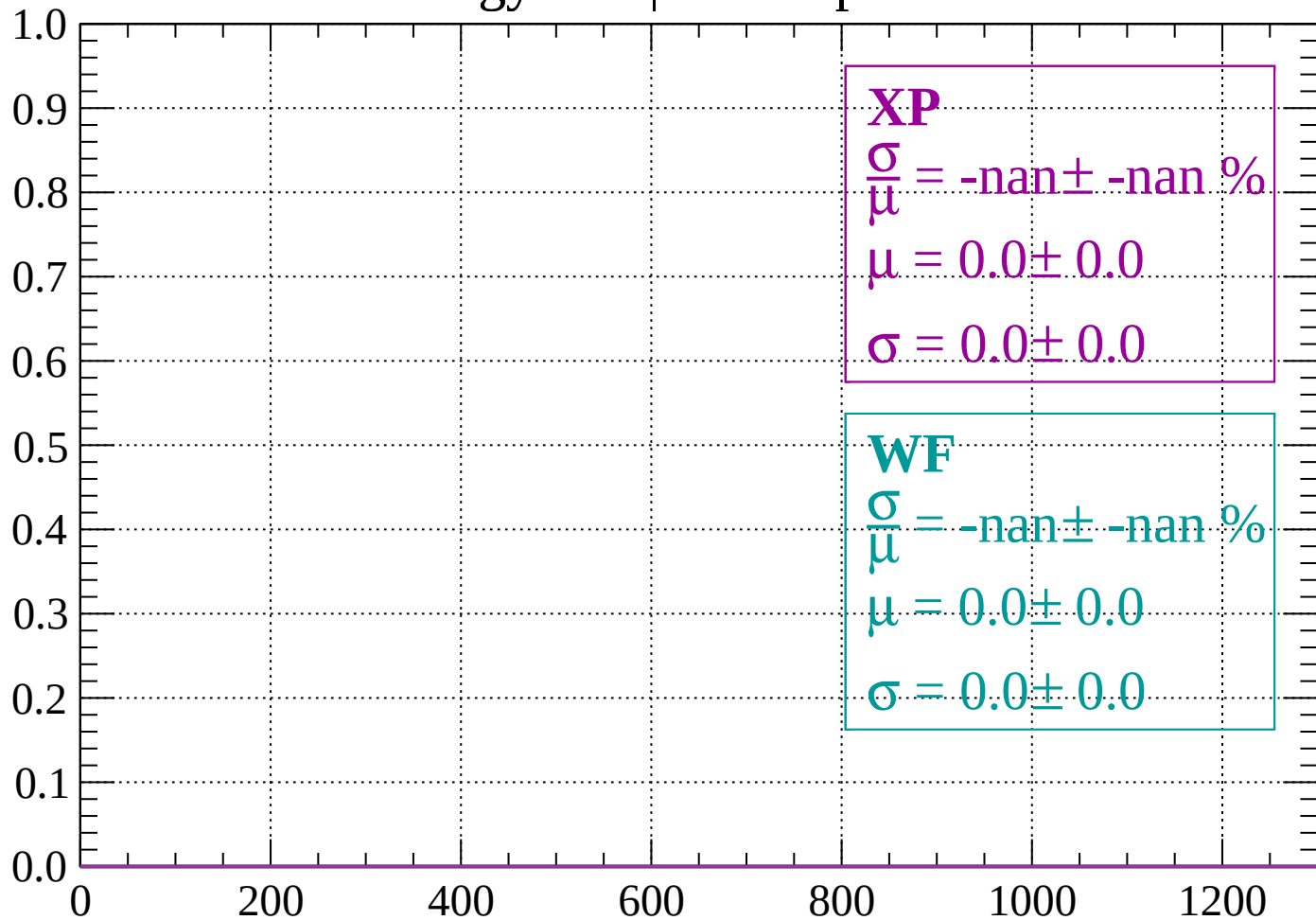
Count



dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

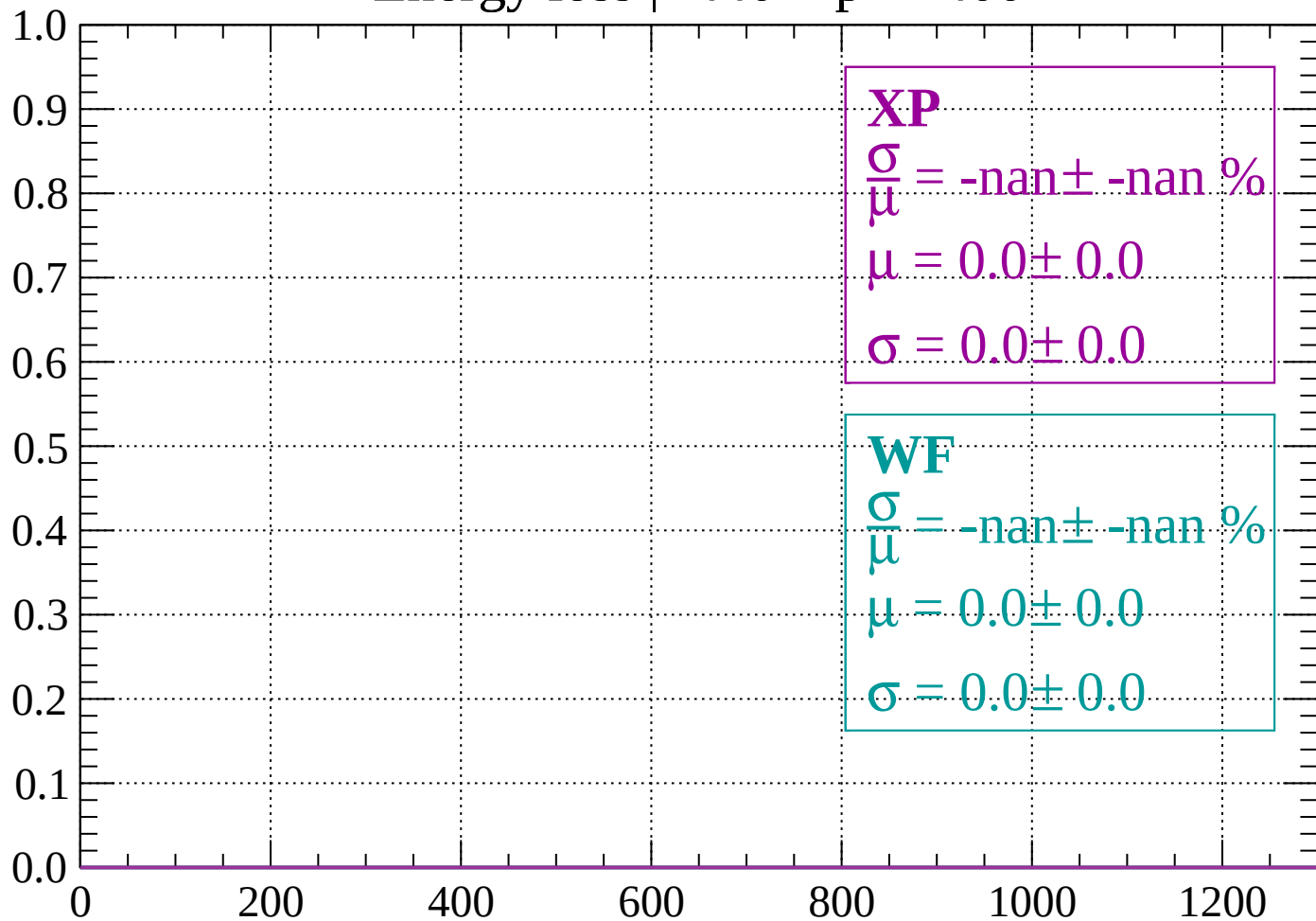
Count



dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

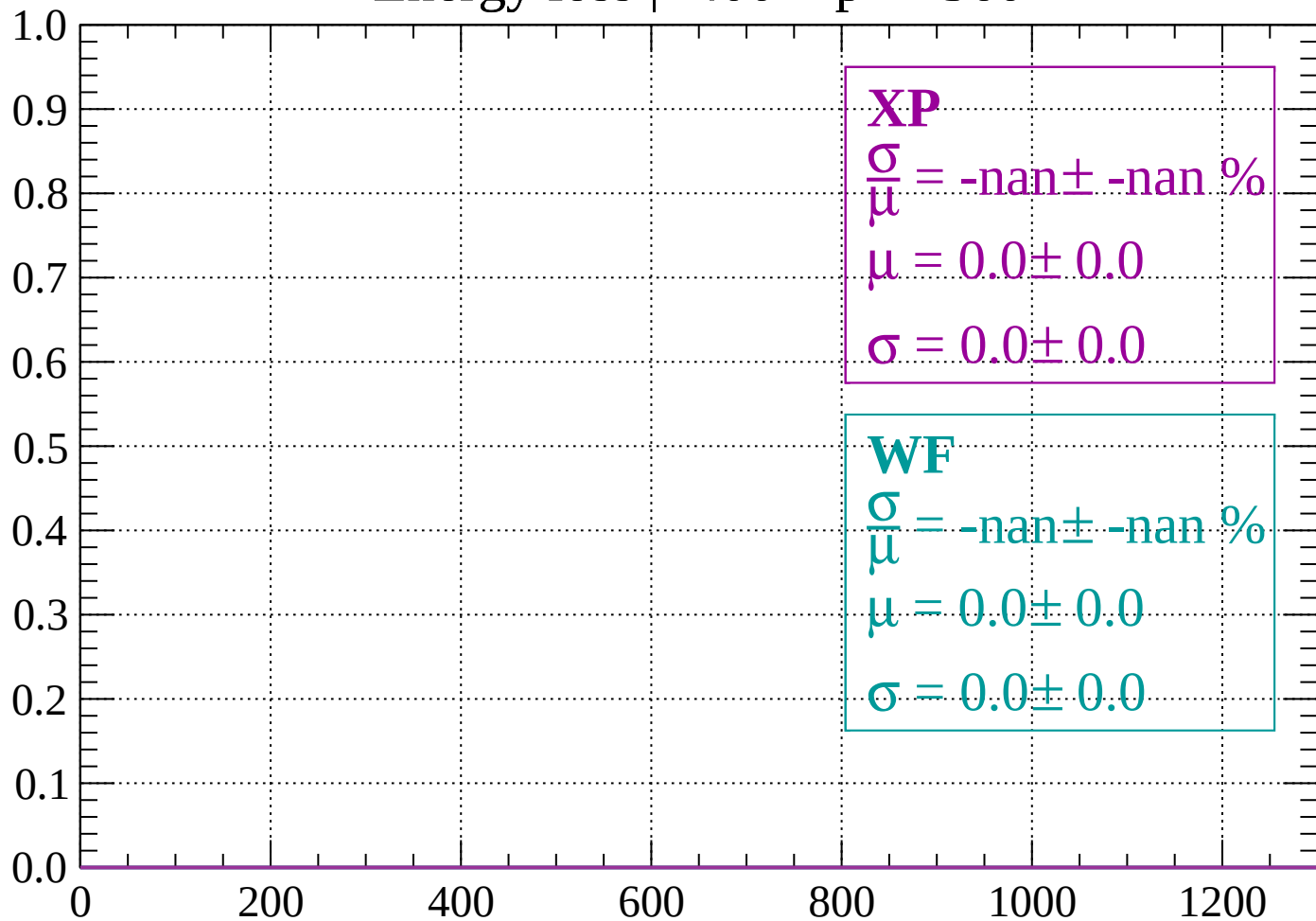
Count



dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

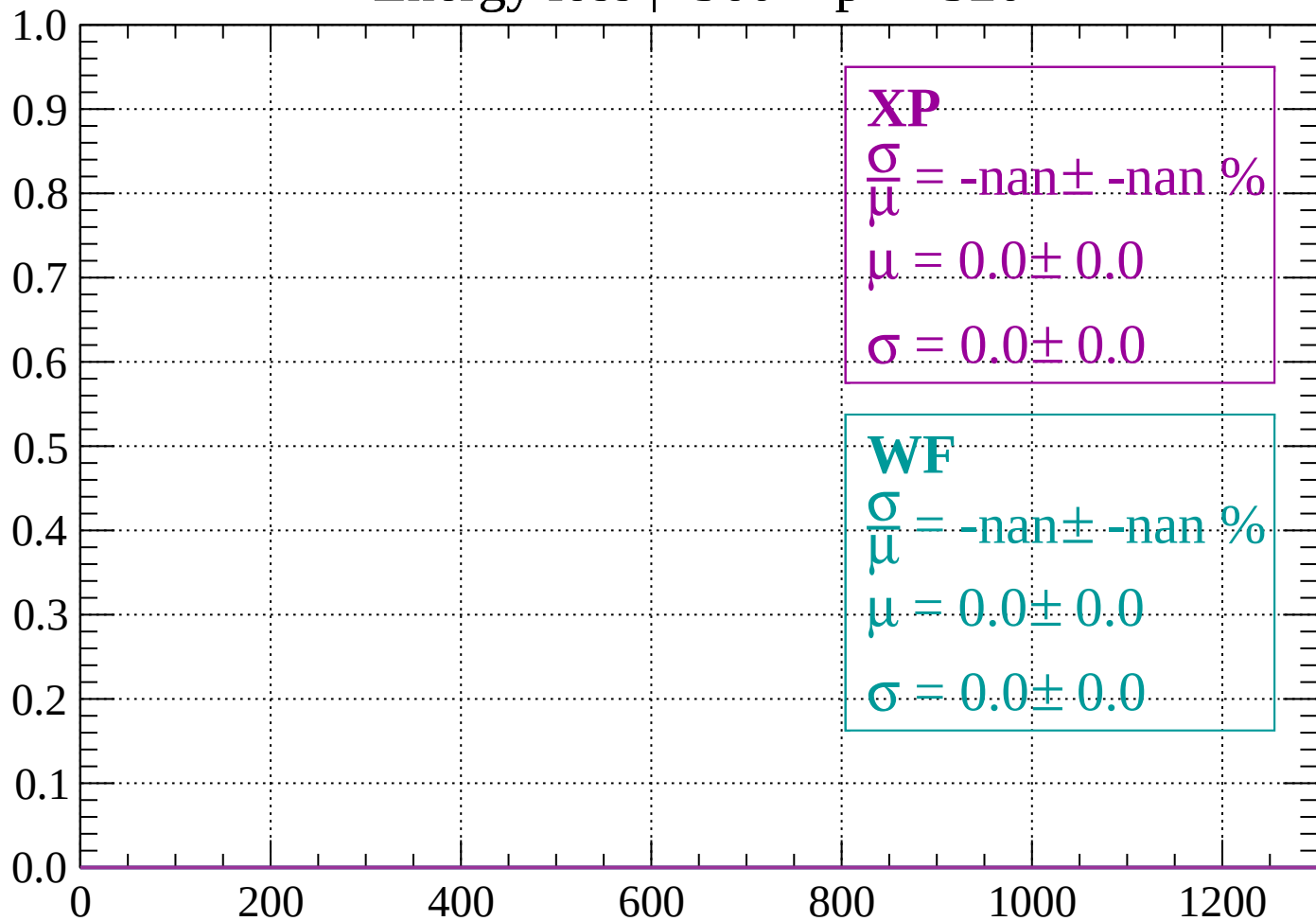
Count



dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

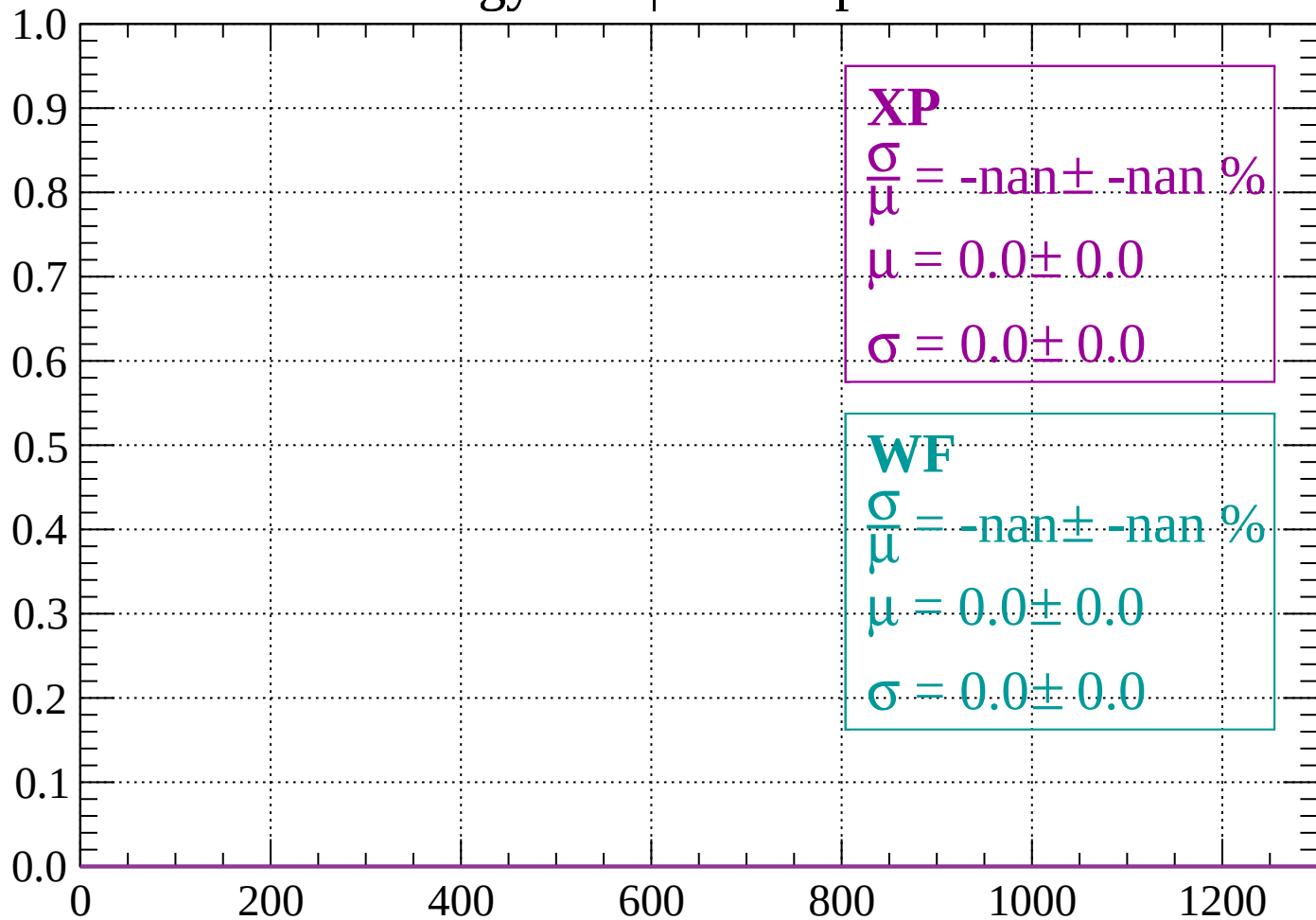
Count



dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

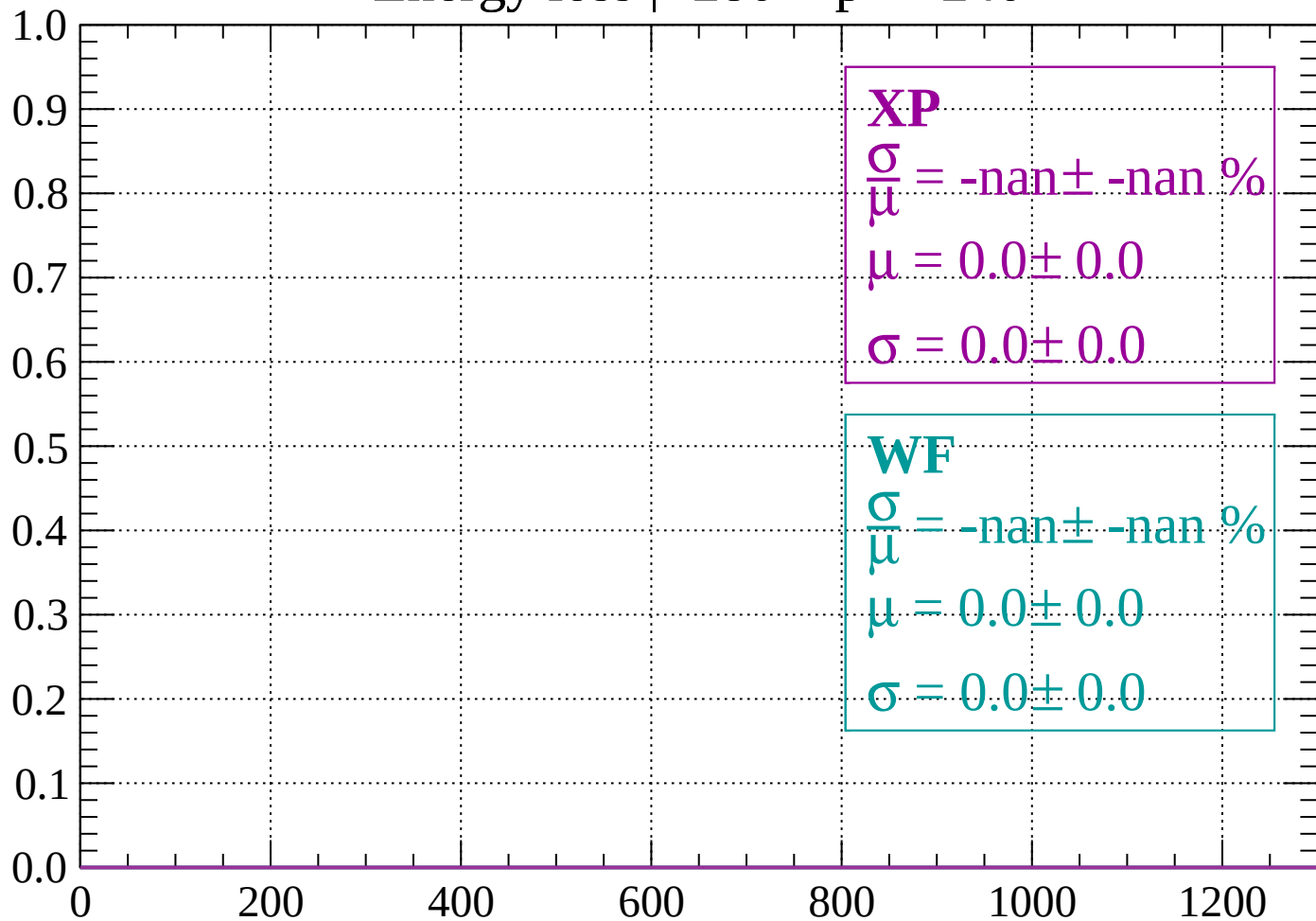
Count



dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

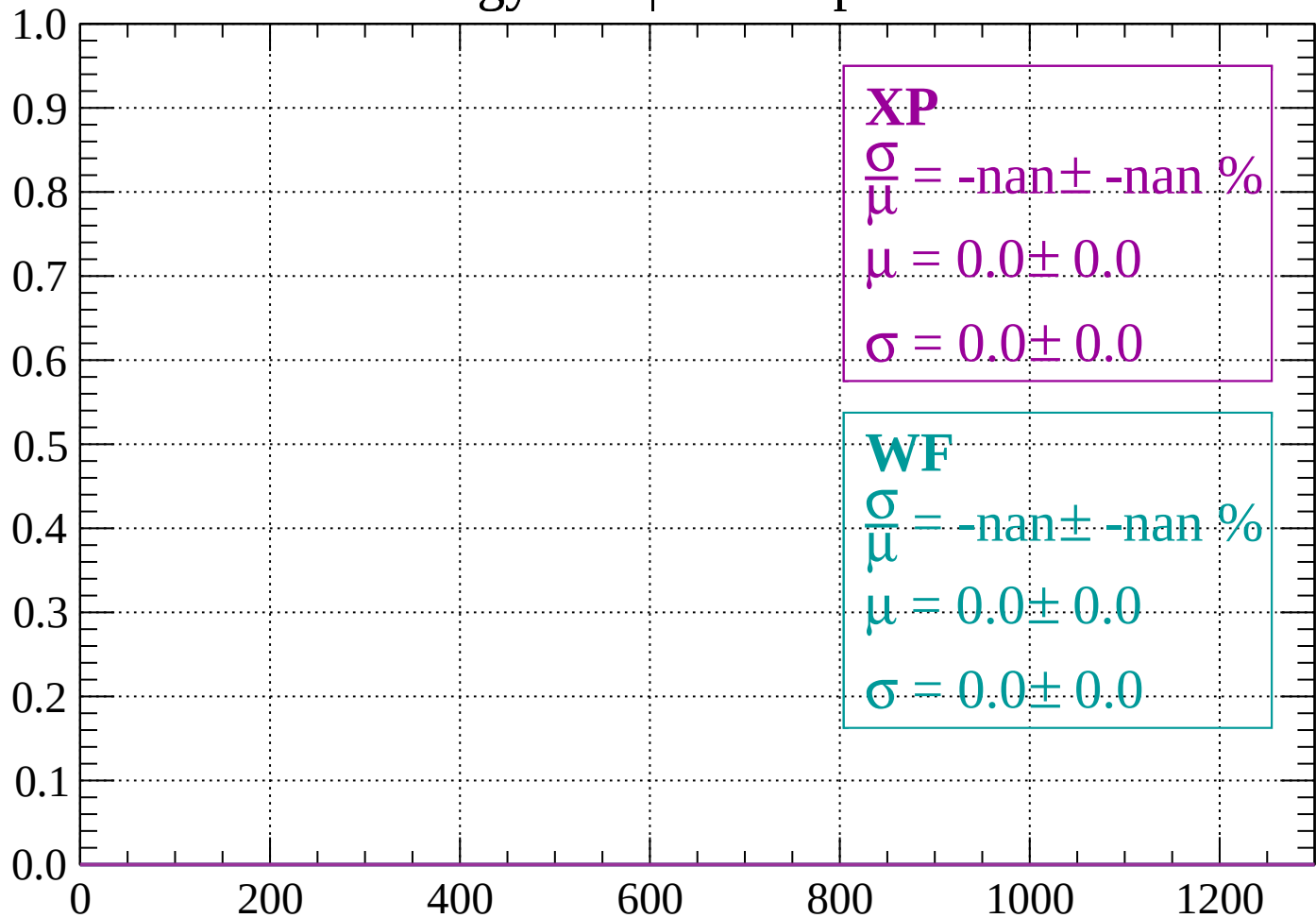
Count



dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

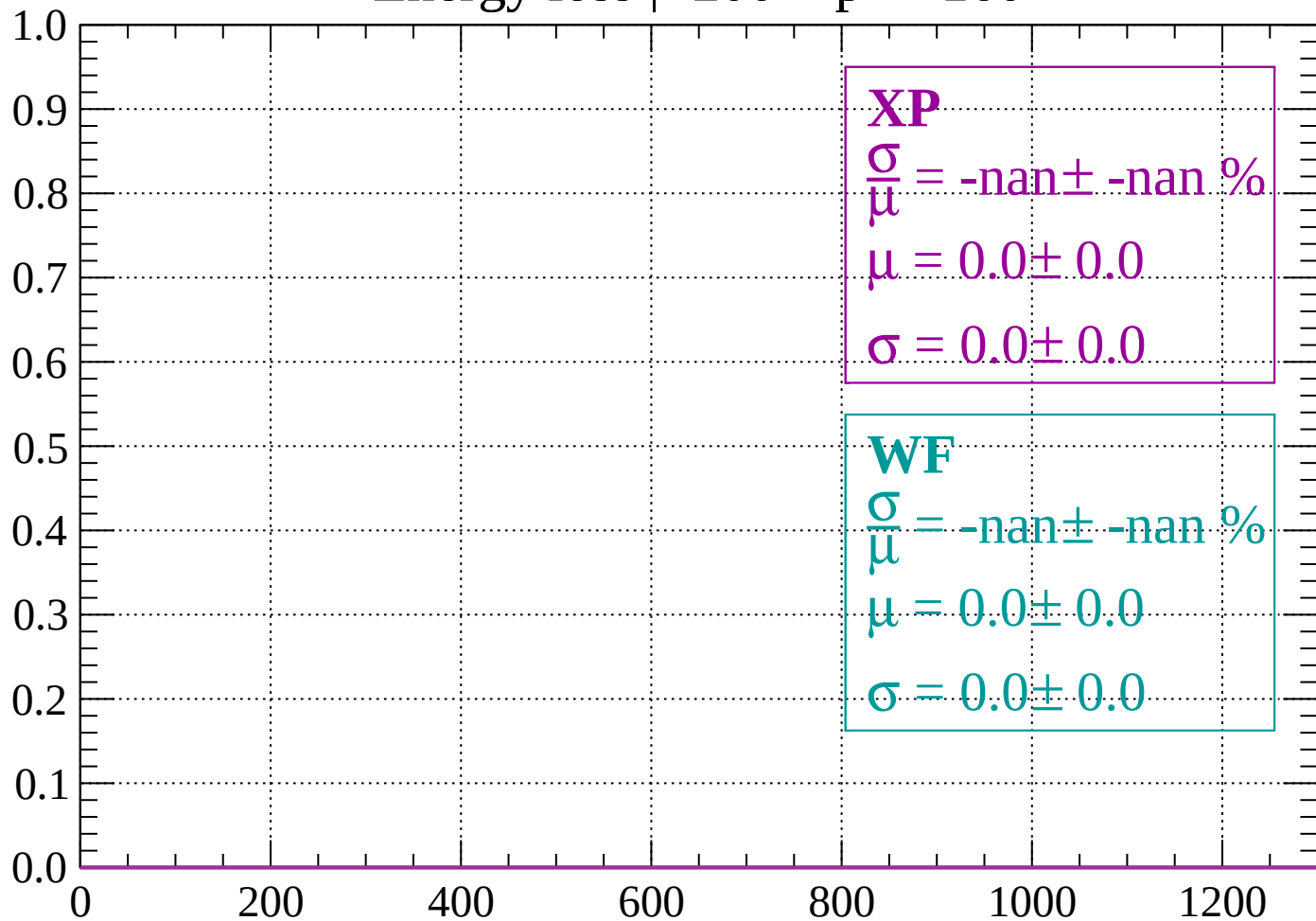
Count



dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

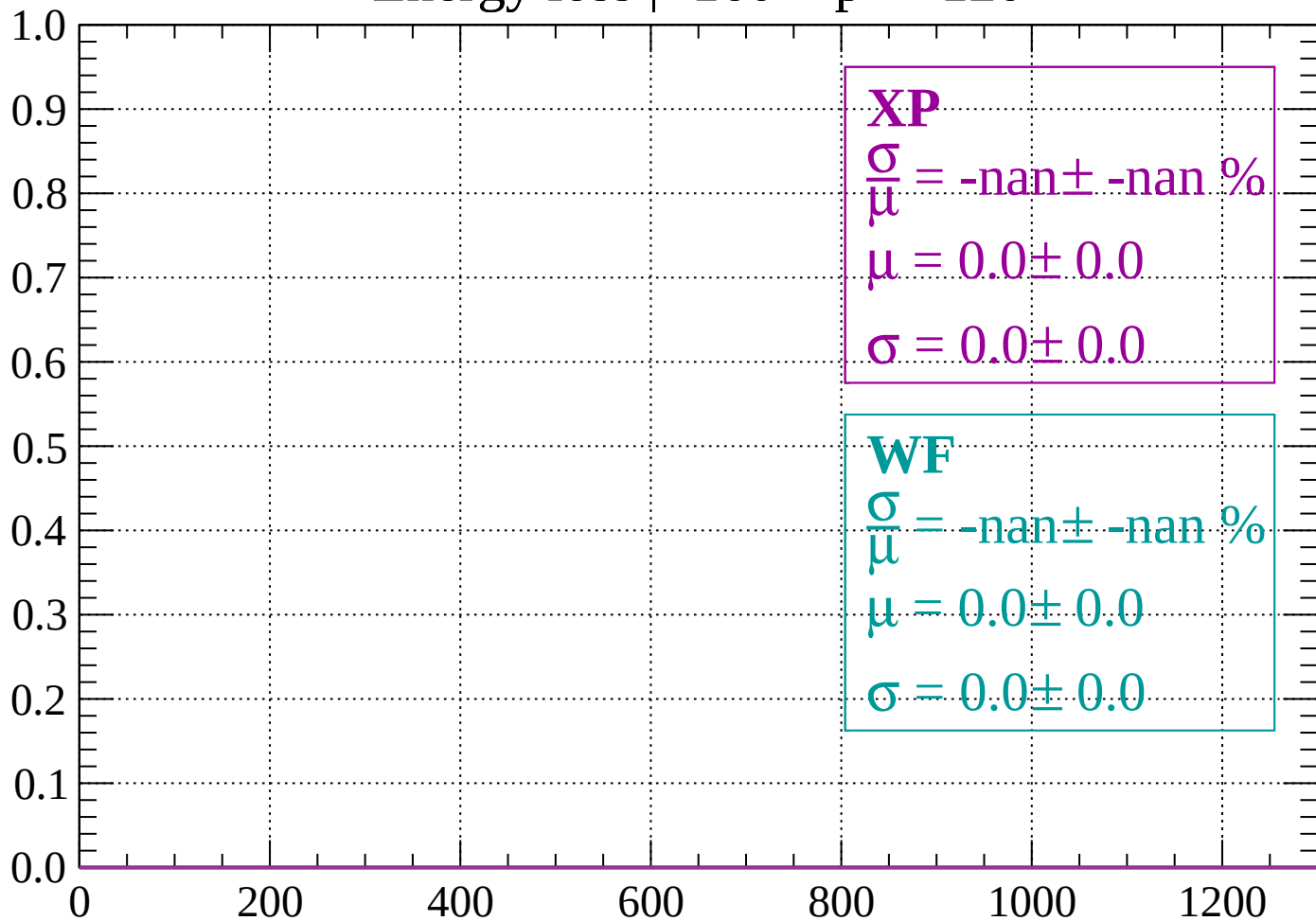
Count



dE/dx (ADC counts/cm)

Energy loss | -160 < p < -120

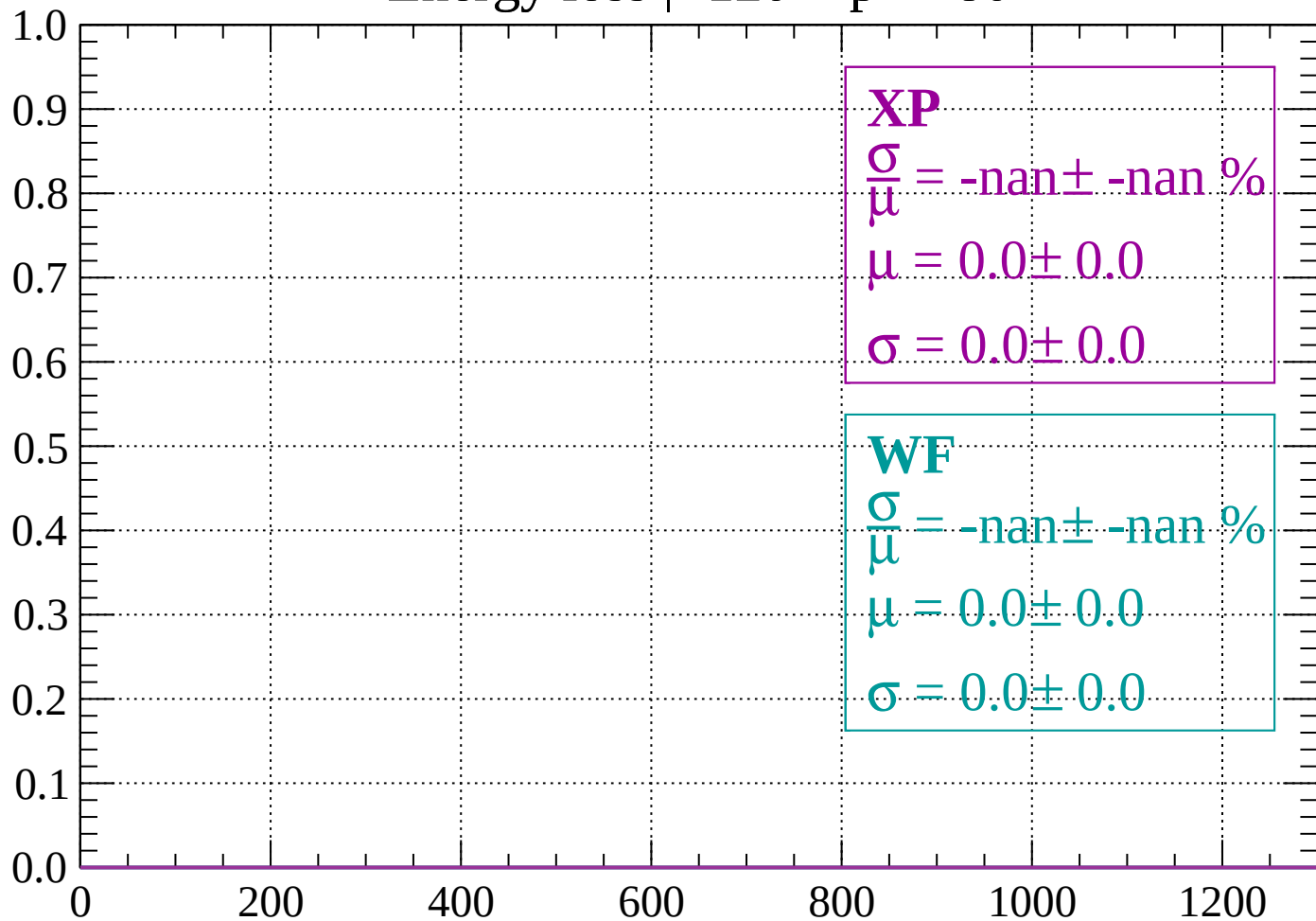
Count



dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -80$

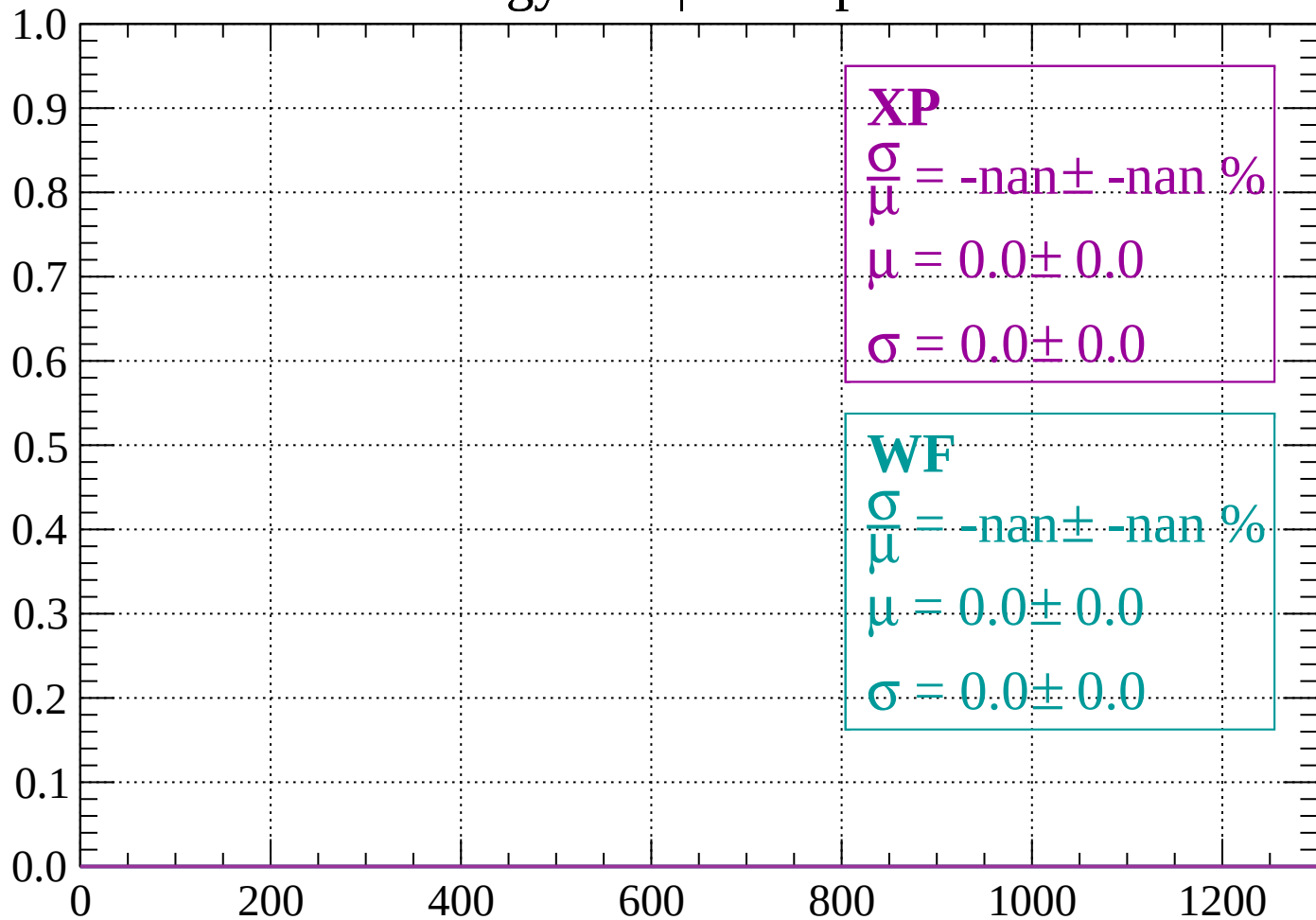
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < p < -40$

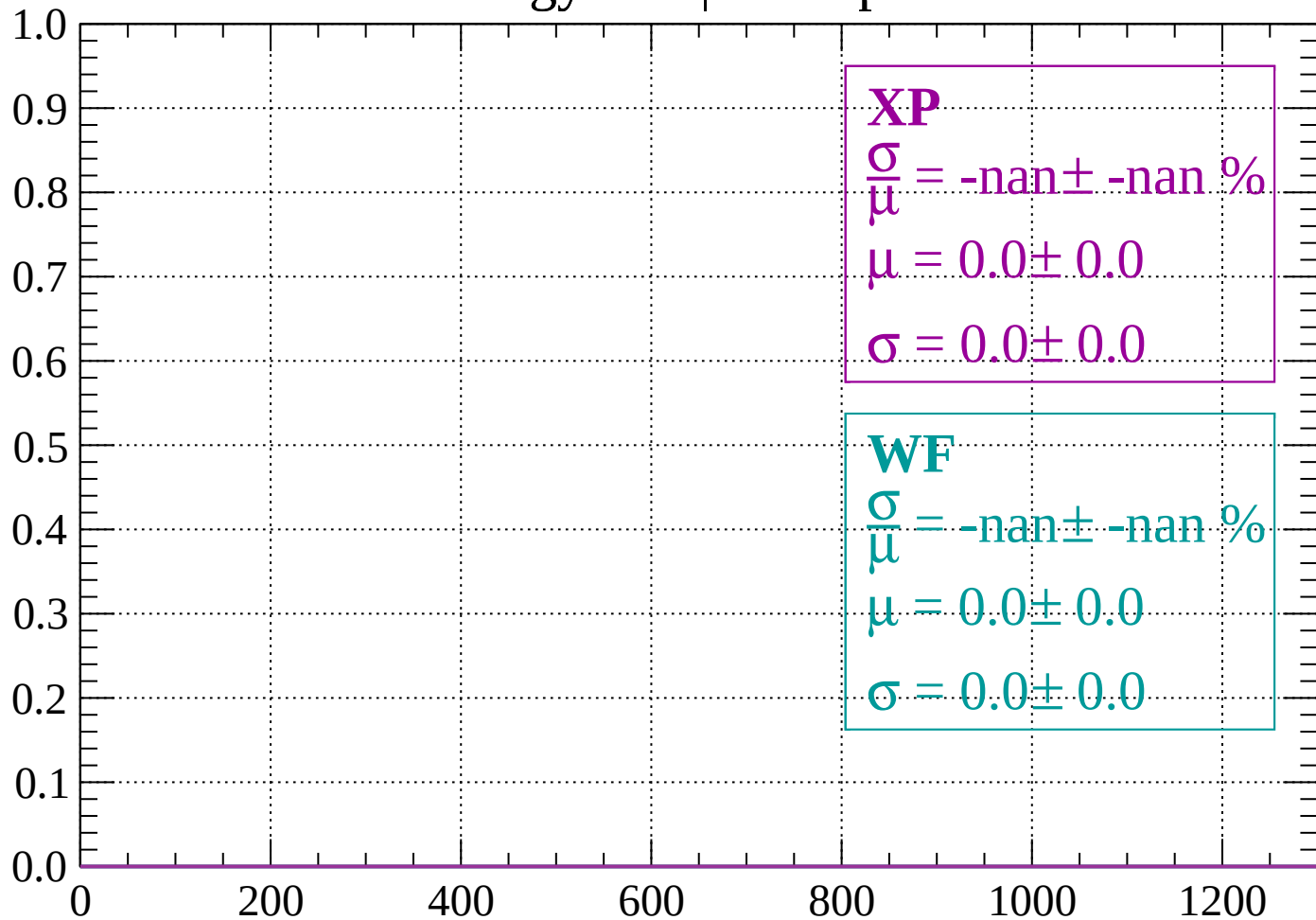
Count



dE/dx (ADC counts/cm)

Energy loss | $-40 < p < 0$

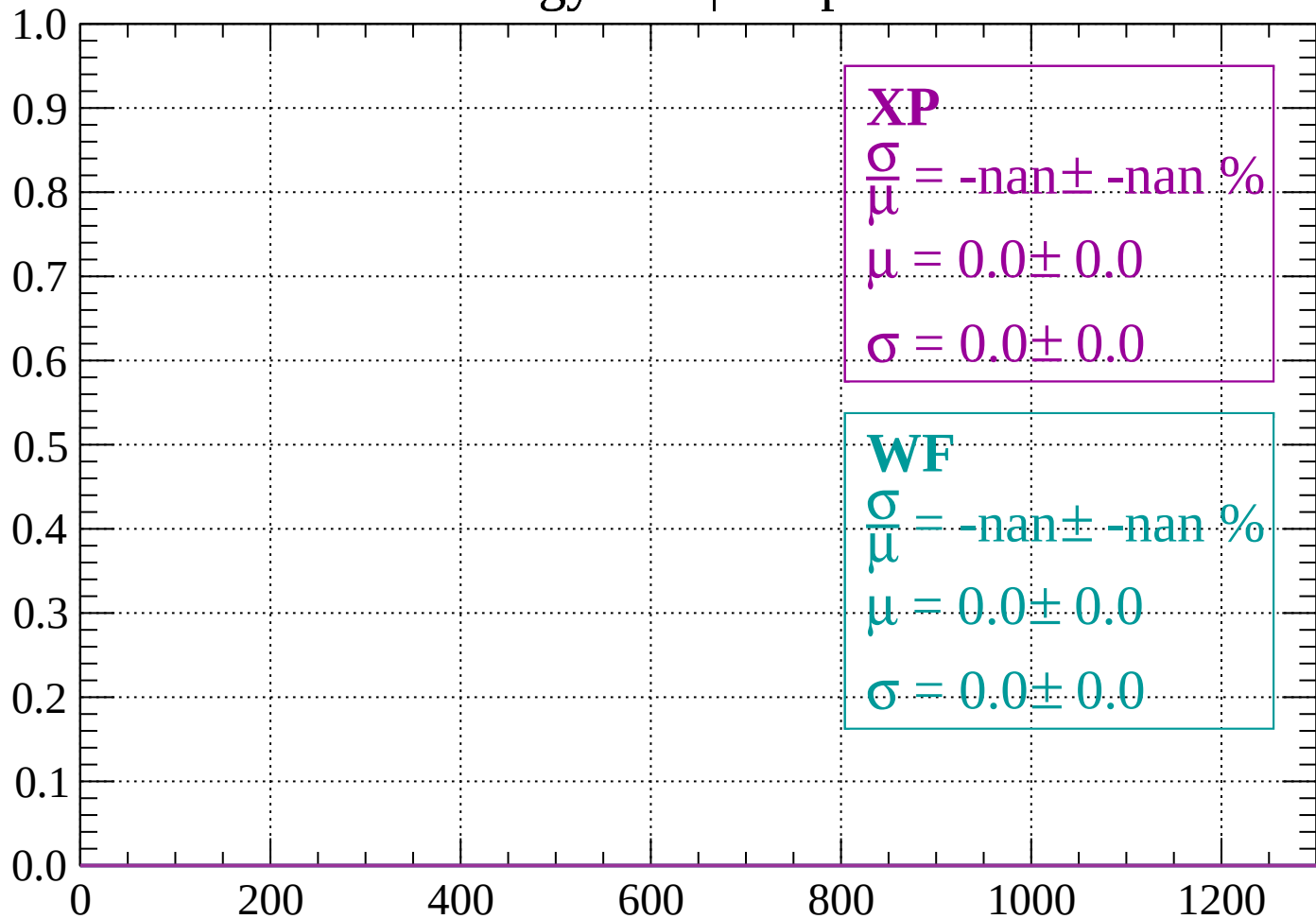
Count



dE/dx (ADC counts/cm)

Energy loss | $0 < p < 40$

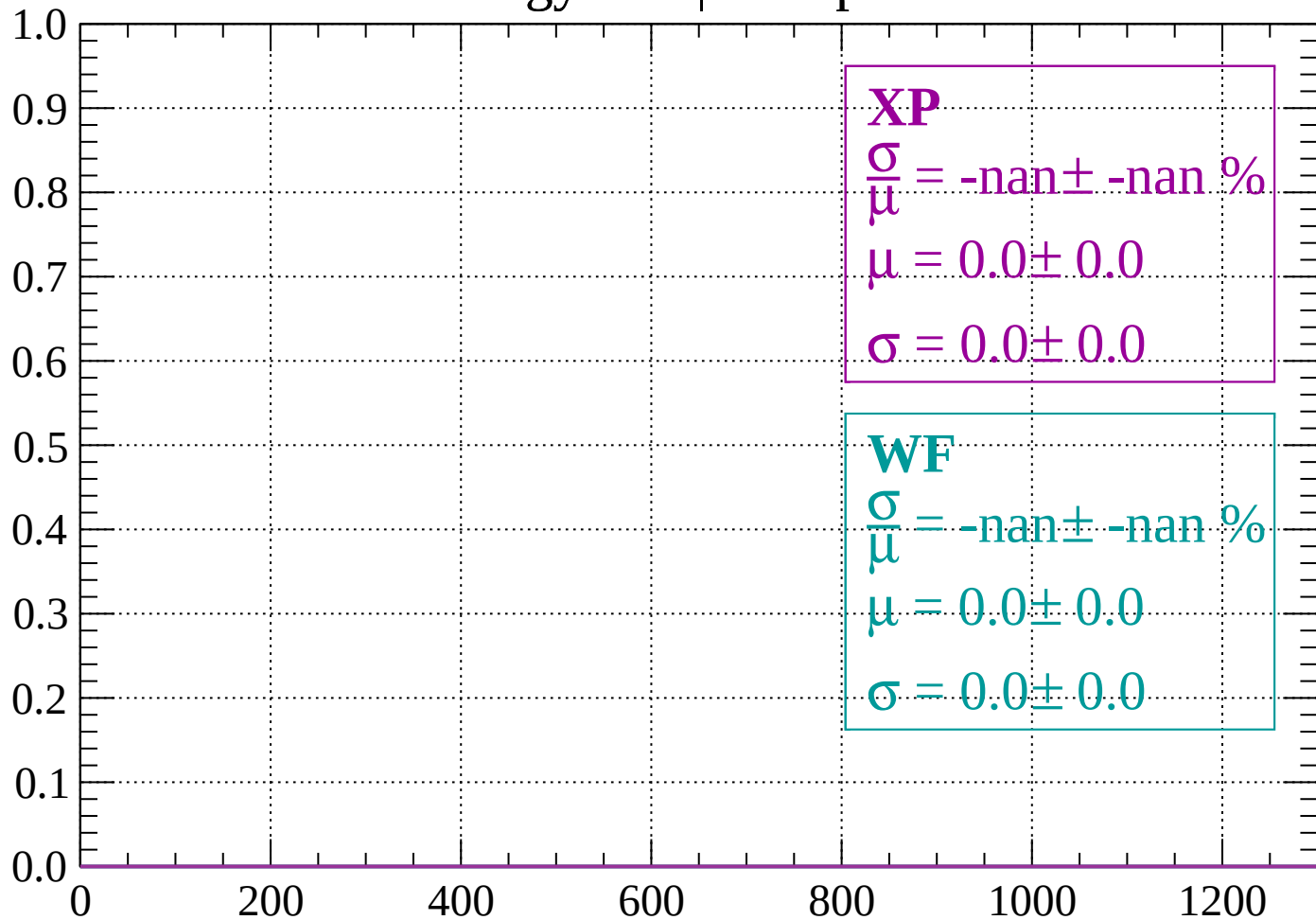
Count



dE/dx (ADC counts/cm)

Energy loss | $40 < p < 80$

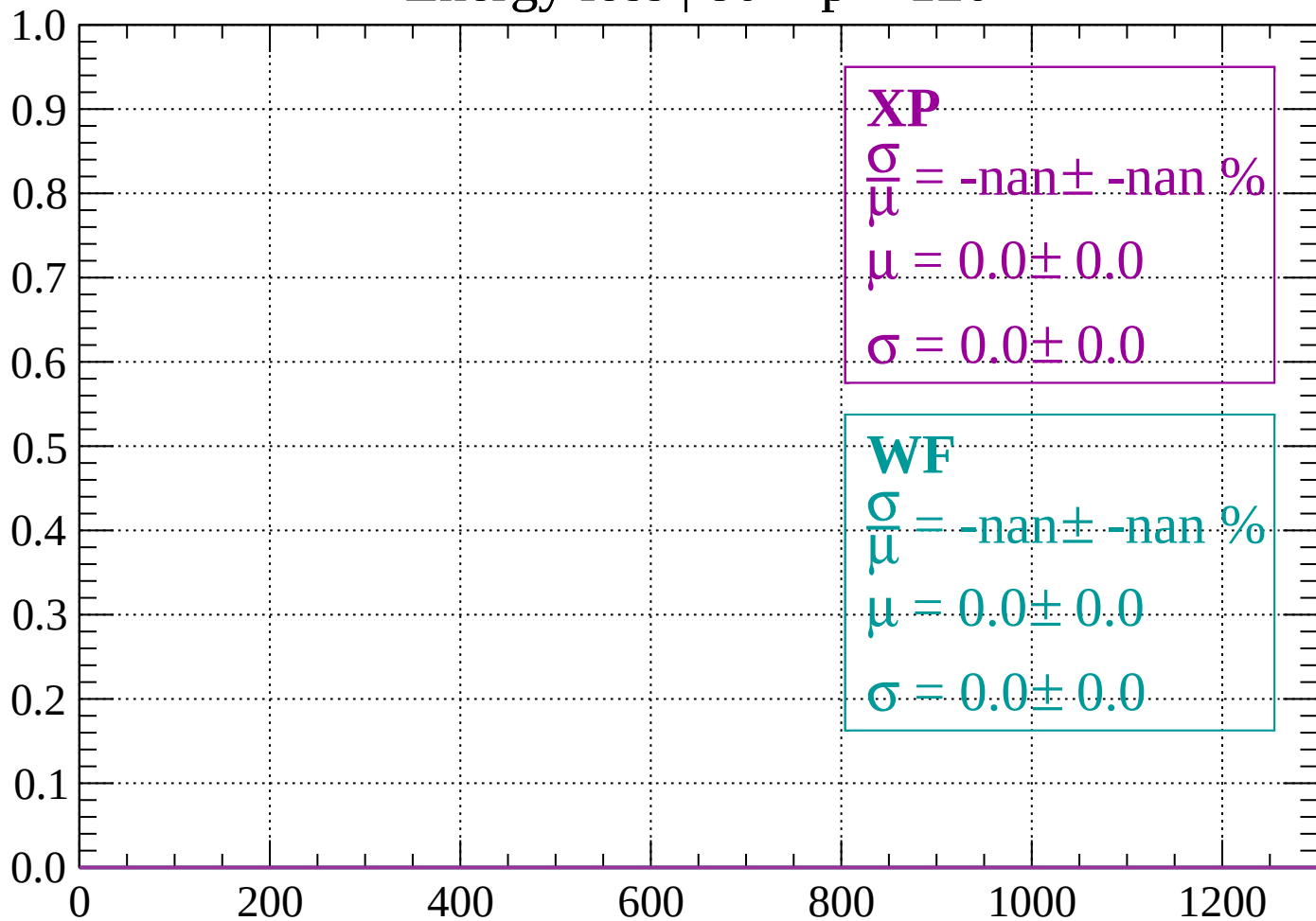
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < p < 120$

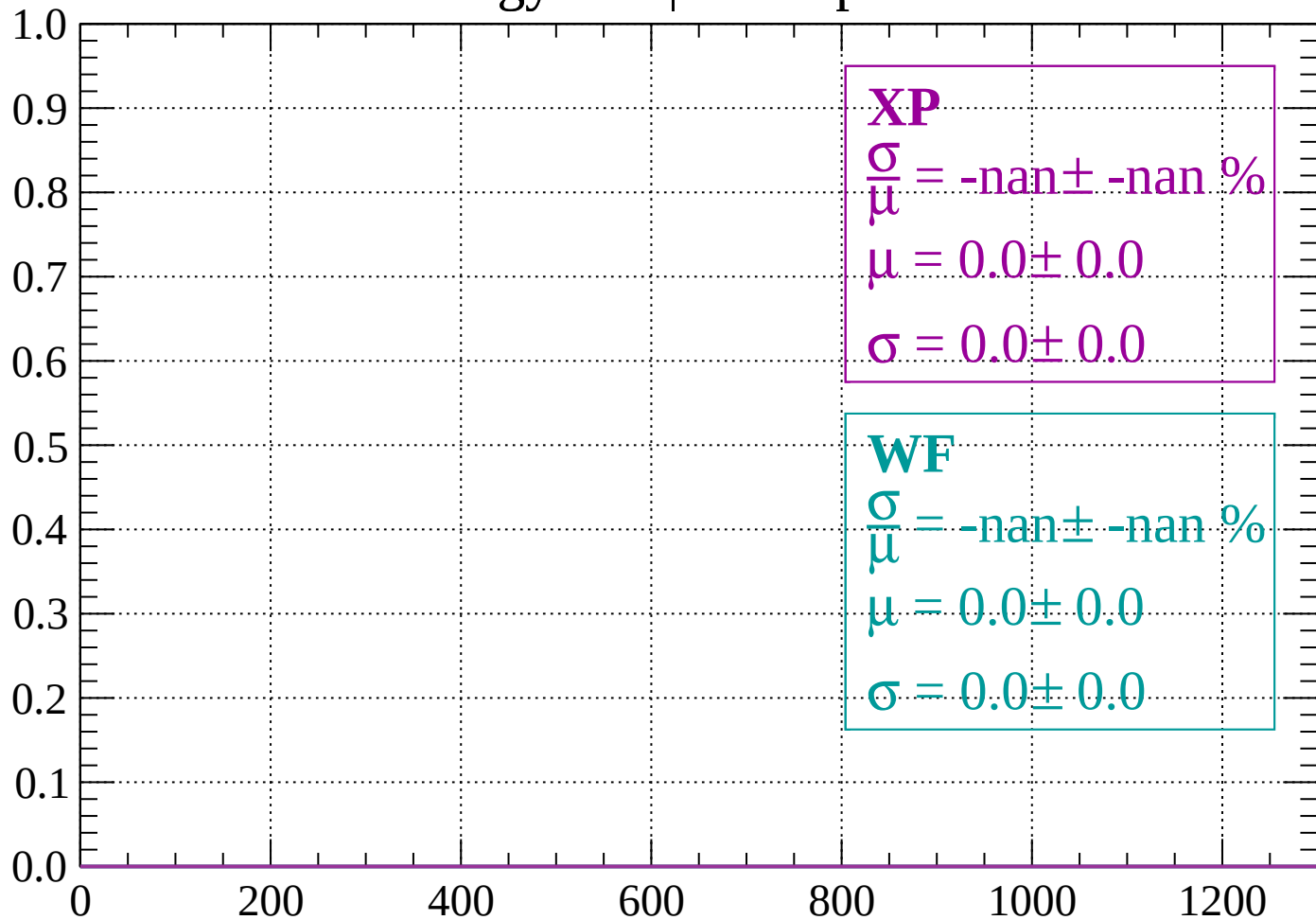
Count



dE/dx (ADC counts/cm)

Energy loss | $120 < p < 160$

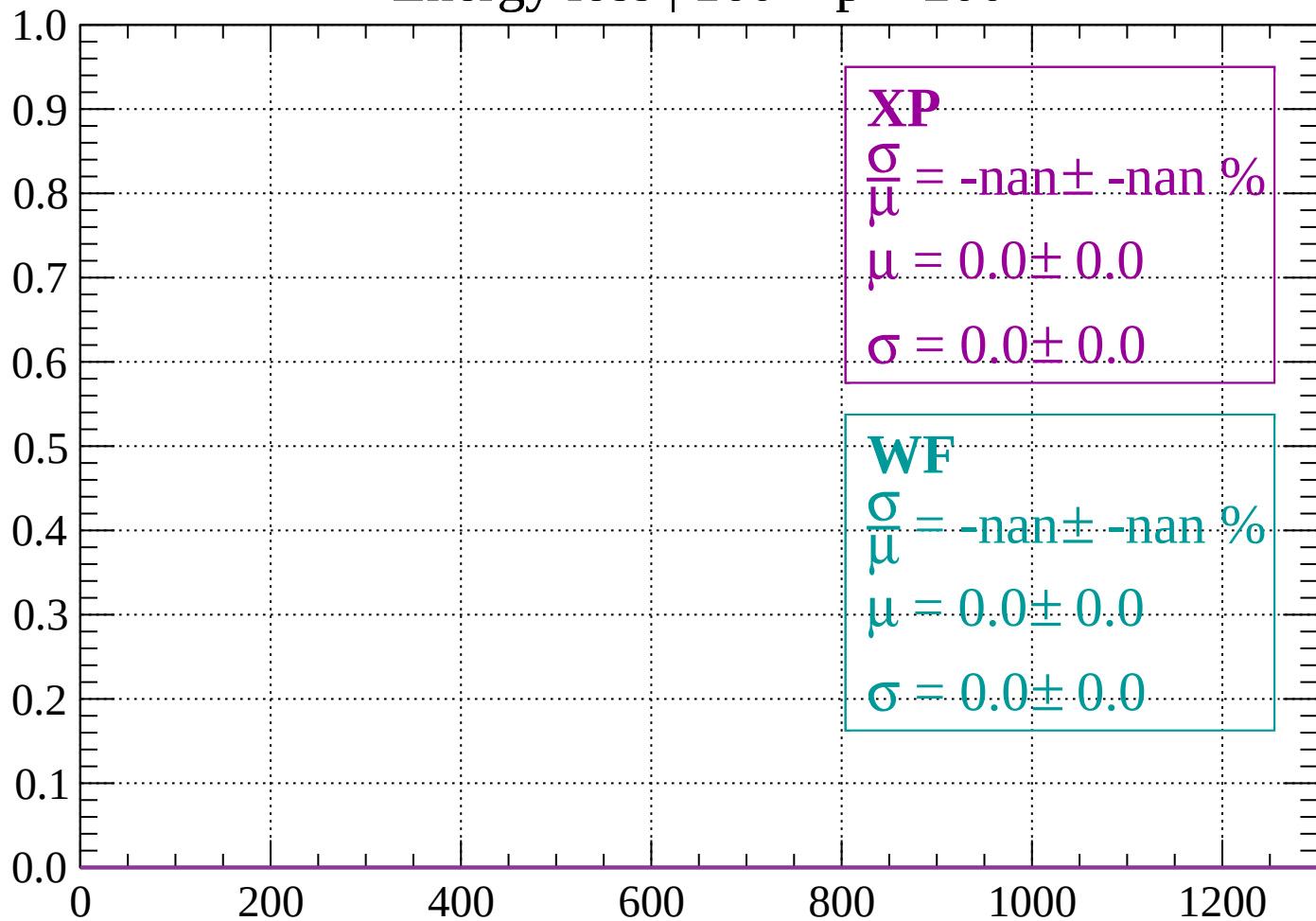
Count



dE/dx (ADC counts/cm)

Energy loss | $160 < p < 200$

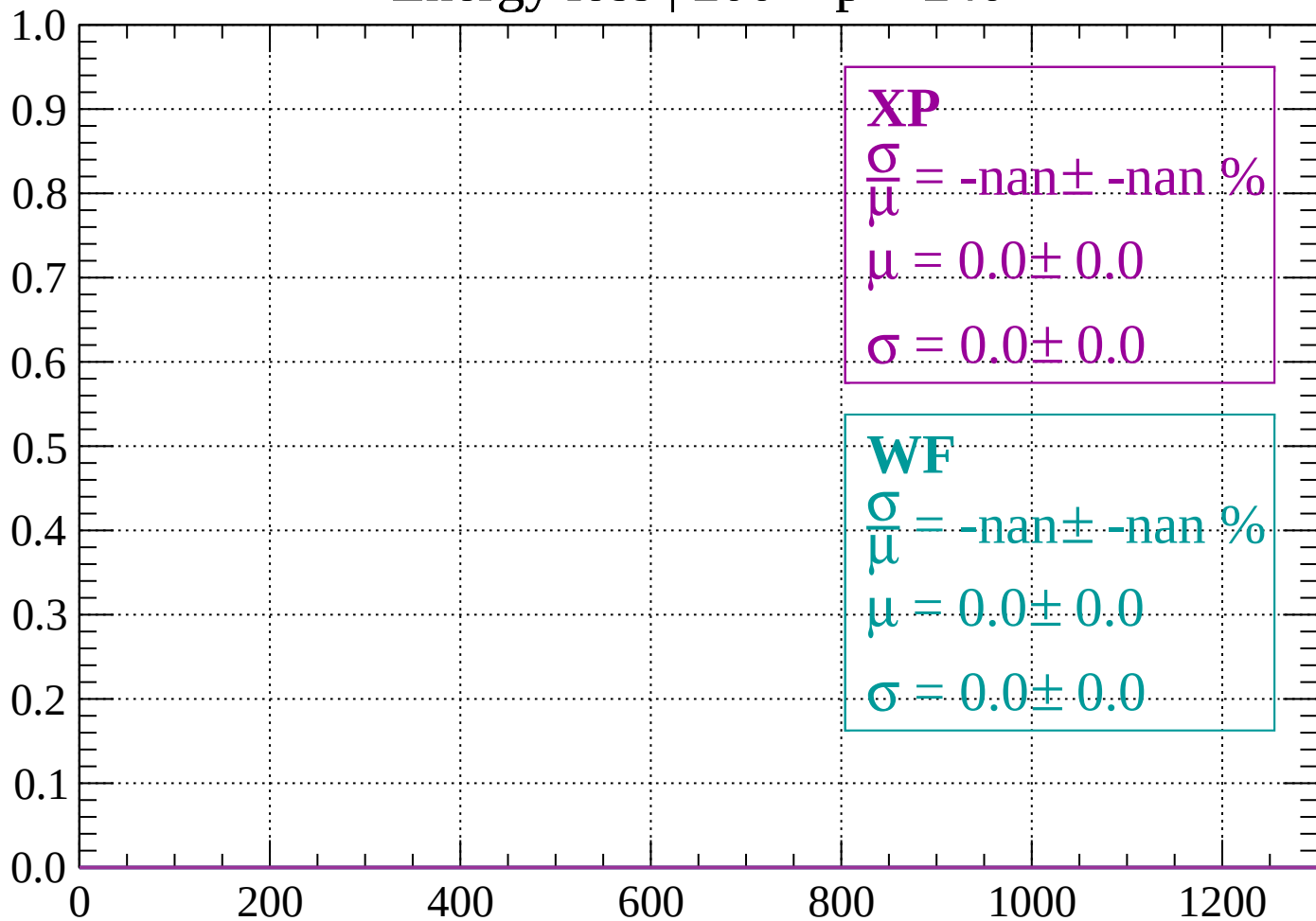
Count



dE/dx (ADC counts/cm)

Energy loss | $200 < p < 240$

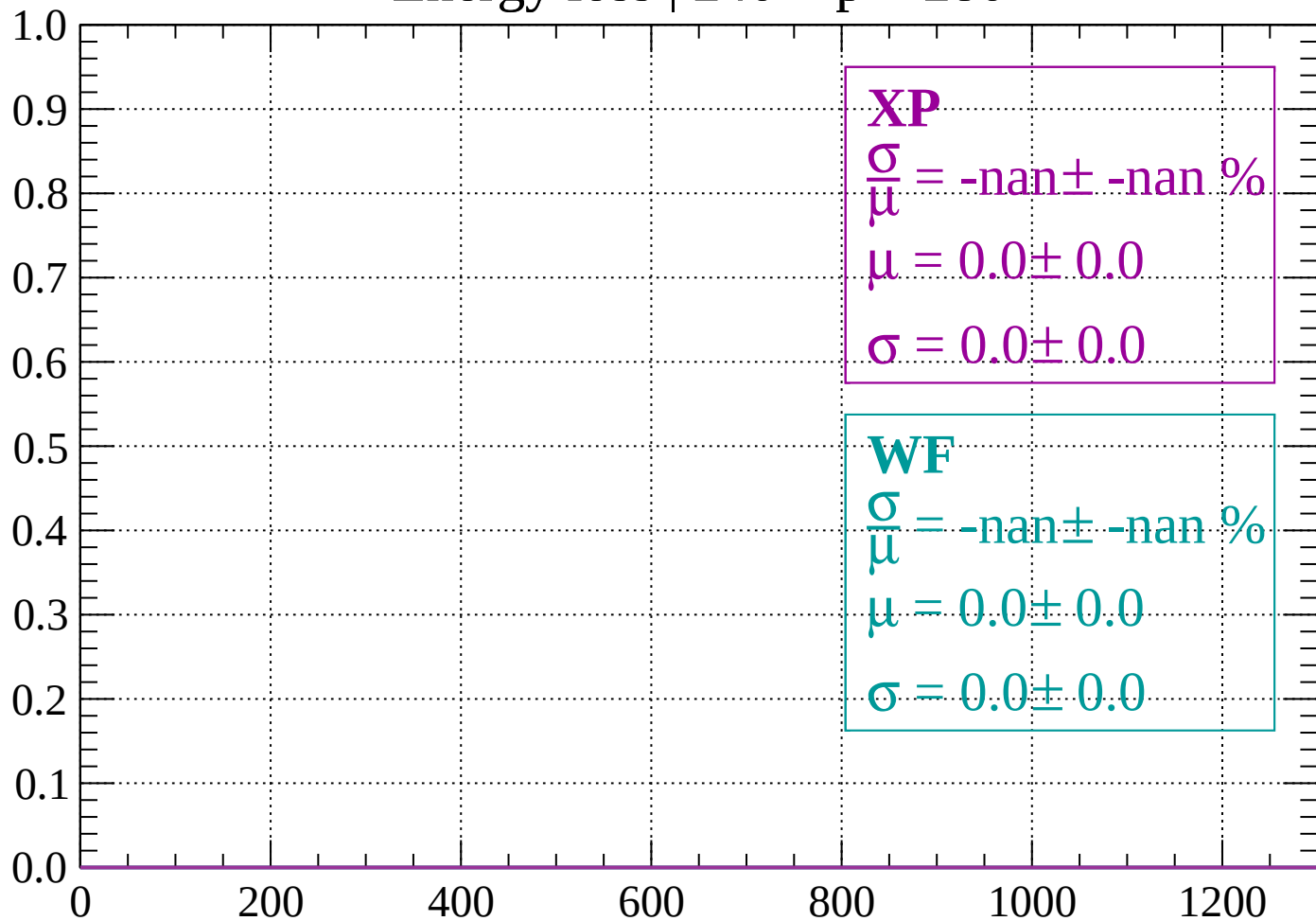
Count



dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

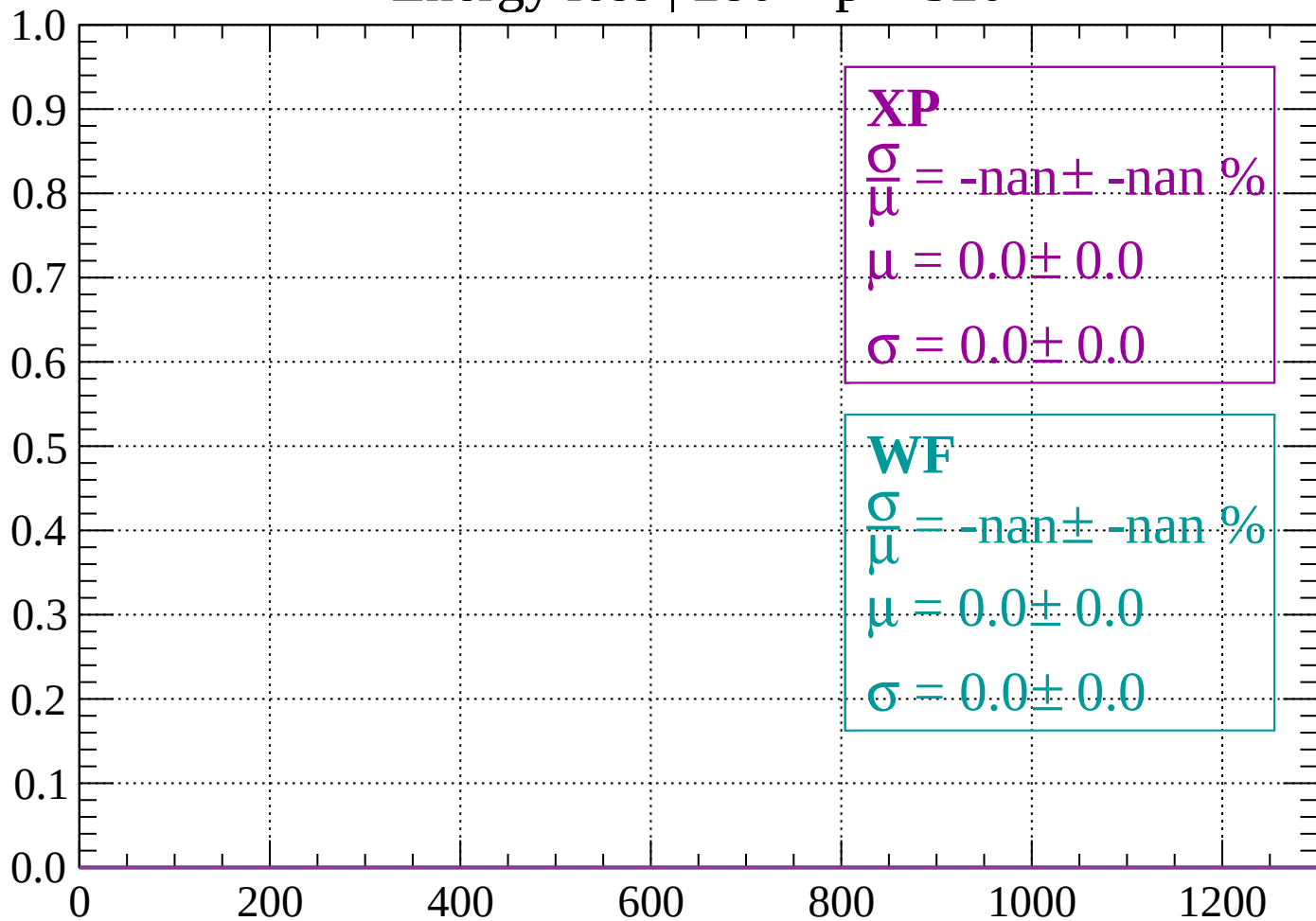
Count



dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

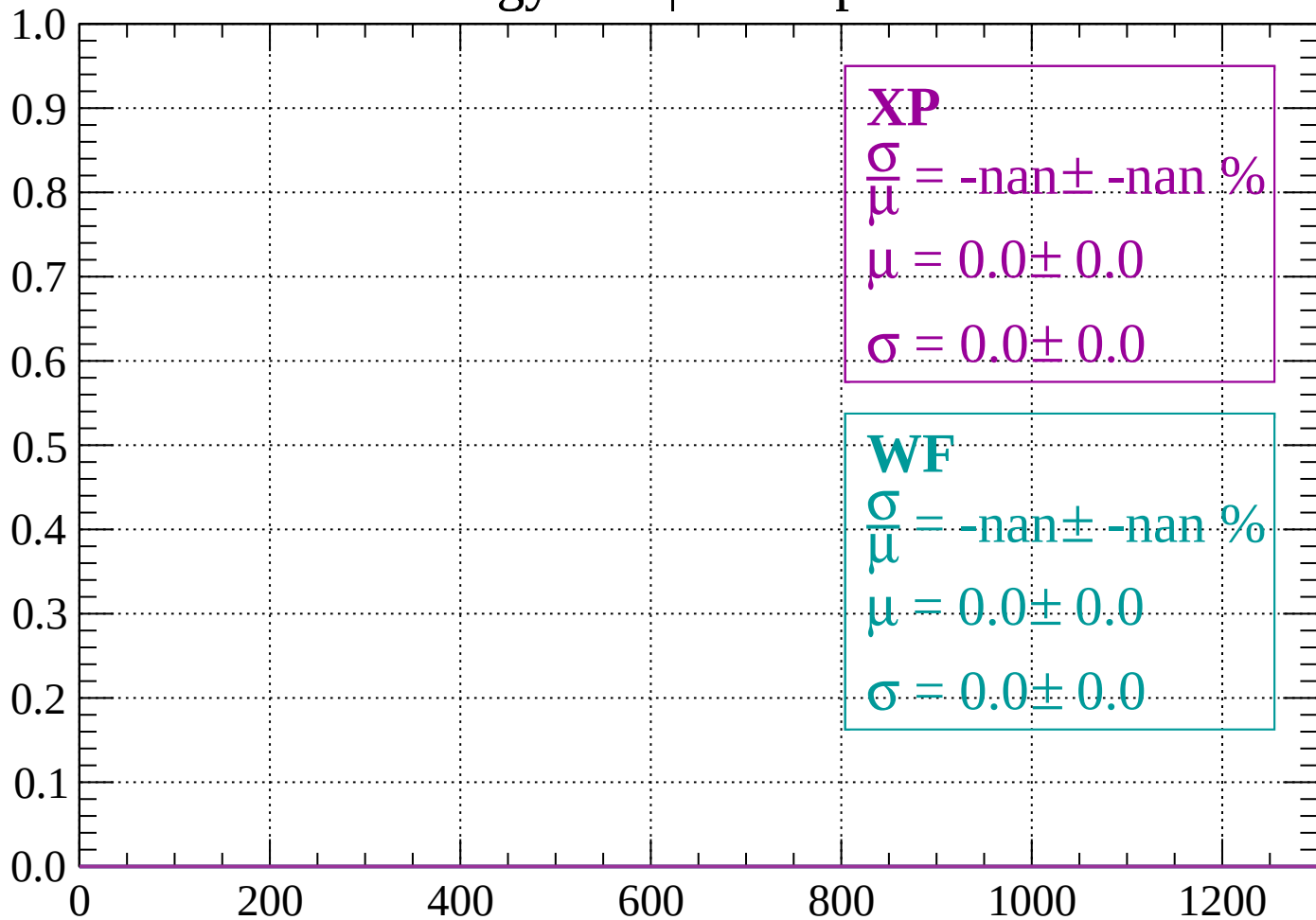
Count



dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

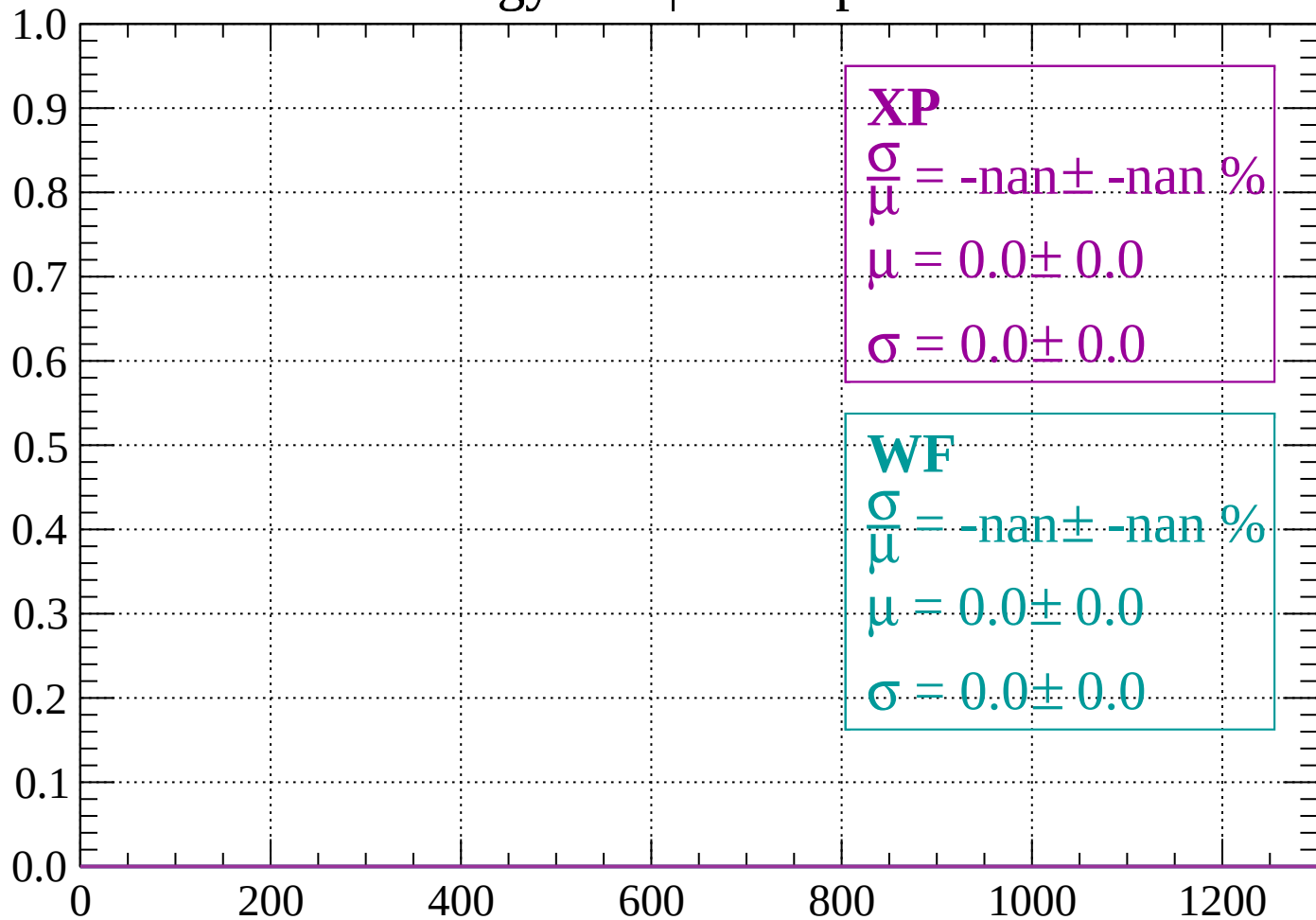
Count



dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

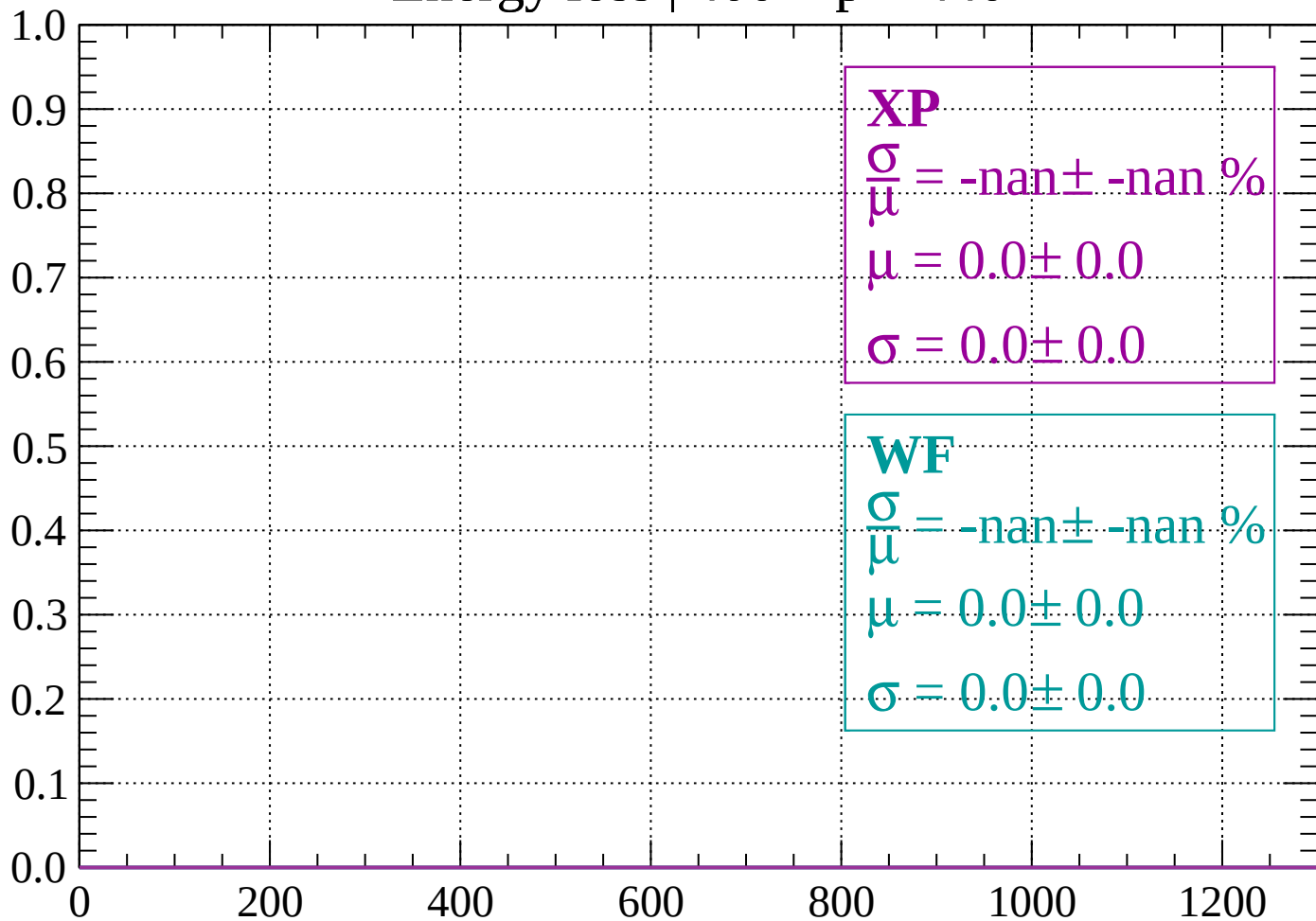
Count



dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

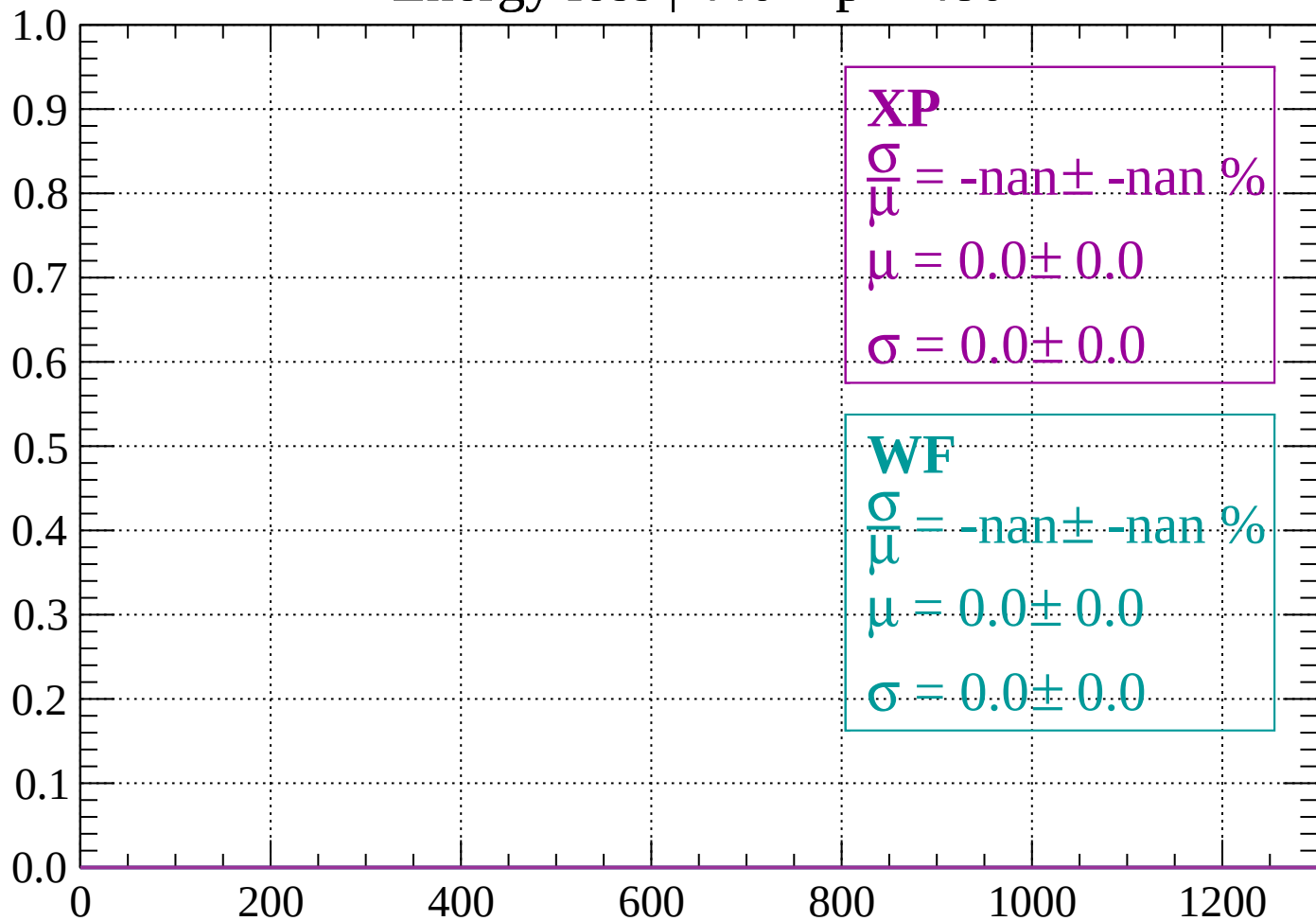
Count



dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

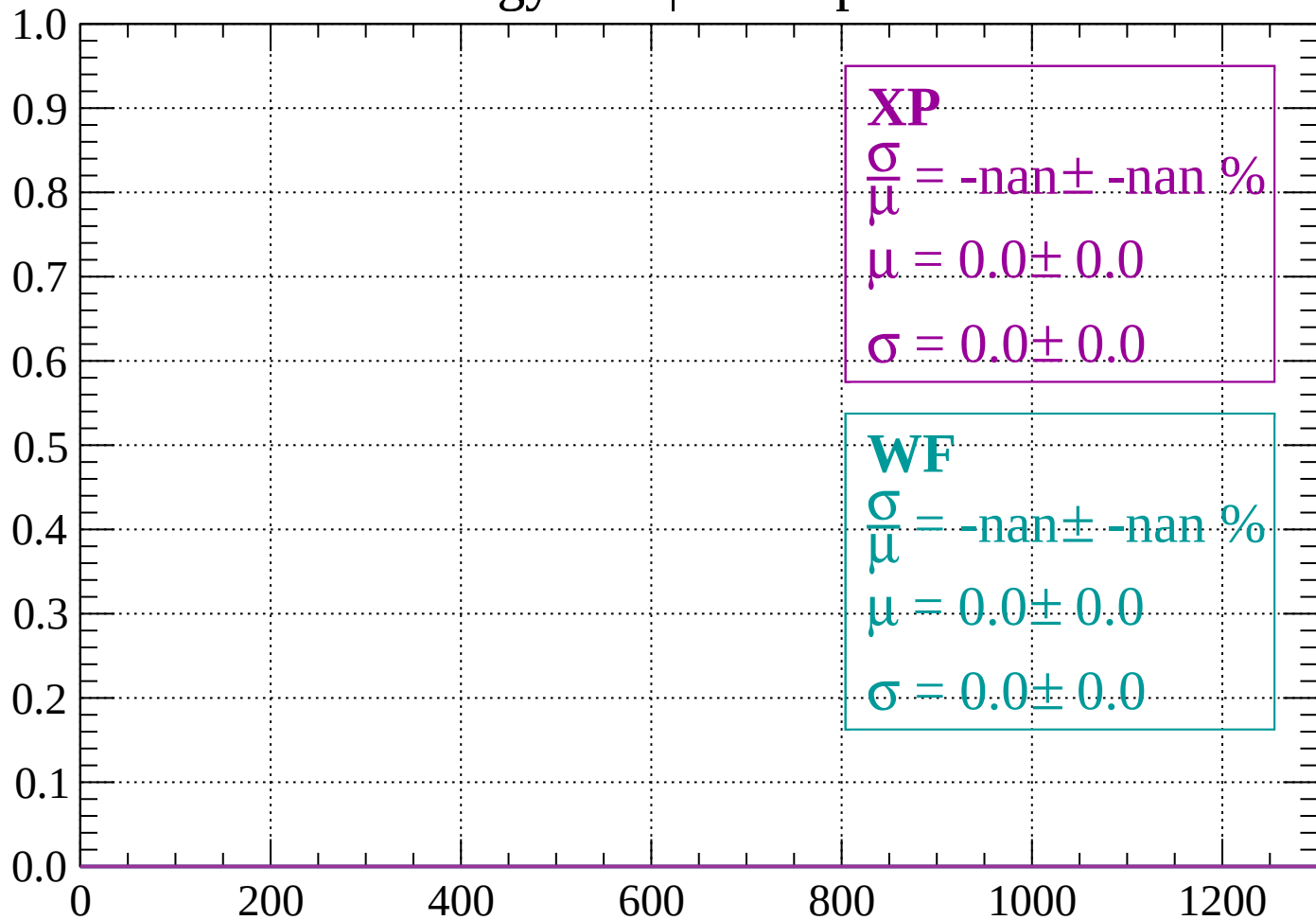
Count



dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

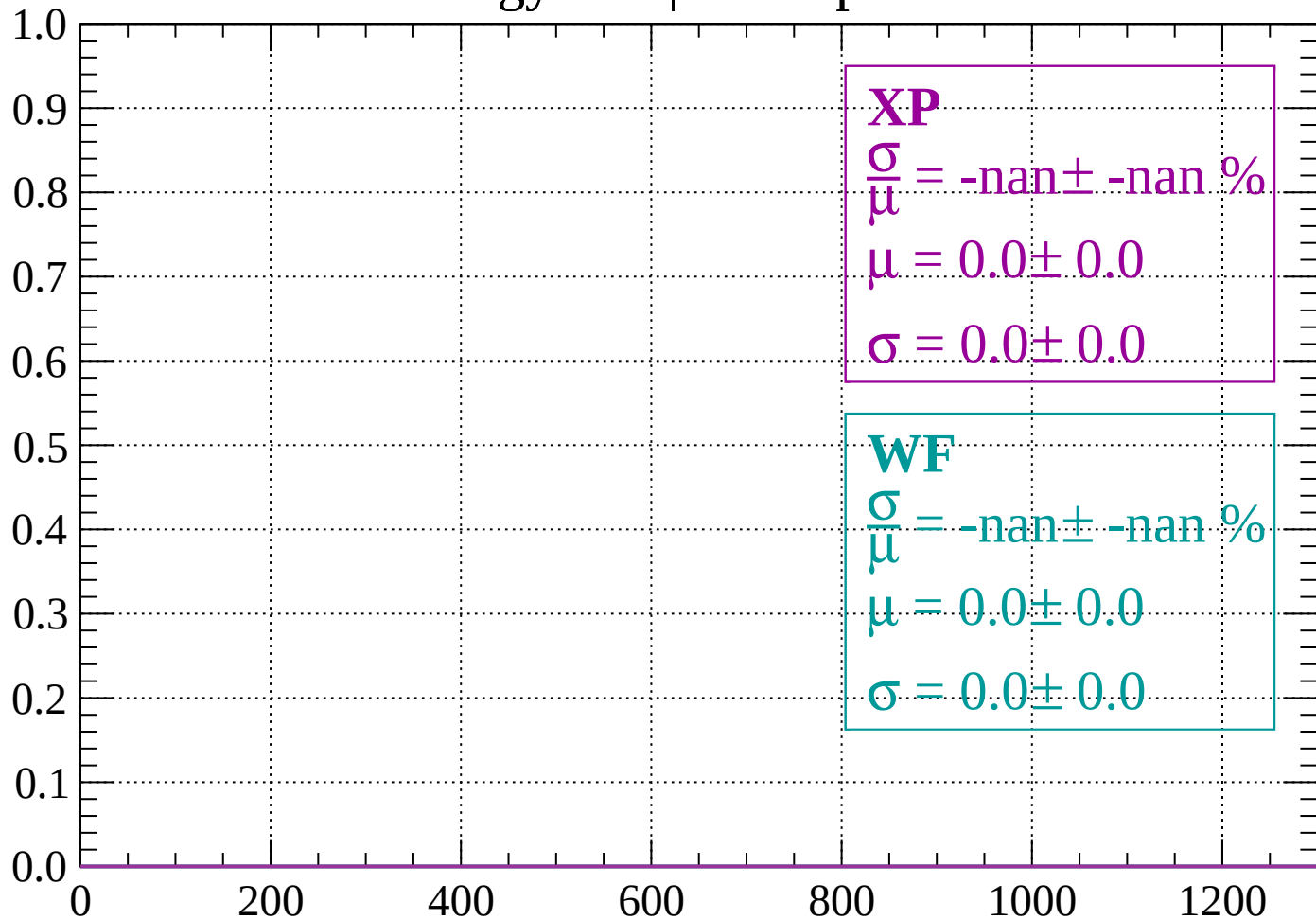
Count



dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

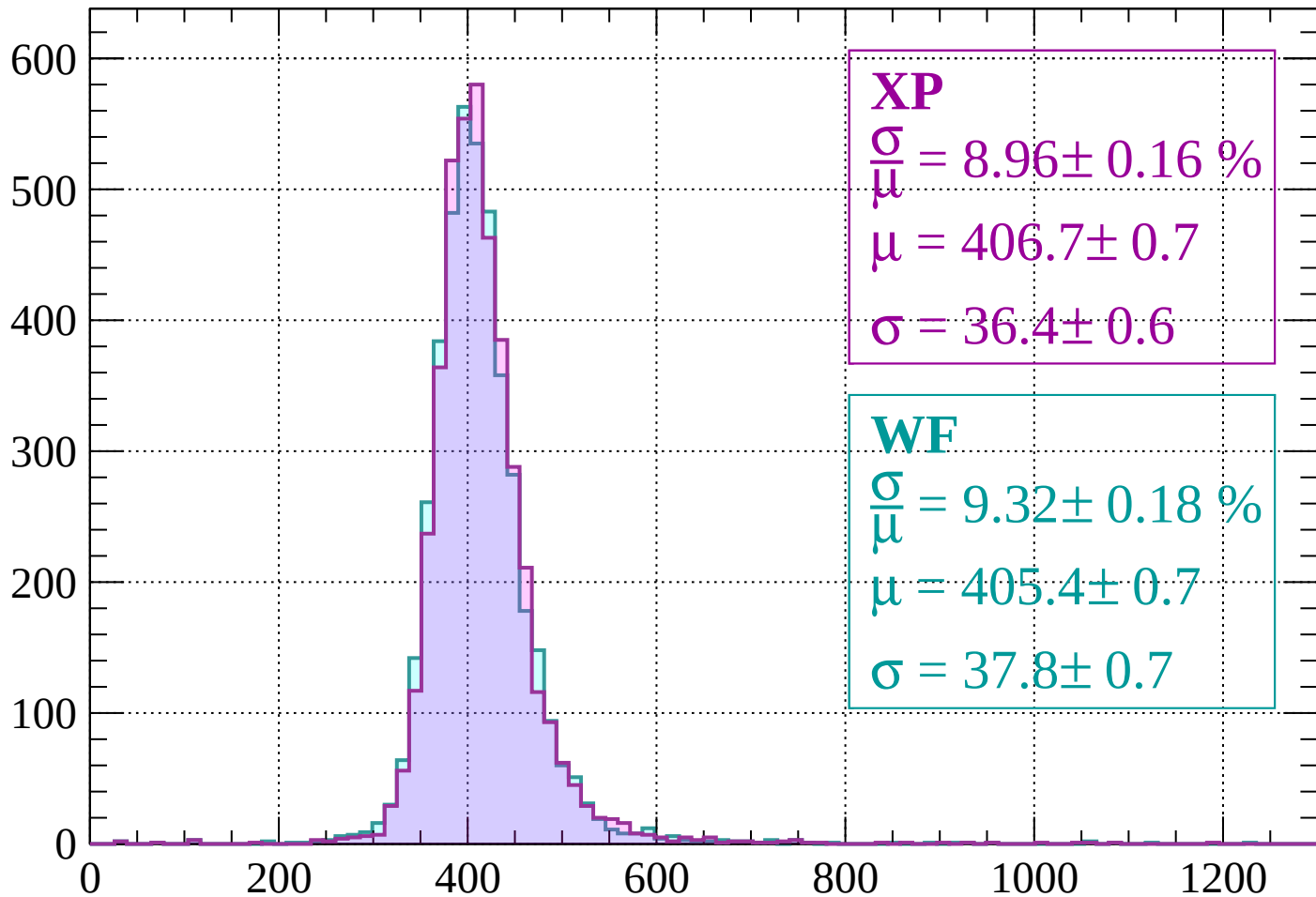
Count



dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 8.96 \pm 0.16 \%$$

$$\mu = 406.7 \pm 0.7$$

$$\sigma = 36.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.32 \pm 0.18 \%$$

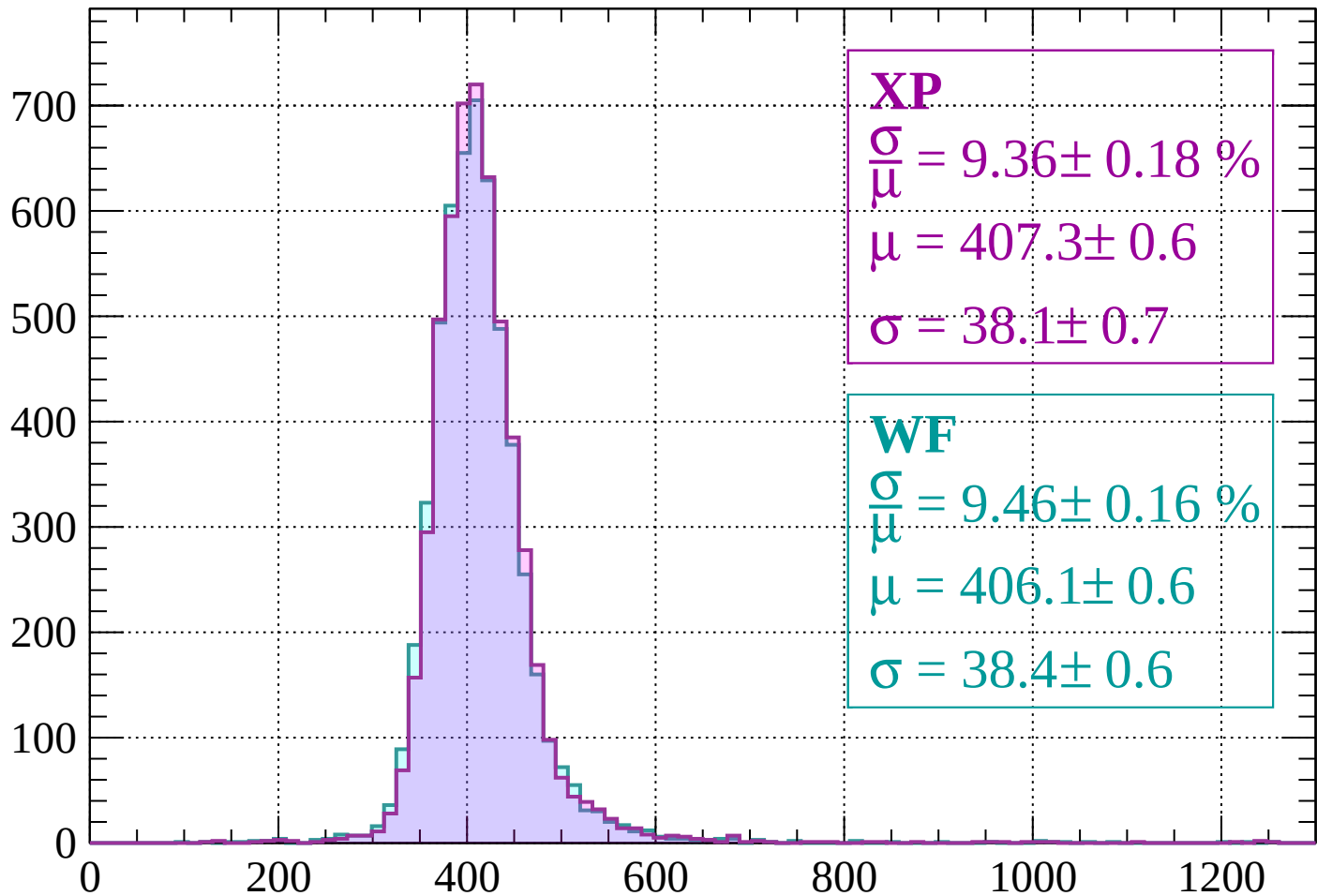
$$\mu = 405.4 \pm 0.7$$

$$\sigma = 37.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 9.36 \pm 0.18 \%$$

$$\mu = 407.3 \pm 0.6$$

$$\sigma = 38.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.46 \pm 0.16 \%$$

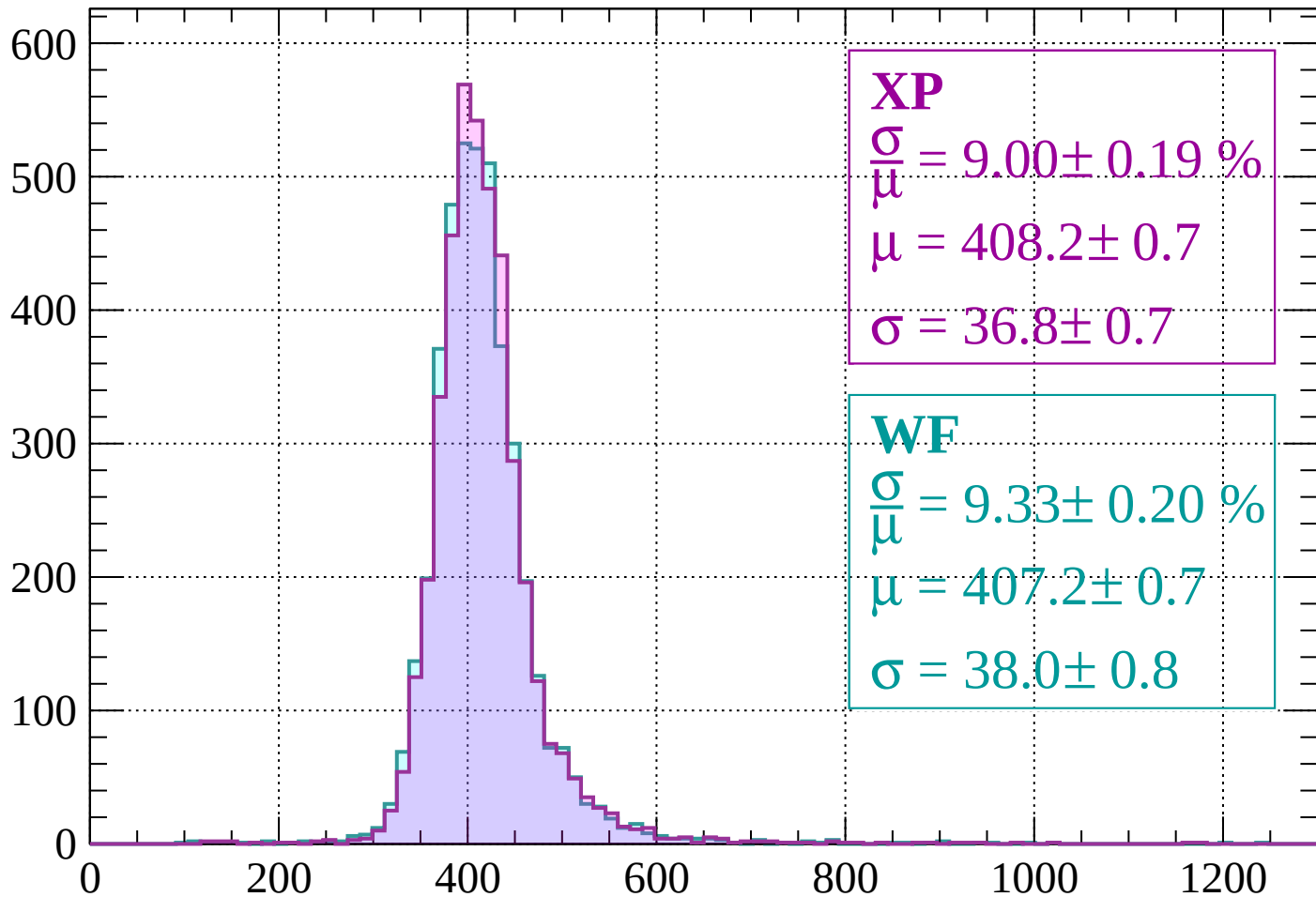
$$\mu = 406.1 \pm 0.6$$

$$\sigma = 38.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\mu} = 9.00 \pm 0.19 \%$$

$$\mu = 408.2 \pm 0.7$$

$$\sigma = 36.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.33 \pm 0.20 \%$$

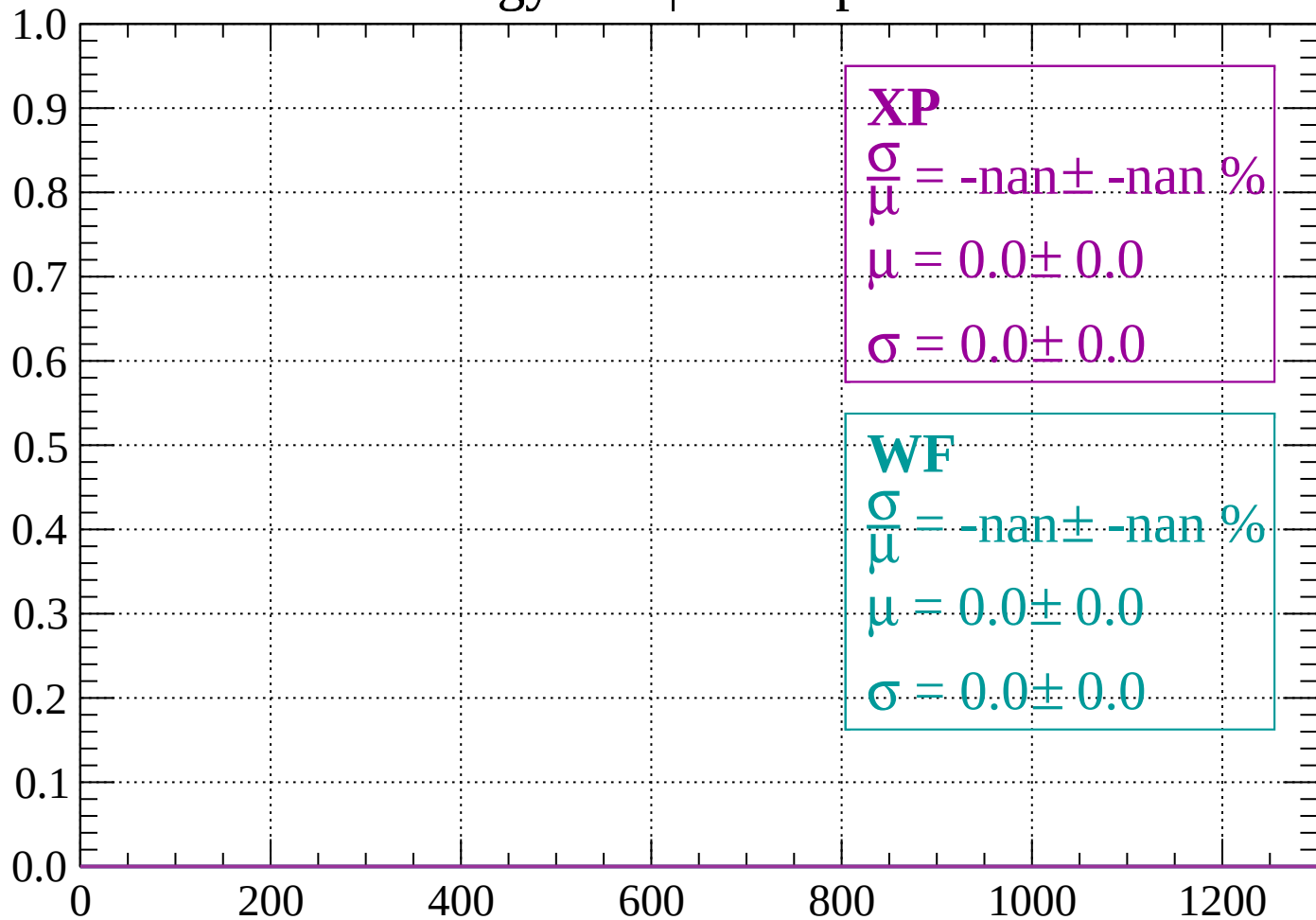
$$\mu = 407.2 \pm 0.7$$

$$\sigma = 38.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $680 < p < 720$

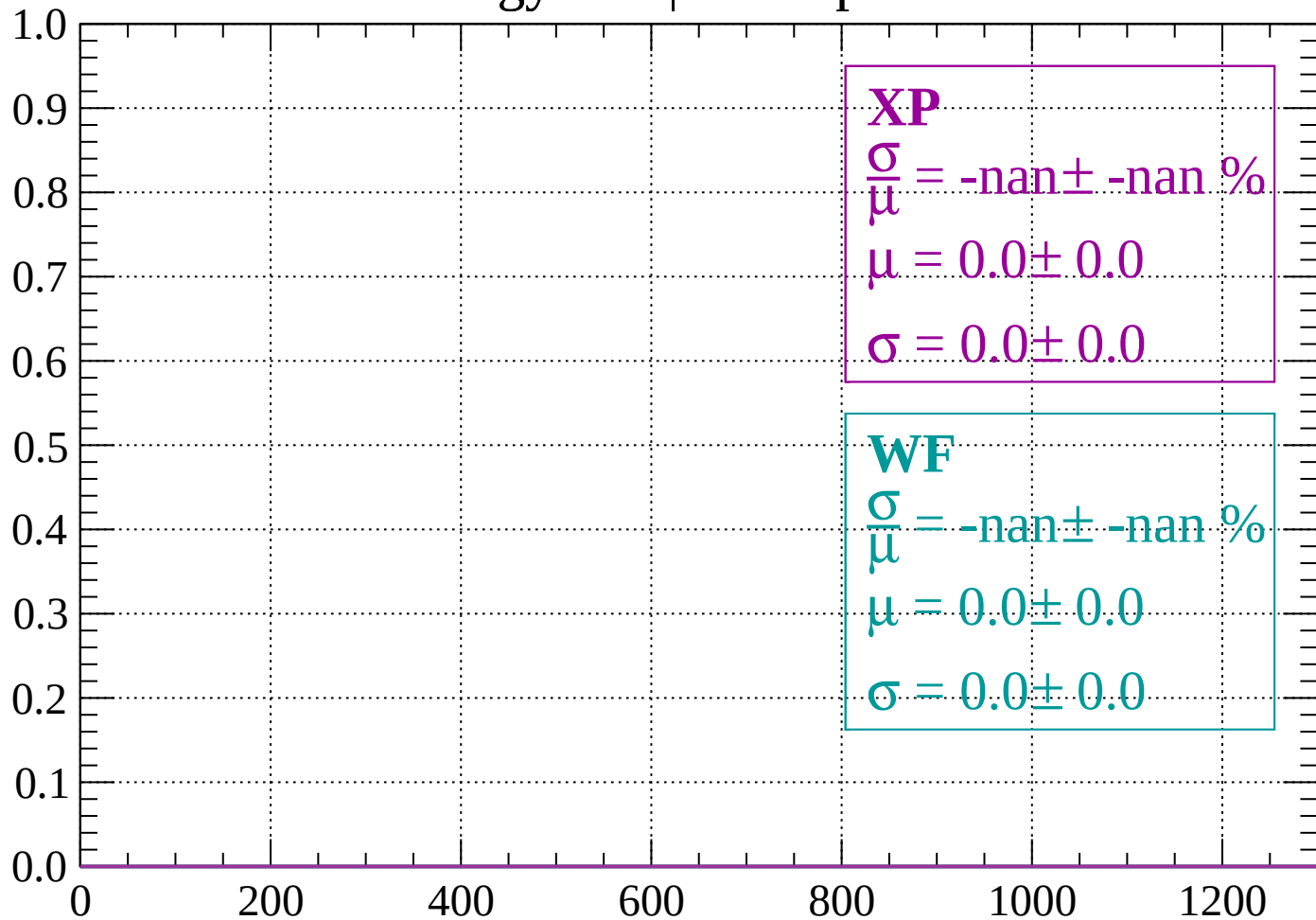
Count



dE/dx (ADC counts/cm)

Energy loss | $720 < p < 760$

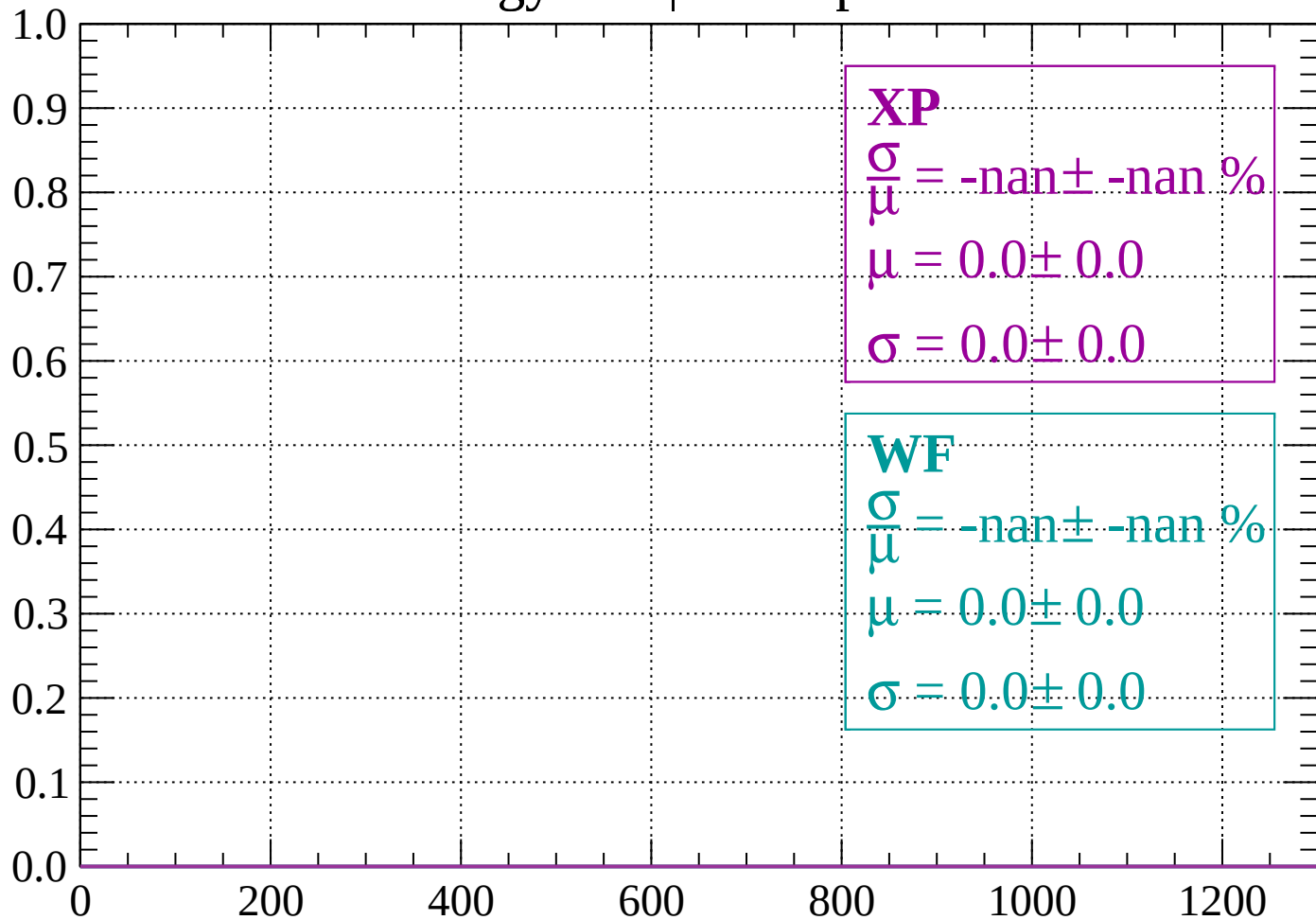
Count



dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

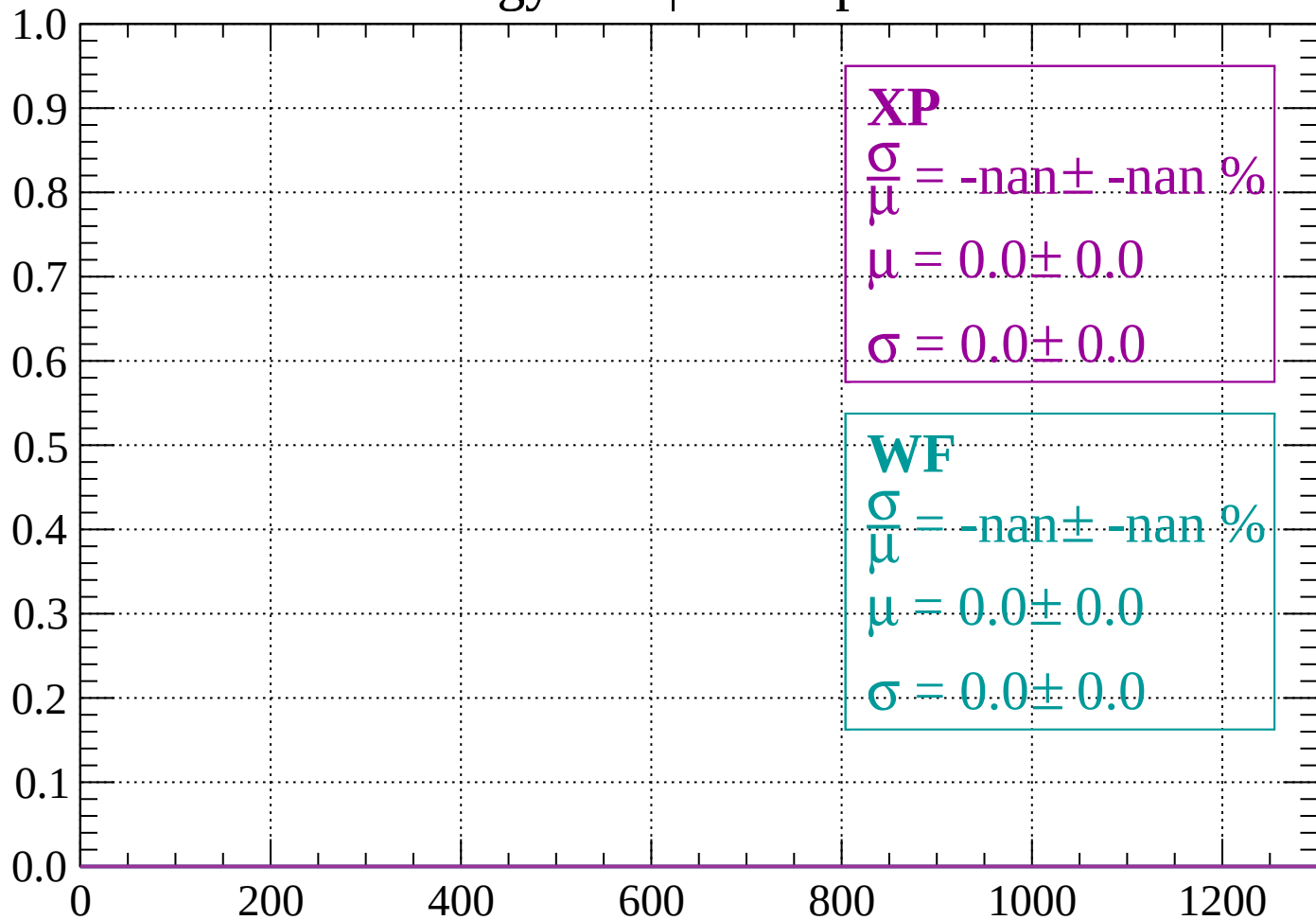
Count



dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

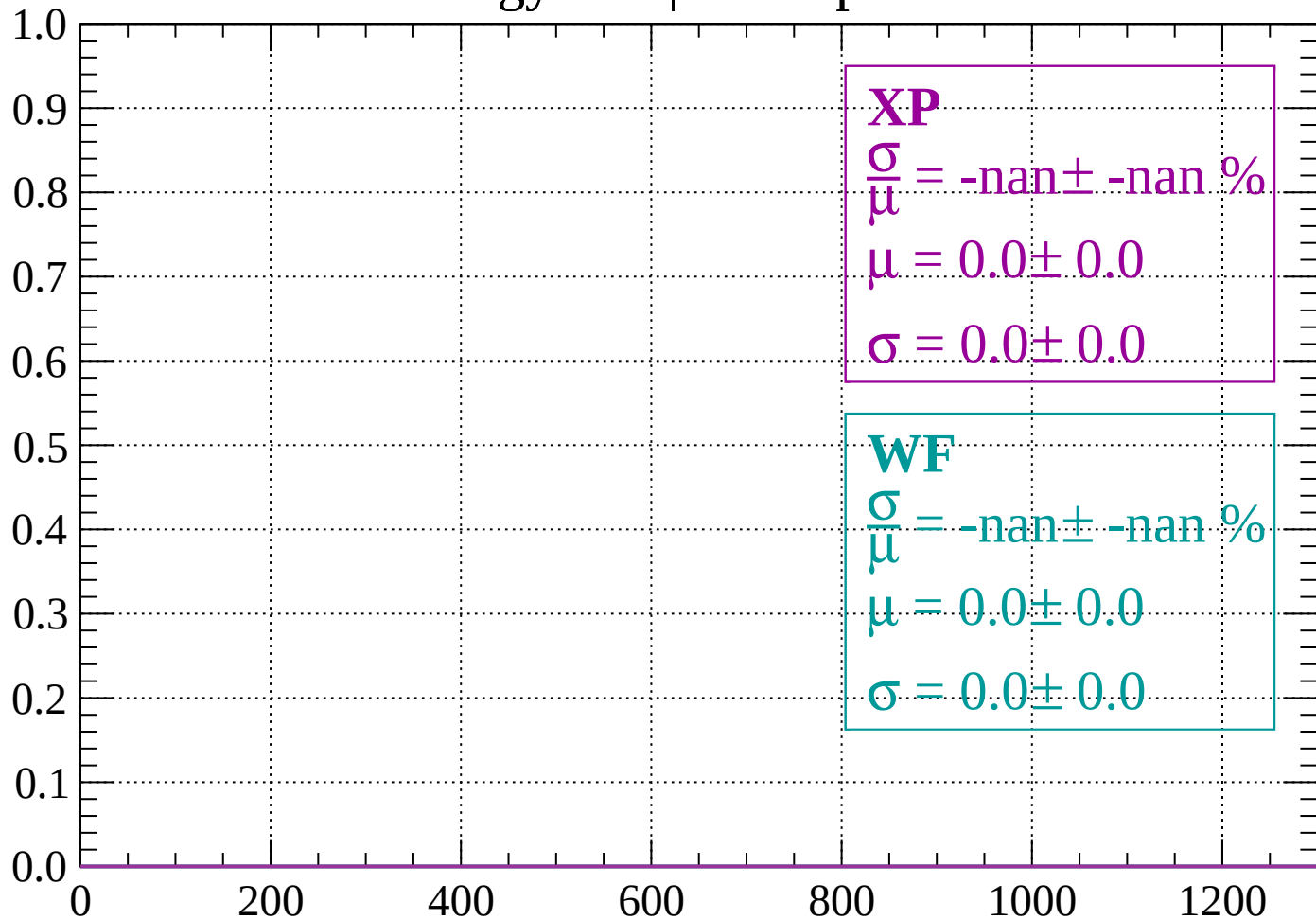
Count



dE/dx (ADC counts/cm)

Energy loss | $840 < p < 880$

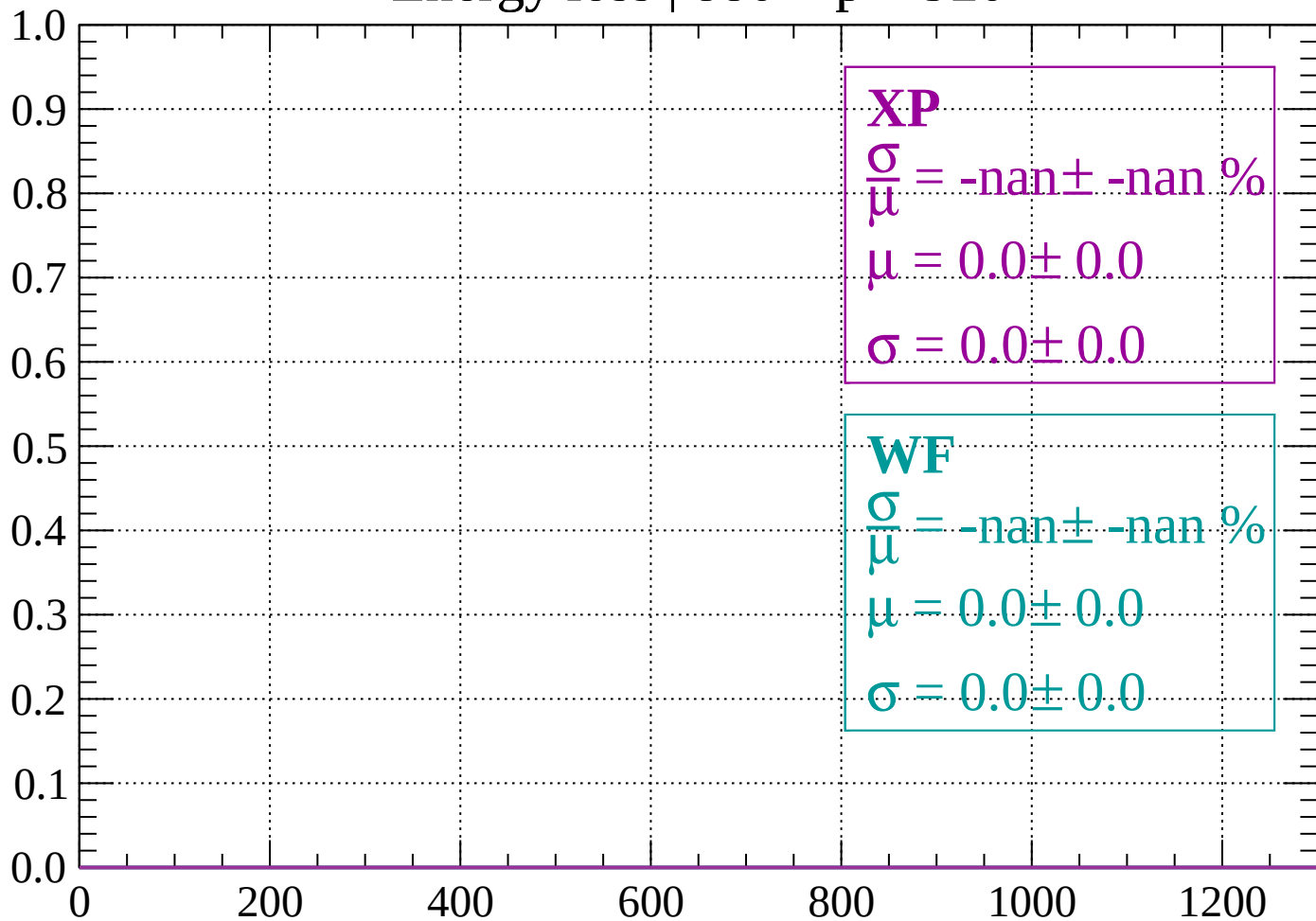
Count



dE/dx (ADC counts/cm)

Energy loss | $880 < p < 920$

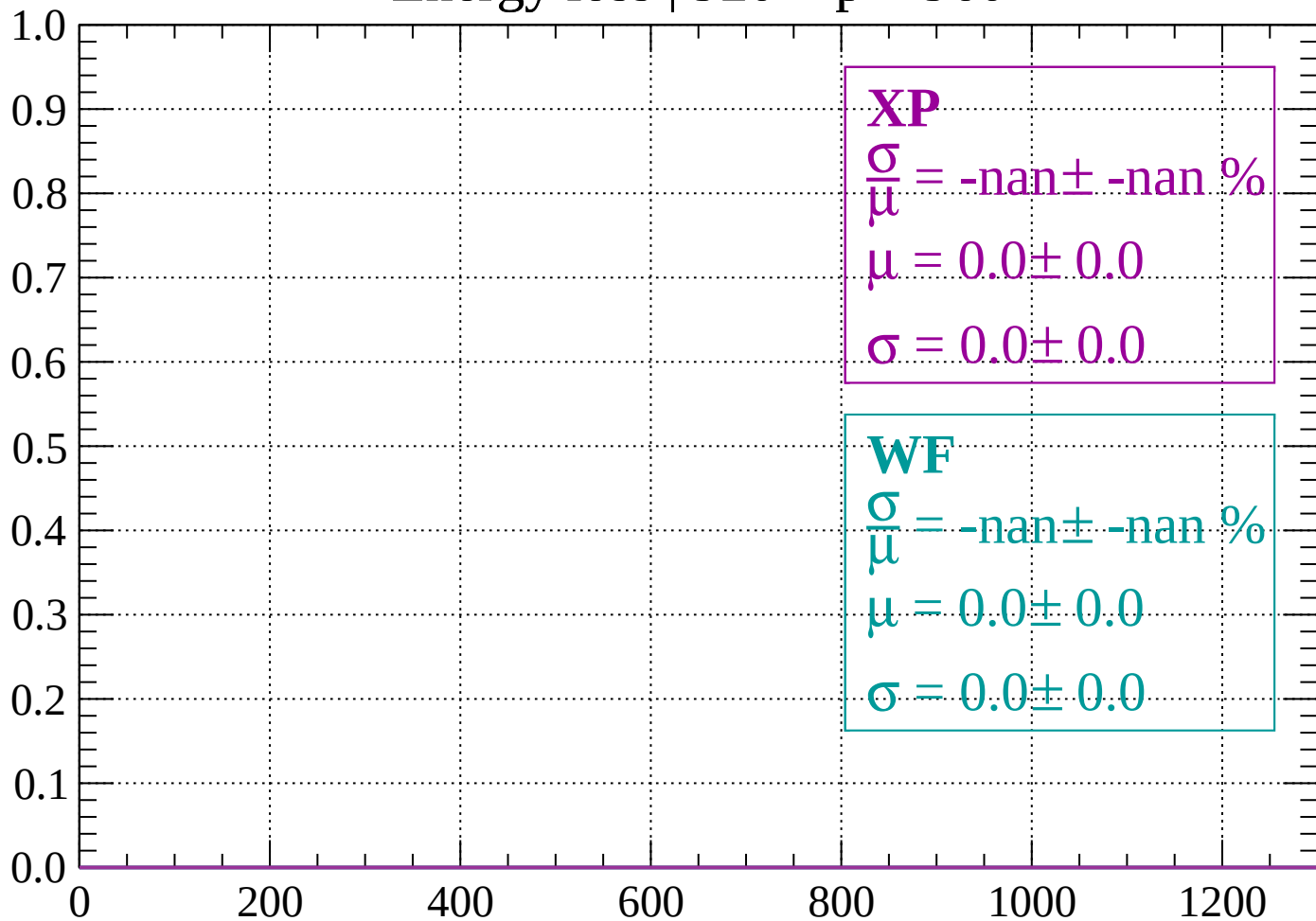
Count



dE/dx (ADC counts/cm)

Energy loss | $920 < p < 960$

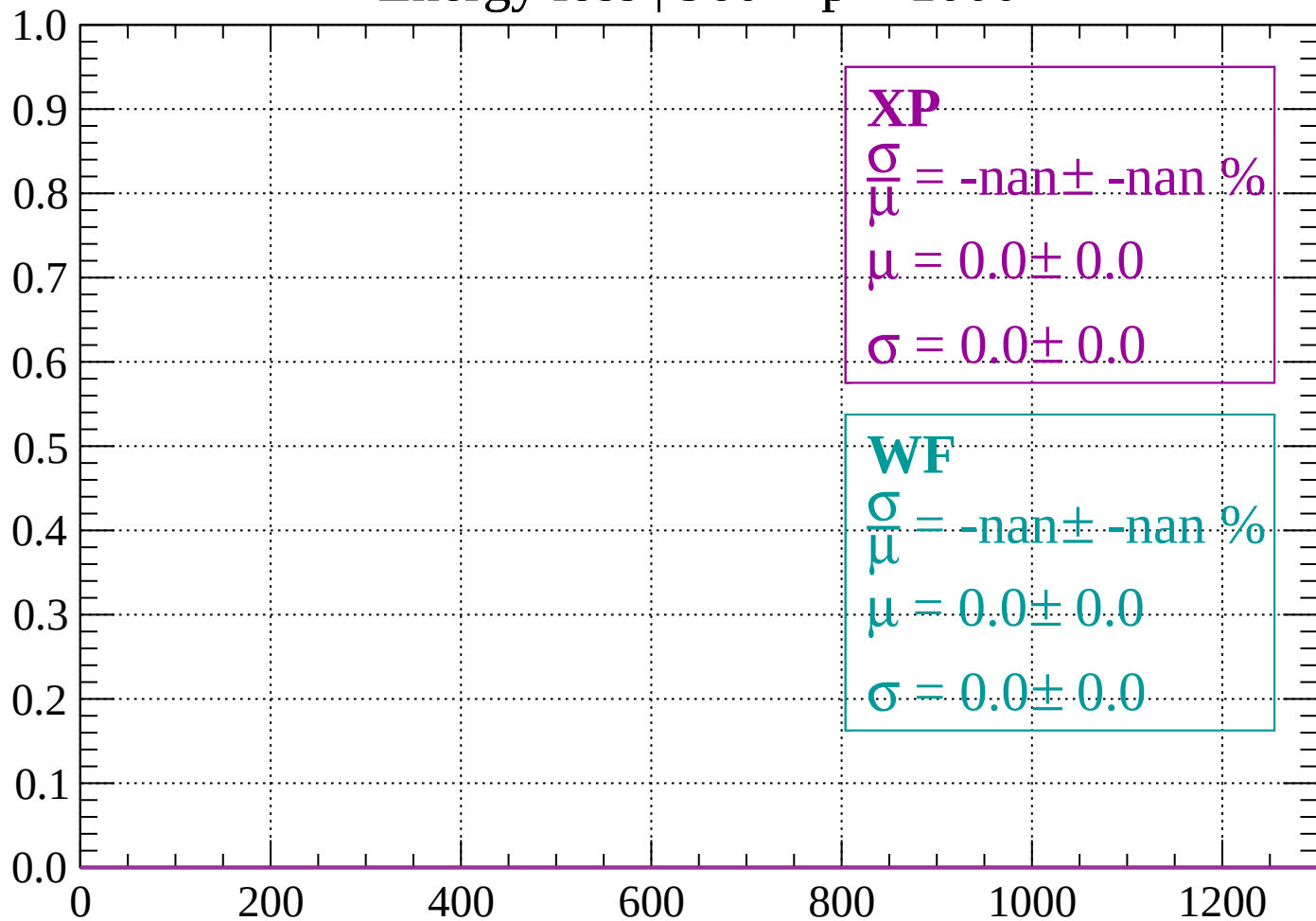
Count



dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

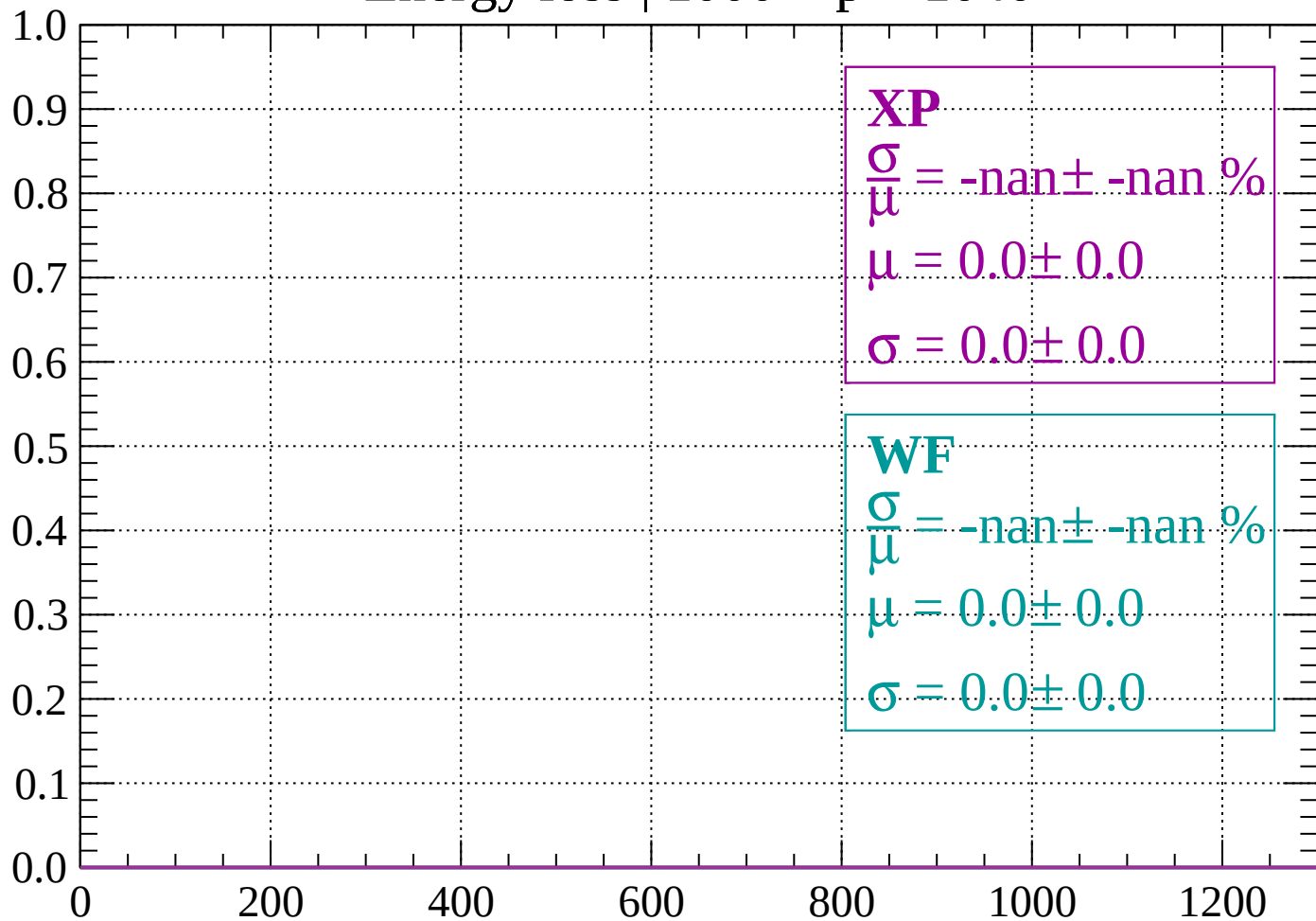
Count



dE/dx (ADC counts/cm)

Energy loss | $1000 < p < 1040$

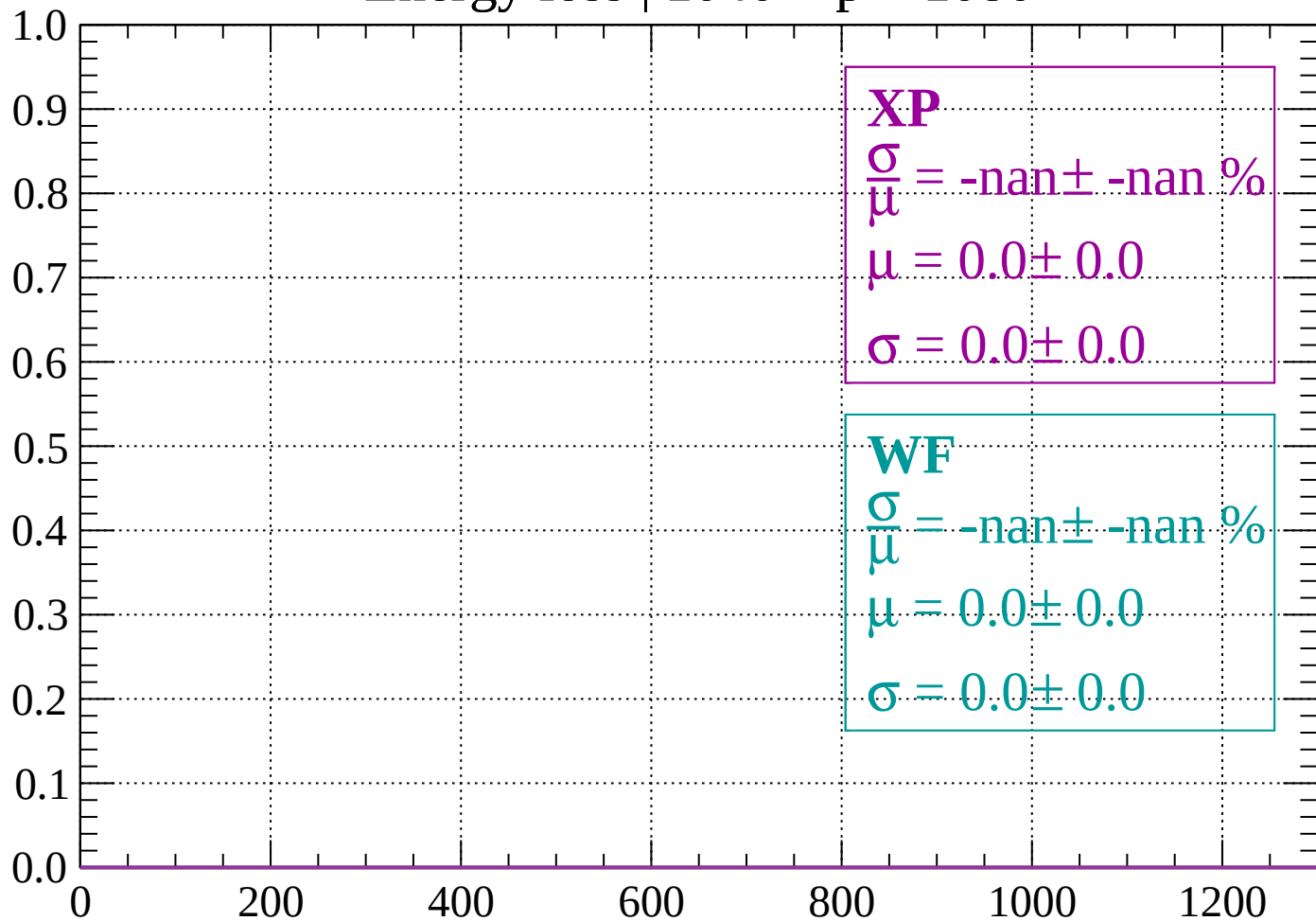
Count



dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1080$

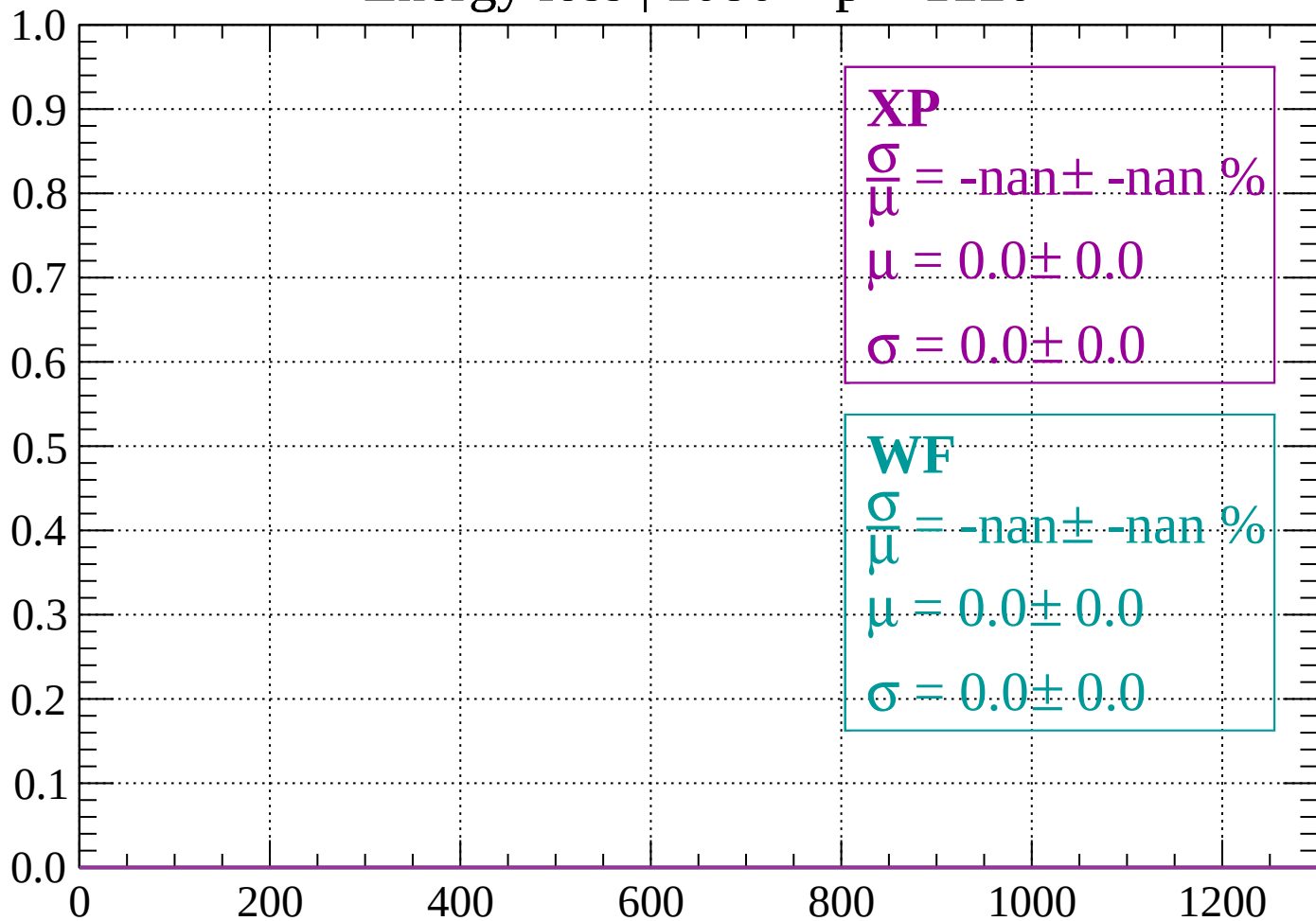
Count



dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1120$

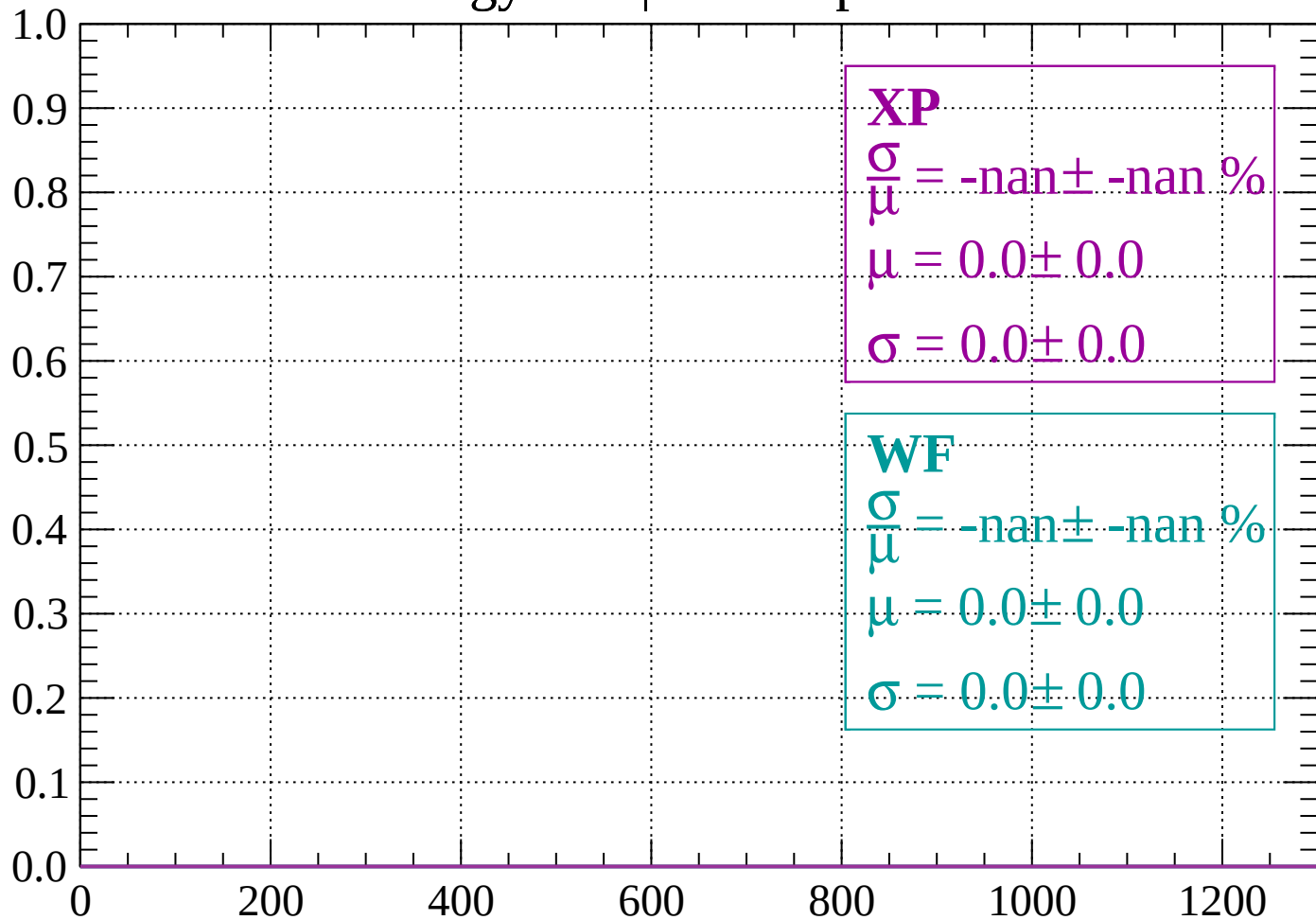
Count



dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1160$

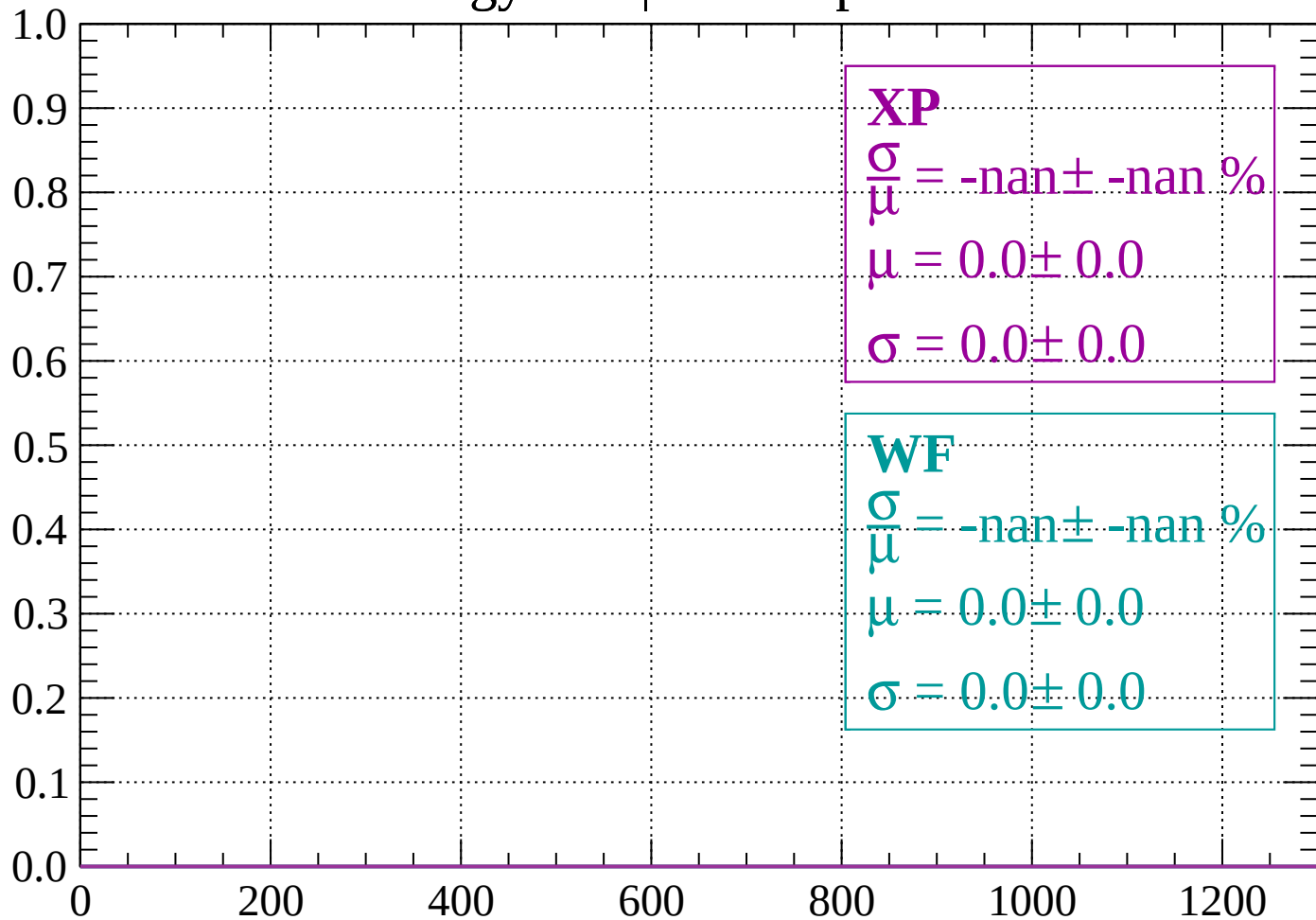
Count



dE/dx (ADC counts/cm)

Energy loss | $1160 < p < 1200$

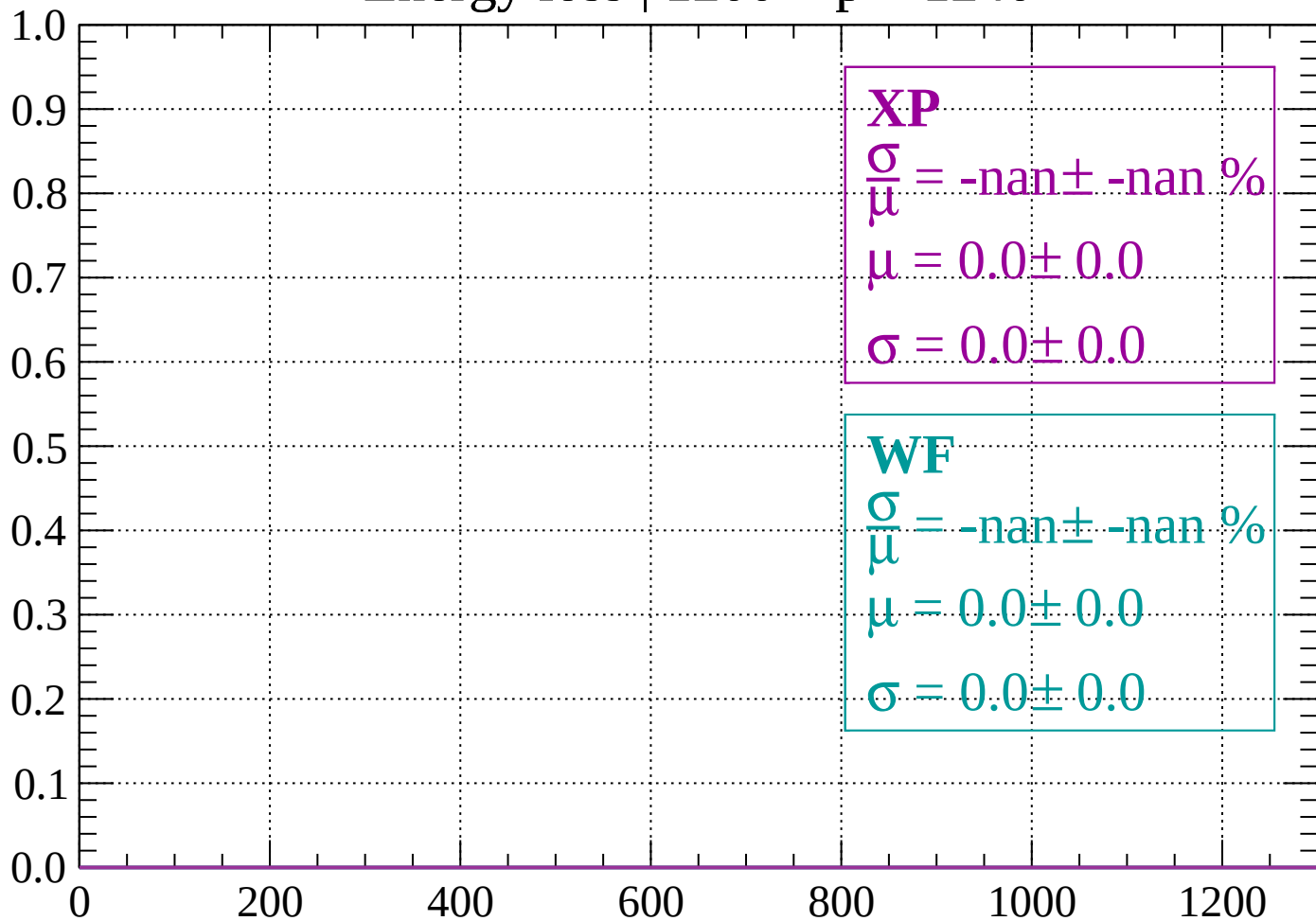
Count



dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

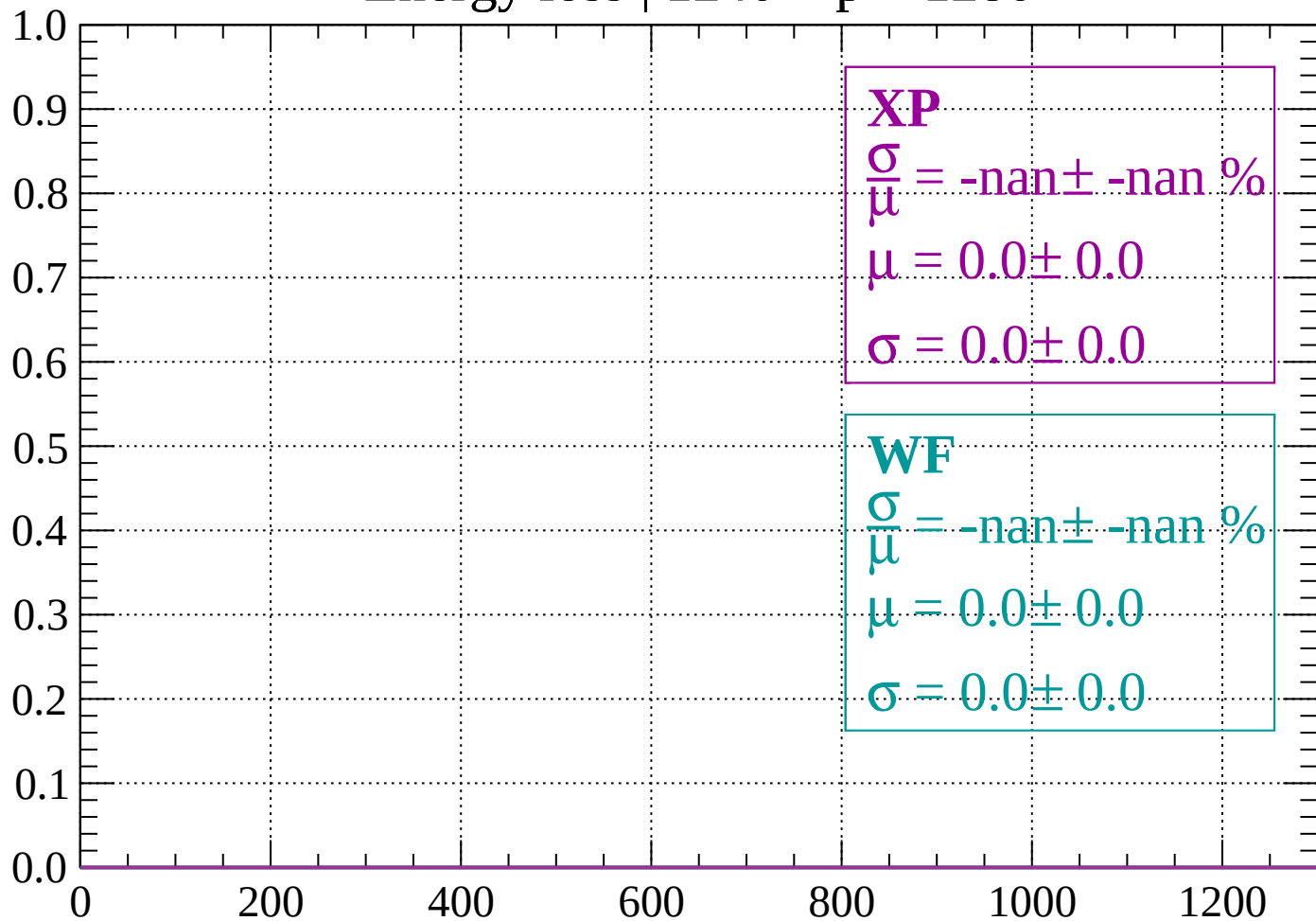
Count



dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

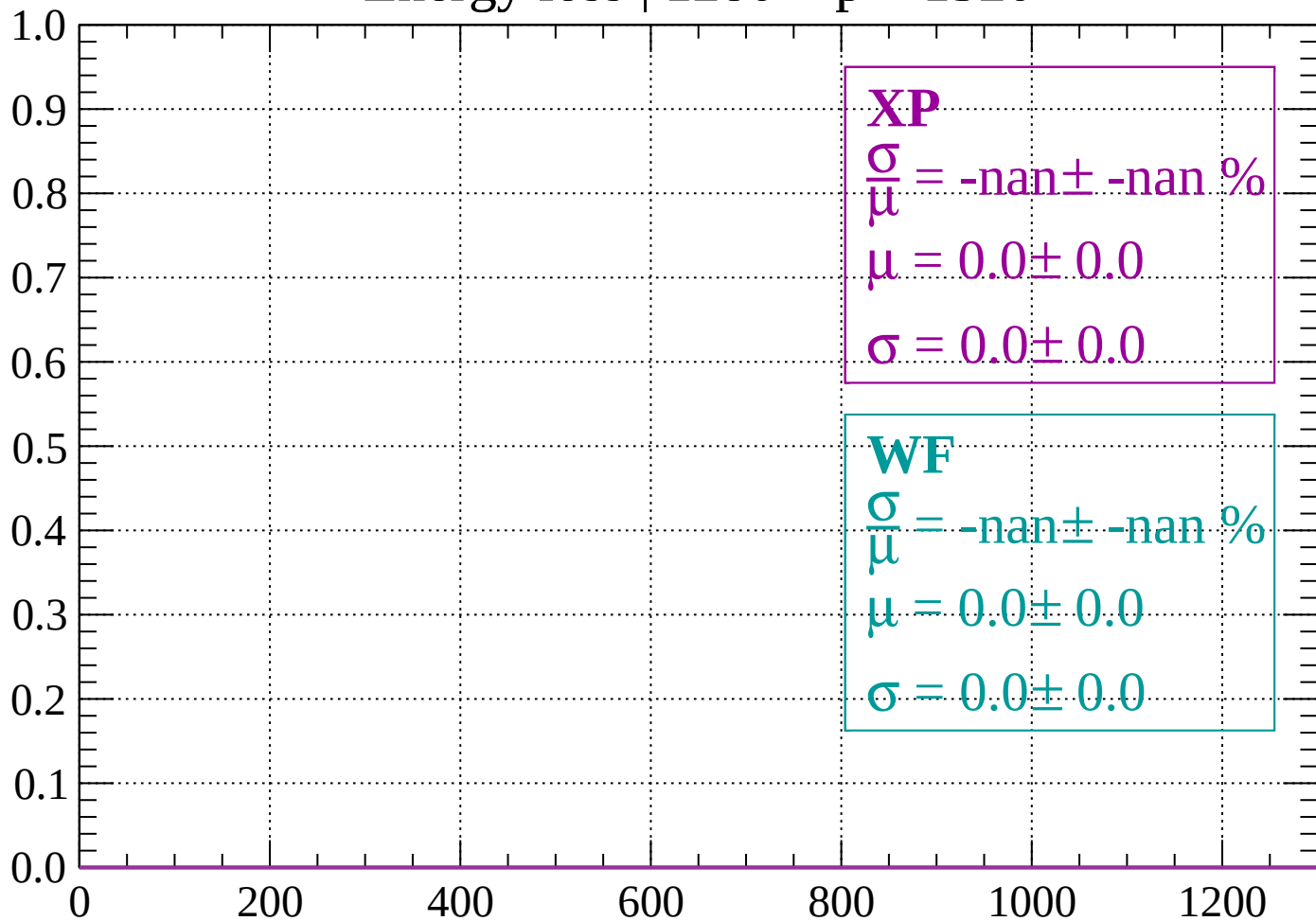
Count



dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

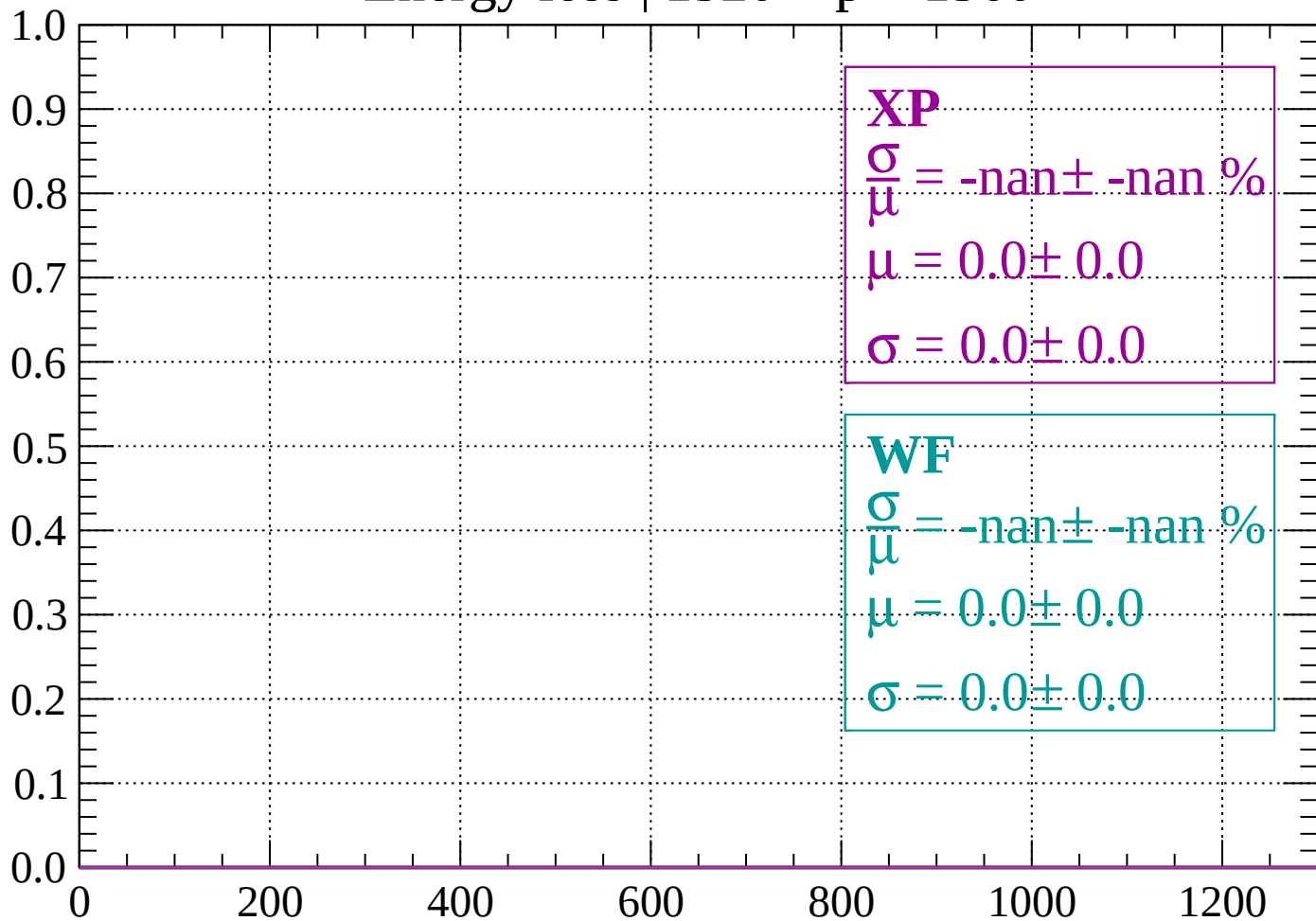
Count



dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1360$

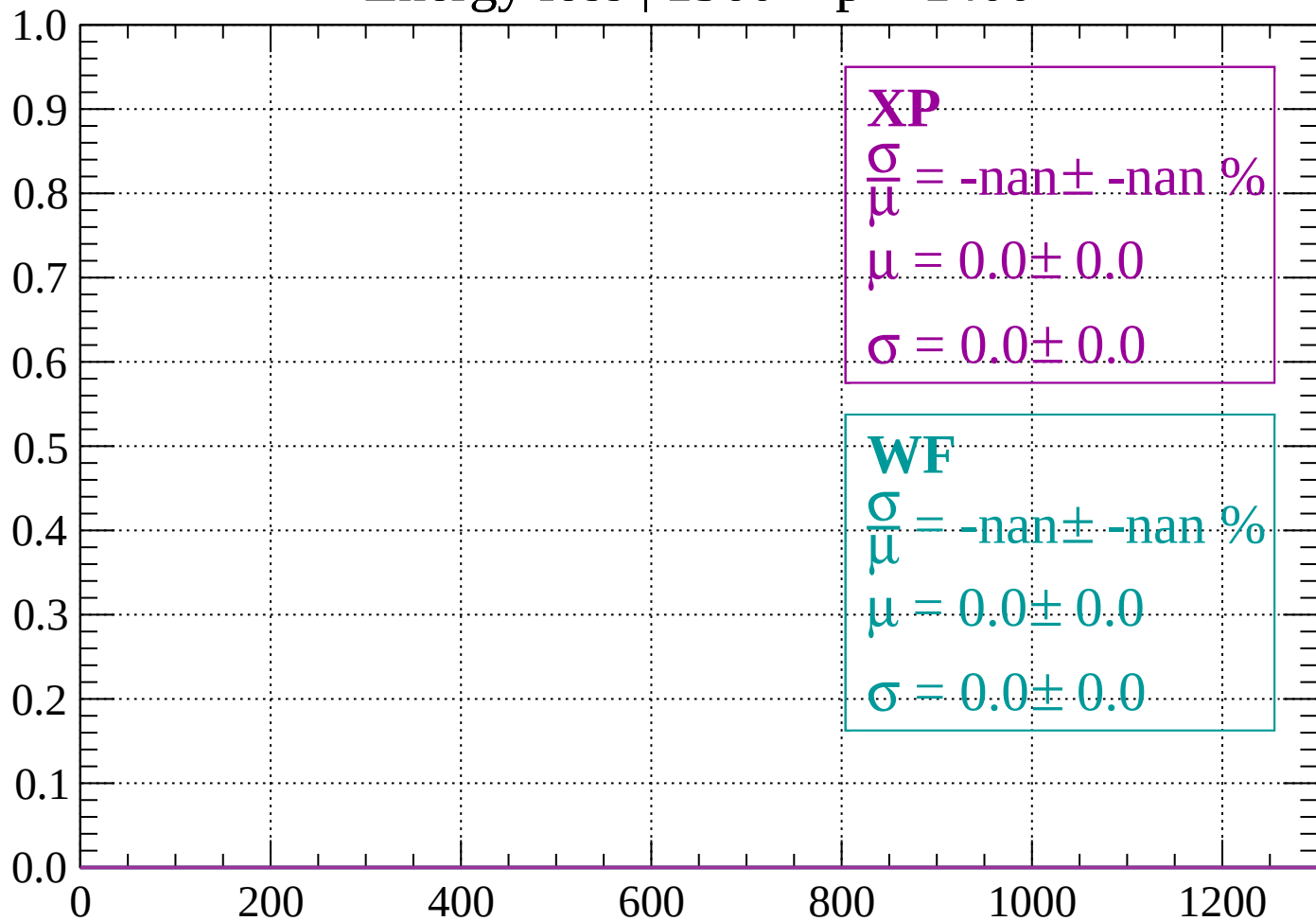
Count



dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1400$

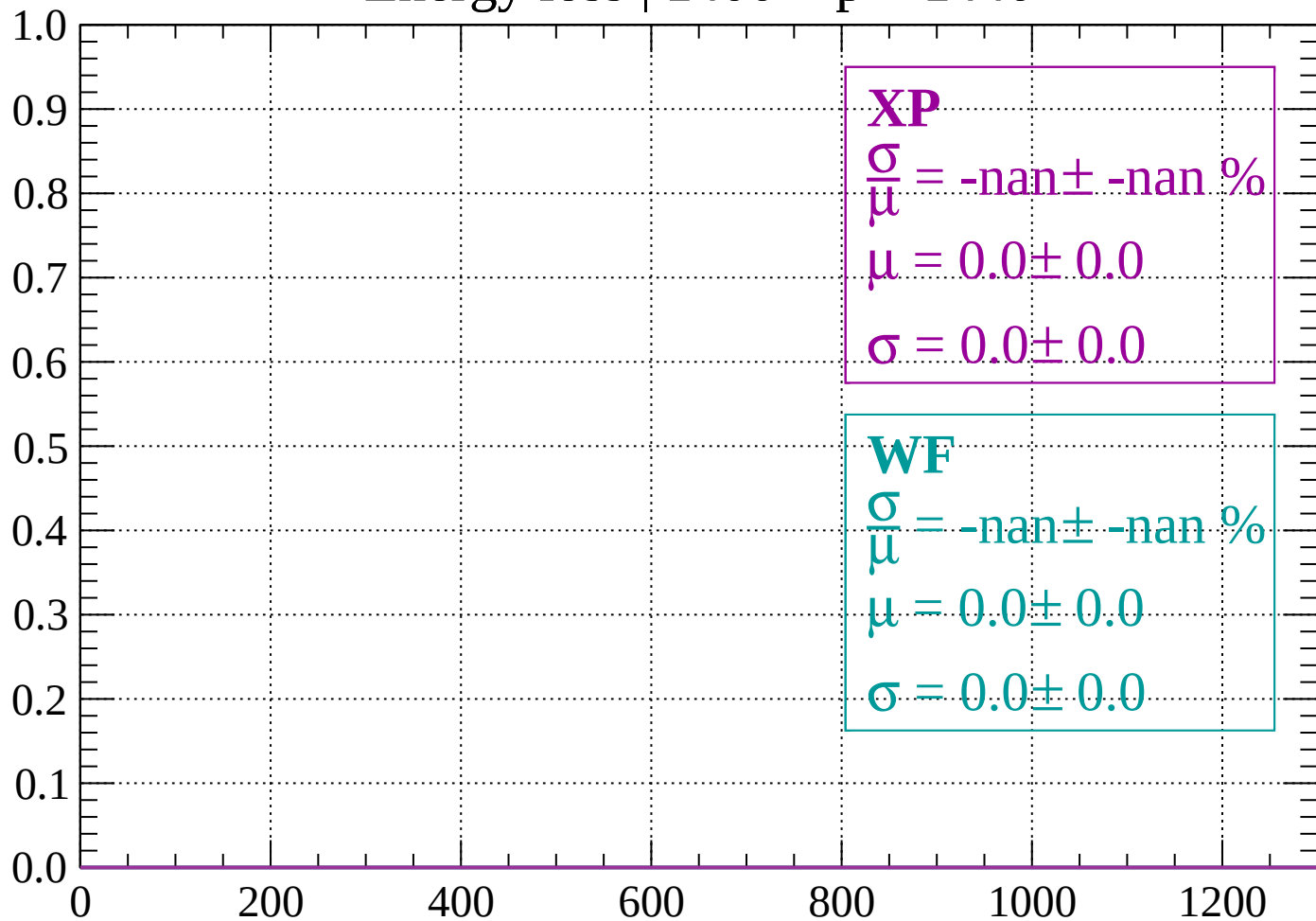
Count



dE/dx (ADC counts/cm)

Energy loss | $1400 < p < 1440$

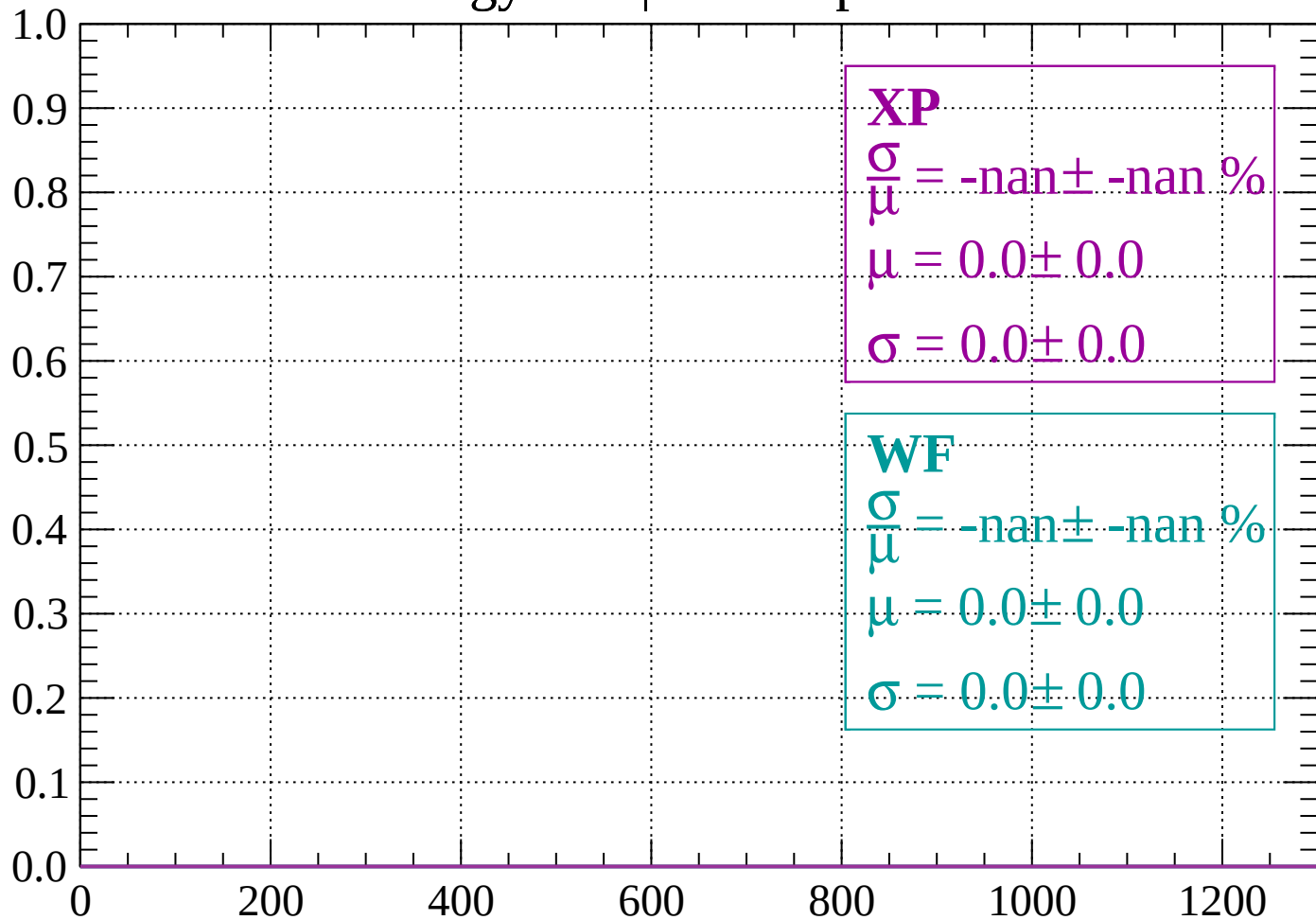
Count



dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

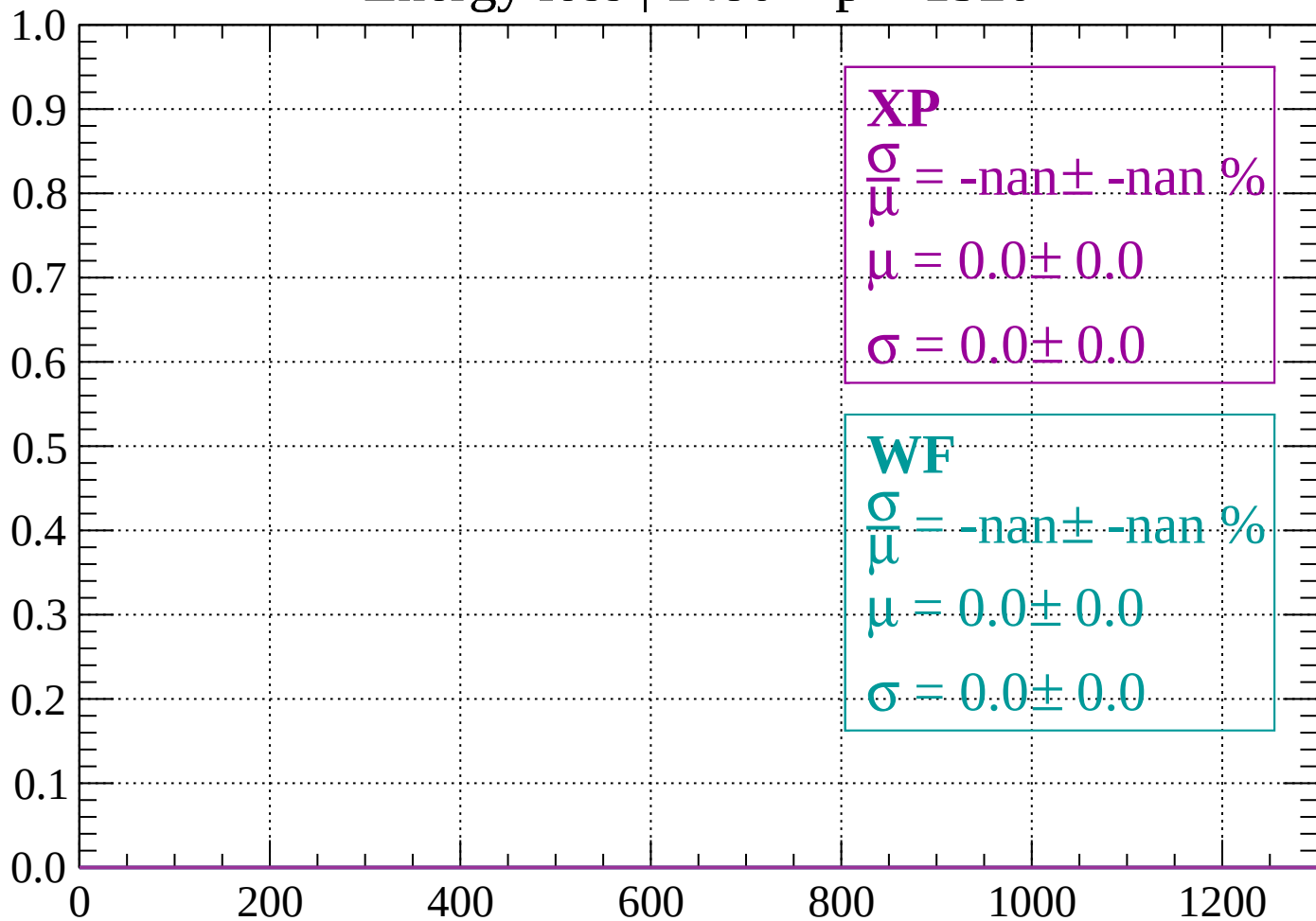
Count



dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$

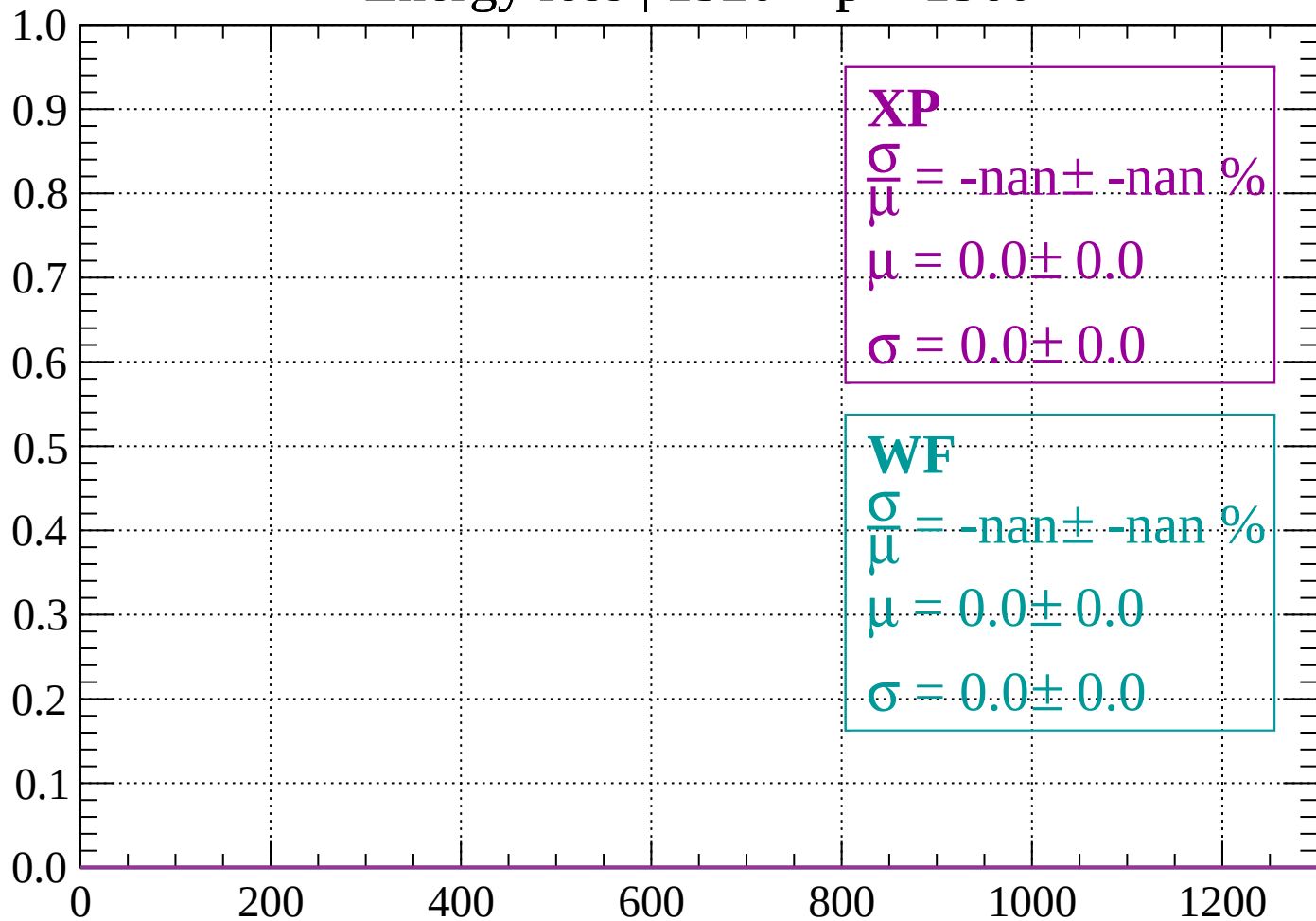
Count



dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$

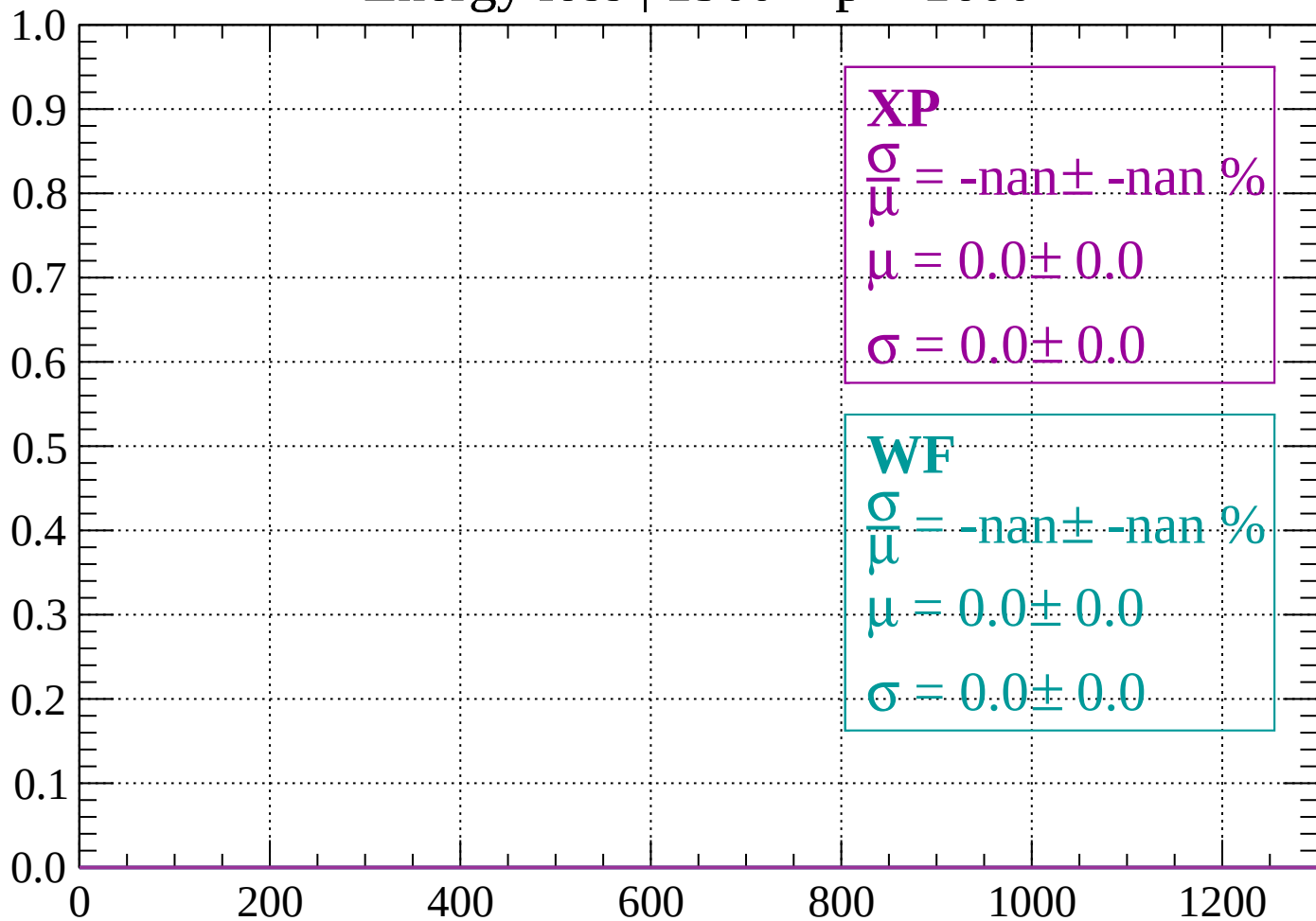
Count



dE/dx (ADC counts/cm)

Energy loss | 1560 < p < 1600

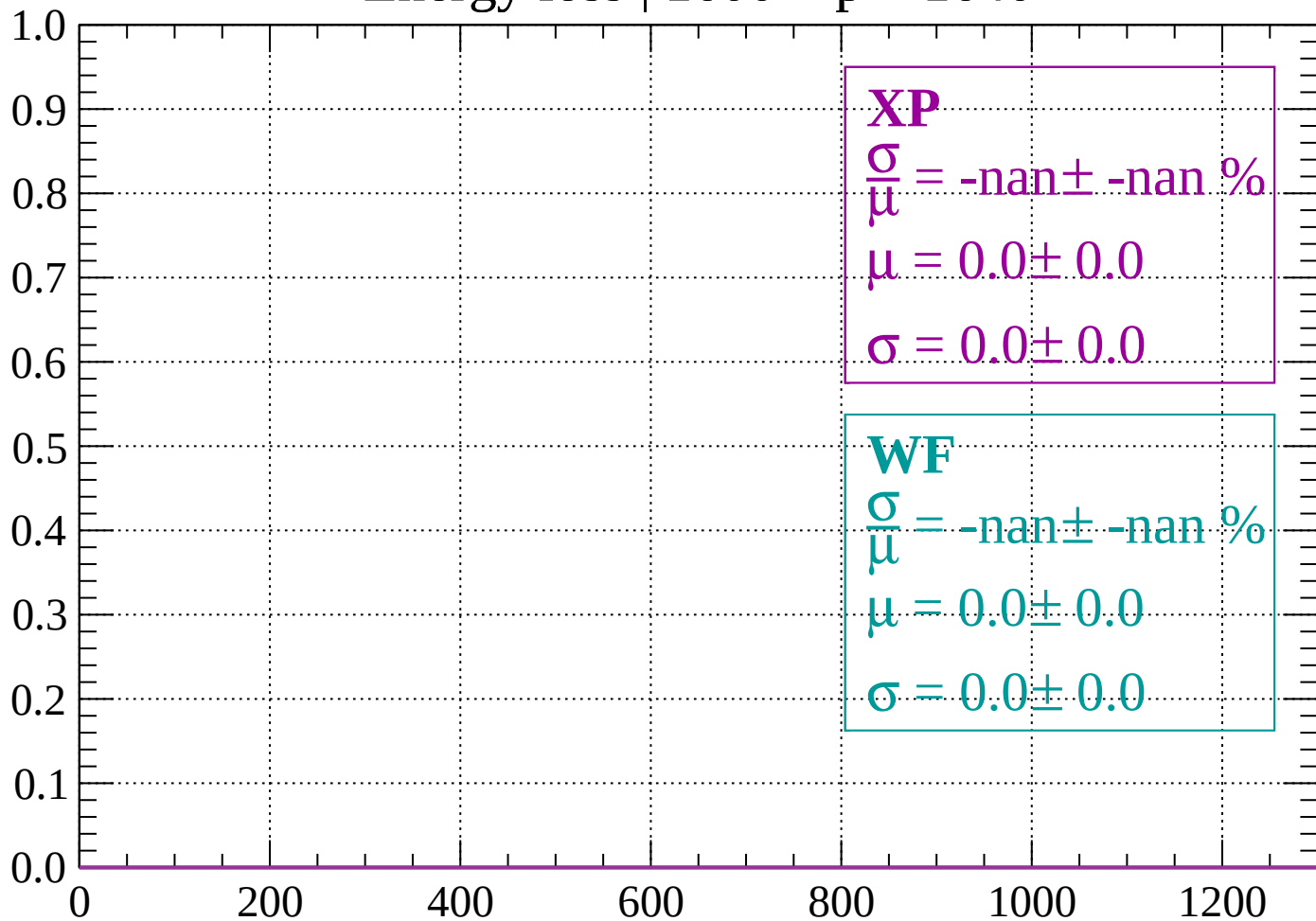
Count



dE/dx (ADC counts/cm)

Energy loss | 1600 < p < 1640

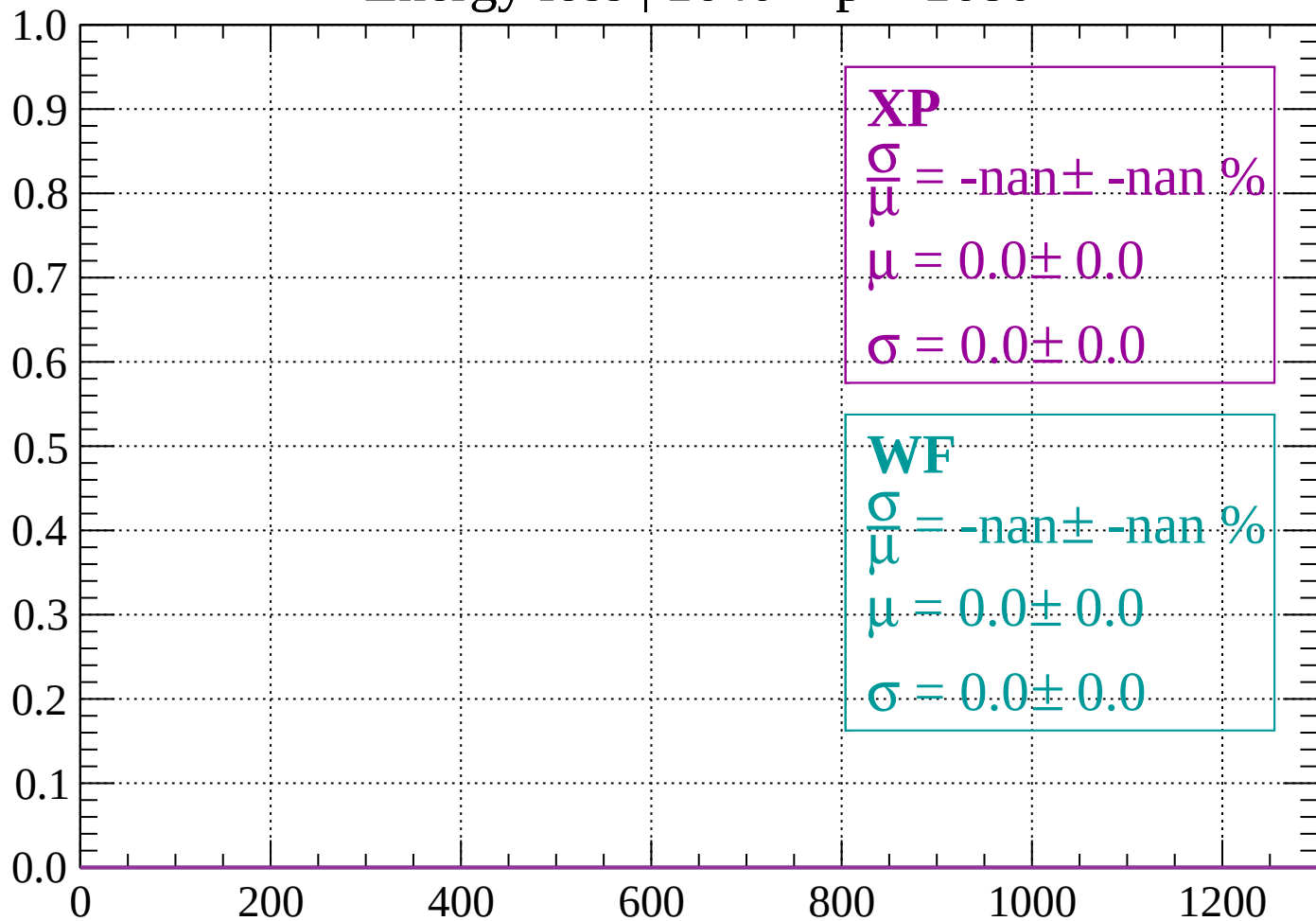
Count



dE/dx (ADC counts/cm)

Energy loss | $1640 < p < 1680$

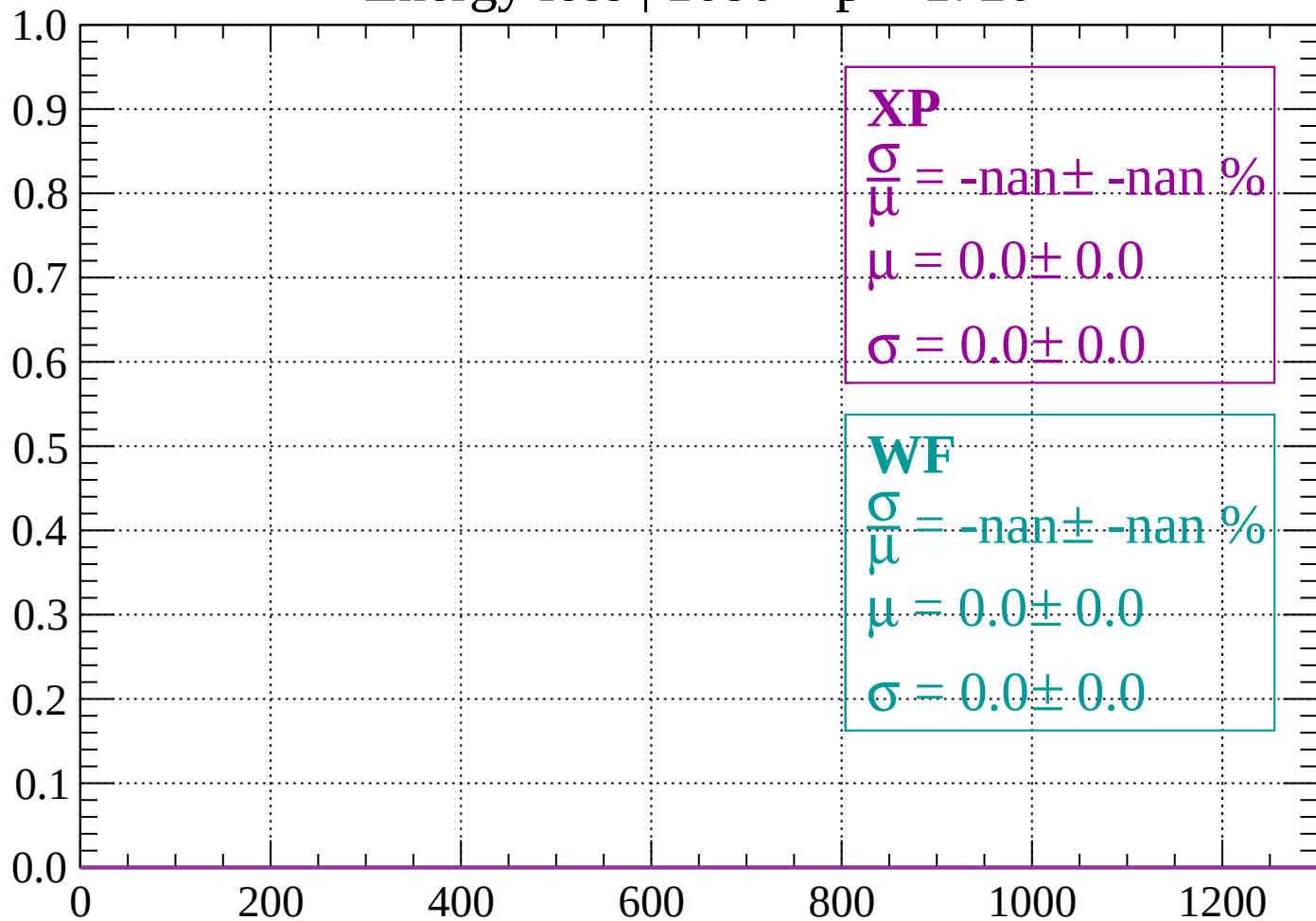
Count



dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1720$

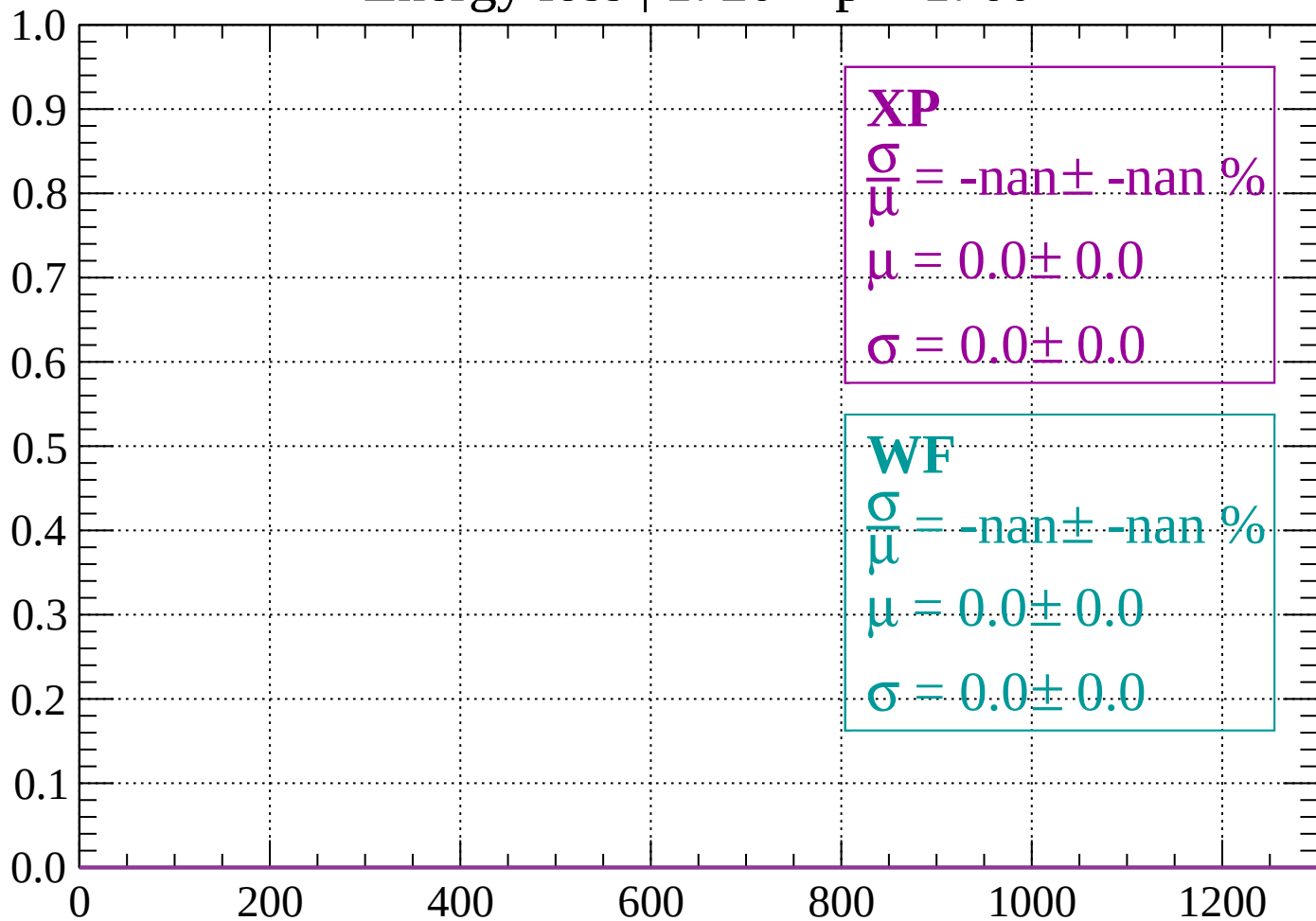
Count



dE/dx (ADC counts/cm)

Energy loss | $1720 < p < 1760$

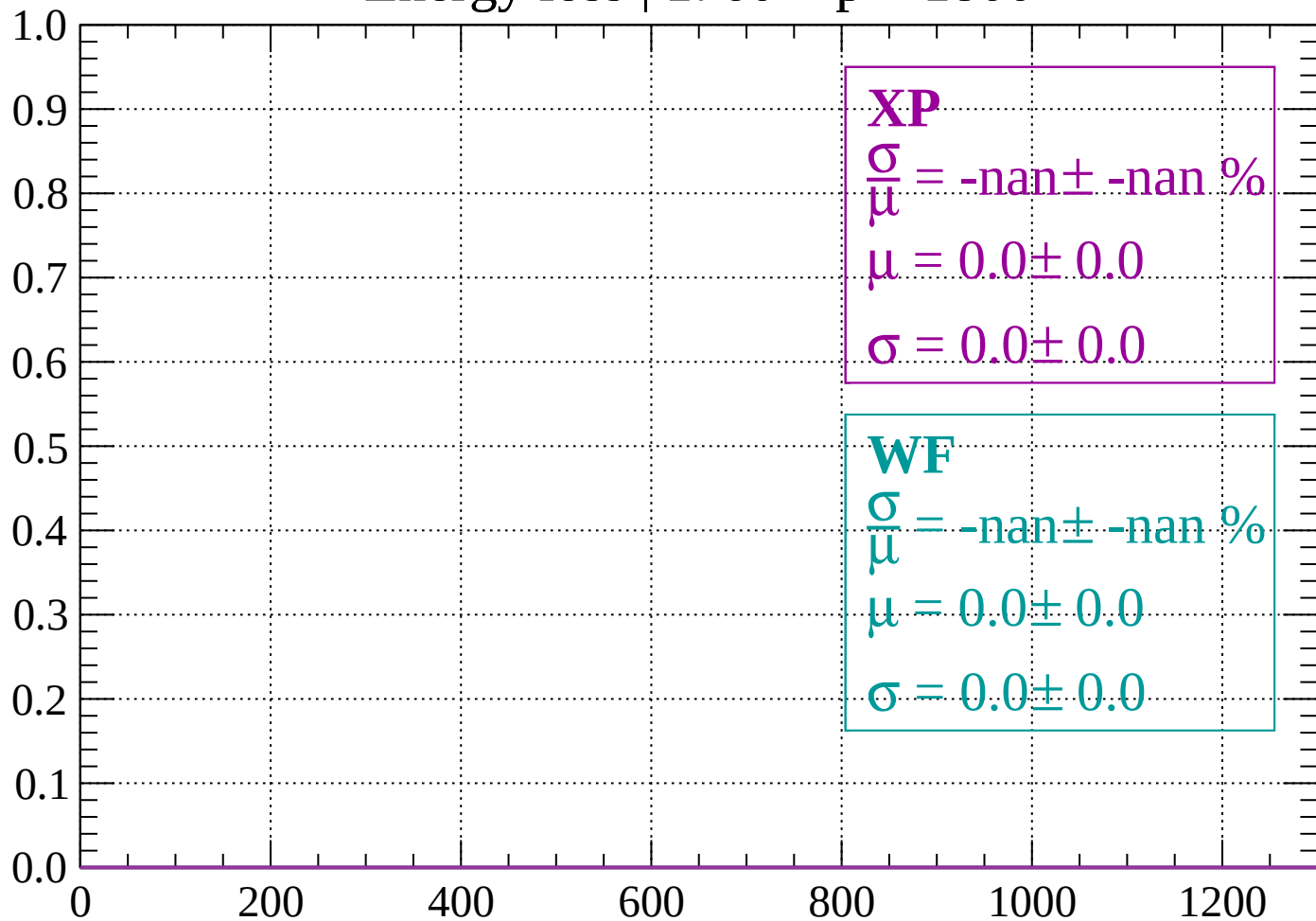
Count



dE/dx (ADC counts/cm)

Energy loss | 1760 < p < 1800

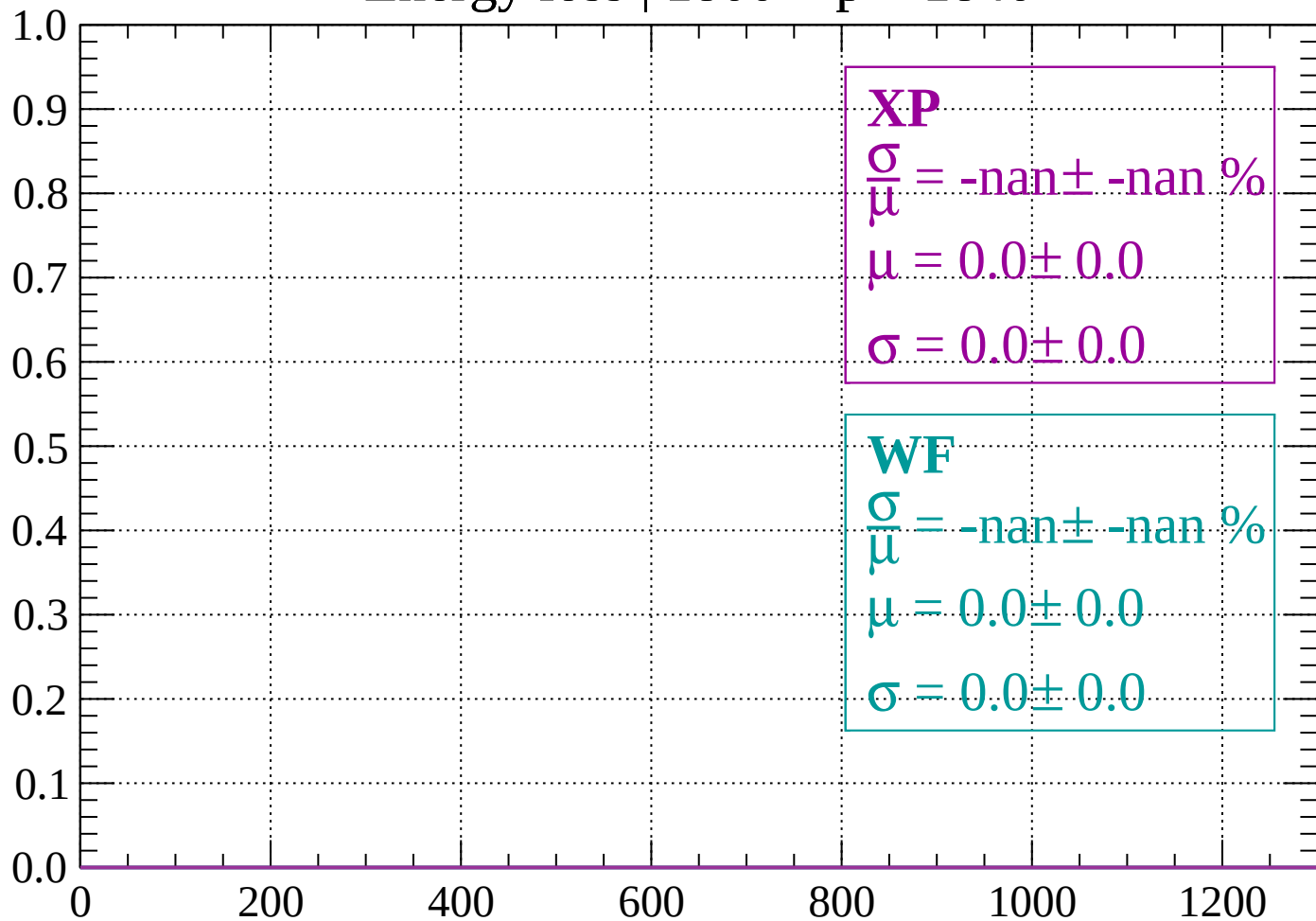
Count



dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1840$

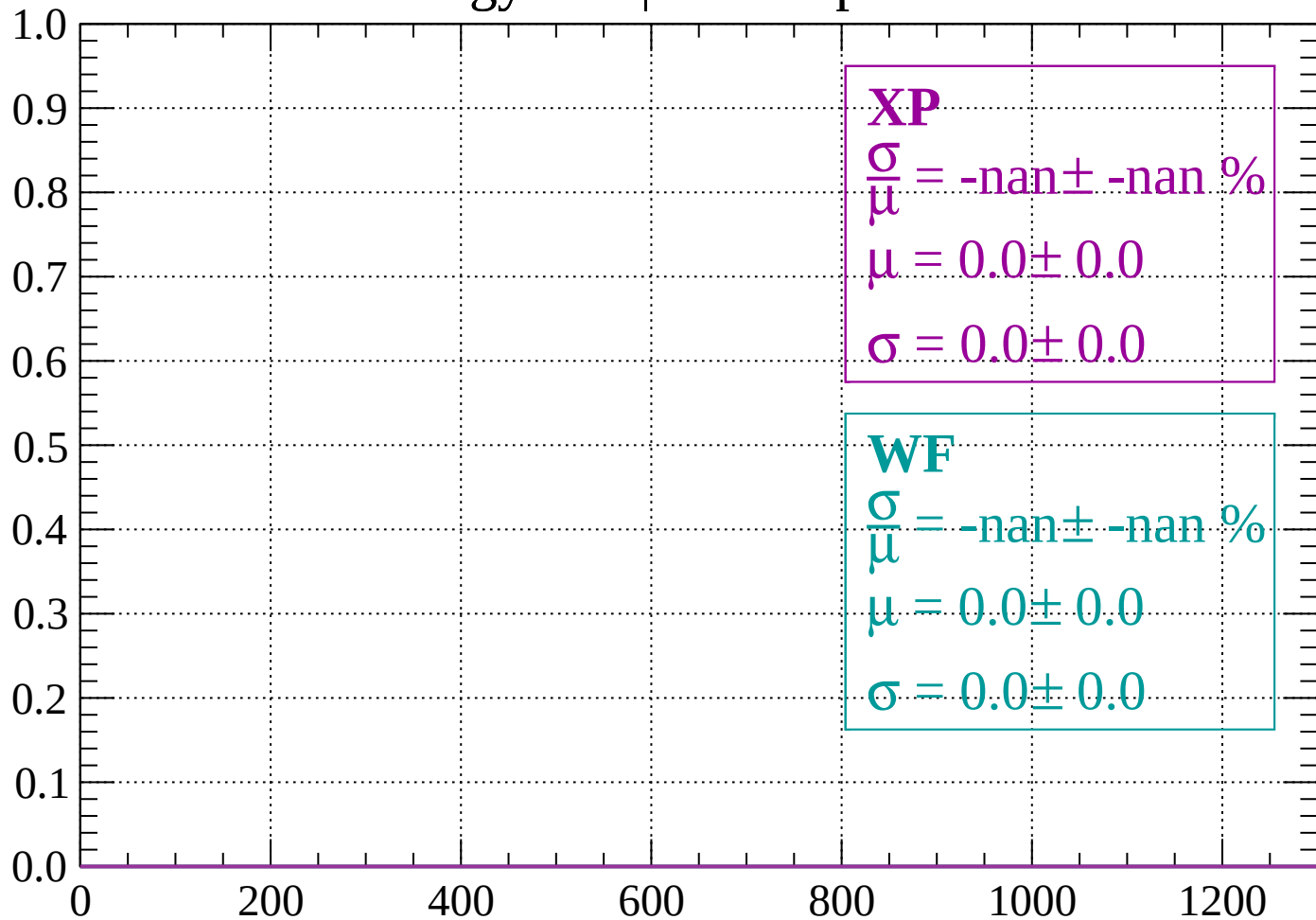
Count



dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1880$

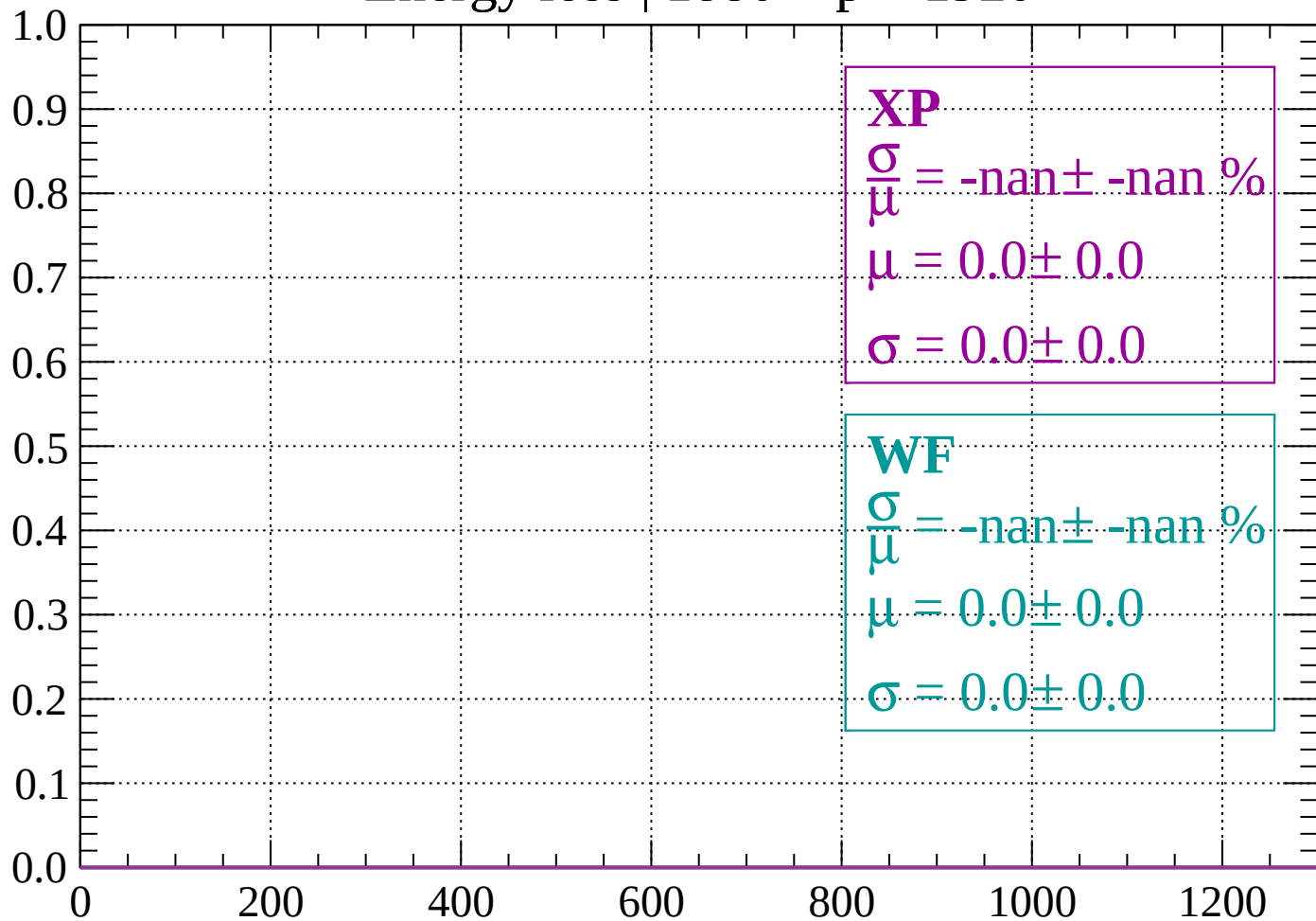
Count



dE/dx (ADC counts/cm)

Energy loss | 1880 < p < 1920

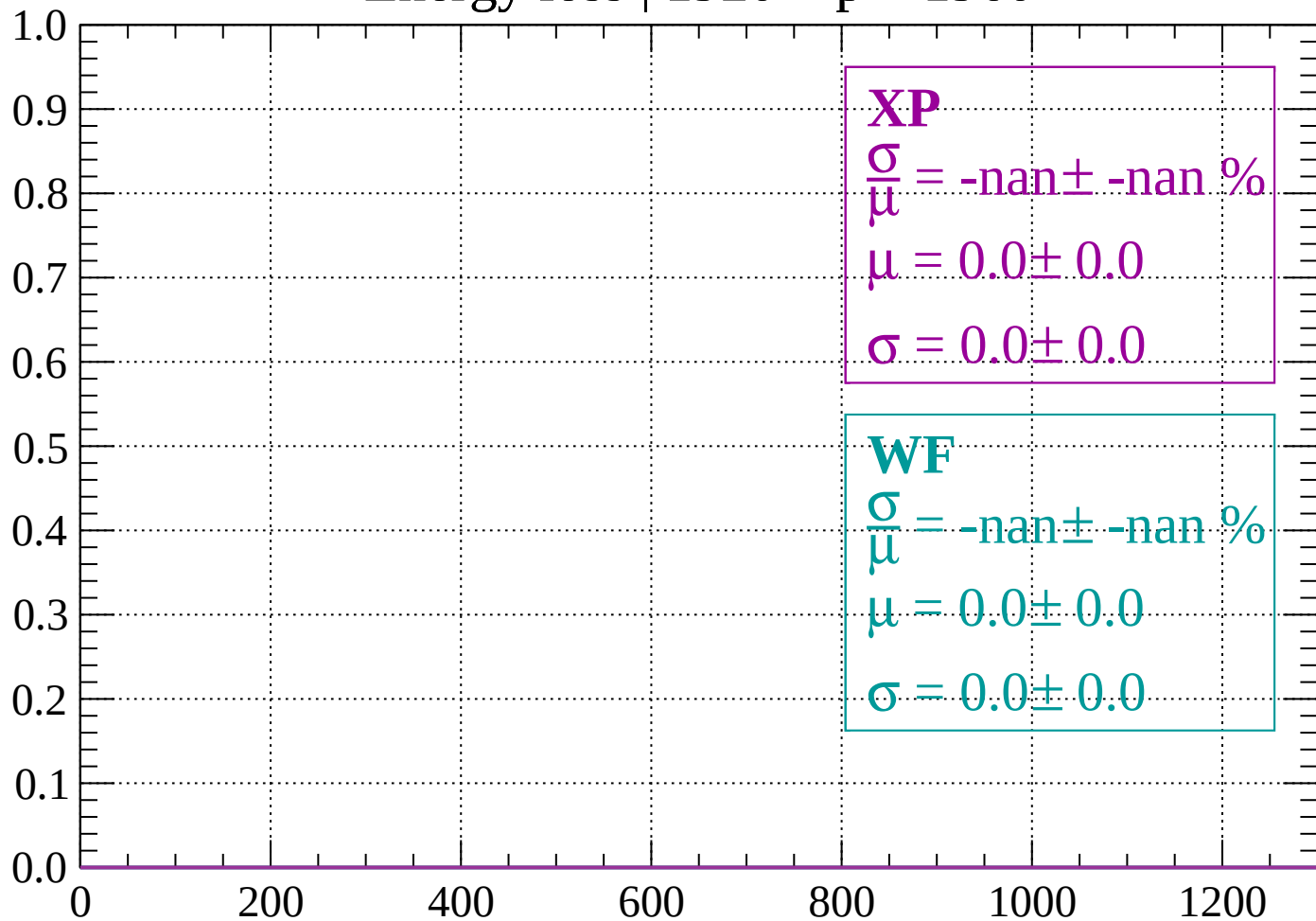
Count



dE/dx (ADC counts/cm)

Energy loss | 1920 < p < 1960

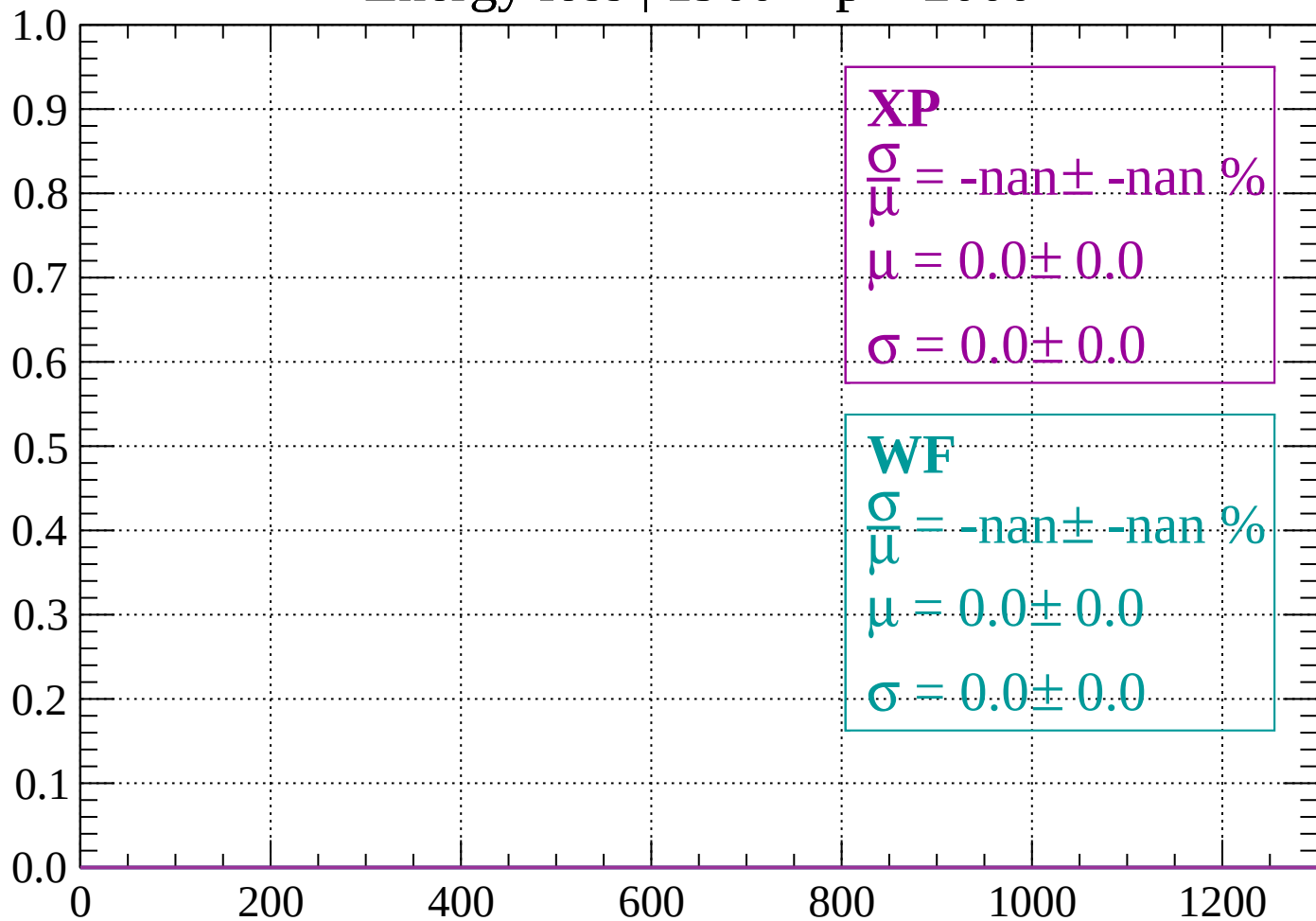
Count



dE/dx (ADC counts/cm)

Energy loss | 1960 < p < 2000

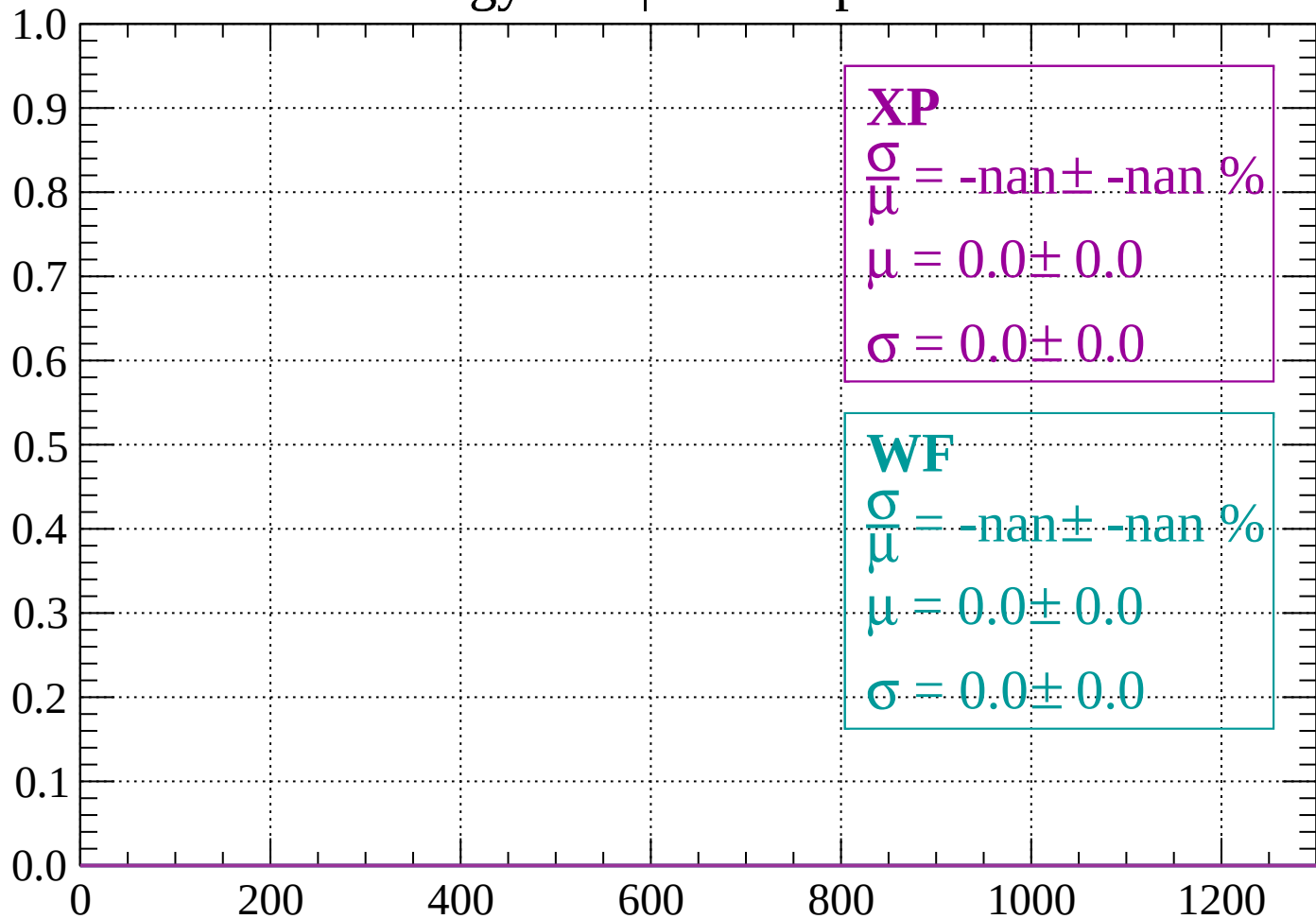
Count



dE/dx (ADC counts/cm)

Energy loss | $2000 < p < 2040$

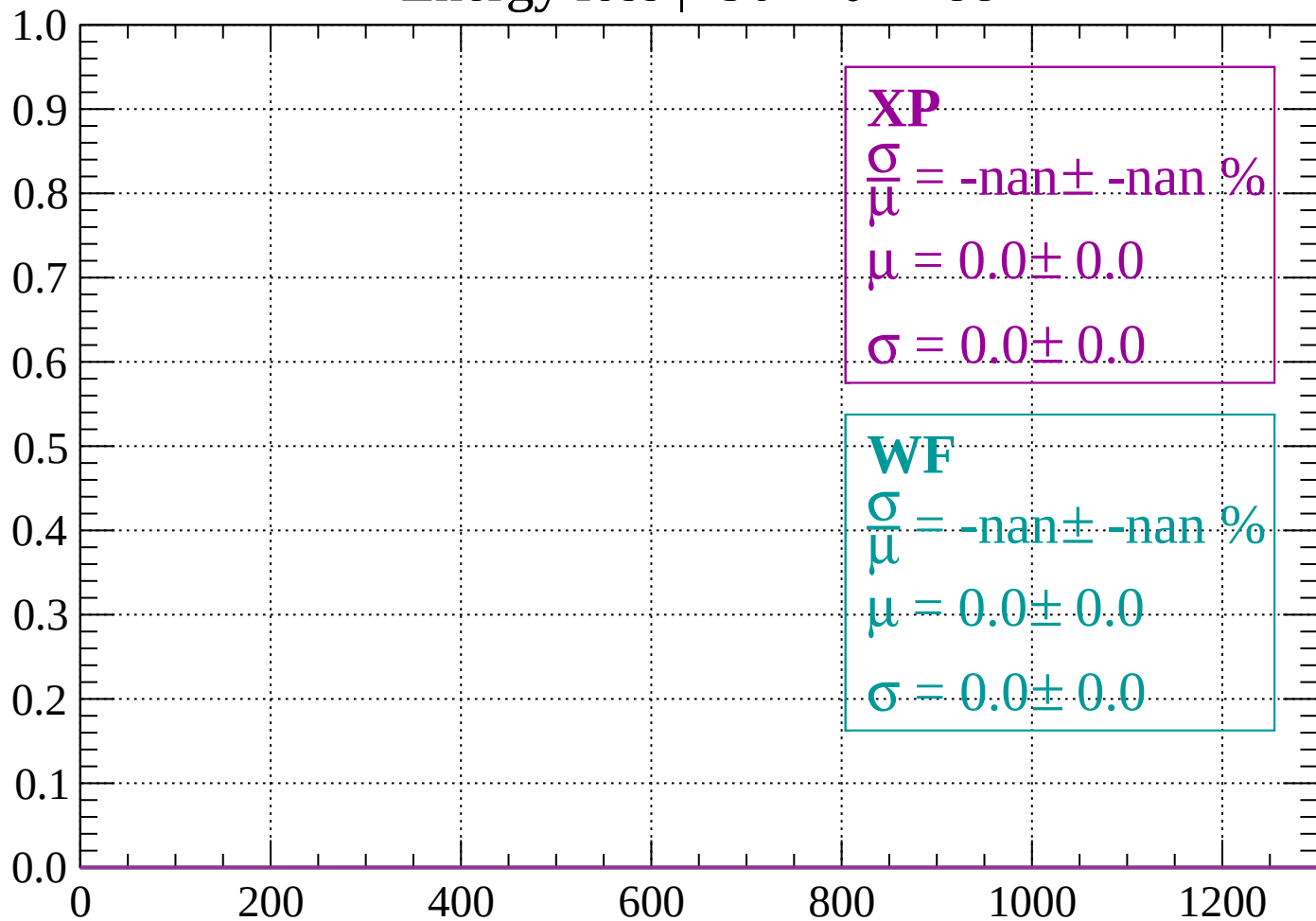
Count



dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

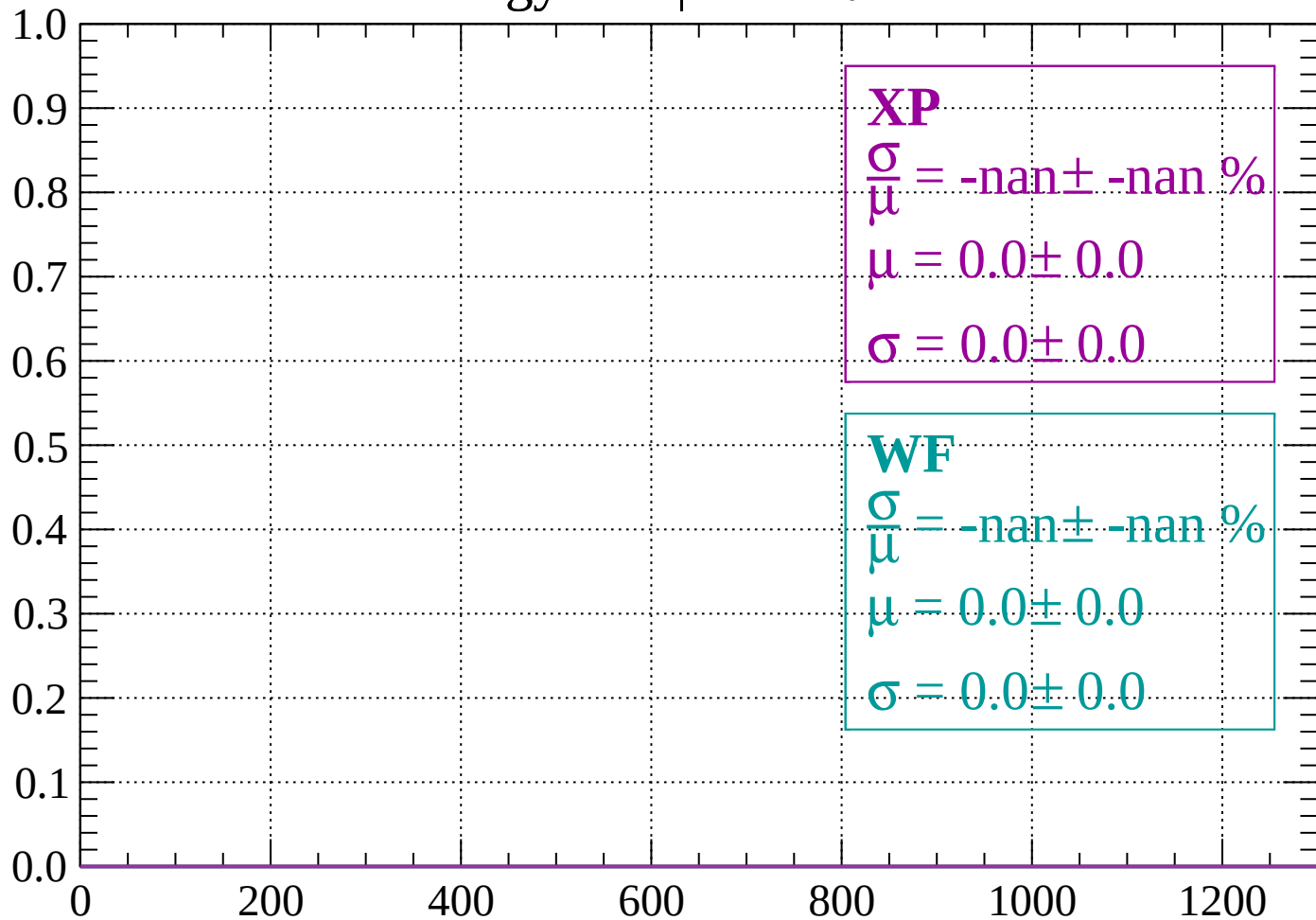
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

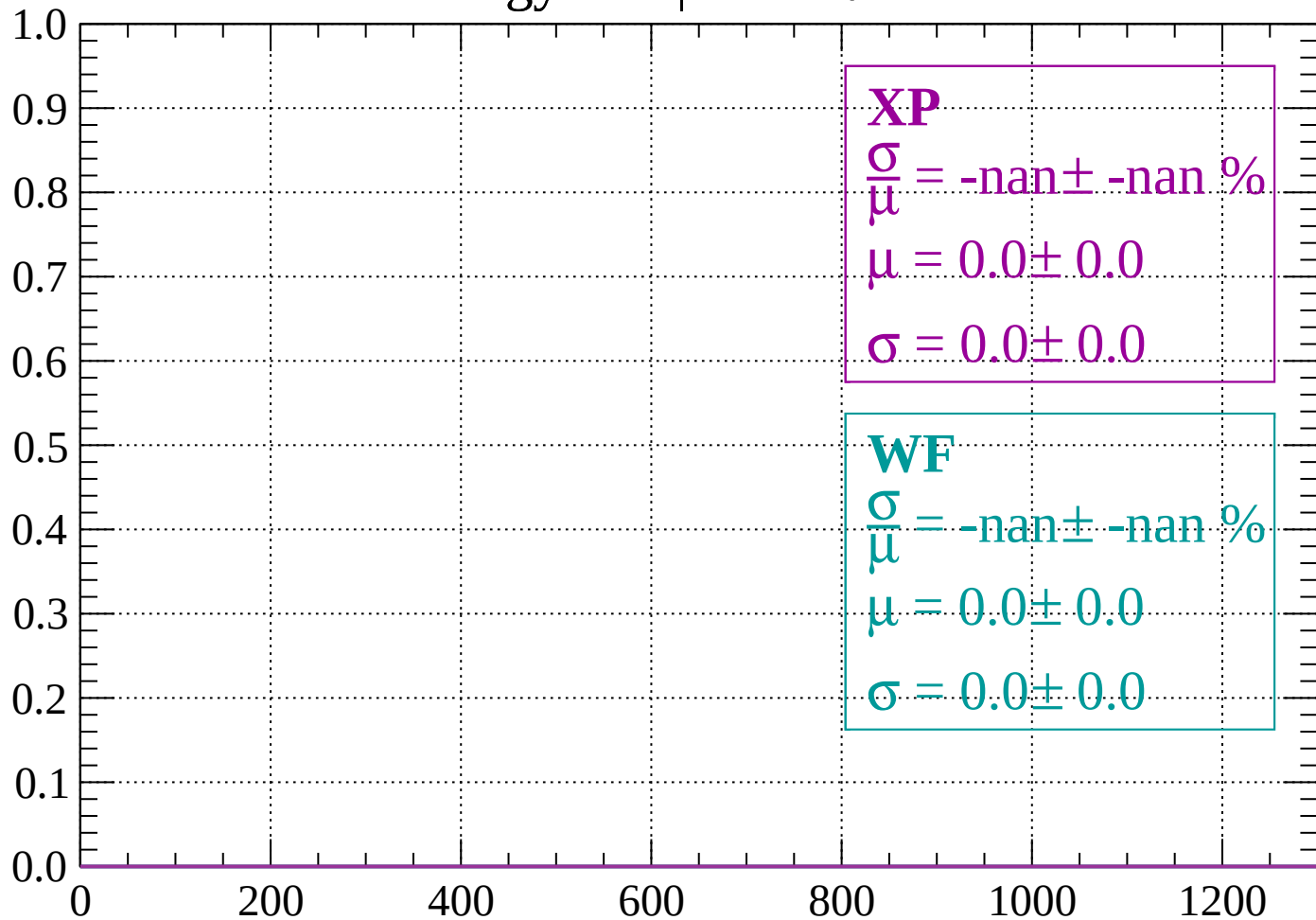
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

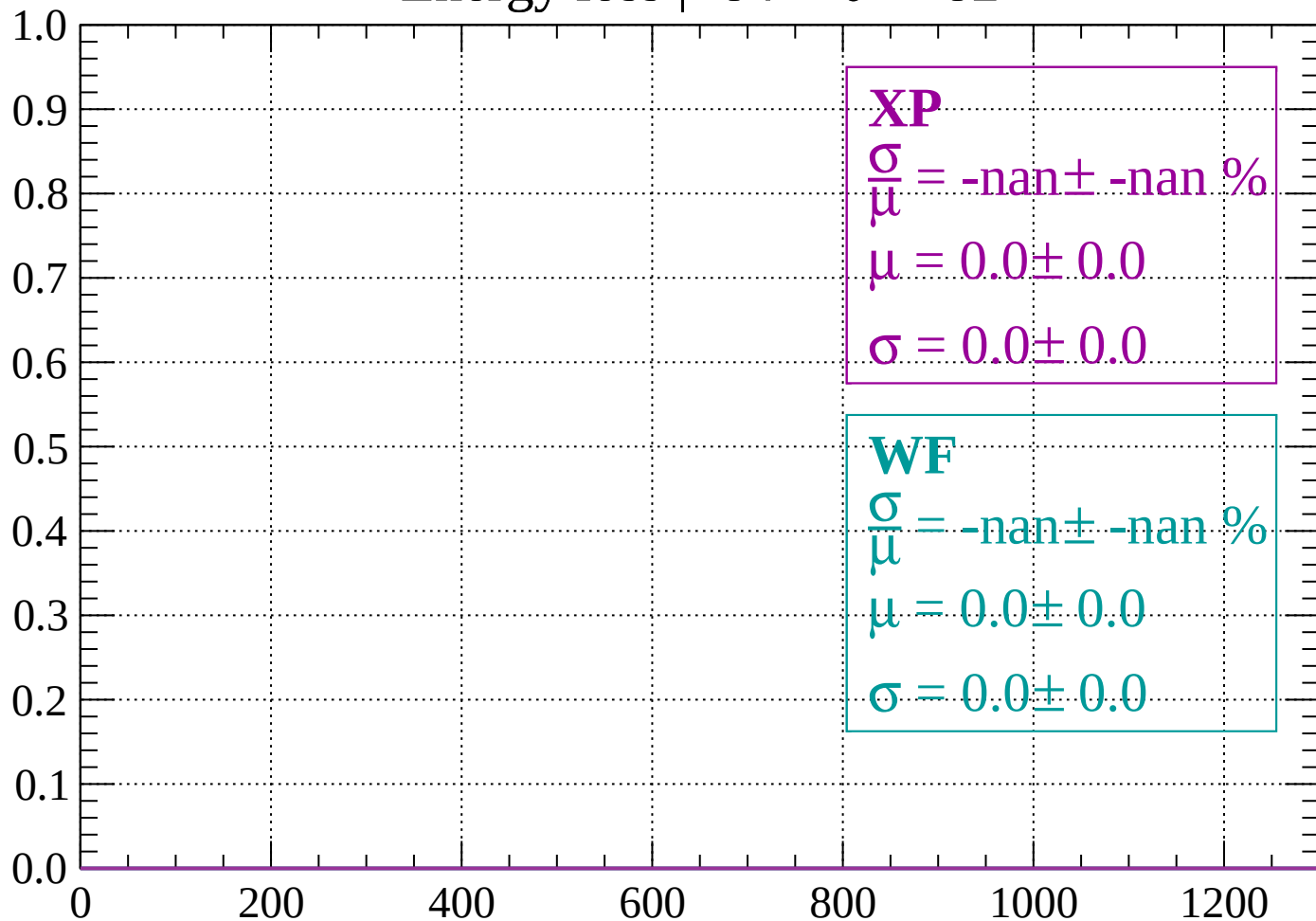
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

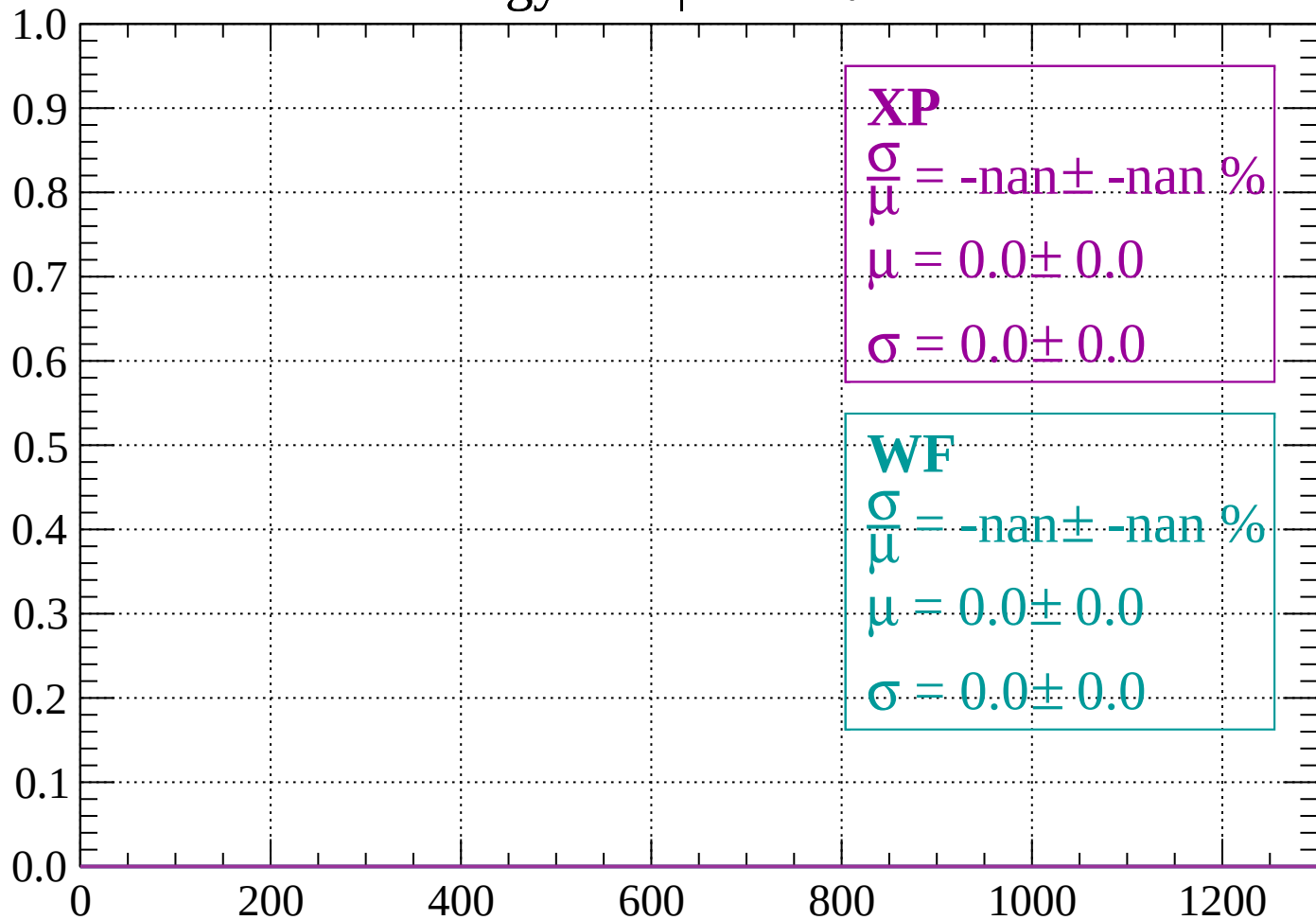
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

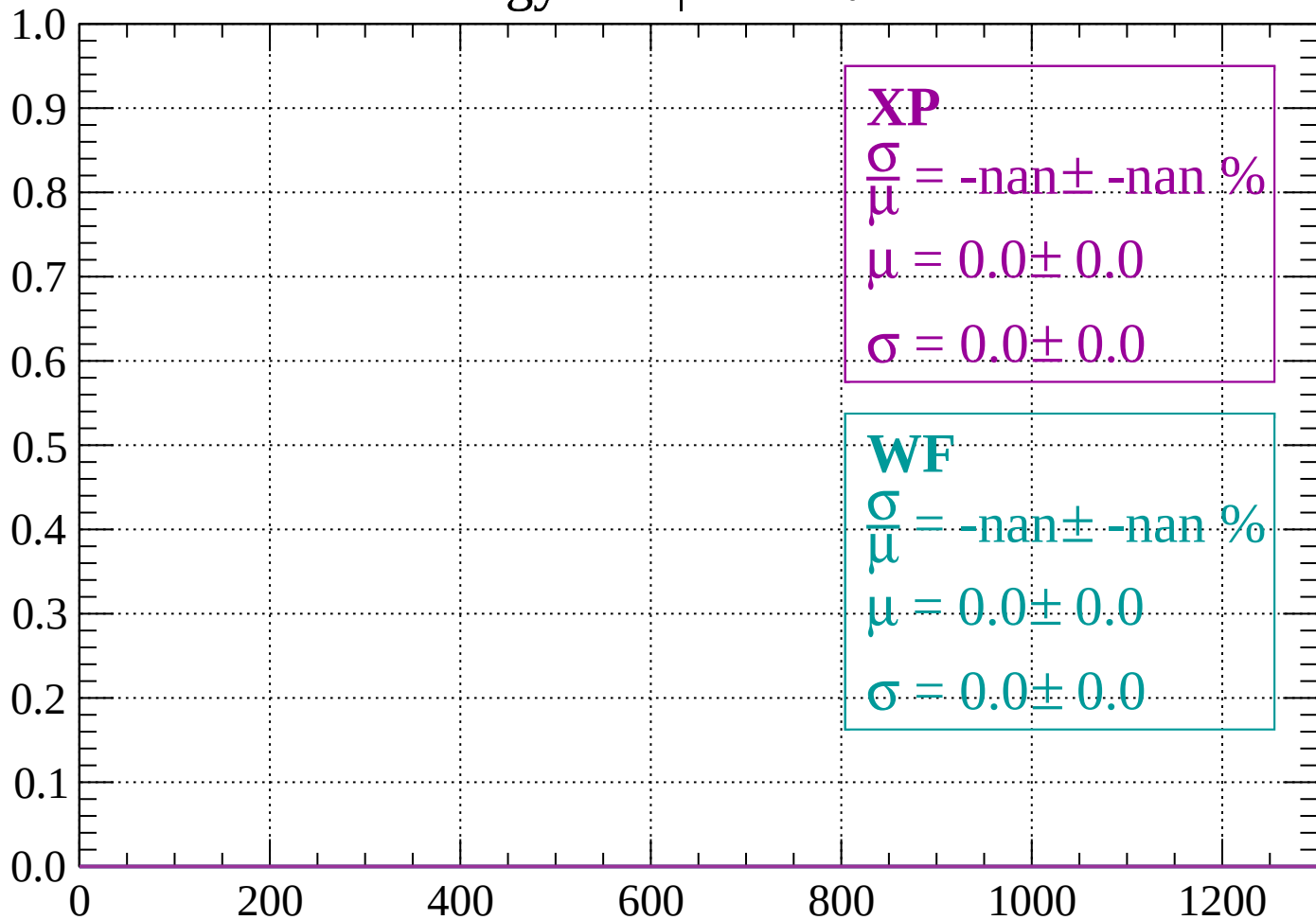
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

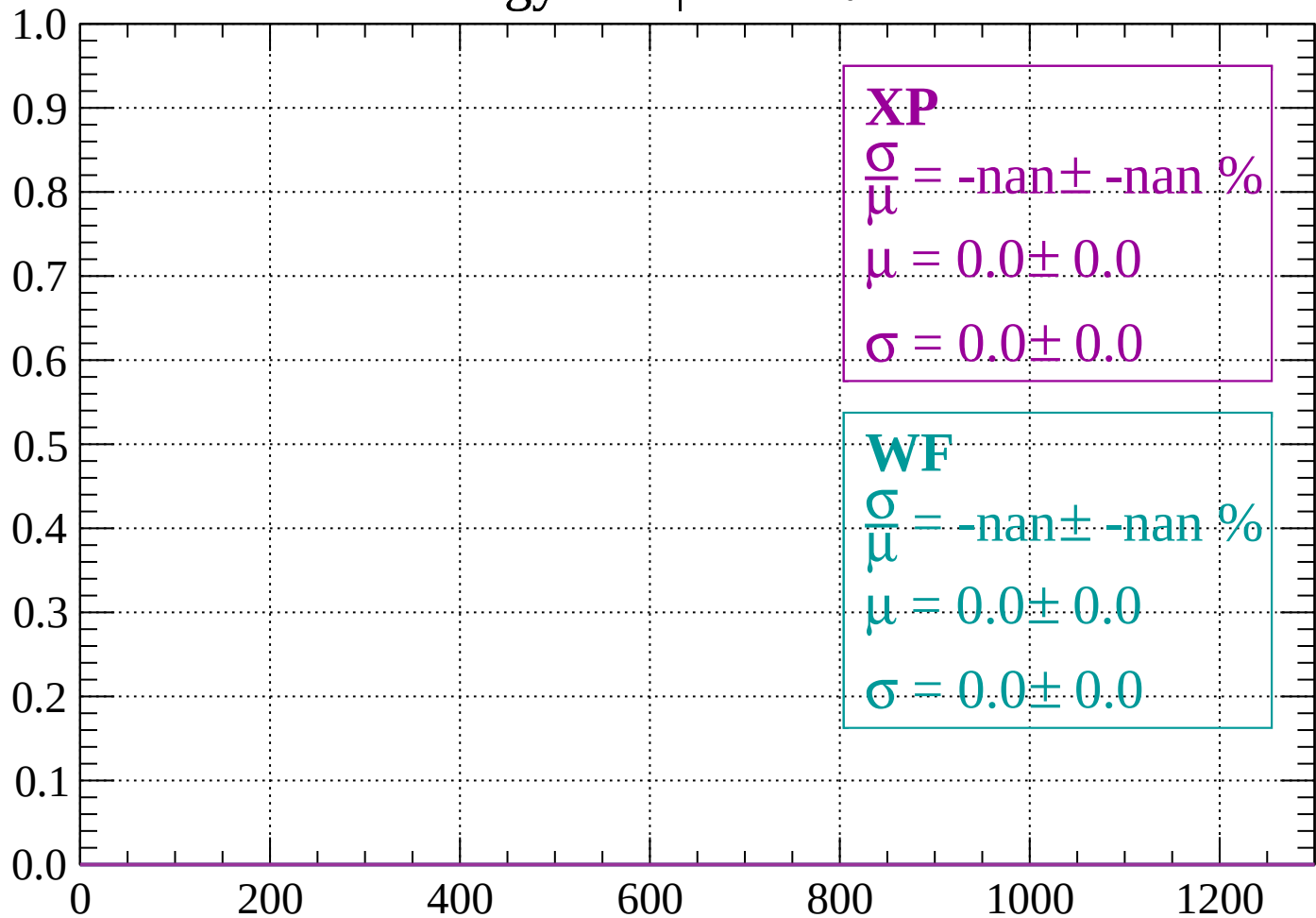
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

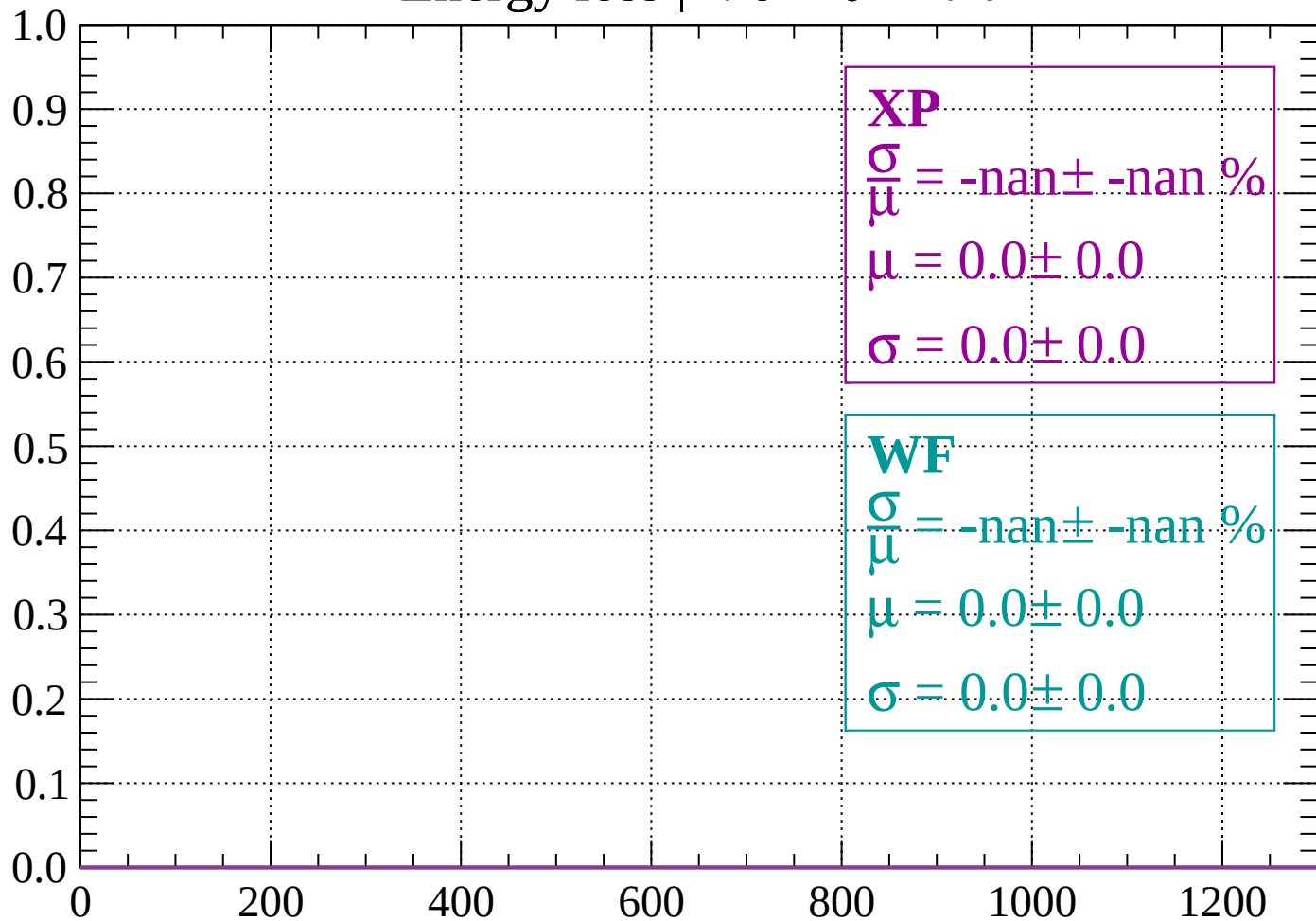
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

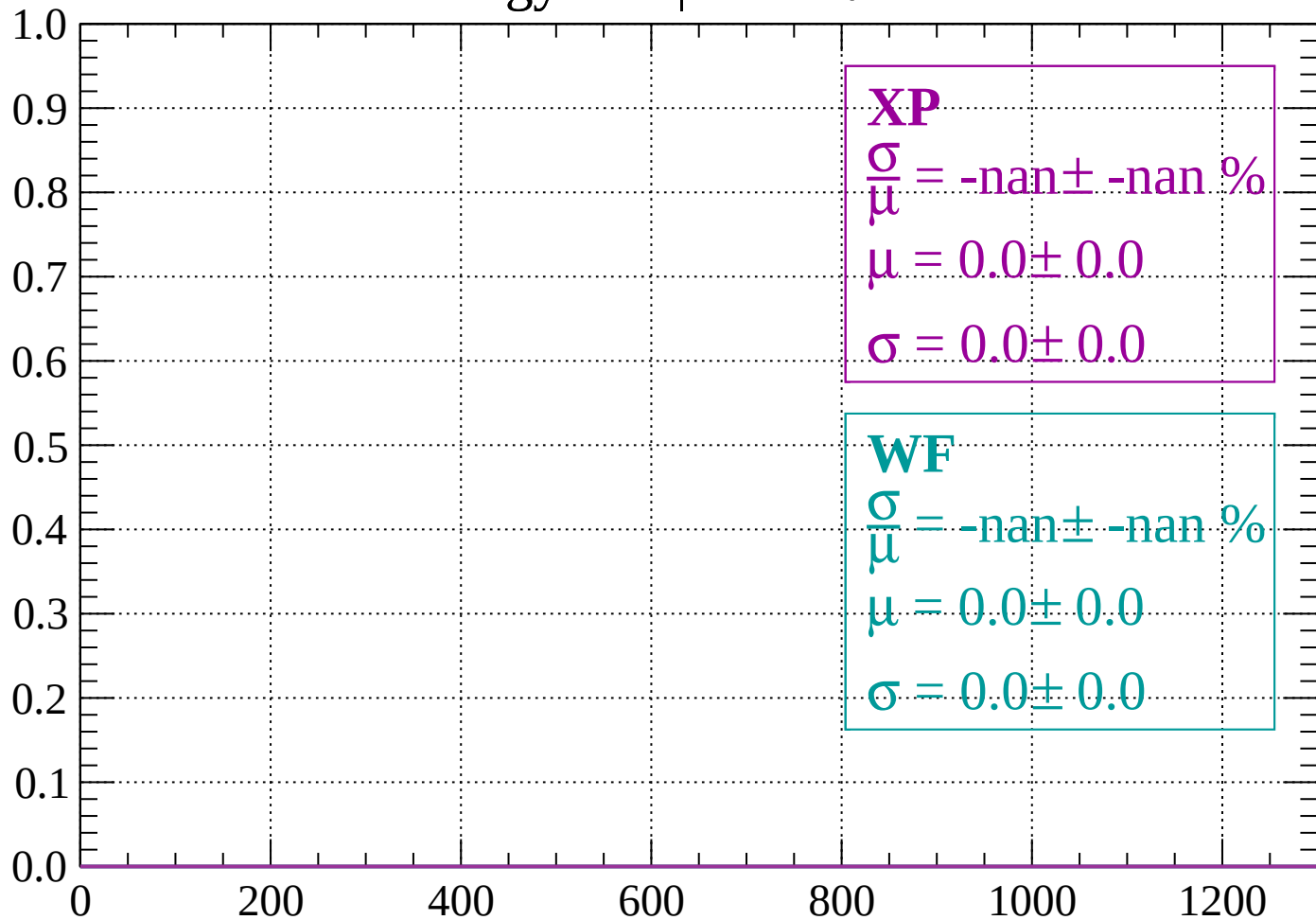
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

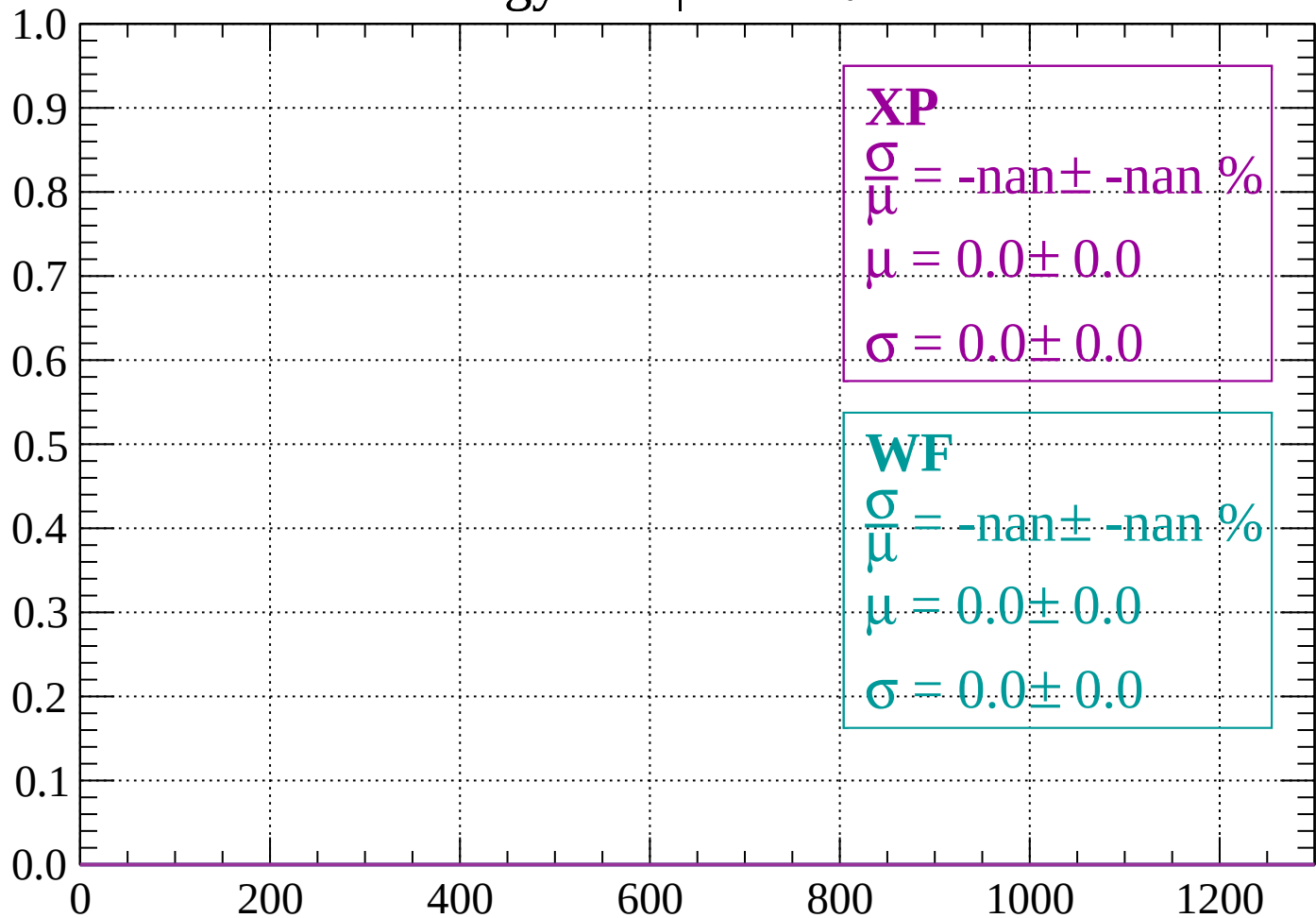
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

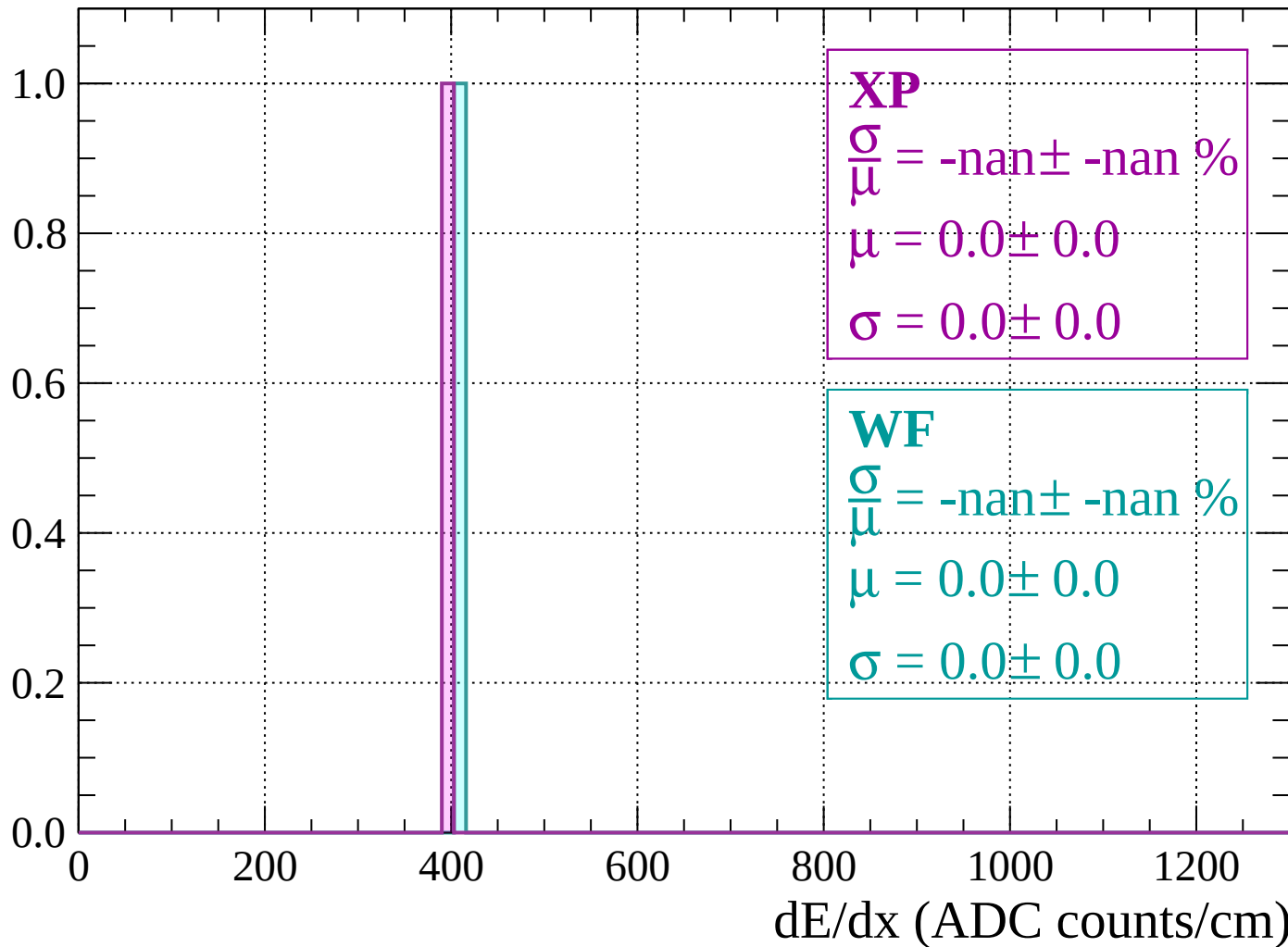
Count



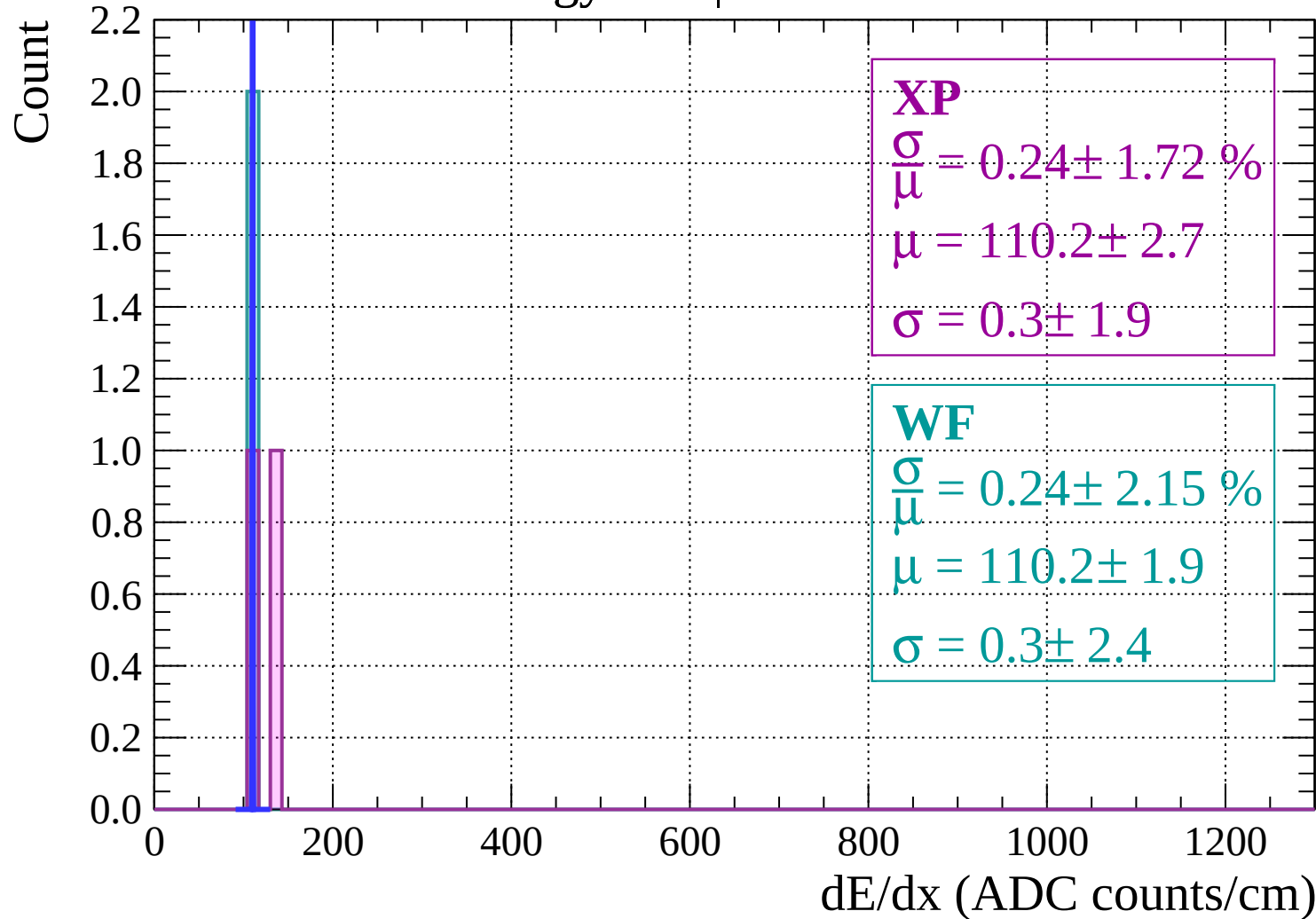
dE/dx (ADC counts/cm)

Energy loss | $-70 < \theta < -68$

Count

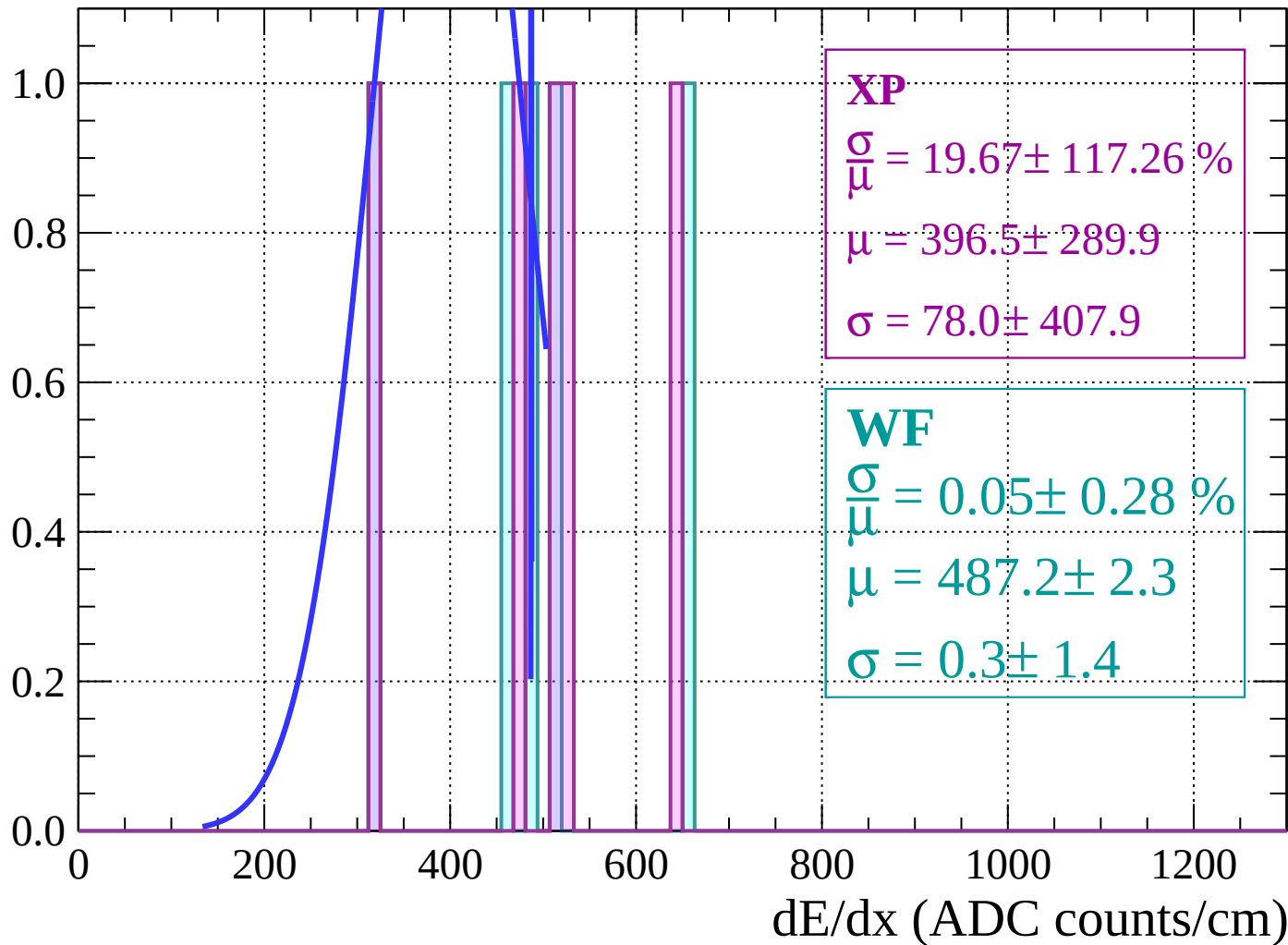


Energy loss | $-68 < \theta < -66$



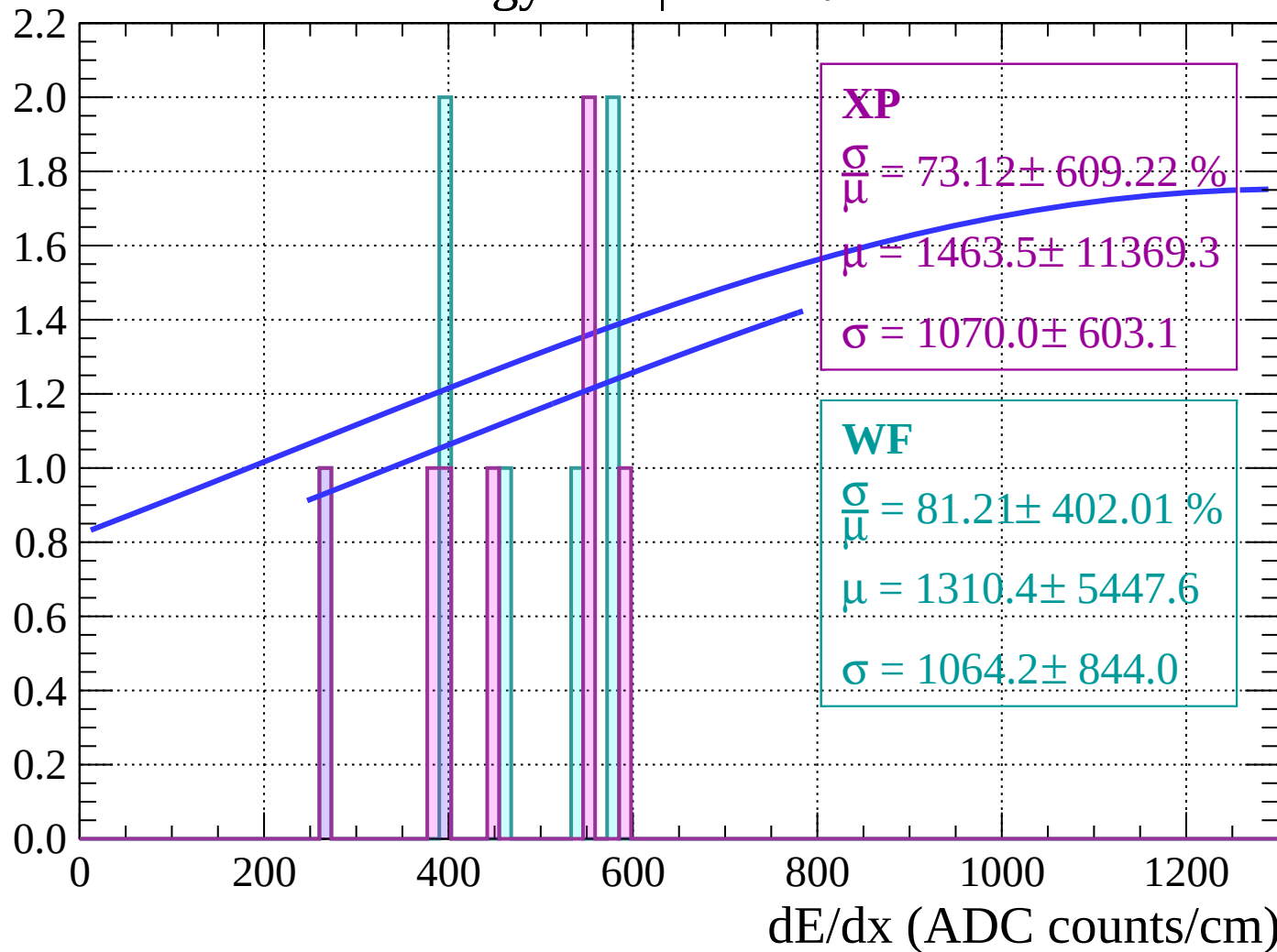
Energy loss | $-66 < \theta < -64$

Count



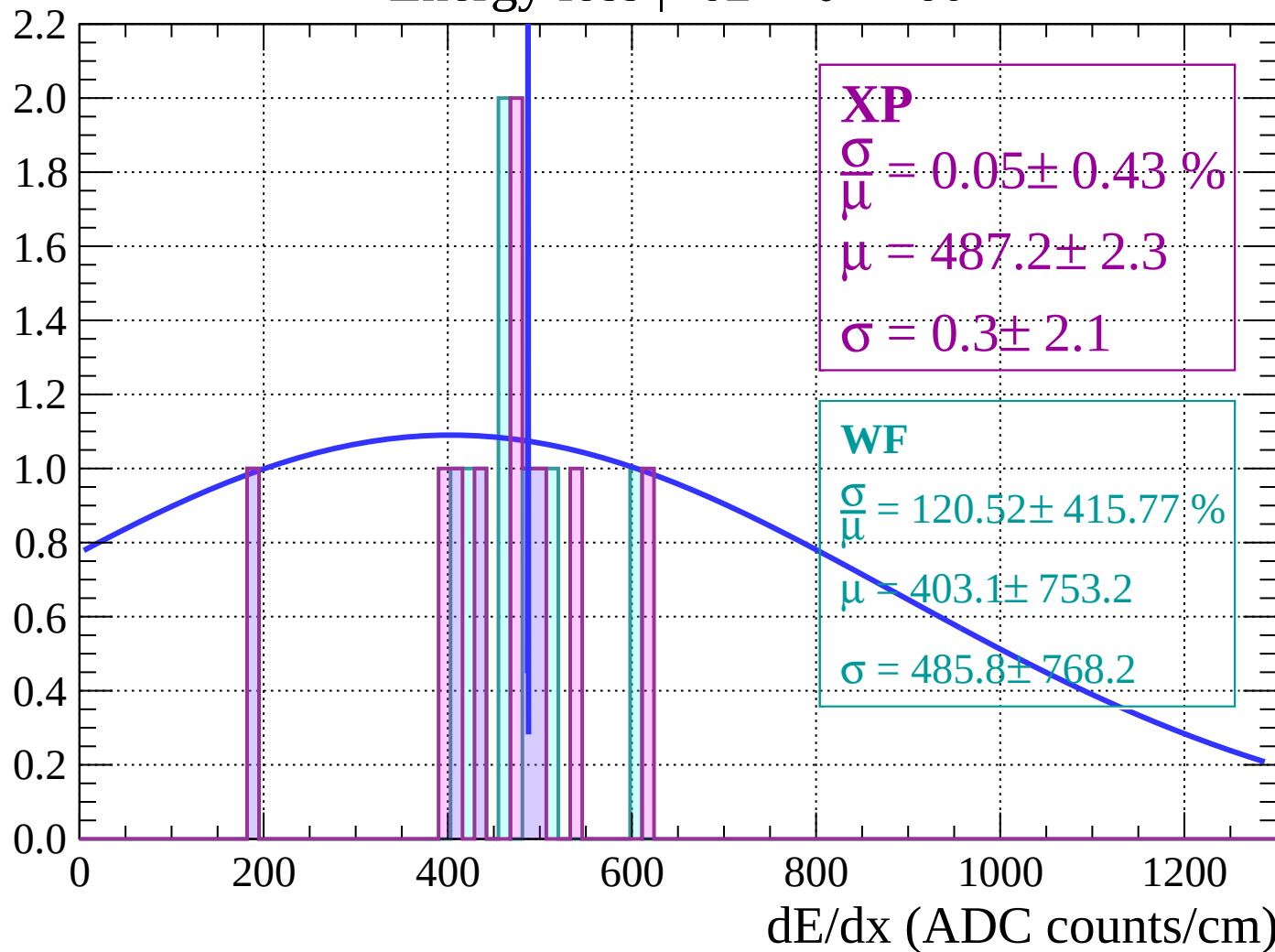
Energy loss | $-64 < \theta < -62$

Count



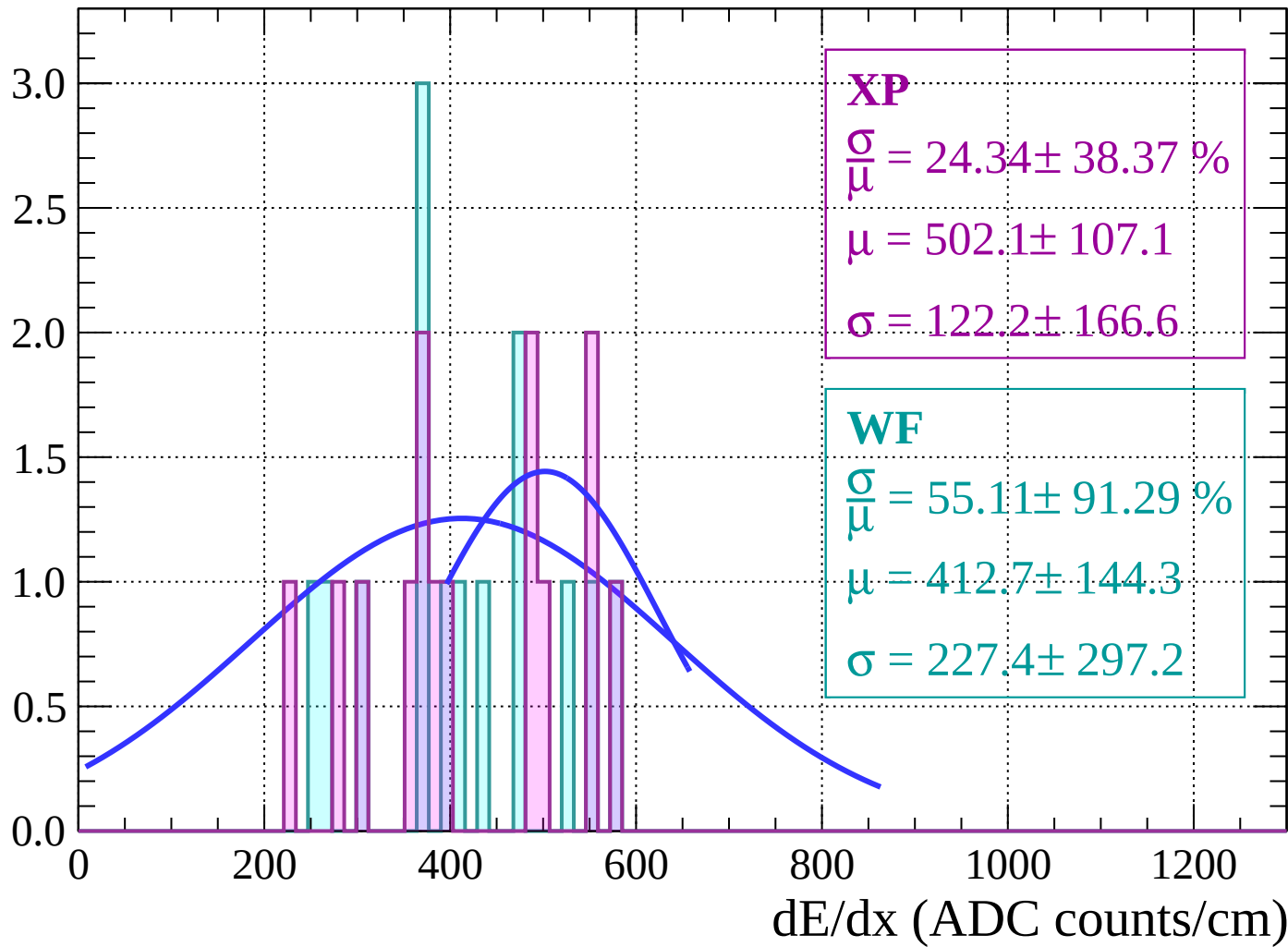
Energy loss | $-62 < \theta < -60$

Count



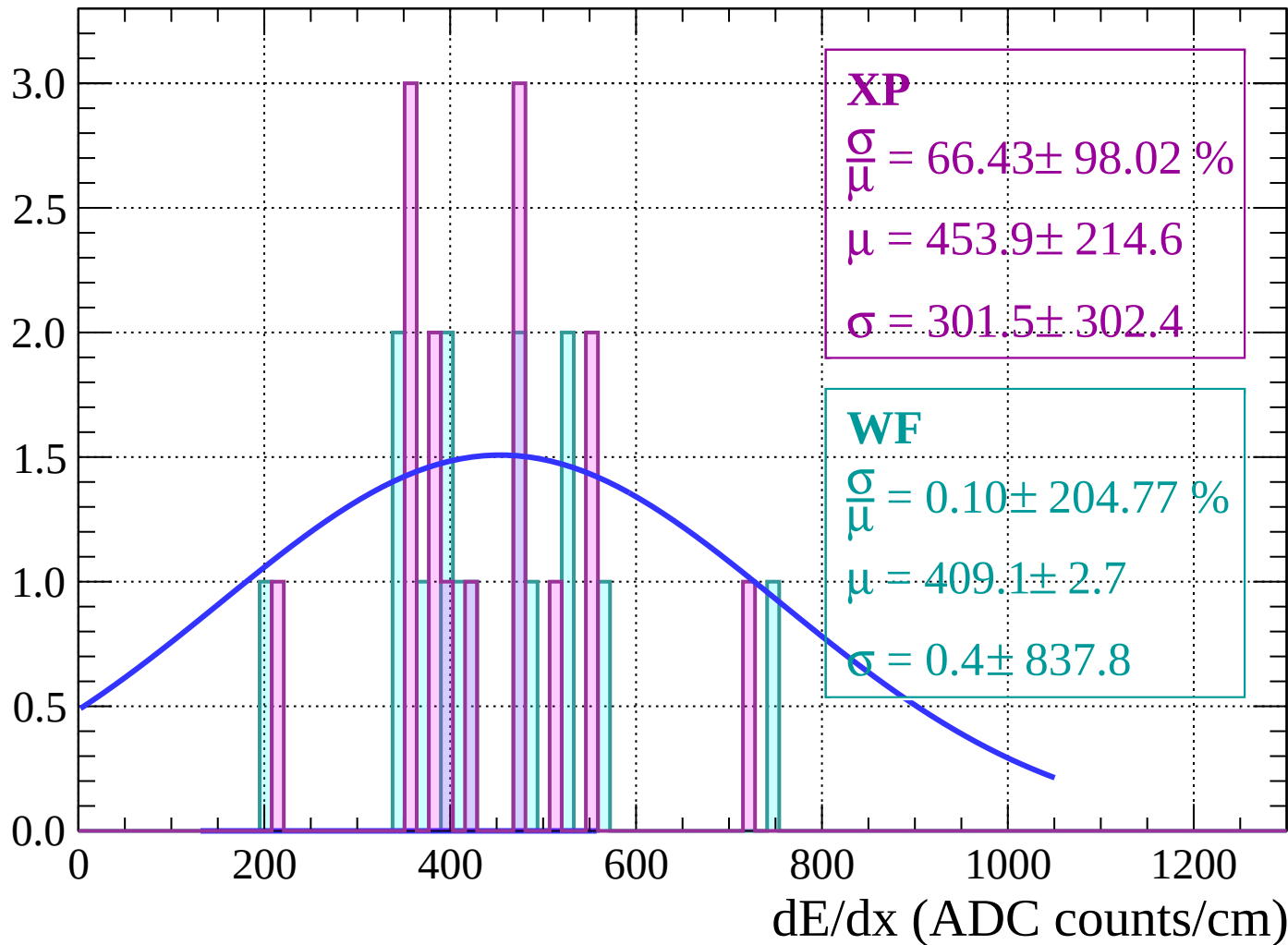
Energy loss | $-60 < \theta < -58$

Count



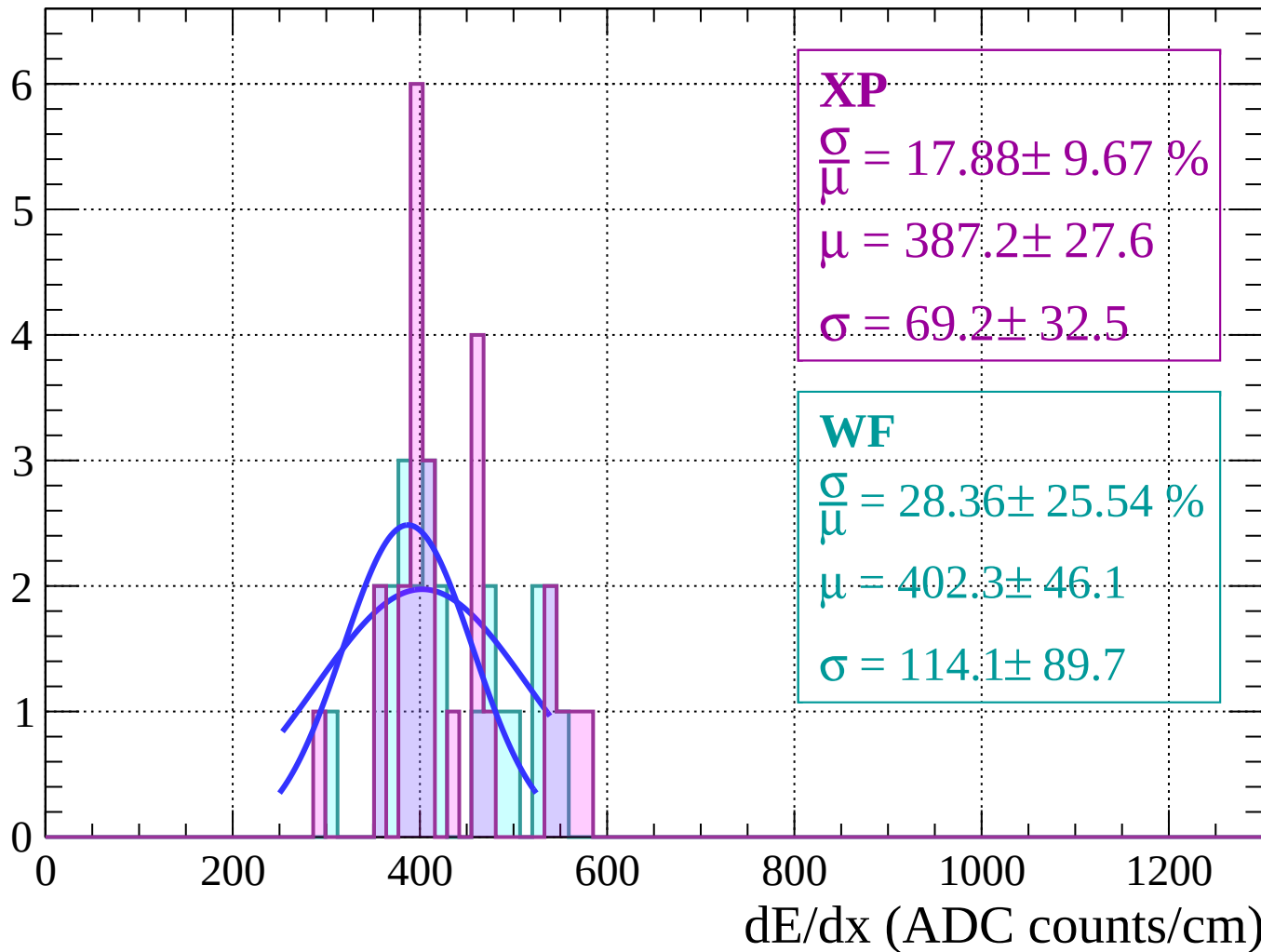
Energy loss | $-58 < \theta < -56$

Count



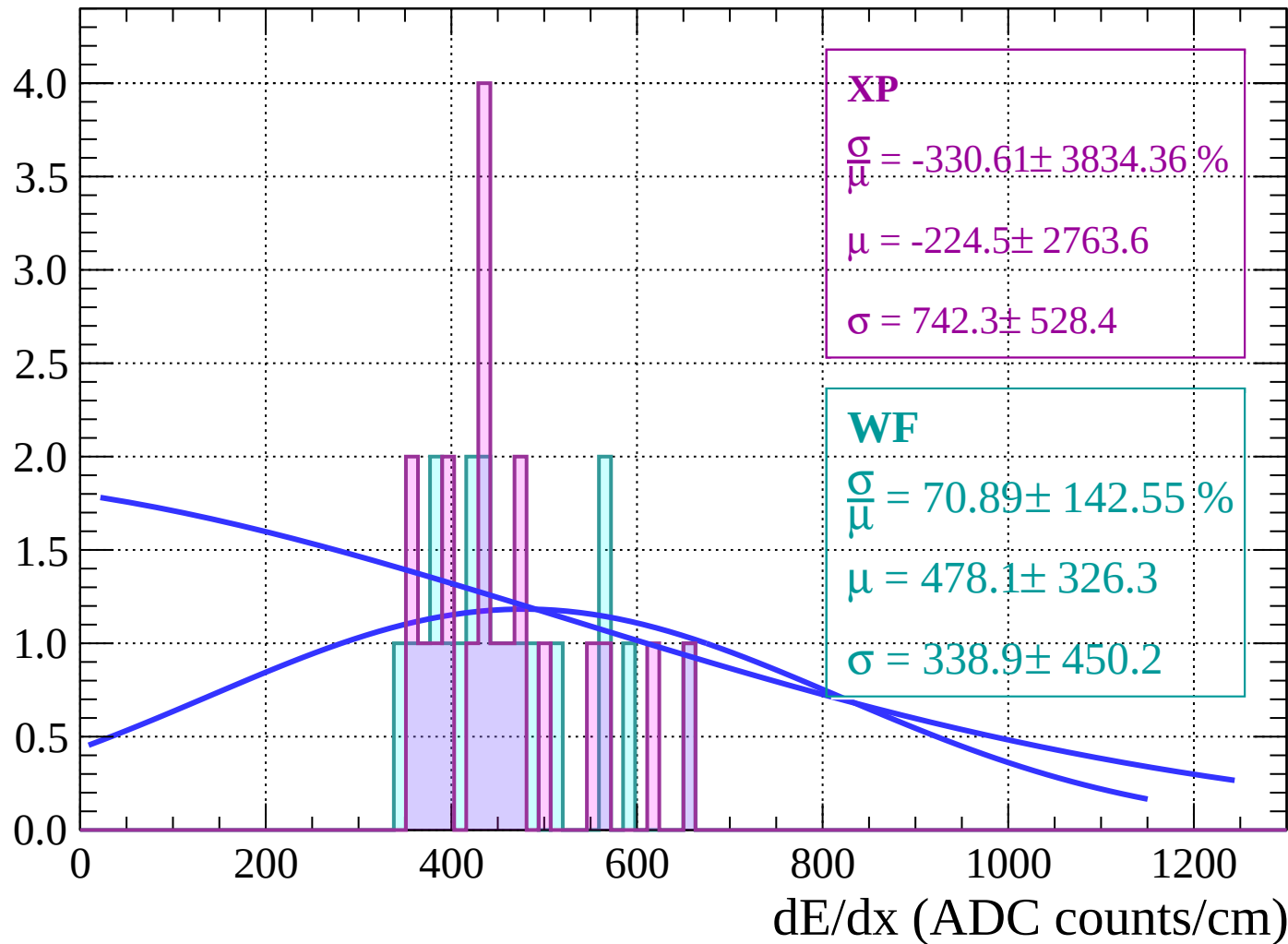
Energy loss | $-56 < \theta < -54$

Count



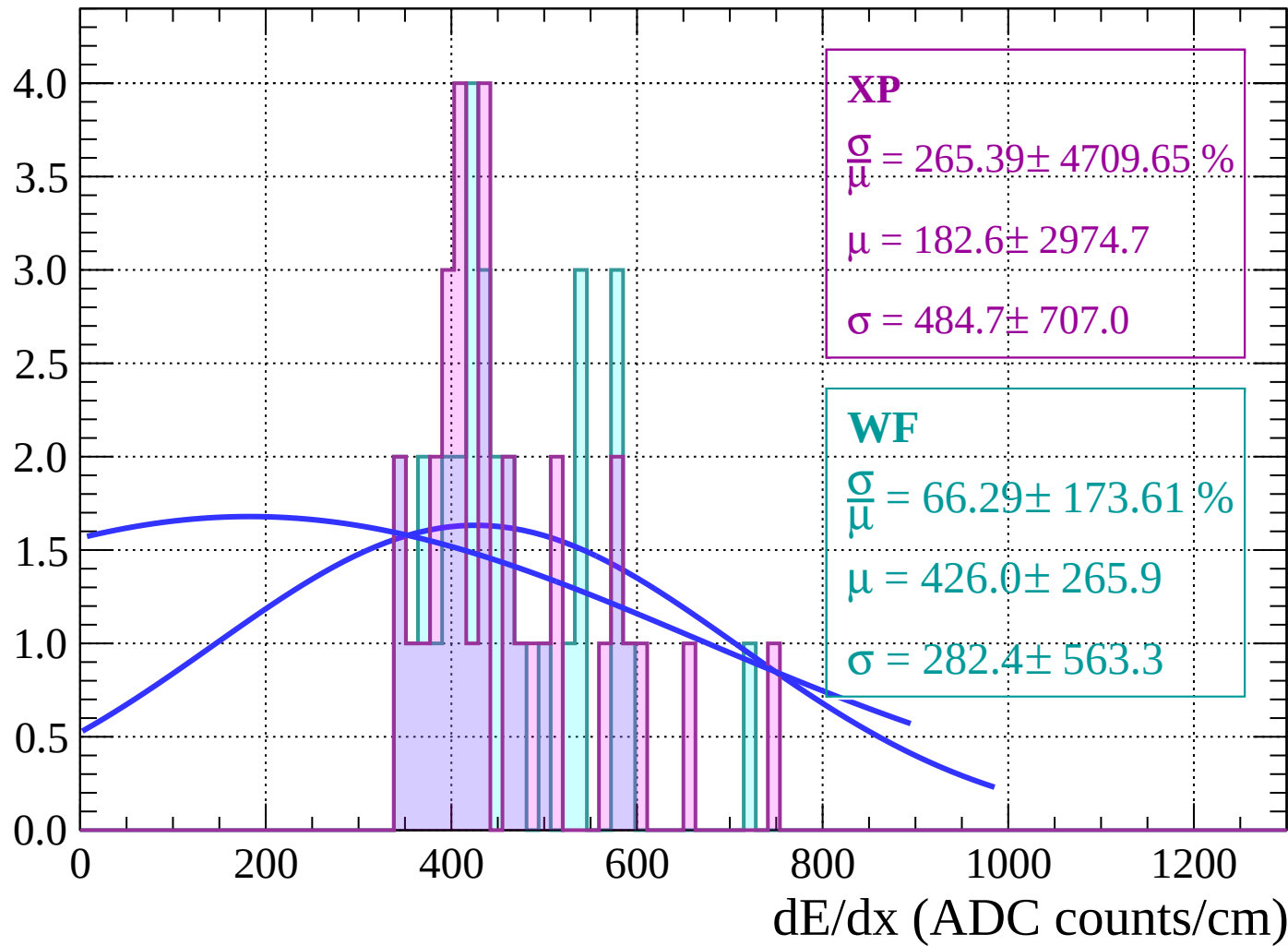
Energy loss | $-54 < \theta < -52$

Count



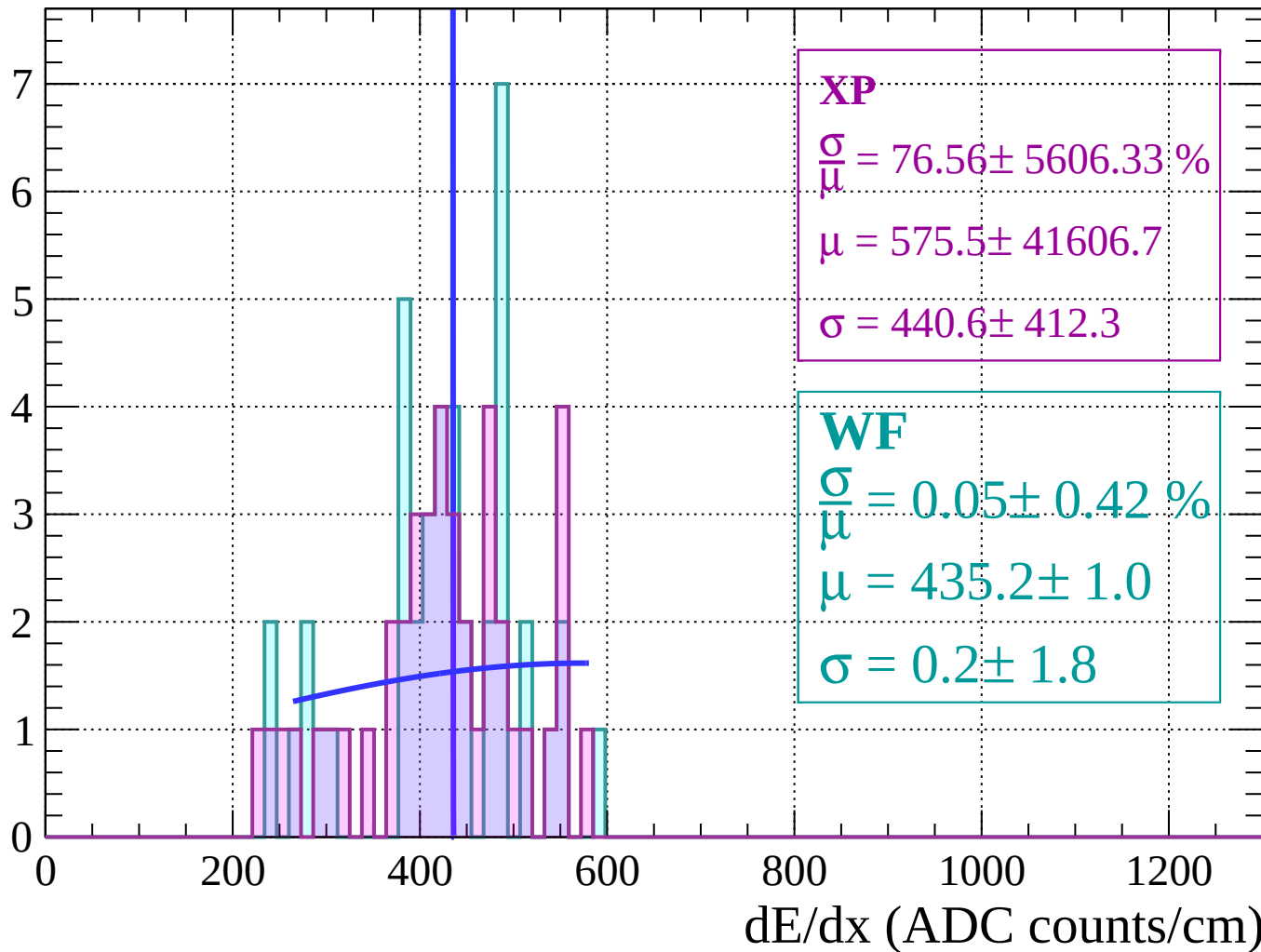
Energy loss | $-52 < \theta < -50$

Count



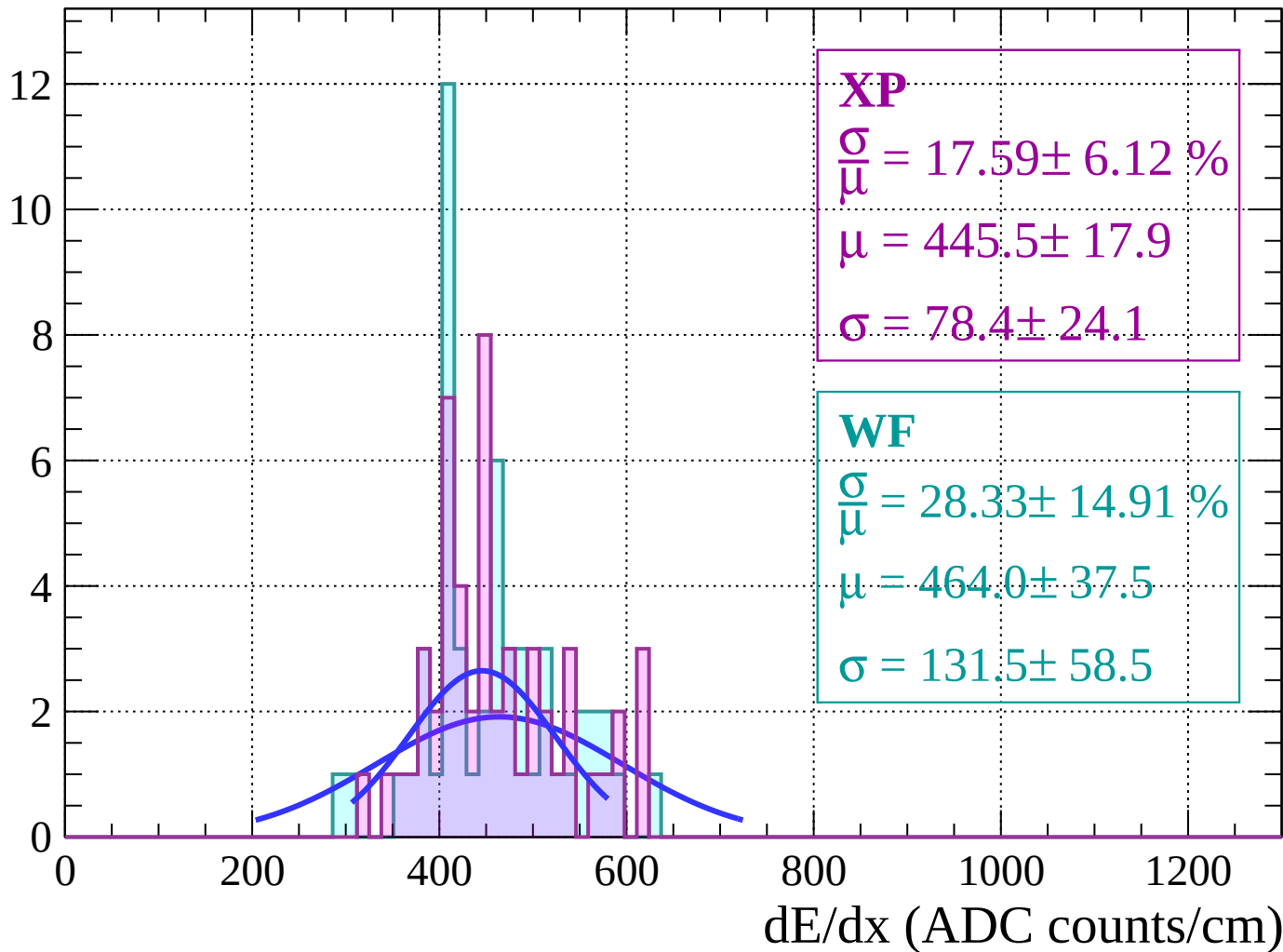
Energy loss | $-50 < \theta < -48$

Count



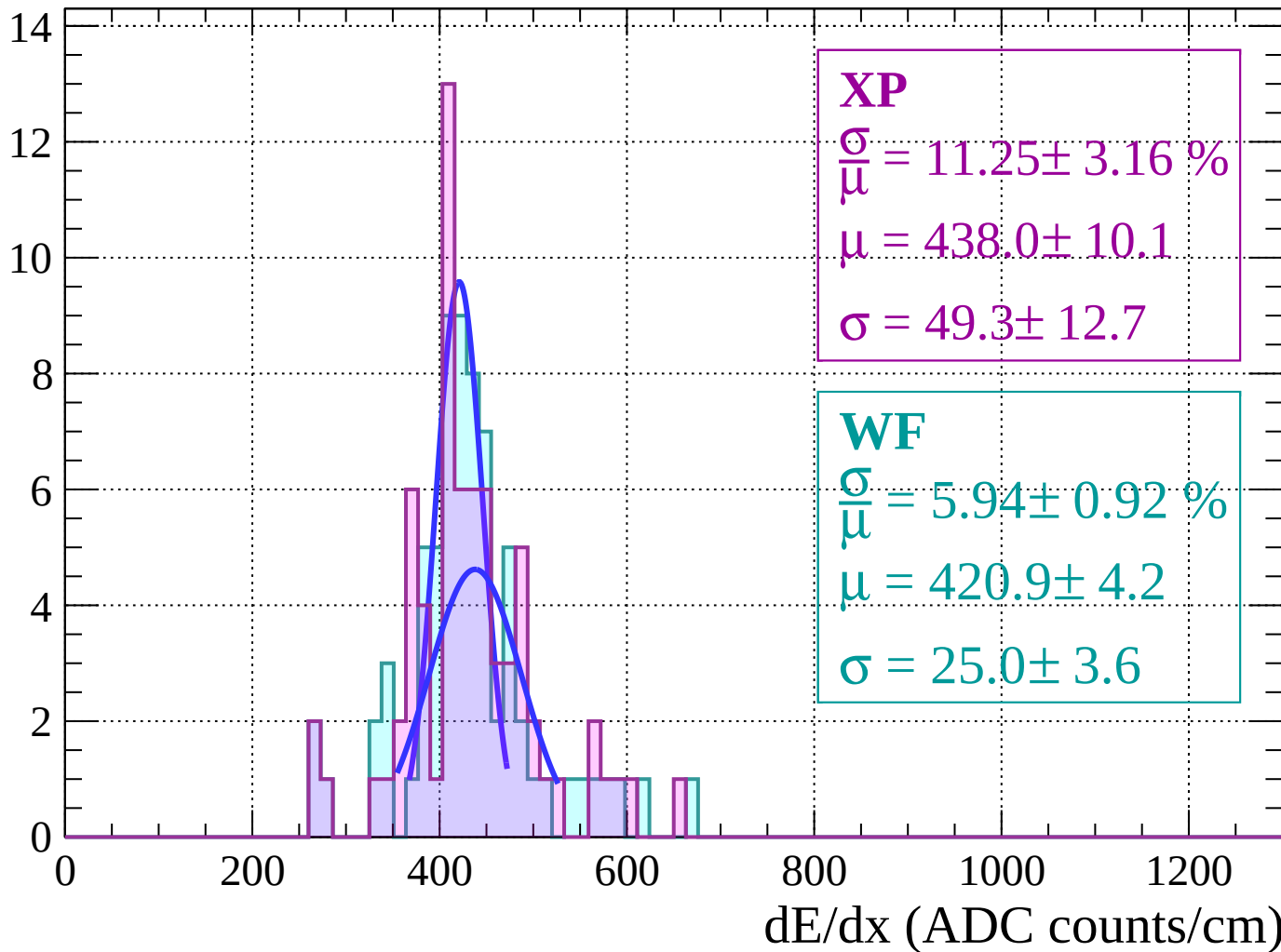
Energy loss | $-48 < \theta < -46$

Count



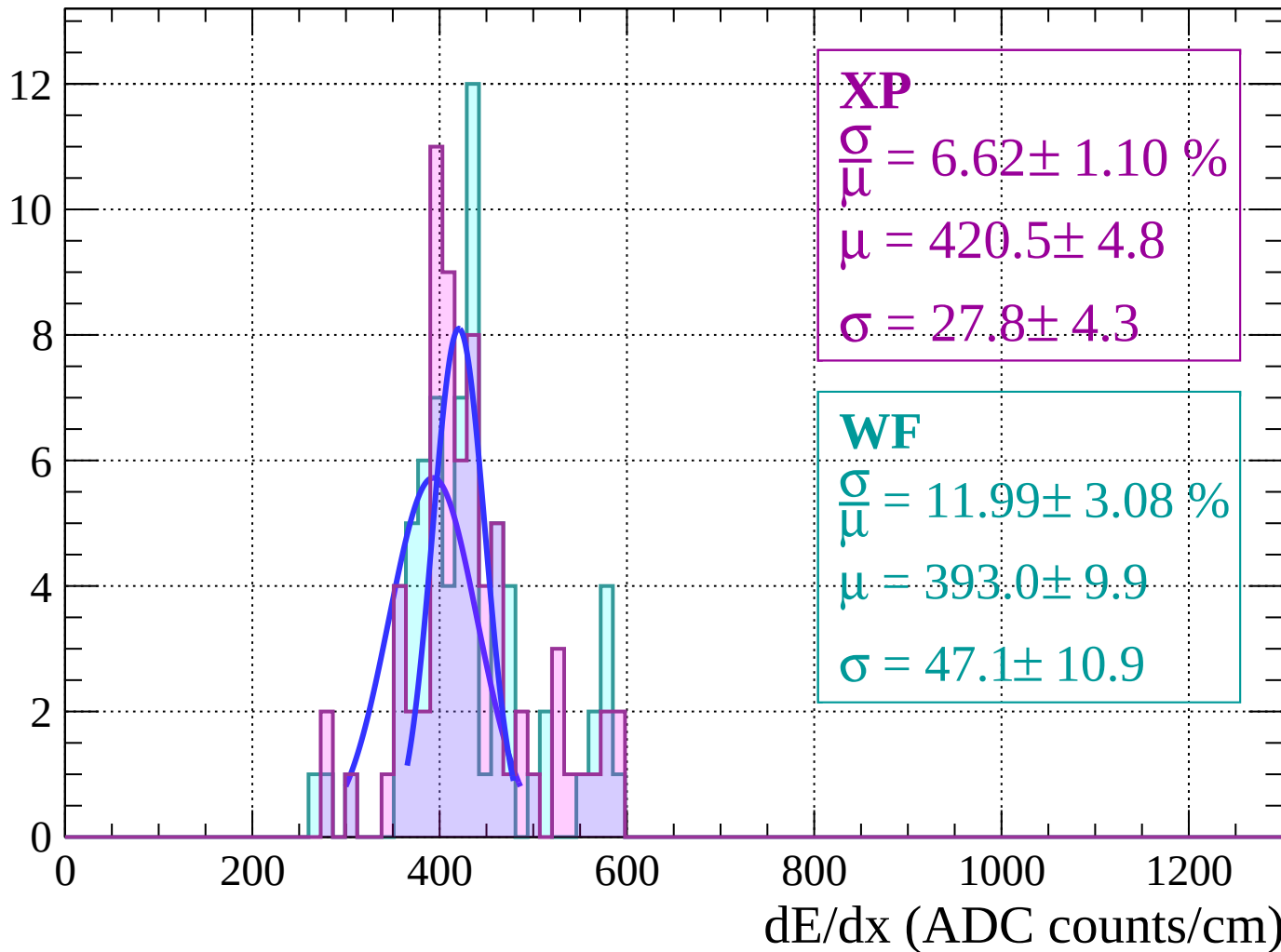
Energy loss | $-46 < \theta < -44$

Count



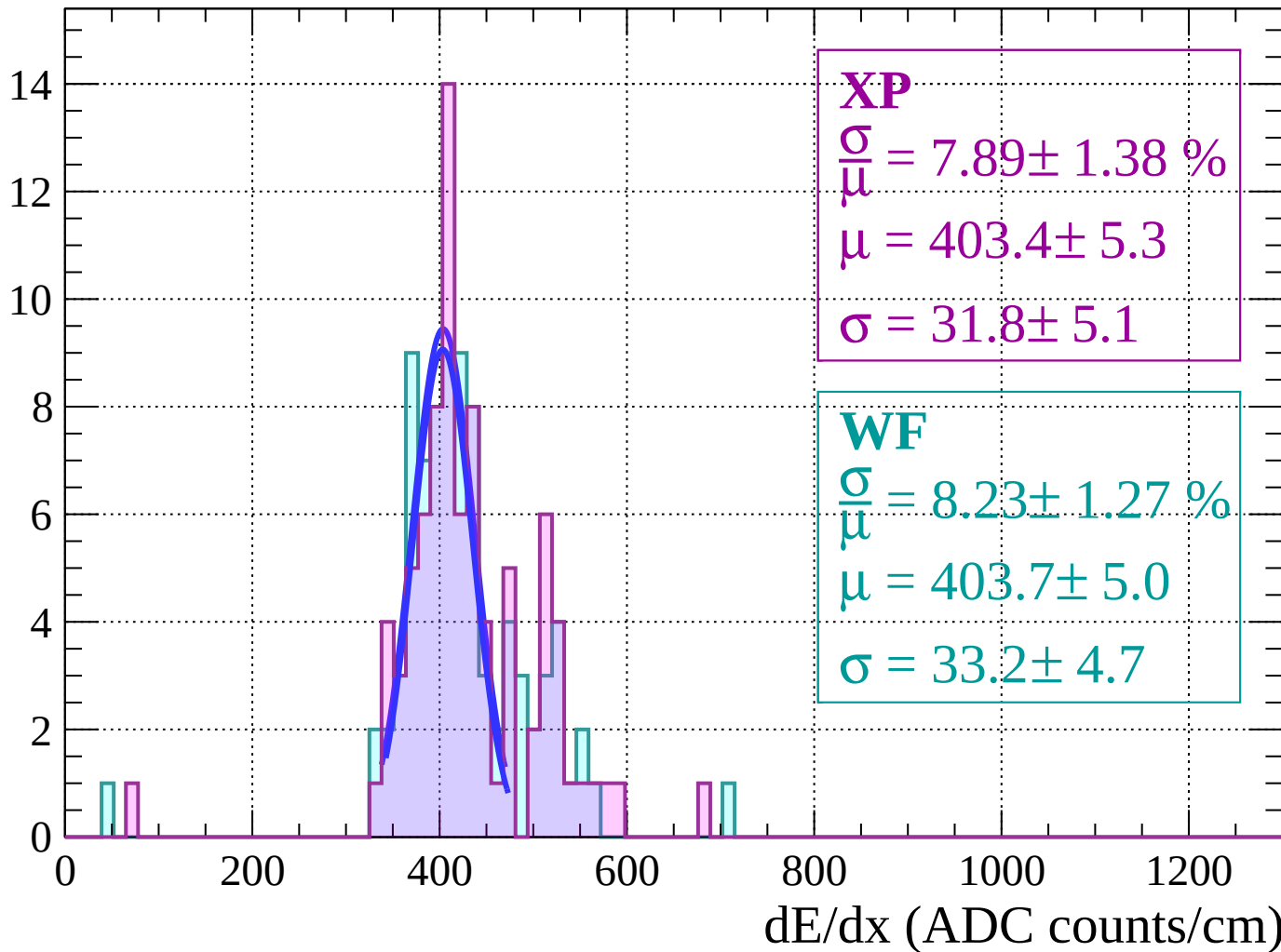
Energy loss | $-44 < \theta < -42$

Count



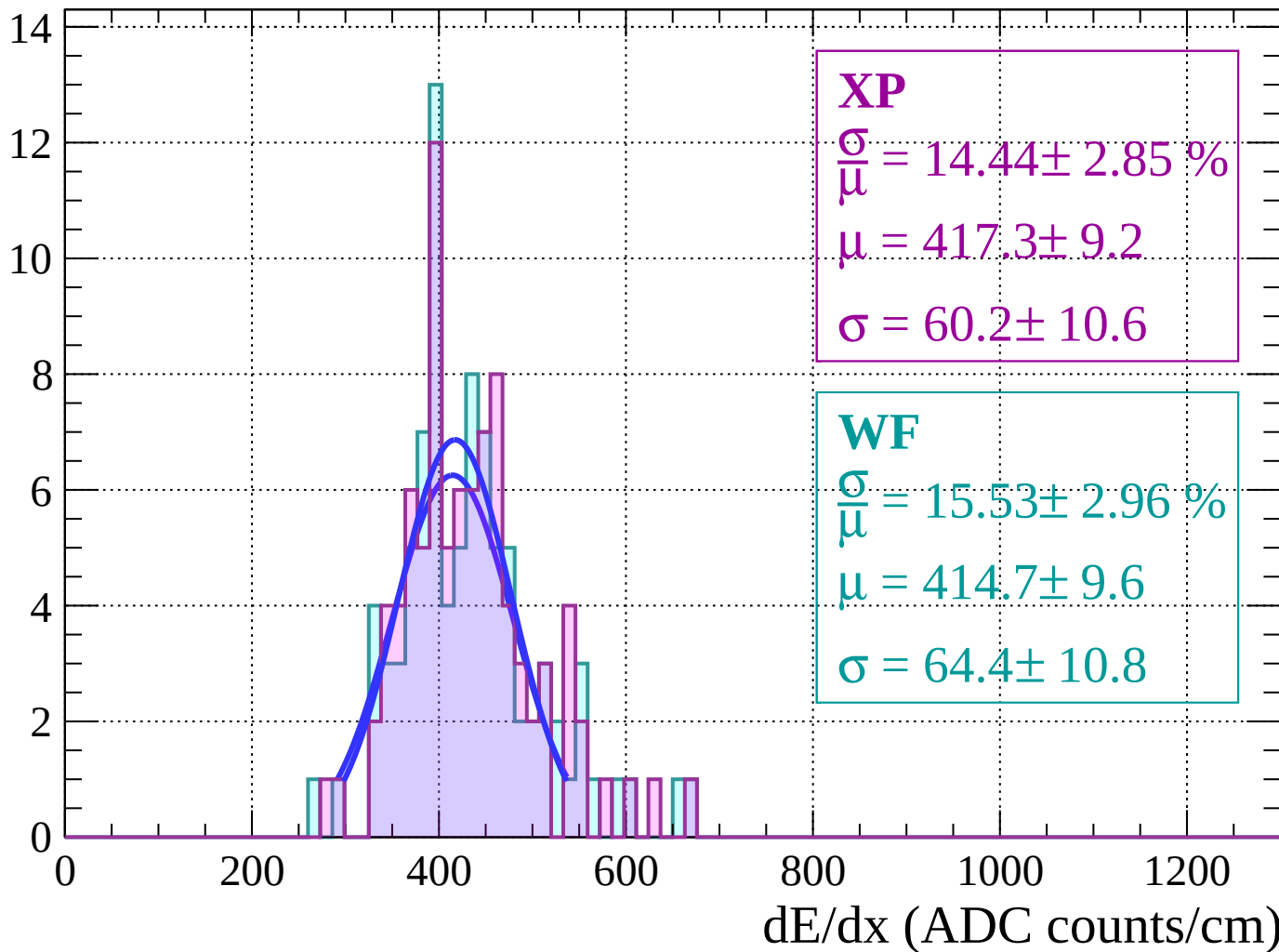
Energy loss | $-42 < \theta < -40$

Count



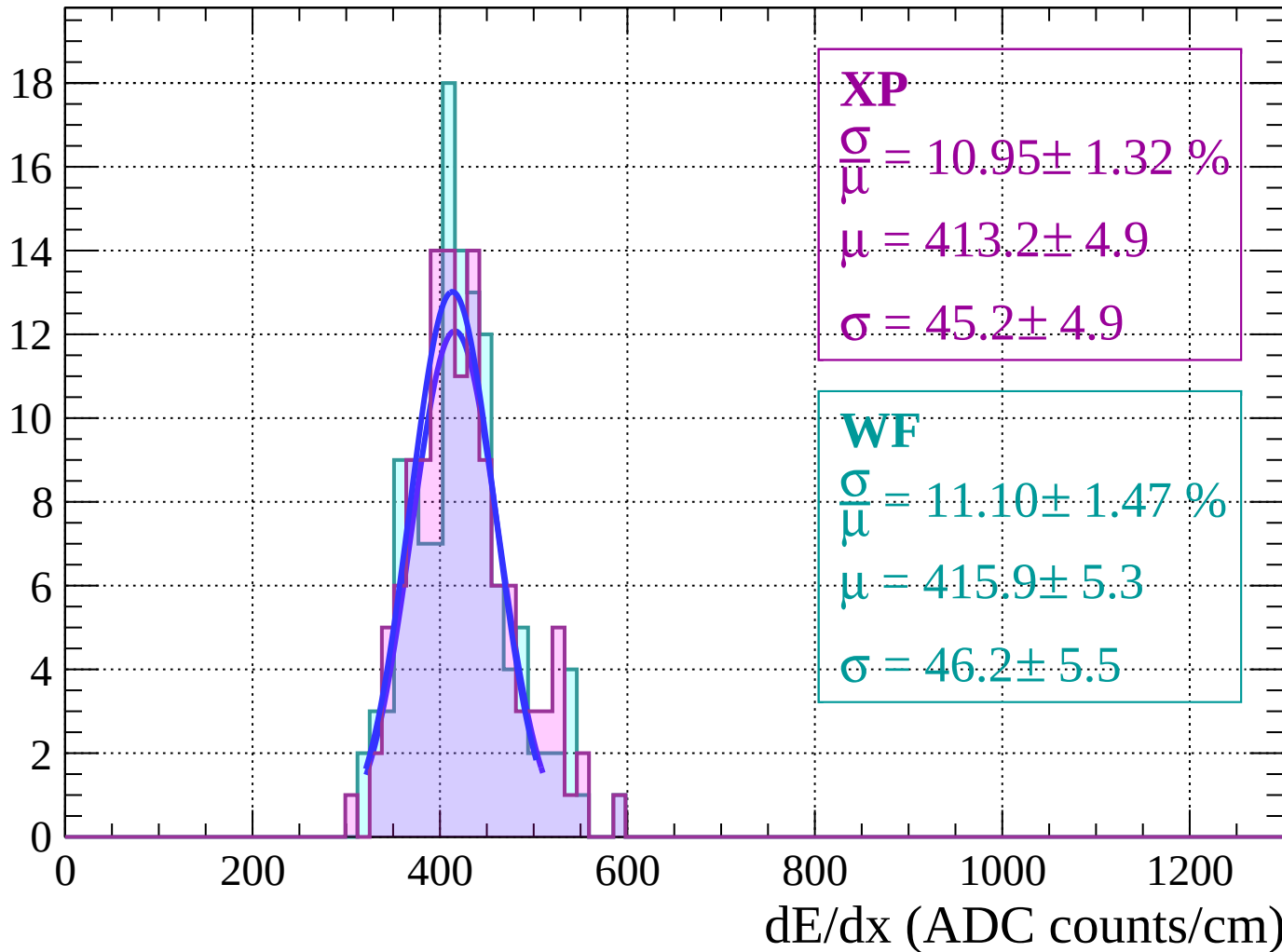
Energy loss | $-40 < \theta < -38$

Count



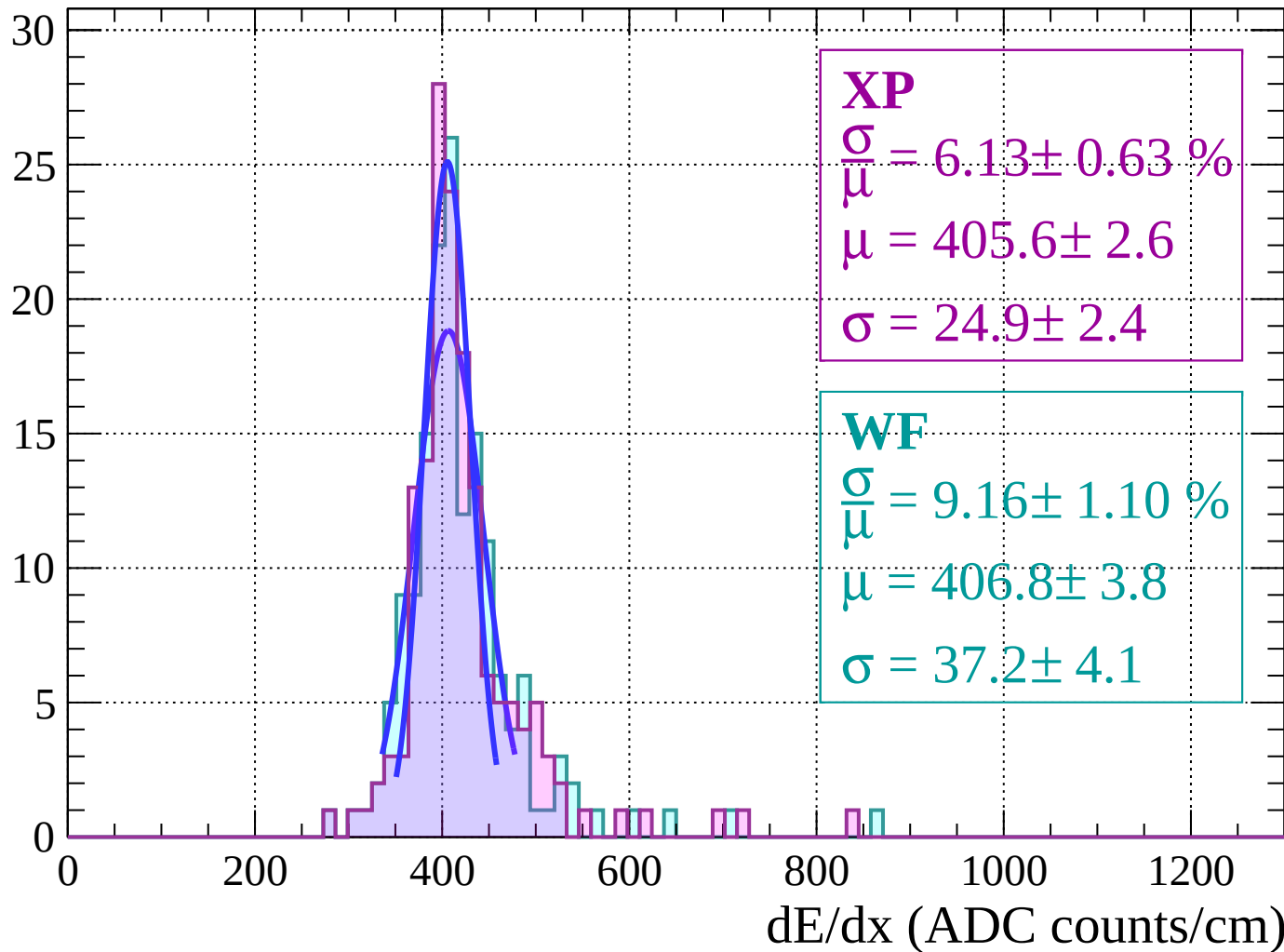
Energy loss | $-38 < \theta < -36$

Count



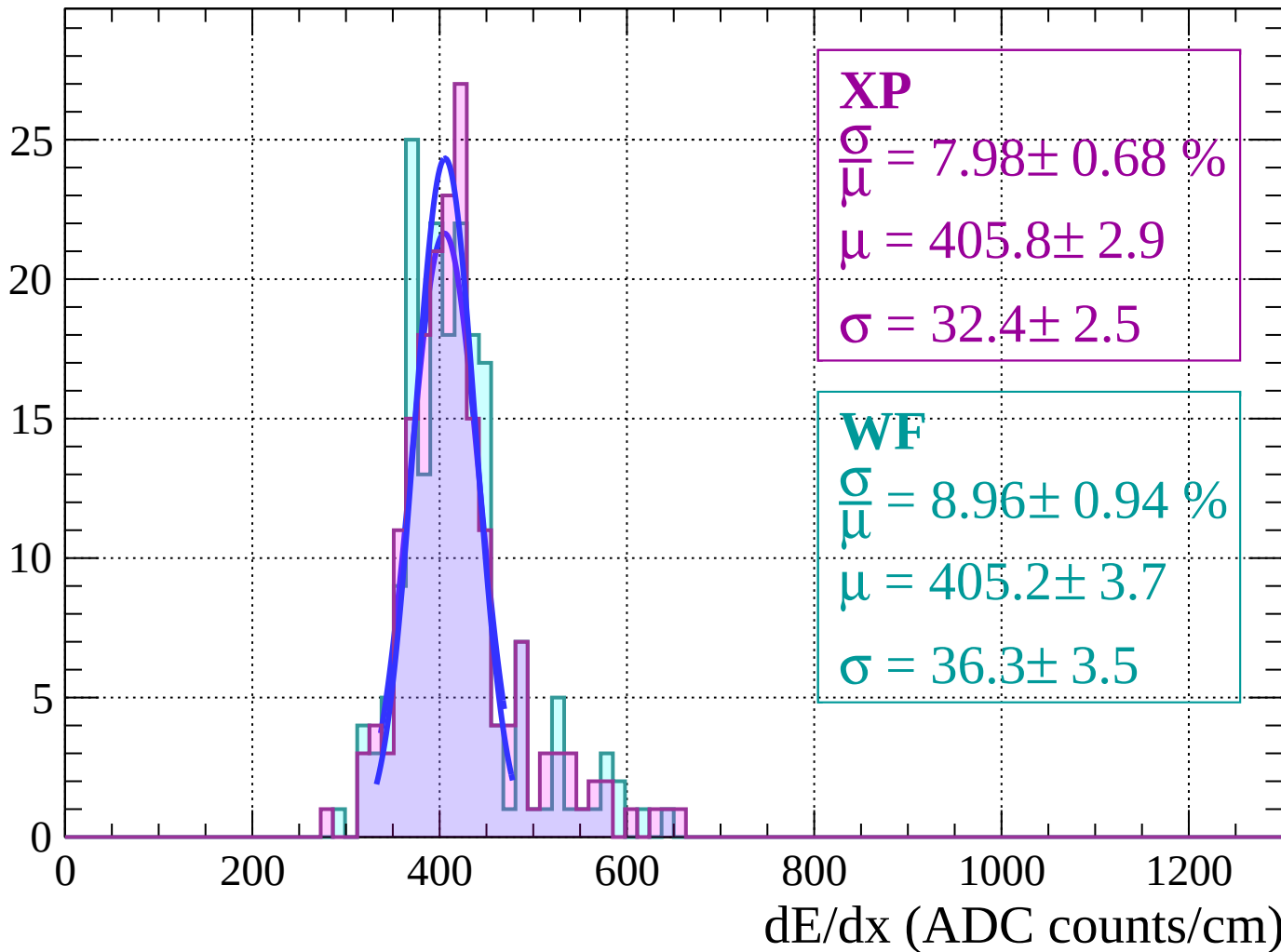
Energy loss | $-36 < \theta < -34$

Count



Energy loss | $-34 < \theta < -32$

Count



Energy loss | $-32 < \theta < -30$

Count

XP

$$\frac{\sigma}{\mu} = 8.28 \pm 0.73 \%$$

$$\mu = 402.2 \pm 2.9$$

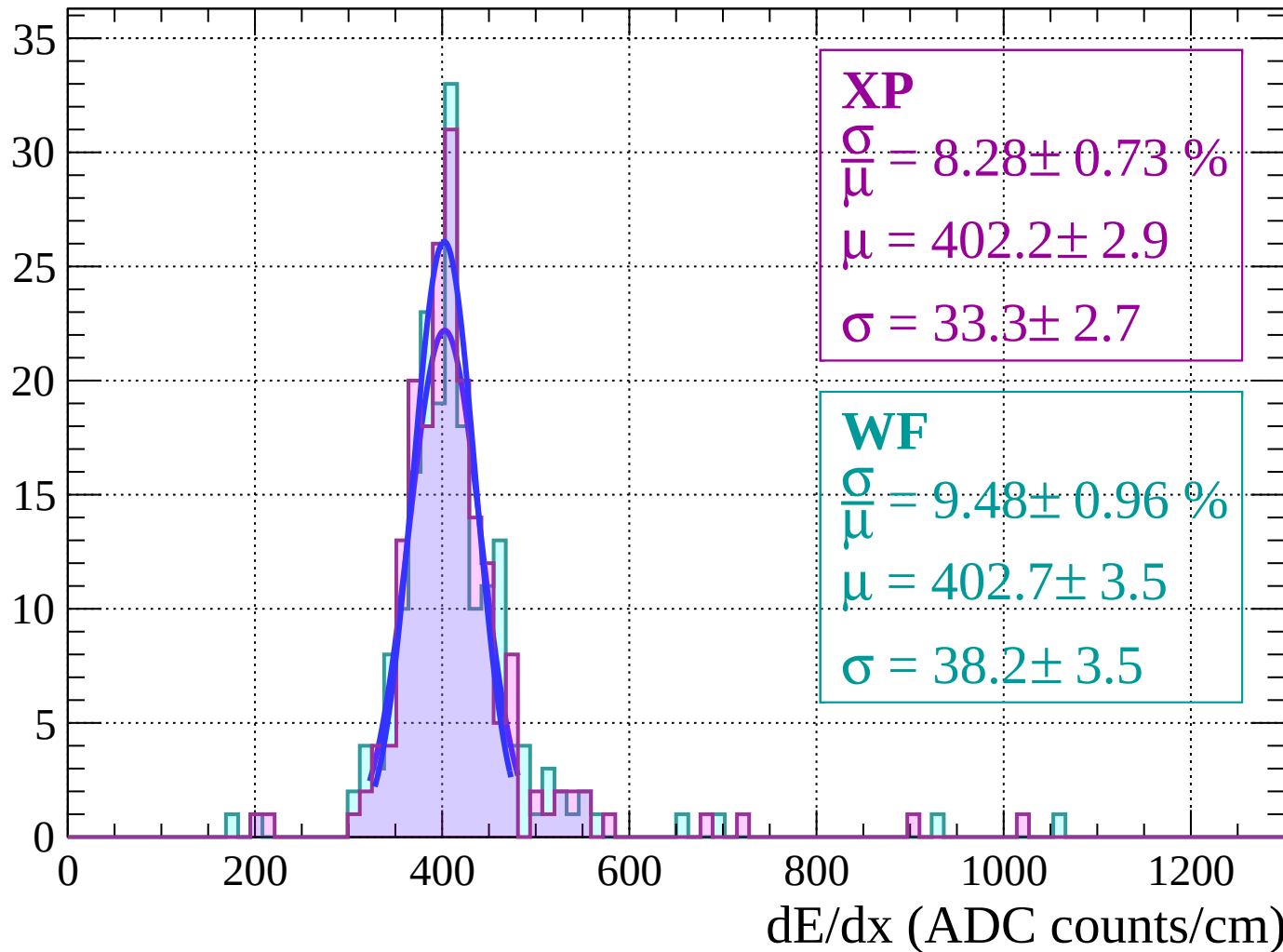
$$\sigma = 33.3 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 9.48 \pm 0.96 \%$$

$$\mu = 402.7 \pm 3.5$$

$$\sigma = 38.2 \pm 3.5$$



Energy loss | $-30 < \theta < -28$

Count

XP

$$\frac{\sigma}{\mu} = 8.66 \pm 0.58 \%$$

$$\mu = 406.8 \pm 2.5$$

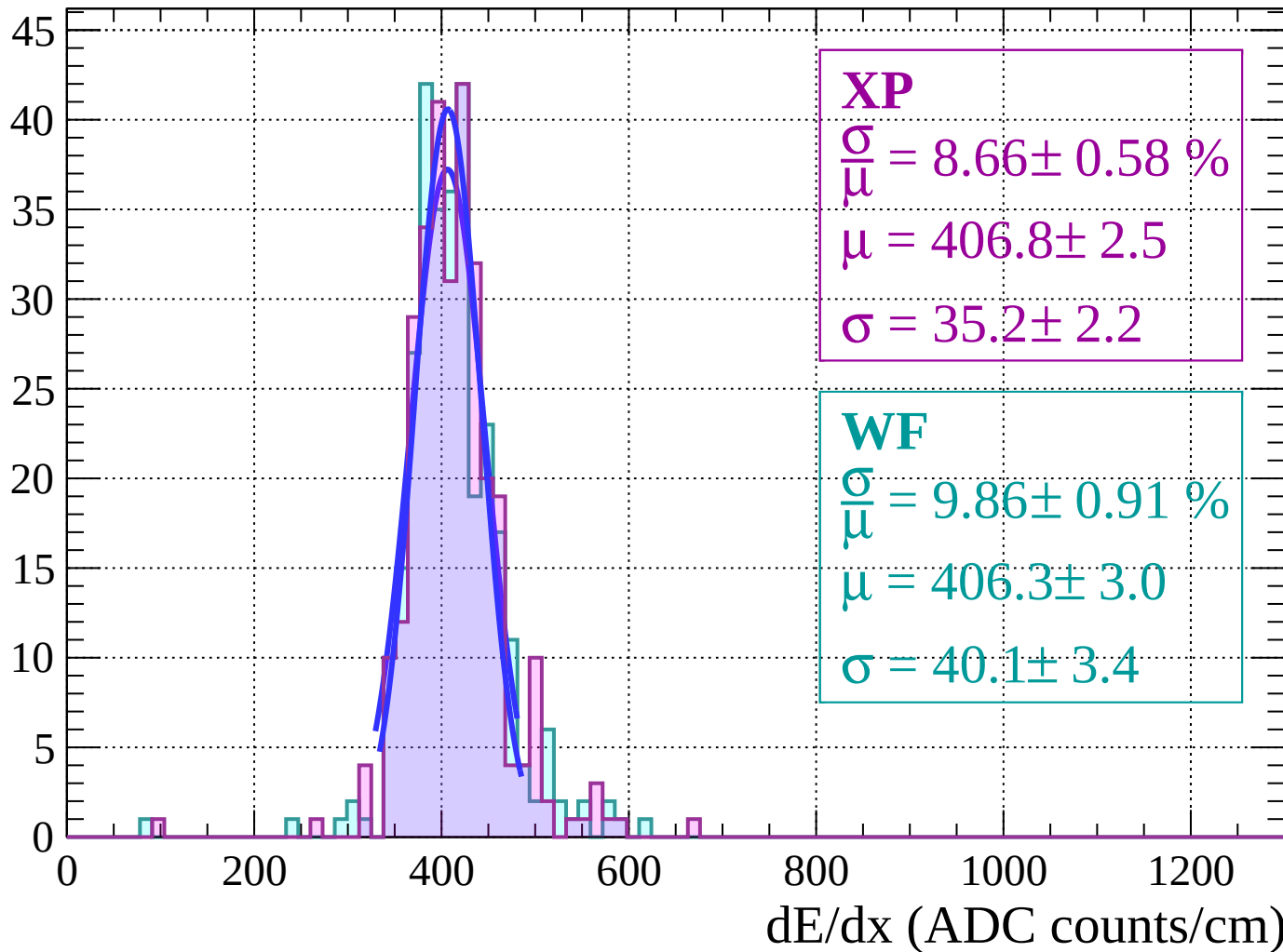
$$\sigma = 35.2 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 9.86 \pm 0.91 \%$$

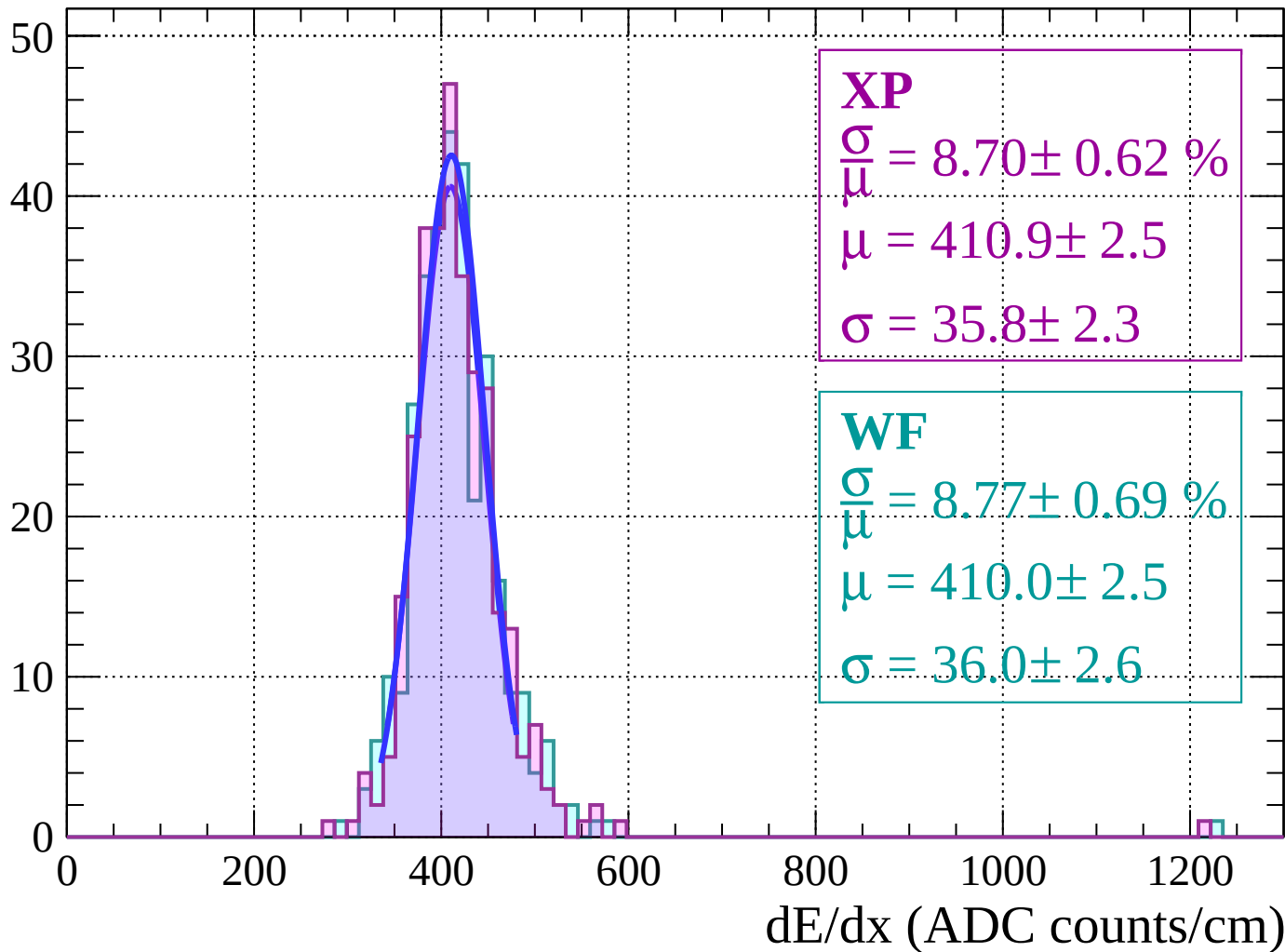
$$\mu = 406.3 \pm 3.0$$

$$\sigma = 40.1 \pm 3.4$$



Energy loss | $-28 < \theta < -26$

Count



Energy loss | $-26 < \theta < -24$

Count

XP

$$\frac{\sigma}{\mu} = 7.67 \pm 0.43 \%$$

$$\mu = 411.9 \pm 2.0$$

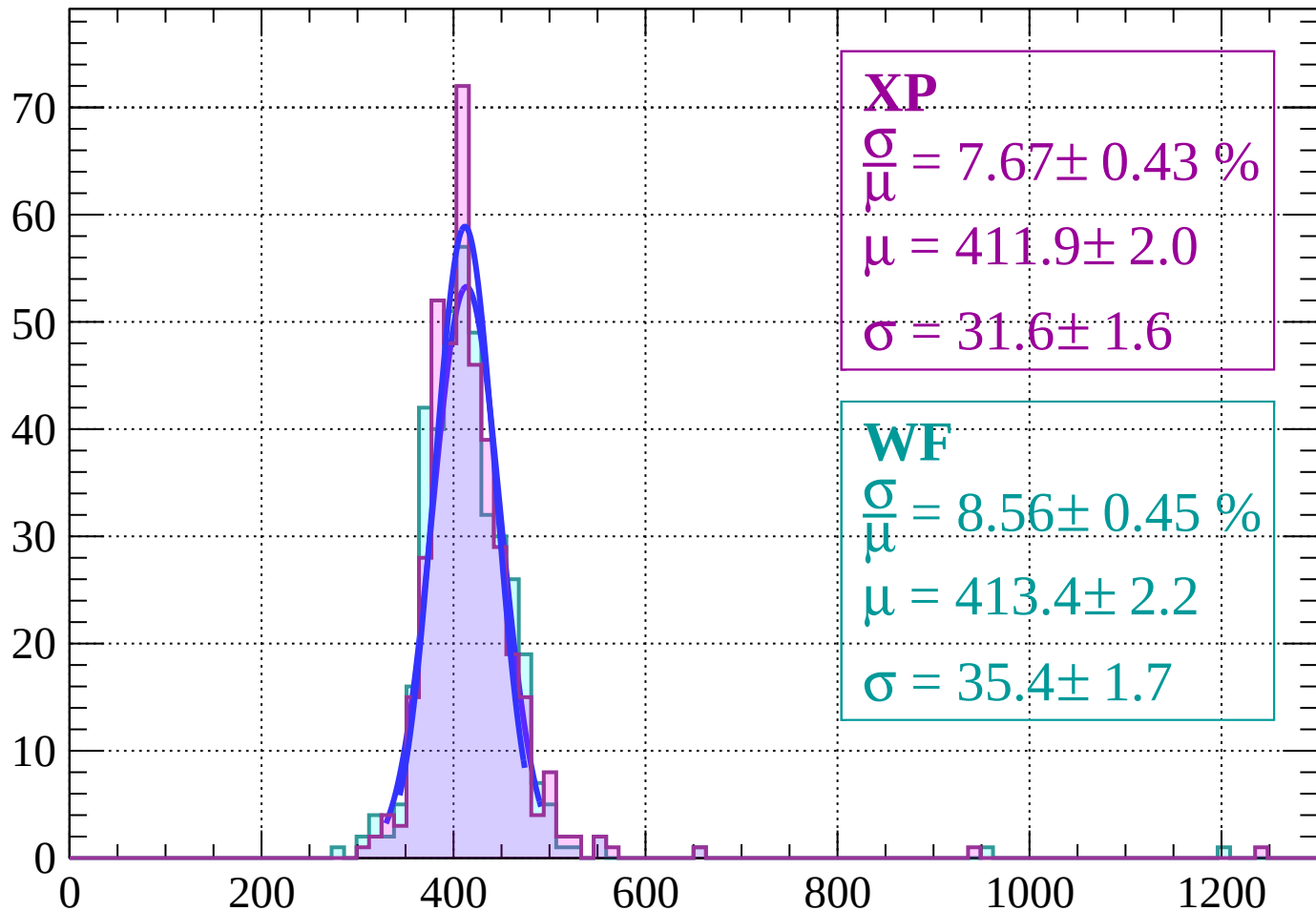
$$\sigma = 31.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.56 \pm 0.45 \%$$

$$\mu = 413.4 \pm 2.2$$

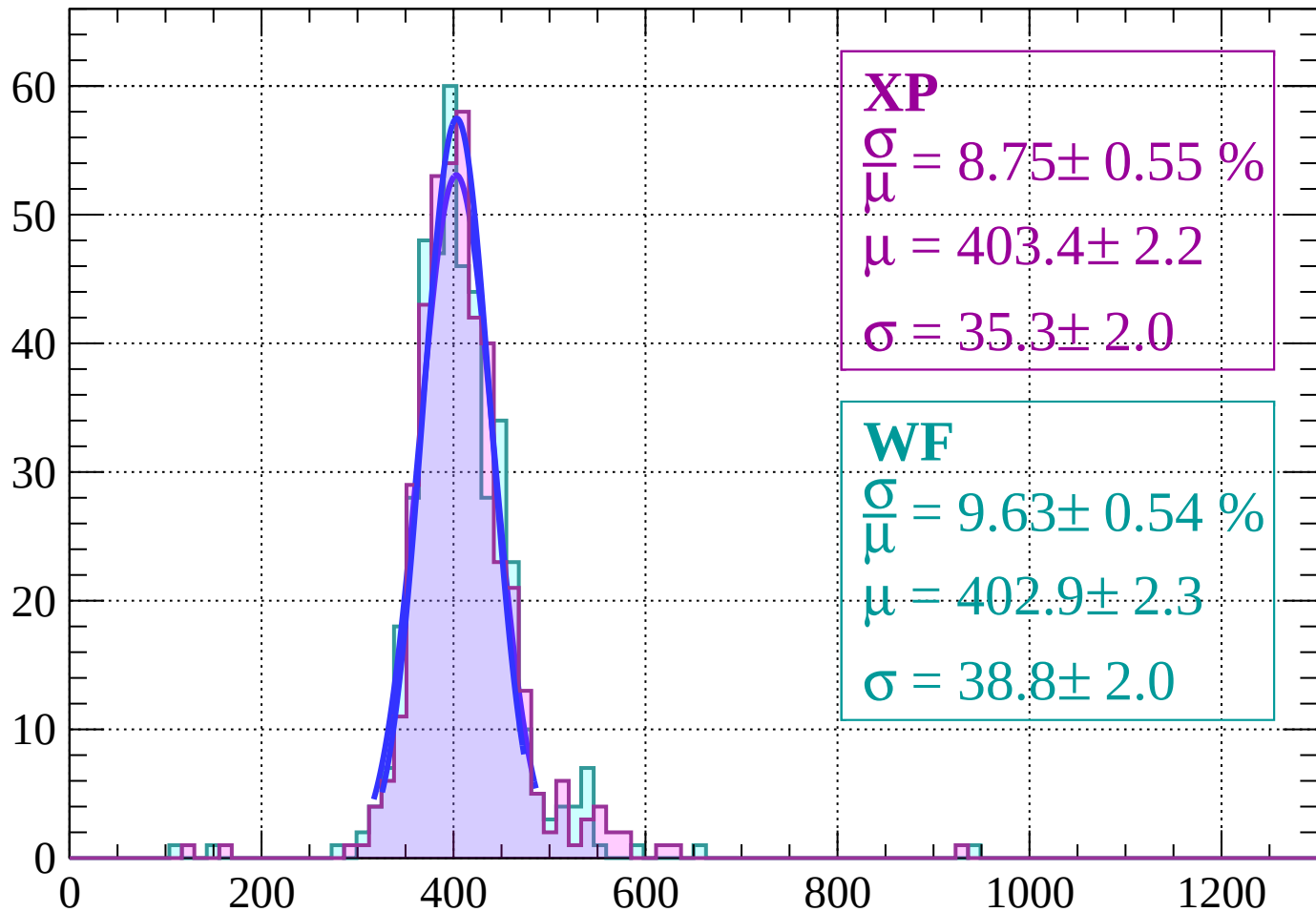
$$\sigma = 35.4 \pm 1.7$$



dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

Count



XP

$$\frac{\sigma}{\mu} = 8.75 \pm 0.55 \%$$

$$\mu = 403.4 \pm 2.2$$

$$\sigma = 35.3 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 9.63 \pm 0.54 \%$$

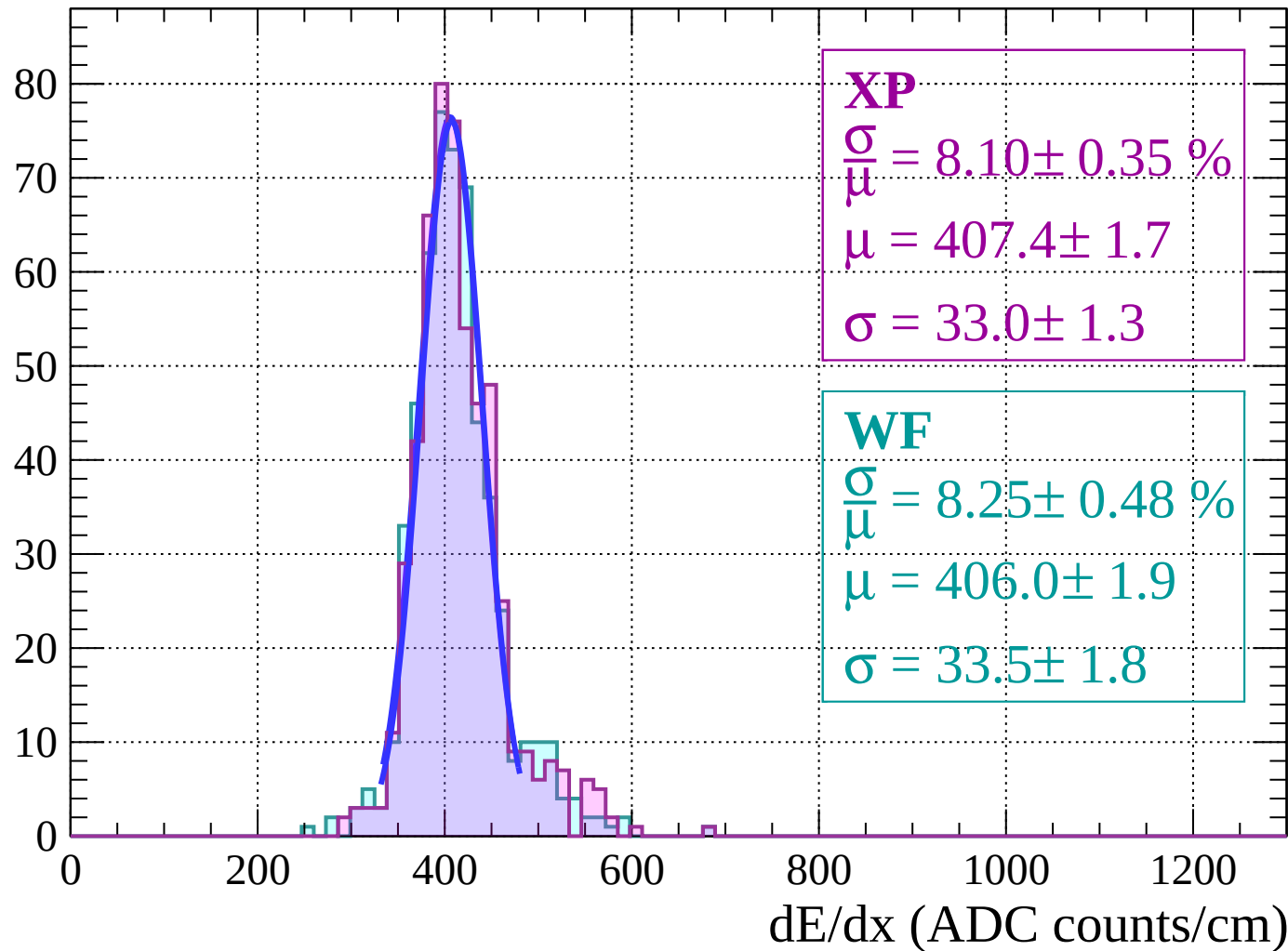
$$\mu = 402.9 \pm 2.3$$

$$\sigma = 38.8 \pm 2.0$$

dE/dx (ADC counts/cm)

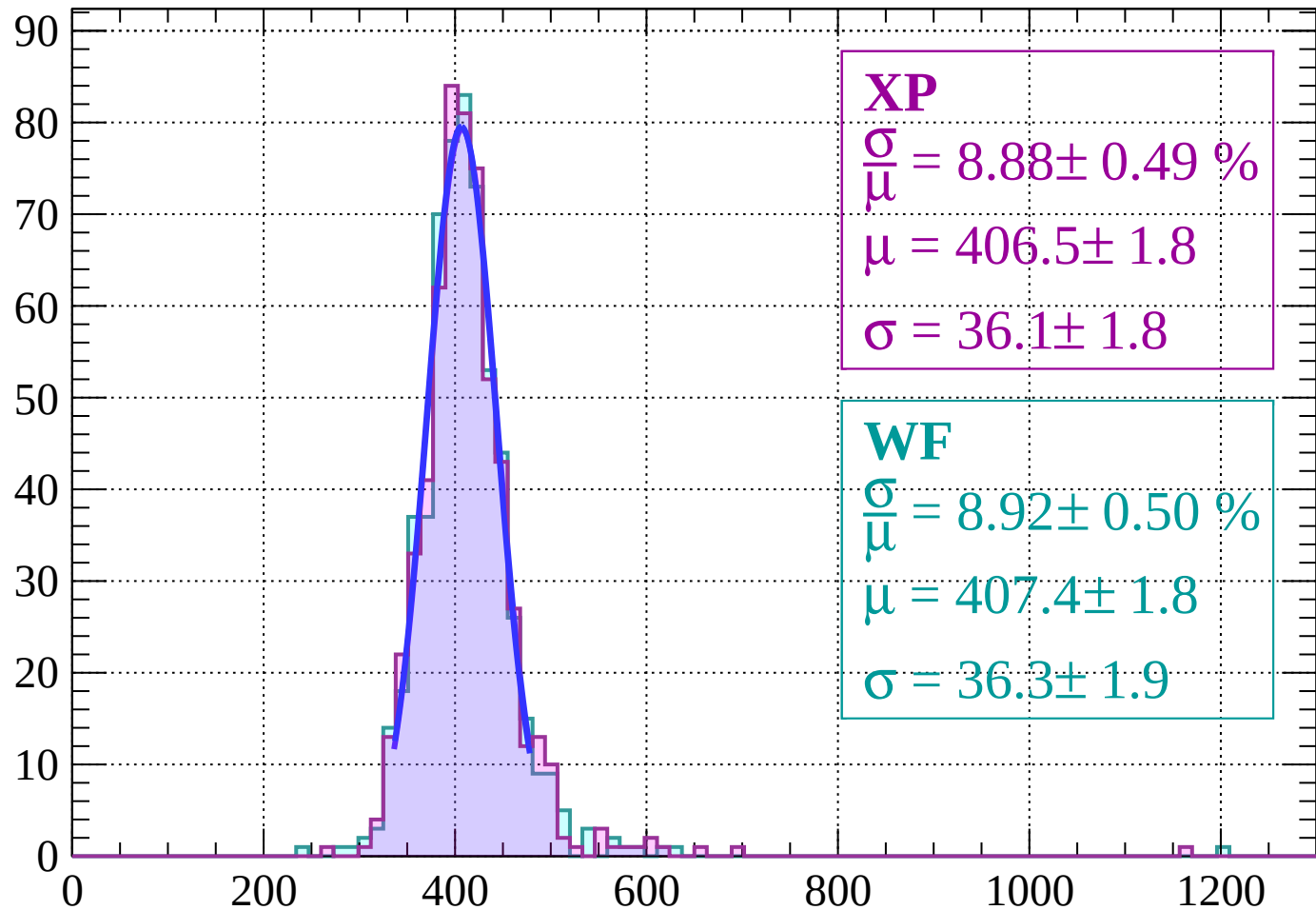
Energy loss | $-22 < \theta < -20$

Count



Energy loss | $-20 < \theta < -18$

Count



XP

$$\frac{\sigma}{\mu} = 8.88 \pm 0.49 \%$$

$$\mu = 406.5 \pm 1.8$$

$$\sigma = 36.1 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.92 \pm 0.50 \%$$

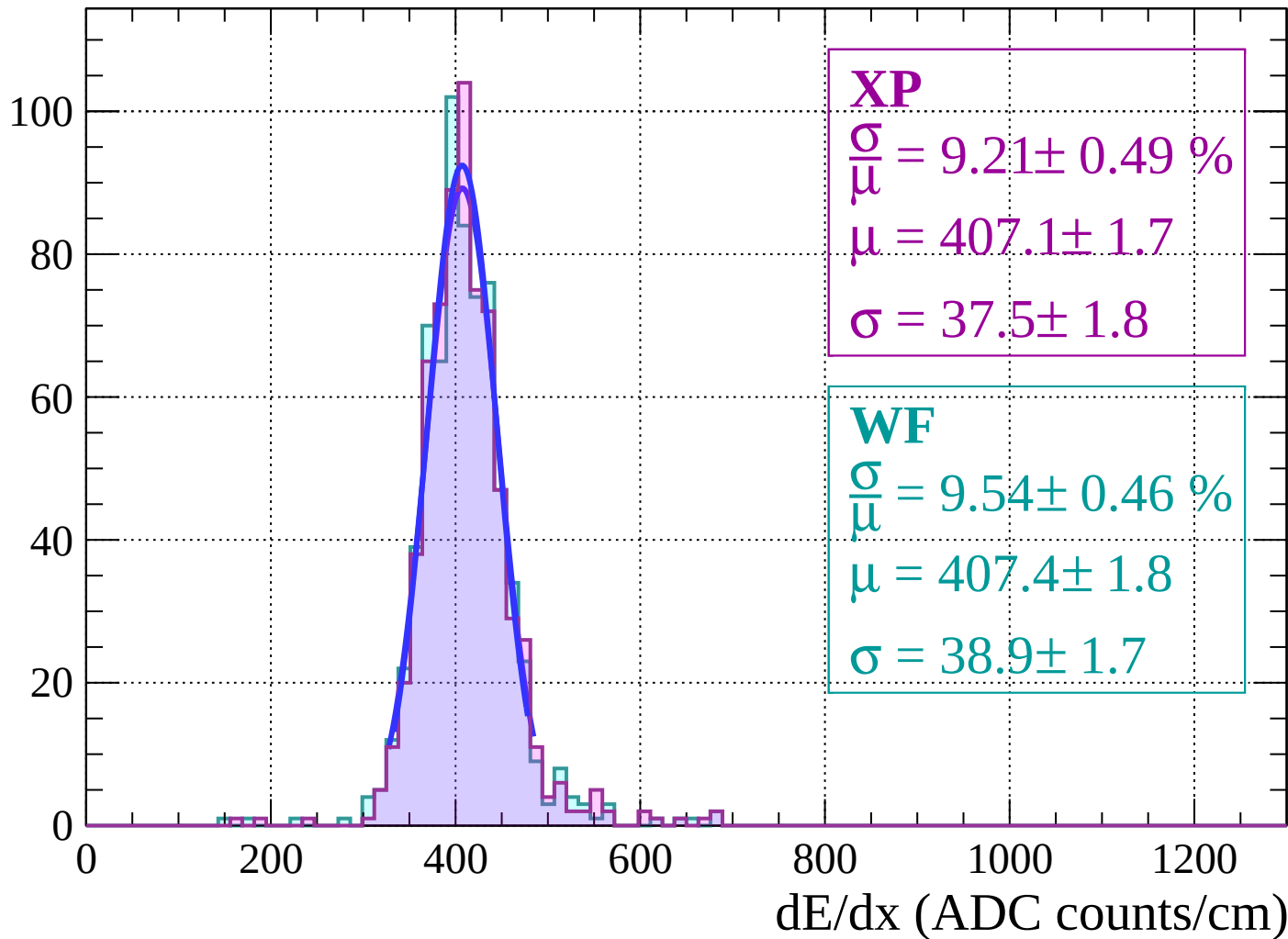
$$\mu = 407.4 \pm 1.8$$

$$\sigma = 36.3 \pm 1.9$$

dE/dx (ADC counts/cm)

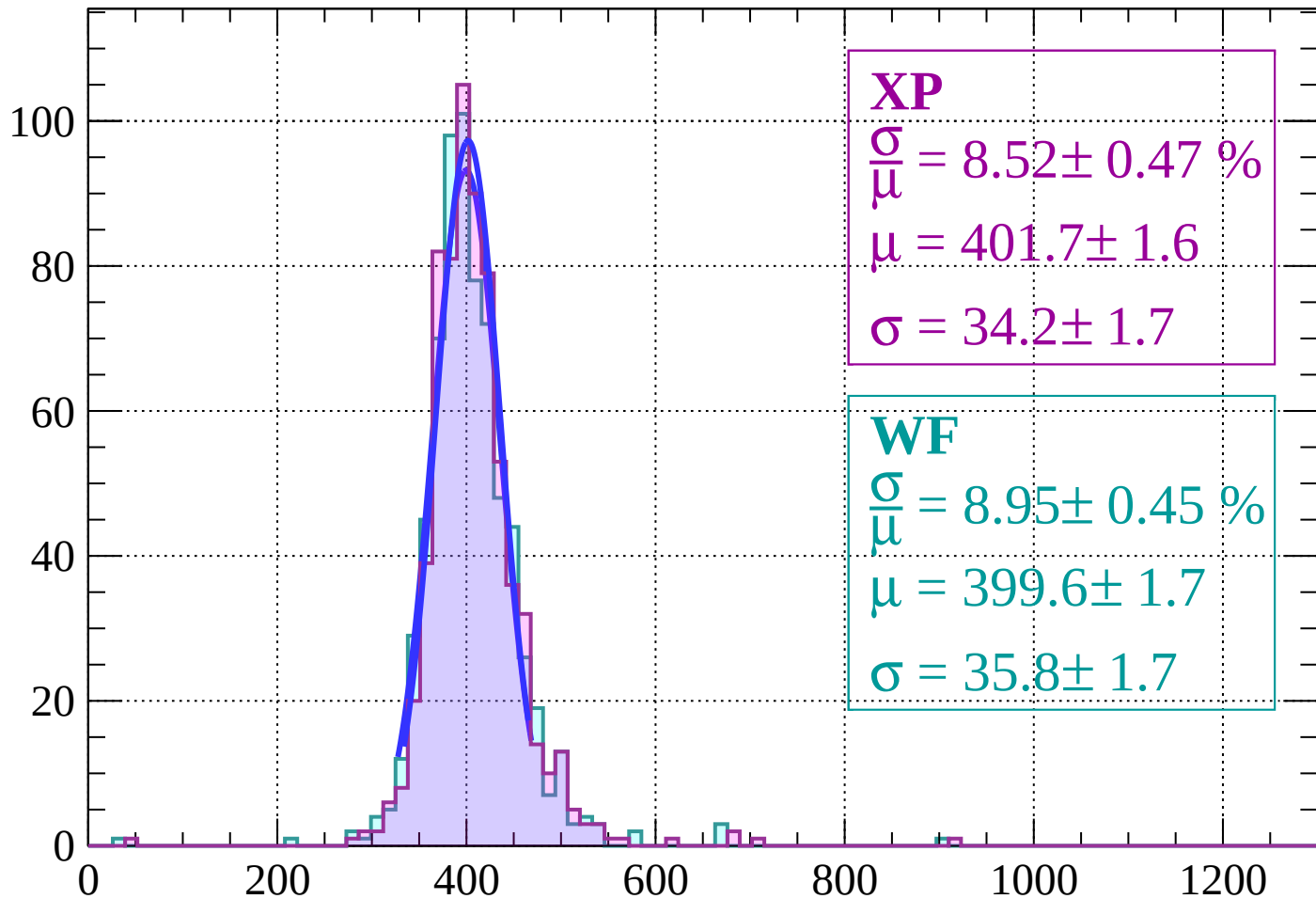
Energy loss $|-18 < \theta < -16$

Count



Energy loss $|-16 < \theta < -14$

Count



XP

$$\frac{\sigma}{\mu} = 8.52 \pm 0.47 \%$$

$$\mu = 401.7 \pm 1.6$$

$$\sigma = 34.2 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 8.95 \pm 0.45 \%$$

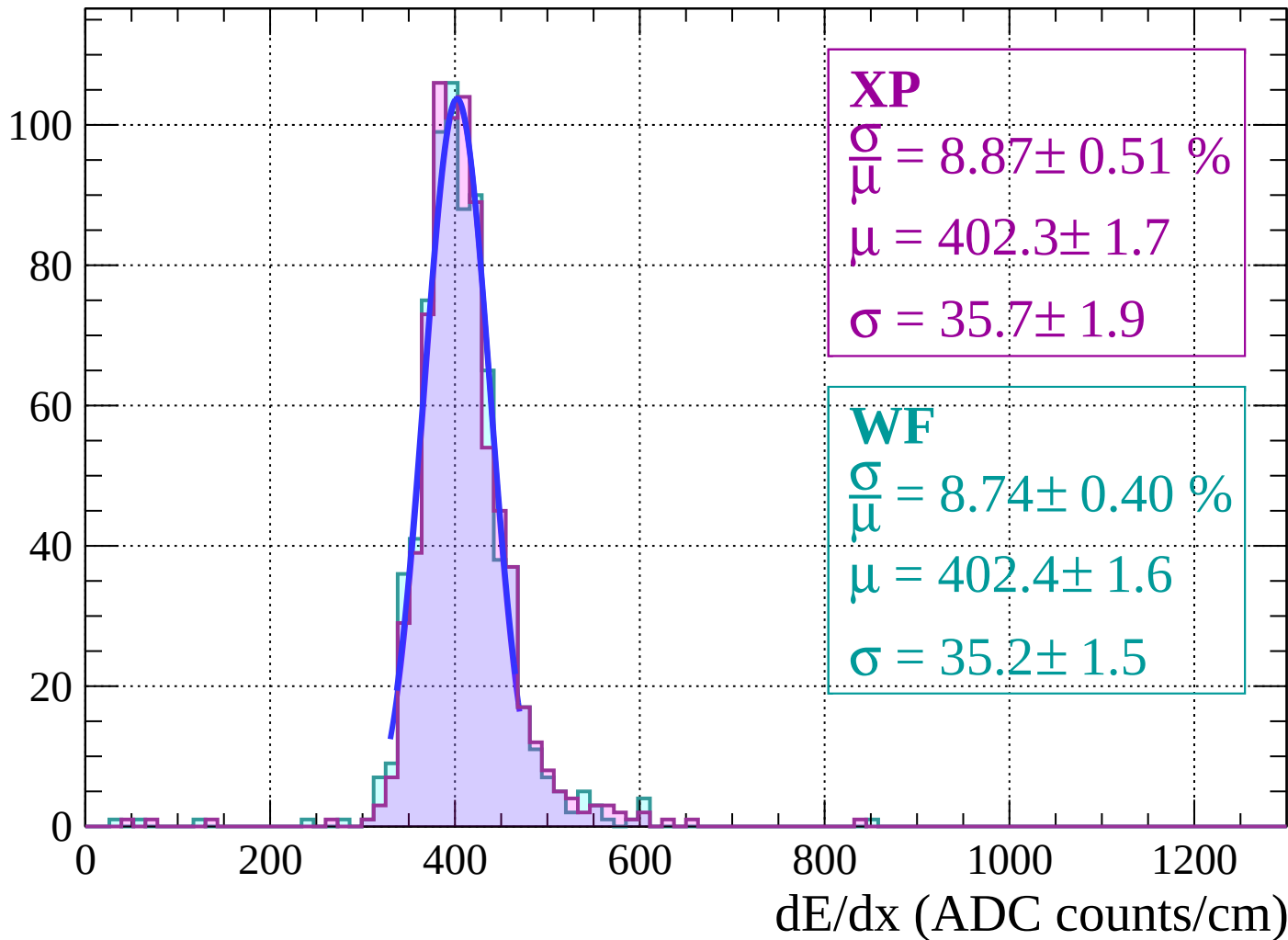
$$\mu = 399.6 \pm 1.7$$

$$\sigma = 35.8 \pm 1.7$$

dE/dx (ADC counts/cm)

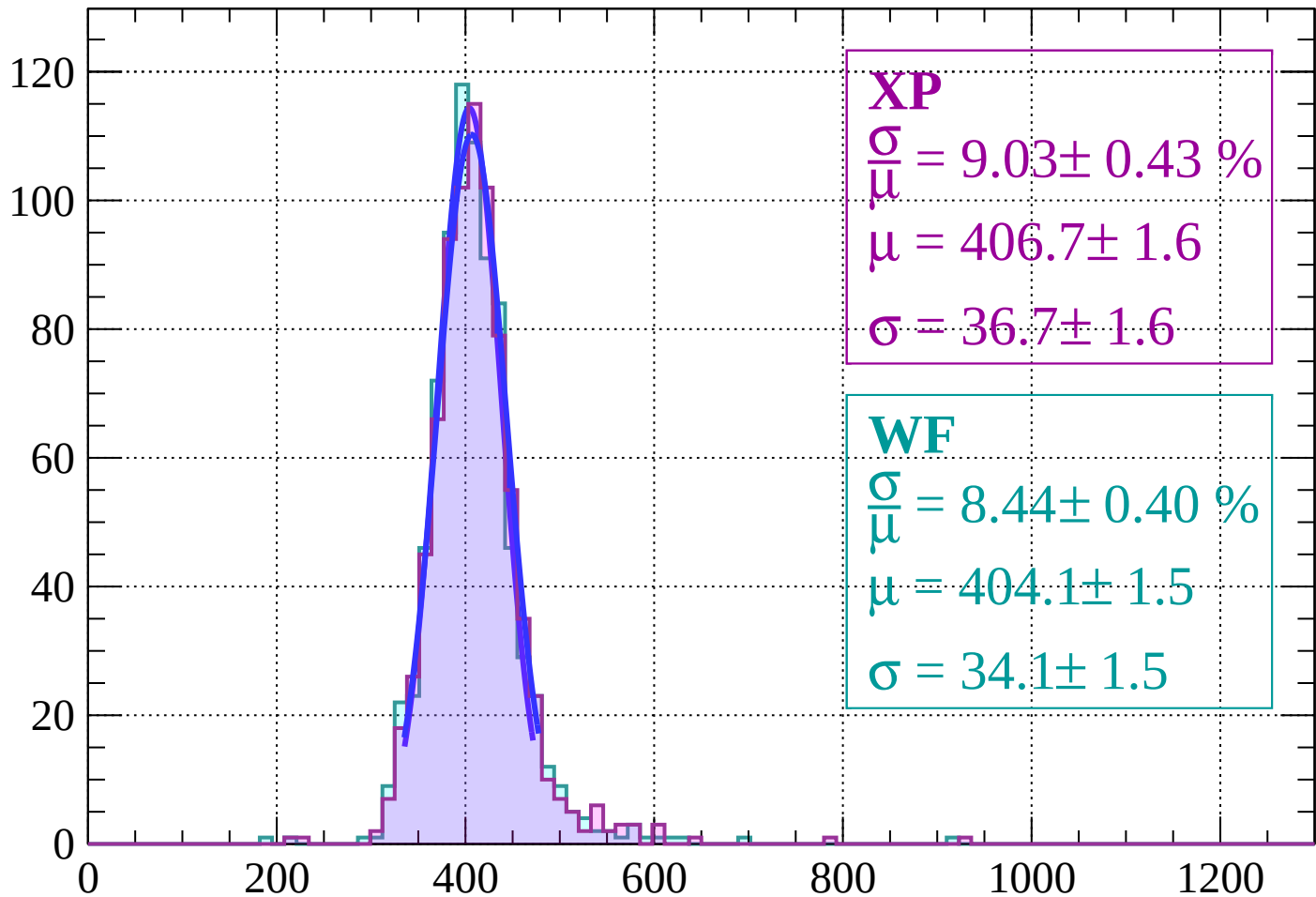
Energy loss $|-14 < \theta < -12$

Count



Energy loss | $-12 < \theta < -10$

Count



XP

$$\frac{\sigma}{\mu} = 9.03 \pm 0.43 \%$$

$$\mu = 406.7 \pm 1.6$$

$$\sigma = 36.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 8.44 \pm 0.40 \%$$

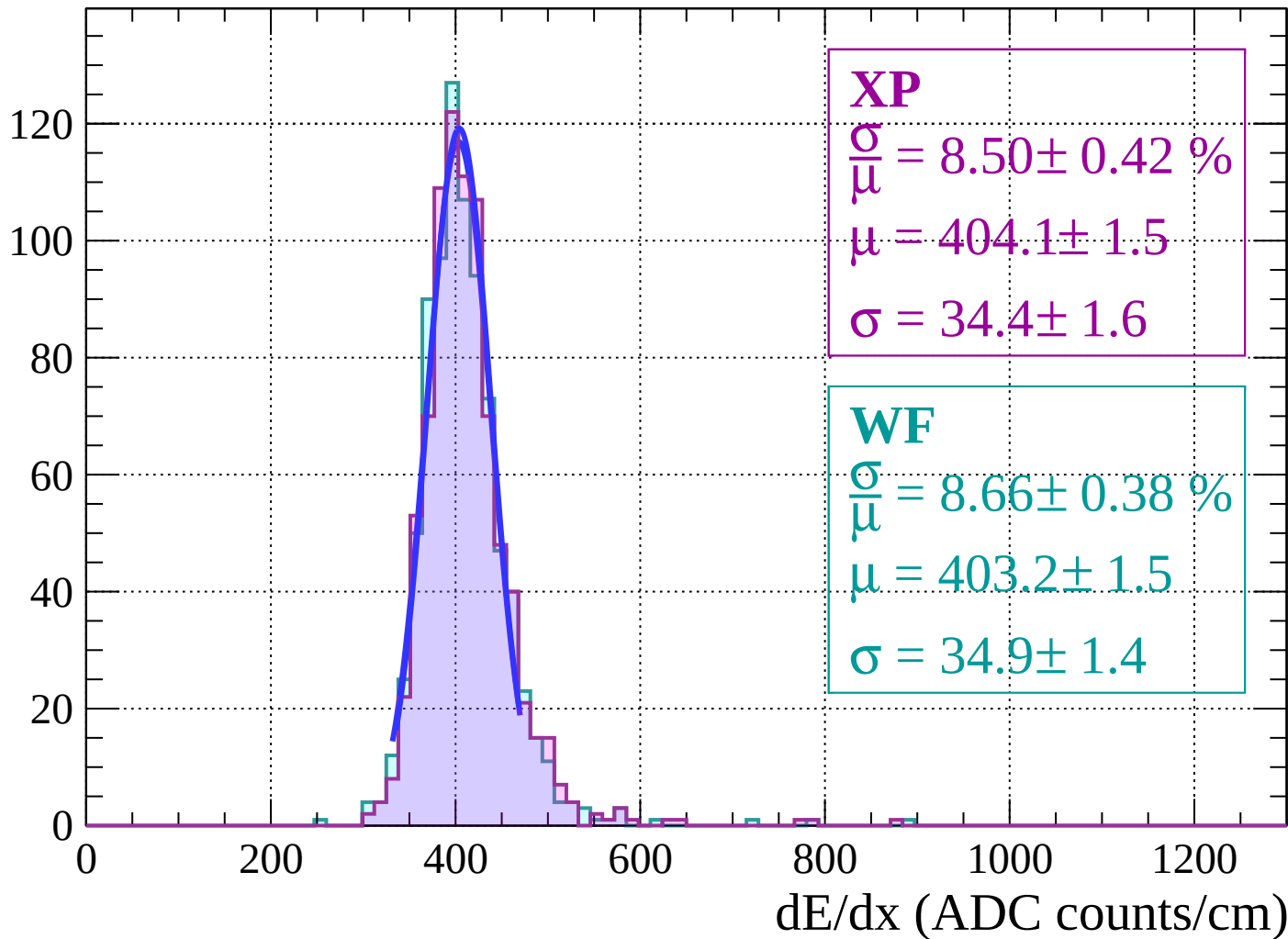
$$\mu = 404.1 \pm 1.5$$

$$\sigma = 34.1 \pm 1.5$$

dE/dx (ADC counts/cm)

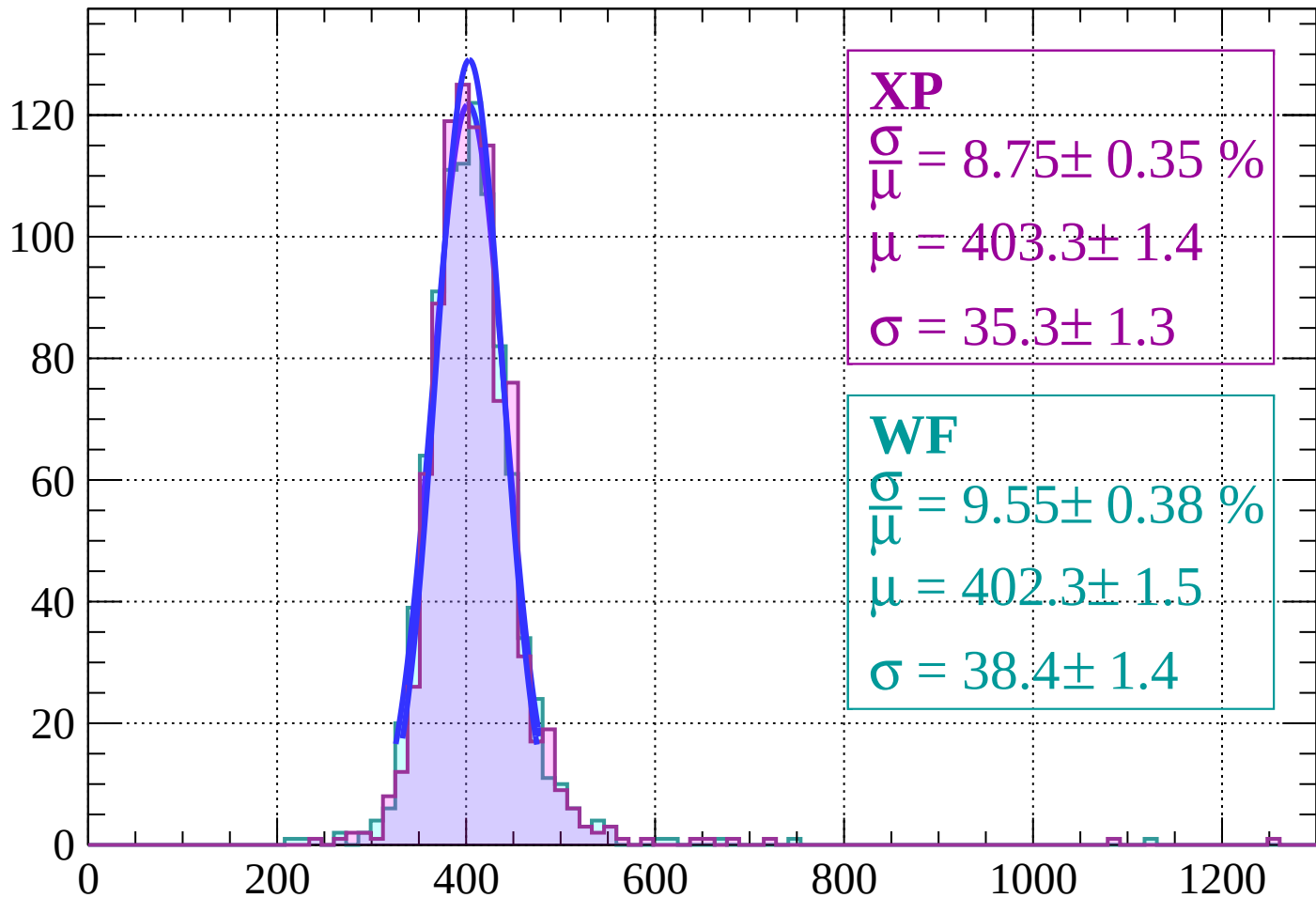
Energy loss | $-10 < \theta < -8$

Count



Energy loss | $-8 < \theta < -6$

Count



XP

$$\frac{\sigma}{\mu} = 8.75 \pm 0.35 \%$$

$$\mu = 403.3 \pm 1.4$$

$$\sigma = 35.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.55 \pm 0.38 \%$$

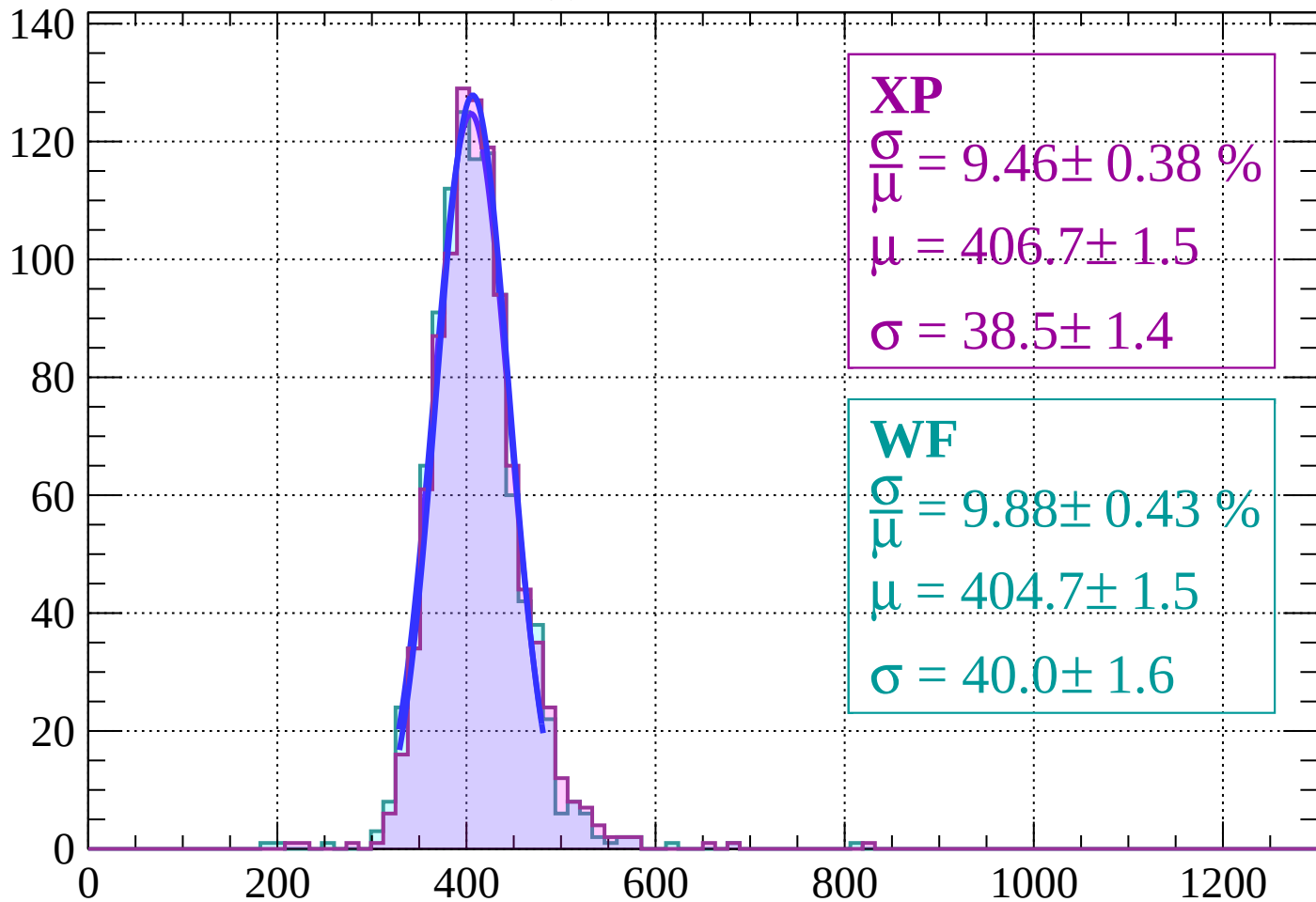
$$\mu = 402.3 \pm 1.5$$

$$\sigma = 38.4 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-6 < \theta < -4$

Count



XP

$$\frac{\sigma}{\mu} = 9.46 \pm 0.38 \%$$

$$\mu = 406.7 \pm 1.5$$

$$\sigma = 38.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.88 \pm 0.43 \%$$

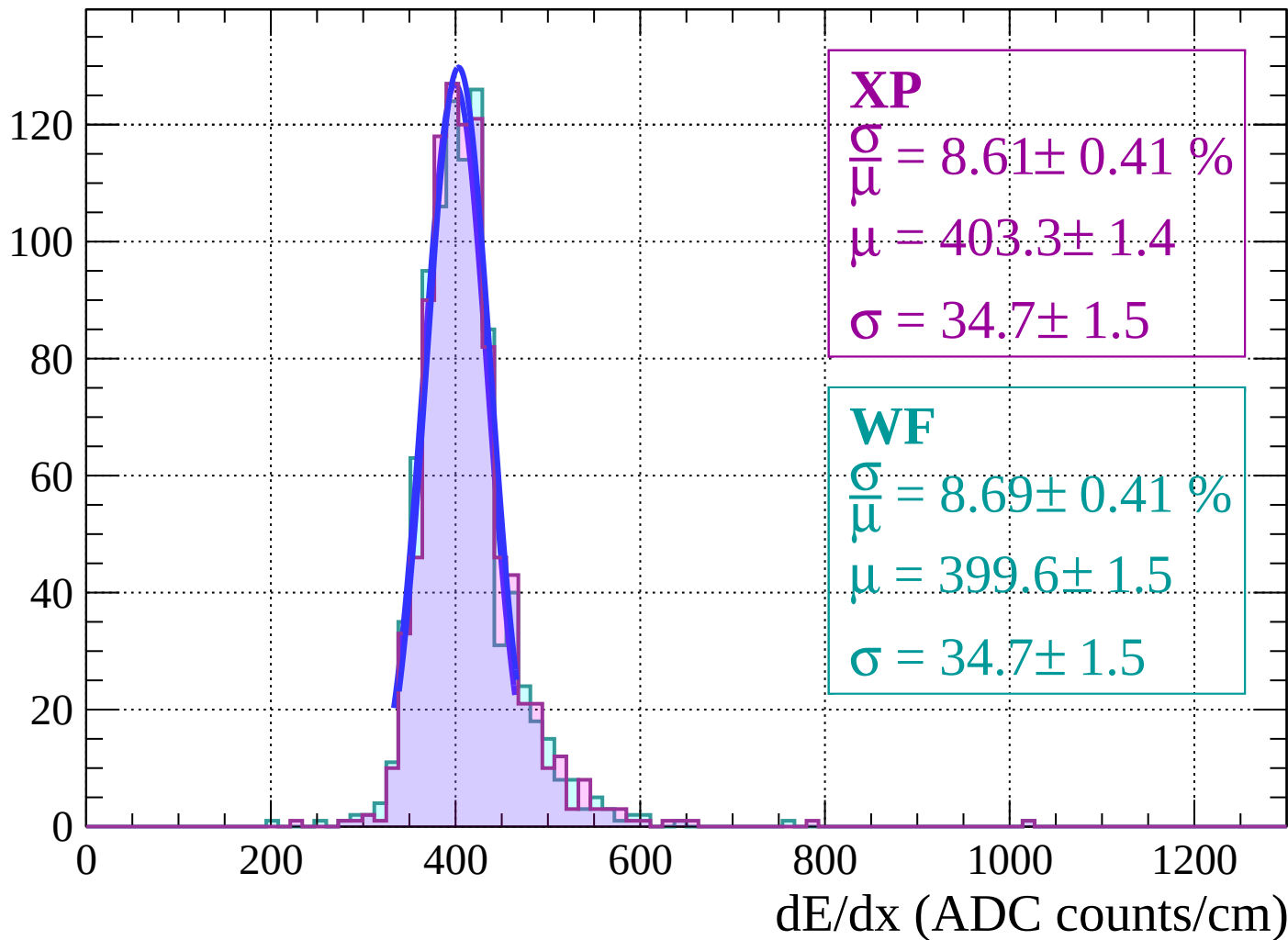
$$\mu = 404.7 \pm 1.5$$

$$\sigma = 40.0 \pm 1.6$$

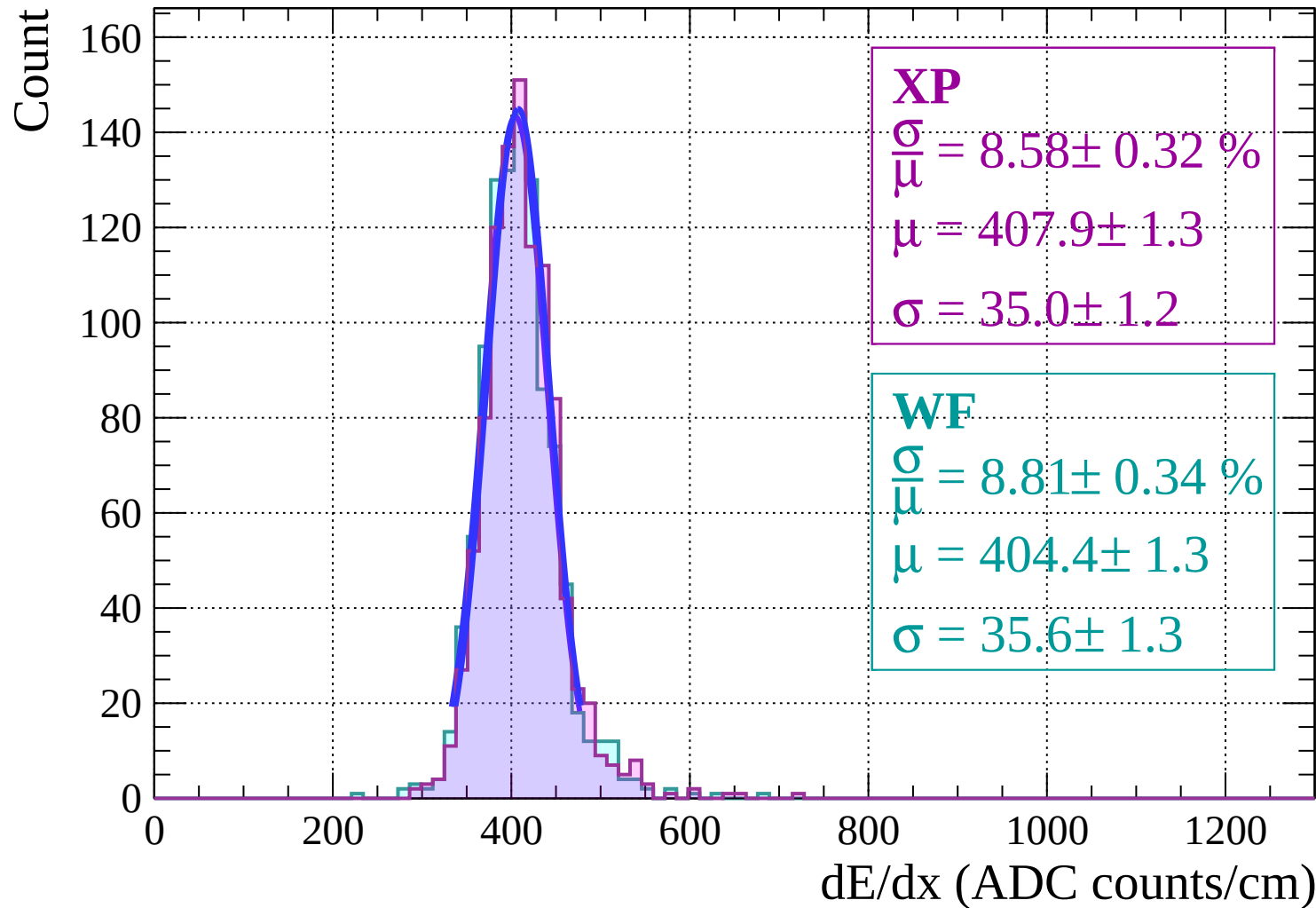
dE/dx (ADC counts/cm)

Energy loss | $-4 < \theta < -2$

Count

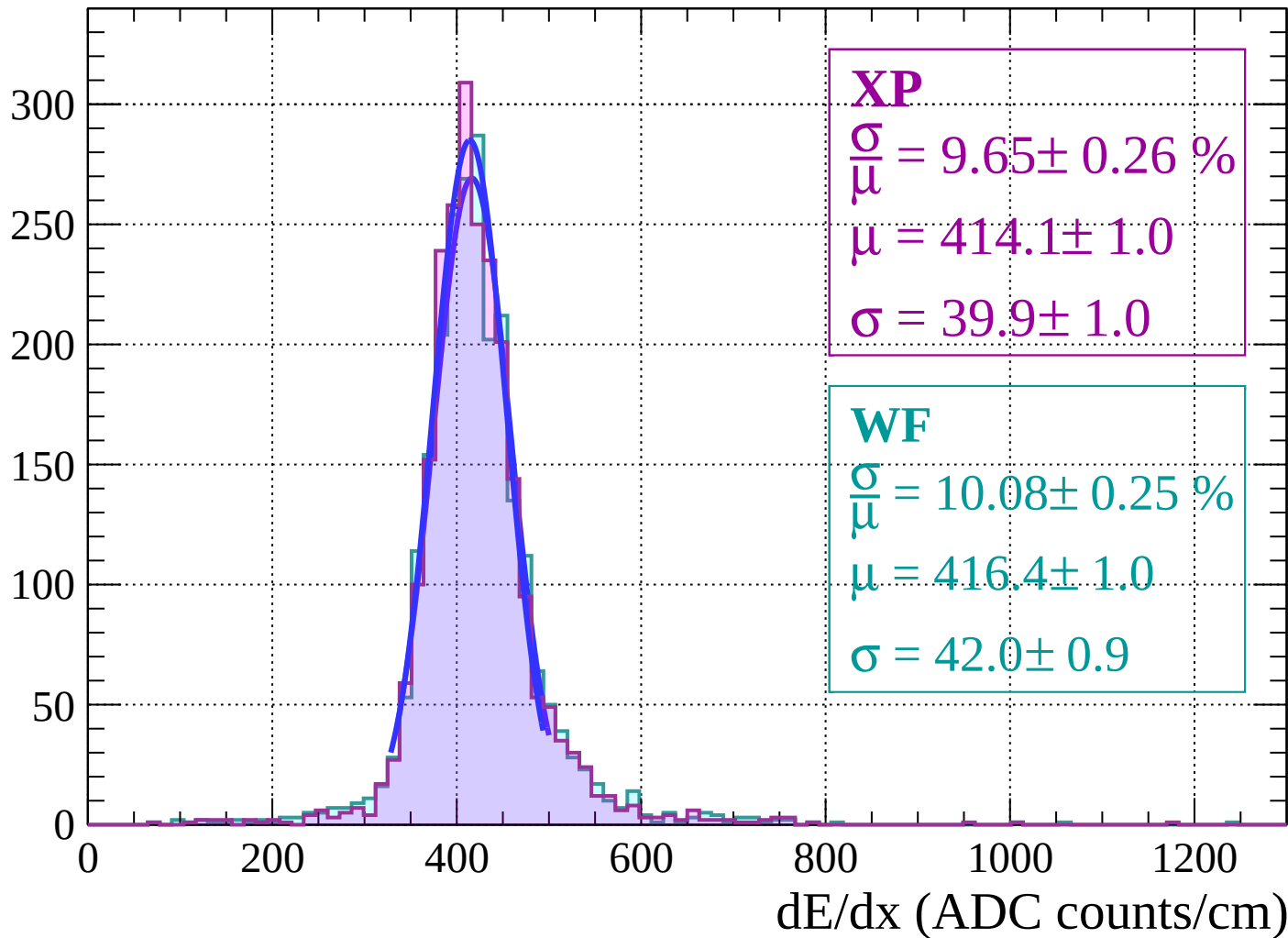


Energy loss | $-2 < \theta < 0$

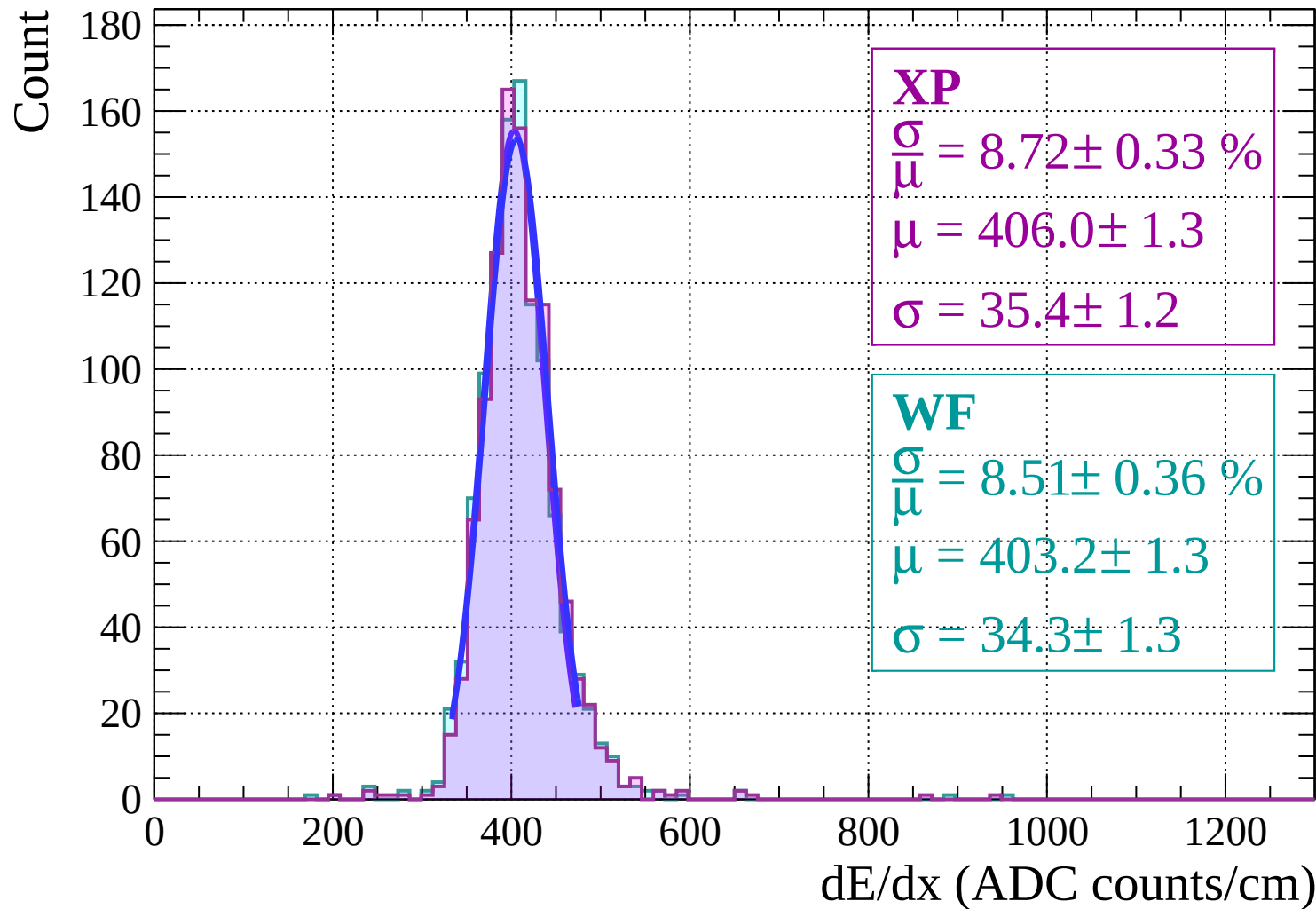


Energy loss | $0 < \theta < 2$

Count



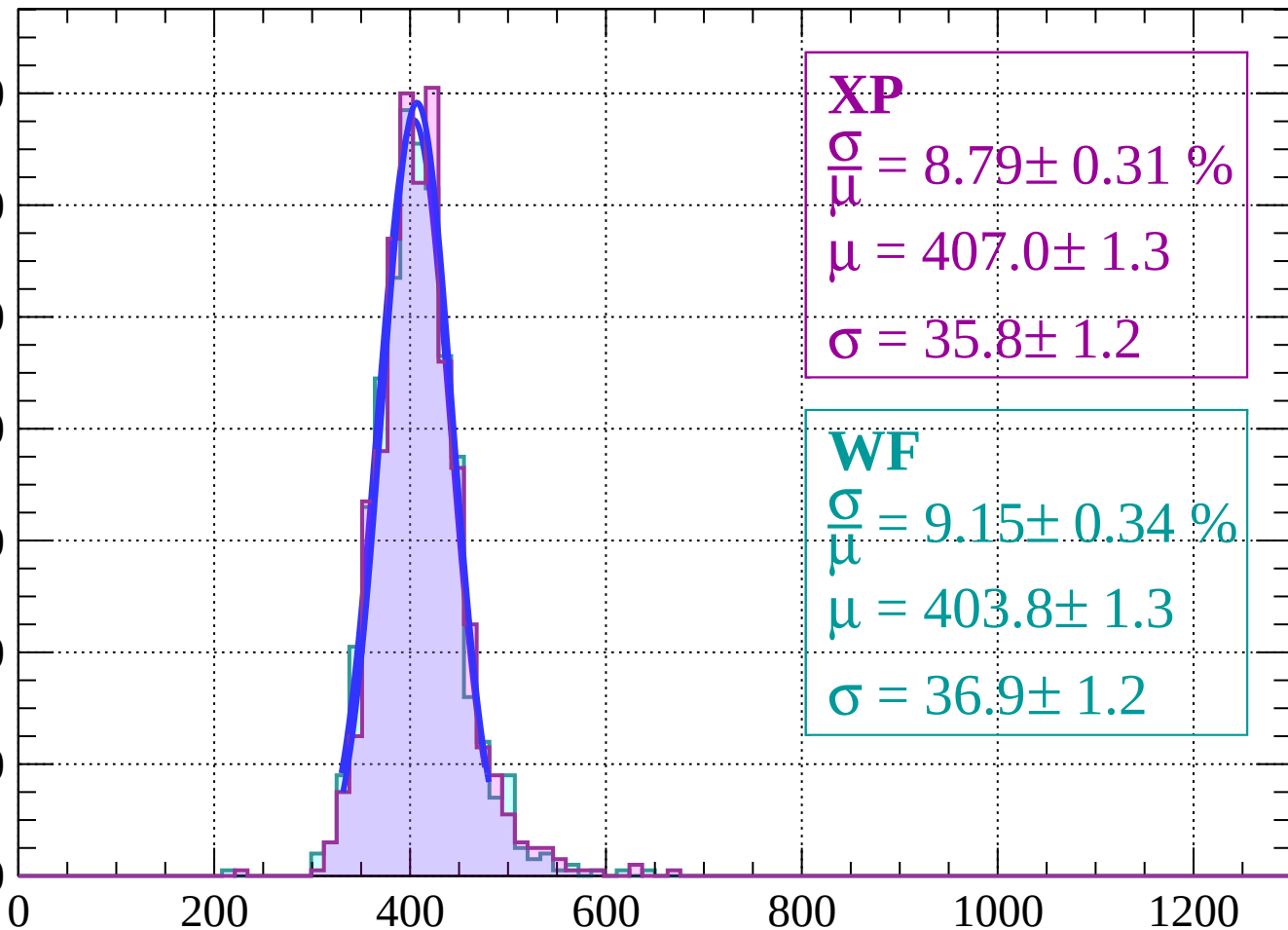
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.79 \pm 0.31 \%$$

$$\mu = 407.0 \pm 1.3$$

$$\sigma = 35.8 \pm 1.2$$

WF

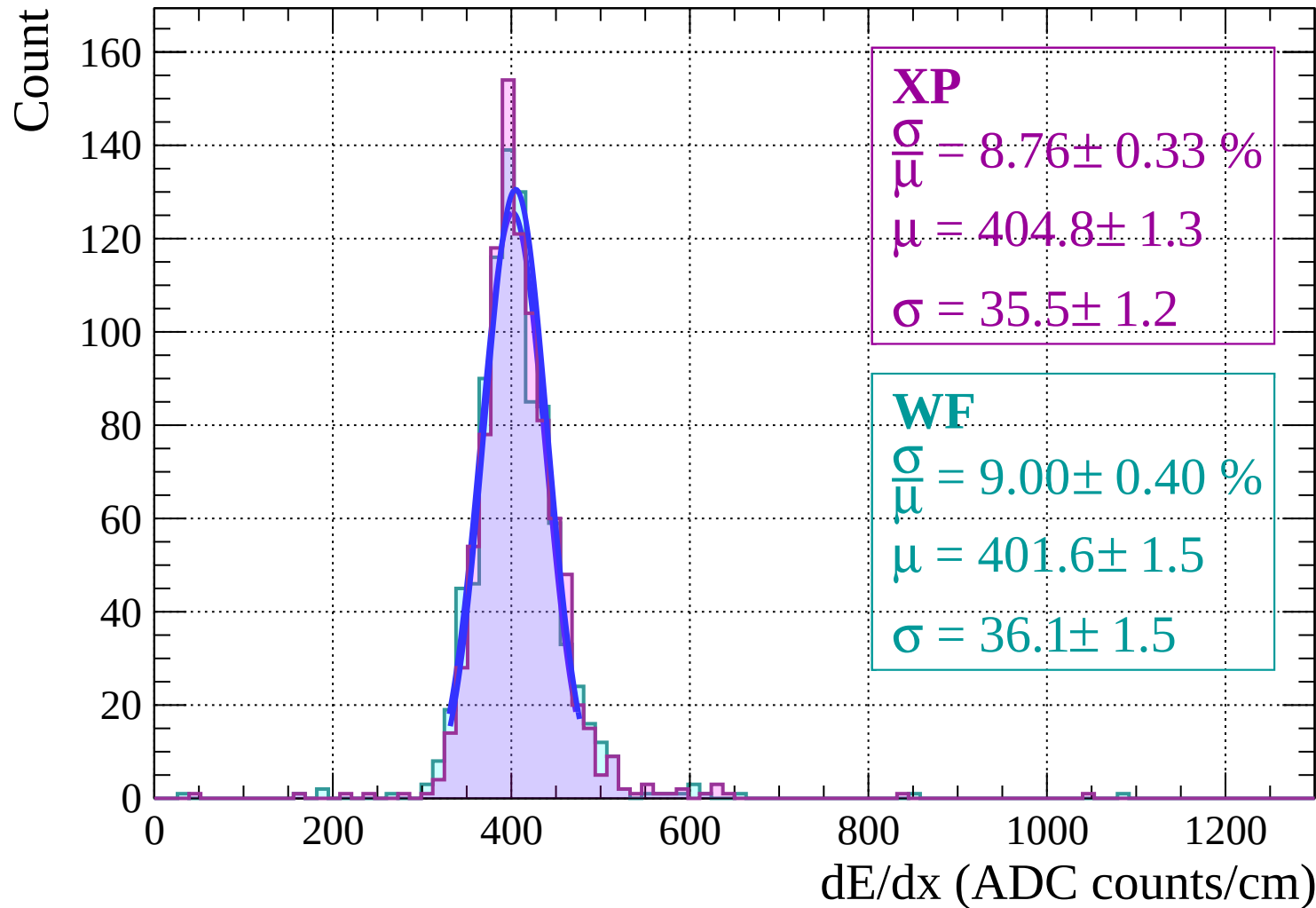
$$\frac{\sigma}{\mu} = 9.15 \pm 0.34 \%$$

$$\mu = 403.8 \pm 1.3$$

$$\sigma = 36.9 \pm 1.2$$

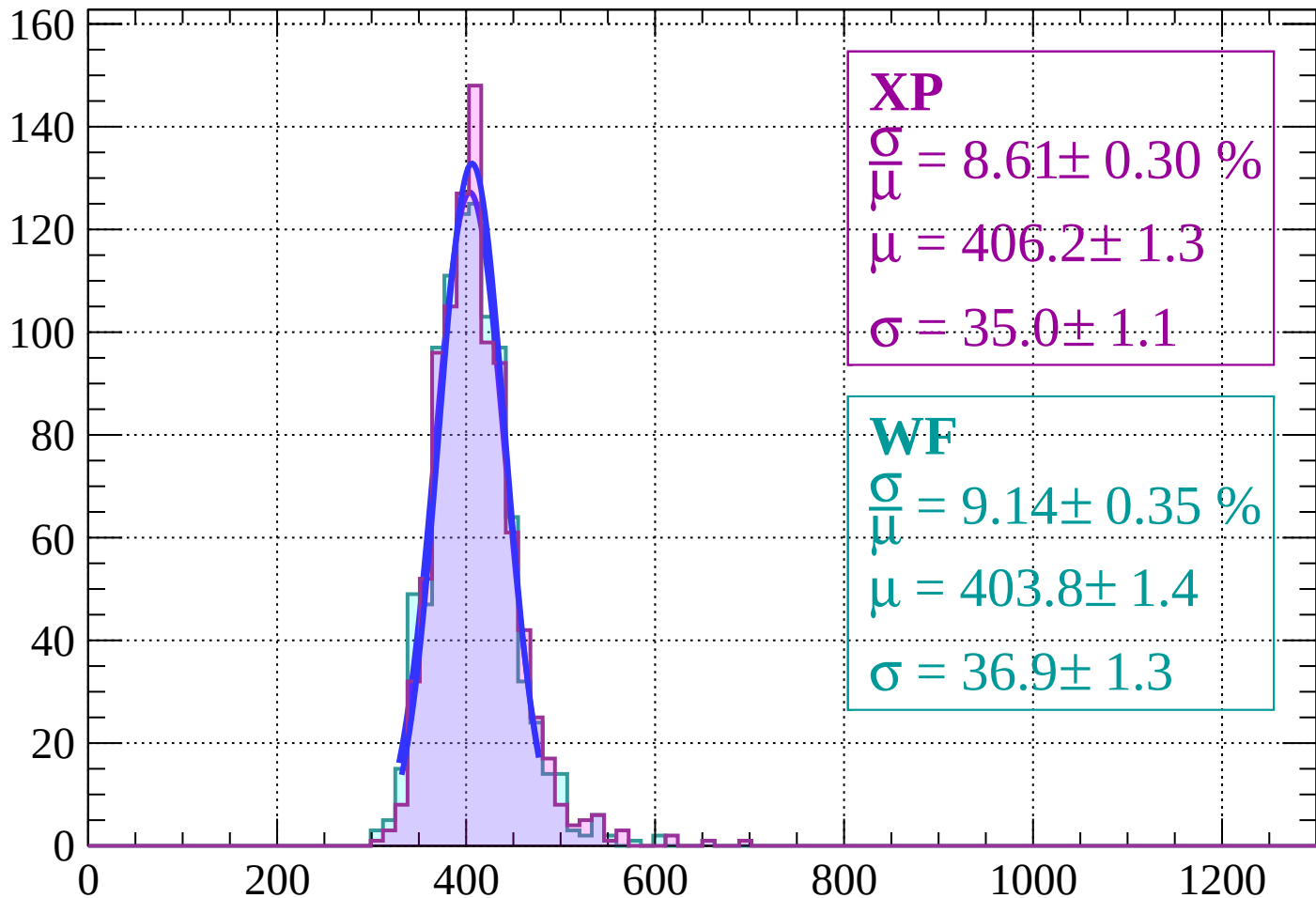
dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 8$



Energy loss | $8 < \theta < 10$

Count



XP

$$\frac{\sigma}{\mu} = 8.61 \pm 0.30 \%$$

$$\mu = 406.2 \pm 1.3$$

$$\sigma = 35.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.14 \pm 0.35 \%$$

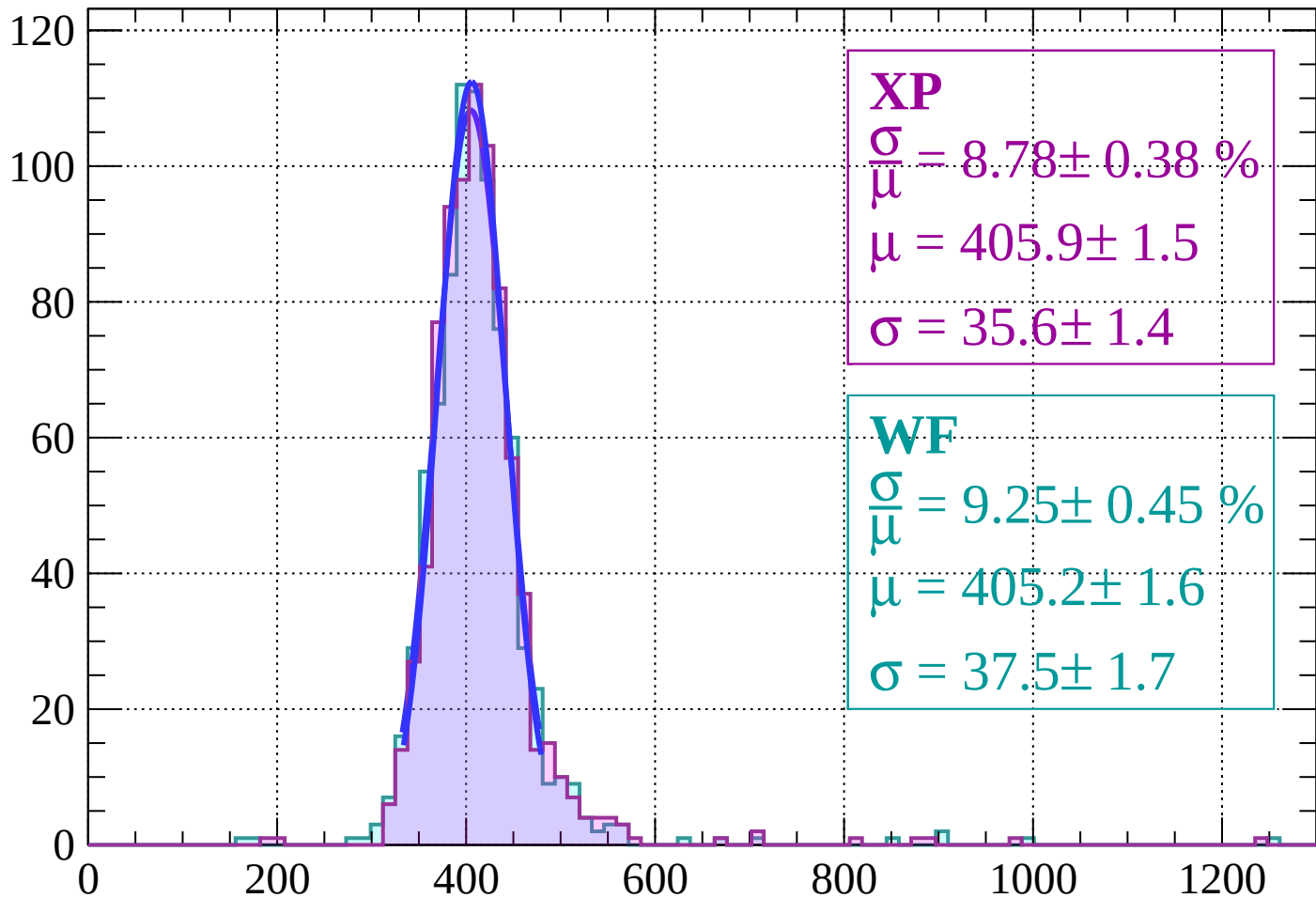
$$\mu = 403.8 \pm 1.4$$

$$\sigma = 36.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $10 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 8.78 \pm 0.38 \%$$

$$\mu = 405.9 \pm 1.5$$

$$\sigma = 35.6 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.25 \pm 0.45 \%$$

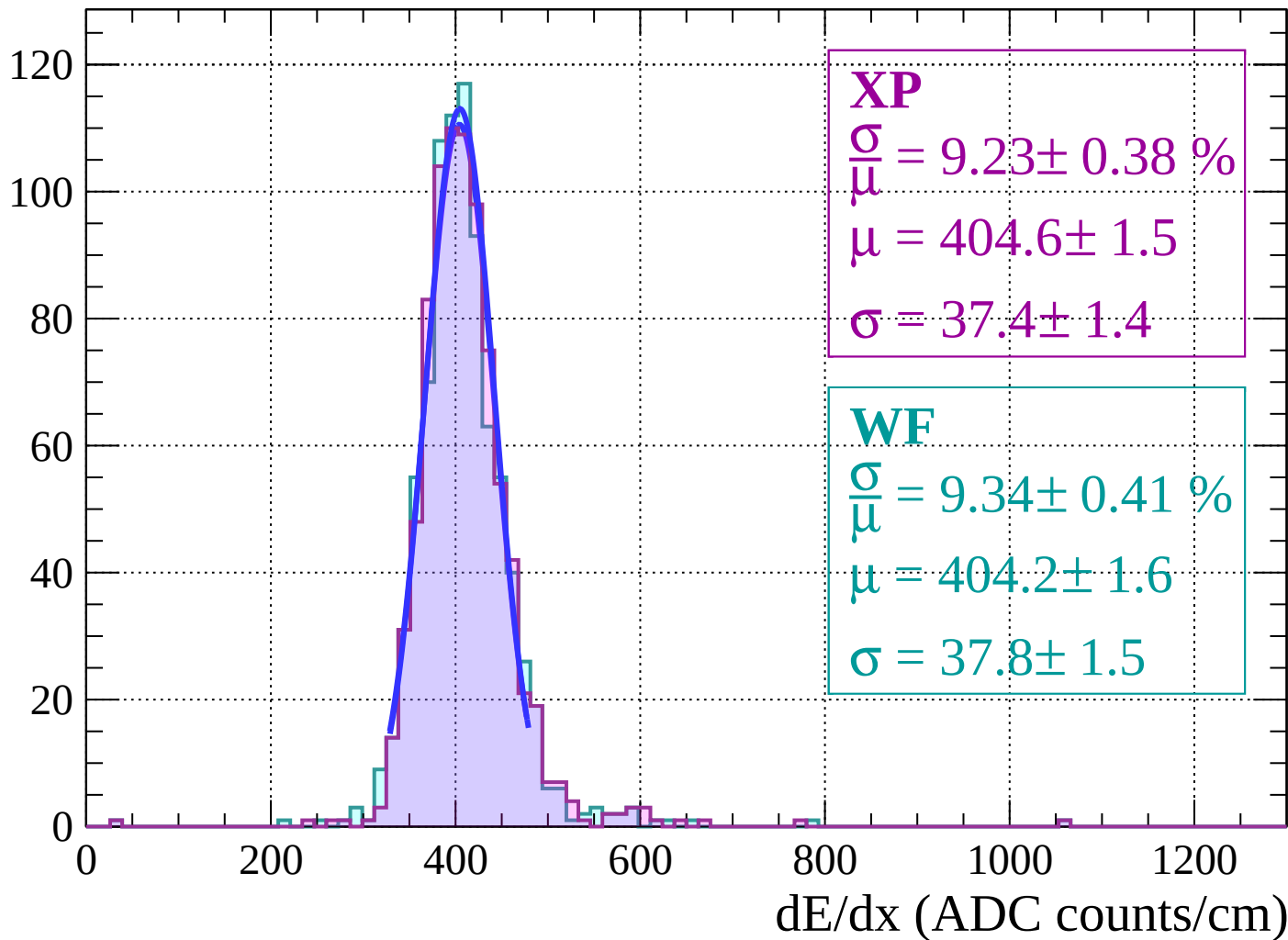
$$\mu = 405.2 \pm 1.6$$

$$\sigma = 37.5 \pm 1.7$$

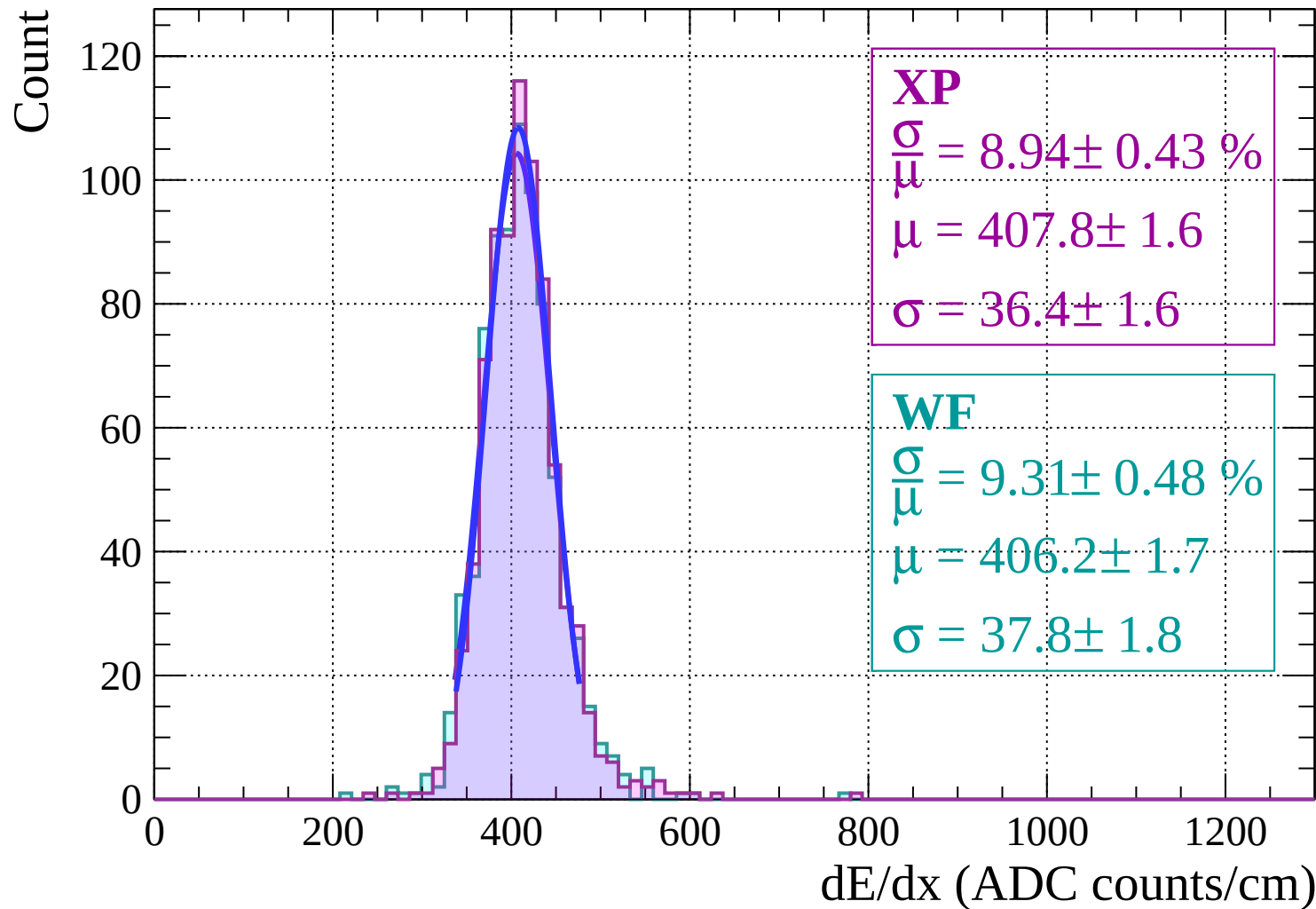
dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$

Count

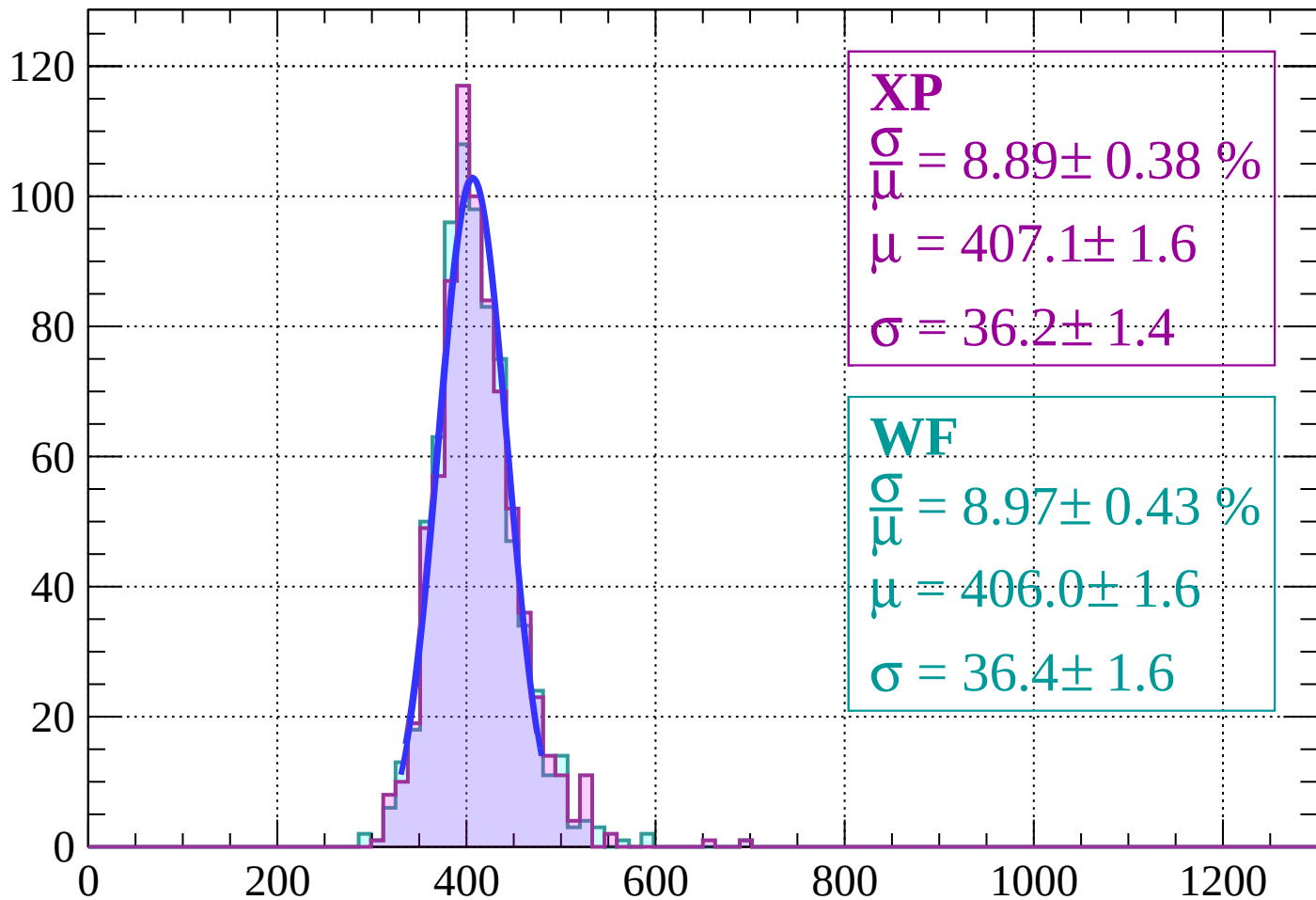


Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.38 \%$$

$$\mu = 407.1 \pm 1.6$$

$$\sigma = 36.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.97 \pm 0.43 \%$$

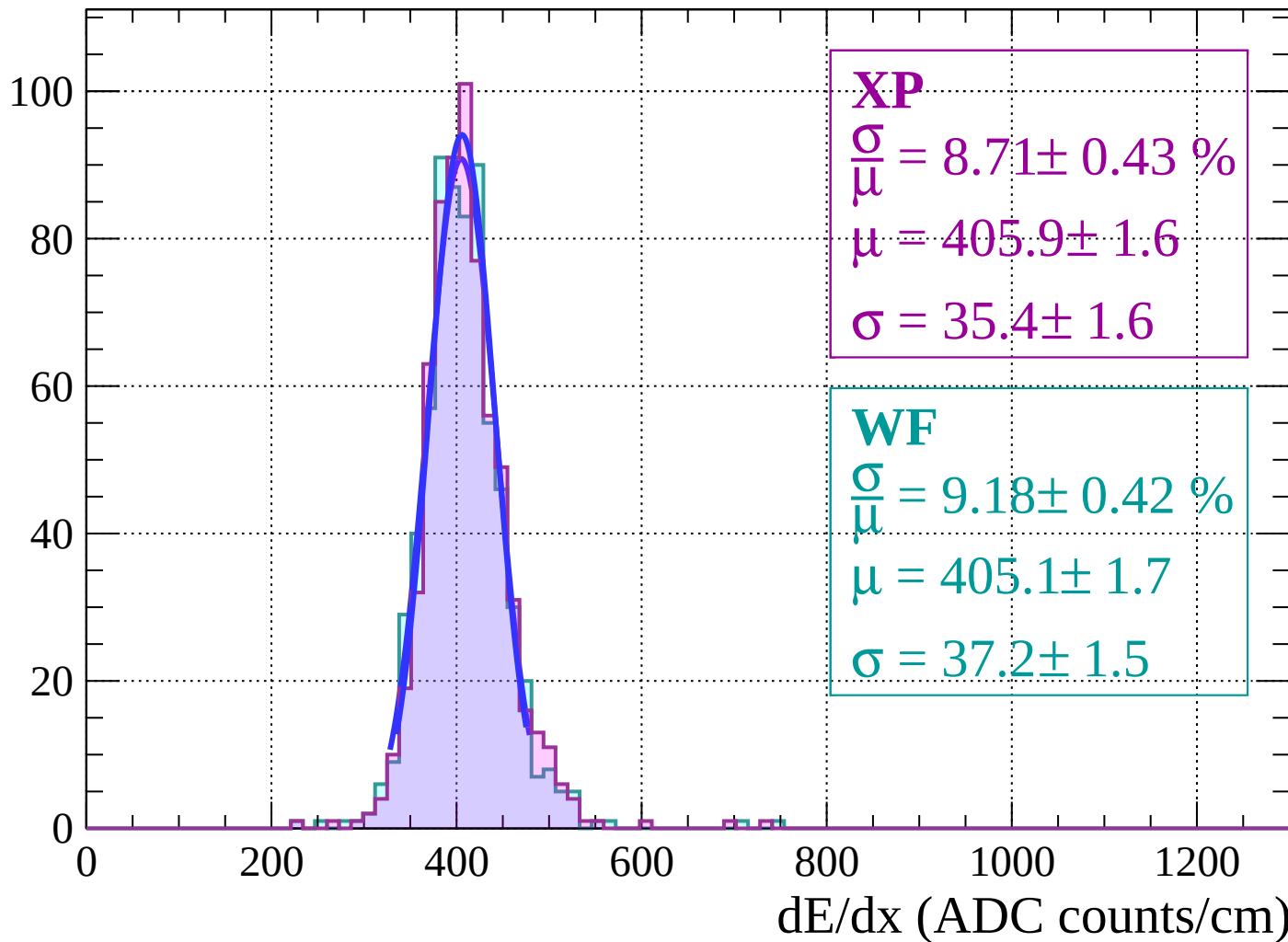
$$\mu = 406.0 \pm 1.6$$

$$\sigma = 36.4 \pm 1.6$$

dE/dx (ADC counts/cm)

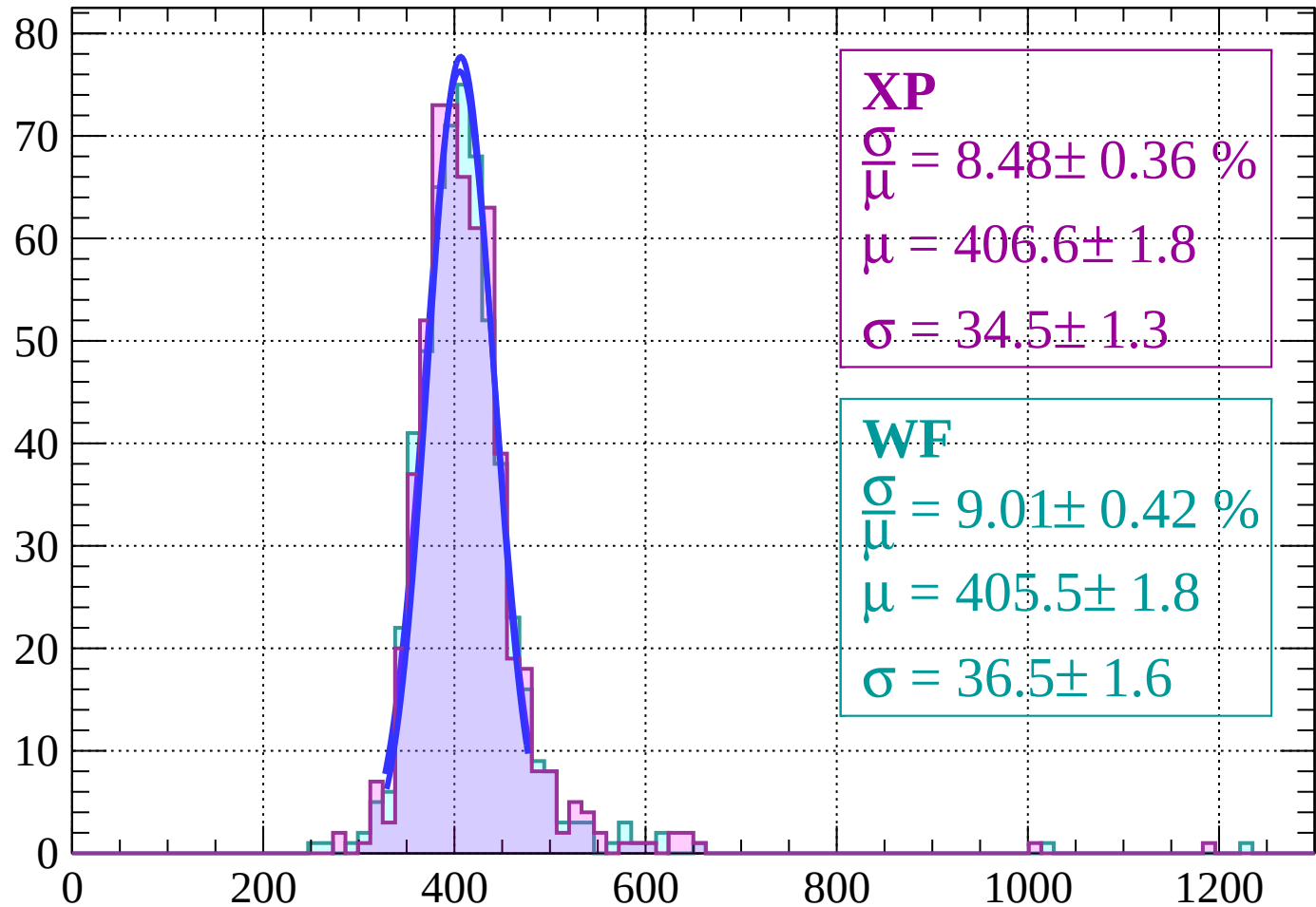
Energy loss | $18 < \theta < 20$

Count



Energy loss | $20 < \theta < 22$

Count



XP

$$\frac{\sigma}{\mu} = 8.48 \pm 0.36 \%$$

$$\mu = 406.6 \pm 1.8$$

$$\sigma = 34.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.01 \pm 0.42 \%$$

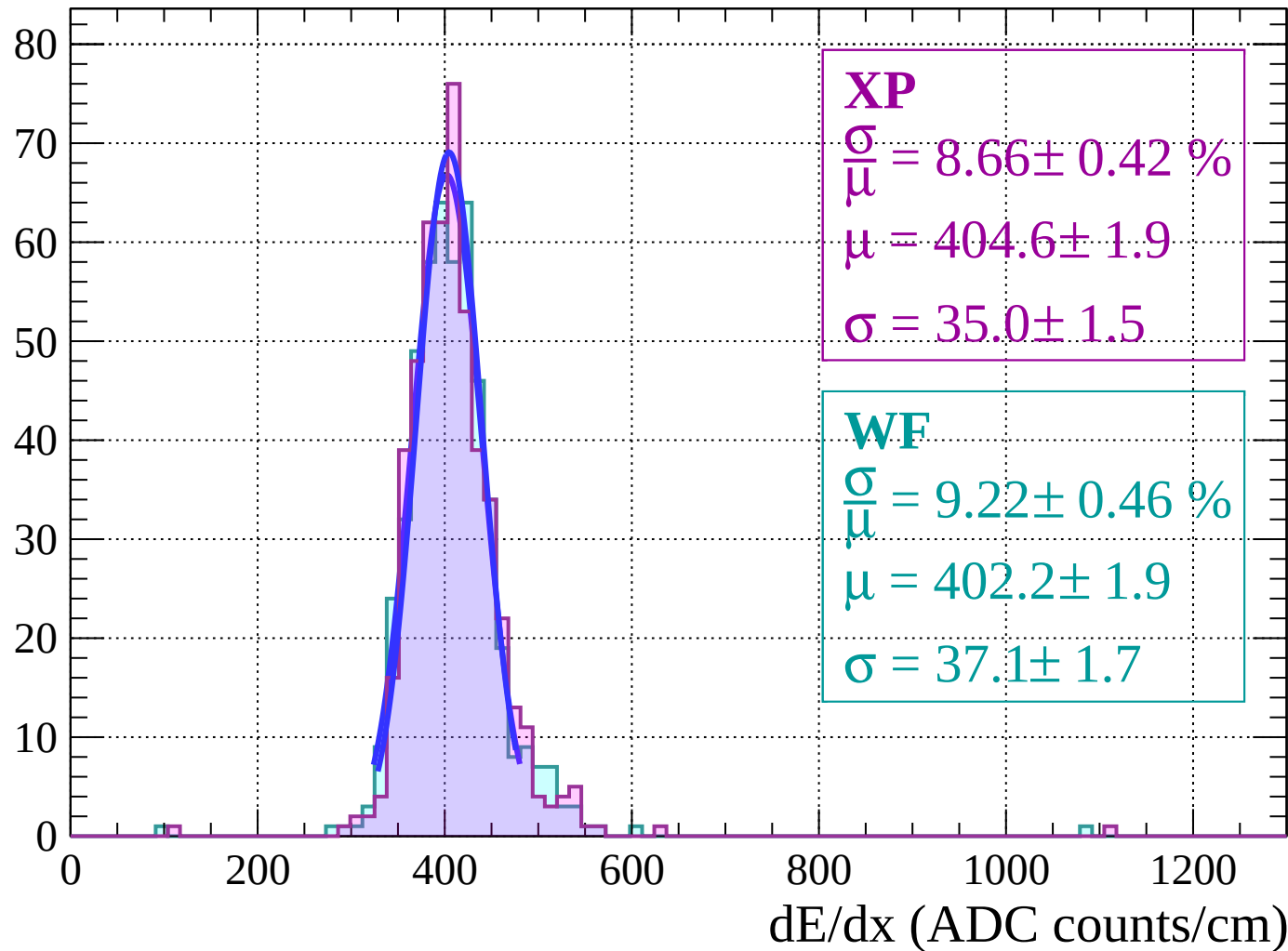
$$\mu = 405.5 \pm 1.8$$

$$\sigma = 36.5 \pm 1.6$$

dE/dx (ADC counts/cm)

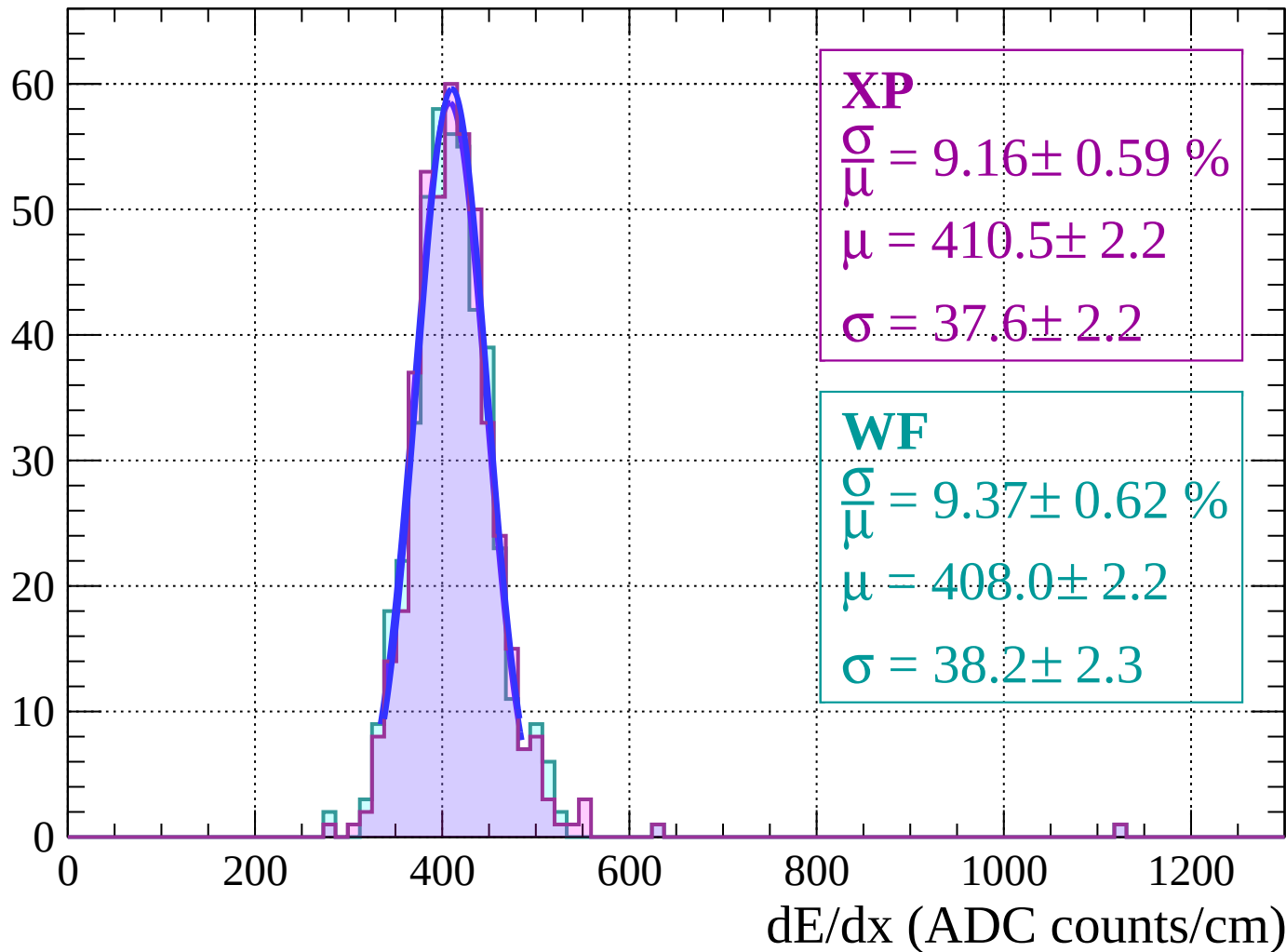
Energy loss | $22 < \theta < 24$

Count



Energy loss | $24 < \theta < 26$

Count



Energy loss | $26 < \theta < 28$

Count

XP

$$\frac{\sigma}{\mu} = 9.38 \pm 0.60 \%$$

$$\mu = 404.3 \pm 2.4$$

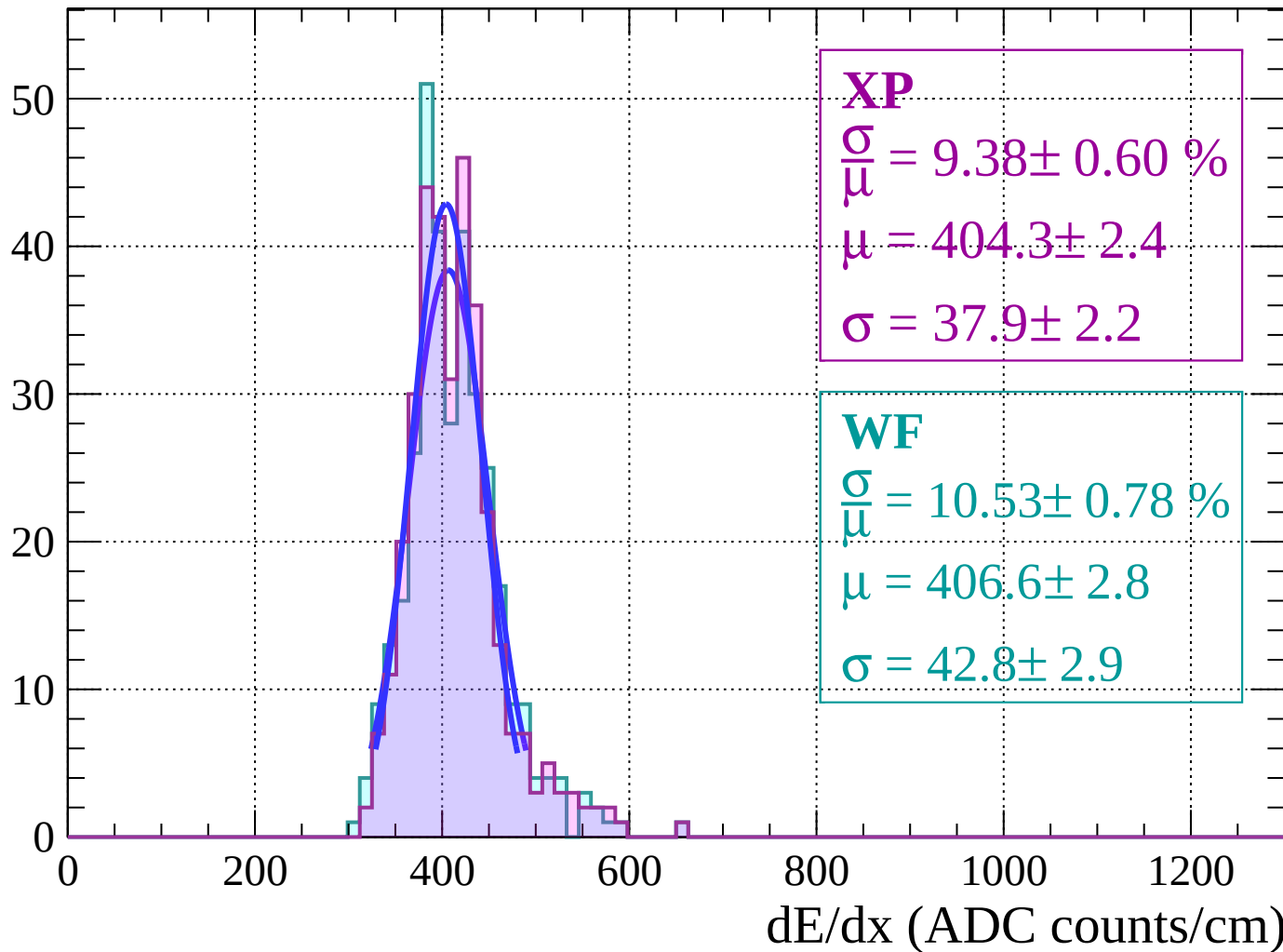
$$\sigma = 37.9 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 10.53 \pm 0.78 \%$$

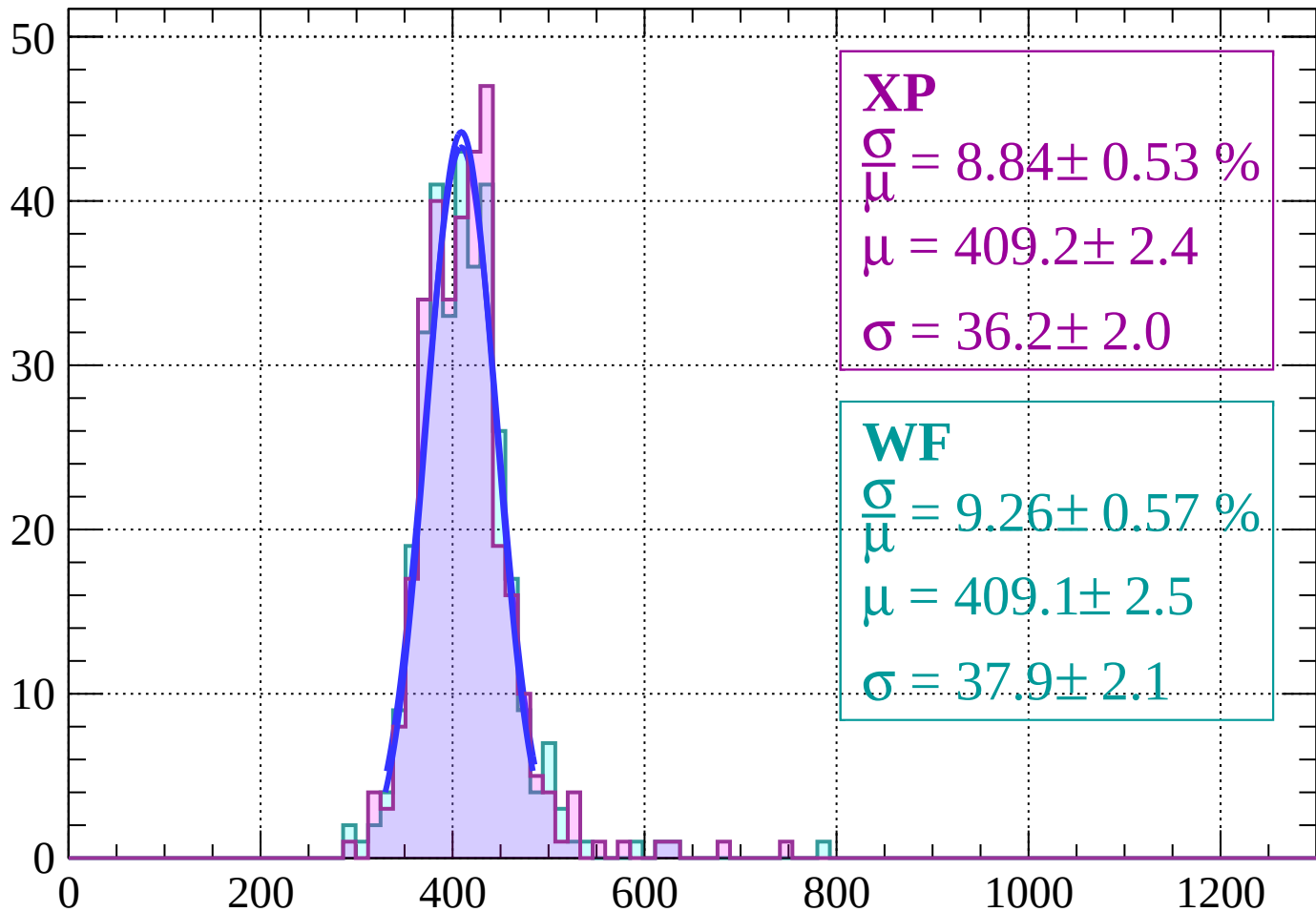
$$\mu = 406.6 \pm 2.8$$

$$\sigma = 42.8 \pm 2.9$$



Energy loss | $28 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 8.84 \pm 0.53 \%$$

$$\mu = 409.2 \pm 2.4$$

$$\sigma = 36.2 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 9.26 \pm 0.57 \%$$

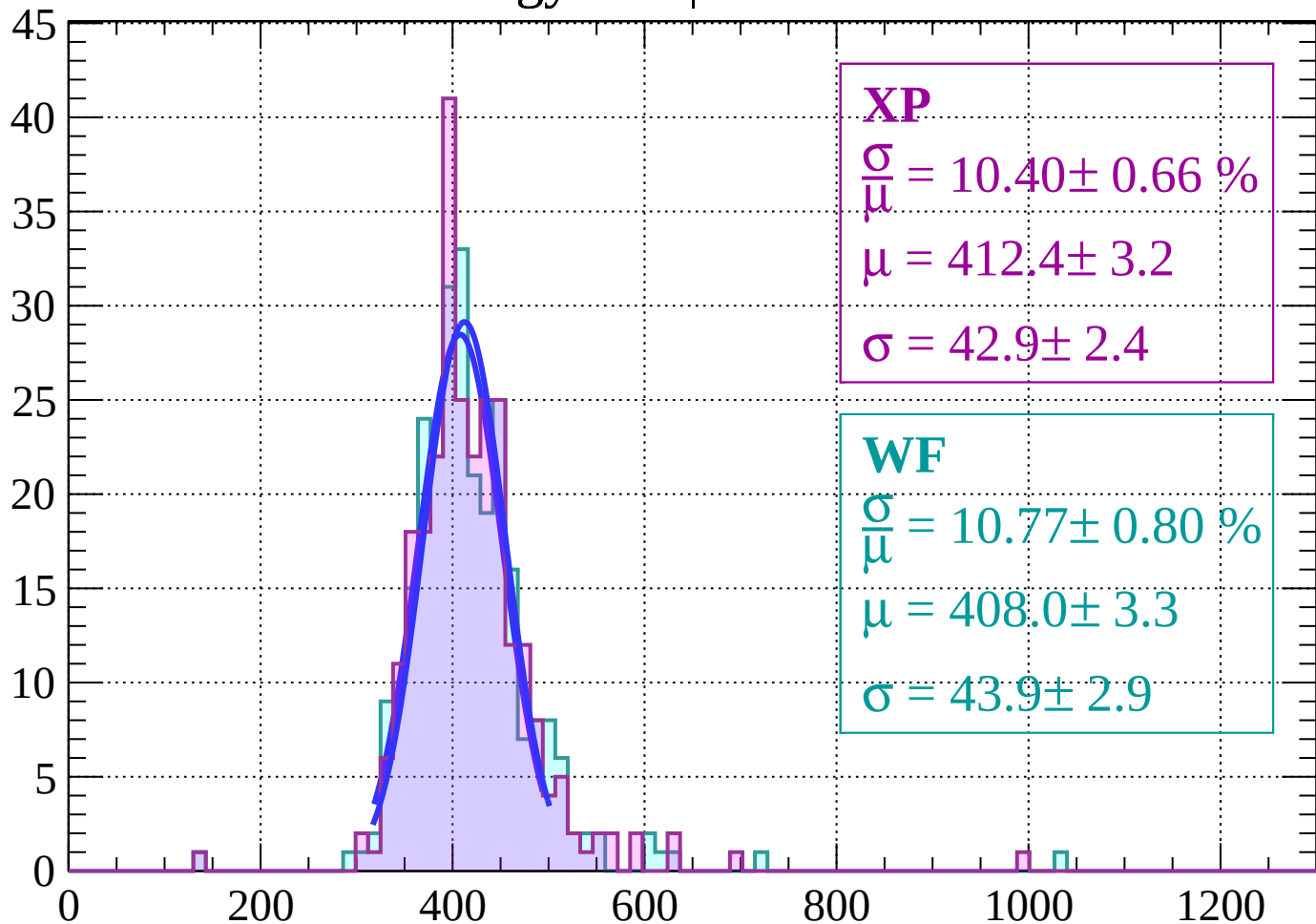
$$\mu = 409.1 \pm 2.5$$

$$\sigma = 37.9 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $30 < \theta < 32$

Count



XP

$$\frac{\sigma}{\mu} = 10.40 \pm 0.66 \%$$

$$\mu = 412.4 \pm 3.2$$

$$\sigma = 42.9 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 10.77 \pm 0.80 \%$$

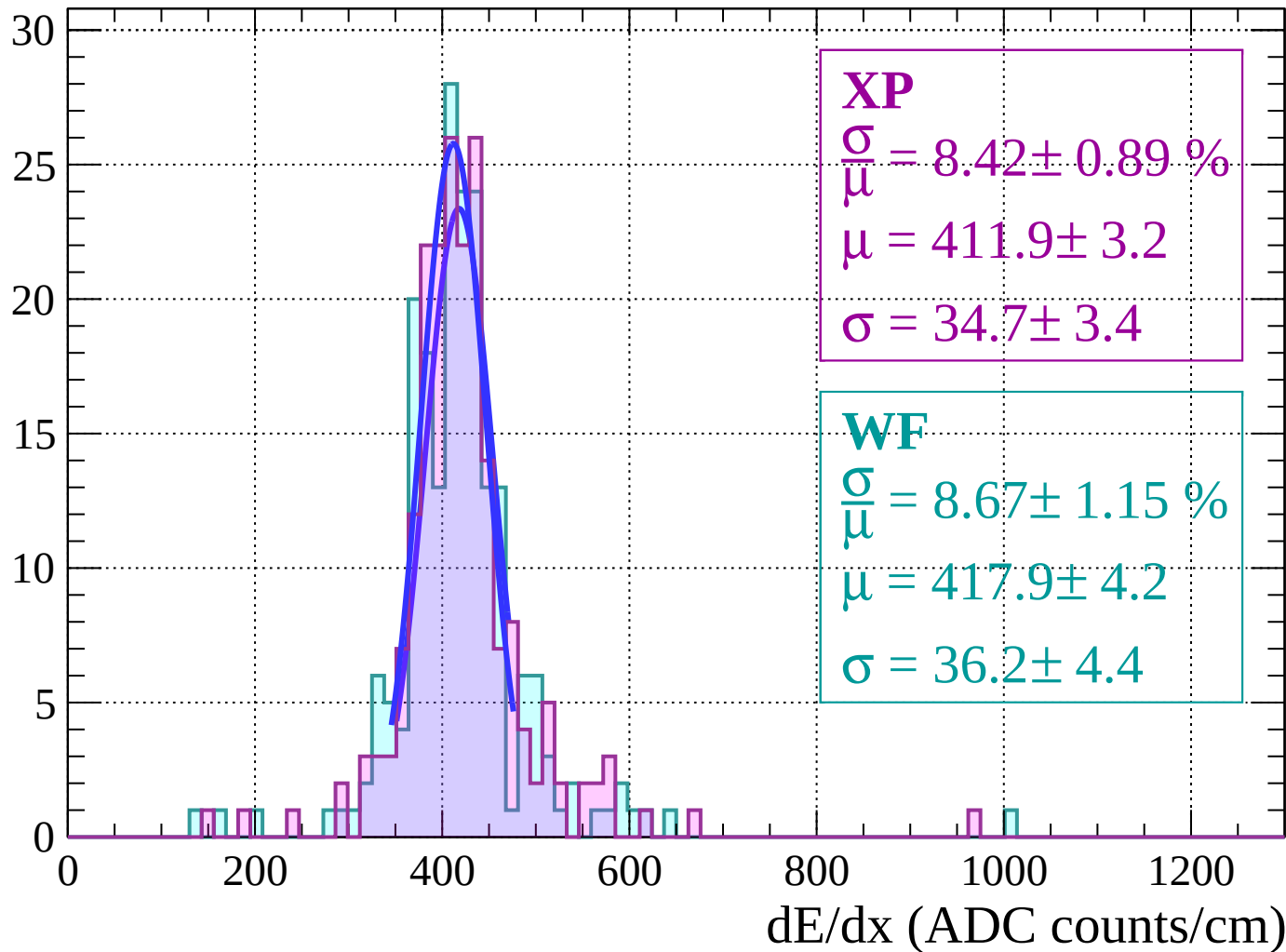
$$\mu = 408.0 \pm 3.3$$

$$\sigma = 43.9 \pm 2.9$$

dE/dx (ADC counts/cm)

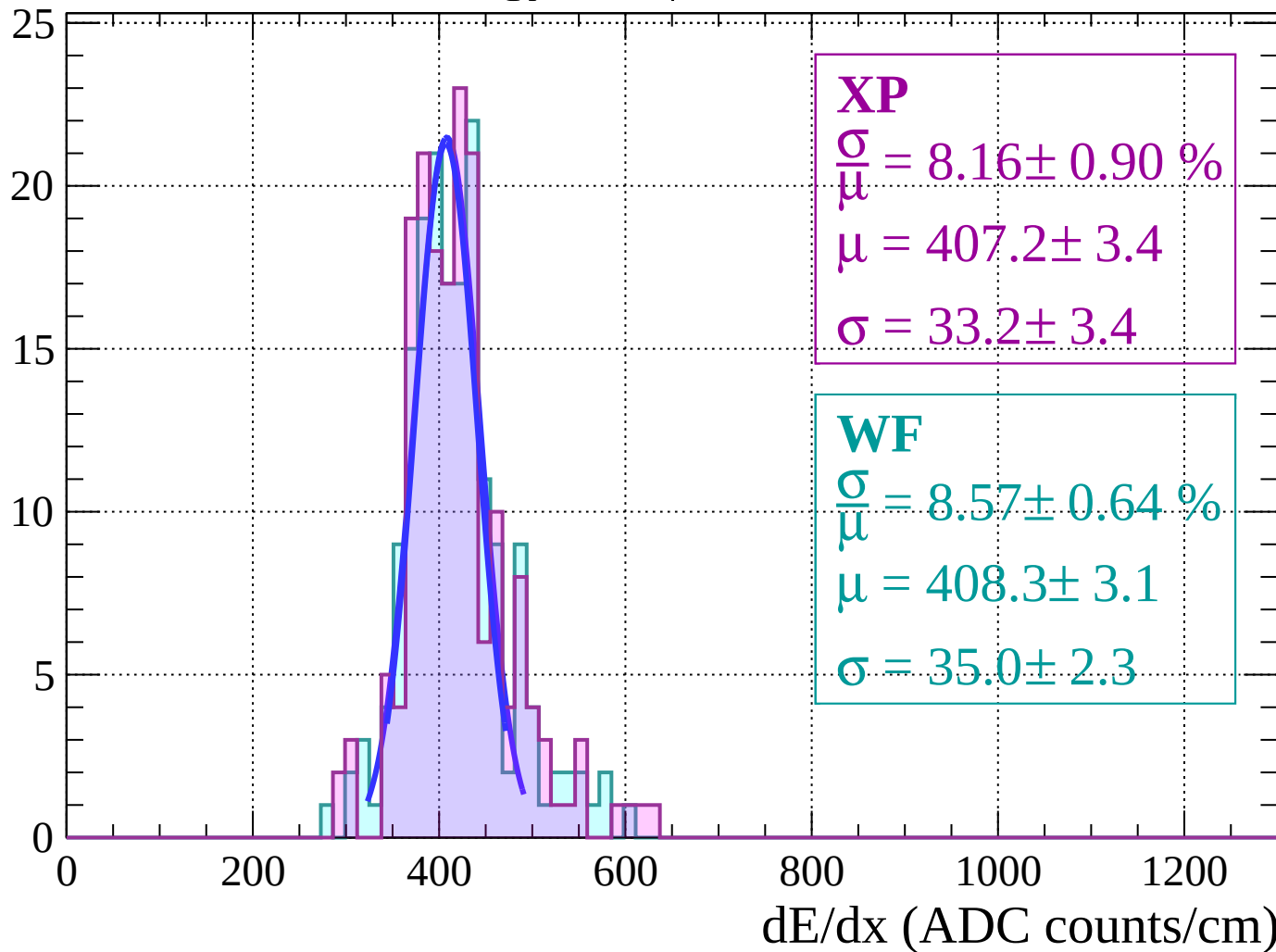
Energy loss | $32 < \theta < 34$

Count



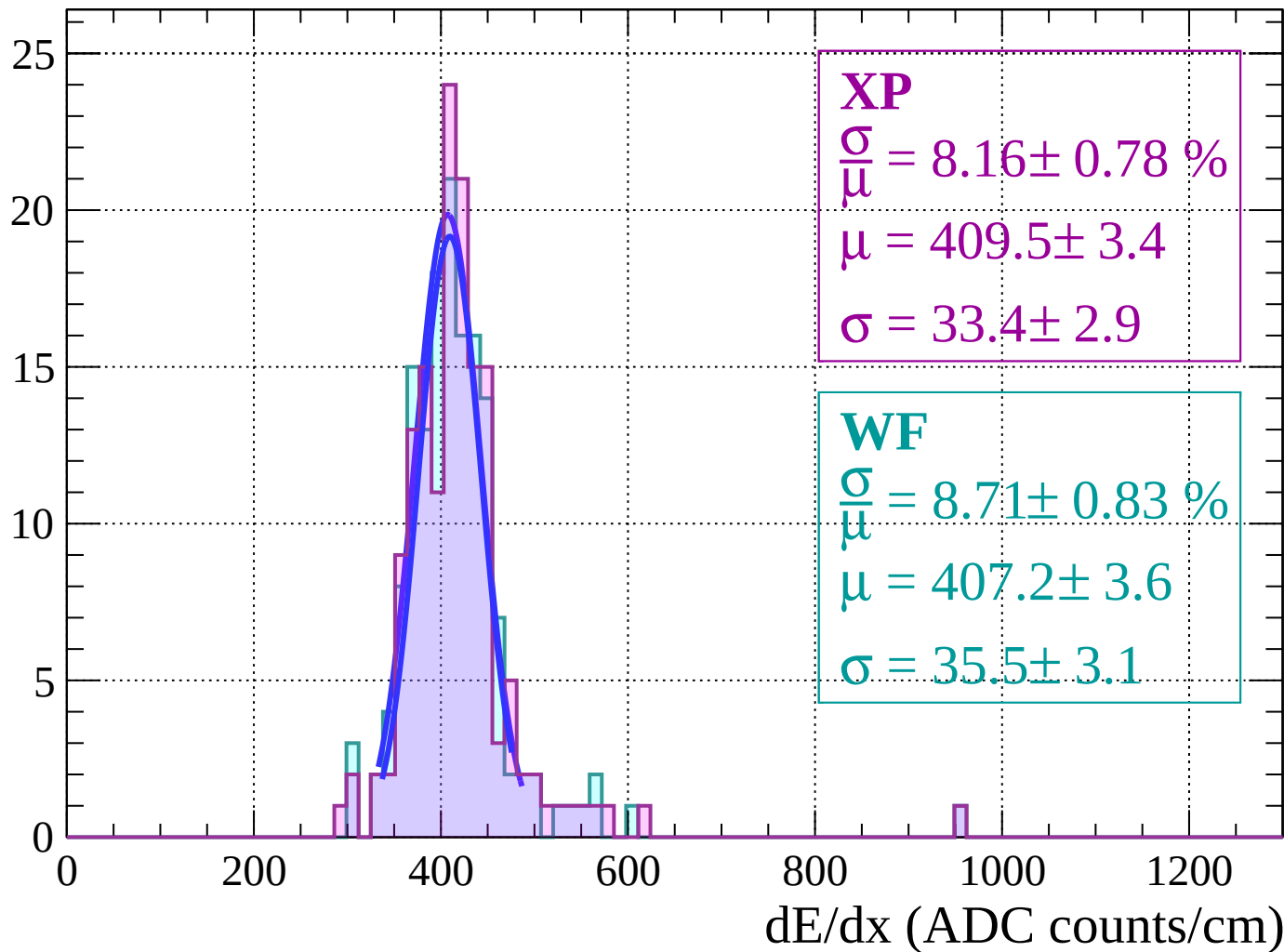
Energy loss | $34 < \theta < 36$

Count



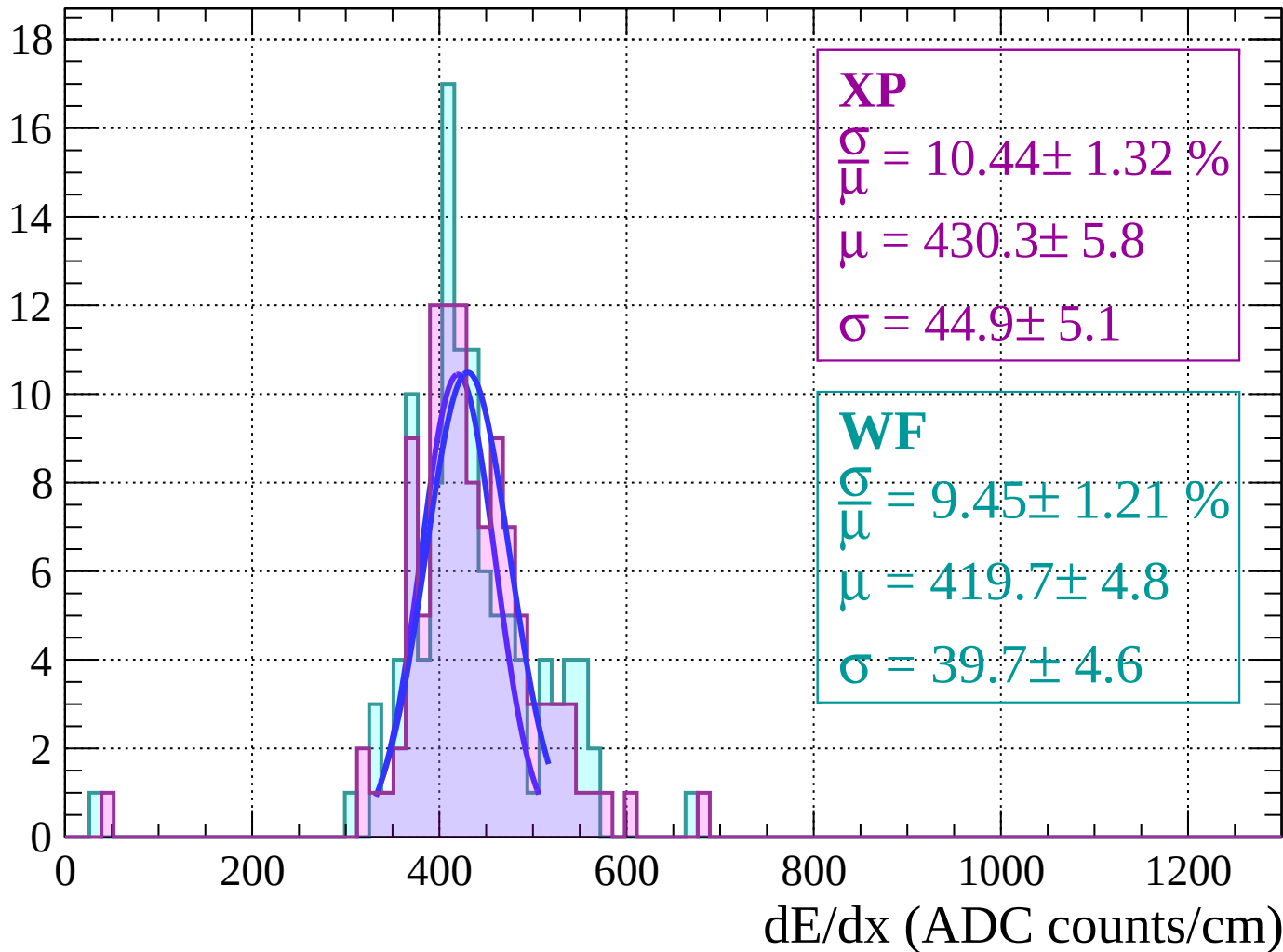
Energy loss | $36 < \theta < 38$

Count



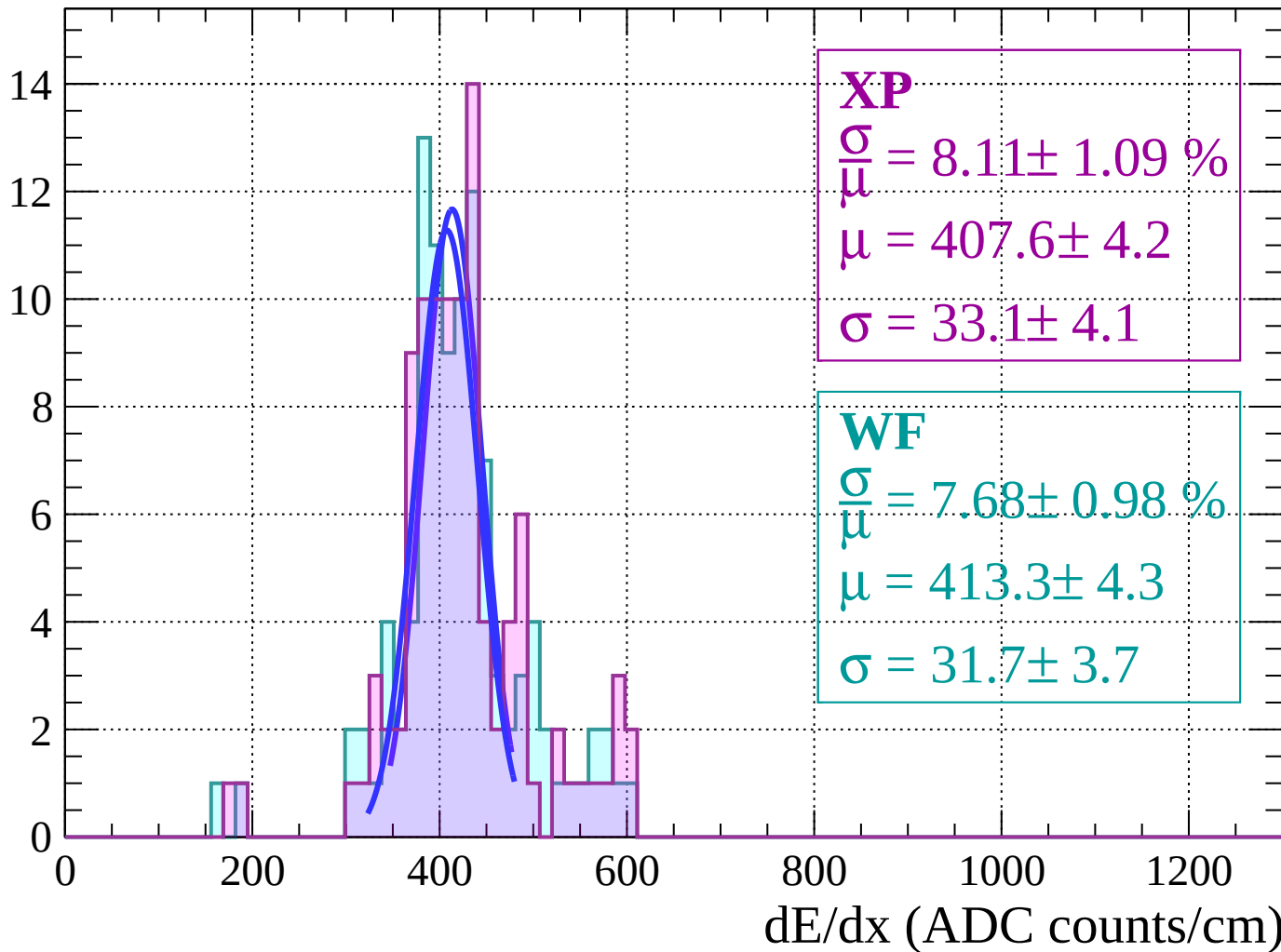
Energy loss | $38 < \theta < 40$

Count



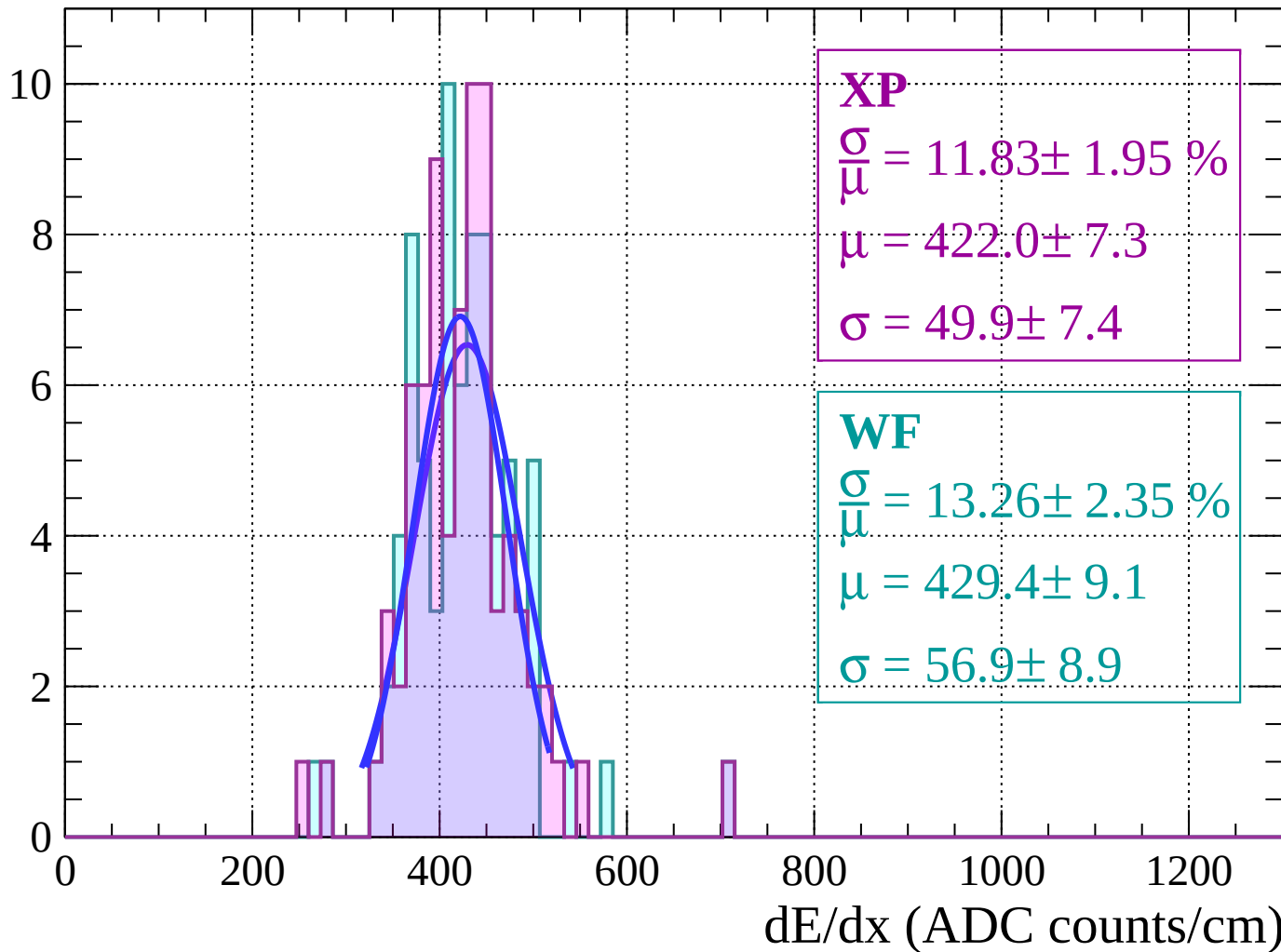
Energy loss | $40 < \theta < 42$

Count



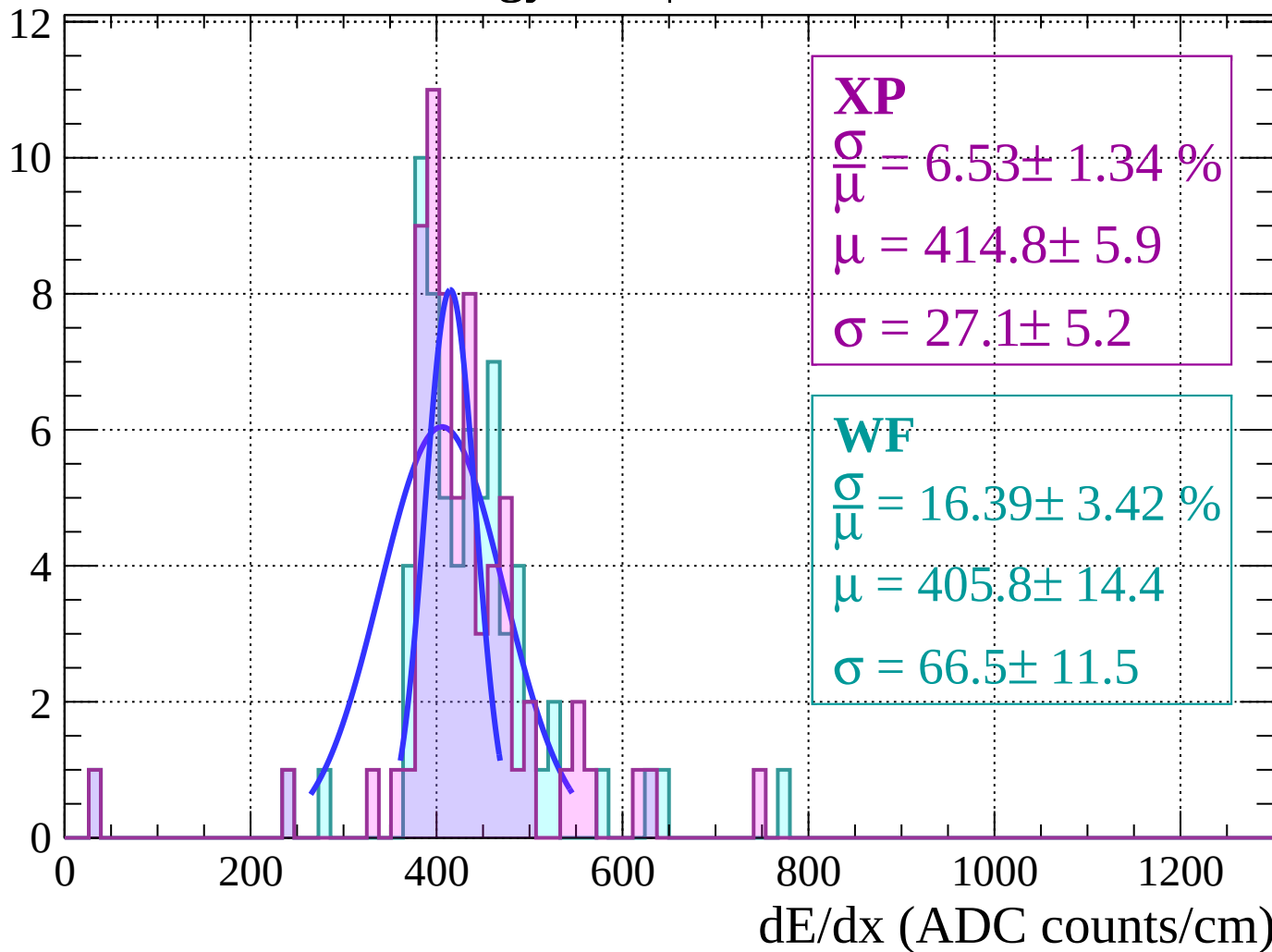
Energy loss | $42 < \theta < 44$

Count



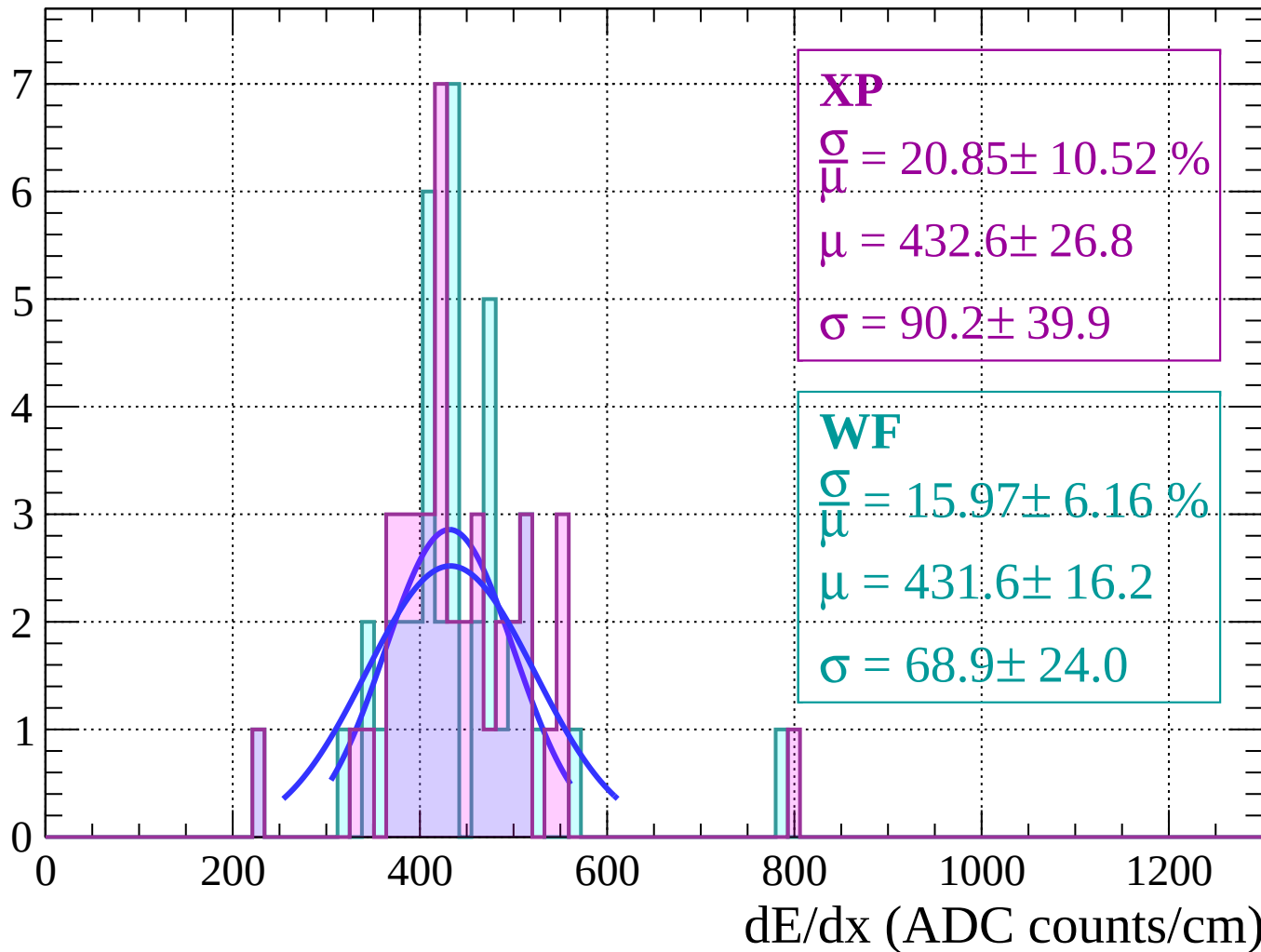
Energy loss | $44 < \theta < 46$

Count



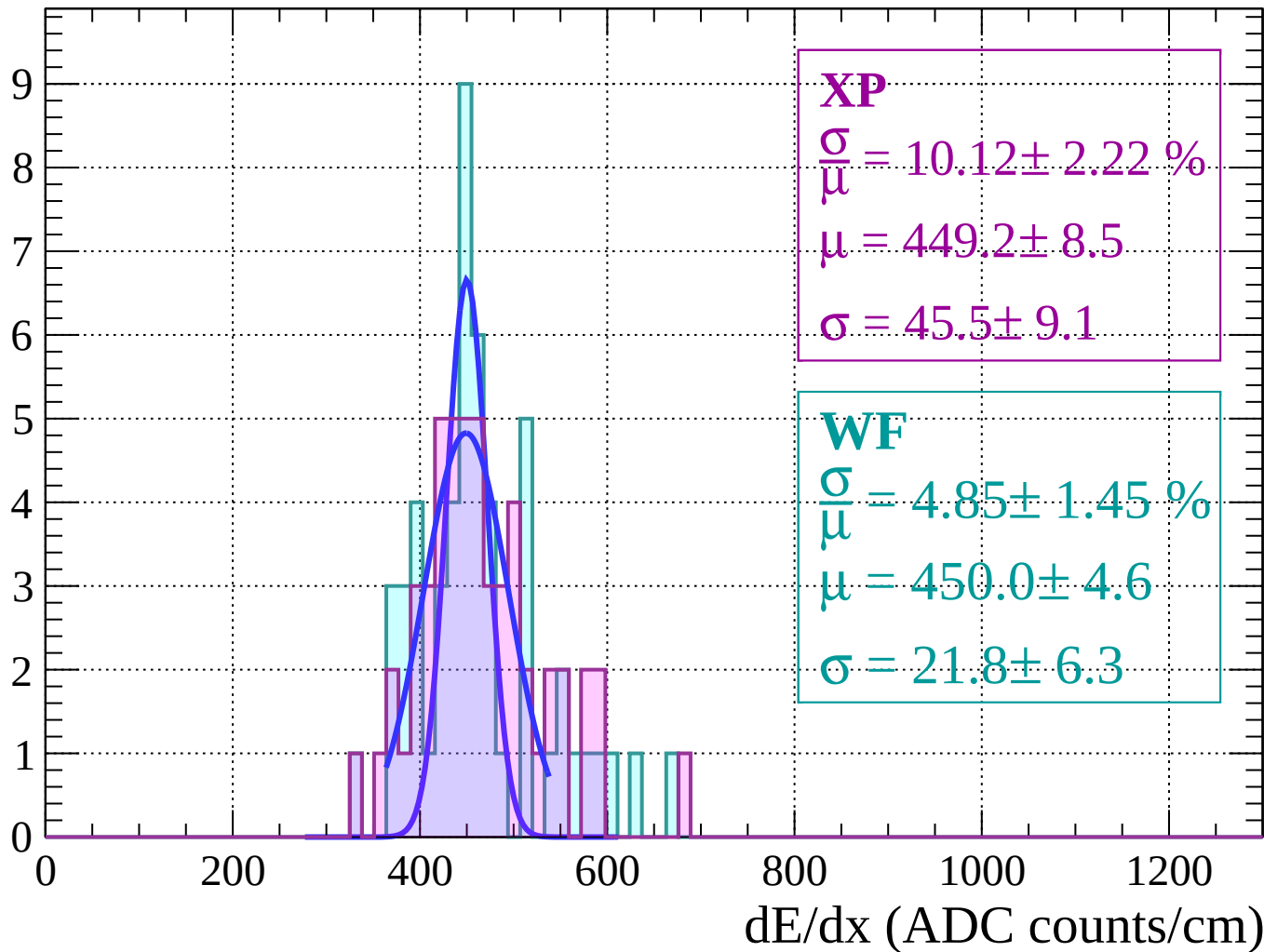
Energy loss | $46 < \theta < 48$

Count



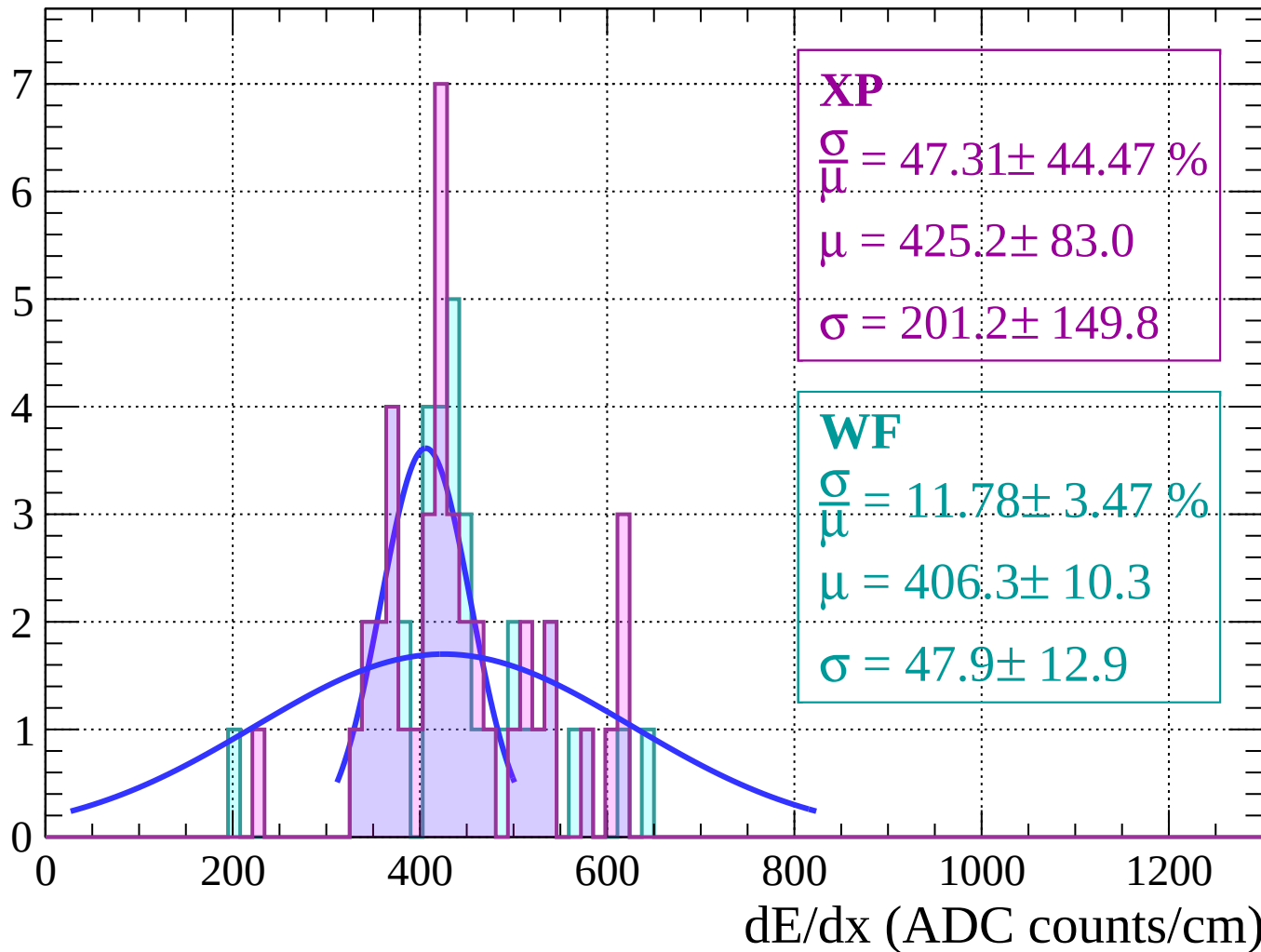
Energy loss | $48 < \theta < 50$

Count



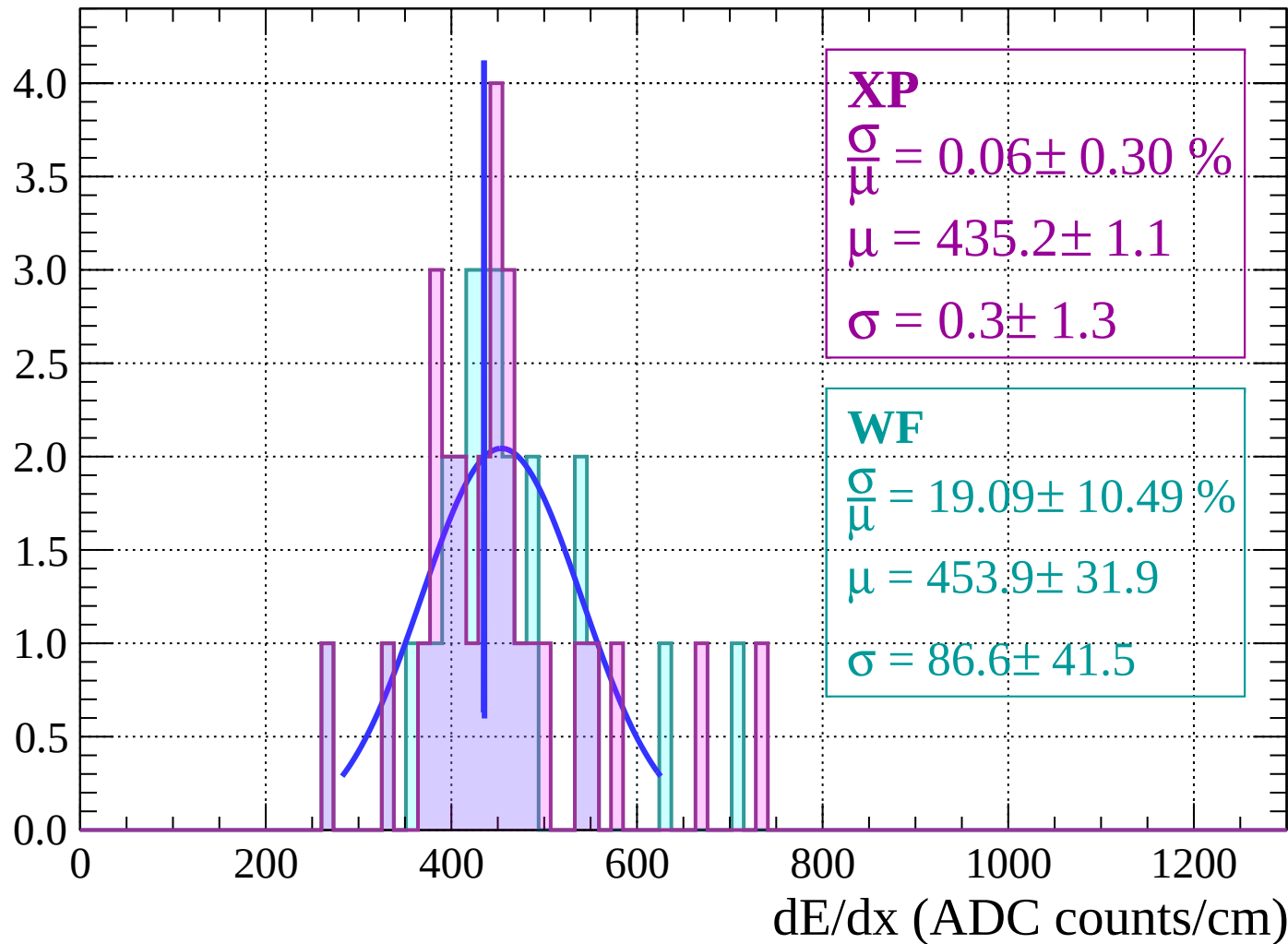
Energy loss | $50 < \theta < 52$

Count



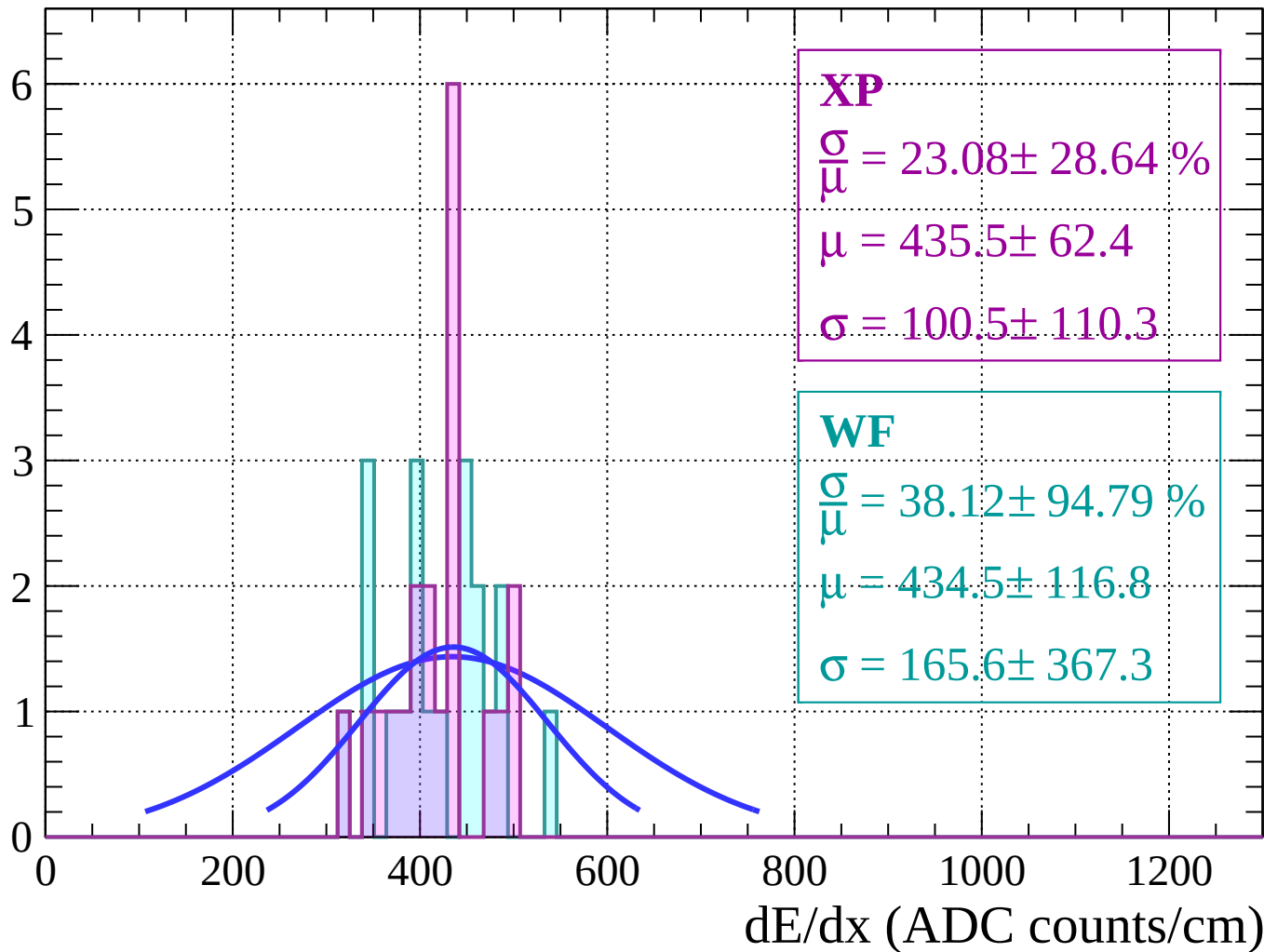
Energy loss | $52 < \theta < 54$

Count



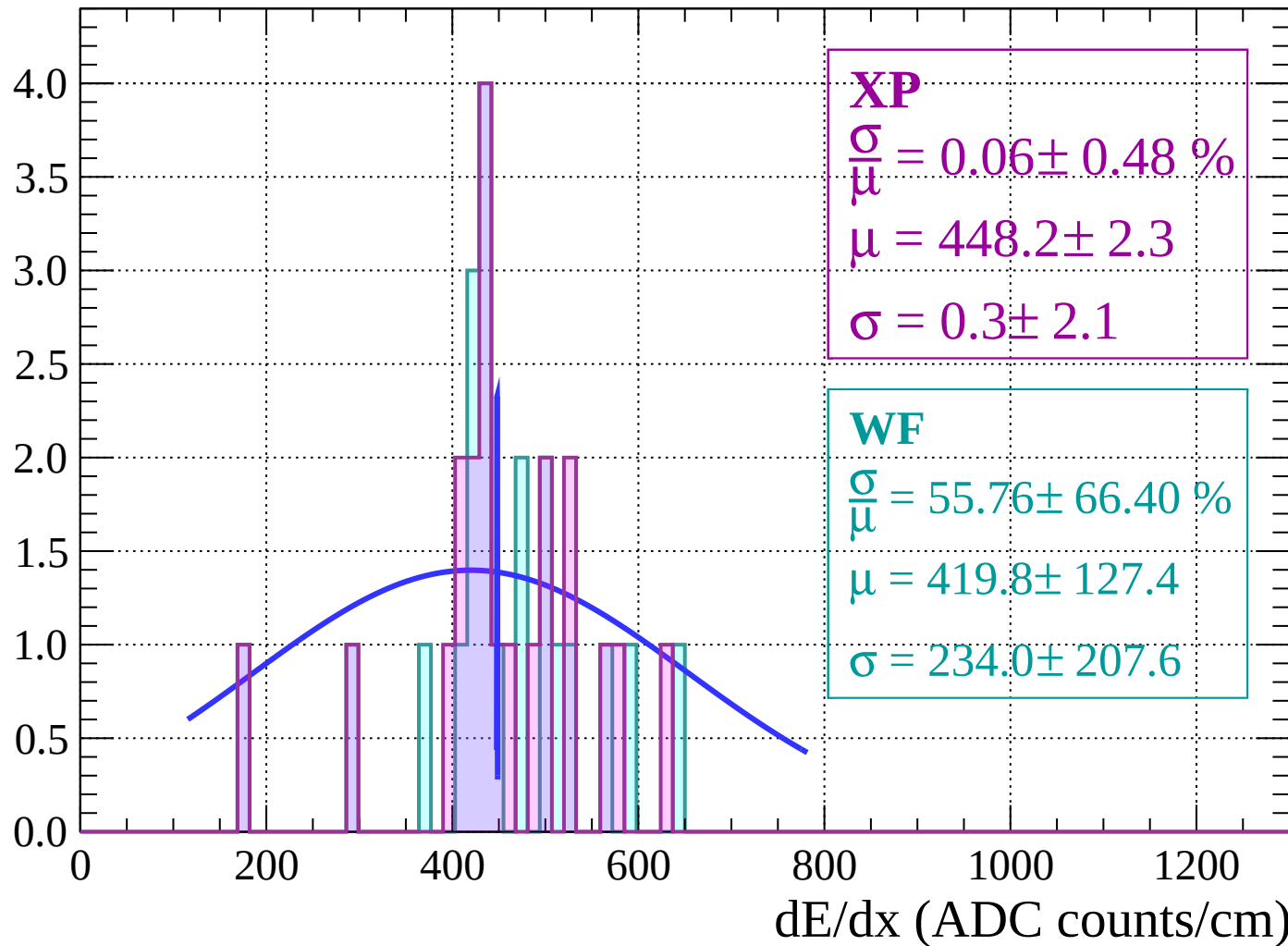
Energy loss | $54 < \theta < 56$

Count



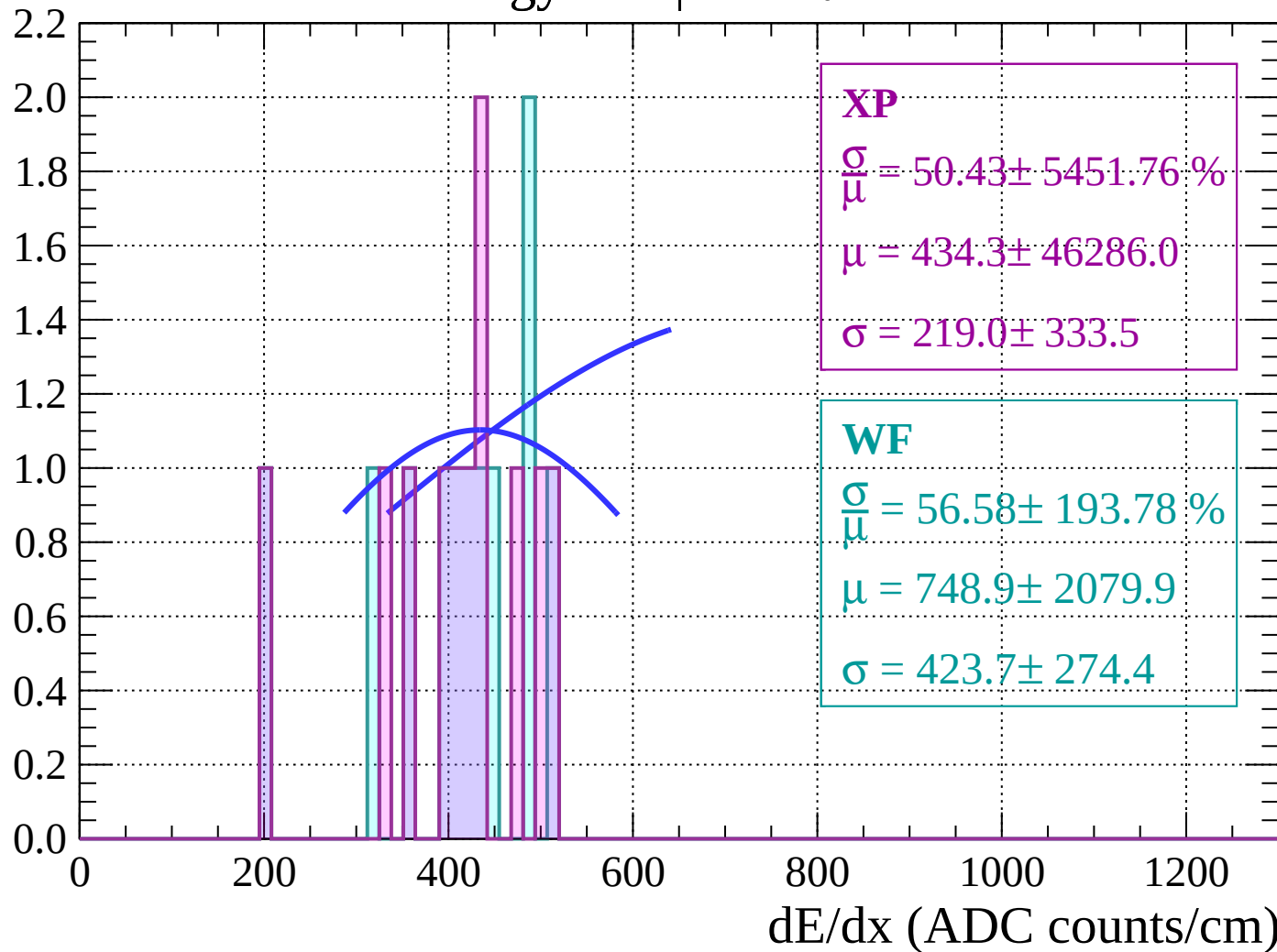
Energy loss | $56 < \theta < 58$

Count



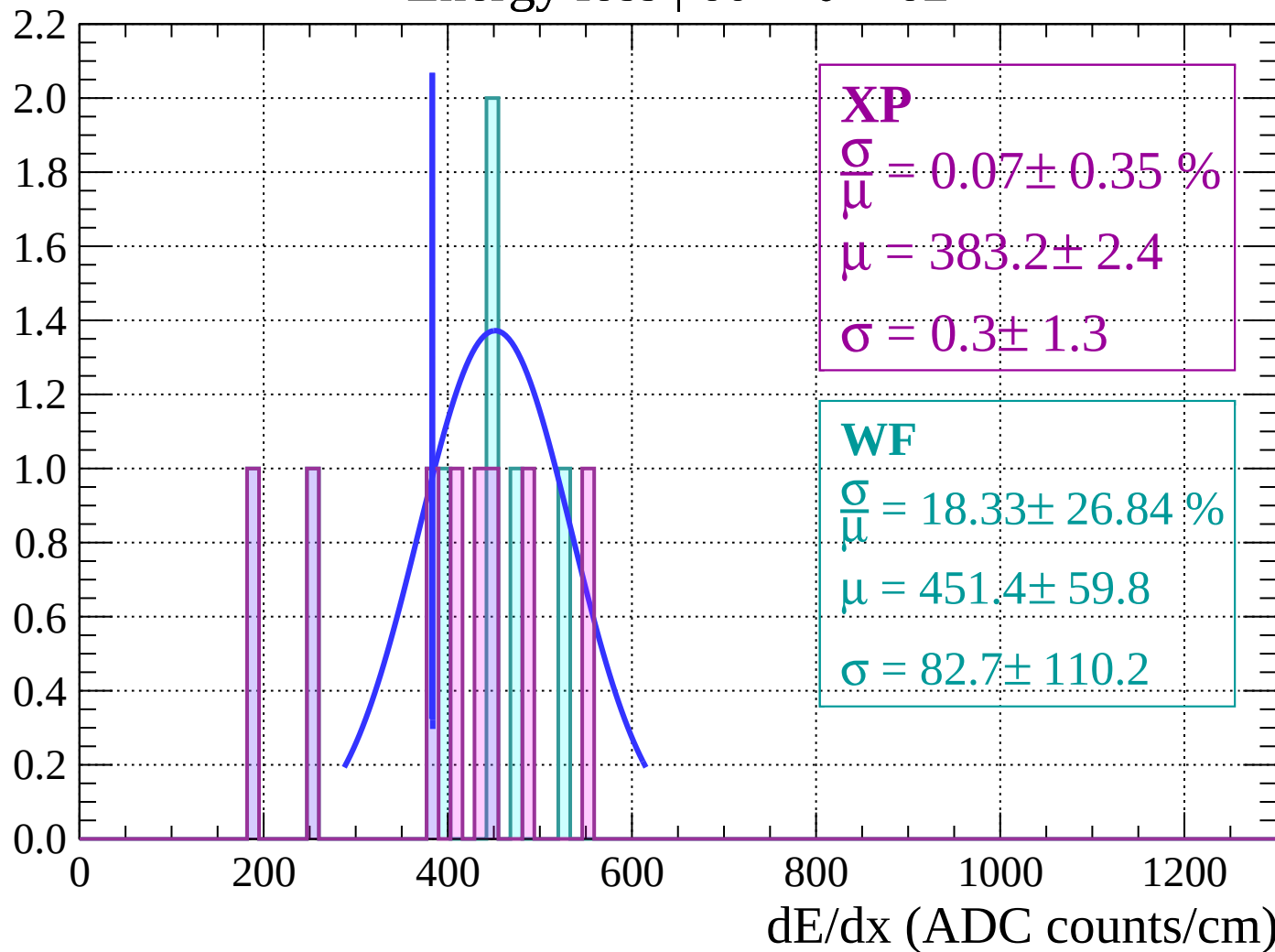
Energy loss | $58 < \theta < 60$

Count



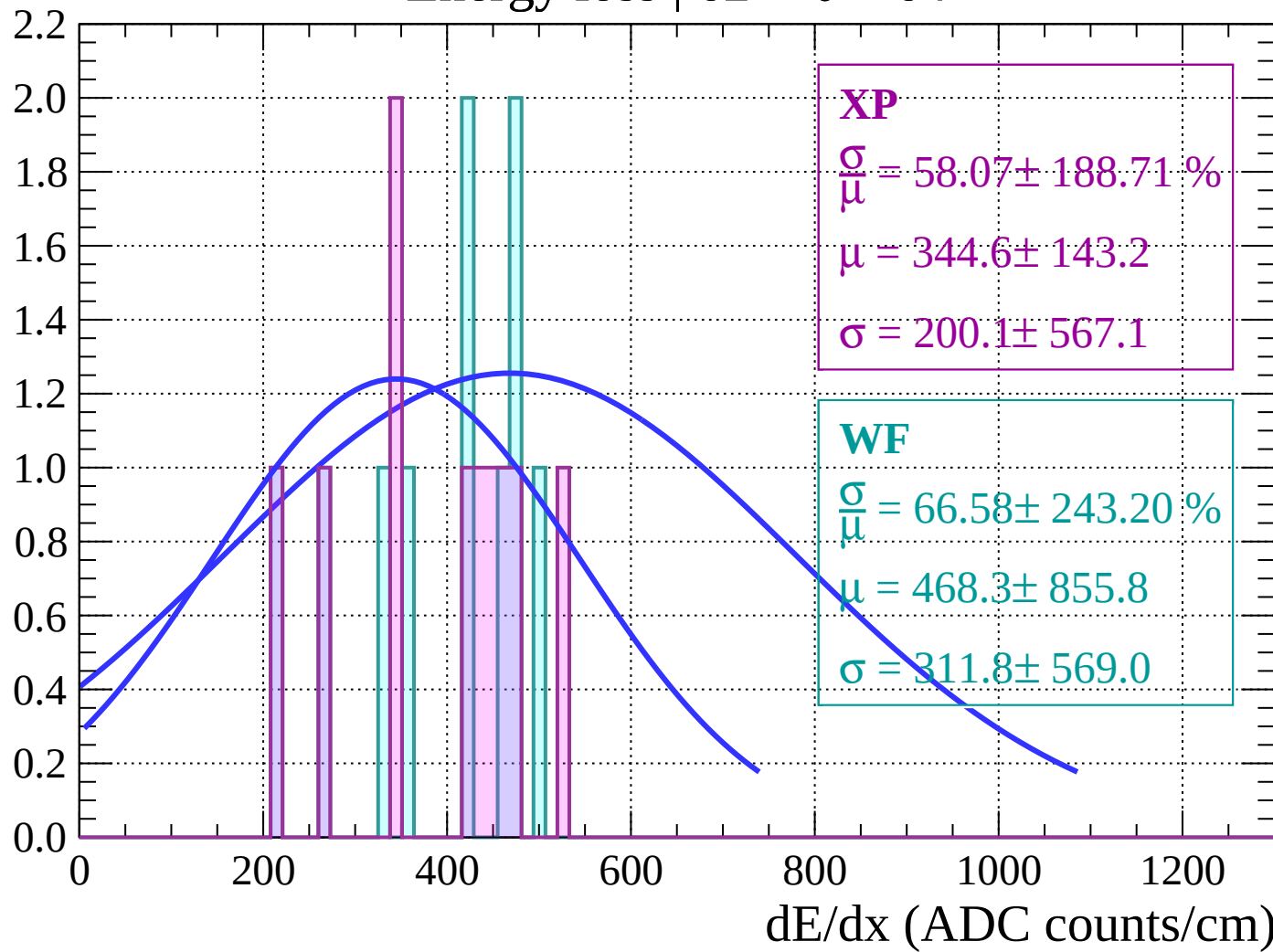
Energy loss | $60 < \theta < 62$

Count



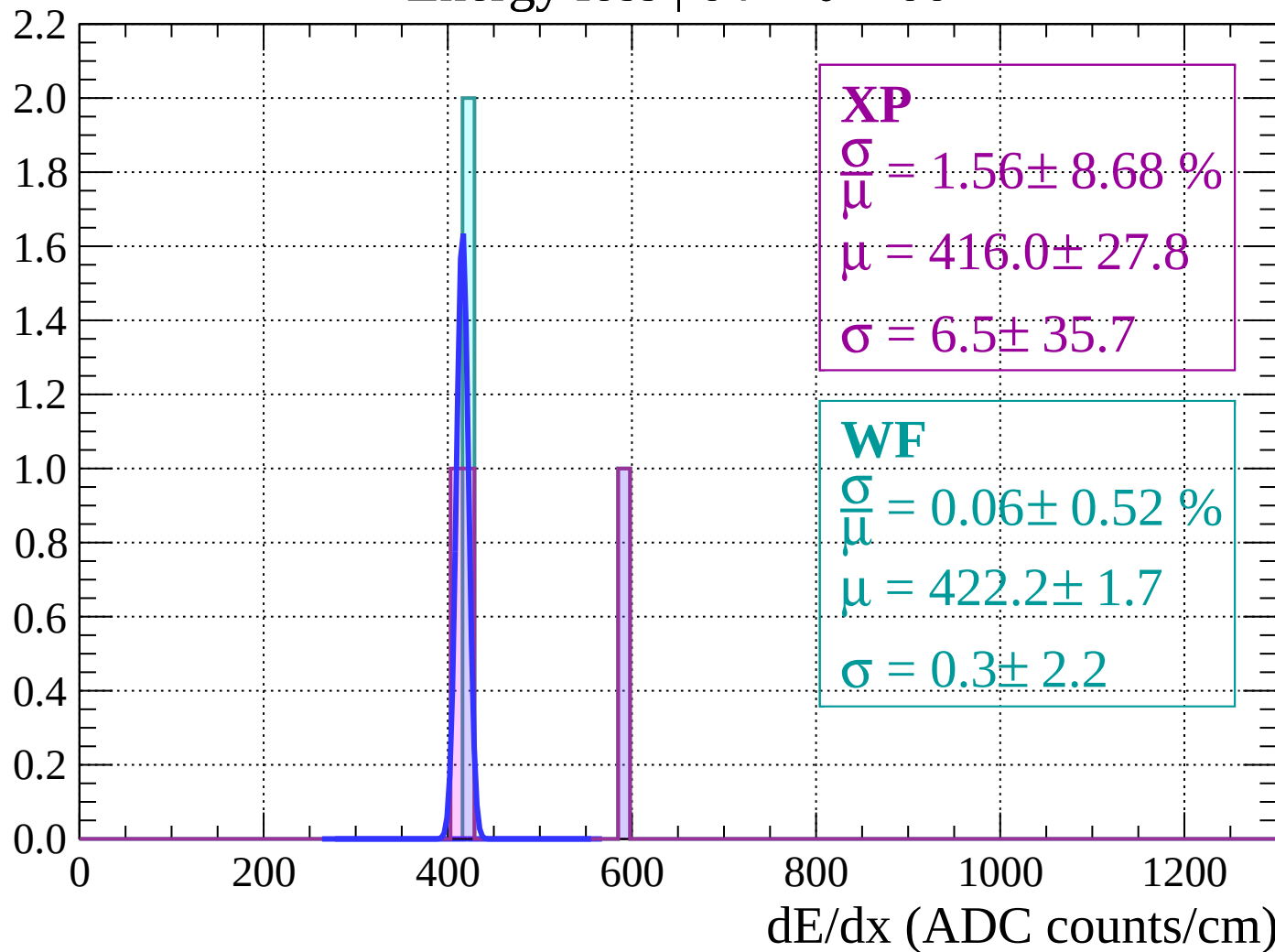
Energy loss | $62 < \theta < 64$

Count



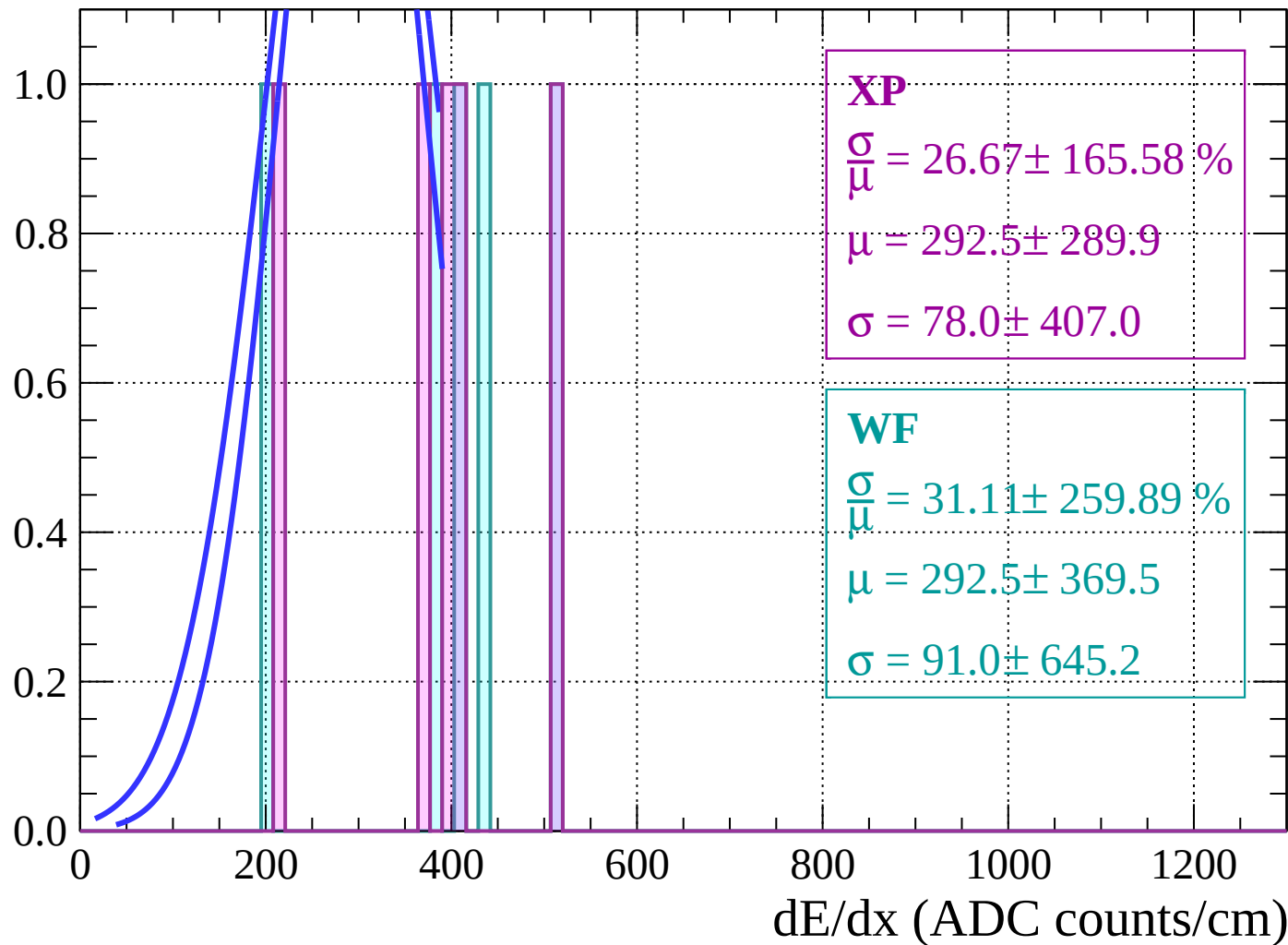
Energy loss | $64 < \theta < 66$

Count



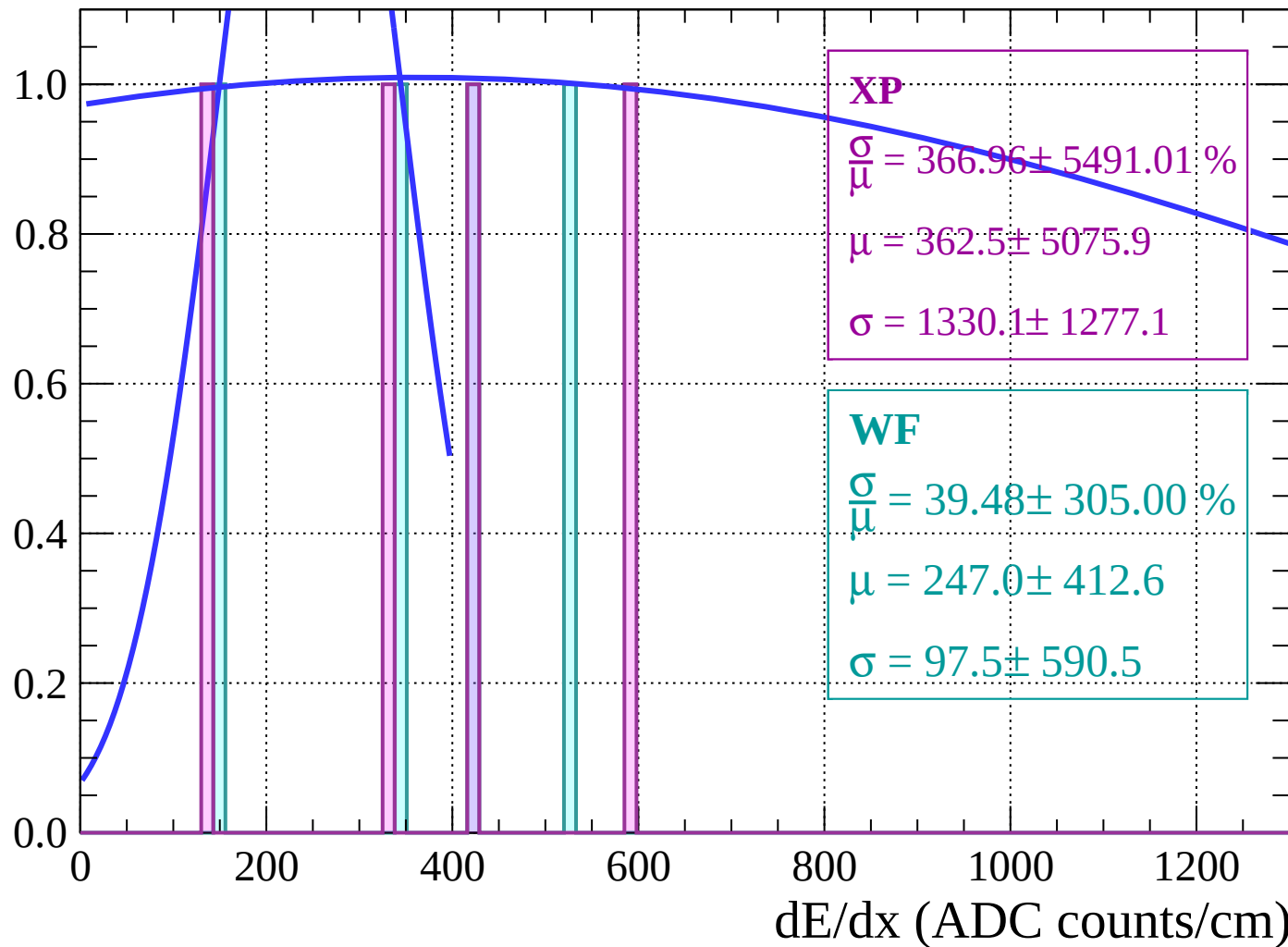
Energy loss | $66 < \theta < 68$

Count



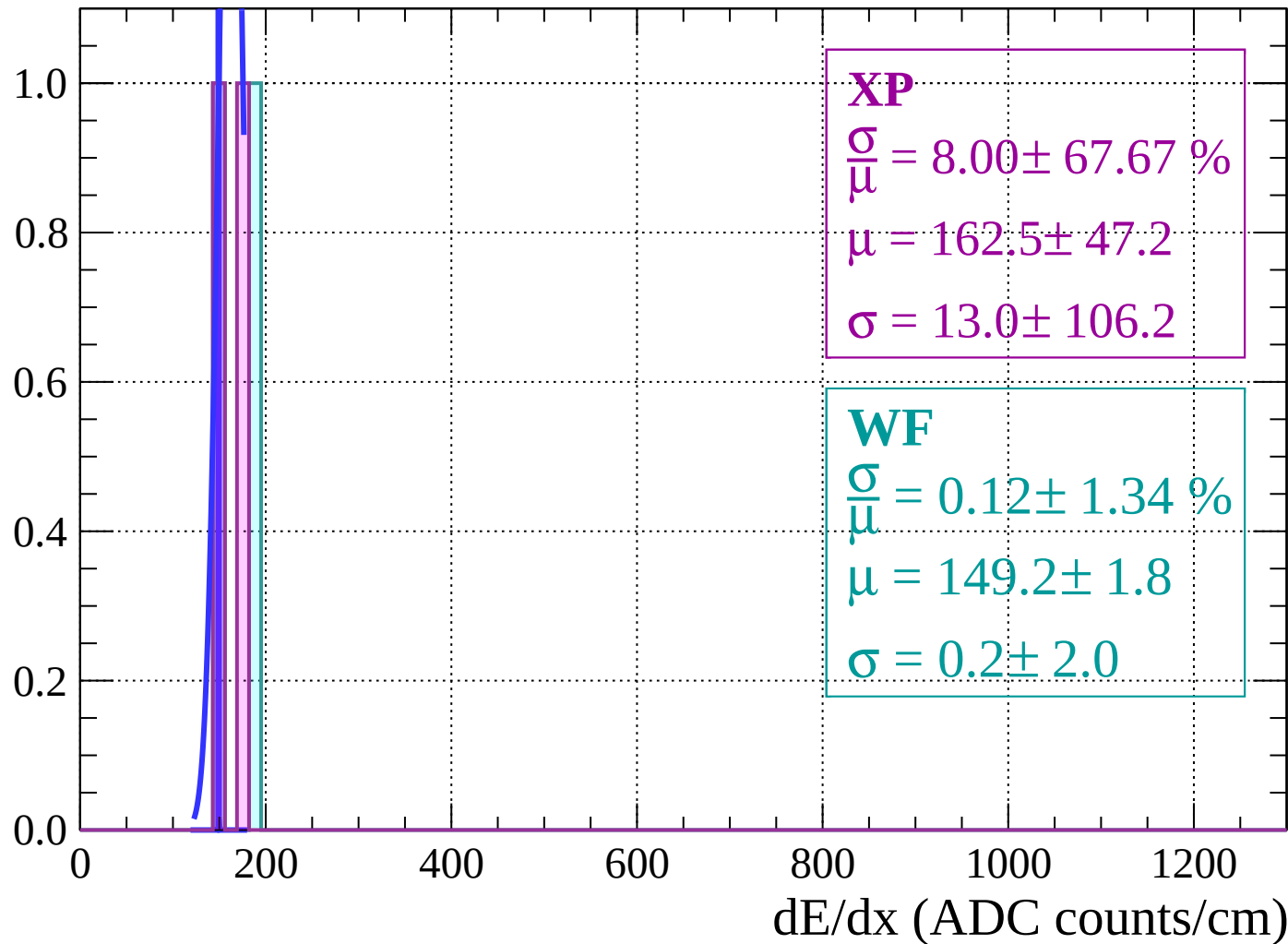
Energy loss | $68 < \theta < 70$

Count



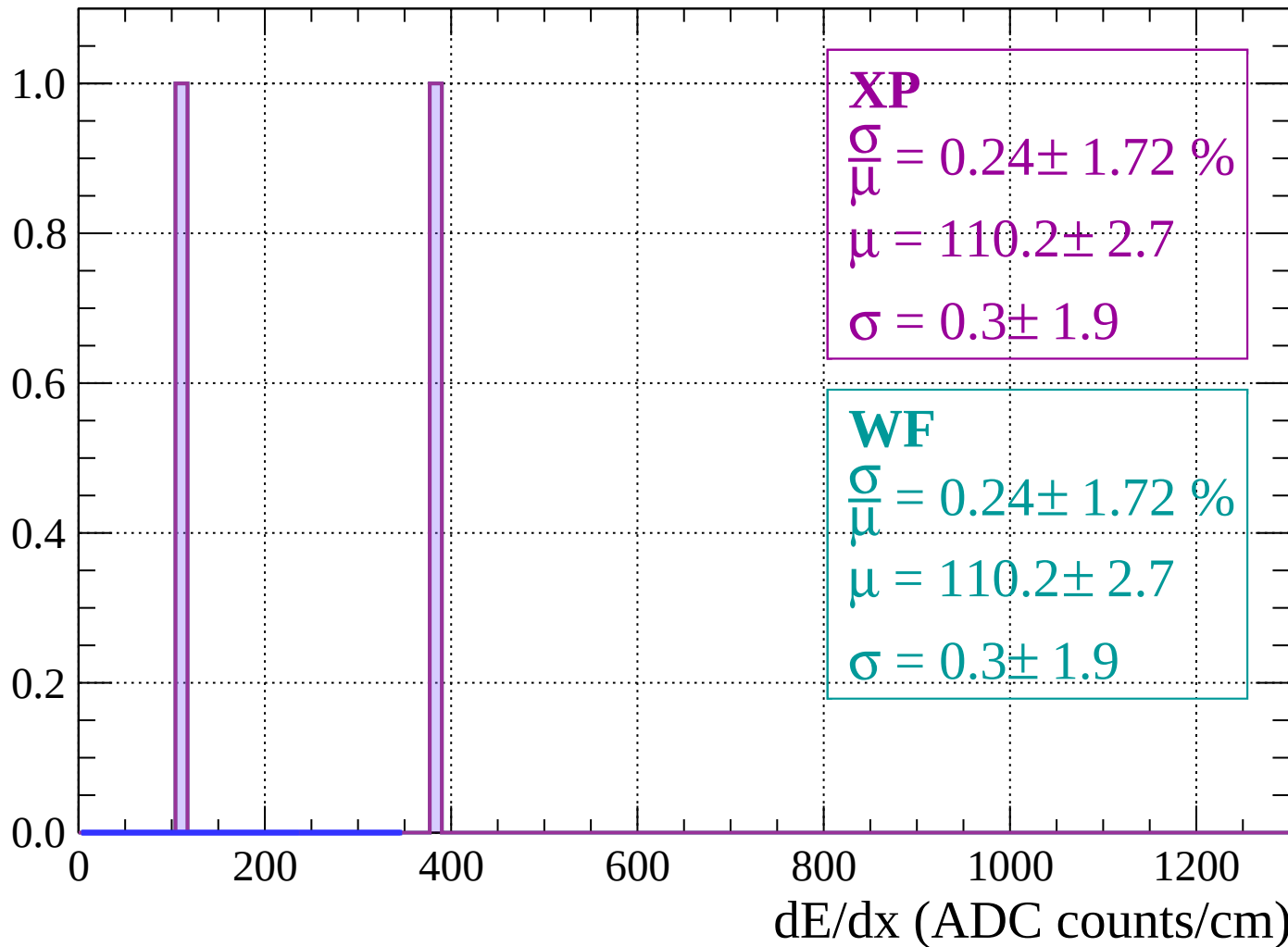
Energy loss | $70 < \theta < 72$

Count



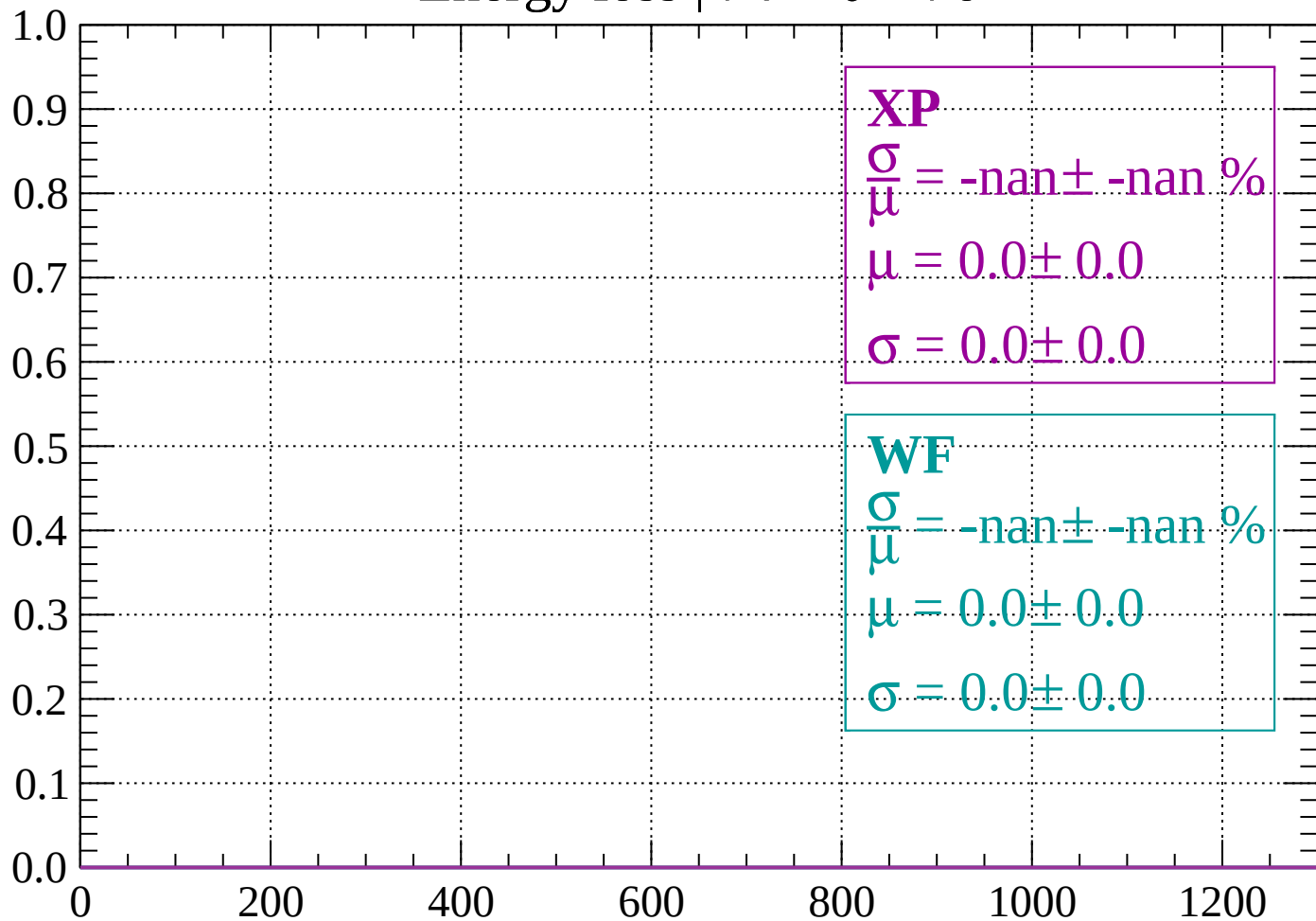
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

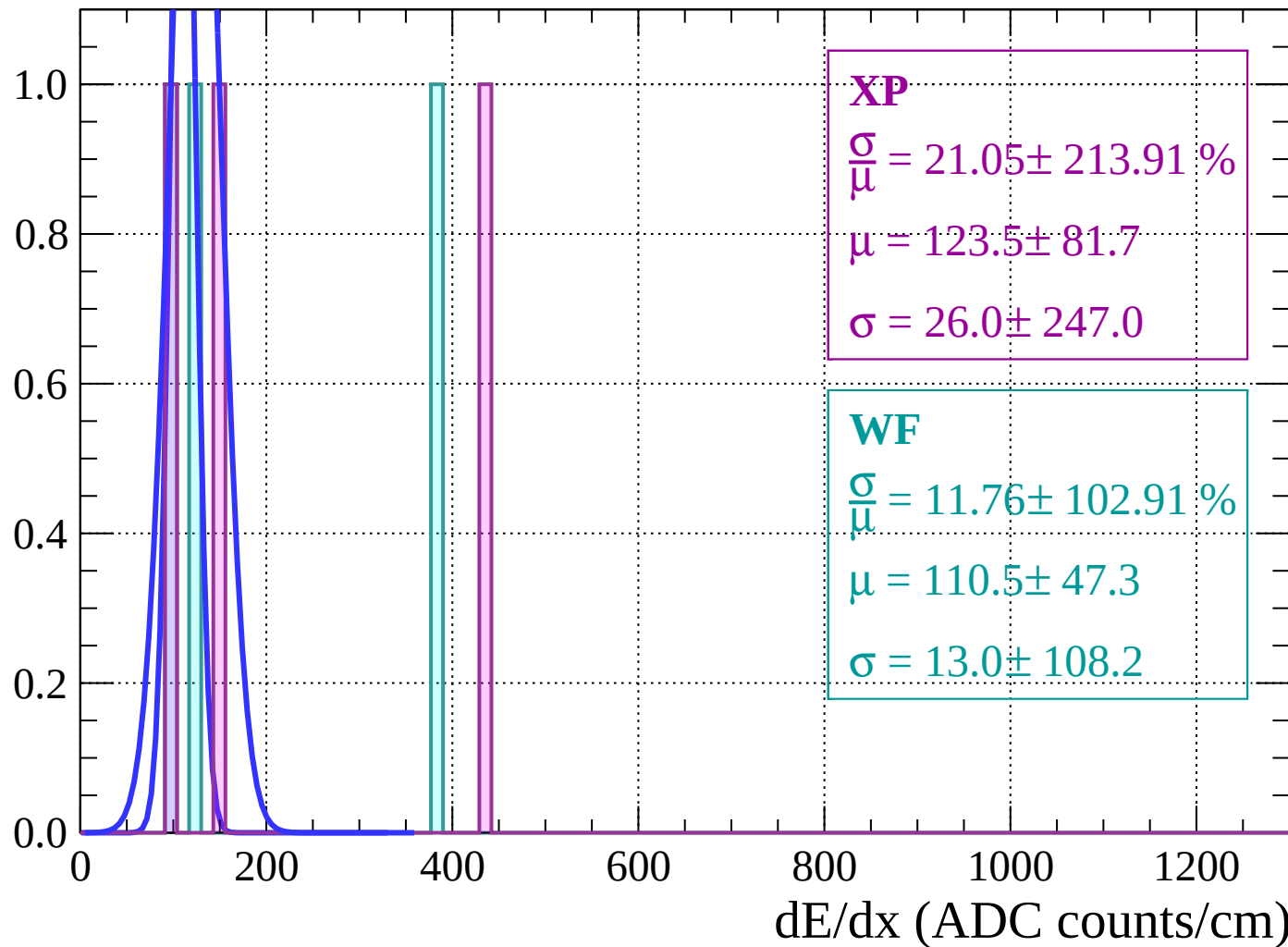
Count



dE/dx (ADC counts/cm)

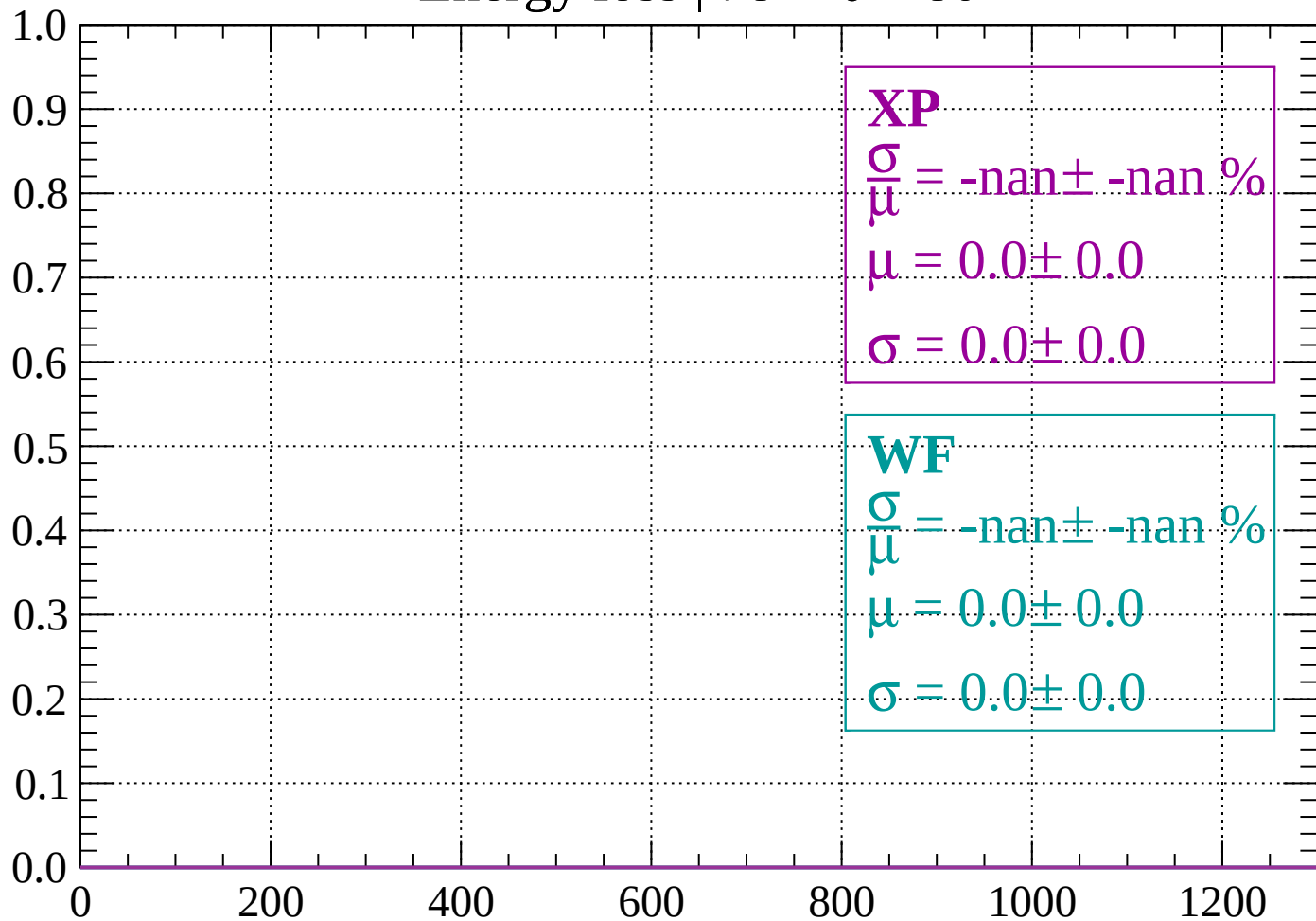
Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

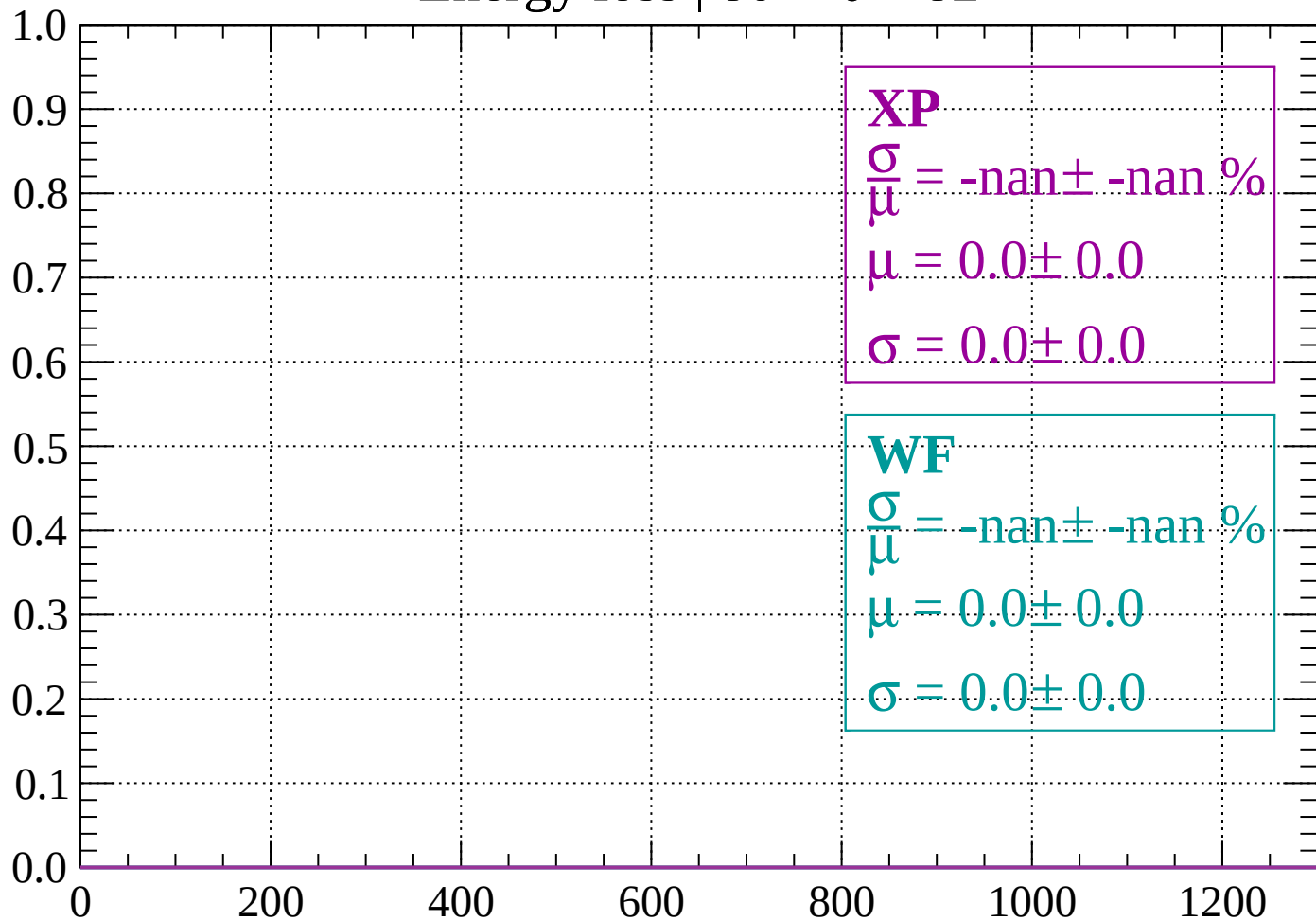
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

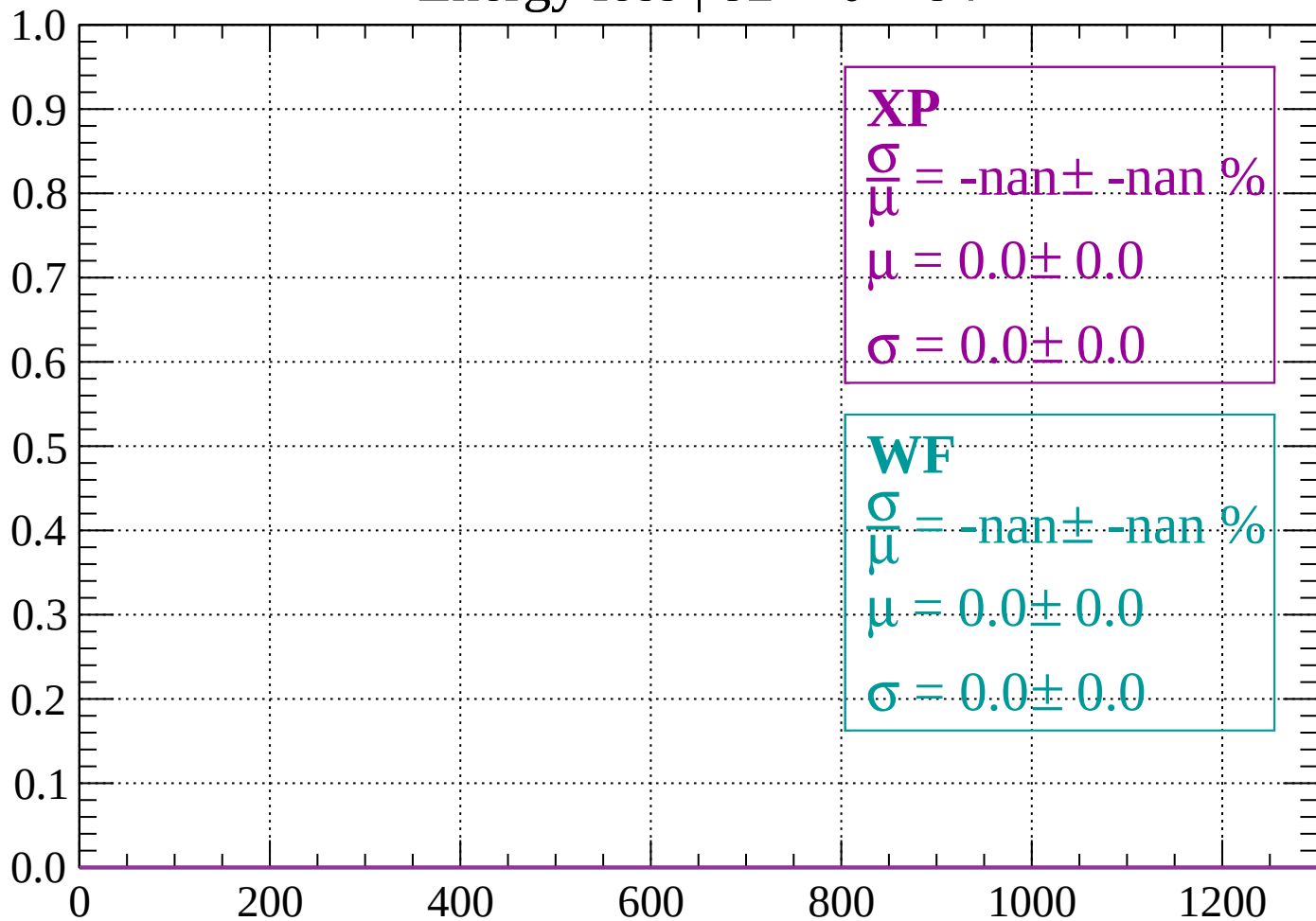
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

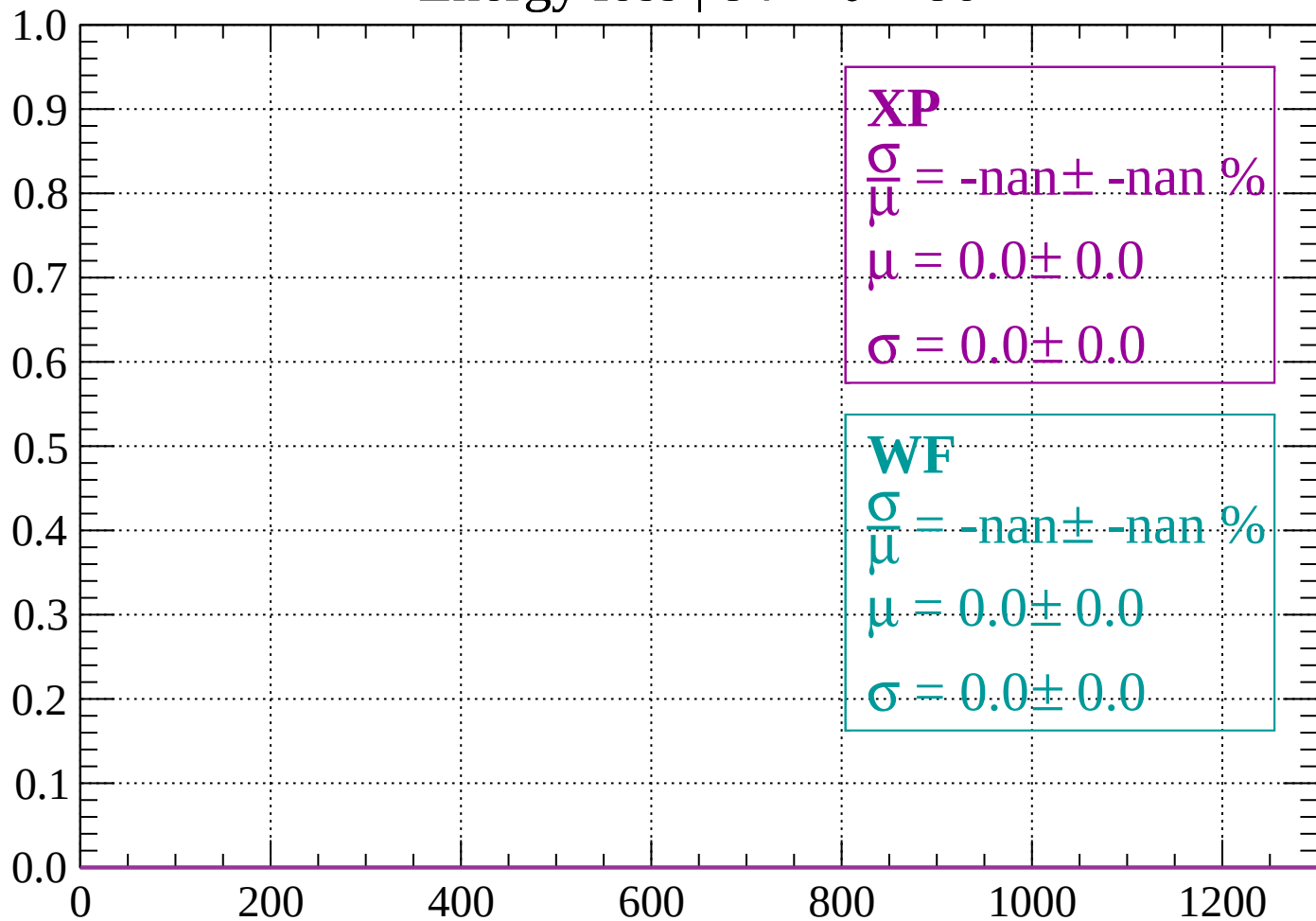
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

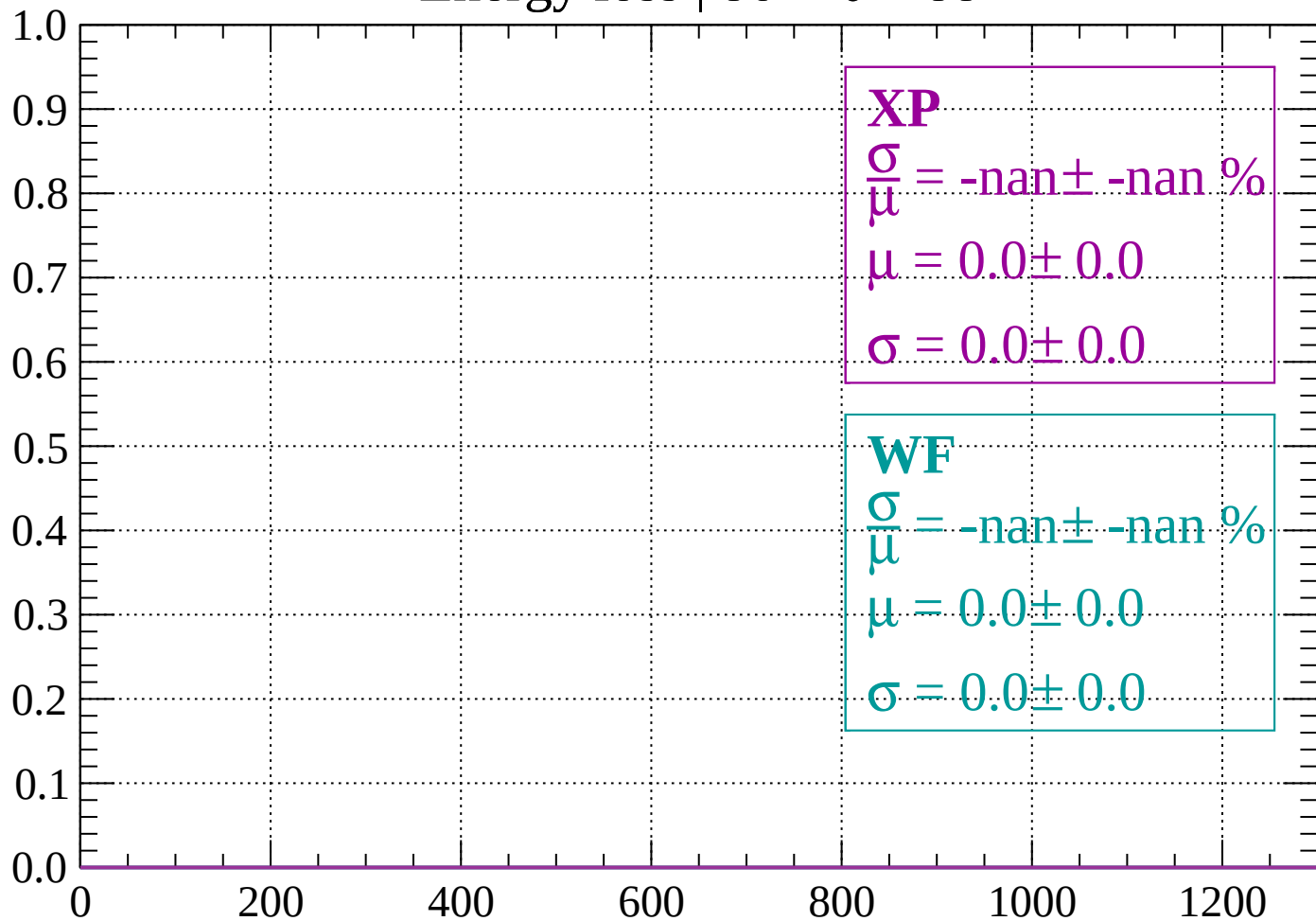
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

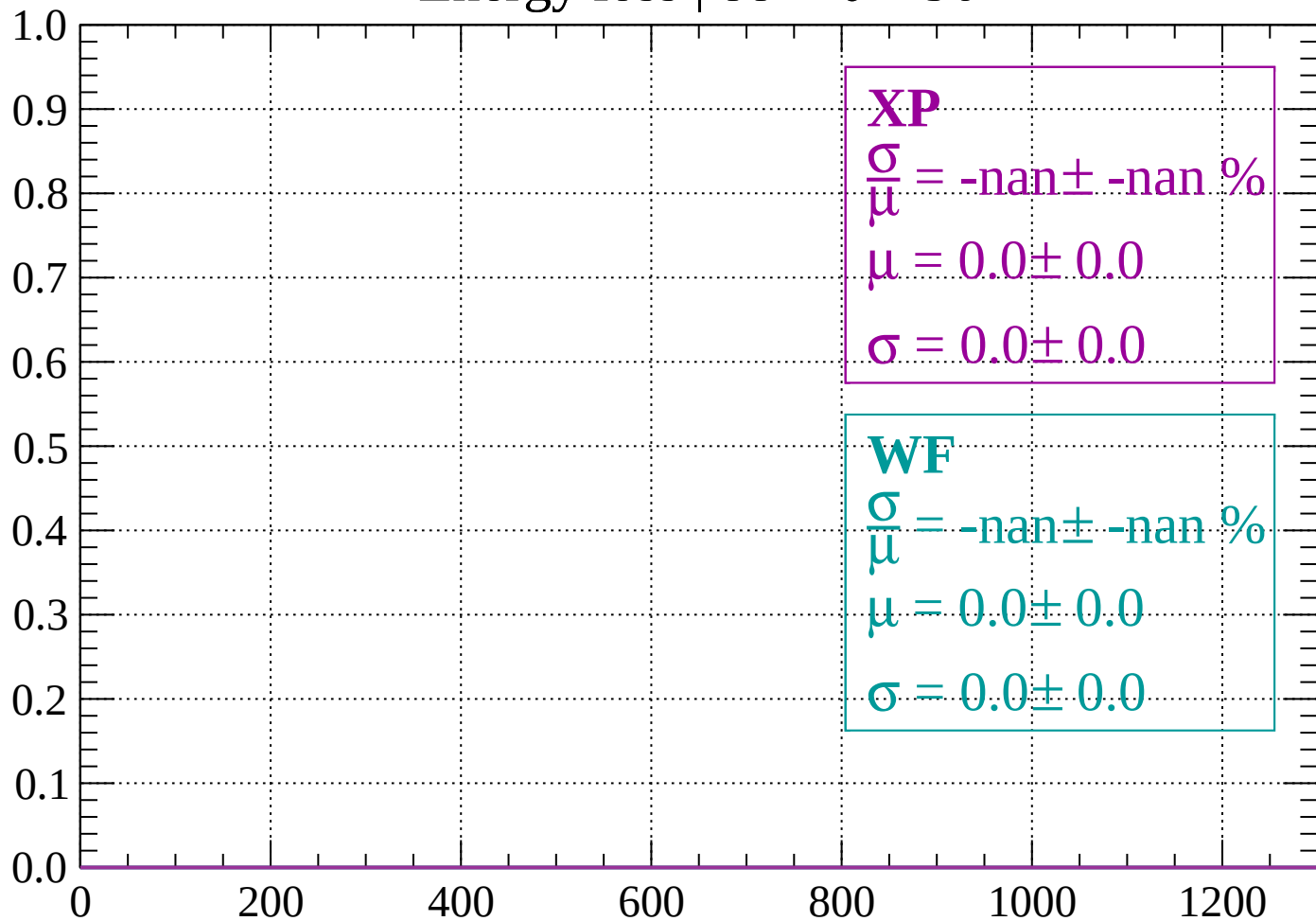
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

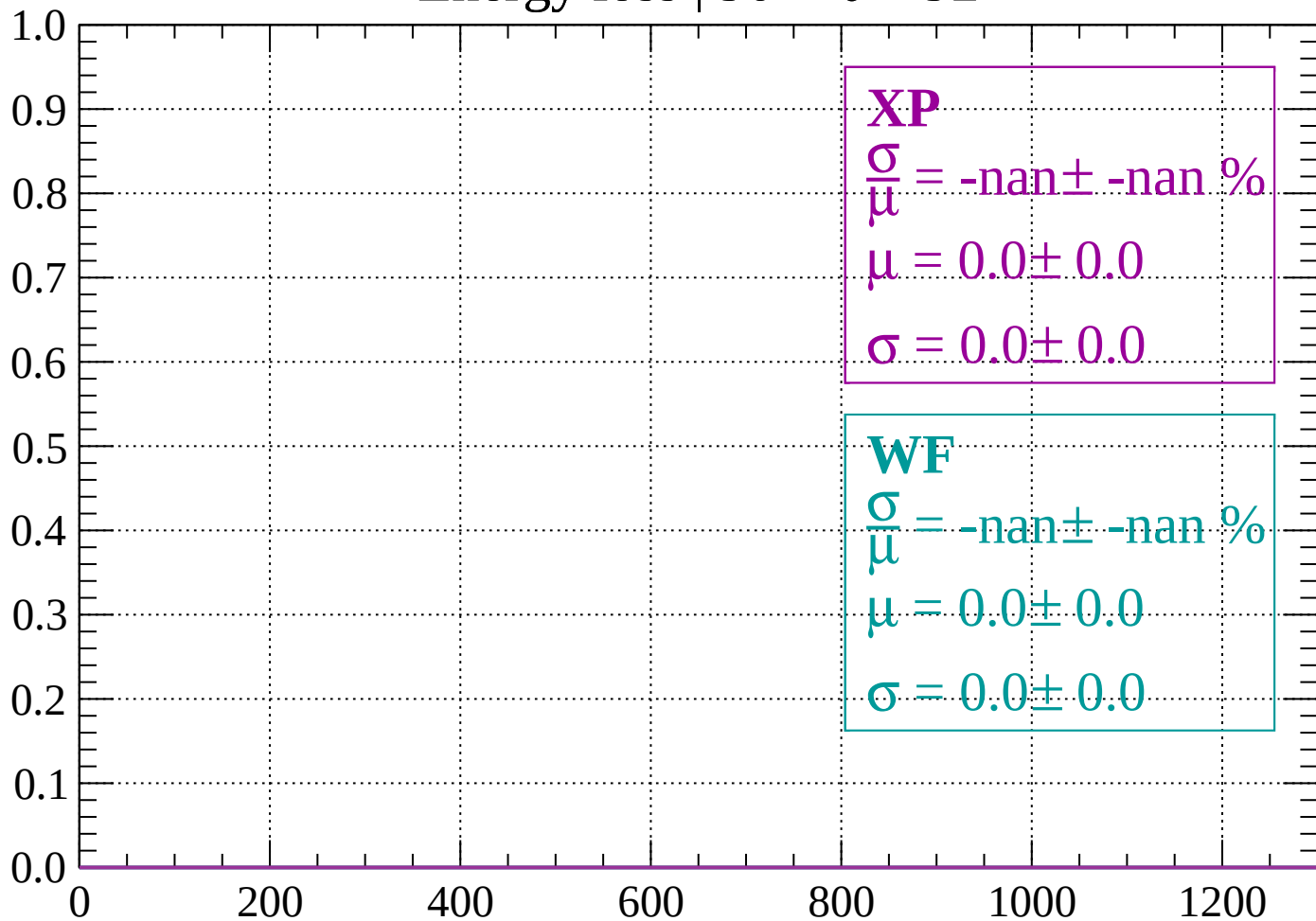
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

